

Skill-IR 语义基线

实际结果复核

完整计划：30 份冻结样例 x 3 次 = 90 次 | 约束契约 v2

运行：baseline-deepseek-v4-flash-max-20260910 | 版本：deepseek-v4-flash-max

传输协议：chat-completions-sse-v1 | 请求传输参数：{"stream": true, "stream_options": {"include_usage": true}}

收到成功响应头后仍未完成的流式请求：5 次。此类 trace 标为 remote_outcome_unknown；试验的 error 是本地传输状态，不能当作模型语义失败。流式与非流式运行按各自身份和协议分别呈现，不混合为同一批次。

本文展示已持久化的模型输出和外置逐事实初审。参考事实已于 2026-09-10 共同确认；本册的模型结果仍待共同复核。结构校验成功、块数变化或一次成功都不是语义正确的证据。

前半册集中显示各样例三次运行的结构状态、修复和 HTTP 统计、差异及逐事实初审。后半册提供已展示试验的实际 CFG、完整边条件、全部操作与输入输出、三层 constraints。参考原文的文件行号对应冻结语料，完整原文见已确认复核材料。

CFG 使用矢量分页邻接图：所有源节点和出边均保留；右框通过真实目标 ID 跨行 / 跨页引用。图不推断连边。操作数中的资源、上下文键、结果与字面量逐项区分。脚本原文仍存于 analysis.json，图和本册不会执行脚本或触发工具。

v1 到 v2 的三层约束字段只属于已确认表达契约迁移；本基线与后续 Prompt 使用相同契约，字段新增不计作优化收益。开发集 22 份、保留集 8 份；人工初审会保留无法判断和原文未决边界。

结构状态计数：{"complete": 85, "error": 5}

逐事实初审计数：冲突 12；遗漏 5；部分保留 119；保留 705；无法判断 50

运行身份与模型返回

有效请求配置来自本批次 `experiment.json`；返回模型名称来自已提交 `trial` 的 `trace`。请求名称和返回名称分别保留，不据名称差异推断等价模型、版本映射或服务器实现。

字段	本批次记录
版本 / 请求模型	deepseek-v4-flash-max / deepseek-v4-flash
Endpoint	https://api.deepseek.com/chat/completions
reasoning_effort	max
读取超时 / HTTP 额外重试	180000 ms / 2
结构修复上限	3

实际返回模型名称

HTTP 返回名称	尝试记录数
deepseek-flash	104
未返回 / 未记录 (null)	0

计数单位是 HTTP 尝试记录，包含结构修复所发请求；不是样例数或三次重复数。缺少返回名称保持未记录，不能视为使用了请求模型。每次生成及 HTTP 的原始名称见试验明细。

存在与请求字符串不同的返回名称：deepseek-flash。本报告不静默改写任一名称。

服务与模型变更来源

变更记录: deepseek-official-v4-flash-max-v1 | 授权日期: 2026-09-10 | 继承旧 trial: 0

这是独立服务/模型基线。与旧批次的比较同时包含模型和服务变化，不能当作仅切换传输方式的收益，也不能合并旧批次补齐三次重复。

来源记录的输入/Prompt 一致性: True; 生产源码一致性: True。

推理模式记录: reasoning_effort=max; official Chat Completions documentation states thinking is enabled by default. The existing client sends no separate thinking override.

请求与返回名称说明: Official GET /models did not list deepseek-v4-flash, but an exact-name completion succeeded with HTTP 200 and returned model deepseek-flash. Keep requesting the user-selected deepseek-v4-flash and preserve the returned model per HTTP attempt. Do not silently rename or assert that listing proves availability.

预检计数边界: One short authentication/model completion is external to the 90 frozen-input trials. Its 44 known tokens are not baseline usage.

变更清单 SHA256: ba186cbaa3475707dbf5e0286f3c14bb6bc78f77c057208a782800da2c38d400

配置 SHA256: c8f88b8b2de3504d72b0a284828e05fac78bfa8869d2b73d34ea8f1bd2966723

关联旧批次: runs/baseline-sol-max-v1
experiment SHA256: 2b4046560059101370a044655c09afe804e7ce928dcca39e99a342e4fb741294
report SHA256: 4a58fc57714e7464256cadfa97bf79e5359c7778260003bdefd026c1b0412e70

关联旧批次: runs/baseline-sol-max-retry-20260910
experiment SHA256: 3f49f4d562c3350e6199651a04ed06af6290125e1eb2829a0997787bfffffeed0
report SHA256: bae2f9d4e6a08383dce7e370a8715f8484ea50684fc41e353a5c55ca001d8adf

关联旧批次: runs/baseline-sol-max-sse-20260910
experiment SHA256: 45a17ada7443a39285263c2c388c1afb449dd76848a3655f2aa3ba29818d1b9d
report SHA256: 31431a06c375b595c9201c0552d799eb2e2d01c048982deb41cf595547c041d8

目录

运行身份与模型返回	2
N01 条件通知与数据发送 ▪ 编号步骤	6
N02 条件通知与数据发送 ▪ 连贯段落	12
N03 条件通知与数据发送 ▪ 表格	17
N04 条件通知与数据发送 ▪ 跨文件混合	22
N05 条件通知与数据发送 ▪ 用途最小差异	27
N06 条件通知与数据发送 ▪ 语义最小差异	32
Q01 查询调用与参数使用 ▪ 编号步骤	37
Q02 查询调用与参数使用 ▪ 连贯段落	42
Q03 查询调用与参数使用 ▪ 表格	47
Q04 查询调用与参数使用 ▪ 跨文件混合	52
Q05 查询调用与参数使用 ▪ 用途最小差异	57
Q06 查询调用与参数使用 ▪ 语义最小差异	62
F01 工具选择与失败回退 ▪ 编号步骤	67
F02 工具选择与失败回退 ▪ 连贯段落	73
F03 工具选择与失败回退 ▪ 表格	78
F04 工具选择与失败回退 ▪ 跨文件混合	83
F05 工具选择与失败回退 ▪ 用途最小差异	89
F06 工具选择与失败回退 ▪ 语义最小差异	94
D01 多文档处理与交付 ▪ 编号步骤	99
D02 多文档处理与交付 ▪ 连贯段落	104
D03 多文档处理与交付 ▪ 表格	110
D04 多文档处理与交付 ▪ 跨文件混合	116

D05 多文档处理与交付 · 用途最小差异 122

D06 多文档处理与交付 · 语义最小差异 127

R01 pdf 133

R02 playwright 138

R03 gh-fix-ci 143

R04 netlify-deploy 149

R05 linear 156

R06 transcribe 162

附录：90 次提取的实际 CFG 与全部操作 169

N01 条件通知与数据发送 · 编号步骤

N01 | 条件通知与数据发送 | 编号步骤 | 开发集 | zh

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	error
块数	6	6	未记录
边数	7	6	未记录
操作数	13	13	未记录
三层约束条数	19	8	未记录
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	1

三次运行差异与初审意见

第1、2次各以1次生成取得6块CFG，边数分别7、6，无结构修复和HTTP重试。前者逐条检查后分支，后者先筛选再迭代选中集合，均保留真实控制与字段数据流。两次都擅自将处理条数确定为选中记录数。第3次因HTTP200期间本地trace替换WinError5结束，没有接受的CFG，不作语义判定。

两份可判定CFG均保留8条核心通知语义；尾部计数行为存在但未保留计数口径的未知边界，F09部分保留。第3次九条事实均不可评估，不能把本地记录失败归结为模型语义缺失。

已接受 CFG 的精确 JSON 有 2 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 N01 的三次明细。

事实表以外的发现

运行 1 | 诊断承认处理条数口径未定义，却实际执行 count_selected_event_records 并按选中且通知的记录数解释；记录歧义不等于保留歧义。

引用: diagnostics, ir_010, ir_012

运行 2 | ir_010 以 selected_event_records 为计数来源，diagnostics 同样选择选中记录数；声明歧义后继续选定策略，仍未保留原文未知口径。

引用: ir_010, ir_012, diagnostics

逐事实初审（参考事实已确认，结果判定待共同复核）

N01-F01 | 行为

读取用户提供的 `events.json` 作为事件记录来源。

归属: operation / 读取事件记录 | 原文: SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_001 实际读取 <code>events.json</code> 产生 <code>event_records</code> , ir_003 消费该结果逐条处理; 用户提供文件的来源完整保留。 引用: ir_001, ir_003
2	保留	ir_001 从用户 <code>events.json</code> 读取 <code>event_records/result_001</code> , ir_003 消费该结果执行筛选, 没有用上下文集合替换原文件来源。 引用: ir_001, ir_003
3	无法判断	本次在 HTTP 200 流接收期间发生本地 <code>trace.json</code> 原子替换 <code>WinError 5</code> , 未获得完整结束响应或接受的 CFG。日志将其包装为 <code>SSE network failure</code> , 但实际证据是本地落盘错误, 不能归为模型语义失败或官方服务不可用。 引用: 无局部引用

N01-F02 | 条件

选中条件为 `(opted_out != true) AND (urgent == true OR value >= 100)`, 退出订阅限制适用于两个可选分支。

归属: block / 记录筛选 | 原文: SKILL.md:L9-L9; SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	ir_006 从当前记录的 <code>opted_out</code> 、 <code>urgent</code> 、 <code>value</code> 结果计算 <code>should_notify</code> , ir_007 消费此判定; edge_004 明确 <code>opted_out</code> 非 <code>true</code> 且 (<code>urgent</code> 为 <code>true</code> 或 <code>value >= 100</code>), edge_005 绕过发送, 复合条件实际决定控制流。 引用: ir_005, ir_006, ir_007, edge_004, edge_005
2	保留	ir_003 将复合条件和 <code>urgent</code> 例外绑定实际集合筛选, 产出 <code>selected_event_records/result_002</code> ; ir_005 只迭代该集合, ir_006 与 edge_003 仅在存在下一条选中记录时进入发送。先筛后逐条循环的等价表达保留条件效力。 引用: ir_003, ir_003.constraints, ir_005, ir_006, edge_003, edge_004
3	无法判断	本次在 HTTP 200 流接收期间发生本地 <code>trace.json</code> 原子替换 <code>WinError 5</code> , 未获得完整结束响应或接受的 CFG。日志将其包装为 <code>SSE network failure</code> , 但实际证据是本地落盘错误, 不能归为模型语义失败或官方服务不可用。 引用: 无局部引用

N01-F03 | 约束

紧急事件可越过数值门槛，但仍受退出订阅限制。

归属：block / 记录筛选 | 原文：SKILL.md:L9-L9; SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	urgent 只豁免数值、不豁免 opted_out 的原文保存在筛选块及 ir_006 约束，edge_004 的括号条件实际维持该例外的局部作用域。 引用：block_003.constraints, ir_006.constraints, edge_004
2	保留	urgent 只豁免数值门槛而不豁免 opted_out 的原句附于实际筛选操作 ir_003，局部作用域明确，没有把例外提升为跳过退出订阅的全局规则。 引用：ir_003.constraints, block_002
3	无法判断	本次在 HTTP 200 流接收期间发生本地 trace.json 原子替换 WinError 5，未获得完整结束响应或接受的 CFG。日志将其包装为 SSE network failure，但实际证据是本地落盘错误，不能归为模型语义失败或官方服务不可用。 引用：无局部引用

N01-F04 | 行为

每条选中记录执行一次 notify.send，接收对象来自该记录的 recipient。

归属：operation / 发送通知 | 原文：SKILL.md:L11-L11

运行	初审	理由与实际输出引用
1	保留	ir_003 逐条产生 current_event_record，ir_005 从当前记录提取 recipient 结果 result_007，ir_008 引用该结果调用 notify.send；选中进入发送，edge_006 返回迭代，无重复发送支路。 引用：ir_003, ir_005, ir_008, edge_004, edge_006
2	保留	ir_005 从选中集合逐条取得 current_event_record，ir_007 从该记录提取 recipient/result_005，ir_008 引用此结果调用 notify.send，edge_005 逐次返回迭代；接收对象来源和每条一次均落实到图。 引用：ir_005, ir_007, ir_008, edge_003, edge_005
3	无法判断	本次在 HTTP 200 流接收期间发生本地 trace.json 原子替换 WinError 5，未获得完整结束响应或接受的 CFG。日志将其包装为 SSE network failure，但实际证据是本地落盘错误，不能归为模型语义失败或官方服务不可用。 引用：无局部引用

N01-F05 | 数据流

`notify.send.body` 直接使用当前选中记录的 `summary` 原值。

归属: `operation` / 发送通知 | 原文: SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	<code>ir_005</code> 的 <code>summary</code> 原值约束绑定实际输出 <code>notification_body/result_008</code> , <code>ir_008</code> 消费该输出作为通知正文, 两个操作之间无摘要生成或转换。 引用: <code>ir_005</code> , <code>ir_005.constraints</code> , <code>ir_008</code>
2	保留	<code>ir_007</code> 实际提取 <code>summary/result_006</code> , <code>ir_008</code> 消费同一结果, 调用约束明确 <code>body</code> 取 <code>summary</code> 原值; 没有从摘要名称推导额外生成步骤。 引用: <code>ir_007</code> , <code>ir_008</code> , <code>ir_008.constraints</code>
3	无法判断	本次在 HTTP 200 流接收期间发生本地 <code>trace.json</code> 原子替换 <code>WinError 5</code> , 未获得完整结束响应或接受的 CFG。日志将其包装为 <code>SSE network failure</code> , 但实际证据是本地落盘错误, 不能归为模型语义失败或官方服务不可用。 引用: 无局部引用

N01-F06 | 条件

未选中的记录没有通知发送行为。

归属: `block` / 未选中分支 | 原文: SKILL.md:L13-L13

运行	初审	理由与实际输出引用
1	保留	<code>edge_005</code> 在 <code>not selected</code> 情况直接返回迭代块, 绕过唯一的 <code>notify.send</code> 操作 <code>ir_008</code> ; 未选中记录没有发送行为。 引用: <code>ir_007</code> , <code>ir_008</code> , <code>edge_004</code> , <code>edge_005</code>
2	保留	<code>ir_003</code> 先筛选得到 <code>selected_event_records</code> , 只有此集合进入 <code>ir_005/ir_008</code> 的循环; 无下一条选中记录时 <code>edge_004</code> 转向计数, 未选中记录不进入唯一通知块。 引用: <code>ir_003</code> , <code>ir_005</code> , <code>ir_008</code> , <code>edge_003</code> , <code>edge_004</code>
3	无法判断	本次在 HTTP 200 流接收期间发生本地 <code>trace.json</code> 原子替换 <code>WinError 5</code> , 未获得完整结束响应或接受的 CFG。日志将其包装为 <code>SSE network failure</code> , 但实际证据是本地落盘错误, 不能归为模型语义失败或官方服务不可用。 引用: 无局部引用

N01-F07 | 约束

整份流程不得把 `access_token` 字段发送给任何接收对象。

归属：skill / 整份通知 Skill | 原文：SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	禁止向任何接收对象发送 <code>access_token</code> 的原句位于 <code>skill.constraints</code> ；通知只引用 <code>recipient</code> 和 <code>summary</code> 对应结果，没有 <code>token</code> 操作数或另设接收对象。 引用：skill.constraints, ir_005, ir_008
2	保留	<code>skill.constraints</code> 完整保留禁止 <code>access_token</code> 发送给任何接收对象的全局原文；实际通知输入仅为工具、 <code>recipient</code> 和 <code>summary</code> ，未出现 <code>token</code> 数据流。 引用：skill.constraints, ir_007, ir_008
3	无法判断	本次在 HTTP 200 流接收期间发生本地 <code>trace.json</code> 原子替换 <code>WinError 5</code> ，未获得完整结束响应或接受的 CFG。日志将其包装为 <code>SSE network failure</code> ，但实际证据是本地落盘错误，不能归为模型语义失败或官方服务不可用。 引用：无局部引用

N01-F08 | 不得凭空增加

不能从字段直接取值要求增加摘要生成、内容改写或格式转换操作。

归属：operation / 发送通知 | 原文：SKILL.md:L15-L15

运行	初审	理由与实际输出引用
1	保留	整图只含事件读取、字段提取、条件判断、通知、计数与本地写入，没有摘要生成、改写或额外格式转换； <code>ir_005</code> 约束再次明确原字段直接取值。 引用：skill.constraints, ir_005, ir_008, ir_010, ir_012
2	保留	完整图没有摘要生成、内容改写或格式转换；字段提取结果直接送入通知， <code>ir_008</code> 约束保留原文对额外操作的否定。 引用：ir_007, ir_008, ir_008.constraints
3	无法判断	本次在 HTTP 200 流接收期间发生本地 <code>trace.json</code> 原子替换 <code>WinError 5</code> ，未获得完整结束响应或接受的 CFG。日志将其包装为 <code>SSE network failure</code> ，但实际证据是本地落盘错误，不能归为模型语义失败或官方服务不可用。 引用：无局部引用

N01-F09 | 行为

处理结束后单独将处理条数写入本地 `count.txt`，保留它对前序记录处理的依赖。

归属：operation / 记录处理条数 | 原文：SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	部分保留	循环结束 <code>edge_003</code> 才进入计数， <code>ir_012</code> 将 <code>ir_010</code> 的结果写入 <code>count.txt</code> ，前序处理与写入依赖保留；但 <code>ir_010</code> 明确为 <code>count_selected_event_records</code> ， <code>diagnostics</code> 也承认并擅自选择选中且已调用通知的记录数，未保留源文未定义的统计口径。 引用： <code>ir_010</code> ， <code>ir_012</code> ， <code>edge_003</code> ， <code>edge_007</code> ， <code>diagnostics</code>
2	部分保留	迭代结束 <code>edge_004</code> 才进入 <code>ir_010</code> 计数，再由 <code>ir_012</code> 写 <code>count.txt</code> ，尾部行为和前序依赖存在；但计数输入实际固定为 <code>selected_event_records/result_002</code> ， <code>diagnostics</code> 也明确采用选中记录数，越过了原文未定义的口径。 引用： <code>ir_010</code> ， <code>ir_012</code> ， <code>edge_004</code> ， <code>edge_006</code> ， <code>diagnostics</code>
3	无法判断	本次在 HTTP 200 流接收期间发生本地 <code>trace.json</code> 原子替换 <code>WinError 5</code> ，未获得完整结束响应或接受的 <code>CFG</code> 。日志将其包装为 <code>SSE network failure</code> ，但实际证据是本地落盘错误，不能归为模型语义失败或官方服务不可用。 引用：无局部引用

N02 条件通知与数据发送 · 连贯段落

N02 | 条件通知与数据发送 | 自然语言段落 | 开发集 | zh

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	6	5	5
边数	7	6	6
操作数	13	16	12
三层约束条数	9	10	17
生成尝试	1	2	1
结构修复	0	1	0
HTTP 尝试	1	2	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三次最终CFG分别6/5/5块、7/6/6边，结构修复0/1/0轮、HTTP重试均0。三次均建立逐记录的条件通知、recipient/summary原值数据流和未选中跳过，均额外确定选中记录计数。第3次还将多条局部原句集中到Skill全局，局部约束只留英文译写；本评定只看各次最终接受的CFG。

核心通知行为和实际条件分支稳定保留，三次计数口径均部分保留；第3次紧急例外的含义存在，但原文约束的局部归属不准确。三次结构complete不构成全部语义完整。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 N02 的三次明细。

事实表以外的发现

运行 1 | 将原文未定义的处理条数解释为触发通知的选中记录数；实际 count_selected_event_records 实现该额外策略，而非保留未知。

引用: ir_010, ir_012, diagnostics

运行 2 | ir_014 直接采用 count_selected_records，未给出未知口径诊断，默认为选中条数；实际统计策略超出原文。

引用: ir_014, ir_015, diagnostics

运行 3 | 原先属于筛选/发送/尾部写入的中文要求被集中进 skill.constraints，再在局部译成英文。实际流程大体正确，但原文约束的层级归属不精确；另以选中记录定义处理条数，未保留未知。

引用: skill.constraints, ir_006.constraints, ir_008.constraints, ir_010.constraints, diagnostics

逐事实初审（参考事实已确认，结果判定待共同复核）

N02-F01 | 行为

读取用户提供的 `events.json` 作为事件记录来源。

归属：operation / 读取事件记录 | 原文：SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_001 从 <code>events.json</code> 读取 <code>events_records/result_001</code> ，ir_003 实际使用该结果迭代；事件来源与用户文件一致。 引用：ir_001, ir_003
2	保留	ir_001 从 <code>events.json</code> 实际读取 <code>event_records</code> ，ir_003 消费该结果逐条处理，文件数据源完整保留。 引用：ir_001, ir_003
3	保留	ir_001 从外部 <code>events.json</code> 读取 <code>events_records/result_001</code> ，后续 ir_003 迭代该来源，没有替换为未声明的 <code>context key</code> 。 引用：ir_001, ir_003

N02-F02 | 条件

选中条件为 `(opted_out != true) AND (urgent == true OR value >= 100)`，退出订阅限制适用于两个可选分支。

归属：block / 记录筛选 | 原文：SKILL.md:L8-L8; SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	ir_005 从当前记录求 <code>selected/result_004</code> ，原复合条件及例外约束附于该实际判定；ir_006 消费判定并由 <code>edge_004/005</code> 选择发送或返回迭代，条件没有仅留在注释中。 引用：ir_005, ir_005.constraints, ir_006, edge_004, edge_005
2	保留	ir_005/006/007 从当前记录提取 <code>opted_out</code> 、 <code>urgent</code> 、 <code>value</code> ，ir_008 产生 <code>record_selected</code> ，ir_009 消费判定并由 <code>edge_004/005</code> 决定发送或跳过，复合条件实际控制执行。 引用：ir_005, ir_006, ir_007, ir_008, ir_009, edge_004, edge_005
3	保留	ir_005 提取原三个筛选字段，ir_006 以明确括号表达 <code>opted_out</code> 非 <code>true</code> 且 <code>(urgent 或 value>=100)</code> 得到 <code>record_selected</code> ，ir_007/edge_004/005 实际分流；条件布尔结构保留。 引用：ir_005, ir_006, ir_007, edge_004, edge_005

N02-F03 | 约束

紧急事件可越过数值门槛，但仍受退出订阅限制。

归属：block / 记录筛选 | 原文：SKILL.md:L8-L8; SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	urgent 只豁免 value、不豁免 opted_out 的原句附于 ir_005 记录筛选，局部作用域保留并决定发送分支。 引用：ir_005.constraints, ir_006, edge_004
2	保留	urgent 只豁免数值、不豁免 opted_out 的完整原文位于实际判断 ir_008.constraints，局部筛选作用域与图中选中/未选中路径一致。 引用：ir_008.constraints, ir_009, edge_004
3	部分保留	筛选 ir_006 的英文约束和实际分支完整保留 urgent 不能豁免 opted_out 的逻辑。partial 专指已确认“原文+作用范围”契约的落点：中文例外原句仅在 skill.constraints，最近的筛选操作只留英文译写，没有同时在局部保留原文；不是筛选逻辑丢失。 引用：skill.constraints, ir_006.constraints, edge_004

N02-F04 | 行为

每条选中记录执行一次 notify.send，接收对象来自该记录的 recipient。

归属：operation / 发送通知 | 原文：SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	ir_003 每轮取得当前记录，选中时 ir_007 提取 recipient/result_005，ir_008 引用该接收对象调用一次 notify.send，edge_006 返回取下一条记录。 引用：ir_003, ir_007, ir_008, edge_004, edge_006
2	保留	当前记录由 ir_003 迭代产生，选中分支 ir_010 提取 recipient/result_008，ir_012 引用该结果调用 notify.send，edge_006 返回下一记录，每条选中记录一次调用保留。 引用：ir_003, ir_010, ir_012, edge_004, edge_006
3	保留	ir_003 逐记录迭代，ir_005 从当前记录提取 recipient/result_004，选中时 ir_008 以 notify.send opcode 和该输入执行一次调用，edge_006 返回循环；资源用 notify 服务名不改变明确的 send 操作。 引用：ir_003, ir_005, ir_008, edge_004, edge_006

N02-F05 | 数据流

notify.send.body 直接使用当前选中记录的 summary 原值。

归属：operation / 发送通知 | 原文：SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	ir_007 产出当前记录 summary/result_006, ir_008 直接引用该结果, 并保留 body 取 summary 原值约束; 没有其他内容生成步骤。 引用: ir_007, ir_008, ir_008.constraints
2	保留	ir_011 明确提取当前记录 summary/result_009, ir_012 将其作为 notification_body 输入并约束为原值; 没有中间改写或转换。 引用: ir_011, ir_012, ir_012.constraints
3	保留	ir_005 产出 event_summary/result_005, ir_008 直接引用该结果, 操作约束明确 body 为当前记录 summary 原值; 来源和消费关系正确。 引用: ir_005, ir_008, ir_008.constraints

N02-F06 | 条件

未选中的记录没有通知发送行为。

归属：block / 未选中分支 | 原文：SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	ir_006 的 not selected 路径 edge_005 直接返回循环, 绕过 ir_008 唯一通知操作; 未选中不发送作为实际路径保留。 引用: ir_006, ir_008, edge_004, edge_005
2	保留	record_selected 为否时 edge_005 直接回到循环, 绕过 ir_012; 未选中不发送的路径明确。 引用: ir_009, ir_012, edge_004, edge_005
3	保留	record_selected 为 false 时 edge_005 回到下一记录, 只有 true 路径 edge_004 进入唯一通知操作 ir_008, 未选中记录没有发送。 引用: ir_007, ir_008, edge_004, edge_005

N02-F07 | 约束

整份流程不得把 access_token 字段发送给任何接收对象。

归属：skill / 整份通知 Skill | 原文：SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	Skill 全局保留禁止将 access_token 发送给任何接收对象的原句, 通知只消费 recipient、summary 两个字段结果, 整图没有令牌发送。 引用: skill.constraints, ir_007, ir_008
2	保留	禁止向任何接收对象发送 access_token 的原文处于 Skill 全局; 实际通知只接收 recipient 和 summary 数据, 没有 token 参数或额外发送路径。 引用: skill.constraints, ir_010, ir_011, ir_012
3	保留	中文禁令完整位于 skill.constraints, 实际发送只引用 recipient 和 summary; 局部英文约束也禁止 access_token, 整图未建立令牌外发。 引用: skill.constraints, ir_005, ir_008

N02-F08 | 不得凭空增加

不能从字段直接取值要求增加摘要生成、内容改写或格式转换操作。

归属：operation / 发送通知 | 原文：SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	原字段提取后直接发送；全局保留不增加摘要生成、改写或格式转换的原句，完整CFG没有这些额外操作。 引用：skill.constraints, ir_007, ir_008, ir_010, ir_012
2	保留	summary 原字段提取后直接进入 notify.send，整图没有生成摘要、内容改写或额外格式转换，原文否定也在 Skill 约束保留。 引用：skill.constraints, ir_011, ir_012, ir_014, ir_015
3	保留	原字段被提取后直接发送，图中没有摘要生成、改写或额外格式转换；中文原限制及发送操作英文限制均保留该否定。 引用：skill.constraints, ir_005, ir_008

N02-F09 | 行为

处理结束后单独将处理条数写入本地 count.txt，保留它对前序记录处理的依赖。

归属：operation / 记录处理条数 | 原文：SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	部分保留	ir_003 确认无更多记录后经 edge_003 进入 ir_010, ir_012 将计数结果写 count.txt，处理完成后的依赖保留；ir_010 明确采用 count_selected_event_records, diagnostics 明确解释为触发通知的选中记录数，超出原文未确定口径。 引用：ir_010, ir_012, edge_003, edge_007, diagnostics
2	部分保留	循环无更多记录后 edge_003 才进入 ir_014/015 的计数和 count.txt 写入，前序依赖保留；但 opcode=count_selected_records 且附有选择条件，已静默确定选中记录计数口径，diagnostics 为空。 引用：ir_014, ir_014.constraints, ir_015, edge_003, diagnostics
3	部分保留	edge_003 在迭代完毕后进入 ir_010/011，计数结果写 count.txt 的数据和时序依赖保留；ir_010 约束直接要求计数满足通知条件的选中记录，diagnostics 也采用该解释，未保留原文未知统计口径。 引用：ir_010, ir_010.constraints, ir_011, edge_003, diagnostics

N03 条件通知与数据发送 · 表格

N03 | 条件通知与数据发送 | 表格 | 开发集 | zh

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	6	6	7
边数	5	7	8
操作数	12	13	14
三层约束条数	10	14	9
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三次最终均1次生成、无修复或HTTP重试，CFG为6/6/7块、5/7/8边。第1次为直线集合筛选与一次通知，第2、3次有实际逐记录选择/跳过/回环。三次计数分别固定选中并发送/选中通知/全部处理记录数。

第2、3次八条核心通知事实保留；第1次逐条条件执行表达不足，F02/F04/F06部分保留。三次F09均因擅自确定计数口径而部分保留。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 N03 的三次明细。

事实表以外的发现

运行 1 | 直线图将选中集合的字段复数结果直接交给一次 notify.send，逐条调用与未选中跳过只保留为文字；还将计数固定为选中并发送记录数。

引用：ir_005, ir_007, ir_009, edge_003, diagnostics

运行 2 | 把未知的处理条数具体化为选中且通知的记录数；诊断声明歧义后仍选择计数策略。

引用：ir_010, ir_012, diagnostics

运行 3 | count_event_records 从全部事件集计数，diagnostics 明确采用全部处理记录数；与前两次采用选中数的差异表明同一未定义边界被不同具体化。

引用：ir_012, ir_013, diagnostics

逐事实初审（参考事实已确认，结果判定待共同复核）

N03-F01 | 行为

读取用户提供的 `events.json` 作为事件记录来源。

归属：operation / 读取事件记录 | 原文：SKILL.md:L10-L10; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	保留	ir_001 从 <code>events.json</code> 读取 <code>events_records</code> , ir_003 实际筛选该结果，原文件来源保留。 引用：ir_001, ir_003
2	保留	ir_001 从 <code>events.json</code> 产生记录集合，ir_003 引用该结果逐条取记录；原外部文件来源完整。 引用：ir_001, ir_003
3	保留	ir_001 读取 <code>events.json</code> , ir_003 从该结果取当前记录，外部文件来源正确。 引用：ir_001, ir_003

N03-F02 | 条件

选中条件为 `(opted_out != true) AND (urgent == true OR value >= 100)`，退出订阅限制适用于两个可选分支。

归属：block / 记录筛选 | 原文：SKILL.md:L11-L11; SKILL.md:L7-L9; SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	部分保留	ir_003 的实际集合筛选保留完整复合条件及 <code>urgent</code> 例外，但筛选后只有无条件直线边，ir_007 用整组 <code>recipients/bodies</code> 调用一次工具。缺少选中集合的逐条或空集合控制，不能仅由条件文字确认按记录决定发送。 引用：ir_003, ir_003.constraints, ir_007, edge_002, edge_003
2	保留	ir_005 从当前记录提取 <code>opted_out/urgent/value</code> , ir_006 依据原复合条件产生 <code>selected</code> , ir_007 与 <code>edge_004/005</code> 实际控制发送或跳过。 引用：ir_005, ir_006, ir_007, edge_004, edge_005
3	保留	ir_005 提取当前记录字段，ir_007 用原筛选式计算 <code>should_notify</code> , ir_008 消费该判定； <code>edge_005</code> 明确带括号的复合条件， <code>edge_006</code> 转向跳过。 引用：ir_005, ir_007, ir_008, edge_005, edge_006

N03-F03 | 约束

紧急事件可越过数值门槛，但仍受退出订阅限制。

归属：block / 记录筛选 | 原文：SKILL.md:L11-L11; SKILL.md:L7-L9; SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	urgent 只豁免 value、不豁免 opted_out 的原句附于实际筛选操作 ir_003，局部原文与作用范围保留；执行路径缺口另在 F02 判定。 引用：ir_003.constraints, block_002
2	保留	urgent 仅豁免数值门槛而不豁免 opted_out 的完整原句位于筛选块及 ir_006，与实际判断一致，局部归属保留。 引用：block_003.constraints, ir_006.constraints, edge_004
3	保留	urgent 仅豁免数值、不豁免 opted_out 的原句保存在 ir_007 约束，选中边 edge_005 明确维持该局部例外的作用范围。 引用：ir_007.constraints, edge_005

N03-F04 | 行为

每条选中记录执行一次 notify.send，接收对象来自该记录的 recipient。

归属：operation / 发送通知 | 原文：SKILL.md:L13-L13; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	部分保留	ir_005 确实从选中记录提取 recipients/result_003，ir_007 消费该来源；但输入是复数组合，图中只有一次无条件发送，逐条一次仅在约束文字，未建立迭代控制。 引用：ir_005, ir_007, ir_007.constraints, edge_003
2	保留	ir_003 逐条取当前记录，ir_005 产出 recipient/result_004，ir_008 引用该来源调用 notify.send，edge_006 返回循环；每条选中记录一次调用保留。 引用：ir_003, ir_005, ir_008, edge_004, edge_006
3	保留	ir_005 从当前记录产出 recipient/result_004，选中时 ir_009 调用 notify_send，block_005 明确每条一次，edge_007 返回迭代；notify 服务资源与 send 操作组合保留指定调用。 引用：ir_003, ir_005, ir_009, block_005.constraints, edge_007

N03-F05 | 数据流

notify.send.body 直接使用当前选中记录的 summary 原值。

归属：operation / 发送通知 | 原文：SKILL.md:L14-L14; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	保留	ir_005 将 summary 原值要求绑定实际 notification_bodies/result_004，ir_007 直接消费该结果，没有正文改写；逐条调度不足已另列 F04。 引用：ir_005, ir_005.constraints, ir_007
2	保留	ir_005 从当前记录产出 summary/result_005，ir_008 消费该结果并约束 body 直接为 summary 原值，没有额外生成步骤。 引用：ir_005, ir_008, ir_008.constraints
3	保留	ir_005 产出 summary/result_005，ir_009 实际消费该结果并保留 body 取 summary 原值约束，没有中间内容变换。 引用：ir_005, ir_009, ir_009.constraints

N03-F06 | 条件

未选中的记录没有通知发送行为。

归属：block / 未选中分支 | 原文：SKILL.md:L15-L15; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	部分保留	选中集合来源和未选中不调用的约束存在，但实际 edge_003 无条件进入唯一通知块，没有逐条跳过或无选中记录时绕过发送的路径。 引用：ir_003, ir_007, ir_007.constraints, edge_003
2	保留	selected 为 false 时 edge_005 直接返回迭代，只有 true 分支进入 ir_008，未选中不发送作为实际控制流保留。 引用：ir_007, ir_008, edge_004, edge_005
3	保留	未选中时 edge_006 进入 block_006，ir_011/edge_008 返回循环，绕过唯一发送 ir_009。 引用：ir_009, block_006.constraints, ir_011, edge_006, edge_008

N03-F07 | 约束

整份流程不得把 access_token 字段发送给任何接收对象。

归属：skill / 整份通知 Skill | 原文：SKILL.md:L16-L16; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	保留	Skill 全局保留禁止 access_token 发送给任何接收对象的原文，实际发送输入是提取出的 recipient 与 summary 结果，没有 token 操作数。 引用：skill.constraints, ir_005, ir_007
2	保留	Skill 全局禁令完整保留，发送只消费 recipient/summary 字段结果，没有 token 输入或额外接收端。 引用：skill.constraints, ir_005, ir_008
3	保留	禁止 access_token 外发的完整原句保留在 Skill 全局，通知仅使用 recipient 和 summary 结果，无 token 输入。 引用：skill.constraints, ir_005, ir_009

N03-F08 | 不得凭空增加

不能从字段直接取值要求增加摘要生成、内容改写或格式转换操作。

归属：operation / 发送通知 | 原文：SKILL.md:L17-L17; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	保留	整图没有摘要生成、改写或格式转换；ir_005 原字段结果直接传递 ir_007，原否定要求在 Skill 约束中保留。 引用：skill.constraints, ir_005, ir_007
2	保留	当前记录字段直接进入通知，完整图没有摘要生成、内容改写或格式转换；全局原文否定要求保留。 引用：skill.constraints, ir_005, ir_008
3	保留	整图无摘要生成、改写或格式转换，原字段输出直接被通知使用，原否定保留。 引用：skill.constraints, ir_005, ir_009

N03-F09 | 行为

处理结束后单独将处理条数写入本地 count.txt，保留它对前序记录处理的依赖。

归属：operation / 记录处理条数 | 原文：SKILL.md:L18-L18; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	部分保留	ir_009/011 位于通知之后，计数结果写入 count.txt 的依赖保留；但 count_selected_records 及 diagnostics 明确选定选中并发送的记录数，超出未定义口径。 引用：ir_009, ir_011, edge_004, edge_005, diagnostics
2	部分保留	迭代结束 edge_003 后 ir_010 才计数，ir_012 引用结果写 count.txt，尾部依赖保留；但 count_selected_event_records 和 diagnostics 明确采用通知选中数，未保留原文口径未知。 引用：ir_010, ir_012, edge_003, edge_007, diagnostics
3	部分保留	无更多记录 edge_003 才进入 ir_012/013 并将计数写 count.txt，处理依赖完整；但 count_event_records 消费全部 event_records，diagnostics 明确统计全部处理记录，擅自确定另一种口径。 引用：ir_012, ir_013, edge_003, diagnostics

N04 条件通知与数据发送 · 跨文件混合

N04 | 条件通知与数据发送 | 跨文件混合 | 开发集 | zh

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	7	5	7
边数	7	4	6
操作数	16	10	14
三层约束条数	11	10	26
生成尝试	2	1	1
结构修复	1	0	0
HTTP 尝试	2	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三次最终CFG分别7/5/7块、7/4/6边，结构修复1/0/0轮、HTTP重试均0。第1次先筛选后逐条循环通知，第2、3次是直线集合通知。三次都把固定 payload.json 声明增加为运行时读取；第2次还由 notify.send 直接产生处理计数，并添加正文默认字段策略。

跨文件流程及主要字段要求被纳入，但三次均出现固定规范被误当运行时读取的额外行为；第2、3次逐条控制不足，第2次正文和recipient只以整集合/元数据表达。三次尾部写入存在，计数口径或来源均被擅自具体化。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 N04 的三次明细。

事实表以外的发现

运行 1 | 包内 payload.json 已固定声明 body_from=summary，与正文规范等价；ir_003 额外把它作为外部运行时配置读取，ir_012 依赖该动态结果，超过已确认的跨文件语义边界。另 ir_007 擅自统计选中记录数。

引用: ir_003, ir_007, ir_012, diagnostics

运行 2 | ir_005 与 diagnostics 明确将固定 payload.json 解释为运行时读取，ir_007 又为其增加 body_field_default=summary 默认策略；同时把未声明返回值的 notify.send 设为 processed_record_count 的生产者。这些行为未由冻结原文规定。

引用: ir_005, ir_007, ir_009, diagnostics

运行 3 | ir_003 增加固定 payload.json 的运行时读取，后续字段提取依赖该动态结果；ir_011 又仅统计选中集合，额外确定了原文未定义的计数口径。

引用: ir_003, ir_007, ir_011, diagnostics

逐事实初审（参考事实已确认，结果判定待共同复核）

N04-F01 | 行为

读取用户提供的 `events.json` 作为事件记录来源。

归属: operation / 读取事件记录 | 原文: references/workflow.md:L3-L3; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_001 实际读取用户 <code>events.json</code> 产生记录，ir_005 消费该结果筛选；根Skill引用的 <code>workflow</code> 被纳入当前流程，事件来源正确。 引用: ir_001, ir_005
2	保留	ir_001 从 <code>events.json</code> 读取 <code>event_records</code> ，ir_003 实际消费此结果；事件文件来源正确。 引用: ir_001, ir_003
3	保留	ir_001 实际读取 <code>events.json</code> ，ir_005 筛选其 <code>event_records</code> 结果，跨文件流程纳入且记录来源正确。 引用: ir_001, ir_005

N04-F02 | 条件

选中条件为 `(opted_out != true) AND (urgent == true OR value >= 100)`，退出订阅限制适用于两个可选分支。

归属: block / 记录筛选 | 原文: references/workflow.md:L6-L6; references/workflow.md:L5-L5; SKILL.md:L8-L8; references/workflow.md:L7-L7

运行	初审	理由与实际输出引用
1	保留	ir_005 将完整复合条件及例外绑定实际集合筛选，ir_009 只迭代 <code>selected_records</code> ；ir_010 与 <code>edge_005/006</code> 根据是否有下一条选中记录进入通知或结束，先筛后循环的等价执行保留。 引用: ir_005, ir_005.constraints, ir_009, ir_010, edge_005, edge_006
2	部分保留	ir_003 保留完整复合条件和例外并产出选中集合；后续只经无条件 <code>edge_003</code> 到 ir_007 通知整组记录，没有逐条或空集合控制，无法完整确认按条件执行通知的控制流。 引用: ir_003, ir_003.constraints, ir_007, edge_003
3	部分保留	ir_005 的集合筛选完整保留复合条件与 <code>urgent</code> 例外；ir_009 之后只接受整组字段结果且控制边均无条件，缺少逐条条件或空集合跳过的控制，不能只据文字条件认定执行语义完整。 引用: ir_005, ir_005.constraints, ir_009, edge_003, edge_004

N04-F03 | 约束

紧急事件可越过数值门槛，但仍受退出订阅限制。

归属：block / 记录筛选 | 原文：references/workflow.md:L6-L6; references/workflow.md:L5-L5; SKILL.md:L8-L8; references/workflow.md:L7-L7

运行	初审	理由与实际输出引用
1	保留	urgent 只豁免数值而不豁免 opted_out 的原句位于 ir_005 实际筛选约束，局部范围准确。 引用：ir_005.constraints, block_003
2	保留	urgent 仅豁免 value 门槛而不豁免 opted_out 的原句处于 ir_003 实际集合筛选约束，局部原文及范围保留。 引用：ir_003.constraints
3	保留	urgent 仅豁免数值而不豁免 opted_out 的完整原句附于筛选块及 ir_005，局部作用范围保留。 引用：block_003.constraints, ir_005.constraints

N04-F04 | 行为

每条选中记录执行一次 notify.send，接收对象来自该记录的 recipient。

归属：operation / 发送通知 | 原文：references/workflow.md:L9-L9; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_009 逐条产出当前选中记录，ir_011 提取 recipient/result_007，ir_013 引用该接收对象执行 notify.send，edge_007 返回循环，每条一次保留。 引用：ir_009, ir_011, ir_013, edge_005, edge_007
2	部分保留	ir_007 消费整组选中记录，recipient_field=recipient 元数据和逐条一次约束保留接收字段意图；但没有从当前记录实际提取 recipient 的结果，也没有逐条迭代路径，一次集合输入不能证明每条记录一次调用。 引用：ir_007, ir_007.constraints, edge_003
3	部分保留	ir_007 从选中集合实际产出 notification_recipients/result_004，ir_009 使用该来源；但逐条一次仅在约束，图为无条件单次集合通知，没有迭代到当前记录。 引用：ir_007, ir_009, ir_009.constraints, edge_004

N04-F05 | 数据流

notify.send.body 直接使用当前选中记录的 summary 原值。

归属: operation / 发送通知 | 原文: references/workflow.md:L11-L11; references/workflow.md:L13-L13; SKILL.md:L8-L8; payload.json:L1-L3

运行	初审	理由与实际输出引用
1	保留	ir_012 从当前选中记录产出 notification_body/result_008, summary 原值要求明确附于此提取操作, ir_013 消费同一结果; 字段来源及直接取值保留。额外把固定 payload.json 读成运行时配置的问题单列 additional_findings。 引用: ir_012, ir_012.constraints, ir_013
2	部分保留	summary 原值要求及 body_field_default=summary 元数据存在; 实际 ir_007 输入只有整组选中记录和运行时 body_from 结果, 未显式产出或引用当前记录 summary 原字段值, 正文数据流仅部分保留。 引用: ir_005, ir_007, ir_007.constraints
3	保留	ir_007 的实际 notification_bodies/result_005 与 summary 原字段约束绑定, ir_009 直接消费该结果, 没有改写; 额外动态读取 payload.json 的问题单列, 逐条调用不足在 F04 另判。 引用: ir_007, ir_007.constraints, ir_009

N04-F06 | 条件

未选中的记录没有通知发送行为。

归属: block / 未选中分支 | 原文: references/workflow.md:L15-L15; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	实际循环只消费 ir_005 的选中集合; edge_006 在无下一条选中记录时直接进入写入, 未选中记录从未进入 ir_013 通知操作。 引用: ir_005, ir_009, ir_013, edge_005, edge_006
2	部分保留	选中集合来源及未选中不调用约束保留, 但 edge_003 无条件进入 ir_007, 没有无选中记录时跳过或逐记录的通知控制路径。 引用: ir_003, ir_007, edge_003
3	部分保留	ir_005 排除了未选中记录且 ir_009 保留不调用原句; 实际 edge_004 仍无条件进通知, 未建立空集合跳过或逐记录判定的发送控制。 引用: ir_005, ir_009, ir_009.constraints, edge_004

N04-F07 | 约束

整份流程不得把 `access_token` 字段发送给任何接收对象。

归属：skill / 整份通知 Skill | 原文：references/workflow.md:L17-L17; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	Skill 全局原文禁止 <code>access_token</code> 发送给任何对象，实际通知只引用 <code>recipient</code> 和 <code>body</code> 字段结果，无 <code>token</code> 数据流。 引用：skill.constraints, ir_011, ir_012, ir_013
2	保留	Skill 级原文保留禁止 <code>access_token</code> 向任何接收对象发送；没有显式 <code>token</code> 外发操作。通知仍使用整记录集合，数据流粒度不足在 F04/F05 另判，不据此猜测未声明的泄露行为。 引用：skill.constraints, ir_007
3	保留	<code>access_token</code> 禁止向任何接收对象外发的全局原句保留，通知实际只使用提取出的 <code>recipients/bodies</code> 结果，不含 <code>token</code> 参数。 引用：skill.constraints, ir_007, ir_009

N04-F08 | 不得凭空增加

不能从字段直接取值要求增加摘要生成、内容改写或格式转换操作。

归属：operation / 发送通知 | 原文：references/workflow.md:L19-L19; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_012 原字段结果直接进入 ir_013，图中没有摘要生成、内容改写或格式转换，原文否定保留。运行时配置读取是另列的额外行为，不是把它误称为内容改写。 引用：skill.constraints, ir_012, ir_013
2	保留	ir_007 明确保留不生成摘要、不改写、不额外格式转换的约束；实际图没有这些转换操作。 引用：ir_007, ir_007.constraints
3	保留	提取出的原字段正文直接用于通知，无摘要生成、内容改写或额外格式转换；原否定在全局和提取操作局部均保留。 引用：skill.constraints, ir_007, ir_007.constraints, ir_009

N04-F09 | 行为

处理结束后单独将处理条数写入本地 `count.txt`，保留它对前序记录处理的依赖。

归属：operation / 记录处理条数 | 原文：references/workflow.md:L21-L21; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	部分保留	ir_015 在迭代结束 <code>edge_006</code> 后写 <code>count.txt</code> ，依赖前序处理；但它消费的 ir_007 计数明确为 <code>count_selected_records</code> ， <code>diagnostics</code> 也选择选中并处理的记录数，未保留原文未知口径。 引用：ir_007, ir_015, edge_006, diagnostics
2	部分保留	ir_009 在通知之后写 <code>count.txt</code> 并依赖 ir_007 的 <code>processed_record_count</code> 结果；但源文未说明 <code>notify.send</code> 返回处理计数，候选直接创造该返回来源，将未定义统计依附于选中通知调用，计数口径和来源均未保留未知。 引用：ir_007, ir_009, edge_004, diagnostics
3	部分保留	通知后 ir_011 计数、ir_013 写 <code>count.txt</code> ，尾部依赖存在；ir_011 只消费 <code>selected_records</code> ， <code>diagnostics</code> 也明确选中记录口径，超出原文未定义边界。 引用：ir_011, ir_013, edge_005, edge_006, diagnostics

N05 条件通知与数据发送 · 用途最小差异

N05 | 条件通知与数据发送 | 最小差异：用途 | 开发集 | zh

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	5	5	5
边数	6	6	6
操作数	11	10	10
三层约束条数	8	7	8
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三次均1次生成、0结构修复、0HTTP重试，均产生5块6边的有效迭代通知CFG。三次都明确只排除第9条历史 count.txt 示例，其余读取、筛选、发送、禁止行为仍执行。第1次在发送块提取通知字段，第2、3次在筛选块提取，来源和后续引用均一致。

27条事实×重复判定均保留。尤其最小对照中的历史范围被正确限定在 count.txt 示例，没有消除当前通知流程；这是对最终实际图逐条复核的结论，不是由结构complete推导。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 N05 的三次明细。

逐事实初审（参考事实已确认，结果判定待共同复核）

N05-F01 | 行为

读取用户提供的 `events.json` 作为事件记录来源。

归属：operation / 读取事件记录 | 原文：SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_001 实际读取 <code>events.json</code> ，ir_003 消费记录结果逐条迭代；未把第9条历史示例误扩展为整份流程不执行。 引用：ir_001, ir_003
2	保留	ir_001 实际读取 <code>events.json</code> ，ir_003 消费记录结果逐条迭代；未把第9条历史示例误扩展为整份流程不执行。 引用：ir_001, ir_003
3	保留	ir_001 实际读取 <code>events.json</code> ，ir_003 消费记录结果逐条迭代；未把第9条历史示例误扩展为整份流程不执行。 引用：ir_001, ir_003

N05-F02 | 条件

选中条件为 `(opted_out != true) AND (urgent == true OR value >= 100)`，退出订阅限制适用于两个可选分支。

归属：block / 记录筛选 | 原文：SKILL.md:L9-L9; SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	ir_005/006 从当前记录字段计算选中判定，ir_007 消费判断并经 <code>edge_004/005</code> 进入发送或跳过；复合条件和 <code>opted_out</code> 对两个可选分支的限制保留。 引用：ir_005, ir_006, ir_007, edge_004, edge_005
2	保留	ir_005/006 从当前记录字段计算选中判定，ir_007 消费判断并经 <code>edge_004/005</code> 进入发送或跳过；复合条件和 <code>opted_out</code> 对两个可选分支的限制保留。 引用：ir_005, ir_006, ir_007, edge_004, edge_005
3	保留	ir_005/006 从当前记录字段计算选中判定，ir_007 消费判断并经 <code>edge_004/005</code> 进入发送或跳过；复合条件和 <code>opted_out</code> 对两个可选分支的限制保留。 引用：ir_005, ir_006, ir_007, edge_004, edge_005

N05-F03 | 约束

紧急事件可越过数值门槛，但仍受退出订阅限制。

归属：block / 记录筛选 | 原文：SKILL.md:L9-L9; SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	urgent 只豁免数值门槛、不豁免 opted_out 的完整原句保存在 ir_006 的筛选约束，且参与实际选中判定，局部作用范围正确。 引用：ir_006.constraints, edge_004
2	保留	urgent 只豁免数值门槛、不豁免 opted_out 的完整原句保存在 ir_006 的筛选约束，且参与实际选中判定，局部作用范围正确。 引用：ir_006.constraints, edge_004
3	保留	urgent 只豁免数值门槛、不豁免 opted_out 的完整原句保存在 ir_006 的筛选约束，且参与实际选中判定，局部作用范围正确。 引用：ir_006.constraints, edge_004

N05-F04 | 行为

每条选中记录执行一次 notify.send，接收对象来自该记录的 recipient。

归属：operation / 发送通知 | 原文：SKILL.md:L11-L11

运行	初审	理由与实际输出引用
1	保留	ir_003 逐记录迭代，ir_008 从当前记录提取 recipient，ir_009 实际引用该接收对象调用 notify.send；edge_006 回到下一记录，每条选中记录一次。 引用：ir_003, ir_008, ir_009, edge_004, edge_006
2	保留	ir_003 逐记录迭代，ir_005 从当前记录提取 recipient，ir_008 实际引用该接收对象调用 notify.send；edge_006 回到下一记录，每条选中记录一次。 引用：ir_003, ir_005, ir_008, edge_004, edge_006
3	保留	ir_003 逐记录迭代，ir_005 从当前记录提取 recipient，ir_008 实际引用该接收对象调用 notify.send；edge_006 回到下一记录，每条选中记录一次。 引用：ir_003, ir_005, ir_008, edge_004, edge_006

N05-F05 | 数据流

notify.send.body 直接使用当前选中记录的 summary 原值。

归属：operation / 发送通知 | 原文：SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	ir_008 产生当前记录 summary 原字段结果，ir_009 引用它并保留 body 直接取 summary 原值要求，无改写步骤。 引用：ir_008, ir_009, ir_009.constraints
2	保留	ir_005 产生当前记录 summary 原字段结果，ir_008 引用它并保留 body 直接取 summary 原值要求，无改写步骤。 引用：ir_005, ir_008, ir_008.constraints
3	保留	ir_005 产生当前记录 summary 原字段结果，ir_008 引用它并保留 body 直接取 summary 原值要求，无改写步骤。 引用：ir_005, ir_008, ir_008.constraints

N05-F06 | 条件

未选中的记录没有通知发送行为。

归属：block / 未选中分支 | 原文：SKILL.md:L13-L13

运行	初审	理由与实际输出引用
1	保留	未选中时 edge_005 直接返回迭代，绕过唯一通知 ir_009；未选中记录不发送的实际路径存在。 引用：ir_007, ir_009, edge_004, edge_005
2	保留	未选中时 edge_005 直接返回迭代，绕过唯一通知 ir_008；未选中记录不发送的实际路径存在。 引用：ir_007, ir_008, edge_004, edge_005
3	保留	未选中时 edge_005 直接返回迭代，绕过唯一通知 ir_008；未选中记录不发送的实际路径存在。 引用：ir_007, ir_008, edge_004, edge_005

N05-F07 | 约束

整份流程不得把 access_token 字段发送给任何接收对象。

归属：skill / 整份通知 Skill | 原文：SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	禁止向任何接收对象发送 access_token 的原句在 Skill 全局保留，ir_009 的实际输入只含工具、recipient 和 summary，没有 token 输入。 引用：skill.constraints, ir_008, ir_009
2	保留	禁止向任何接收对象发送 access_token 的原句在 Skill 全局保留，ir_008 的实际输入只含工具、recipient 和 summary，没有 token 输入。 引用：skill.constraints, ir_005, ir_008
3	保留	禁止向任何接收对象发送 access_token 的原句在 Skill 全局保留，ir_008 的实际输入只含工具、recipient 和 summary，没有 token 输入。 引用：skill.constraints, ir_005, ir_008

N05-F08 | 不得凭空增加

不能从字段直接取值要求增加摘要生成、内容改写或格式转换操作。

归属：operation / 发送通知 | 原文：SKILL.md:L15-L15

运行	初审	理由与实际输出引用
1	保留	完整图只有读取、字段提取、条件判断、通知与返回；ir_008 字段结果直接进入 ir_009，无摘要生成、改写或额外格式转换。 引用：skill.constraints, ir_008, ir_009
2	保留	完整图只有读取、字段提取、条件判断、通知与返回；ir_005 字段结果直接进入 ir_008，无摘要生成、改写或额外格式转换。 引用：skill.constraints, ir_005, ir_008
3	保留	完整图只有读取、字段提取、条件判断、通知与返回；ir_005 字段结果直接进入 ir_008，无摘要生成、改写或额外格式转换。 引用：skill.constraints, ir_005, ir_008

N05-F09 | 不得凭空增加

写入 count.txt 仅是明确不执行的历史示例，当前流程不新增该写入行为。

归属：operation / 历史写入示例 | 原文：SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	保留	diagnostics 明确只排除第9条历史 count.txt 示例；实际图没有计数或写 count.txt，迭代完成后经 edge_003 进入 ir_011 返回，其余8条通知要求保持激活。 引用：diagnostics, ir_001, ir_009, ir_011, edge_003
2	保留	diagnostics 明确只排除第9条历史 count.txt 示例；实际图没有计数或写 count.txt，迭代完成后经 edge_003 进入 ir_010 返回，其余8条通知要求保持激活。 引用：diagnostics, ir_001, ir_008, ir_010, edge_003
3	保留	diagnostics 明确只排除第9条历史 count.txt 示例；实际图没有计数或写 count.txt，迭代完成后经 edge_003 进入 ir_010 返回，其余8条通知要求保持激活。 引用：diagnostics, ir_001, ir_008, ir_010, edge_003

N06 条件通知与数据发送 · 语义最小差异

N06 | 条件通知与数据发送 | 最小差异：语义 | 开发集 | zh

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	6	5	4
边数	5	6	3
操作数	12	13	9
三层约束条数	9	9	9
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三次均1次生成、无结构修复或HTTP重试；最终CFG为6/5/4块、5/6/3边。第2次明确逐记录迭代与record_id提取，第1/3次为集合字段或载荷直线通知。三次实际正文来源都改为record_id原值，没有沿用summary；计数分别采用选中/全部/选中接收记录数。

正文最小对照字段在三次都落实到实际提取或结果来源；第1/3次条件与逐条调用控制不足，三次仍擅自具体化未定义计数口径。不能把结构complete视为全部语义通过。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 N06 的三次明细。

事实表以外的发现

运行 1 | ir_009 仅统计 selected_records, diagnostics 将口径解释为选中且通知的记录数；同时逐条执行仅有直线图中的文字约束。

引用: ir_009, ir_011, diagnostics

运行 2 | ir_011 从全部 event_records 统计，未记录口径歧义；与第1次采用选中数不同，都是额外确定原文未知的策略。

引用: ir_011, ir_012, diagnostics

运行 3 | ir_007 以 selected_recipients 为计数输入，确定了选中记录口径；与第2次全部记录计数不同，都未保留原文未知边界。第3次逐条通知仍只在直线CFG的约束文字中。

引用: ir_005, ir_007, ir_008, edge_002, diagnostics

逐事实初审（参考事实已确认，结果判定待共同复核）

N06-F01 | 行为

读取用户提供的 `events.json` 作为事件记录来源。

归属：operation / 读取事件记录 | 原文：SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_001 实际读取 <code>events.json</code> ，ir_003 筛选其 <code>events_records</code> 结果，记录来源保留。 引用：ir_001, ir_003
2	保留	ir_001 从 <code>events.json</code> 读取记录，ir_003 引用该结果迭代，原文件来源正确。 引用：ir_001, ir_003
3	保留	ir_001 读取 <code>events.json</code> 产出 <code>event_records</code> ，ir_003 实际筛选这个结果，记录来源保留。 引用：ir_001, ir_003

N06-F02 | 条件

选中条件为 `(opted_out != true) AND (urgent == true OR value >= 100)`，退出订阅限制适用于两个可选分支。

归属：block / 记录筛选 | 原文：SKILL.md:L9-L9; SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	部分保留	ir_003 实际筛选保留原复合条件与 <code>urgent</code> 例外，后续仅有无条件直线边通向一次集合通知；没有逐条或空集合分支，按条件执行发送的控制流不完整。 引用：ir_003, ir_003.constraints, ir_007, edge_003
2	保留	ir_005 根据当前记录和完整复合筛选约束求 <code>is_selected</code> ，ir_006 消费结果， <code>edge_004/005</code> 分别进入发送或返回循环，条件实际控制执行。 引用：ir_005, ir_005.constraints, ir_006, edge_004, edge_005
3	部分保留	ir_003 保留实际筛选及完整复合条件、 <code>urgent</code> 例外，并产出选中字段；但 ir_004 无判定输入， <code>edge_002</code> 无条件进入一次集合通知，缺少逐条/空集合控制。 引用：ir_003, ir_003.constraints, ir_004, ir_005, edge_002

N06-F03 | 约束

紧急事件可越过数值门槛，但仍受退出订阅限制。

归属：block / 记录筛选 | 原文：SKILL.md:L9-L9; SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	<code>urgent</code> 只豁免数值、不豁免 <code>opted_out</code> 的原句附于 ir_003 筛选操作，局部语义归属保留。 引用：ir_003.constraints
2	保留	<code>urgent</code> 不能豁免 <code>opted_out</code> 的完整原句附于筛选判断 ir_005，紧急例外局部作用范围与实际分支一致。 引用：ir_005.constraints, edge_004
3	保留	<code>urgent</code> 仅豁免数值、不豁免 <code>opted_out</code> 的完整原文附于 ir_003 实际筛选操作，局部归属保留。 引用：ir_003.constraints

N06-F04 | 行为

每条选中记录执行一次 `notify.send`，接收对象来自该记录的 `recipient`。

归属：operation / 发送通知 | 原文：SKILL.md:L11-L11

运行	初审	理由与实际输出引用
1	部分保留	ir_005 从选中集合提取 <code>notification_payloads</code> ，约束保留 <code>recipient</code> 字段来源，ir_007 消费此结果；但实际图没有当前记录迭代，每条选中记录一次只在约束中。 引用：ir_005, ir_005.constraints, ir_007, ir_007.constraints, edge_003
2	保留	ir_003 逐条取当前记录，选中时 ir_007 提取 <code>recipient/result_005</code> ，ir_009 引用其调用 <code>notify.send</code> ，edge_006 返回迭代，每条选中记录一次。 引用：ir_003, ir_007, ir_009, edge_004, edge_006
3	部分保留	ir_003 显式产出 <code>selected_recipients/result_002</code> ，ir_005 消费该接收对象来源；但当前记录迭代没有出现，每条选中记录一次仅是调用约束，图只有一次无条件集合通知。 引用：ir_003, ir_005, ir_005.constraints, edge_002

N06-F05 | 数据流

`notify.send.body` 直接使用当前选中记录的 `record_id` 原值。

归属：operation / 发送通知 | 原文：SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	ir_005 实际产生 <code>notification_payloads</code> ，并把 <code>body</code> 直接取 <code>record_id</code> 原值的要求绑定于该提取操作；ir_007 消费这一载荷结果，未沿用对照包的 <code>summary</code> 或增加转换。 引用：ir_005, ir_005.constraints, ir_007
2	保留	ir_008 明确从当前记录提取 <code>record_id/result_006</code> ，ir_009 实际引用该结果并保留 <code>body</code> 原值约束；最小对照字段变化已落实为数据流，不是只改描述。 引用：ir_008, ir_009, ir_009.constraints
3	保留	ir_003 实际产出 <code>selected_record_ids/result_003</code> ，ir_005 将同一结果用作 <code>body</code> 且约束直接取 <code>record_id</code> 原值；未沿用 <code>summary</code> ，字段最小变化反映到实际数据来源。 引用：ir_003, ir_005, ir_005.constraints

N06-F06 | 条件

未选中的记录没有通知发送行为。

归属：block / 未选中分支 | 原文：SKILL.md:L13-L13

运行	初审	理由与实际输出引用
1	部分保留	选中集合和未选中不调用的原句保留，但实际 edge_003 无条件进入发送操作，没有逐记录跳过或空选中集合绕过调用的路径。 引用：ir_003, ir_007, ir_007.constraints, edge_003
2	保留	未选中时 edge_005 直接返回迭代，不经过唯一通知 ir_009，未选中不发送的路径完整。 引用：ir_006, ir_009, edge_004, edge_005
3	部分保留	筛选结果及未选中不调用原句保留，实际 edge_002 仍无条件通向 ir_005，未选中或空选中结果如何绕过通知没有控制流表达。 引用：ir_003, ir_005, ir_005.constraints, edge_002

N06-F07 | 约束

整份流程不得把 `access_token` 字段发送给任何接收对象。

归属：skill / 整份通知 Skill | 原文：SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	禁止 <code>access_token</code> 发送给任何接收对象的原句在 Skill 全局保留，载荷只按 <code>recipient/record_id</code> 字段提取，没有 <code>token</code> 外发操作。 引用：skill.constraints, ir_005, ir_007
2	保留	Skill 级禁止 <code>access_token</code> 外发的原句保留，通知实际只消费 <code>recipient</code> 与 <code>record_id</code> 字段结果，没有 <code>token</code> 输入。 引用：skill.constraints, ir_007, ir_008, ir_009
3	保留	Skill 全局保留 <code>access_token</code> 禁止向任何对象发送的原句，通知实际仅使用 <code>recipient</code> 和 <code>record_id</code> 结果，无 <code>token</code> 数据流。 引用：skill.constraints, ir_003, ir_005

N06-F08 | 不得凭空增加

不能从字段直接取值要求增加摘要生成、内容改写或格式转换操作。

归属：operation / 发送通知 | 原文：SKILL.md:L15-L15

运行	初审	理由与实际输出引用
1	保留	载荷提取将 <code>record_id</code> 原值直接传入通知，完整图没有摘要生成、内容改写或额外格式转换，原否定要求保留。 引用：skill.constraints, ir_005, ir_007
2	保留	<code>record_id</code> 原值提取后直接作为通知输入，整图没有摘要生成、改写或额外格式转换；原文否定在 Skill 约束保留。 引用：skill.constraints, ir_008, ir_009
3	保留	原 <code>record_id</code> 字段直接进入通知，完整图只有读取、筛选字段、通知、计数与写文件，没有摘要生成、改写或额外格式转换。 引用：skill.constraints, ir_003, ir_005, ir_007, ir_008

N06-F09 | 行为

处理结束后单独将处理条数写入本地 count.txt，保留它对前序记录处理的依赖。

归属：operation / 记录处理条数 | 原文：SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	部分保留	通知之后 ir_009 产生计数，再由 ir_011 写 count.txt，尾部依赖保留；ir_009 实际只消费 selected_records，diagnostics 也采用选中并通知记录数，越过源文未定义统计口径。 引用：ir_009, ir_011, edge_004, edge_005, diagnostics
2	部分保留	迭代全部结束后 edge_003 才进入 ir_011/012，计数写入 count.txt 依赖前序处理；但 ir_011 消费全部 event_records 集合，实际固定为全部记录数，diagnostics 为空，未保留未知口径。 引用：ir_011, ir_012, edge_003, diagnostics
3	部分保留	ir_007/008 位于通知之后，计数结果写 count.txt 的依赖保留；ir_007 实际计数 selected_recipients/result_002，静默固定为选中接收记录数，diagnostics 没有保留源文未知口径。 引用：ir_007, ir_008, edge_003, diagnostics

Q01 查询调用与参数使用 · 编号步骤

Q01 | 查询调用与参数使用 | 编号步骤 | 保留验证集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	7	7	13
边数	8	9	12
操作数	12	15	37
三层约束条数	10	11	7
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

第1/2次以存在分支配合唯一search的局部省略规则表达可选参数；第3次明确展开四个互斥search路径。三次均保留读文件、查询、写total和原样items返回。依冻结可接受变体，调用附着参数规则本身可接受；静态可选result列表的消费边界另列为复核点。

逐事实依据冻结原文与本批次三次已提交CFG独立初审；结构complete不代表语义通过，结论仍待共同复核。参数格式和能力说明均未凭空新增执行行为。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 Q01 的三次明细。

事实表以外的发现

运行 1, 2 | 唯一search的静态inputs仍列条件性产生的from_date/limit结果，缺失路径不产生这些值。原文omit规则确已附着于调用，按冻结可接受变体认定语义规则保留；本初审不把它解释为已验证的运行时可选操作数机制。第3次四条路径的实际inputs区分更明确。

引用：block_006

逐事实初审（参考事实已确认，结果判定待共同复核）

Q01-F01 | 行为

Read request.json as the source of term and optional from_date and limit.

归属: operation / Read query request | 原文: SKILL.md:L9-L9

运行	初审	理由与实际输出引用
1	保留	文件读取直接产生term、请求内容及可选字段存在标记，源为外部request.json。 引用: ir_001
2	保留	读取request.json后只按存在分支提取可选值；后续查询输入均追溯该文件。 引用: ir_001, ir_003, ir_004, ir_005, ir_007, ir_009
3	保留	外部request.json经字段提取产生term和两项存在标记，没有虚构运行上下文。 引用: ir_001, ir_003, ir_004, ir_005

Q01-F02 | 行为

Invoke index.search exactly once with request.term unchanged as query.

归属: operation / Search index | 原文: SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	唯一search节点以result_001作为query来源，并明确exactly once与term unchanged。 引用: ir_009
2	保留	单一search调用消费request_term，整体约束exactly once，调用约束保持query原值。 引用: ir_012, skill.constraints
3	保留	四项互斥存在组合各进入一次search，均消费同一个request_term；节点数量不是运行时调用次数。 引用: ir_014, ir_016, ir_018, ir_020, edge_002, edge_003, edge_004, edge_005, skill.constraints

Q01-F03 | 约束

A present from_date is passed unchanged and must have the documented YYYY-MM-DD format.

归属: operation / Search index: from_date argument | 原文: SKILL.md:L11-L11

运行	初审	理由与实际输出引用
1	保留	存在时从请求提取from_date；search局部约束保留原值和YYYY-MM-DD。 引用: ir_004, ir_009.constraints
2	保留	from_date存在分支产生原值，调用约束明确原样传递及YYYY-MM-DD格式。 引用: ir_007, ir_012.constraints
3	保留	两条from_date存在路径实际传相应提取结果，命名参数的原值/日期格式规则在Skill约束中保留。 引用: ir_007, ir_010, ir_014, ir_016, skill.constraints

Q01-F04 | 条件

If from_date is absent, omit that argument rather than send an empty or null value.

归属: operation / Search index: from_date argument | 原文: SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	缺失路径跳过取值, search明确缺失时省略from_date; 按照冻结可接受变体允许规则附着于调用, 未添加null/空串参数。 引用: ir_009.constraints, edge_003
2	保留	缺失分支绕过取值, 调用同时接收存在标记并明确省略参数, 未填null或空串。 引用: ir_012.constraints, edge_003
3	保留	缺少from_date的两条互斥路径调用输入实际不含from_date值。 引用: ir_018, ir_020, edge_004, edge_005

Q01-F05 | 数据流

A present limit is passed unchanged as the limit argument.

归属: operation / Search index: limit argument | 原文: SKILL.md:L13-L13

运行	初审	理由与实际输出引用
1	保留	存在路径从请求读取limit并传到search, 原值传递约束保留。 引用: ir_007, ir_009.constraints
2	保留	请求limit经存在分支提取, 调用绑定该result并保留unchanged。 引用: ir_010, ir_012.constraints
3	保留	limit存在的两类路径读取并使用其实际值, 整体参数约束禁止改写。 引用: ir_008, ir_012, ir_014, ir_018, skill.constraints

Q01-F06 | 条件

If limit is absent, omit the limit argument; do not supply an invented default.

归属: operation / Search index: limit argument | 原文: SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	limit缺失分支跳过提取, 调用明确省略; 没有虚构默认值。 引用: ir_009.constraints, edge_006
2	保留	两个limit缺失分支最终进入同一调用, 调用保留omit规则及存在标记, 没有默认常量。 引用: ir_012.constraints, edge_005, edge_007
3	保留	limit缺失两条search调用的输入实际不含limit值或默认常量。 引用: ir_016, ir_020, edge_003, edge_005

Q01-F07 | 数据流

Return the response.items value unchanged.

归属: operation / Return query result | 原文: SKILL.md:L18-L18

运行	初审	理由与实际输出引用
1	保留	return直接消费该search产生的response.items，并保留unchanged约束。 引用: ir_009, ir_012
2	保留	return使用单次search输出的search_items并明确保持原值。 引用: ir_012, ir_015
3	保留	四条路径分别从各自search_response提取items并直接返回，没有跨路径拼接或转换。 引用: ir_024, ir_025, ir_028, ir_029, ir_032, ir_033, ir_036, ir_037

Q01-F08 | 约束

The prohibition on index.delete applies to the entire workflow.

归属: skill / Entire query Skill | 原文: SKILL.md:L15-L15

运行	初审	理由与实际输出引用
1	保留	index.delete禁令位于整份Skill约束，整个CFG没有delete调用。 引用: skill.constraints
2	保留	index.delete禁令位于整份Skill约束，整个CFG没有delete调用。 引用: skill.constraints
3	保留	index.delete禁令位于整份Skill约束，整个CFG没有delete调用。 引用: skill.constraints

Q01-F09 | 不得凭空增加

Fuzzy-matching capability alone does not add a fuzzy-expansion operation.

归属: operation / Search index | 原文: SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	模糊匹配仅作能力说明/约束，实际search前后均没有扩展步骤。 引用: skill.constraints, ir_009
2	保留	能力说明保持非执行含义，实际操作序列没有模糊扩展。 引用: skill.constraints, ir_012
3	保留	诊断明确能力说明非执行，四条实际search路径均不添加模糊扩展。 引用: diagnostics, ir_014, ir_016, ir_018, ir_020

Q01-F10 | 不得凭空增加

Date format and parameter requirements do not add validation or normalization operations.

归属: operation / Search argument preparation | 原文: SKILL.md:L11-L11; SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	保留	仅有存在判断及原字段取值；日期要求保留为调用约束，没有日期校验/规范化。 引用: skill.constraints, ir_004, ir_007, ir_009
2	保留	存在检查服务于显式省略规则，未增加格式校验或内容归一化步骤。 引用: skill.constraints, ir_004, ir_005, ir_007, ir_010, ir_012
3	保留	存在检查与字段提取只落实参数分支；未新增日期解析校验、标准化或模糊处理。 引用: skill.constraints, ir_004, ir_005, ir_007, ir_010, ir_012

Q01-F11 | 行为

After the search and before returning items, separately write response.total to local count.txt.

归属: operation / Write result count | 原文: SKILL.md:L17-L17; SKILL.md:L18-L18

运行	初审	理由与实际输出引用
1	保留	search的response.total通过result_007写入count.txt，同一块中写入先于items返回。 引用: ir_009, ir_011, ir_012, edge_008
2	保留	唯一search之后write_total消费search_total写count.txt，再执行return。 引用: ir_012, ir_014, ir_015, edge_009
3	保留	四条互斥路径各将本次response.total取出并写count.txt，随后返回同次items。 引用: ir_022, ir_023, ir_026, ir_027, ir_030, ir_031, ir_034, ir_035

Q02 查询调用与参数使用 · 连贯段落

Q02 | 查询调用与参数使用 | 自然语言段落 | 保留验证集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	8	8	9
边数	9	9	10
操作数	16	18	19
三层约束条数	9	11	9
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三次均为唯一search及显式字段存在分支。第2次把存在标记一并交给search；第1/3次用调用约束保留省略条件。三次actual result均可追溯request.json和同一次search响应，未新增模糊扩展、格式验证或归一化。

逐事实依据冻结原文与本批次三次已提交CFG独立初审；结构complete不代表语义通过，结论仍待共同复核。参数格式和能力说明均未凭空新增执行行为。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 Q02 的三次明细。

事实表以外的发现

运行 1, 2, 3 | search的固定inputs同时列两项可选值，缺失路径绕过取值；局部constraints明确省略，第二次另传存在标记。冻结规范允许调用附着规则，但条件性result消费机制仍是表达边界，不等同实际执行已通过。

引用：block_007

逐事实初审（参考事实已确认，结果判定待共同复核）

Q02-F01 | 行为

Read request.json as the source of term and optional from_date and limit.

归属: operation / Read query request | 原文: SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	外部request.json被读取并解析为term及可选字段存在标记，未换用context。 引用: ir_001, ir_003
2	保留	读取request.json，再按存在情况提取两项可选字段。 引用: ir_001, ir_003, ir_005, ir_007, ir_009, ir_011
3	保留	读取外部request.json，并以其result作为term及可选字段取值来源。 引用: ir_001, ir_003, ir_005, ir_007, ir_009, ir_011

Q02-F02 | 行为

Invoke index.search exactly once with request.term unchanged as query.

归属: operation / Search index | 原文: SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	唯一search实际消费request_term，约束明确exactly once与query unchanged。 引用: ir_011
2	保留	唯一search调用使用request_term，调用局部约束保留次数和query原值。 引用: ir_013
3	保留	单一search消费request_term；次数在Skill约束、query不变在调用约束中保留。 引用: ir_013, skill.constraints

Q02-F03 | 约束

A present from_date is passed unchanged and must have the documented YYYY-MM-DD format.

归属: operation / Search index: from_date argument | 原文: SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	存在分支读取from_date原值并用于search，YYYY-MM-DD与unchanged位于调用约束。 引用: ir_006, ir_011.constraints
2	保留	实际from_date结果传入search，调用附着原样/格式要求。 引用: ir_007, ir_013.constraints
3	保留	存在时提取from_date并传给search，格式约束未转化为校验步骤。 引用: ir_007, ir_013.constraints

Q02-F04 | 条件

If from_date is absent, omit that argument rather than send an empty or null value.

归属: operation / Search index: from_date argument | 原文: SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	from_date缺失分支跳过取值；调用约束明确omit，未传空串/null。冻结规范允许条件规则附着在调用。 引用: ir_011.constraints, edge_004
2	保留	缺失路径跳过from_date提取，调用有存在标记和omit约束，无空值替代。 引用: ir_013.constraints, edge_004
3	保留	缺失条件边与调用omit约束一致，无新造的null或空串。 引用: ir_013.constraints, edge_004

Q02-F05 | 数据流

A present limit is passed unchanged as the limit argument.

归属: operation / Search index: limit argument | 原文: SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	limit存在分支从请求提取并传至search，unchanged明确保留。 引用: ir_009, ir_011.constraints
2	保留	实际limit取值result进入search，调用约束保持原值。 引用: ir_011, ir_013.constraints
3	保留	limit从request提取并绑定search，原值规则保留。 引用: ir_011, ir_013.constraints

Q02-F06 | 条件

If limit is absent, omit the limit argument; do not supply an invented default.

归属: operation / Search index: limit argument | 原文: SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	limit缺失时绕过取值，调用保留omit且未增加默认常量。 引用: ir_011.constraints, edge_007
2	保留	缺失条件分支与调用omit规则保留，没有默认值。 引用: ir_013.constraints, edge_007
3	保留	缺失路径直接到search，调用约束明确省略limit，没有默认常量。 引用: ir_013.constraints, edge_007

Q02-F07 | 数据流

Return the response.items value unchanged.

归属: operation / Return query result | 原文: SKILL.md:L18-L18

运行	初审	理由与实际输出引用
1	保留	从该search_response提取items，return直接消费该items结果；没有转换。 引用: ir_014, ir_016
2	保留	items从单次search_response取出，return用同一result并保留块级unchanged。 引用: ir_016, ir_018, block_008.constraints
3	保留	从单次response取出items后原结果直接return，无改写步骤。 引用: ir_018, ir_019

Q02-F08 | 约束

The prohibition on index.delete applies to the entire workflow.

归属: skill / Entire query Skill | 原文: SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	整份Skill保留index.delete禁令，实际CFG不存在delete调用。 引用: skill.constraints
2	保留	整份Skill保留index.delete禁令，实际CFG不存在delete调用。 引用: skill.constraints
3	保留	整份Skill保留index.delete禁令，实际CFG不存在delete调用。 引用: skill.constraints

Q02-F09 | 不得凭空增加

Fuzzy-matching capability alone does not add a fuzzy-expansion operation.

归属: operation / Search index | 原文: SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	能力说明未变成执行步骤，search没有fuzzy参数/扩展操作。 引用: skill.constraints, ir_011
2	保留	fuzzy能力仅为非执行说明，没有新增扩展操作。 引用: skill.constraints, ir_013
3	保留	能力说明被保留，实际search链不增加模糊扩展。 引用: skill.constraints, ir_013

Q02-F10 | 不得凭空增加

Date format and parameter requirements do not add validation or normalization operations.

归属: operation / Search argument preparation | 原文: SKILL.md:L10-L10; SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	保留	JSON字段解析与存在判断落实读取/参数省略; 未加入日期合法性校验或归一化。 引用: skill.constraints, ir_003, ir_006, ir_009
2	保留	仅判断字段是否存在并保持原值, 不把YYYY-MM-DD变成验证步骤。 引用: skill.constraints, ir_005, ir_009, ir_013
3	保留	存在检查落实明确条件, 格式仍为调用要求; 无新增校验/规范化。 引用: skill.constraints, ir_005, ir_009, ir_013

Q02-F11 | 行为

After the search and before returning items, separately write response.total to local count.txt.

归属: operation / Write result count | 原文: SKILL.md:L16-L16; SKILL.md:L18-L18

运行	初审	理由与实际输出引用
1	保留	从同一search_response取total, 写入count.txt后返回items。 引用: ir_013, ir_015, ir_016
2	保留	total与items都出自该search_response, 先写count.txt再return。 引用: ir_015, ir_017, ir_018
3	保留	搜索后单独提取total写count.txt, 再沿后继返回items。 引用: ir_015, ir_016, ir_019, edge_009, edge_010

Q03 查询调用与参数使用 · 表格

Q03 | 查询调用与参数使用 | 表格 | 保留验证集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	error	complete
块数	7	未记录	3
边数	8	未记录	2
操作数	17	未记录	6
三层约束条数	7	未记录	7
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	1	0

三次运行差异与初审意见

第1次显式存在分支加search局部规则，第3次直接将可选参数规则附着唯一search；两次均保留读请求、写total、返回items。第2次HTTP 200后IncompleteRead断流，未接受CFG，全部事实无法判断；不把基础设施失败算语义错误或成功。

基于本批次已提交产物的外置初审。第1/3次表格参数语义在调用约束中保留，按冻结可接受变体认可；静态可选操作数表示另列边界。第2次无完整输出，不能推断语义。所有结论待共同复核。

已接受 CFG 的精确 JSON 有 2 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 Q03 的三次明细。

事实表以外的发现

运行 1, 3 | search静态inputs包含两项可选结果，省略条件附着于该调用；第1次另有取值存在分支，第3次没有分支。冻结规范允许调用附着规则，因此保留参数要求，但本初审不把它等同于已验证的运行时可选操作数机制。

引用：无局部引用

逐事实初审（参考事实已确认，结果判定待共同复核）

Q03-F01 | 行为

Read request.json as the source of term and optional from_date and limit.

归属: operation / Read query request | 原文: SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	读取外部request.json，并从同一结果取得term、可选字段值和存在情况。 引用: ir_001, ir_003, ir_004, ir_006, ir_008, ir_010
2	无法判断	该次生成在收到HTTP 200后发生IncompleteRead断流，未形成接受的CFG，无法判定此事实；不能以传输失败推断理解错误。 引用: diagnostics
3	保留	实际读取request.json并产出term/from_date/limit，未改为运行上下文。 引用: ir_001

Q03-F02 | 行为

Invoke index.search exactly once with request.term unchanged as query.

归属: operation / Search index | 原文: SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	唯一index.search调用消费request的term结果，局部约束明确exactly once及query unchanged。 引用: ir_012
2	无法判断	该次生成在收到HTTP 200后发生IncompleteRead断流，未形成接受的CFG，无法判定此事实；不能以传输失败推断理解错误。 引用: diagnostics
3	保留	单一index.search节点消费request_term，Skill约束及调用约束保留次数与query原值。 引用: ir_003, skill.constraints

Q03-F03 | 约束

A present from_date is passed unchanged and must have the documented YYYY-MM-DD format.

归属: operation / Search index: from_date argument | 原文: SKILL.md:L16-L16; SKILL.md:L13-L15

运行	初审	理由与实际输出引用
1	保留	from_date存在分支提取请求值，调用附着原样传递和YYYY-MM-DD格式要求。 引用: ir_006, ir_012.constraints
2	无法判断	该次生成在收到HTTP 200后发生IncompleteRead断流，未形成接受的CFG，无法判定此事实；不能以传输失败推断理解错误。 引用: diagnostics
3	保留	调用使用实际request_from_date，命名约束明确存在时原样传递和YYYY-MM-DD，未增加格式转换。 引用: ir_003.constraints

Q03-F04 | 条件

If from_date is absent, omit that argument rather than send an empty or null value.

归属: operation / Search index: from_date argument | 原文: SKILL.md:L16-L16; SKILL.md:L13-L15

运行	初审	理由与实际输出引用
1	保留	缺失条件边绕过from_date取值，调用约束明确omit，无null或空串；冻结可接受变体允许规则附着调用。 引用: ir_012.constraints, edge_003
2	无法判断	该次生成在收到HTTP 200后发生IncompleteRead断流，未形成接受的CFG，无法判定此事实；不能以传输失败推断理解错误。 引用: diagnostics
3	保留	from_date缺失时省略的规则完整附着实际search调用，未增加null/空串字面量。冻结可接受变体允许此种调用约束表达。 引用: ir_003.constraints

Q03-F05 | 数据流

A present limit is passed unchanged as the limit argument.

归属: operation / Search index: limit argument | 原文: SKILL.md:L17-L17; SKILL.md:L13-L15

运行	初审	理由与实际输出引用
1	保留	limit存在路径提取请求值并供search使用，unchanged约束保留。 引用: ir_010, ir_012.constraints
2	无法判断	该次生成在收到HTTP 200后发生IncompleteRead断流，未形成接受的CFG，无法判定此事实；不能以传输失败推断理解错误。 引用: diagnostics
3	保留	search消费request_limit结果，明确存在时原样作为limit传递。 引用: ir_003.constraints

Q03-F06 | 条件

If limit is absent, omit the limit argument; do not supply an invented default.

归属: operation / Search index: limit argument | 原文: SKILL.md:L17-L17; SKILL.md:L13-L15

运行	初审	理由与实际输出引用
1	保留	缺失limit路径绕过取值，调用保留omit规则且没有默认常量。 引用: ir_012.constraints, edge_006
2	无法判断	该次生成在收到HTTP 200后发生IncompleteRead断流，未形成接受的CFG，无法判定此事实；不能以传输失败推断理解错误。 引用: diagnostics
3	保留	limit缺失时omit完整附着调用，实际输入没有虚构的默认字面量。 引用: ir_003.constraints

Q03-F07 | 数据流

Return the response.items value unchanged.

归属: operation / Return query result | 原文: SKILL.md:L25-L25

运行	初审	理由与实际输出引用
1	保留	从同次search_response提取items并直接return，没有改写操作。 引用: ir_015, ir_017
2	无法判断	该次生成在收到HTTP 200后发生IncompleteRead断流，未形成接受的CFG，无法判定此事实；不能以传输失败推断理解错误。 引用: diagnostics
3	保留	search的items结果直接被return消费，未增加转换。 引用: ir_003, ir_006

Q03-F08 | 约束

The prohibition on index.delete applies to the entire workflow.

归属: skill / Entire query Skill | 原文: SKILL.md:L19-L19

运行	初审	理由与实际输出引用
1	保留	index.delete禁令在整份Skill，实际CFG没有delete操作。 引用: skill.constraints
2	无法判断	该次生成在收到HTTP 200后发生IncompleteRead断流，未形成接受的CFG，无法判定此事实；不能以传输失败推断理解错误。 引用: diagnostics
3	保留	index.delete禁令在整份Skill，实际CFG没有delete操作。 引用: skill.constraints

Q03-F09 | 不得凭空增加

Fuzzy-matching capability alone does not add a fuzzy-expansion operation.

归属: operation / Search index | 原文: SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	fuzzy能力保持非执行说明，实际操作序列没有模糊扩展。 引用: skill.constraints, ir_012
2	无法判断	该次生成在收到HTTP 200后发生IncompleteRead断流，未形成接受的CFG，无法判定此事实；不能以传输失败推断理解错误。 引用: diagnostics
3	保留	能力说明未成为额外执行步骤或fuzzy参数。 引用: skill.constraints, ir_003

Q03-F10 | 不得凭空增加

Date format and parameter requirements do not add validation or normalization operations.

归属: operation / Search argument preparation | 原文: SKILL.md:L16-L16; SKILL.md:L13-L15; SKILL.md:L21-L21

运行	初审	理由与实际输出引用
1	保留	存在检查落实显式可选规则，没有日期校验、规范化或额外转换。 引用: skill.constraints, ir_004, ir_008, ir_012
2	无法判断	该次生成在收到HTTP 200后发生IncompleteRead断流，未形成接受的CFG，无法判定此事实；不能以传输失败推断理解错误。 引用: diagnostics
3	保留	参数格式仅在调用约束中，无验证、规范化步骤。 引用: skill.constraints, ir_003

Q03-F11 | 行为

After the search and before returning items, separately write response.total to local count.txt.

归属: operation / Write result count | 原文: SKILL.md:L23-L23; SKILL.md:L25-L25

运行	初审	理由与实际输出引用
1	保留	search后取同次response.total写count.txt，然后返回同次items。 引用: ir_014, ir_016, ir_017, edge_008
2	无法判断	该次生成在收到HTTP 200后发生IncompleteRead断流，未形成接受的CFG，无法判定此事实；不能以传输失败推断理解错误。 引用: diagnostics
3	保留	search产生total和items，write消费total写count.txt，随后return消费items。 引用: ir_003, ir_005, ir_006, edge_002

Q04 查询调用与参数使用 · 跨文件混合

Q04 | 查询调用与参数使用 | 跨文件混合 | 保留验证集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	13	11	7
边数	12	13	8
操作数	29	22	14
三层约束条数	34	25	14
生成尝试	1	2	1
结构修复	0	1	0
HTTP 尝试	1	2	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

第1次四个互斥search及独立后处理，第2次四类参数集合经select后调用唯一search，第3次为存在分支和调用约束。三次正确合并跨文件规则，query.yaml未成为额外运行时读取。

逐事实依据冻结原文与本批次三次已提交CFG独立初审；结构complete不代表语义通过，结论仍待共同复核。参数格式和能力说明均未凭空新增执行行为。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 Q04 的三次明细。

事实表以外的发现

运行 3 | 第3次唯一search的inputs列两个条件性pass结果；缺失省略保留在调用约束，不能据此宣称实际运行时可选操作数机制已经得到验证。

引用：ir_011

运行 1, 2, 3 | 逐块核对未见query.yaml的运行时read；它只作为固定缺失参数规则进入constraints，符合跨文件静态配置边界。

引用：skill.constraints

逐事实初审（参考事实已确认，结果判定待共同复核）

Q04-F01 | 行为

Read request.json as the source of term and optional from_date and limit.

归属: operation / Read query request | 原文: references/workflow.md:L5-L5; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	跨文件引用的当前workflow落实为读取外部request.json及各字段/存在标记。 引用: ir_001, ir_003, ir_004, ir_005
2	保留	request.json读取结果是term、from_date、limit及存在标记的唯一来源。 引用: ir_001, ir_003
3	保留	读取外部request.json并产出term与两项可选字段，未增加配置文件运行时读取。 引用: ir_001

Q04-F02 | 行为

Invoke index.search exactly once with request.term unchanged as query.

归属: operation / Search index | 原文: references/workflow.md:L7-L7; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	四条互斥参数组合每次只选一条search，均使用request_term原值。 引用: ir_010, ir_016, ir_022, ir_026, edge_002, edge_003, edge_004, edge_005
2	保留	四条参数构造路径选择当前参数集合后进入唯一search，query追溯request_term并保留unchanged。 引用: ir_007, ir_009, ir_011, ir_013, ir_015, ir_017
3	保留	单一search实际使用request_term；一次调用与原值规则均保留。 引用: ir_011, skill.constraints

Q04-F03 | 约束

A present from_date is passed unchanged and must have the documented YYYY-MM-DD format.

归属: operation / Search index: from_date argument | 原文: references/workflow.md:L10-L10;
references/workflow.md:L9-L9; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	from_date存在的两条search传实际提取值，保持YYYY-MM-DD及unchanged。 引用: ir_010.constraints, ir_016.constraints
2	保留	含from_date的参数集合消费请求原值并保留格式要求，最终search再次承接同一规则。 引用: ir_007.constraints, ir_009.constraints, ir_017.constraints
3	保留	from_date的pass操作仅原值传递，日期格式附着search。 引用: ir_005, ir_011.constraints

Q04-F04 | 条件

If from_date is absent, omit that argument rather than send an empty or null value.

归属: operation / Search index: from_date argument | 原文: references/workflow.md:L11-L11;
references/workflow.md:L9-L9; references/workflow.md:L15-L15; SKILL.md:L8-L8; query.yaml:L2-L2; query.yaml:L1-L1

运行	初审	理由与实际输出引用
1	保留	缺少from_date两条search的操作数实际不含该参数，query.yaml的omit规则被应用。 引用: ir_022, ir_026
2	保留	from_date缺失路径构造的参数集合不含该值，search保留omit规则。 引用: ir_011, ir_013, ir_017.constraints
3	保留	缺失分支跳过pass，调用约束明确omit；固定query.yaml按静态参数规则处理。 引用: ir_011.constraints, edge_003

Q04-F05 | 数据流

A present limit is passed unchanged as the limit argument.

归属: operation / Search index: limit argument | 原文: references/workflow.md:L12-L12;
references/workflow.md:L9-L9; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	limit存在路径提取请求值，两个search均明确unchanged。 引用: ir_008, ir_020, ir_010.constraints, ir_022.constraints
2	保留	存在路径参数集合包含request_limit原值，最终调用无转换。 引用: ir_007, ir_011, ir_017.constraints
3	保留	limit经原值pass进入search，保持unchanged。 引用: ir_009, ir_011.constraints

Q04-F06 | 条件

If limit is absent, omit the limit argument; do not supply an invented default.

归属: operation / Search index: limit argument | 原文: references/workflow.md:L13-L13;
references/workflow.md:L9-L9; references/workflow.md:L15-L15; SKILL.md:L8-L8; query.yaml:L3-L3; query.yaml:L1-L1

运行	初审	理由与实际输出引用
1	保留	limit缺失的两条实际search输入无limit或默认常量，静态配置omit得到保留。 引用: ir_016, ir_026
2	保留	缺失limit的参数集合不含该值，search明确omit，无新默认常量。 引用: ir_009, ir_013, ir_017.constraints
3	保留	limit缺失路径跳过pass且调用明确omit，无默认值。 引用: ir_011.constraints, edge_006

Q04-F07 | 数据流

Return the response.items value unchanged.

归属: operation / Return query result | 原文: references/workflow.md:L23-L23; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	四个互斥后处理分支直接返回各自search产生的items，并有unchanged约束。 引用: ir_013, ir_019, ir_025, ir_029
2	保留	从唯一search_response取items并直接return，块约束保持原样。 引用: ir_021, ir_022, block_011.constraints
3	保留	search_items result直接被return消费，块约束保留unchanged。 引用: ir_011, ir_014, block_007.constraints

Q04-F08 | 约束

The prohibition on index.delete applies to the entire workflow.

归属: skill / Entire query Skill | 原文: references/workflow.md:L17-L17; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	跨文件workflow中的index.delete禁令提升为整份Skill约束，所有路径均无delete。 引用: skill.constraints
2	保留	跨文件workflow中的index.delete禁令提升为整份Skill约束，所有路径均无delete。 引用: skill.constraints
3	保留	跨文件workflow中的index.delete禁令提升为整份Skill约束，所有路径均无delete。 引用: skill.constraints

Q04-F09 | 不得凭空增加

Fuzzy-matching capability alone does not add a fuzzy-expansion operation.

归属: operation / Search index | 原文: references/workflow.md:L3-L3; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	引用块中的能力说明没有新增fuzzy扩展或查询参数。 引用: skill.constraints, ir_010, ir_016, ir_022, ir_026
2	保留	fuzzy能力仍非执行说明，实际search禁止增加扩展，未生成扩展步骤。 引用: skill.constraints, ir_017.constraints
3	保留	能力说明转为不执行模糊扩展的约束，实际CFG无该操作。 引用: skill.constraints, ir_011

Q04-F10 | 不得凭空增加

Date format and parameter requirements do not add validation or normalization operations.

归属: operation / Search argument preparation | 原文: references/workflow.md:L10-L10;
references/workflow.md:L9-L9; SKILL.md:L8-L8; references/workflow.md:L19-L19

运行	初审	理由与实际输出引用
1	保留	存在检查服务于显式参数分支；未因日期格式或YAML增加验证/标准化操作。 引用: skill.constraints, ir_004, ir_005
2	保留	参数构造和当前路径选择落实可选字段要求，未加入值规范化或格式校验。 引用: skill.constraints, ir_007, ir_009, ir_011, ir_013, ir_017
3	保留	from_date/limit仅原值pass，YYYY-MM-DD保留为调用约束，没有校验/转换。 引用: ir_011.constraints, ir_005, ir_009

Q04-F11 | 行为

After the search and before returning items, separately write response.total to local count.txt.

归属: operation / Write result count | 原文: references/workflow.md:L21-L21; SKILL.md:L8-L8;
references/workflow.md:L23-L23

运行	初审	理由与实际输出引用
1	保留	四条实际路径分别用同次search_total写count.txt，然后返回对应items。 引用: ir_012, ir_018, ir_024, ir_028
2	保留	唯一search之后提取total写count.txt，写入先于items返回。 引用: ir_019, ir_020, ir_022
3	保留	search产出total与items，先用total写count.txt，再return items。 引用: ir_011, ir_013, ir_014

Q05 查询调用与参数使用 · 用途最小差异

Q05 | 查询调用与参数使用 | 最小差异：用途 | 保留验证集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	9	9	7
边数	10	8	8
操作数	17	18	14
三层约束条数	9	16	11
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三次均保留真实读request、调用search和返回items，只移除不执行的历史count写入。第2次四个互斥调用路径明确省略参数；第1/3次为唯一调用加局部规则。没有把历史示例的非执行性扩大到整个流程。

逐事实依据冻结原文与本批次三次已提交CFG独立初审；结构complete不代表语义通过，结论仍待共同复核。参数格式和能力说明均未凭空新增执行行为。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 Q05 的三次明细。

事实表以外的发现

运行 1, 3 | 第1/3次使用唯一search配合局部省略约束，静态inputs仍列条件性产生的可选值；记录该表示边界，不把缺少值默认为null，也不强制增加原文未要求的验证。

引用：无局部引用

逐事实初审（参考事实已确认，结果判定待共同复核）

Q05-F01 | 行为

Read request.json as the source of term and optional from_date and limit.

归属: operation / Read query request | 原文: SKILL.md:L9-L9

运行	初审	理由与实际输出引用
1	保留	历史count示例没有删除当前读文件行为，实际读取request.json并提取字段。 引用: ir_001, ir_003, ir_004, ir_005
2	保留	实际读取request.json并产生term、可选值及存在标记；没有将整个流程当作示例。 引用: ir_001
3	保留	真实读取request.json，存在检查从请求对象读取字段情况。 引用: ir_001, ir_003, ir_007

Q05-F02 | 行为

Invoke index.search exactly once with request.term unchanged as query.

归属: operation / Search index | 原文: SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	保留唯一search调用与request_term输入，exactly once和query不变规则均在。 引用: ir_014, skill.constraints
2	保留	四条互斥条件路径每次执行一个search，全部消费原始request_term。 引用: ir_003, ir_005, ir_007, ir_009, skill.constraints, edge_001, edge_002, edge_003, edge_004
3	保留	唯一search使用request_term，整份Skill要求一次调用且query原样。 引用: ir_011, skill.constraints

Q05-F03 | 约束

A present from_date is passed unchanged and must have the documented YYYY-MM-DD format.

归属: operation / Search index: from_date argument | 原文: SKILL.md:L11-L11

运行	初审	理由与实际输出引用
1	保留	存在时提取from_date并绑定search，格式和原值规则在调用约束。 引用: ir_007, ir_014.constraints
2	保留	from_date存在的两条search实际消费请求值并要求YYYY-MM-DD和unchanged。 引用: ir_003.constraints, ir_005.constraints
3	保留	原值pass产生from_date_argument，格式要求在pass与search保留，无日期转换。 引用: ir_005.constraints, ir_011.constraints

Q05-F04 | 条件

If from_date is absent, omit that argument rather than send an empty or null value.

归属: operation / Search index: from_date argument | 原文: SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	缺失条件进入omit命名块且调用保留省略规则；不新增null或空串。 引用: ir_014.constraints, edge_003, block_004
2	保留	from_date缺失的两个search实际没有from_date输入。 引用: ir_007, ir_009
3	保留	缺失边明确omit且调用规则一致，没有空值替代。 引用: ir_011.constraints, edge_003

Q05-F05 | 数据流

A present limit is passed unchanged as the limit argument.

归属: operation / Search index: limit argument | 原文: SKILL.md:L13-L13

运行	初审	理由与实际输出引用
1	保留	limit存在路径提取原字段并传入search，unchanged规则明确。 引用: ir_011, ir_014.constraints
2	保留	limit存在的两个search消费原始request.limit，局部约束保持不变。 引用: ir_003, ir_007
3	保留	limit原值pass到调用，未引入转换或新默认值。 引用: ir_009.constraints, ir_011.constraints

Q05-F06 | 条件

If limit is absent, omit the limit argument; do not supply an invented default.

归属: operation / Search index: limit argument | 原文: SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	limit缺失路径进入omit块且调用保持omit，无默认常量。 引用: ir_014.constraints, edge_007, block_007
2	保留	limit缺失两个调用输入不含limit，且约束明确omit。 引用: ir_005, ir_009
3	保留	缺失边与调用共同保留omit语义，未新增默认值。 引用: ir_011.constraints, edge_006

Q05-F07 | 数据流

Return the response.items value unchanged.

归属: operation / Return query result | 原文: SKILL.md:L18-L18

运行	初审	理由与实际输出引用
1	保留	从实际search_response提取items并按原值return，历史示例不影响真实结果返回。 引用: ir_016, ir_017
2	保留	四条路径从各自search_response取items再原样return，无跨路径混合。 引用: ir_011, ir_012, ir_013, ir_014, ir_015, ir_016, ir_017, ir_018
3	保留	search_response的items提取结果直接return并明确unchanged。 引用: ir_013, ir_014

Q05-F08 | 约束

The prohibition on index.delete applies to the entire workflow.

归属: skill / Entire query Skill | 原文: SKILL.md:L15-L15

运行	初审	理由与实际输出引用
1	保留	index.delete全流程禁令保留，所有路径均未生成delete。 引用: skill.constraints
2	保留	index.delete全流程禁令保留，所有路径均未生成delete。 引用: skill.constraints
3	保留	index.delete全流程禁令保留，所有路径均未生成delete。 引用: skill.constraints

Q05-F09 | 不得凭空增加

Fuzzy-matching capability alone does not add a fuzzy-expansion operation.

归属: operation / Search index | 原文: SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	诊断明确fuzzy为能力说明，实际操作无扩展步骤。 引用: diagnostics, ir_014
2	保留	四条查询分支没有扩展操作；能力说明非执行。 引用: skill.constraints, ir_003, ir_005, ir_007, ir_009
3	保留	能力说明的非执行边界保留，未增加模糊扩展。 引用: diagnostics, skill.constraints, ir_011

Q05-F10 | 不得凭空增加

Date format and parameter requirements do not add validation or normalization operations.

归属: operation / Search argument preparation | 原文: SKILL.md:L11-L11; SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	保留	存在判断仅落实可选参数规则，不等于格式校验；实际无规范化操作。 引用: skill.constraints, ir_004, ir_005, ir_014
2	保留	直接按存在标记分支调用，未添加日期校验或标准化步骤。 引用: skill.constraints, ir_001, ir_003, ir_005
3	保留	pass操作仅传递原值，未因参数要求新增转换/校验。 引用: skill.constraints, ir_005, ir_009

Q05-F11 | 不得凭空增加

Writing response.total to count.txt is an explicitly inactive historical example, not a current action.

归属: operation / Historical count example | 原文: SKILL.md:L17-L17

运行	初审	理由与实际输出引用
1	保留	诊断明确count.txt写入为不执行的历史示例，完整CFG没有total提取或文件写入，真实search和return仍保留。 引用: diagnostics, ir_014, ir_017
2	保留	历史count写入被排除，四条实际路径只提取items并return，没有写count.txt。 引用: diagnostics, ir_012, ir_014, ir_016, ir_018
3	保留	历史count写入未进入CFG；流程仍真实执行查询并返回items。 引用: diagnostics, ir_011, ir_014

Q06 查询调用与参数使用 · 语义最小差异

Q06 | 查询调用与参数使用 | 最小差异：语义 | 保留验证集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	10	14	10
边数	11	13	13
操作数	22	38	19
三层约束条数	10	24	18
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三次均识别limit缺失时传10；第1次有显式default与resolve，第2/3次在缺失分支直接传字面量10。第3次汇合后的write/return同时列四套互斥结果，缺少当前响应选择，F07/F11判部分保留。

逐事实依据冻结原文与本批次三次已提交CFG独立初审；结构complete不代表语义通过，结论仍待共同复核。第3次response结果汇合存在实际数据引用边界。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 Q06 的三次明细。

事实表以外的发现

运行 3 | 四条互斥search输出汇入一个写入和return，操作数同时列全部分支值而未提供当前路径选择。该问题影响F07/F11的单一response数据绑定，不能由约束原文或结构校验通过消除。

引用：ir_018, ir_019

逐事实初审（参考事实已确认，结果判定待共同复核）

Q06-F01 | 行为

Read request.json as the source of term and optional from_date and limit.

归属: operation / Read query request | 原文: SKILL.md:L9-L9

运行	初审	理由与实际输出引用
1	保留	外部request.json提供term与可选字段，不因新增默认limit而改写输入来源。 引用: ir_001, ir_003, ir_004, ir_008
2	保留	读外部request.json，term和存在标记均从该文件取得。 引用: ir_001, ir_003, ir_005, ir_006
3	保留	从request.json提取term与字段存在标记，未更换为context。 引用: ir_001, ir_003

Q06-F02 | 行为

Invoke index.search exactly once with request.term unchanged as query.

归属: operation / Search index | 原文: SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	唯一search使用request_term原值，次数及query不变约束完整。 引用: ir_016, skill.constraints
2	保留	四个互斥组合每次只进入一次search，各调用保持query原值。 引用: ir_011, ir_015, ir_019, ir_021, edge_003, edge_004, edge_005, edge_006
3	保留	控制流依据两个存在条件选择唯一search，实际输入保持request_term。 引用: ir_010, ir_012, ir_014, ir_016, skill.constraints

Q06-F03 | 约束

A present from_date is passed unchanged and must have the documented YYYY-MM-DD format.

归属: operation / Search index: from_date argument | 原文: SKILL.md:L11-L11

运行	初审	理由与实际输出引用
1	保留	from_date存在分支提取并传给search，原值和YYYY-MM-DD保留。 引用: ir_006, ir_016.constraints
2	保留	含from_date的两条调用使用实际提取值，原值/格式要求保留。 引用: ir_011.constraints, ir_015.constraints
3	保留	from_date存在路径提取原值并供对应search使用，格式留为约束。 引用: ir_005, ir_010.constraints, ir_012.constraints

Q06-F04 | 条件

If from_date is absent, omit that argument rather than send an empty or null value.

归属: operation / Search index: from_date argument | 原文: SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	from_date缺失分支跳过取值，调用明确omit，无空值替代。 引用: ir_016.constraints, edge_003
2	保留	from_date缺失两条调用的输入实际不包含该字段。 引用: ir_019, ir_021
3	保留	from_date缺失两条search实际省略该参数，无null或空串。 引用: ir_014, ir_016

Q06-F05 | 数据流

A present limit is passed unchanged as the limit argument.

归属: operation / Search index: limit argument | 原文: SKILL.md:L13-L13

运行	初审	理由与实际输出引用
1	保留	存在时提取request_limit，resolve依据limit_present选择参数，调用保持原值。 引用: ir_010, ir_014, ir_016.constraints
2	保留	limit存在的两条路径实际取请求值传给search，没有改用默认10。 引用: ir_009, ir_017, ir_011, ir_019
3	保留	limit存在路径提取原值，并分别在有/无from_date的调用中使用。 引用: ir_008, ir_010, ir_014

Q06-F06 | 条件

If limit is absent, explicitly pass 10 as the limit argument.

归属: operation / Search index: limit argument | 原文: SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	limit缺失条件进入use_default_limit，literal 10经resolve_limit_argument到search，正确保留最小差异。 引用: ir_012, ir_014, ir_016, edge_006
2	保留	两条limit缺失search均直接以literal 10作为limit输入，没有沿用Q01的省略规则。 引用: ir_015, ir_021, edge_004, edge_006
3	保留	两个limit缺失分支直接传literal 10，正确区别于普通样例的omit。 引用: ir_012, ir_016, edge_005, edge_007

Q06-F07 | 数据流

Return the response.items value unchanged.

归属: operation / Return query result | 原文: SKILL.md:L18-L18

运行	初审	理由与实际输出引用
1	保留	从唯一search_response提取items并直接return，没有转换。 引用: ir_021, ir_022
2	保留	四条互斥路径分别取本次response.items原样return，绑定不跨路径。 引用: ir_025, ir_026, ir_029, ir_030, ir_033, ir_034, ir_037, ir_038
3	部分保留	return保留items unchanged文字，但inputs同时消费四个互斥search_items；没有当前路径选择或合并result，不能确认只返回本次单一response.items。 引用: ir_019

Q06-F08 | 约束

The prohibition on index.delete applies to the entire workflow.

归属: skill / Entire query Skill | 原文: SKILL.md:L15-L15

运行	初审	理由与实际输出引用
1	保留	整份Skill保留index.delete禁令，实际路径没有该调用。 引用: skill.constraints
2	保留	整份Skill保留index.delete禁令，实际路径没有该调用。 引用: skill.constraints
3	保留	整份Skill保留index.delete禁令，实际路径没有该调用。 引用: skill.constraints

Q06-F09 | 不得凭空增加

Fuzzy-matching capability alone does not add a fuzzy-expansion operation.

归属: operation / Search index | 原文: SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	fuzzy仍是能力说明，实际单次search未新增模糊扩展。 引用: skill.constraints, ir_016
2	保留	四条实际调用只处理原参数与明确默认10，没有fuzzy扩展。 引用: skill.constraints, ir_011, ir_015, ir_019, ir_021
3	保留	能力说明未扩张为操作，四个search节点均无新增模糊处理。 引用: skill.constraints, ir_010, ir_012, ir_014, ir_016

Q06-F10 | 不得凭空增加

Date format and parameter requirements do not add validation or normalization operations.

归属: operation / Search argument preparation | 原文: SKILL.md:L11-L11; SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	保留	默认10和参数选择落实明确limit规则，不是额外验证/标准化；实际没有日期校验。 引用: skill.constraints, ir_012, ir_014
2	保留	存在检查与字面量10直接落实原要求，没有预调用格式校验或归一化。 引用: skill.constraints, ir_005, ir_006
3	保留	仅检查存在与取字段，未增添日期校验、转换或默认值推断。 引用: skill.constraints, ir_003, ir_008

Q06-F11 | 行为

After the search and before returning items, separately write response.total to local count.txt.

归属: operation / Write result count | 原文: SKILL.md:L17-L17; SKILL.md:L18-L18

运行	初审	理由与实际输出引用
1	保留	search后提取total写count.txt，随后返回同次items，写入顺序独立明确。 引用: ir_018, ir_019, ir_022, edge_010, edge_011
2	保留	四条互斥路径各从当前search_response取total写count.txt，再return对应items。 引用: ir_023, ir_024, ir_027, ir_028, ir_031, ir_032, ir_035, ir_036
3	部分保留	count写入确在search之后且return之前，但write inputs同时列四条互斥search_total，未选择当前路径结果；来源和顺序部分保留，单次response.total绑定不充分。 引用: ir_018, ir_019

F01 工具选择与失败回退 · 编号步骤

F01 | 工具选择与失败回退 | 编号步骤 | 开发集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	14	13	10
边数	15	14	11
操作数	28	26	20
三层约束条数	8	37	19
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三次均1次模型生成、0结构修复、0抽取HTTP重试。实际业务CFG均表达密钥门控、仅首次瞬态错误的一次fast重试、archive单次回退和四个最终返回出口。第1/2次在结果分支内追加success/failure状态，第3次直接引用相应调用的status结果追加；这些是Skill业务语义，不与抽取服务重试次数混淆。

30条事实×重复判定均保留：实际结果数据流、受限重试、回退、成功后停止fetch、archive失败返回以及每个出口前的状态追加均有CFG证据。FAST_KEY全局禁令和archive唯一参数限制得到保留。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 F01 的三次明细。

逐事实初审（参考事实已确认，结果判定待共同复核）

F01-F01 | 行为

Read source_id from the request and FAST_KEY from the environment.

归属: operation / Read fetch inputs | 原文: SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_001 从请求 source_id 读取 result_001, ir_003 从环境 FAST_KEY 读取 result_002; 来源由实际上下文操作数和读取操作明确区分。 引用: ir_001, ir_003
2	保留	ir_001/003 分别从请求source_id、环境FAST_KEY读取实际结果，两种来源没有混淆或用常量替代。 引用: ir_001, ir_003
3	保留	ir_001 使用user_request.source_id, ir_003使用environment.FAST_KEY，各自产出结果供后续调用；请求与环境来源显式区分。 引用: ir_001, ir_003

F01-F02 | 条件

FAST_KEY presence gates fast.fetch; prefer it when present and go directly to archive.fetch when absent.

归属: block / Preferred tool selection | 原文: SKILL.md:L9-L9

运行	初审	理由与实际输出引用
1	保留	ir_005 读取实际 fast_key 结果产生存在判定, ir_006 与 edge_003/004 决定先 fast.fetch 或直接 archive.fetch; 未补充未定义的具体 presence 检测规则。 引用: ir_005, ir_006, edge_003, edge_004
2	保留	ir_005/006 根据读取后的fast_key存在性分流, edge_003先进入fast, edge_004在缺失时直接进入archive，首选条件实际存在。 引用: ir_005, ir_006, edge_003, edge_004
3	保留	ir_005 从fast_key结果计算存在性, ir_006/edge_003/004分别进入首次fast或直接archive；没有新增空字符串等具体存在规则。 引用: ir_005, ir_006, edge_003, edge_004

F01-F03 | 数据流

`fast.fetch` receives `request.source_id` and the `FAST_KEY` read from the environment.

归属: operation / Preferred fetch | 原文: SKILL.md:L8-L8; SKILL.md:L9-L9

运行	初审	理由与实际输出引用
1	保留	ir_007 首次调用与 ir_015 唯一重试都实际消费请求读取的 source_id/result_001 和环境读取的 fast_key/result_002, 参数来源完整。 引用: ir_001, ir_003, ir_007, ir_015
2	保留	首次 ir_007 和重试 ir_011 都使用相同的source_id/result_001、fast_key/result_002, 来自对应读取操作。 引用: ir_001, ir_003, ir_007, ir_011
3	保留	ir_007 首次fast与ir_009重试均实际引用source_id/result_001、fast_key/result_002, 参数来源与源文一致。 引用: ir_001, ir_003, ir_007, ir_009

F01-F04 | 条件

Retry `fast.fetch` once only after a transient first failure; a non-transient first failure is not retried.

归属: block / Preferred fetch retry | 原文: SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	首次失败后 ir_013 检查命名的 transient_error, edge_008 只将瞬态失败送入唯一重试 ir_015; 非瞬态走 edge_009 archive, 无回到重试的边。 引用: ir_013, ir_014, ir_015, ir_015.constraints, edge_008, edge_009
2	保留	ir_009 对首次fast返回分类, edge_006仅瞬态错误进入ir_011唯一重试; edge_007非瞬态直接 archive, 重试块不存在再次重试的回环。 引用: ir_009, ir_010, ir_011, ir_011.constraints, edge_006, edge_007
3	保留	首次fast状态驱动edge_006仅在瞬态失败时进入ir_009唯一重试, edge_007非瞬态直达archive; 重试失败后不再经过首次或重试节点。 引用: ir_007, ir_008, ir_009, ir_009.constraints, edge_006, edge_007, edge_009

F01-F05 | 条件

A non-transient first failure or any failed retry causes archive.fetch fallback.

归属: block / Fallback after preferred failure | 原文: SKILL.md:L11-L11

运行	初审	理由与实际输出引用
1	保留	非瞬态首次失败 edge_009 与重试失败 edge_012 都进入同一个 archive.fetch 操作 ir_021, 不会再次调用 fast.fetch。 引用: ir_021, edge_009, edge_012
2	保留	首次非瞬态失败edge_007和任意重试失败edge_010均汇入ir_015 archive.fetch; 重试成功走自己的返回出口。 引用: ir_015, edge_007, edge_010, edge_011
3	保留	首次非瞬态失败edge_007和任意重试失败edge_009均进入ir_011 archive; 图中不存在第三次fast尝试。 引用: ir_011, edge_007, edge_009

F01-F06 | 约束

archive.fetch executes at most once, with source_id as its only argument.

归属: operation / Archive fetch | 原文: SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	ir_021 仅接收 archive.fetch 服务标识和 source_id/result_001, 不接收 FAST_KEY; 局部原文限制最多一次, 实际图无任何 archive 回环。 引用: ir_021, ir_021.constraints, edge_004, edge_009, edge_012
2	保留	ir_015 的实际参数只有source_id, 局部原文保留archive最多一次且只传source_id; 缺key、初次非瞬态、重试失败共用该单次节点且无回环。 引用: ir_015, ir_015.constraints, edge_004, edge_007, edge_010
3	保留	ir_011 archive只使用source_id/result_001, 局部保留最多一次原句, 所有入口共用且无重试回边, 不传FAST_KEY。 引用: ir_011, ir_011.constraints, edge_004, edge_007, edge_009

F01-F07 | 数据流

A successful fetch returns that same response.body unchanged and ends further fetch calls.

归属: block / Successful fetch exit | 原文: SKILL.md:L13-L13

运行	初审	理由与实际输出引用
1	保留	三个成功出口 ir_012/020/026 分别直接返回相应 fast首次/重试/archive产生的 body/result_004/008/011; 成功后仅追加状态并返回, 没有进一步fetch。 引用: ir_007, ir_015, ir_021, ir_012, ir_020, ir_026, edge_006, edge_011, edge_014
2	保留	ir_020/022/024 分别直接引用首次fast、重试fast、archive产生的body/result_005/008/011返回; 成功路径只追加状态再返回, 无后续fetch。 引用: ir_007, ir_011, ir_015, ir_020, ir_022, ir_024, edge_008, edge_011, edge_013
3	保留	三个成功出口ir_014/016/018分别直接返回fast首次body/result_004、重试body/result_006、archivebody/result_008; 成功后的路径不再fetch。 引用: ir_007, ir_009, ir_011, ir_014, ir_016, ir_018, edge_005, edge_008, edge_010

F01-F08 | 条件

An archive.fetch failure returns that error and stops without any archive retry.

归属: block / Archive failure exit | 原文: SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	archive失败 edge_015 进入最终失败块, ir_028 返回同次 archive_error/result_012; 没有 archive 重试或其他fetch出口。 引用: ir_021, ir_028, edge_015
2	保留	archive结果判断失败经edge_014进入ir_025/026, 先状态append后直接返回同次 archive_error/result_012, 图中无archive重试。 引用: ir_015, ir_017, ir_026, edge_014
3	保留	archive失败edge_011进入最后的状态追加与ir_020返回, 返回值是该次archive_error/result_009, 没有archive重试。 引用: ir_011, ir_019, ir_020, edge_011

F01-F09 | 约束

FAST_KEY must never flow to archive.fetch or diagnostic output anywhere in the Skill.

归属: skill / Entire fetch Skill | 原文: SKILL.md:L15-L15

运行	初审	理由与实际输出引用
1	保留	Skill全局完整保留 FAST_KEY 不得进入 archive 或 diagnostic 的原句; 实际 key 只流入两次 fast.fetch, archive和状态写入均无key参数。 引用: skill.constraints, ir_003, ir_007, ir_015, ir_021, ir_027
2	保留	FAST_KEY禁入archive和diagnostics的完整原文既在Skill全局也在archive局部, 实际archive和日志均不消费FAST_KEY结果。 引用: skill.constraints, ir_003, ir_015, ir_015.constraints, ir_019, ir_025
3	保留	禁止FAST_KEY进入archive或diagnostics的原句位于Skill全局并附于archive调用; key结果只被存在判断和fast调用使用, 未进入archive或status写入。 引用: skill.constraints, ir_003, ir_007, ir_009, ir_011, ir_013, ir_019

F01-F10 | 行为

Before returning from each success or failure path, separately append its final status to local status.txt.

归属: operation / Append final status | 原文: SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	保留	四个最终出口均先由 ir_011/019/025/027 append到本地status.txt，再执行对应return。成功/失败文字由实际工具结果分支控制，分别与最终结果对应；不是先return后补写。 引用: ir_011, ir_019, ir_025, ir_027, ir_012, ir_020, ir_026, ir_028, edge_006, edge_011, edge_014, edge_015
2	保留	四个成功/失败终点分别由ir_019/021/023/025先append对应状态，再由ir_020/022/024/026返回；每个日志值与进入该终点的真实结果分支一致，所有返回路径均覆盖。 引用: ir_019, ir_021, ir_023, ir_025, ir_020, ir_022, ir_024, ir_026, edge_008, edge_011, edge_013, edge_014
3	保留	ir_013/015/017/019在四个最终出口分别引用对应fast首次status/result_005、重试status/result_007或archivestatus/result_010追加status.txt，然后返回；时序与最终状态数据依赖均显式保留。 引用: ir_013, ir_015, ir_017, ir_019, ir_014, ir_016, ir_018, ir_020

F02 工具选择与失败回退 · 连贯段落

F02 | 工具选择与失败回退 | 自然语言段落 | 开发集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	error
块数	10	13	未记录
边数	11	14	未记录
操作数	20	26	未记录
三层约束条数	25	29	未记录
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	1

三次运行差异与初审意见

第1、2次各1次生成、无修复或HTTP重试，最终分别10块11边、13块14边；两份均建立完整业务重试与回退图。
第3次HTTP200后IncompleteRead，没有接受CFG，十条事实不可评估。
两个可判定结果的20条事实均保留；第三次十条不可评估，不能以传输故障推断模型语义表现。成功正文与失败错误都引用相应响应结果，所有最终出口先追加状态。

已接受 CFG 的精确 JSON 有 2 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 F02 的三次明细。

逐事实初审（参考事实已确认，结果判定待共同复核）

F02-F01 | 行为

Read source_id from the request and FAST_KEY from the environment.

归属: operation / Read fetch inputs | 原文: SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_001 从用户请求读取source_id, ir_003从环境读取FAST_KEY，分别产生真实结果供后续消费。 引用: ir_001, ir_003
2	保留	ir_001读取请求source_id, ir_003读取环境FAST_KEY并分别产出结果，源文的两个来源完整。 引用: ir_001, ir_003
3	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用: 无局部引用

F02-F02 | 条件

`FAST_KEY` presence gates `fast.fetch`; prefer it when present and go directly to `archive.fetch` when absent.

归属: block / Preferred tool selection | 原文: SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_005/006 以已读取的key存在性分流，edge_003进入fast，edge_004缺key直接archive，门控不依赖推测的具体空值测试。 引用：ir_005, ir_006, edge_003, edge_004
2	保留	ir_005判断读取的key是否存在，ir_006经edge_003进入fast或edge_004直接archive，未凭空规定key存在测试细节。 引用：ir_005, ir_006, edge_003, edge_004
3	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用

F02-F03 | 数据流

`fast.fetch` receives `request.source_id` and the `FAST_KEY` read from the environment.

归属: operation / Preferred fetch | 原文: SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_007首次与ir_009重试都显式引用请求source_id/result_001和环境fast_key/result_002。 引用：ir_001, ir_003, ir_007, ir_009
2	保留	ir_007首次fast和ir_011重试都引用相同source_id/result_001及fast_key/result_002，参数来自实际读取。 引用：ir_001, ir_003, ir_007, ir_011
3	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用

F02-F04 | 条件

Retry fast.fetch once only after a transient first failure; a non-transient first failure is not retried.

归属: block / Preferred fetch retry | 原文: SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	首次fast状态由ir_008分流，只有edge_006瞬态失败进入唯一重试ir_009，非瞬态edge_007直接archive，图中没有第二次重试。 引用: ir_008, ir_009, ir_009.constraints, edge_006, edge_007
2	保留	ir_009从首次响应得出success与transient判断，ir_010/edge_007只把首次瞬态失败送到ir_011唯一重试，edge_008非瞬态直达archive。 引用: ir_009, ir_010, ir_011, block_006.constraints, edge_007, edge_008
3	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用: 无局部引用

F02-F05 | 条件

A non-transient first failure or any failed retry causes archive.fetch fallback.

归属: block / Fallback after preferred failure | 原文: SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	非瞬态首次失败edge_007和重试失败edge_009都到ir_011 archive.fetch，不再回到fast。 引用: ir_011, edge_007, edge_009
2	保留	edge_008首次非瞬态和edge_011任意重试失败均到ir_015 archive.fetch；重试成功另走返回，不追加其它fast调用。 引用: ir_015, edge_008, edge_011, edge_010
3	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用: 无局部引用

F02-F06 | 约束

archive.fetch executes at most once, with source_id as its only argument.

归属: operation / Archive fetch | 原文: SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	ir_011仅使用source_id数据参数，block_006原文限制最多一次；所有入口共用一个archive节点且没有回边。 引用: ir_011, block_006.constraints, edge_004, edge_007, edge_009
2	保留	ir_015仅传source_id，局部完整原句限制archive最多一次，图中无archive回环或第二个archive节点。 引用: ir_015, ir_015.constraints
3	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用: 无局部引用

F02-F07 | 数据流

A successful fetch returns that same response.body unchanged and ends further fetch calls.

归属: block / Successful fetch exit | 原文: SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	ir_014/016/018分别直接返回首fast、重试fast、archive产生的body/result_005/007/009，三个成功路径只追加状态后返回，无后续fetch。 引用: ir_007, ir_009, ir_011, ir_014, ir_016, ir_018, edge_005, edge_008, edge_010
2	保留	ir_009/013/017分别从各自实际响应提取body，ir_020/022/024直接引用result_007/010/013返回；成功后不再fetch，没有内容改写。 引用: ir_009, ir_013, ir_017, ir_020, ir_022, ir_024
3	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用: 无局部引用

F02-F08 | 条件

An archive.fetch failure returns that error and stops without any archive retry.

归属: block / Archive failure exit | 原文: SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	archive失败经edge_011到最终失败出口，ir_020返回同次archive_error/result_010，没有archive重试路径。 引用: ir_011, ir_020, edge_011
2	保留	ir_017从archive响应提取error/result_014，失败edge_014最终由ir_026返回该错误，只有状态追加而无重试。 引用: ir_017, ir_025, ir_026, edge_014
3	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用: 无局部引用

F02-F09 | 约束

FAST_KEY must never flow to archive.fetch or diagnostic output anywhere in the Skill.

归属: skill / Entire fetch Skill | 原文: SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	全局保留FAST_KEY禁入archive/diagnostic原句，ir_011不接收key，四次状态追加也只消费相应status结果。 引用: skill.constraints, ir_011, ir_013, ir_015, ir_017, ir_019
2	保留	FAST_KEY全局禁令保留，archive实际参数无key，状态写入只有success/failure值并附有不得包含FAST_KEY的限制，没有诊断外发key。 引用: skill.constraints, ir_015, ir_019, ir_021, ir_023, ir_025
3	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用: 无局部引用

F02-F10 | 行为

Before returning from each success or failure path, separately append its final status to local status.txt.

归属: operation / Append final status | 原文: SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	保留	四个返回出口均先执行ir_013/015/017/019追加本地status.txt，各自引用对应工具的最终status/result_004/006/008，然后返回，状态来源与时序显式保留。 引用: ir_013, ir_015, ir_017, ir_019, ir_014, ir_016, ir_018, ir_020
2	保留	四个最终分支内ir_019/021/023/025均先append对应success/failure到status.txt，然后才return；日志值与实际进入的成功或失败终点对应，所有终点覆盖。 引用: ir_019, ir_021, ir_023, ir_025, ir_020, ir_022, ir_024, ir_026, edge_006, edge_010, edge_013, edge_014
3	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用: 无局部引用

F03 工具选择与失败回退 · 表格

F03 | 工具选择与失败回退 | 表格 | 开发集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	error	error	complete
块数	未记录	未记录	11
边数	未记录	未记录	12
操作数	未记录	未记录	22
三层约束条数	未记录	未记录	8
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	1	1	0

三次运行差异与初审意见

第1、2次均在HTTP200后发生IncompleteRead，各无接受CFG。第3次以1次生成取得11块12边，无结构修复或HTTP重试；实际包含一次fast业务重试、单次archive和四个终点。

第3次10条事实均保留；前两次20条不可评估。业务重试与提取服务断流分别记账，未用唯一成功结果补写失败重复。

已接受 CFG 的精确 JSON 有 1 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 F03 的三次明细。

逐事实初审（参考事实已确认，结果判定待共同复核）

F03-F01 | 行为

Read source_id from the request and FAST_KEY from the environment.

归属：operation / Read fetch inputs | 原文：SKILL.md:L10-L10; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
2	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
3	保留	ir_001读取用户请求source_id，ir_003读取环境FAST_KEY，实际结果分别供工具和key门控使用。 引用：ir_001, ir_003

F03-F02 | 条件

`FAST_KEY presence gates fast.fetch; prefer it when present and go directly to archive.fetch when absent.`

归属: block / Preferred tool selection | 原文: SKILL.md:L11-L11; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
2	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
3	保留	ir_005/006以key存在性判定控制edge_003先fast或edge_004缺key直接archive，保留命名条件而未增补具体空值检查。 引用：ir_005, ir_006, edge_003, edge_004

F03-F03 | 数据流

`fast.fetch receives request.source_id and the FAST_KEY read from the environment.`

归属: operation / Preferred fetch | 原文: SKILL.md:L10-L10; SKILL.md:L7-L9; SKILL.md:L11-L11

运行	初审	理由与实际输出引用
1	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
2	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
3	保留	ir_007首次fast和ir_011唯一重试都消费source_id/result_001、fast_key/result_002，来源与参数一致。 引用：ir_001, ir_003, ir_007, ir_011

F03-F04 | 条件

Retry fast.fetch once only after a transient first failure; a non-transient first failure is not retried.

归属: block / Preferred fetch retry | 原文: SKILL.md:L12-L12; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
2	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
3	保留	首次失败edge_006后ir_009检查transient，edge_007只瞬态进入ir_011一次重试，edge_008非瞬态去archive，重试之后没有再次重试回边。 引用：ir_009, ir_010, ir_011, ir_011.constraints, edge_006, edge_007, edge_008

F03-F05 | 条件

A non-transient first failure or any failed retry causes archive.fetch fallback.

归属: block / Fallback after preferred failure | 原文: SKILL.md:L13-L13; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
2	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
3	保留	首次非瞬态失败edge_008和任意重试失败edge_010都汇入ir_013 archive，未将retry失败再次分类后重试。 引用：ir_013, edge_008, edge_010

F03-F06 | 约束

`archive.fetch` executes at most once, with `source_id` as its only argument.

归属: operation / Archive fetch | 原文: SKILL.md:L14-L14; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
2	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
3	保留	ir_013只接收source_id，局部原句完整限制archive最多一次；共享archive节点无回边，不接收FAST_KEY。 引用：ir_013, ir_013.constraints

F03-F07 | 数据流

A successful fetch returns that same response.body unchanged and ends further fetch calls.

归属: block / Successful fetch exit | 原文: SKILL.md:L15-L15; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
2	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
3	保留	ir_016/018/020分别直接返回首fastbody/result_004、重试body/result_007、archivebody/result_009，值来自相应实际工具响应；成功只append状态然后return，无后续fetch。 引用：ir_007, ir_011, ir_013, ir_016, ir_018, ir_020, edge_005, edge_009, edge_011

F03-F08 | 条件

An archive.fetch failure returns that error and stops without any archive retry.

归属: block / Archive failure exit | 原文: SKILL.md:L16-L16; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
2	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
3	保留	archive失败edge_012进入末尾错误出口，ir_022直接返回同次archive_error/result_010，无archive重试。 引用：ir_013, ir_022, block_011.constraints, edge_012

F03-F09 | 约束

FAST_KEY must never flow to archive.fetch or diagnostic output anywhere in the Skill.

归属: skill / Entire fetch Skill | 原文: SKILL.md:L17-L17; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
2	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
3	保留	全局原文禁止FAST_KEY进入archive或diagnostic，key只用于门控和两次fast；archive及四个status追加操作均无key数据输入。 引用：skill.constraints, ir_003, ir_007, ir_011, ir_013, ir_015, ir_021

F03-F10 | 行为

Before returning from each success or failure path, separately append its final status to local status.txt.

归属: operation / Append final status | 原文: SKILL.md:L18-L18; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
2	无法判断	本次抽取在HTTP 200后的SSE接收中以IncompleteRead结束，没有完整响应或接受的CFG；不能把传输中断判为该条业务语义缺失，也不能用其它重复的结果补齐。 引用：无局部引用
3	保留	虽然部分块名用Return... and append字样，实际指令顺序在四个终点均为ir_015/017/019/021先append相应success/failure，然后ir_016/018/020/022返回；以实际图确认写入前置与最终状态关联。 引用：ir_015, ir_017, ir_019, ir_021, ir_016, ir_018, ir_020, ir_022

F04 工具选择与失败回退 · 跨文件混合

F04 | 工具选择与失败回退 | 跨文件混合 | 开发集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	9	11	13
边数	10	12	14
操作数	18	22	26
三层约束条数	10	24	11
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三次均1次生成、无结构修复或抽取HTTP重试。最终CFG分别9/11/13块、10/12/14边，业务重试、回退和四个最终出口均完整。三次都将retry.yaml作为固定规则表达，没有增加运行时配置获取；第2、3次diagnostics还明确说明这一处理。

30条事实×重复判定均保留。跨文件引用没有导致遗漏当前流程或新增配置读取，fast两次尝试上限、archive单次/唯一参数、body原值返回、全局密钥禁令和每条最终路径前的状态追加均有实际图证据。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 F04 的三次明细。

逐事实初审（参考事实已确认，结果判定待共同复核）

F04-F01 | 行为

Read source_id from the request and FAST_KEY from the environment.

归属: operation / Read fetch inputs | 原文: references/workflow.md:L3-L3; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_001从user_request.source_id读取输入，ir_003从environment.FAST_KEY读取密钥，跨文件workflow的来源要求被纳入。 引用: ir_001, ir_003
2	保留	ir_001从user_request提取source_id，ir_003从环境FAST_KEY读取密钥，后续工具使用相应结果；workflow引用被纳入。 引用: ir_001, ir_003
3	保留	ir_001读取请求source_id，ir_003读取环境FAST_KEY，两个结果供后续使用；未添加retry.yaml运行时读取。 引用: ir_001, ir_003, diagnostics

F04-F02 | 条件

FAST_KEY presence gates fast.fetch; prefer it when present and go directly to archive.fetch when absent.

归属: block / Preferred tool selection | 原文: references/workflow.md:L5-L5; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_004消费实际fast_key，edge_002存在时先fast、edge_003缺失时直达archive；无额外具体presence规则。 引用: ir_003, ir_004, edge_002, edge_003
2	保留	ir_005/006根据读取后的key存在性分流，edge_003先fast，edge_004缺key直接archive，首选门控保留。 引用: ir_005, ir_006, edge_003, edge_004
3	保留	ir_005/006以实际key存在性控制edge_003先fast或edge_004直接archive；命名条件保留而无凭空细化。 引用: ir_005, ir_006, edge_003, edge_004

F04-F03 | 数据流

`fast.fetch` receives `request.source_id` and the `FAST_KEY` read from the environment.

归属: operation / Preferred fetch | 原文: references/workflow.md:L3-L3; SKILL.md:L8-L8;
references/workflow.md:L5-L5

运行	初审	理由与实际输出引用
1	保留	ir_005首次fast与ir_007重试均引用source_id/result_001及fast_key/result_002，原数据来源保留。 引用: ir_001, ir_003, ir_005, ir_007
2	保留	ir_007首次和ir_011重试均实际引用source_id/result_001、FAST_KEY/result_002，参数来源不变。 引用: ir_001, ir_003, ir_007, ir_011
3	保留	ir_007首次fast和ir_013唯一重试均引用source_id/result_001与fast_key/result_002。 引用: ir_001, ir_003, ir_007, ir_013

F04-F04 | 条件

Retry `fast.fetch` once only after a transient first failure; a non-transient first failure is not retried.

归属: block / Preferred fetch retry | 原文: references/workflow.md:L8-L8; references/workflow.md:L7-L7;
references/workflow.md:L11-L11; SKILL.md:L8-L8; retry.yaml:L1-L3

运行	初审	理由与实际输出引用
1	保留	ir_006按首次状态分支，edge_005仅瞬态失败进入ir_007唯一重试，edge_006非瞬态直达archive；原文及YAML两次上限附于重试，没有第三次调用。 引用: ir_005, ir_006, ir_007, ir_007.constraints, edge_005, edge_006
2	保留	首次错误经ir_009分类，edge_007仅transient进入ir_011唯一重试，edge_008非瞬态走archive；YAML作为固定约束保留，未增加配置读取。 引用: ir_009, ir_010, ir_011, ir_011.constraints, edge_007, edge_008, diagnostics
3	保留	ir_009分类首次结果，ir_010经edge_007只把首次瞬态失败送至ir_013重试；edge_008非瞬态直接archive，无重复重试回边，固定YAML规则与图一致。 引用: ir_009, ir_010, ir_013, block_005.constraints, edge_007, edge_008, diagnostics

F04-F05 | 条件

A non-transient first failure or any failed retry causes archive.fetch fallback.

归属: block / Fallback after preferred failure | 原文: references/workflow.md:L9-L9; references/workflow.md:L7-L7; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	edge_006首次非瞬态失败和edge_008重试失败共用ir_009 archive, 失败重试不再进入fast。 引用: ir_009, edge_006, edge_008
2	保留	首次非瞬态edge_008与重试失败edge_010进入ir_013 archive, 重试成功另行返回, 没有额外fast调用。 引用: ir_013, edge_008, edge_010, edge_009
3	保留	首次非瞬态edge_008和任意重试失败edge_011均到ir_019 archive; 重试成功走返回, 不再调用fast。 引用: ir_019, edge_008, edge_011, edge_010

F04-F06 | 约束

archive.fetch executes at most once, with source_id as its only argument.

归属: operation / Archive fetch | 原文: references/workflow.md:L13-L13; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_009仅传source_id, 局部原文与archive_fetch:max_attempts:1限制一致; 单个archive节点无回环, FAST_KEY未传入。 引用: ir_009, ir_009.constraints
2	保留	ir_013只引用source_id, 局部原文和YAML单次限制保留, 图无archive回边或第二个archive操作。 引用: ir_013, ir_013.constraints
3	保留	ir_019只传source_id, 所在block_010完整保留最多一次且唯一参数原文; 单一archive节点无回环。 引用: ir_019, block_010.constraints

F04-F07 | 数据流

A successful fetch returns that same response.body unchanged and ends further fetch calls.

归属: block / Successful fetch exit | 原文: references/workflow.md:L15-L15; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_012/014/016分别返回相应首次fast、重试fast、archive的body/result_003/005/007, 成功路径只先append状态再返回, 不再fetch。 引用: ir_005, ir_007, ir_009, ir_012, ir_014, ir_016, edge_004, edge_007, edge_009
2	保留	三个成功返回ir_016/018/020分别使用对应工具产生的body/result_004/007/009, 不改写且不再fetch。 引用: ir_007, ir_011, ir_013, ir_016, ir_018, ir_020
3	保留	ir_012/018/024分别直接返回首fastbody/result_004、重试body/result_007、archivebody/result_010; 成功后仅追加状态再结束fetch。 引用: ir_007, ir_013, ir_019, ir_012, ir_018, ir_024

F04-F08 | 条件

An archive.fetch failure returns that error and stops without any archive retry.

归属: block / Archive failure exit | 原文: references/workflow.md:L17-L17; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	archive失败edge_010最终由ir_018返回本次error/result_008，无archive重试；先前仅追加最终状态。 引用: ir_009, ir_017, ir_018, edge_010
2	保留	archive失败edge_012到ir_021状态追加和ir_022错误返回，引用本次archive_error/result_010，不重试archive。 引用: ir_013, ir_021, ir_022, edge_012
3	保留	archive失败edge_014经ir_025追加状态后ir_026返回同次error/result_011，失败出口无重试。 引用: ir_019, ir_025, ir_026, block_011.constraints, edge_014

F04-F09 | 约束

FAST_KEY must never flow to archive.fetch or diagnostic output anywhere in the Skill.

归属: skill / Entire fetch Skill | 原文: references/workflow.md:L19-L19; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	全局原文禁止FAST_KEY进入archive/diagnostic，实际密钥只传入fast，archive和状态追加不含key操作数。 引用: skill.constraints, ir_005, ir_007, ir_009, ir_011, ir_017
2	保留	Skill级完整禁令保留，archive参数只有source_id，日志操作只有status常量且附禁令，无key进入archive/diagnostic。 引用: skill.constraints, ir_013, ir_015, ir_017, ir_019, ir_021
3	保留	全局FAST_KEY禁入archive/diagnostic原文保留；archive只有source_id，状态追加使用分类status结果而非密钥。 引用: skill.constraints, ir_019, ir_011, ir_017, ir_023, ir_025

F04-F10 | 行为

Before returning from each success or failure path, separately append its final status to local status.txt.

归属: operation / Append final status | 原文: references/workflow.md:L21-L21; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	四个终点ir_011/013/015/017分别使用对应工具status结果追加本地status.txt，再由ir_012/014/016/018返回，状态来源和前置顺序明确。 引用: ir_011, ir_013, ir_015, ir_017, ir_012, ir_014, ir_016, ir_018
2	保留	四个最终结果分支均先由ir_015/017/019/021追加对应success/failure到本地status.txt，后执行return；控制依赖使状态与最终路径相符。 引用: ir_015, ir_017, ir_019, ir_021, ir_016, ir_018, ir_020, ir_022
3	保留	ir_011/017/023/025分别引用首次、重试、archive的分类结果status追加本地status.txt，紧接对应return，四个成功/失败出口前置写入及数据依赖齐全。 引用: ir_009, ir_015, ir_021, ir_011, ir_017, ir_023, ir_025, ir_012, ir_018, ir_024, ir_026

F05 工具选择与失败回退 · 用途最小差异

F05 | 工具选择与失败回退 | 最小差异：用途 | 开发集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	10	13	13
边数	11	14	14
操作数	16	22	22
三层约束条数	10	15	17
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三次均1次生成、无结构修复或抽取HTTP重试，最终CFG分别10/13/13块、11/14/14边。三次正确保留当前取数/有界重试/回退/返回，只排除第9条status.txt历史示例，没有清空整个 workflow。

30条事实×重复判定均保留。历史示例作用范围准确，成功body与失败error实际返回链完整，archive唯一参数和全局密钥禁令仍生效，所有当前路径无额外状态文件写入。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 F05 的三次明细。

逐事实初审（参考事实已确认，结果判定待共同复核）

F05-F01 | 行为

Read source_id from the request and FAST_KEY from the environment.

归属：operation / Read fetch inputs | 原文：SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_001从请求读取source_id，ir_003从环境读取FAST_KEY，实际读取和后续执行保留，没有把历史示例扩大到整个流程。 引用：ir_001，ir_003
2	保留	ir_001读取请求source_id，ir_003读取环境FAST_KEY，各自产出实际结果，历史排除不影响当前取输入行为。 引用：ir_001，ir_003
3	保留	ir_001从user_request提取source_id，ir_003读取环境FAST_KEY，两个来源及当前 workflow 实际读取保留。 引用：ir_001，ir_003

F05-F02 | 条件

`FAST_KEY` presence gates `fast.fetch`; prefer it when present and go directly to `archive.fetch` when absent.

归属: block / Preferred tool selection | 原文: SKILL.md:L9-L9

运行	初审	理由与实际输出引用
1	保留	ir_005/006基于key存在性让edge_003先fast、edge_004缺失时直达archive，原门控保留。 引用: ir_005, ir_006, edge_003, edge_004
2	保留	ir_005/006以FAST_KEY存在性判断控制edge_003先fast或edge_004缺key直接archive，没有增补具体存在规则。 引用: ir_005, ir_006, edge_003, edge_004
3	保留	ir_005/006判断key存在性，edge_003优先fast、edge_004缺key直接archive，没有先调用fast再检查密钥。 引用: ir_005, ir_006, edge_003, edge_004

F05-F03 | 数据流

`fast.fetch` receives `request.source_id` and the `FAST_KEY` read from the environment.

归属: operation / Preferred fetch | 原文: SKILL.md:L8-L8; SKILL.md:L9-L9

运行	初审	理由与实际输出引用
1	保留	ir_007首次和ir_009重试都消费source_id/result_001和fast_key/result_002，来自对应读取步骤。 引用: ir_001, ir_003, ir_007, ir_009
2	保留	ir_007首次fast与ir_012重试均使用读取后的source_id/result_001和fast_key/result_002。 引用: ir_001, ir_003, ir_007, ir_012
3	保留	ir_007首次和ir_012重试均引用请求source_id/result_001和环境fast_key/result_002，参数链正确。 引用: ir_001, ir_003, ir_007, ir_012

F05-F04 | 条件

Retry `fast.fetch` once only after a transient first failure; a non-transient first failure is not retried.

归属: block / Preferred fetch retry | 原文: SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	首fast结果经ir_008与edge_006仅在瞬态失败时进入ir_009唯一重试，edge_007非瞬态直接archive，重试后没有fast回环。 引用: ir_008, ir_009, ir_009.constraints, edge_006, edge_007
2	保留	ir_009从首响应分类，ir_010经edge_007仅首次瞬态失败进入ir_012唯一重试，edge_008非瞬态直达archive，图无第二次重试回边。 引用: ir_009, ir_010, ir_012, ir_012.constraints, edge_007, edge_008
3	保留	ir_009分类首次响应，ir_010经edge_007只在首次瞬态失败时进入ir_012重试；edge_008非瞬态直达archive，图无第二重试循环。 引用: ir_009, ir_010, ir_012, block_007.constraints, edge_007, edge_008

F05-F05 | 条件

A non-transient first failure or any failed retry causes archive.fetch fallback.

归属: block / Fallback after preferred failure | 原文: SKILL.md:L11-L11

运行	初审	理由与实际输出引用
1	保留	非瞬态首次失败edge_007和任意重试失败edge_009共用ir_011 archive节点，不再重试fast。 引用: ir_011, edge_007, edge_009
2	保留	首次非瞬态edge_008与任意重试失败edge_011均到ir_016 archive，没有继续fast的失败路径。 引用: ir_016, edge_008, edge_011
3	保留	首次非瞬态edge_008和重试失败edge_011都进入ir_017 archive，任意重试失败后无进一步fast调用。 引用: ir_017, edge_008, edge_011

F05-F06 | 约束

archive.fetch executes at most once, with source_id as its only argument.

归属: operation / Archive fetch | 原文: SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	ir_011只传source_id且局部原句限制archive最多一次，图无archive重试路径，无FAST_KEY输入。 引用: ir_011, ir_011.constraints
2	保留	ir_016只消费source_id数据参数，局部完整原句限制最多一次，所有回退入口共享单个archive节点无回环。 引用: ir_016, ir_016.constraints
3	保留	ir_017实际只有source_id参数，局部原句明确archive最多一次，不传FAST_KEY，无archive回环。 引用: ir_017, ir_017.constraints

F05-F07 | 数据流

A successful fetch returns that same response.body unchanged and ends further fetch calls.

归属: block / Successful fetch exit | 原文: SKILL.md:L13-L13

运行	初审	理由与实际输出引用
1	保留	三个成功出口ir_013/014/015直接返回对应首次fast、重试fast、archive的body/result_005/007/009，未改写且不再fetch。 引用: ir_007, ir_009, ir_011, ir_013, ir_014, ir_015
2	保留	ir_009/014/018从各自响应提取body，ir_011/020/021引用result_008/012/016原值返回，无后续fetch或摘要生成。 引用: ir_009, ir_014, ir_018, ir_011, ir_020, ir_021
3	保留	ir_011/016/021分别返回首fast、重试fast、archive产生的body/result_005/008/011，成功后直接停止fetch且不做内容变换。 引用: ir_007, ir_012, ir_017, ir_011, ir_016, ir_021

F05-F08 | 条件

An archive.fetch failure returns that error and stops without any archive retry.

归属: block / Archive failure exit | 原文: SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	archive失败edge_011直接到ir_016返回同次archive_error/result_010，原禁止重试的局部约束保留。 引用: ir_011, ir_016, block_010.constraints, edge_011
2	保留	archive失败edge_014最终由ir_022返回ir_018提取的archive_error/result_017，局部禁止archive重试且图中确无回边。 引用: ir_018, ir_022, block_013.constraints, edge_014
3	保留	archive失败经edge_014由ir_022返回该调用的error/result_012，原句局部限制停止且不重试，实际图无再fetch。 引用: ir_017, ir_022, ir_022.constraints, edge_014

F05-F09 | 约束

FAST_KEY must never flow to archive.fetch or diagnostic output anywhere in the Skill.

归属: skill / Entire fetch Skill | 原文: SKILL.md:L15-L15

运行	初审	理由与实际输出引用
1	保留	FAST_KEY禁入archive/diagnostics的全局原文保留，key仅进入首fast与唯一重试，archive没有该参数，图无日志外发。 引用: skill.constraints, ir_003, ir_007, ir_009, ir_011
2	保留	Skill全局完整保留key禁入archive和diagnostic原文，ir_016仅传source_id，无任何key日志输出或文件写入。 引用: skill.constraints, ir_003, ir_007, ir_012, ir_016
3	保留	全局禁止FAST_KEY进入archive/diagnostic原文保留，archive只传source_id，没有日志步骤可接收密钥。 引用: skill.constraints, ir_003, ir_007, ir_012, ir_017

F05-F10 | 不得凭空增加

The final-status append is an explicitly inactive historical example and adds no current file write.

归属: operation / Historical status example | 原文: SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	保留	diagnostics明确只排除第9条历史status.txt示例，所有当前fetch/回退/返回仍存在，四个return前无状态追加或任何文件写入。 引用: diagnostics, ir_007, ir_009, ir_011, ir_013, ir_014, ir_015, ir_016
2	保留	diagnostics正确限定第9条是历史且不执行；图仍有全部fetch/重试/回退，四个成功失败出口直接返回，没有status.txt追加。 引用: diagnostics, ir_007, ir_012, ir_016, ir_011, ir_020, ir_021, ir_022
3	保留	diagnostics只排除第9条历史status写入；实际图仍执行所有取数与有界回退，四个返回出口没有status.txt或其它文件追加。 引用: diagnostics, ir_007, ir_012, ir_017, ir_011, ir_016, ir_021, ir_022

F06 工具选择与失败回退 · 语义最小差异

F06 | 工具选择与失败回退 | 最小差异：语义 | 开发集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	10	13	13
边数	11	14	14
操作数	20	26	26
三层约束条数	24	24	31
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三次均1次生成、无结构修复或抽取HTTP重试，最终CFG分别10/13/13块、11/14/14边。三次所有成功出口都实际引用该次响应summary，未沿用基础包的body；业务门控、有界重试、单次archive与前置status追加均完整。

30条事实×重复判定均保留。最小对照中的summary返回字段得到实际数据流支持，不是通过摘要生成替代；成功停止fetch、archive失败返回原错误和全局密钥禁令均保持。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 F06 的三次明细。

逐事实初审（参考事实已确认，结果判定待共同复核）

F06-F01 | 行为

Read source_id from the request and FAST_KEY from the environment.

归属：operation / Read fetch inputs | 原文：SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_001从user_request读取source_id，ir_003从环境读取FAST_KEY，两个来源产生实际结果。 引用：ir_001，ir_003
2	保留	ir_001读取请求source_id，ir_003读取环境FAST_KEY，参数来源完整分离。 引用：ir_001，ir_003
3	保留	ir_001实际读取请求source_id，ir_003读取环境FAST_KEY并产出结果，来源正确。 引用：ir_001，ir_003

F06-F02 | 条件

`FAST_KEY presence gates fast.fetch; prefer it when present and go directly to archive.fetch when absent.`

归属: block / Preferred tool selection | 原文: SKILL.md:L9-L9

运行	初审	理由与实际输出引用
1	保留	ir_005/006通过key存在性控制edge_003先fast或edge_004缺key直接archive，保留命名条件无额外测试策略。 引用: ir_005, ir_006, edge_003, edge_004
2	保留	ir_005/006根据读取key存在性分流，edge_003先fast、edge_004缺key直达archive，不额外规定presence测试。 引用: ir_005, ir_006, edge_003, edge_004
3	保留	ir_005/006使用key存在判定，edge_003优先fast、edge_004缺失时archive；原门控和命名条件保留。 引用: ir_005, ir_006, edge_003, edge_004

F06-F03 | 数据流

`fast.fetch receives request.source_id and the FAST_KEY read from the environment.`

归属: operation / Preferred fetch | 原文: SKILL.md:L8-L8; SKILL.md:L9-L9

运行	初审	理由与实际输出引用
1	保留	ir_007首次fast及ir_009重试都实际消费source_id/result_001与fast_key/result_002，来自对应读取操作。 引用: ir_001, ir_003, ir_007, ir_009
2	保留	ir_007首次fast与ir_011重试都引用source_id/result_001和fast_key/result_002，两项来自实际读取。 引用: ir_001, ir_003, ir_007, ir_011
3	保留	首次ir_007和唯一重试ir_011均传请求source_id/result_001及环境fast_key/result_002，数据来源一致。 引用: ir_001, ir_003, ir_007, ir_011

F06-F04 | 条件

Retry fast.fetch once only after a transient first failure; a non-transient first failure is not retried.

归属: block / Preferred fetch retry | 原文: SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	首次结果由ir_008分支, edge_006仅瞬态失败进入ir_009唯一重试; 非瞬态edge_007进入archive, 图无第三次fast尝试。 引用: ir_008, ir_009, ir_009.constraints, edge_006, edge_007
2	保留	ir_009分类首fast响应, ir_010经edge_007仅瞬态失败进入ir_011唯一重试, edge_008非瞬态转archive, 重试后不回到fast。 引用: ir_009, ir_010, ir_011, ir_011.constraints, edge_007, edge_008
3	保留	ir_009分类首fast响应, ir_010经edge_007仅在首次瞬态失败时进入ir_011重试, edge_008非瞬态直接archive, 图无再次重试。 引用: ir_009, ir_010, ir_011, block_005.constraints, edge_007, edge_008

F06-F05 | 条件

A non-transient first failure or any failed retry causes archive.fetch fallback.

归属: block / Fallback after preferred failure | 原文: SKILL.md:L11-L11

运行	初审	理由与实际输出引用
1	保留	edge_007首次非瞬态和edge_009任意重试失败都到ir_011 archive, 不再回到fast。 引用: ir_011, edge_007, edge_009
2	保留	非瞬态首次失败edge_008与任意重试失败edge_011共用ir_015archive, 回退条件实际存在。 引用: ir_015, edge_008, edge_011
3	保留	首次非瞬态edge_008和重试失败edge_011均到ir_015archive, 失败重试后不再调用fast。 引用: ir_015, edge_008, edge_011

F06-F06 | 约束

archive.fetch executes at most once, with source_id as its only argument.

归属: operation / Archive fetch | 原文: SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	ir_011archive仅传source_id, 局部完整原句限制最多一次, 图中共享单个archive节点且无重试回边。 引用: ir_011, ir_011.constraints
2	保留	ir_015只使用source_id数据参数, 局部原文限制archive最多一次且不重试, 图无archive循环或额外节点。 引用: ir_015, ir_015.constraints
3	保留	ir_015只有source_id参数, 原文最多一次限制附于操作及所在块, 图无archive回环, 不向其传FAST_KEY。 引用: ir_015, ir_015.constraints, block_008.constraints

F06-F07 | 数据流

A successful fetch returns that same response.summary unchanged and ends further fetch calls.

归属: block / Successful fetch exit | 原文: SKILL.md:L13-L13

运行	初审	理由与实际输出引用
1	保留	ir_007/009/011实际产出首fast、重试fast、archive的summary/result_005/008/011，ir_014/016/018分别引用这些原值返回；没有沿用body或新增摘要生成，成功后停止fetch。 引用: ir_007, ir_009, ir_011, ir_014, ir_016, ir_018, ir_014.constraints
2	保留	ir_009/013/017从对应工具响应实际提取summary/result_006/009/012，ir_020/022/024直接返回这些值，最小对照由body切为summary落实为数据来源而非描述替换。 引用: ir_009, ir_013, ir_017, ir_020, ir_022, ir_024
3	保留	首fast、重试fast、archive实际产生summary/result_004/007/010，ir_020/022/024分别直接返回对应值，未使用body且无新增摘要生成，成功出口无后续fetch。 引用: ir_007, ir_011, ir_015, ir_020, ir_022, ir_024, block_010.constraints

F06-F08 | 条件

An archive.fetch failure returns that error and stops without any archive retry.

归属: block / Archive failure exit | 原文: SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	archive失败edge_011最终由ir_020返回同次archive_error/result_012，除前置status追加外无进一步fetch或archive重试。 引用: ir_011, ir_019, ir_020, ir_020.constraints, edge_011
2	保留	archive失败edge_014到最终错误出口，ir_026返回ir_017提取的同次archive_error/result_013，之后无重试。 引用: ir_017, ir_025, ir_026, edge_014
3	保留	archive失败经edge_014在状态追加后由ir_026返回同次error/result_011，原禁止archive重试约束与图一致。 引用: ir_015, ir_025, ir_026, block_013.constraints, edge_014

F06-F09 | 约束

FAST_KEY must never flow to archive.fetch or diagnostic output anywhere in the Skill.

归属: skill / Entire fetch Skill | 原文: SKILL.md:L15-L15

运行	初审	理由与实际输出引用
1	保留	FAST_KEY禁入archive或diagnostic的全局原文保留，archive仅接收source_id，四个状态追加不含key且附禁令。 引用: skill.constraints, ir_011, ir_013, ir_015, ir_017, ir_019
2	保留	全局和archive局部均保留FAST_KEY禁入archive/diagnostic原文，实际archive和status追加不消费密钥结果。 引用: skill.constraints, ir_015, ir_015.constraints, ir_019, ir_025
3	保留	Skill及archive局部完整保留FAST_KEY不得流向archive或diagnostic的原句，实际密钥只被门控和fast使用，没有日志/回退外发。 引用: skill.constraints, ir_007, ir_011, ir_015, ir_015.constraints, ir_019, ir_025

F06-F10 | 行为

Before returning from each success or failure path, separately append its final status to local status.txt.

归属: operation / Append final status | 原文: SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	保留	四个结果分支均先由ir_013/015/017/019追加对应success/failure到status.txt再return，实际路径保证状态值对应最终结果并覆盖成功与失败。 引用: ir_013, ir_015, ir_017, ir_019, ir_014, ir_016, ir_018, ir_020
2	保留	ir_019/021/023/025分别引用首faststatus/result_005、重试status/result_008、archivestatus/result_011追加本地status.txt，再执行对应return；状态数据依赖与时序完整。 引用: ir_009, ir_013, ir_017, ir_019, ir_021, ir_023, ir_025, ir_020, ir_022, ir_024, ir_026
3	保留	四个终点先由ir_019/021/023/025使用对应分类status/result_006/009/012追加status.txt，再由ir_020/022/024/026返回；每条最终路径的状态来源及前置依赖保留。 引用: ir_009, ir_013, ir_017, ir_019, ir_021, ir_023, ir_025, ir_020, ir_022, ir_024, ir_026

D01 多文档处理与交付 · 编号步骤

D01 | 多文档处理与交付 | 编号步骤 | 开发集 | zh

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	7	7	7
边数	6	6	6
操作数	14	14	14
三层约束条数	6	6	5
生成尝试	2	1	1
结构修复	1	0	0
HTTP 尝试	2	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三轮均保留读取、转换黑盒、打包、交付及收据，且为无回边的七块顺序图。第 2 轮把请求读取放在 paths 提取之前，数据依赖仍成立。三轮逐个转换都仅留在批次操作命名/说明中。第 1、2 轮将原顺序备注错误提升到 Skill 全局，第 3 轮保留范围未知诊断。

结构接受不等于逐项执行语义完整：主要问题是文档集合直接进入单次转换操作，以及前两轮把未知顺序范围具体化。其余冻结行为、黑盒脚本 I/O、结果依赖及明确约束均有实际落点。本文件为来源对照的初审意见，仍供共同复核。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 D01 的三次明细。

逐事实初审（参考事实已确认，结果判定待共同复核）

D01-F01 | 行为

读取 `manifest.json.paths` 作为文档路径来源，并读取请求中的 `output_dir` 和 `delivery_path`。

归属: operation / 读取清单与请求 | 原文: SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_001 读取 <code>manifest.json</code> 得到 <code>result_001</code> ，ir_003 从该结果提取 <code>paths</code> ；ir_005 从请求读取 <code>output_dir</code> 、 <code>delivery_path</code> ，分别产出 <code>result_003/result_004</code> 。 引用: ir_001, ir_003, ir_005
2	保留	ir_001 读取 <code>manifest.json</code> ；ir_005 消费其 <code>result_001</code> 提取 <code>paths</code> 为 <code>result_004</code> 。ir_003 独立读取请求的 <code>output_dir</code> 与 <code>delivery_path</code> 为 <code>result_002/result_003</code> ；读取顺序变化没有丢失来源。 引用: ir_001, ir_003, ir_005
3	保留	ir_001 读取 <code>manifest.json</code> ，ir_003 从 <code>result_001</code> 提取 <code>document_paths</code> 为 <code>result_002</code> ；ir_005 实际读取请求的两个路径并产出后续使用的结果。 引用: ir_001, ir_003, ir_005

D01-F02 | 行为

逐个以文档路径和 `output_dir` 调用 `scripts/convert.py`；每次调用读取该 UTF-8 源文，按需创建输出目录，将内容写入 `output_dir` 下的 `<源文主名>.txt`，并把标准输出的该路径作为转换产物路径。

归属: operation / 文档转换调用 | 原文: SKILL.md:L9-L9; scripts/convert.py:L5-L10

运行	初审	理由与实际输出引用
1	部分保留	ir_007 消费整份 <code>document_paths result_002</code> 与 <code>output_dir result_003</code> ，保留单文档命令及逐字匹配的 <code>convert.py</code> ，故 UTF-8 读取、建目录、 <code><stem>.txt</code> 写出与 <code>stdout</code> 路径可由黑盒代码核实。但 CFG 只有一次批次转换和顺序出边，未绑定当前文档或表达外层逐个调用/收集；块名中的 <code>each</code> 不能替代这部分控制流。此处记录逐项路径及每次 <code>stdout</code> 绑定的可验证粒度不足，逐项意图并未完全丢失；不要求固定的 <code>loop opcode</code> ，待共同复核。 引用: ir_007, block_004, edge_004
2	部分保留	ir_007 的名称说明 <code>for_each_document</code> ，输入是整份 <code>result_004</code> 路径列表和 <code>result_002</code> 输出目录，输出为转换路径列表；原脚本逐字保留其单文档 I/O。然而只有一次批次操作，无逐项参数绑定、循环或外层收集控制流，逐个调用仅在命名中保留。此处记录逐项路径及每次 <code>stdout</code> 绑定的可验证粒度不足，逐项意图并未完全丢失；不要求固定的 <code>loop opcode</code> ，待共同复核。 引用: ir_007, block_004, edge_004
3	部分保留	ir_007 保留脚本原文、单文档命令和 <code>stdout</code> 路径集合 <code>result_005</code> ，消费路径列表 <code>result_002</code> 与目录 <code>result_003</code> 。但七块顺序图没有逐个文档的控制流/参数绑定， <code>execute_convert_script_for_each_document</code> 的名字只描述了迭代意图。此处记录逐项路径及每次 <code>stdout</code> 绑定的可验证粒度不足，逐项意图并未完全丢失；不要求固定的 <code>loop opcode</code> ，待共同复核。 引用: ir_007, block_004, edge_004

D01-F03 | 约束

保留原文标题仅约束转换操作，不约束汇总标题。

归属：operation / 文档转换调用 | 原文：SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	标题原文完整放在 ir_007.constraints，所属操作为转换调用；未挂到汇总、打包或整份 Skill。 引用：ir_007.constraints
2	保留	ir_007.constraints 原样保留“仅限转换，不限制汇总标题”，位置限定在转换调用。 引用：ir_007.constraints
3	保留	ir_007.constraints 保留转换标题要求及其局部限定，没有扩展到打包、交付或收据。 引用：ir_007.constraints

D01-F04 | 不得凭空增加

convert.py 按黑盒调用保留参数、文件输入、输出产物和标准输出路径，不把内部变量、循环或控制流拆成额外流程步骤，也不重复执行内部处理。

归属：operation / 文档转换调用 | 原文：SKILL.md:L9-L9；SKILL.md:L11-L11；scripts/convert.py:L5-L10

运行	初审	理由与实际输出引用
1	保留	block_004 只有转换黑盒 ir_007 和 dispatch，ir_007 绑定脚本、路径集合、输出目录并产出转换路径，script_content 与冻结 convert.py 一致；文件读写和 stdout 留在该调用内，没有拆内部变量、mkdir 或 write_text 成主流程步骤。 引用：block_004, ir_007
2	保留	ir_007 以一个黑盒调用保留两个参数来源、转换路径输出和完整 convert.py，约束也明确不重复内部处理。除外层逐项表达不足已在 F02 记录外，没有展开脚本内部操作。 引用：block_004, ir_007, ir_007.constraints
3	保留	ir_007 保留脚本原文、输入来源及 stdout 路径输出，局部约束要求主流程只用输出路径；没有把脚本内部路径变量、建目录或文件读写另拆执行。 引用：block_004, ir_007, ir_007.constraints

D01-F05 | 行为

必须执行被明确采用的参考打包命令，以转换产物路径列表为 `--paths` 输入；调用读取各产物文件并写入由 `--output` 指定的 `bundle.zip`，压缩包内使用各文件的文件名。

归属：operation / 打包转换产物 | 原文：SKILL.md:L12-L12; SKILL.md:L21-L23; scripts/package.py:L7-L13

运行	初审	理由与实际输出引用
1	保留	ir_009 消费转换路径 result_005 与 bundle.zip 常量，command 原样采用 <code>--paths/--output</code> 参考命令；逐字保留的 package.py 明确读取产物并以 path.name 写入 ZIP，输出 result_006 供交付。 引用：ir_009, ir_009.constraints
2	保留	ir_009 调用 scripts/package.py，消费 result_005 和 bundle.zip，局部约束明确必须采用参考命令并传入 <code>--paths</code> ；完整脚本保留 <code>--output</code> 、ZIP 写出及 <code>arcname=path.name</code> ，产出 bundle_zip_path。 引用：ir_009, ir_009.constraints
3	保留	ir_009 的 reference_command 保留指定打包命令，实际输入为 result_005、bundle.zip 与脚本资源；原脚本保留 ZIP 写出及文件名入包，result_006 表示生成压缩包。 引用：ir_009, ir_009.constraints

D01-F06 | 数据流

将打包生成的 `bundle.zip` 交付到先前从请求读取的 `delivery_path`，交付操作消费该读取结果，无需再次读取。

归属：operation / 交付压缩包 | 原文：SKILL.md:L8-L8; SKILL.md:L13-L13

运行	初审	理由与实际输出引用
1	保留	ir_011 消费打包产物 result_006 和 ir_005 先前读取的 delivery_path result_004；没有重新读取请求目标。 引用：ir_005, ir_009, ir_011
2	保留	ir_011 消费 ir_009 的 bundle_zip_path result_006 和 ir_003 已读取的 resolved_delivery_path result_003，交付依赖明确。 引用：ir_003, ir_009, ir_011
3	保留	ir_011 将 ir_009 生成的 bundle_zip result_006 交付到 ir_005 的 requested_delivery_path result_004，沿用请求读取结果。 引用：ir_005, ir_009, ir_011

D01-F07 | 行为

单独将转换产物路径列表写入本地 `receipt.txt`，并保留对转换产物路径的依赖。

归属：operation / 写入交付收据 | 原文：SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	ir_013 单独写本地 receipt.txt，实际消费转换产物路径 result_005，未改用 ZIP 路径或无来源常量。 引用：ir_007, ir_013
2	保留	ir_013 独立写 receipt.txt，数据输入为 ir_007 产出的 converted_artifact_paths result_005。 引用：ir_007, ir_013
3	保留	ir_013 以转换路径列表 result_005 为内容写 receipt.txt，保留附加收据行为及数据依赖。 引用：ir_007, ir_013

D01-F08 | 约束

不得修改输入原文文件的禁止要求作用于整份 Skill。

归属：skill / 整份多文档 Skill | 原文：SKILL.md:L15-L15

运行	初审	理由与实际输出引用
1	保留	Skill constraints 原样包含整份流程不得修改输入原文文件，保留全局禁止范围。 引用：skill.constraints
2	保留	不得修改输入原文文件原文放在 Skill constraints，未缩小到转换操作。 引用：skill.constraints
3	保留	整份流程不得修改输入原文文件保留在 Skill constraints。 引用：skill.constraints

D01-F09 | 约束

不得向外部服务上传文档的禁止要求作用于整份 Skill。

归属：skill / 整份多文档 Skill | 原文：SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	保留	整份流程禁止把文档上传到外部服务保留在 Skill constraints，范围为整份流程。 引用：skill.constraints
2	保留	Skill constraints 原样保留外部上传禁令，没有局限于某一步。 引用：skill.constraints
3	保留	外部服务上传禁令位于 Skill constraints，原文及全局范围均保留。 引用：skill.constraints

D01-F10 | 约束

“必要时保留原顺序”的作用范围不明确，应待共同确认，不能自动提升为全局规则。

归属：unresolved / 原文未指明转换、打包或收据中的哪一处 | 原文：SKILL.md:L17-L17

运行	初审	理由与实际输出引用
1	冲突	Skill constraints 增加“必要时保留原顺序”，diagnostics 为空；冻结事实明确条件/对象未定，不能把该备注自动提升为全局规则。 引用：skill.constraints, diagnostics
2	冲突	未知范围的“必要时保留原顺序”被放入 Skill constraints，且没有范围未定诊断，违背冻结的 unresolved 边界。 引用：skill.constraints, diagnostics
3	保留	diagnostics 明确说明备注作用域不明确、未提升为 Skill-wide 约束；各层 constraints 未加入这条未知范围要求。 引用：diagnostics

D02 多文档处理与交付 · 连贯段落

D02 | 多文档处理与交付 | 自然语言段落 | 开发集 | zh

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	7	7	7
边数	6	6	6
操作数	14	14	14
三层约束条数	5	6	6
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三轮保留七块顺序主流程，逐项转换均压缩为单个消费路径列表的操作。第 1 轮以诊断保留原顺序范围未知，第 2、3 轮提升为 Skill 全局；第 2 轮还把转换命名为 in_order。第 1、3 轮打包无结果输出，后续沿同一固定 bundle.zip 资源/路径交付，已读 delivery_path 的结果依赖仍明确。

外层逐文档调用的控制流与未知顺序范围是主要不足；黑盒脚本、请求来源、参考打包、交付目标及收据依赖均保留。固定文件副作用可以沿资源路径衔接，不因没有为 bundle.zip 制造结果输出而判缺失。第 3 轮黑盒限制放在全局的落点另列说明。本文件为初审意见，供共同复核。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 D02 的三次明细。

事实表以外的发现

运行 3 | “转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。”被挂到 Skill constraints，虽然句子仍明确指向转换、实际调用也未展开内部步骤，但层级与原文转换归属不一致，应共同复核该约束落点；不据此推断实际发生了重复执行。

引用: skill.constraints, ir_007

逐事实初审（参考事实已确认，结果判定待共同复核）

D02-F01 | 行为

读取 `manifest.json.paths` 作为文档路径来源，并读取请求中的 `output_dir` 和 `delivery_path`。

归属：operation / 读取清单与请求 | 原文：SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	ir_001 从 <code>manifest.json</code> 读取结果，ir_003 从该结果提取 <code>paths</code> ；ir_005 从请求实际读取 <code>output_dir</code> 、 <code>delivery_path</code> ，分别产出 <code>result_003/result_004</code> 。 引用：ir_001, ir_003, ir_005
2	保留	ir_001/ir_003 保留 <code>manifest.json</code> 到 <code>document_paths</code> <code>result_002</code> 的来源链；ir_005 读取请求的两项路径并产出 <code>result_003/result_004</code> 。 引用：ir_001, ir_003, ir_005
3	保留	ir_001 读取清单，ir_003 提取 <code>paths</code> ；ir_005 明确读取请求 <code>output_dir</code> 和 <code>delivery_path</code> ，后续分别使用 <code>result_003/result_004</code> 。 引用：ir_001, ir_003, ir_005

D02-F02 | 行为

逐个以文档路径和 `output_dir` 调用 `scripts/convert.py`；每次调用读取该 UTF-8 源文，按需创建输出目录，将内容写入 `output_dir` 下的 `<源文主名>.txt`，并把标准输出的该路径作为转换产物路径。

归属：operation / 文档转换调用 | 原文：SKILL.md:L8-L8；scripts/convert.py:L5-L10

运行	初审	理由与实际输出引用
1	部分保留	ir_007 保留原样 <code>convert.py</code> 、单文档命令、输入目录和 <code>stdout</code> 产物路径集合，黑盒文件读写有代码证据；但实际消费整份 <code>result_002</code> ，CFG 没有逐项绑定或迭代。诊断还明确把 <code>per-document loop</code> 压缩为 <code>batch operation</code> ，因未给循环变量而省去外层执行语义不充分。此处记录逐项路径及每次 <code>stdout</code> 绑定的可验证粒度不足，逐项意图并未完全丢失；不要求固定的 <code>loop opcode</code> ，待共同复核。 引用：ir_007, block_004, edge_004, diagnostics
2	部分保留	ir_007 的调用内容和原脚本保留单文档读写及产物路径，输入为完整 <code>result_002</code> 路径列表和 <code>result_003</code> 。只有一次转换操作，没有外层逐项绑定/回边；名称 <code>in_order</code> 也不能证明逐个调用或来源未确定的顺序要求。此处记录逐项路径及每次 <code>stdout</code> 绑定的可验证粒度不足，逐项意图并未完全丢失；不要求固定的 <code>loop opcode</code> ，待共同复核。 引用：ir_007, block_004, edge_004
3	部分保留	ir_007 保留 <code>convert.py</code> 、单文档 <code>command</code> 及 <code>side_effects</code> 描述，消费整份路径列表并产生转换路径集合。主流程没有逐文档迭代或当前路径参数绑定，因此单个 <code>for_each_path</code> 名称没有完整落实逐项执行。此处记录逐项路径及每次 <code>stdout</code> 绑定的可验证粒度不足，逐项意图并未完全丢失；不要求固定的 <code>loop opcode</code> ，待共同复核。 引用：ir_007, block_004, edge_004

D02-F03 | 约束

保留原文标题仅约束转换操作，不约束汇总标题。

归属：operation / 文档转换调用 | 原文：SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	ir_007.constraints 完整保留标题要求及“仅限转换，不限制汇总标题”，落在转换操作。 引用：ir_007.constraints
2	保留	标题原文位于 ir_007.constraints，仅限制转换，没有提升到 Skill 全局。 引用：ir_007.constraints
3	保留	转换标题要求及排除汇总标题的措辞均位于 ir_007.constraints，作用范围明确。 引用：ir_007.constraints

D02-F04 | 不得凭空增加

convert.py 按黑盒调用保留参数、文件输入、输出产物和标准输出路径，不把内部变量、循环或控制流拆成额外流程步骤，也不重复执行内部处理。

归属：operation / 文档转换调用 | 原文：SKILL.md:L8-L8; SKILL.md:L10-L10; scripts/convert.py:L5-L10

运行	初审	理由与实际输出引用
1	保留	ir_007 保留完整且逐字匹配的 convert.py、文档/目录参数来源和转换路径输出；内部 mkdir、变量与读写没有拆进主流程，局部约束明确只使用脚本输出路径。 引用：block_004, ir_007, ir_007.constraints
2	保留	转换 block 只有一个黑盒 ir_007，绑定脚本、文档集合与目录并返回产物路径；原始代码与不重复内部处理的局部约束都在，未展开内部控制流。 引用：block_004, ir_007, block_004.constraints
3	保留	ir_007 以单个黑盒保留脚本原文、参数和产物/stdout 路径，未新增内部处理步骤。只使用输出路径的文字被放在 Skill constraints 的层级差异另列 additional finding；这里依据实际黑盒调用判断未重复执行。 引用：block_004, ir_007, skill.constraints

D02-F05 | 行为

必须执行被明确采用的参考打包命令，以转换产物路径列表为 `--paths` 输入；调用读取各产物文件并写入由 `--output` 指定的 `bundle.zip`，压缩包内使用各文件的文件名。

归属：operation / 打包转换产物 | 原文：SKILL.md:L12-L12; SKILL.md:L20-L22; scripts/package.py:L7-L13

运行	初审	理由与实际输出引用
1	保留	ir_009 消费转换路径 result_005, command 明确 <code>--paths</code> 与 <code>--output bundle.zip</code> ; 逐字保留的 package.py 读取产物并以 <code>path.name</code> 入 ZIP。bundle.zip 作为写出资源，纯文件副作用无需额外返回值。 引用：ir_009, ir_009.constraints
2	保留	ir_009 采用 reference_command 指定的 package.py 命令，消费 result_005 与 bundle.zip; 原始脚本保留 ZIP 写出和文件名入包, result_006 为打包路径结果。 引用：ir_009, ir_009.constraints
3	保留	ir_009 的 command、输入 result_005 与 bundle.zip 及原始 package.py 一起保留强制打包行为。脚本明确读取各文件并写出 ZIP、 <code>arcname=path.name</code> ; 无 outputs 不抹去已明确的文件副作用。 引用：ir_009, ir_009.constraints

D02-F06 | 数据流

将打包生成的 `bundle.zip` 交付到先前从请求读取的 `delivery_path`，交付操作消费该读取结果，无需再次读取。

归属：operation / 交付压缩包 | 原文：SKILL.md:L8-L8; SKILL.md:L12-L12

运行	初审	理由与实际输出引用
1	保留	ir_009 明确写出固定 bundle.zip, edge_005 后 ir_011 消费同名文件资源和 ir_005 已读取的 result_004 delivery_path。实际交付文件及请求目标均保留，不要求纯文件写出额外伪造返回结果。 引用：ir_005, ir_009, ir_011, edge_005
2	保留	ir_011 消费打包结果 result_006 及先前读取的 delivery_path result_004, 目标来源和产物依赖明确。 引用：ir_005, ir_009, ir_011
3	保留	ir_009 写出命令的固定目标是 bundle.zip, 后续 ir_011 以同一字面路径交付，目标消费 ir_005 读取的 delivery_target_path result_004。固定文件副作用和顺序连边已衔接产物，无重复读取请求。 引用：ir_005, ir_009, ir_011, edge_005

D02-F07 | 行为

单独将转换产物路径列表写入本地 receipt.txt，并保留对转换产物路径的依赖。

归属：operation / 写入交付收据 | 原文：SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	ir_013 独立写 receipt.txt，内容来自 ir_007 的转换产物路径 result_005。 引用：ir_007, ir_013
2	保留	ir_013 单独将 conversion_artifact_paths result_005 写入 receipt.txt，保留附加行为及真实结果依赖。 引用：ir_007, ir_013
3	保留	ir_013 消费 converted_artifact_paths result_005 写入本地 receipt.txt，没有把收据误作历史说明。 引用：ir_007, ir_013

D02-F08 | 约束

不得修改输入原文文件的禁止要求作用于整份 Skill。

归属：skill / 整份多文档 Skill | 原文：SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	Skill constraints 保留整份流程不得修改输入原文文件的禁止要求。 引用：skill.constraints
2	保留	不得修改输入原文文件位于 Skill constraints，原文和全局归属保留。 引用：skill.constraints
3	保留	Skill constraints 原样包含输入原文禁止修改要求，范围没有缩小。 引用：skill.constraints

D02-F09 | 约束

不得向外部服务上传文档的禁止要求作用于整份 Skill。

归属：skill / 整份多文档 Skill | 原文：SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	保留	Skill constraints 保留整份流程禁止向外部服务上传文档。 引用：skill.constraints
2	保留	文档外部上传禁令原样保留在 Skill constraints。 引用：skill.constraints
3	保留	外部服务上传禁令位于 Skill constraints，仍覆盖整份流程。 引用：skill.constraints

D02-F10 | 约束

“必要时保留原顺序”的作用范围不明确，应待共同确认，不能自动提升为全局规则。

归属：unresolved / 原文未指明转换、打包或收据中的哪一处 | 原文：SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	保留	diagnostics 明确原顺序备注没有指定操作或列表，未提升为全局约束；各层约束均没有擅自指定其范围。 引用：diagnostics
2	冲突	Skill constraints 将“备注：必要时保留原顺序”提升为全局，diagnostics 为空；转换操作又命名为 in_order，未保留原文作用对象不明的边界。 引用：skill.constraints, ir_007, diagnostics
3	冲突	未知范围的原顺序备注进入 Skill constraints，没有诊断说明范围不明，自动全局化违背冻结事实。 引用：skill.constraints, diagnostics

D03 多文档处理与交付 · 表格

D03 | 多文档处理与交付 | 表格 | 开发集 | zh

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	7	7	8
边数	6	6	7
操作数	14	14	16
三层约束条数	5	5	5
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三轮均保留表格中的来源、转换、打包、交付与收据；第 3 轮将请求两项读取拆成两个真实来源块，形成八块，其余为七块。三轮都在 diagnostics 保留原顺序范围未知。三轮转换仍是单个批次黑盒操作，未将表格“逐个”落实为外层控制流。第 1 轮标题限制放在仅含转换调用的基本块，实际作用对象仍局限于转换。

表格的行为与约束主体能够识别，未知顺序范围三轮均保留。主要不足是外层逐项调用仅由命名或命令模板描述，实际图消费的是路径集合。黑盒脚本内部读写保留在代码中并可核验，不因未拆内部步骤扣分。本文件为初审意见，供共同复核。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 D03 的三次明细。

逐事实初审（参考事实已确认，结果判定待共同复核）

D03-F01 | 行为

读取 `manifest.json.paths` 作为文档路径来源，并读取请求中的 `output_dir` 和 `delivery_path`。

归属：operation / 读取清单与请求 | 原文：SKILL.md:L10-L10; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	保留	ir_001 读取 <code>manifest.json</code> , ir_003 从 <code>result_001</code> 提取 <code>paths</code> ; ir_005 读取请求中的 <code>output_dir</code> 与 <code>delivery_path</code> , 产出 <code>result_003/result_004</code> 。 引用：ir_001, ir_003, ir_005
2	保留	ir_001/ir_003 保留清单到 <code>document_paths</code> 的来源链, ir_005 实际读取请求两项路径, 后续消费其结果。 引用：ir_001, ir_003, ir_005
3	保留	ir_001 读取 <code>manifest.json</code> , ir_007 消费其 <code>result_001</code> 提取 <code>paths</code> ; ir_003 与 ir_005 分别从请求读取 <code>output_dir</code> 、 <code>delivery_path</code> 为 <code>result_002/result_003</code> 。两项读取拆块没有丢失来源。 引用：ir_001, ir_003, ir_005, ir_007

D03-F02 | 行为

逐个以文档路径和 `output_dir` 调用 `scripts/convert.py`; 每次调用读取该 UTF-8 源文，按需创建输出目录，将内容写入 `output_dir` 下的 `<源文主名>.txt`，并把标准输出的该路径作为转换产物路径。

归属：operation / 文档转换调用 | 原文：SKILL.md:L11-L11; SKILL.md:L7-L9; scripts/convert.py:L5-L10

运行	初审	理由与实际输出引用
1	部分保留	ir_007 保留逐字匹配的 <code>convert.py</code> 、单文档 <code>reference_command</code> 及目录/路径输入，代码证明 UTF-8 读写、建目录与 <code>stdout</code> 路径；但实际输入是整份 <code>result_002</code> 路径列表，只有一次转换操作，无外层逐项绑定或迭代。此处记录逐项路径及每次 <code>stdout</code> 绑定的可验证粒度不足，逐项意图并未完全丢失；不要求固定的 <code>loop opcode</code> ，待共同复核。 引用：ir_007, block_004, edge_004
2	部分保留	ir_007 保留单文档命令、原始 <code>convert.py</code> 和输出路径集合，绑定 <code>result_002</code> 路径列表与 <code>result_003</code> 目录。块名虽称 <code>each</code> ，七块顺序图没有逐项调用控制流或当前文档参数绑定。此处记录逐项路径及每次 <code>stdout</code> 绑定的可验证粒度不足，逐项意图并未完全丢失；不要求固定的 <code>loop opcode</code> ，待共同复核。 引用：ir_007, block_004, edge_004
3	部分保留	ir_009 保留 <code>convert.py</code> 及单文档 <code>command_template</code> ，实际消费 <code>document_paths</code> <code>result_004</code> 与目录 <code>result_002</code> ，产出路径列表 <code>result_005</code> 。 <code>for_each_document</code> 仅在命名中，八块顺序图没有外层逐项执行/收集控制流。此处记录逐项路径及每次 <code>stdout</code> 绑定的可验证粒度不足，逐项意图并未完全丢失；不要求固定的 <code>loop opcode</code> ，待共同复核。 引用：ir_009, block_005, edge_005

D03-F03 | 约束

保留原文标题仅约束转换操作，不约束汇总标题。

归属：operation / 文档转换调用 | 原文：SKILL.md:L12-L12; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	保留	标题要求原文放在 block_004.constraints; 该块仅有转换黑盒 ir_007 和 dispatch, 未包含汇总或其他业务操作, 实际归属仍限于转换。记录块级落点, 不因其与单一转换业务范围一致而判扩大。 引用: block_004.constraints, ir_007
2	保留	ir_007.constraints 完整保留“仅限转换, 不限制汇总标题”, 标题约束限定于转换操作。 引用: ir_007.constraints
3	保留	ir_009.constraints 原样保留转换期间标题要求与不约束汇总标题的限定, 位置属于转换调用。 引用: ir_009.constraints

D03-F04 | 不得凭空增加

convert.py 按黑盒调用保留参数、文件输入、输出产物和标准输出路径, 不把内部变量、循环或控制流拆成额外流程步骤, 也不重复执行内部处理。

归属：operation / 文档转换调用 | 原文：SKILL.md:L11-L11; SKILL.md:L7-L9; SKILL.md:L13-L13;
scripts/convert.py:L5-L10

运行	初审	理由与实际输出引用
1	保留	block_004 内仅有转换黑盒 ir_007 和 dispatch, 代码逐字匹配 convert.py, 输入/输出及 stdout 路径保留; 局部约束说明主流程只使用输出路径, 没有拆内部变量或重复文件处理。 引用: block_004, ir_007, block_004.constraints
2	保留	ir_007 保留脚本参数来源、完整代码与转换路径结果; 约束明确黑盒及不重复处理, 主图没有展开脚本内部读写或控制流。 引用: block_004, ir_007, ir_007.constraints
3	保留	ir_009 以单个黑盒保留文档/目录输入、完整 convert.py 与路径输出, 局部约束禁止重复内部处理; 没有把 mkdir、write_text 或局部变量拆进主流程。 引用: block_005, ir_009, ir_009.constraints

D03-F05 | 行为

必须执行被明确采用的参考打包命令，以转换产物路径列表为 `--paths` 输入；调用读取各产物文件并写入由 `--output` 指定的 `bundle.zip`，压缩包内使用各文件的文件名。

归属: operation / 打包转换产物 | 原文: SKILL.md:L14-L14; SKILL.md:L7-L9; SKILL.md:L23-L25; scripts/package.py:L7-L13

运行	初审	理由与实际输出引用
1	保留	ir_009 消费转换路径 result_005 和 bundle.zip, reference_command 及块约束明确采用参考命令。原 package.py 保留 --paths/--output、ZIP 写出和 arcname=path.name, 未把已采用命令当成非执行示例。 引用: ir_009, block_005.constraints
2	保留	ir_009 的 command 明确 --paths <转换产物路径列表> --output bundle.zip, 实际消费 result_005; 逐字匹配的 package.py 保留各产物文件读取与文件名入包, 并产出 result_006。 引用: ir_009, ir_009.constraints
3	保留	ir_011 消费转换路径 result_005 和 bundle.zip 资源, command 原样采用参考打包命令; 原始 package.py 保留 ZIP 写出及文件名入包, 无 outputs 不取消文件副作用。 引用: ir_011, ir_011.constraints

D03-F06 | 数据流

将打包生成的 `bundle.zip` 交付到先前从请求读取的 `delivery_path`，交付操作消费该读取结果，无需再次读取。

归属: operation / 交付压缩包 | 原文: SKILL.md:L10-L10; SKILL.md:L7-L9; SKILL.md:L15-L15

运行	初审	理由与实际输出引用
1	保留	ir_009 通过命令和黑盒代码写固定 bundle.zip; 后继 ir_011 消费同名文件资源与 ir_005 读取的 result_004 delivery_path, 文件副作用和顺序边衔接交付。 引用: ir_005, ir_009, ir_011, edge_005
2	保留	ir_011 同时消费打包结果 result_006 和 ir_005 已读取的 requested_delivery_path result_004, 交付目标没有重复获取或改来源。 引用: ir_005, ir_009, ir_011
3	保留	ir_011 写固定 bundle.zip; ir_013 随后消费同名资源并交付到 ir_005 已读取的 delivery_path_value result_003。固定文件资源衔接保留打包产物, 实际请求目标使用结果引用。 引用: ir_005, ir_011, ir_013, edge_006

D03-F07 | 行为

单独将转换产物路径列表写入本地 receipt.txt，并保留对转换产物路径的依赖。

归属：operation / 写入交付收据 | 原文：SKILL.md:L16-L16; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	保留	ir_013 单独消费转换产物路径 result_005 写 receipt.txt，保留表格额外收据行为。 引用：ir_007, ir_013
2	保留	ir_013 独立写本地 receipt.txt，内容来自转换输出 result_005，而非无来源文字。 引用：ir_007, ir_013
3	保留	ir_015 以 ir_009 产出的 converted_artifact_paths result_005 写 receipt.txt，保留独立收据动作和路径依赖。 引用：ir_009, ir_015

D03-F08 | 约束

不得修改输入原文文件的禁止要求作用于整份 Skill。

归属：skill / 整份多文档 Skill | 原文：SKILL.md:L17-L17; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	保留	整份流程不得修改输入原文文件原样位于 Skill constraints。 引用：skill.constraints
2	保留	Skill constraints 保留不得修改输入原文文件的全局禁止范围。 引用：skill.constraints
3	保留	不得修改输入原文文件保留在 Skill constraints，没有缩成局部约束。 引用：skill.constraints

D03-F09 | 约束

不得向外部服务上传文档的禁止要求作用于整份 Skill。

归属：skill / 整份多文档 Skill | 原文：SKILL.md:L18-L18; SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	保留	Skill constraints 原样保留整份流程禁止把文档上传到外部服务。 引用：skill.constraints
2	保留	外部服务上传禁令原文位于 Skill constraints，范围为整份流程。 引用：skill.constraints
3	保留	整份流程的文档外部上传禁令保留在 Skill constraints。 引用：skill.constraints

D03-F10 | 约束

“必要时保留原顺序”的作用范围不明确，应待共同确认，不能自动提升为全局规则。

归属：unresolved / 原文未指明转换、打包或收据中的哪一处 | 原文：SKILL.md:L19-L19；SKILL.md:L7-L9

运行	初审	理由与实际输出引用
1	保留	diagnostics 明确第 10 行备注范围不明，未建成约束；原顺序文字没有进入 Skill、block 或 instruction constraints，保留 unresolved。 引用：diagnostics
2	保留	diagnostics 指明原顺序备注 scope unclear 且未提升为 constraint，各层没有擅自落实它的对象或范围。 引用：diagnostics
3	保留	diagnostics 明确第 10 行备注范围不明确、未提升为 Skill 级约束；各层 constraints 只保留其他有明确来源的要求。 引用：diagnostics

D04 多文档处理与交付 · 跨文件混合

D04 | 多文档处理与交付 | 跨文件混合 | 开发集 | zh

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	7	7	7
边数	6	6	6
操作数	14	14	14
三层约束条数	7	6	9
生成尝试	2	2	1
结构修复	1	1	0
HTTP 尝试	2	2	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三轮均为 7 块、6 边，实际主流程均采用跨文件要求。第 1 轮先读请求再取 paths，后两轮相反，不影响数据来源；标题规则第 2 轮放在仅含转换调用的块内。原顺序未知范围三轮都被确定：第 1、3 轮全局，第 2 轮转换块。逐项转换的集合粒度问题三轮一致。

主要读取、黑盒、打包、交付与回执行行为保留。三轮转换逐项绑定均为待共同复核的部分保留；三轮未知范围备注均发生无依据的范围赋值。结构有效不等于本样例语义全部正确。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 D04 的三次明细。

逐事实初审（参考事实已确认，结果判定待共同复核）

D04-F01 | 行为

读取 `manifest.json.paths` 作为文档路径来源，并读取请求中的 `output_dir` 和 `delivery_path`。

归属：operation / 读取清单与请求 | 原文：references/workflow.md:L3-L3; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	实际读取 <code>manifest.json</code> 、提取 <code>paths</code> ，并从请求读取 <code>output_dir</code> 与 <code>delivery_path</code> ；转换与交付分别使用这些读取结果，跨文件流程引用已进入当前 CFG。 引用：ir_001, ir_003, ir_005
2	保留	实际读取 <code>manifest.json</code> 、提取 <code>paths</code> ，并从请求读取 <code>output_dir</code> 与 <code>delivery_path</code> ；转换与交付分别使用这些读取结果，跨文件流程引用已进入当前 CFG。 引用：ir_001, ir_003, ir_005
3	保留	实际读取 <code>manifest.json</code> 、提取 <code>paths</code> ，并从请求读取 <code>output_dir</code> 与 <code>delivery_path</code> ；转换与交付分别使用这些读取结果，跨文件流程引用已进入当前 CFG。 引用：ir_001, ir_003, ir_005

D04-F02 | 行为

逐个以文档路径和 `output_dir` 调用 `scripts/convert.py`；每次调用读取该 UTF-8 源文，按需创建输出目录，将内容写入 `output_dir` 下的 `<源文主名>.txt`，并把标准输出的该路径作为转换产物路径。

归属：operation / 文档转换调用 | 原文：references/workflow.md:L6-L6; references/workflow.md:L5-L5; SKILL.md:L8-L8; scripts/convert.py:L5-L10

运行	初审	理由与实际输出引用
1	部分保留	转换指令的逐文档名称、单文档命令及完整 <code>convert.py</code> 保留了逐个意图、UTF-8 读取、创建目录、写入主名.txt 与 <code>stdout</code> 路径。但指令实际消费整份路径集合并一次产出转换列表，没有明确每个 <code>path</code> 与每次 <code>stdout</code> 的结构绑定。这是可验证表达粒度边界，待共同复核；不要求固定 <code>loop opcode</code> ，也不判定逐个意图完全丢失。 引用：ir_007
2	部分保留	转换指令的逐文档名称、单文档命令及完整 <code>convert.py</code> 保留了逐个意图、UTF-8 读取、创建目录、写入主名.txt 与 <code>stdout</code> 路径。但指令实际消费整份路径集合并一次产出转换列表，没有明确每个 <code>path</code> 与每次 <code>stdout</code> 的结构绑定。这是可验证表达粒度边界，待共同复核；不要求固定 <code>loop opcode</code> ，也不判定逐个意图完全丢失。 引用：ir_007
3	部分保留	转换指令的逐文档名称、单文档命令及完整 <code>convert.py</code> 保留了逐个意图、UTF-8 读取、创建目录、写入主名.txt 与 <code>stdout</code> 路径。但指令实际消费整份路径集合并一次产出转换列表，没有明确每个 <code>path</code> 与每次 <code>stdout</code> 的结构绑定。这是可验证表达粒度边界，待共同复核；不要求固定 <code>loop opcode</code> ，也不判定逐个意图完全丢失。 引用：ir_007

D04-F03 | 约束

保留原文标题仅约束转换操作，不约束汇总标题。

归属：operation / 文档转换调用 | 原文：references/workflow.md:L7-L7; references/workflow.md:L5-L5; SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	原文标题保留规则及“仅限转换、不限制汇总标题”的范围限制均保留在转换操作上。 引用：ir_007, ir_007.constraints
2	保留	标题规则位于 block_004，该块只有转换调用和分派，实际范围仍仅限转换；原文明确排除汇总标题，未被提升到全局。 引用：ir_007, block_004.constraints
3	保留	原文标题保留规则及“仅限转换、不限制汇总标题”的范围限制均保留在转换操作上。 引用：ir_007, ir_007.constraints

D04-F04 | 不得凭空增加

convert.py 按黑盒调用保留参数、文件输入、输出产物和标准输出路径，不把内部变量、循环或控制流拆成额外流程步骤，也不重复执行内部处理。

归属：operation / 文档转换调用 | 原文：references/workflow.md:L6-L6; references/workflow.md:L5-L5; SKILL.md:L8-L8; references/workflow.md:L9-L9; scripts/convert.py:L5-L10

运行	初审	理由与实际输出引用
1	保留	convert.py 为一次黑盒脚本调用，metadata.script_content 与冻结完整脚本文本一致；内部读写没有拆成额外流程操作。黑盒约束落在转换所属范围。 引用：ir_007, ir_007.constraints
2	保留	convert.py 为一次黑盒脚本调用，metadata.script_content 与冻结完整脚本文本一致；内部读写没有拆成额外流程操作。黑盒约束落在转换所属范围。 引用：ir_007, block_004.constraints
3	保留	convert.py 为一次黑盒脚本调用，metadata.script_content 与冻结完整脚本文本一致；内部读写没有拆成额外流程操作。黑盒约束落在转换所属范围。 引用：ir_007, ir_007.constraints

D04-F05 | 行为

必须执行被明确采用的参考打包命令，以转换产物路径列表为 `--paths` 输入；调用读取各产物文件并写入由 `--output` 指定的 `bundle.zip`，压缩包内使用各文件的文件名。

归属：operation / 打包转换产物 | 原文：references/workflow.md:L11-L11; SKILL.md:L8-L8; references/workflow.md:L15-L17; scripts/package.py:L7-L13

运行	初审	理由与实际输出引用
1	保留	被当前流程明确采用的打包示例已成为 <code>package.py</code> 调用，消费转换列表、输出 <code>bundle.zip</code> ；完整脚本保留按文件 <code>basename</code> 写入 ZIP 的逻辑。 引用：ir_007, ir_009
2	保留	被当前流程明确采用的打包示例已成为 <code>package.py</code> 调用，消费转换列表、输出 <code>bundle.zip</code> ；完整脚本保留按文件 <code>basename</code> 写入 ZIP 的逻辑。 引用：ir_007, ir_009
3	保留	被当前流程明确采用的打包示例已成为 <code>package.py</code> 调用，消费转换列表、输出 <code>bundle.zip</code> ；完整脚本保留按文件 <code>basename</code> 写入 ZIP 的逻辑。 引用：ir_007, ir_009

D04-F06 | 数据流

将打包生成的 `bundle.zip` 交付到先前从请求读取的 `delivery_path`，交付操作消费该读取结果，无需再次读取。

归属：operation / 交付压缩包 | 原文：references/workflow.md:L3-L3; SKILL.md:L8-L8; references/workflow.md:L19-L19

运行	初审	理由与实际输出引用
1	保留	打包生成的 <code>bundle.zip</code> 结果与先前请求读取的 <code>delivery_path</code> 一起进入交付指令，交付目的地有实际数据引用。 引用：ir_003, ir_009, ir_011
2	保留	打包生成的 <code>bundle.zip</code> 结果与先前请求读取的 <code>delivery_path</code> 一起进入交付指令，交付目的地有实际数据引用。 引用：ir_005, ir_009, ir_011
3	保留	打包生成的 <code>bundle.zip</code> 结果与先前请求读取的 <code>delivery_path</code> 一起进入交付指令，交付目的地有实际数据引用。 引用：ir_005, ir_009, ir_011

D04-F07 | 行为

单独将转换产物路径列表写入本地 receipt.txt，并保留对转换产物路径的依赖。

归属：operation / 写入交付收据 | 原文：references/workflow.md:L21-L21；SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	额外回执步骤属于当前流程并独立执行；回执操作消费转换产物列表，写入 receipt.txt，未被打包或交付步骤吞掉。 引用：ir_007, ir_013
2	保留	额外回执步骤属于当前流程并独立执行；回执操作消费转换产物列表，写入 receipt.txt，未被打包或交付步骤吞掉。 引用：ir_007, ir_013
3	保留	额外回执步骤属于当前流程并独立执行；回执操作消费转换产物列表，写入 receipt.txt，未被打包或交付步骤吞掉。 引用：ir_007, ir_013

D04-F08 | 约束

不得修改输入原文文件的禁止要求作用于整份 Skill。

归属：skill / 整份多文档 Skill | 原文：references/workflow.md:L23-L23；SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	不得修改输入文件的原文禁止事项保留为整份 Skill 的约束，未被缩小到某个操作。 引用：skill.constraints
2	保留	不得修改输入文件的原文禁止事项保留为整份 Skill 的约束，未被缩小到某个操作。 引用：skill.constraints
3	保留	不得修改输入文件的原文禁止事项保留为整份 Skill 的约束，未被缩小到某个操作。 引用：skill.constraints

D04-F09 | 约束

不得向外部服务上传文档的禁止要求作用于整份 Skill。

归属：skill / 整份多文档 Skill | 原文：references/workflow.md:L25-L25；SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	不得上传到外部服务的原文禁止事项保留为整份 Skill 的约束；本地交付不被凭空改写为外部上传。 引用：skill.constraints, ir_011
2	保留	不得上传到外部服务的原文禁止事项保留为整份 Skill 的约束；本地交付不被凭空改写为外部上传。 引用：skill.constraints, ir_011
3	保留	不得上传到外部服务的原文禁止事项保留为整份 Skill 的约束；本地交付不被凭空改写为外部上传。 引用：skill.constraints, ir_011

D04-F10 | 约束

“必要时保留原顺序”的作用范围不明确，应待共同确认，不能自动提升为全局规则。

归属：unresolved / 原文未指明转换、打包或收据中的哪一处 | 原文：references/workflow.md:L27-L27；SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	冲突	原文的原顺序备注作用范围未知，结果却将其放入 skill.constraints，擅自提升为全局规则。现有诊断讨论循环和回执顺序，未消除这一全局落点。 引用：skill.constraints, ir_007
2	冲突	原文未说明“必要时保留原顺序”约束谁；结果直接将其放入转换块 block_004.constraints，擅自确定了转换范围。未提升为全局仍不符合保留未知作用范围的约定。 引用：block_004.constraints
3	冲突	原文的原顺序备注作用范围未知，结果却将其放入 skill.constraints，擅自提升为全局规则。转换块名称还增加了 in order 表述。 引用：skill.constraints, ir_007

D05 多文档处理与交付 · 用途最小差异

D05 | 多文档处理与交付 | 最小差异：用途 | 开发集 | zh

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	6	6	6
边数	5	5	5
操作数	12	12	12
三层约束条数	5	5	5
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三轮均为 6 块、5 边，历史回执均未进入执行路径；读取、转换、打包和交付保持一致。第 1 轮标题规则位于转换块，后两轮位于转换操作，实际范围一致。原顺序备注三轮均保留未知范围；逐文档转换仍采用集合输入的抽象。

历史说明与当前指令的最小差异三轮均识别；未知作用范围三轮均妥善保留。唯一逐项部分保留点是转换的逐文档输入与 stdout 绑定表达粒度，需共同确认其可接受程度。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 D05 的三次明细。

逐事实初审（参考事实已确认，结果判定待共同复核）

D05-F01 | 行为

读取 `manifest.json.paths` 作为文档路径来源，并读取请求中的 `output_dir` 和 `delivery_path`。

归属：operation / 读取清单与请求 | 原文：SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	读取 <code>manifest.json</code> 、提取 <code>paths</code> ，并读取请求中的 <code>output_dir</code> 和 <code>delivery_path</code> ；转换及交付消费相应结果，数据来源明确。 引用：ir_001, ir_003, ir_005
2	保留	读取 <code>manifest.json</code> 、提取 <code>paths</code> ，并读取请求中的 <code>output_dir</code> 和 <code>delivery_path</code> ；转换及交付消费相应结果，数据来源明确。 引用：ir_001, ir_003, ir_005
3	保留	读取 <code>manifest.json</code> 、提取 <code>paths</code> ，并读取请求中的 <code>output_dir</code> 和 <code>delivery_path</code> ；转换及交付消费相应结果，数据来源明确。 引用：ir_001, ir_003, ir_005

D05-F02 | 行为

逐个以文档路径和 `output_dir` 调用 `scripts/convert.py`；每次调用读取该 UTF-8 源文，按需创建输出目录，将内容写入 `output_dir` 下的 `<源文主名>.txt`，并把标准输出的该路径作为转换产物路径。

归属：operation / 文档转换调用 | 原文：SKILL.md:L9-L9；scripts/convert.py:L5-L10

运行	初审	理由与实际输出引用
1	部分保留	逐文档转换意图、单文档命令和原始 <code>convert.py</code> 全文均保留，包括 UTF-8 读取、按需建目录、写入主名.txt 和打印路径。但实际指令仍以整份 <code>paths</code> 为输入、转换列表为输出，每个文档与每次 <code>stdout</code> 的绑定未在结构中明确。这是表达粒度的待共同复核点，不以缺少固定循环 <code>opcode</code> 作为判断理由。 引用：ir_007
2	部分保留	逐文档转换意图、单文档命令和原始 <code>convert.py</code> 全文均保留，包括 UTF-8 读取、按需建目录、写入主名.txt 和打印路径。但实际指令仍以整份 <code>paths</code> 为输入、转换列表为输出，每个文档与每次 <code>stdout</code> 的绑定未在结构中明确。这是表达粒度的待共同复核点，不以缺少固定循环 <code>opcode</code> 作为判断理由。 引用：ir_007
3	部分保留	逐文档转换意图、单文档命令和原始 <code>convert.py</code> 全文均保留，包括 UTF-8 读取、按需建目录、写入主名.txt 和打印路径。但实际指令仍以整份 <code>paths</code> 为输入、转换列表为输出，每个文档与每次 <code>stdout</code> 的绑定未在结构中明确。这是表达粒度的待共同复核点，不以缺少固定循环 <code>opcode</code> 作为判断理由。 引用：ir_007

D05-F03 | 约束

保留原文标题仅约束转换操作，不约束汇总标题。

归属：operation / 文档转换调用 | 原文：SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	保留原文标题且仅约束转换、不限制汇总标题的原文要求落在转换范围内。第 1 轮使用只有转换调用与分派的 block_004，实际范围等价。 引用：ir_007, block_004.constraints
2	保留	保留原文标题且仅约束转换、不限制汇总标题的原文要求落在转换范围内。未扩展为 Skill 全局规则。 引用：ir_007, ir_007.constraints
3	保留	保留原文标题且仅约束转换、不限制汇总标题的原文要求落在转换范围内。未扩展为 Skill 全局规则。 引用：ir_007, ir_007.constraints

D05-F04 | 不得凭空增加

convert.py 按黑盒调用保留参数、文件输入、输出产物和标准输出路径，不把内部变量、循环或控制流拆成额外流程步骤，也不重复执行内部处理。

归属：operation / 文档转换调用 | 原文：SKILL.md:L9-L9; SKILL.md:L11-L11; scripts/convert.py:L5-L10

运行	初审	理由与实际输出引用
1	保留	转换作为一次黑盒脚本调用，完整脚本与冻结来源逐字一致；内部读取、建目录、写入及打印没有膨胀成主流程子步骤。 引用：ir_007, block_004.constraints
2	保留	转换作为一次黑盒脚本调用，完整脚本与冻结来源逐字一致；内部读取、建目录、写入及打印没有膨胀成主流程子步骤。 引用：ir_007, ir_007.constraints
3	保留	转换作为一次黑盒脚本调用，完整脚本与冻结来源逐字一致；内部读取、建目录、写入及打印没有膨胀成主流程子步骤。 引用：ir_007, ir_007.constraints

D05-F05 | 行为

必须执行被明确采用的参考打包命令，以转换产物路径列表为 `--paths` 输入；调用读取各产物文件并写入由 `--output` 指定的 `bundle.zip`，压缩包内使用各文件的文件名。

归属: operation / 打包转换产物 | 原文: SKILL.md:L12-L12; SKILL.md:L21-L23; scripts/package.py:L7-L13

运行	初审	理由与实际输出引用
1	保留	打包示例被明确采用，实际 <code>package.py</code> 调用消费转换列表并生成 <code>bundle.zip</code> ；保留完整打包脚本与 <code>archive basename</code> 逻辑。 引用: <code>ir_007</code> , <code>ir_009</code>
2	保留	打包示例被明确采用，实际 <code>package.py</code> 调用消费转换列表并生成 <code>bundle.zip</code> ；保留完整打包脚本与 <code>archive basename</code> 逻辑。 引用: <code>ir_007</code> , <code>ir_009</code>
3	保留	打包示例被明确采用，实际 <code>package.py</code> 调用消费转换列表并生成 <code>bundle.zip</code> ；保留完整打包脚本与 <code>archive basename</code> 逻辑。 引用: <code>ir_007</code> , <code>ir_009</code>

D05-F06 | 数据流

将打包生成的 `bundle.zip` 交付到先前从请求读取的 `delivery_path`，交付操作消费该读取结果，无需再次读取。

归属: operation / 交付压缩包 | 原文: SKILL.md:L8-L8; SKILL.md:L13-L13

运行	初审	理由与实际输出引用
1	保留	交付操作实际消费 <code>bundle.zip</code> 结果及请求读取的 <code>delivery_path</code> ，打包到交付的数据关系可追踪。 引用: <code>ir_005</code> , <code>ir_009</code> , <code>ir_011</code>
2	保留	交付操作实际消费 <code>bundle.zip</code> 结果及请求读取的 <code>delivery_path</code> ，打包到交付的数据关系可追踪。 引用: <code>ir_005</code> , <code>ir_009</code> , <code>ir_011</code>
3	保留	交付操作实际消费 <code>bundle.zip</code> 结果及请求读取的 <code>delivery_path</code> ，打包到交付的数据关系可追踪。 引用: <code>ir_005</code> , <code>ir_009</code> , <code>ir_011</code>

D05-F07 | 不得凭空增加

写入 `receipt.txt` 是明确不执行的历史示例，当前流程不增加该写入。

归属: operation / 历史收据示例 | 原文: SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	本版本把回执改为历史示例。诊断明确说明排除历史回执，实际 <code>CFG</code> 在交付后结束，没有创建 <code>receipt.txt</code> 的操作，识别了文本用途的最小差异。 引用: <code>diagnostics</code> , <code>ir_011</code> , <code>ir_012</code>
2	保留	本版本把回执改为历史示例。诊断明确说明排除历史回执，实际 <code>CFG</code> 在交付后结束，没有创建 <code>receipt.txt</code> 的操作，识别了文本用途的最小差异。 引用: <code>diagnostics</code> , <code>ir_011</code> , <code>ir_012</code>
3	保留	本版本把回执改为历史示例。诊断明确说明排除历史回执，实际 <code>CFG</code> 在交付后结束，没有创建 <code>receipt.txt</code> 的操作，识别了文本用途的最小差异。 引用: <code>diagnostics</code> , <code>ir_011</code> , <code>ir_012</code>

D05-F08 | 约束

不得修改输入原文文件的禁止要求作用于整份 Skill。

归属：skill / 整份多文档 Skill | 原文：SKILL.md:L15-L15

运行	初审	理由与实际输出引用
1	保留	不得修改输入文件保持为 Skill 全局原文约束，作用范围与来源一致。 引用：skill.constraints
2	保留	不得修改输入文件保持为 Skill 全局原文约束，作用范围与来源一致。 引用：skill.constraints
3	保留	不得修改输入文件保持为 Skill 全局原文约束，作用范围与来源一致。 引用：skill.constraints

D05-F09 | 约束

不得向外部服务上传文档的禁止要求作用于整份 Skill。

归属：skill / 整份多文档 Skill | 原文：SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	保留	不得向外部服务上传保持为 Skill 全局原文约束，未据此制造新的调用或上传行为。 引用：skill.constraints
2	保留	不得向外部服务上传保持为 Skill 全局原文约束，未据此制造新的调用或上传行为。 引用：skill.constraints
3	保留	不得向外部服务上传保持为 Skill 全局原文约束，未据此制造新的调用或上传行为。 引用：skill.constraints

D05-F10 | 约束

“必要时保留原顺序”的作用范围不明确，应待共同确认，不能自动提升为全局规则。

归属：unresolved / 原文未指明转换、打包或收据中的哪一处 | 原文：SKILL.md:L17-L17

运行	初审	理由与实际输出引用
1	保留	原顺序备注在 diagnostics 中保留，明确作用范围未确定；未擅自加入全局、基本块或操作约束。冻结约定要求未知范围留在诊断，因此这里采用诊断证据。 引用：diagnostics
2	保留	原顺序备注在 diagnostics 中保留，明确作用范围未确定；未擅自加入全局、基本块或操作约束。冻结约定要求未知范围留在诊断，因此这里采用诊断证据。 引用：diagnostics
3	保留	原顺序备注在 diagnostics 中保留，明确作用范围未确定；未擅自加入全局、基本块或操作约束。冻结约定要求未知范围留在诊断，因此这里采用诊断证据。 引用：diagnostics

D06 多文档处理与交付 · 语义最小差异

D06 | 多文档处理与交付 | 最小差异：语义 | 开发集 | zh

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	8	7	8
边数	7	6	7
操作数	16	14	16
三层约束条数	7	5	5
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

第 1、3 轮为 8 块、7 边，第 2 轮合并读取 manifest.paths 而为 7 块、6 边；三轮都额外读取 archive_path 并实际用于交付，仍保留未使用的 delivery_path 读取。前两轮用 bundle.zip 文件副作用衔接交付，第 3 轮使用显式打包结果，两者均有可追踪文件与目的地。第 1 轮错误将原顺序备注全局化，后两轮保持未知范围。

交付目的地变化三轮均识别，黑盒、打包与独立回执行为保留。逐项转换绑定三轮均记为待共同复核的部分保留；第 1 轮另外存在未知范围被全局化的语义偏差。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 D06 的三次明细。

逐事实初审（参考事实已确认，结果判定待共同复核）

D06-F01 | 行为

读取 `manifest.json.paths` 作为文档路径来源，并读取请求中的 `output_dir` 和 `delivery_path`。

归属：operation / 读取清单与请求 | 原文：SKILL.md:L8-L8

运行	初审	理由与实际输出引用
1	保留	<code>manifest.json</code> 的 <code>paths</code> 是文档路径来源，同时实际读取请求的 <code>output_dir</code> 与 <code>delivery_path</code> ；即使 <code>delivery_path</code> 在此最小差异版本不再用于交付，该读取行为仍保留。 引用：ir_001, ir_003, ir_005
2	保留	<code>manifest.json</code> 的 <code>paths</code> 是文档路径来源，同时实际读取请求的 <code>output_dir</code> 与 <code>delivery_path</code> ；即使 <code>delivery_path</code> 在此最小差异版本不再用于交付，该读取行为仍保留。 引用：ir_001, ir_003
3	保留	<code>manifest.json</code> 的 <code>paths</code> 是文档路径来源，同时实际读取请求的 <code>output_dir</code> 与 <code>delivery_path</code> ；即使 <code>delivery_path</code> 在此最小差异版本不再用于交付，该读取行为仍保留。 引用：ir_001, ir_003, ir_005

D06-F02 | 行为

逐个以文档路径和 `output_dir` 调用 `scripts/convert.py`；每次调用读取该 UTF-8 源文，按需创建输出目录，将内容写入 `output_dir` 下的 `<源文主名>.txt`，并把标准输出的该路径作为转换产物路径。

归属：operation / 文档转换调用 | 原文：SKILL.md:L9-L9; scripts/convert.py:L5-L10

运行	初审	理由与实际输出引用
1	部分保留	逐文档意图、单文档命令及完整转换脚本保留了 UTF-8 读取、创建目录、写入主名.txt 和 stdout 路径。实际转换操作消费路径集合并产出列表，每个输入 path 与各次 stdout 的绑定未明确。此项为可验证表达粒度的部分保留，待共同复核；不要求某个固定循环 opcode。 引用：ir_007
2	部分保留	逐文档意图、单文档命令及完整转换脚本保留了 UTF-8 读取、创建目录、写入主名.txt 和 stdout 路径。实际转换操作消费路径集合并产出列表，每个输入 path 与各次 stdout 的绑定未明确。此项为可验证表达粒度的部分保留，待共同复核；不要求某个固定循环 opcode。 引用：ir_005
3	部分保留	逐文档意图、单文档命令及完整转换脚本保留了 UTF-8 读取、创建目录、写入主名.txt 和 stdout 路径。实际转换操作消费路径集合并产出列表，每个输入 path 与各次 stdout 的绑定未明确。此项为可验证表达粒度的部分保留，待共同复核；不要求某个固定循环 opcode。 引用：ir_007

D06-F03 | 约束

保留原文标题仅约束转换操作，不约束汇总标题。

归属：operation / 文档转换调用 | 原文：SKILL.md:L10-L10

运行	初审	理由与实际输出引用
1	保留	标题保留及其仅作用于转换、不限制汇总标题的原文要求关联到转换调用，没有扩展到汇总或整份 Skill。 引用：ir_007, ir_007.constraints
2	保留	标题保留及其仅作用于转换、不限制汇总标题的原文要求关联到转换调用，没有扩展到汇总或整份 Skill。 引用：ir_005, ir_005.constraints
3	保留	标题保留及其仅作用于转换、不限制汇总标题的原文要求关联到转换调用，没有扩展到汇总或整份 Skill。 引用：ir_007, ir_007.constraints

D06-F04 | 不得凭空增加

convert.py 按黑盒调用保留参数、文件输入、输出产物和标准输出路径，不把内部变量、循环或控制流拆成额外流程步骤，也不重复执行内部处理。

归属：operation / 文档转换调用 | 原文：SKILL.md:L9-L9; SKILL.md:L11-L11; scripts/convert.py:L5-L10

运行	初审	理由与实际输出引用
1	保留	转换保持黑盒调用并附完整原始 convert.py，脚本文本已与冻结来源核对一致；未展开其内部读取和写入步骤。 引用：ir_007, ir_007.constraints
2	保留	转换保持黑盒调用并附完整原始 convert.py，脚本文本已与冻结来源核对一致；未展开其内部读取和写入步骤。 引用：ir_005, ir_005.constraints
3	保留	转换保持黑盒调用并附完整原始 convert.py，脚本文本已与冻结来源核对一致；未展开其内部读取和写入步骤。 引用：ir_007, ir_007.constraints

D06-F05 | 行为

必须执行被明确采用的参考打包命令，以转换产物路径列表为 `--paths` 输入；调用读取各产物文件并写入由 `--output` 指定的 `bundle.zip`，压缩包内使用各文件的文件名。

归属：operation / 打包转换产物 | 原文：SKILL.md:L12-L12; SKILL.md:L21-L23; scripts/package.py:L7-L13

运行	初审	理由与实际输出引用
1	保留	实际采用 <code>package.py</code> 示例打包转换列表，输出文件为 <code>bundle.zip</code> ；完整脚本保留按 <code>basename</code> 归档的行为。这里以文件写入副作用表达产物，未强制新增结果值。 引用：ir_007, ir_009
2	保留	实际采用 <code>package.py</code> 示例打包转换列表，输出文件为 <code>bundle.zip</code> ；完整脚本保留按 <code>basename</code> 归档的行为。这里以文件写入副作用表达产物，未强制新增结果值。 引用：ir_005, ir_007
3	保留	实际采用 <code>package.py</code> 示例打包转换列表，输出文件为 <code>bundle.zip</code> ；完整脚本保留按 <code>basename</code> 归档的行为。打包另有显式产物结果。 引用：ir_007, ir_009

D06-F06 | 数据流

从请求读取 `archive_path`，并将打包生成的 `bundle.zip` 交付到该读取结果指定的位置；先前明确要求读取的 `delivery_path` 仍保留，但不再用于交付。

归属：operation / 交付压缩包 | 原文：SKILL.md:L8-L8; SKILL.md:L13-L13

运行	初审	理由与实际输出引用
1	保留	交付前单独读取请求的 <code>archive_path</code> ，并把这一读取结果用作 <code>bundle.zip</code> 的交付目的地；先前 <code>delivery_path</code> 读取仍存在但不再使用。 <code>bundle.zip</code> 的路径与前序黑盒写出的同一文件一致，文件副作用衔接成立。数据改动已进入实际操作数，而非仅写在诊断中。 引用：ir_003, ir_009, ir_011, ir_013
2	保留	交付前单独读取请求的 <code>archive_path</code> ，并把这一读取结果用作 <code>bundle.zip</code> 的交付目的地；先前 <code>delivery_path</code> 读取仍存在但不再使用。 <code>bundle.zip</code> 的路径与前序黑盒写出的同一文件一致，文件副作用衔接成立。数据改动已进入实际操作数，而非仅写在诊断中。 引用：ir_003, ir_007, ir_009, ir_011
3	保留	交付前单独读取请求的 <code>archive_path</code> ，并把这一读取结果用作 <code>bundle.zip</code> 的交付目的地；先前 <code>delivery_path</code> 读取仍存在但不再使用。交付同时引用打包生成的产物结果。数据改动已进入实际操作数，而非仅写在诊断中。 引用：ir_005, ir_009, ir_011, ir_013

D06-F07 | 行为

单独将转换产物路径列表写入本地 receipt.txt，并保留对转换产物路径的依赖。

归属：operation / 写入交付收据 | 原文：SKILL.md:L14-L14

运行	初审	理由与实际输出引用
1	保留	回执仍是当前独立行为，消费转换产物列表并写入 receipt.txt；更改交付目的地未删除这一行为或改用打包内容作为输入。 引用：ir_007, ir_015
2	保留	回执仍是当前独立行为，消费转换产物列表并写入 receipt.txt；更改交付目的地未删除这一行为或改用打包内容作为输入。 引用：ir_005, ir_013
3	保留	回执仍是当前独立行为，消费转换产物列表并写入 receipt.txt；更改交付目的地未删除这一行为或改用打包内容作为输入。 引用：ir_007, ir_015

D06-F08 | 约束

不得修改输入原文文件的禁止要求作用于整份 Skill。

归属：skill / 整份多文档 Skill | 原文：SKILL.md:L15-L15

运行	初审	理由与实际输出引用
1	保留	不得修改输入文件的原文要求保持 Skill 全局范围。 引用：skill.constraints
2	保留	不得修改输入文件的原文要求保持 Skill 全局范围。 引用：skill.constraints
3	保留	不得修改输入文件的原文要求保持 Skill 全局范围。 引用：skill.constraints

D06-F09 | 约束

不得向外部服务上传文档的禁止要求作用于整份 Skill。

归属：skill / 整份多文档 Skill | 原文：SKILL.md:L16-L16

运行	初审	理由与实际输出引用
1	保留	不得上传外部服务的原文要求保持 Skill 全局范围，交付仍对应本地请求路径。 引用：skill.constraints, ir_013
2	保留	不得上传外部服务的原文要求保持 Skill 全局范围，交付仍对应本地请求路径。 引用：skill.constraints, ir_011
3	保留	不得上传外部服务的原文要求保持 Skill 全局范围，交付仍对应本地请求路径。 引用：skill.constraints, ir_013

D06-F10 | 约束

“必要时保留原顺序”的作用范围不明确，应待共同确认，不能自动提升为全局规则。

归属：unresolved / 原文未指明转换、打包或收据中的哪一处 | 原文：SKILL.md:L17-L17

运行	初审	理由与实际输出引用
1	冲突	未知范围的“必要时保留原顺序”被直接放进 skill.constraints，擅自提升为全局规则；关于 archive_path 和未使用 delivery_path 的诊断没有解决这一范围错误。 引用：skill.constraints
2	保留	diagnostics 明确保留原顺序备注的作用范围不明，未把它分配给全局或某个操作。这与冻结约定中未知范围留待共同确认的表达一致。 引用：diagnostics
3	保留	diagnostics 明确保留原顺序备注的作用范围不明，未把它分配给全局或某个操作。这与冻结约定中未知范围留待共同确认的表达一致。 引用：diagnostics

R01 pdf

R01 | 真实 Skill | 原始快照 | 开发集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	23	17	20
边数	27	18	31
操作数	45	34	40
三层约束条数	21	12	28
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三次均保留主要PDF工作流和质量/路径约束。第1次把Python依赖安装另成组件且丢缺失前提，第2次将替代安装命令无条件拼接，第3次较完整保留依赖分支但render缺输出前缀操作数。第2次还遗漏文字提取不可证明版式的局部约束。

本批次三次实际CFG的逐事实初审；结构complete不意味着条件、数据流与约束完整。未执行被审Skill脚本或工具，结论待共同复核。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 R01 的三次明细。

事实表以外的发现

运行 1 | 生成/阅读/提取分支汇合后summary同时消费仅文字提取分支产生的extracted_text/text_checks；未说明创建或单纯视觉复核路径的缺失结果处理。另有依据agent默认提示加入summary的诊断，不把此元数据当作独立用户任务。

引用：ir_033

运行 3 | 初次与更新后的PNG结果同时列作检查输入，没有显式最新版本选择；虽有正确复核回边，当前页面版本的消费边界值得共同核对。

引用：ir_031, ir_037

运行 1, 2, 3 | 中间文件清理保留delete when done及after final approval两种原文表述；未擅自消除冻结开放问题。

引用：skill.constraints

逐事实初审（参考事实已确认，结果判定待共同复核）

R01-F01 | 行为

阅读或复核版式时，将 PDF 页面渲染为 PNG 并逐页检查；新建文档时采用 reportlab 生成 PDF。

归属：block / Workflow | 原文：SKILL.md:L14-L20

运行	初审	理由与实际输出引用
1	保留	创建分支使用reportlab；阅读/复核路径渲染实际PDF到PNG并消费PNG做视觉检查。 引用：ir_003, ir_013, ir_023, edge_001, edge_002
2	保留	按任务类型选择阅读复核或创建；reportlab生成及两组PNG渲染/视觉检查均为实际操作。 引用：ir_009, ir_011, ir_017, ir_023, ir_025, edge_006, edge_007
3	保留	新建条件使用reportlab；渲染产生PNG并进入视觉检查，保留主要实际行为。 引用：ir_023, ir_029, ir_031, edge_019, edge_021

R01-F02 | 条件

有 pdftoppm 时使用它；不可用时安装 Poppler 或请用户本地复核。无法安装则说明缺失依赖和本地安装方法。

归属：block / Workflow / dependency fallback | 原文：SKILL.md:L14-L20；SKILL.md:L47-L47

运行	初审	理由与实际输出引用
1	部分保留	pdftoppm可用/不可用及安装或本地复核分支存在；但没有安装不可行时说明缺失依赖及安装方法的操作，安装分支直接进入渲染。 引用：ir_011, ir_017, ir_019, ir_021, edge_008, edge_009, edge_010
2	部分保留	有一般缺依赖安装和失败后请用户本地复核，但未保留pdftoppm专属可用优先规则和说明缺失依赖/安装方法的失败输出。 引用：ir_003, ir_005, ir_033, edge_002, edge_005
3	部分保留	有pdftoppm检查、安装/本地复核分支与失败返回；通知操作的实际输入仅泛称Missing dependency并无依赖名称或安装方法，完整要求仅留在块约束。 引用：ir_009, ir_017, ir_019, edge_009, edge_010, edge_011

R01-F03 | 约束

提取文字和快速检查可用 pdfplumber 或 pypdf；“do not rely on it for layout fidelity”约束文字提取操作，不能由文字提取成功推断版式已经合格。

归属：operation / text extraction | 原文：SKILL.md:L19-L20

运行	初审	理由与实际输出引用
1	保留	文字提取实际使用pdfplumber/pypdf和PDF，局部块约束明确不据此推断版式保真。 引用：ir_005, block_003.constraints
2	部分保留	确有pdfplumber_or_pypdf文字提取，但全局、块、操作约束均未保留do not rely on it for layout fidelity；文字检查不能替代该局部限制。 引用：ir_013, ir_021
3	保留	文字提取局部块明确pdfplumber或pypdf且不可依赖它证明版式，后续另有PNG检查。 引用：ir_025, block_013.constraints

R01-F04 | 数据流

渲染命令的 INPUT_PDF 是待渲染 PDF，OUTPUT_PREFIX 决定 PNG 输出前缀；输入 PDF 与输出页面图像必须作为操作数据保留。

归属: operation / pdftoppm render | 原文: SKILL.md:L15-L17; SKILL.md:L52-L55

运行	初审	理由与实际输出引用
1	保留	render实际输入为所选PDF及output_prefix，输出rendered_pngs，脚本亦保留INPUT_PDF/OUTPUT_PREFIX对应。 引用: ir_009, ir_013
2	保留	实际读取INPUT_PDF/OUTPUT_PREFIX到result，两个render操作都消费对应PDF与prefix并产出PNG。 引用: ir_001, ir_009, ir_023
3	部分保留	PDF输入和PNG输出在操作数中保留，但两次render的OUTPUT_PREFIX只存在script_content，没有作为输入数据暴露。 引用: ir_027, ir_029, ir_035

R01-F05 | 条件

每次实质更新后重新渲染并检查对齐、间距和可读性；交付前最新 PNG 检查必须无视觉或格式缺陷。

归属: block / update and delivery gate | 原文: SKILL.md:L20-L20; SKILL.md:L64-L66

运行	初审	理由与实际输出引用
1	保留	有检查缺陷、更新并重新渲染、最新检查结果汇合以及最终检查通过才交付的条件；更新后对齐/间距/可读性要求在局部约束中保留。 引用: ir_025, ir_027, ir_031, edge_017, edge_021, edge_022, block_014.constraints
2	部分保留	创建路径有缺陷后更新重渲染循环，全局有交付门槛；阅读复核路径的final check后直接return，且更新后对齐/间距/可读性要求未完整保留。 引用: ir_027, ir_029, edge_016, edge_017, edge_018, skill.constraints
3	保留	缺陷触发更新与重渲染后回到PNG检查，最终检查通过才离开；更新后对齐/间距/可读性与交付门槛保留。 引用: ir_031, ir_033, ir_035, ir_037, edge_026, edge_027, edge_029, edge_030, edge_031

R01-F06 | 约束

中间文件放 tmp/pdfs/，最终成品在本仓库放 output/pdf/，文件名稳定且有描述性；这些路径约束对应文件写出。

归属: skill / artifact paths | 原文: SKILL.md:L22-L25

运行	初审	理由与实际输出引用
1	保留	全局准确保留tmp/pdfs/中间文件、output/pdf/最终成品和稳定描述性文件名要求；清理时点的两种原文表述未强行合并。 引用: skill.constraints
2	保留	全局准确保留tmp/pdfs/中间文件、output/pdf/最终成品和稳定描述性文件名要求；清理时点的两种原文表述未强行合并。 引用: skill.constraints
3	保留	全局准确保留tmp/pdfs/中间文件、output/pdf/最终成品和稳定描述性文件名要求；清理时点的两种原文表述未强行合并。 引用: skill.constraints

R01-F07 | 不得凭空增加

依赖安装有“install if missing”前提，uv 不可用才采用 pip；macOS 和 Ubuntu 命令分别是平台分支，不应无条件逐条安装。

归属: block / Dependencies | 原文: SKILL.md:L27-L45

运行	初审	理由与实际输出引用
1	部分保留	uv/pip互斥和macOS/Ubuntu安装分支存在，但Python安装独立组件无install-if-missing条件，不能据uv可用便安装。 引用: ir_035, ir_037, ir_039, edge_023, edge_024, edge_011, edge_012
2	冲突	虽有dependencies missing前提，黑盒脚本却将uv pip、python3 pip、brew、apt四条命令无条件串联，丢uv不可用及平台互斥条件。 引用: ir_005, edge_002
3	保留	先检查缺失，缺失才安装；uv与pip、macOS与Ubuntu分支分开，不把替代命令全部执行。 引用: ir_001, ir_003, ir_005, ir_007, ir_013, ir_015, edge_001, edge_002, edge_003, edge_004, edge_012, edge_013

R01-F08 | 约束

整份 PDF 要避免裁切、重叠、破损表格和不可读字符；页眉页脚、页码与章节转换需复核。这是交付质量约束，不凭空制造图表内容。

归属：skill / PDF quality and final checks | 原文：SKILL.md:L57-L62；SKILL.md:L64-L67

运行	初审	理由与实际输出引用
1	保留	整份PDF质量约束保留裁切、重叠、表格/字形、页眉页脚页码章节等要求，没有将质量说明扩成无依据图表内容。 引用：skill.constraints
2	保留	整份PDF质量约束保留裁切、重叠、表格/字形、页眉页脚页码章节等要求，没有将质量说明扩成无依据图表内容。 引用：skill.constraints
3	保留	整份PDF质量约束保留裁切、重叠、表格/字形、页眉页脚页码章节等要求，没有将质量说明扩成无依据图表内容。 引用：skill.constraints

R02 playwright

R02 | 真实 Skill | 原始快照 | 开发集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	11	12	10
边数	9	12	11
操作数	22	23	20
三层约束条数	31	27	32
生成尝试	1	1	1
结构修复	0	0	0
HTTP 尝试	1	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

第2次实际snapshot/交互/刷新循环最完整；第1次刷新线性串接，第3次核心浏览器流程仅藏于通用wrapper块约束。三次都保留原wrapper脚本内的session优先级，未把命令目录串成全量执行；工具例外和配置/工作目录补充存在不同程度遗漏。

本批次三次实际CFG的逐事实初审；结构complete不意味着条件、数据流与约束完整。未执行被审Skill脚本或工具，结论待共同复核。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 R02 的三次明细。

事实表以外的发现

运行 1 | wrapper参数读取与执行另成独立组件，未与主流程的wrapper资源引用连边；保留黑盒源码可见会话语义，但整体连通关系仍需复核。

引用：block_010, block_011

运行 3 | 两个参考文档needed条件可能同时为真，当前CFG却各自直接汇入会话步骤，未说明需要同时阅读两份参考时的控制流。

引用：ir_011, edge_006, edge_007

逐事实初审（参考事实已确认，结果判定待共同复核）

R02-F01 | 约束

默认优先 `bundled wrapper`；仓库已采用全局安装标准或用户偏好全局安装时，可以使用全局 `CLI`。用户没有明确要求测试文件时，不切换到 `@playwright/test`。

归属：skill / tool choice | 原文：SKILL.md:L9-L10；SKILL.md:L122-L130；SKILL.md:L57-L62

运行	初审	理由与实际输出引用
1	部分保留	<code>wrapper</code> 优先与未经用户要求不切test均保留；全局安装例外表述为CLI已安装，未完整保留用户偏好/仓库标准两个来源条件。 引用：skill.constraints, block_005.constraints
2	部分保留	<code>wrapper</code> 优先、仓库全局安装例外及不切test保留，但用户偏好全局安装这一例外未明确保留。 引用：skill.constraints, ir_013.constraints
3	部分保留	<code>wrapper</code> 优先和test禁令保留，例外仅写CLI已全局安装，未完整保留用户偏好或仓库标准。 引用：skill.constraints

R02-F02 | 条件

提出命令前检查 `npx`；缺失则暂停并让用户安装 `Node.js/npm`，提供原文安装步骤；`npx` 存在后继续 `wrapper`，全局安装为可选。

归属：block / prerequisite gate | 原文：SKILL.md:L12-L32

运行	初审	理由与实际输出引用
1	保留	命令前检查npx，缺失时展示Node/npm及安装CLI步骤后return；可用分支才解析wrapper，整体说明全局安装可选。 引用：ir_001, ir_003, ir_004, edge_001, edge_002
2	保留	npx检查为入口门槛，缺失分支提示安装步骤后停止，可用路径进入wrapper配置。 引用：ir_001, ir_003, ir_004, edge_001, edge_002
3	保留	npx检查后缺失分支携带原步骤列表并结束，正常路径才继续wrapper。 引用：ir_003, ir_005, ir_006, edge_002, edge_003

R02-F03 | 数据流

打开目标页面后 snapshot 获取元素引用，交互消费最新 snapshot 的引用；导航或显著 DOM 变化后重新 snapshot。

归属：block / Core workflow | 原文：SKILL.md:L63-L69

运行	初审	理由与实际输出引用
1	部分保留	open后snapshot产refs且交互消费它，但refresh是无条件后继，updated_element_refs无后续消费；显著变化条件仅在块文字中。 引用：ir_009, ir_011, ir_013, ir_015, edge_005, edge_006, edge_007
2	保留	打开后snapshot产生实际refs供交互，交互结果用于是否刷新判断；导航/DOM变化等条件回到snapshot，再以新refs交互。 引用：ir_013, ir_015, ir_017, ir_019, edge_007, edge_008, edge_010
3	部分保留	核心open/snapshot/interact/刷新步骤仅写在黑盒调用的块约束；实际只有通用wrapper操作，未暴露snapshot产生refs及交互消费refs的数据流。 引用：ir_019, block_010.constraints

R02-F04 | 条件

引用丢失或过期导致命令失败时，重新 snapshot 后重试；该恢复规则只针对引用失效，不能扩展为任意失败都重试。原文未规定最大重试次数，不能凭空增加次数上限。

归属：block / stale reference recovery | 原文：SKILL.md:L80-L89；SKILL.md:L132-L137；
references/workflows.md:L91-L95

运行	初审	理由与实际输出引用
1	部分保留	全局文字保留ref失败snapshot后retry，未扩大为任意失败；实际CFG没有ref失败分支或回到交互的重试边。 引用：skill.constraints, ir_013, ir_015
2	保留	仅missing/stale ref失败触发对应恢复，条件回到snapshot后再交互；没有任意错误重试或虚构次数上限。 引用：ir_019.constraints, edge_010
3	部分保留	保留refs过期重新snapshot的泛化约束，但缺少ref失败后的明确retry要求和恢复数据流。 引用：skill.constraints, block_010.constraints

R02-F05 | 约束

尚无新 snapshot 时只能使用 eX 等占位引用并解释；禁止通过 run-code 绕过引用。eval/run-code 仅在需要时使用，不能当作默认动作。

归属: operation / element interaction | 原文: SKILL.md:L139-L147

运行	初审	理由与实际输出引用
1	保留	没有fresh snapshot时eX占位并说明、禁止run-code绕过refs以及eval/run-code非默认全部保留；未新增这些调用。 引用: skill.constraints
2	保留	eX占位解释、禁止绕过refs和只在需要时eval/run-code在交互局部完整保留。 引用: ir_017.constraints
3	保留	该交互黑盒局部保留eX占位说明、禁止run-code绕过和显式命令优先规则。 引用: block_010.constraints

R02-F06 | 数据流

wrapper 使用 npx 启动 @playwright/cli 并透传调用参数；只有调用未显式给 --session 且环境变量非空时，才注入 PLAYWRIGHT_CLI_SESSION。

归属: operation / wrapper invocation | 原文: SKILL.md:L124-L130; scripts/playwright_cli.sh:L9-L25

运行	初审	理由与实际输出引用
1	保留	黑盒边界明确cli_args/session输入与cli_output输出；保留原脚本，实际条件为无显式--session且环境非空才注入，然后追加完整\$@以npx运行。 引用: ir_019, ir_021
2	保留	wrapper原脚本完整保留在实际调用metadata，含--session/--session=*扫描、环境非空条件及\$@透传；未展开内部循环符合允许黑盒变体。 引用: ir_013
3	保留	实际wrapper黑盒保留原脚本及会话输入；源码里的显式session优先、非空环境注入及\$@透传均未改写。 引用: ir_019

R02-F07 | 不得凭空增加

CLI reference 按 Core / Navigation / Keyboard 等列出的命令是可用调用形式；不能把 open、close、全部点击与键盘命令接成一次无条件流程。

归属: block / CLI reference catalogue | 原文: SKILL.md:L132-L137; references/cli.md:L19-L49

运行	初审	理由与实际输出引用
1	保留	命令参考与示例没有被串成全部调用；实际只有任务相关open、snapshot、交互及产物流程。 引用: diagnostics, ir_009, ir_013
2	保留	reference示例被当文档，没有固定执行全部导航/键盘/关闭命令。 引用: diagnostics, ir_013, ir_017
3	保留	具体参考命令未变成无条件操作，实际由任务输入驱动wrapper黑盒。 引用: diagnostics, ir_019

R02-F08 | 约束

仓库中捕获的产物放 `output/playwright/`; `workflow` 对此补充以 `output/playwright/<label>/` 为工作目录保持产物集中。

归属: `block / artifact capture` | 原文: `SKILL.md:L146-L146`; `SKILL.md:L132-L137`; `references/workflows.md:L3-L5`

运行	初审	理由与实际输出引用
1	保留	产物 <code>output/playwright/</code> 以及工作目录 <code>output/playwright/<label>/</code> 均完整保留。 引用: <code>skill.constraints</code> , <code>block_009.constraints</code>
2	部分保留	保留 <code>output/playwright/</code> 产物目录, 但 <code>references/workflows.md</code> 所给 <code><label></code> 工作目录补充被遗漏。 引用: <code>skill.constraints</code> , <code>ir_021.constraints</code>
3	保留	全局含 <code>output/playwright/<label>/</code> 工作目录, 捕获产物局部保留 <code>output/playwright/</code> 。 引用: <code>skill.constraints</code> , <code>block_010.constraints</code>

R02-F09 | 不得凭空增加

`playwright-cli.json` 默认从当前目录读取, `--config` 可覆盖路径; 后接 JSON 标为 `Minimal example`, 不能把 `headless=false` 或 `1280×720` 视作每次执行的固定参数。

归属: `operation / CLI configuration` | 原文: `SKILL.md:L132-L137`; `references/workflows.md:L72-L89`

运行	初审	理由与实际输出引用
1	部分保留	没有把 <code>headless=false</code> 或 <code>1280x720</code> 固化, 但当前目录 <code>playwright-cli.json</code> 及 <code>--config</code> 覆盖语义未进入结果。 引用: <code>diagnostics</code>
2	部分保留	未固化示例配置, 却同时省略默认当前目录配置和 <code>--config</code> 覆盖规则。 引用: <code>diagnostics</code>
3	保留	明确默认当前目录 <code>playwright-cli.json</code> 及 <code>--config</code> 覆盖, 示例参数没有被固定使用。 引用: <code>skill.constraints</code> , <code>diagnostics</code>

R03 gh-fix-ci

R03 | 真实 Skill | 原始快照 | 开发集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	14	13	26
边数	14	13	29
操作数	28	26	52
三层约束条数	17	30	14
生成尝试	2	2	3
结构修复	1	1	2
HTTP 尝试	2	2	3
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

第 1、2 轮将手动回退压成一条批次来源，条件重试与跨调用绑定主要停留在约束/命令文字；第 3 轮显式区分缺省 PR、Actions/external 和 pending 日志，仍缺手动字段重试与 job id/写文件语义。三轮认证、黑盒脚本及实施审批门保留，非零退出均未抹去脚本报告；失败摘要主要供内部计划，面向用户交付不完整。第 2 轮源码是 510 项行数组（多一尾空行），第 1、3 轮为逐字匹配的原字符串。

真实包多分支语义可部分保留，完整黑盒脚本为字段兼容、日志回退及退出码的有效证据；不能拿脚本分支替手动分支补齐。结构成功不代表条件路径、结果可用性或对用户交付语义完整。逐事实理由和 additional findings 为初审意见，供共同复核。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 R03 的三次明细。

事实表以外的发现

运行 1 | 脚本和手动检查由互斥分支选择，但合流处 ir_013 同时读取 script_exit_code result_006 与 manual_failing_checks_found result_008，ir_015 又同时消费两分支的 inspection_results result_005/result_007。实际每条路径只产生其中一组，未见当前路径选择或合流值绑定；结构校验接受不证明路径上两个结果同时可用。这里要求明确适用结果的选择语义，不要求固定 phi opcode。

引用：ir_009, ir_011, ir_013, ir_015, edge_004, edge_005, edge_006, edge_007

运行 2 | metadata.script_content 使用 510 项字符串行数组而非原代码字符串。逐行比较与冻结 inspect_pr_checks.py 的差别仅为末尾多一空行，语义内容完整可恢复；记录表示形式差异，不将其描述成原始字符串逐字相同，也不据此宣称代码行为缺失。

引用：ir_008

运行 2 | edge_005 在尚未调用首选脚本的 block_004 就以 “preferred method fails” 进入手动回退；此时没有产生首选调用的失败结果，ir_008 执行后又只前往选择块。脚本不可用的选择意图可理解，实际执行失败

后的回退路径未表达，且不能以所有非零退出码推定没有可用分析结果。

引用: ir_006, ir_008, ir_010, edge_004, edge_005

运行 3 | ir_032 无条件消费 job_logs result_018，但它只由 pending 分支 ir_030 产生；logs available 的 edge_018 绕过该调用直接进入总结，因此该路径没有 job_logs 结果。没有对这一可选日志结果的选择/缺省绑定，结构接受不能证明路径上可用；不要求固定 phi opcode。

引用: ir_030, ir_032, edge_017, edge_018, edge_019

逐事实初审（参考事实已确认，结果判定待共同复核）

R03-F01 | 条件

先在仓库验证 gh auth status；未认证时请用户登录且具备所需 scope，之后才继续。

归属: block / authentication gate | 原文: SKILL.md:L29-L31

运行	初审	理由与实际输出引用
1	保留	ir_003 在 repo_path 下读取 gh auth 状态，edge_002 未认证分支进入 ir_005 请求 gh auth login 与 repo/workflow scope 后 return；只有 edge_003 authenticated 才继续查询。 引用: ir_003, ir_005, edge_002, edge_003
2	保留	ir_003 在 repo_path 中执行 gh auth status，edge_002 未认证即 ir_005 返回登录与 repo/workflow scopes 提示，edge_003 仅认证后继续。 引用: ir_003, ir_005, edge_002, edge_003
3	保留	ir_003 在 repo_path 中检查 gh auth 状态；未认证的 edge_002 进入 ir_005 登录指导与 ir_006 停止，要求 repo/workflow scope，只有认证分支继续。 引用: ir_003, ir_005, ir_006, edge_002, edge_003

R03-F02 | 数据流

用户提供的 PR 号或 URL 直接作为 PR 标识；省略时读取当前分支 PR，后续 failing-check 查询消费这个解析结果。

归属: block / resolve PR then inspect | 原文: SKILL.md:L16-L20; SKILL.md:L32-L38

运行	初审	理由与实际输出引用
1	部分保留	ir_001 实际读取可选 pr 并传给 bundled ir_009 和手动 ir_011；ir_009 原始脚本在 resolve_pr 中保留显式 PR 直用、缺省 gh pr view 的来源链。但手动路径直接使用可缺省 pr_input，commands 中没有 gh pr view，也没有解析当前分支 PR 的结果生产者。 引用: ir_001, ir_009, ir_011, edge_004, edge_005
2	部分保留	ir_001 读取用户 PR，ir_006 产出 pr_value result_004，bundled 和手动检查均消费它；显式 PR 直用、缺省 gh pr view 已在 ir_006.constraints 保留。但 ir_006 被无条件建成 external 来源，图未区分只有省略 PR 时才需当前分支查询；条件意图有文字，条件读取路径仍不充分。 引用: ir_001, ir_006, ir_006.constraints, ir_008, ir_010
3	保留	ir_001 读取可选 pr，block_004 按 provided/not provided 分支，缺省才由 ir_009 获取当前分支 PR；ir_011 select_pr_value 产出 resolved_pr result_005，bundled ir_014 与手动 ir_016 均消费该选择结果。 引用: ir_001, ir_009, ir_011, ir_014, ir_016, edge_004, edge_005

R03-F03 | 行为

首选运行 `bundled inspect_pr_checks.py` 取得失败检查、GitHub Actions 日志和失败片段；可通过 `--json` 获取供总结的机器输出，脚本作为一个调用边界保留。

归属: `operation / inspect_pr_checks.py` | 原文: SKILL.md:L35-L38; SKILL.md:L60-L69

运行	初审	理由与实际输出引用
1	保留	ir_009 在首选 <code>bundled</code> 分支以一个黑盒保留完整且逐字匹配的 <code>inspect_pr_checks.py</code>，输入 <code>repo/pr</code> 与 <code>gh/git</code> 资源，输出分析结果 <code>result_005</code> 及退出码 <code>result_006</code>。原代码保留 <code>--json</code>、失败检查、日志和片段输出，未展开内部处理成额外图步骤。 引用: <code>block_004.constraints</code>, <code>ir_009</code>
2	保留	ir_008 是首选 <code>bundled script</code> 黑盒调用，消费 <code>repo</code> 与已解析 <code>PR</code>，产出分析报告和退出码，约束保留 <code>--json</code>。<code>script_content</code> 虽为行数组，逐行比较只比冻结文件多一个末尾空行，代码行为、参数与日志片段输出完整可恢复，没有展开内部函数/循环。表示形式差异另列 <code>finding</code>。 引用: <code>ir_008</code>, <code>ir_008.constraints</code>
3	保留	ir_014 在首选分支以单个黑盒调用 <code>inspect_pr_checks.py</code>，实际绑定 <code>repo</code> 和 <code>resolved_pr</code>，并分别产出报告与退出码。<code>script_content</code> 与冻结脚本本字符串逐字一致，包含 <code>--json</code>、失败检查、日志及片段；未把代码内部函数/循环拆进主图。 引用: <code>ir_014</code>, <code>edge_008</code>

R03-F04 | 条件

手动回退查询若 `JSON` 字段被 `gh` 拒绝，按 `gh` 返回的可用字段重试；脚本同样用 `available_fields` 筛选 `fallback_fields`，无可字段或再失败则返回。

归属: `block / gh field compatibility fallback` | 原文: SKILL.md:L39-L44; `scripts/inspect_pr_checks.py`:L182-L215

运行	初审	理由与实际输出引用
1	部分保留	<code>bundled ir_009</code> 的原代码保留 <code>available_fields</code> 筛选、无可字段/再失败停止。手动 <code>ir_011</code> 只在 <code>block_006.constraints</code> 写字段被拒绝时按可用字段重跑，<code>commands</code> 中只有首次查询，无字段结果生产/重试路径；诊断明确把这些条件压进单操作，手动条件只保留了文字意图。 引用: <code>ir_009</code>, <code>ir_011</code>, <code>block_006.constraints</code>, <code>diagnostics</code>
2	部分保留	ir_008 保留的脚本行数组完整包含 <code>available_fields</code>、<code>fallback_fields</code> 与失败停止逻辑。手动 <code>ir_010</code> 将字段拒绝后的重跑仅放 <code>constraints</code>，<code>metadata</code> 仍是首次命令，未产生可用字段结果或分支/重试调用，无法以脚本分支证据代替手动分支。 引用: <code>ir_008</code>, <code>ir_010</code>, <code>ir_010.constraints</code>, <code>diagnostics</code>
3	部分保留	ir_014 原始脚本保留 <code>available_fields</code> 过滤、无可字段或再失败停止。手动 <code>ir_016</code> 只在 <code>constraints</code> 保留字段拒绝时重跑，下一步直接过滤检查，没有字段错误判定、可用字段结果或实际重试调用。脚本正确不能填补手动分支。 引用: <code>ir_014</code>, <code>ir_016</code>, <code>ir_016.constraints</code>, <code>edge_011</code>

R03-F05 | 数据流

从失败检查的 `detailsUrl` 提取 `run id` 并读取对应 `run` 元信息和日志；`run` 日志报告仍在运行中且有 `job id` 时回退至 `job logs`。手动 `gh api` 回退将日志重定向写入指定的 `<path>`；脚本回退返回内存中的日志内容及状态，失败或尚未完成分别保留状态。

归属：block / log retrieval | 原文：SKILL.md:L42-L46；scripts/inspect_pr_checks.py:L333-L355；scripts/inspect_pr_checks.py:L366-L377

运行	初审	理由与实际输出引用
1	部分保留	ir_009 原代码保留 <code>run/job id</code> 提取、 <code>run</code> 元信息与日志读取、 <code>pending</code> 且有 <code>job_id</code> 才回退、内存日志和状态返回。手动 <code>ir_011.commands</code> 也保留 <code>gh api > <path></code> 的文件写出文本，但 <code>run_id/job_id/owner/repo/path</code> 只有占位符，没有从失败检查提取并传递的结果，也没有日志 <code>pending</code> 回退路径，不能借脚本分支补全。 引用：ir_009, ir_011, block_006.constraints, diagnostics
2	部分保留	bundled ir_008 代码保留 <code>run/job</code> 标识、 <code>pending</code> 且有 <code>job id</code> 的回退以及内存日志/状态。手动 <code>ir_010</code> 的约束和 <code>manual_commands</code> 保留 <code>run</code> 查询与 <code>gh api</code> 重定向文本，但占位 <code>run_id/job_id/path</code> 没有对应生产/传递结果，逐检查及日志回退条件没有实际控制流。 引用：ir_008, ir_010, ir_010.constraints, diagnostics
3	部分保留	手动 <code>ir_022</code> 已从 <code>Actions</code> 检查提取 <code>run_ids</code> ， <code>ir_024/026</code> 消费该结果读取元信息/日志， <code>ir_028</code> 与 <code>edge_017</code> 保留 <code>pending</code> 回退，较前两轮更完整。但 <code>ir_030</code> 只消费 <code>pending_logs</code> ，无 <code>job_id</code> 提取/有 <code>job id</code> 条件，也没有手动 <code>gh api > <path></code> 的指定文件写出；正常日志路径仍消费仅回退产生的 <code>job_logs</code> ，另列 <code>finding</code> 。bundled <code>ir_014</code> 原代码保留内存日志与状态边界。 引用：ir_014, ir_022, ir_024, ir_026, ir_028, ir_030, ir_032, edge_017, edge_018

R03-F06 | 约束

非 GitHub Actions 检查标为 `external`，只报告 `URL`；禁止进入 `Buildkite` 等外部供应商的调查，不能把供应商能力说明转换成调用。

归属：block / external provider branch | 原文：SKILL.md:L47-L49；scripts/inspect_pr_checks.py:L244-L257

运行	初审	理由与实际输出引用
1	部分保留	ir_009 原始脚本在无 GitHub Actions <code>run id</code> 时返回 <code>status=external</code> ；Skill 及手动块约束禁止调查 <code>Buildkite</code> ，总结约束要求外部检查仅报告 <code>URL</code> 。手动批次 <code>ir_011</code> 没有产生 <code>external</code> 分类或限制后续 <code>run</code> 查询的分支，分类与仅 <code>URL</code> 范围在该路径只保留为约束文字。 引用：ir_009, ir_011, skill.constraints, block_006.constraints, block_009.constraints
2	部分保留	bundled ir_008 代码实际标 <code>external</code> 并返回 <code>URL</code> ；手动 <code>block_006</code> 也保留非 GitHub Actions 仅 <code>URL</code> 和禁止供应商调查的限制。但手动调用仍是单批次命令，无 <code>external</code> 分类结果或限制 <code>run/job</code> 查询的路径，只有文字范围未完整落实。 引用：ir_008, ir_010, block_005.constraints, block_006.constraints
3	保留	ir_020 将失败检查实际分成 <code>gha_checks result_011</code> 与 <code>external_checks result_012</code> ，只有前者进入 <code>run ID/日志</code> 查询；外部检查流入总结，局部约束明确标 <code>external</code> 、仅 <code>URL</code> 且不调查其他供应商。脚本 <code>ir_014</code> 也保留同样外部分类。 引用：ir_014, ir_020, ir_020.constraints, ir_022, ir_032

R03-F07 | 数据流

向用户汇报失败检查名、run URL 和简短日志片段，缺失日志必须显式标出；之后提出修复计划，获得明确批准后才应用计划。

归属: block / summary and approval gate | 原文: SKILL.md:L50-L58

运行	初审	理由与实际输出引用
1	部分保留	ir_015 消费实际分析结果并生成 summary，块约束保留检查名、run URL、片段及缺失日志说明；ir_017/ir_018 提计划请求批准，ir_020 与 approved/not approved 边确保仅批准后实施。但 failure_summary result_009 只流入计划生成，没有单独向用户交付失败摘要的操作/返回；块名保留面向用户的意图，实际交付表达仍不充分。 引用: ir_015, block_009.constraints, ir_017, ir_018, ir_020, ir_022, edge_012, edge_013
2	部分保留	ir_012 先选择检查结果供 ir_014 总结，ir_014.constraints 保留检查名、run URL、片段及缺失日志说明；ir_017 请求批准，ir_019 和 approved 边确保批准后才执行 ir_022。但 failure_summary result_009 只流向计划生成，未单独交付用户，面向用户意图仅在块名与约束中。 引用: ir_012, ir_014, ir_014.constraints, ir_017, ir_019, ir_022, edge_011, edge_012
3	部分保留	检查结果流经 ir_032/034 到计划，ir_036-042 按 plan skill 可用性选计划；ir_044 向用户请求批准并取得 decision，仅 approved 边进入 ir_047，审批门保留。但失败摘要只流向计划，没有向用户交付检查名、run URL 和片段的单独操作/返回；手动摘要仅明确缺失日志要求，输出给用户的内容仍表达不足。 引用: ir_032, ir_034, ir_038, ir_040, ir_044, ir_047, edge_027, edge_028

R03-F08 | 不得凭空增加

修复后原文只要求建议重跑相关测试和 gh pr checks；不能把该建议或询问开 PR 改写成已自动运行测试或已打开 PR。

归属: block / after approved changes | 原文: SKILL.md:L55-L58

运行	初审	理由与实际输出引用
1	保留	批准后 ir_024 仅 ask_about_opening_pr，ir_026 仅建议重新运行相关测试与 gh pr checks，返回同样是建议文字；没有自动开 PR 或运行测试的操作。 引用: ir_024, ir_026, block_013
2	保留	ir_023 仅询问是否开 PR，ir_025 生成重新运行测试/gh pr checks 的建议并由 ir_026 返回；没有自动开 PR 或执行测试。 引用: ir_023, ir_025, ir_026
3	保留	ir_049 仅询问是否打开 PR，ir_051 只建议重跑相关测试与 gh pr checks，并返回建议结果；没有实际开 PR 或执行测试。 引用: ir_049, ir_051, ir_052

R03-F09 | 条件

脚本检测不到失败检查时打印消息并以 0 退出；存在失败检查时输出结果后以 1 退出。非零返回码不能直接被理解为没有产生可用分析结果。

归属：operation / inspect_pr_checks.py result | 原文：SKILL.md:L64-L64; scripts/inspect_pr_checks.py:L110-L135

运行	初审	理由与实际输出引用
1	保留	ir_009 的完整代码在没有失败检查时打印并返回 0，有失败检查时先输出结果再返回 1；图同时保留结果 result_005 和退出码 result_006，后续 failure 分支仍用结果总结，没有把非零直接当成无分析输出。 引用：ir_009, block_005.constraints, ir_013, ir_015, edge_008, edge_009
2	保留	ir_008 保留的代码在无失败检查时打印消息并返回 0，存在失败时先打印结果再返回 1；报告与退出码分开产出，报告继续进入选择与总结，未把 nonzero 等同于没有结果。 引用：ir_008, ir_008.constraints, ir_012, ir_014
3	保留	ir_014 原始代码在无失败检查时输出消息并返回 0，有失败时先输出报告再返回 1；图独立保留报告 result_007 和退出码 result_008，报告继续被选择使用，没有以非零抹去可用分析结果。 引用：ir_014, ir_014.constraints, ir_034

R04 netlify-deploy

R04 | 真实 Skill | 原始快照 | 开发集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	18	23	18
边数	26	28	21
操作数	40	43	35
三层约束条数	29	14	24
生成尝试	2	1	1
结构修复	1	0	0
HTTP 尝试	2	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三轮保留 status→关联判断→Git URL→link/init→部署的主线。第 2 轮认证含 API token 与失败结束最完整；第 1 轮认证循环无失败结束。第 1 轮有网络故障重跑和一般故障检查，第 2、3 轮排除了故障恢复。三轮均未保留 bundled 文档的新站直接生产与 Always Preview First 冲突，且部署类型选择消费不是所有路径产生的建站/关联结果。

提取较稳定地保留认证/关联和部署类型主干，但明确的故障恢复、配置与秘密数据链被压成约束或误当指导，来源冲突也未保留。需区分真正非执行能力条目与被明确采用的参考流程；不能用习惯补全发布顺序。所有权限/部署文字仅作为静态语义审查，本轮未实际执行。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 R04 的三次明细。

事实表以外的发现

运行 1 | ir_025 同时消费 link_result result_007 和 init_result result_008：已关联路径均不产生，link 成功路径不产生 init_result，非 Git 直接 init 路径不产生 link_result。诊断承认有意在合流后引用分支结果，但没有当前路径选择或缺省值绑定；结构可接受不证明这些值同时可用。

引用: ir_013, ir_015, ir_025, edge_005, edge_010, edge_012, diagnostics

运行 1 | ir_035 及 ir_039 同时消费 preview、production、escalated 三组日志/URL。三种调用并不每次都执行：普通 preview 成功路径没有 production/escalated 结果，production 同理；报告/错误处理未见适用结果选择，不能靠合流之后位置认为全部可用。

引用: ir_027, ir_029, ir_031, ir_035, ir_039, edge_019, edge_022, edge_025

运行 1 | package.json 的框架识别原文限定 if possible，也支持 Static HTML；图却对所有关联成功路径无条件读取 package.json，没有文件不存在/不可检测时的来源边界。

引用: ir_017, ir_019, skill.constraints

运行 2 | ir_034 无条件消费 new_site_created result_014, 但该结果只有 init 的 ir_026 产生; already linked 和 link succeeded 路径绕过 init 后仍汇合进入部署类型判定, 缺少那些路径上的站点状态结果/选择绑定。结构接受不证明该值在每条路径可用, 不要求固定 phi opcode。

引用: ir_026, ir_034, edge_011, edge_013, edge_019, edge_021

运行 2 | 原文要求 “if possible” 从 package.json 判断框架, 并支持没有构建命令的 Static HTML; 图在所有成功关联路径无条件执行 cat package.json, 未保留文件不存在/不可检测时的路径边界。记录这一额外条件范围问题, 不能假定每个部署项目都有 package.json。

引用: ir_028, edge_011, edge_013, edge_019, edge_021

运行 3 | ir_026 消费 site_creation_result result_010, 但该结果只在 init 的 ir_018 产生; 原已 linked 和 link succeeds 两条路径绕过 init 后仍进入 mode 判定, 未见这些路径上的状态选择或缺省绑定。

引用: ir_018, ir_026, edge_008, edge_012, edge_014

运行 3 | ir_020 的约束写 if possible, 但实际对全部关联成功路径从 package.json 读取, 未表达文件不存在/不可检测的条件路径; 原文支持 Static HTML, 不能默认一定有 package.json。

引用: ir_020, ir_020.constraints

逐事实初审（参考事实已确认，结果判定待共同复核）

R04-F01 | 条件

检查 Netlify 登录状态; 未认证时指导 OAuth 登录并等待用户完成, 再查 status; 浏览器认证不可用可改用 NETLIFY_AUTH_TOKEN, 仍不能建立认证时结束。

归属: block / authentication | 原文: SKILL.md:L30-L36; SKILL.md:L52-L66

运行	初审	理由与实际输出引用
1	部分保留	ir_001/003 查认证, 未认证经 ir_005 login, block_003 明确等待用户, 再由回边重查 status。但认证仍失败只会再次进入 login, 没有结束路径; API token 仅全局约束, 没有浏览器不可用时的条件替代。诊断称源文未指定切换谓词, 与明确 browser authentication unavailable 不符。 引用: ir_001, ir_003, ir_005, block_003.constraints, edge_003, edge_004, skill.constraints, diagnostics
2	保留	ir_003/ir_005 从 status 判断认证; edge_003/004 按浏览器可用性选择 login 或 NETLIFY_AUTH_TOKEN 指导, block_004.constraints 明确等待用户完成。ir_011 再查 status, 仍未认证经 edge_010 到 ir_015 结束。 引用: ir_003, ir_005, ir_007, block_004.constraints, ir_009, ir_011, ir_015, edge_003, edge_004, edge_010
3	部分保留	ir_003/005 查 status, 未认证去 ir_007 login, 其约束保留等待用户后重查; edge_005 回 status, edge_006 无法认证时 ir_009 结束。浏览器不可用时 NETLIFY_AUTH_TOKEN 的替代指导/路径未出现在 CFG 或约束, 认证事实只保留 OAuth 部分。 引用: ir_003, ir_005, ir_007, ir_007.constraints, ir_009, edge_005, edge_006

R04-F02 | 数据流

从 status 输出取得站点关联状态；已关联则跳过关联步骤，未关联且为 Git 仓库则读取 origin remote URL 传给 netlify link --git-remote-url，关联失败时 init。

归属: block / site linking | 原文: SKILL.md:L70-L98

运行	初审	理由与实际输出引用
1	保留	ir_007 从 status 获取关联状态, already linked 跳过关联; 否则 ir_009 读取 origin, ir_011 提取 remote_url result_005, 仅 Git 路径 ir_013 用该 URL link, 失败经 edge_011 执行 init。 引用: ir_007, ir_009, ir_011, ir_013, ir_015, edge_005, edge_008, edge_011
2	保留	ir_005/ir_013 从 status 产出关联状态, block_009/010 已关联路径跳过关联; 未关联才 ir_018 读 origin, ir_020 提取 URL result_010, ir_022 消费 URL 进行 link, 失败经 edge_020 到 init。 引用: ir_005, ir_013, ir_018, ir_020, ir_022, ir_026, edge_011, edge_013, edge_016, edge_020
3	保留	ir_010 从 status 解析关联状态, linked 路径跳过关联; 未关联才 ir_012 读 origin, ir_014 提取 URL, Git 分支 ir_016 消费 remote_url result_007 进行 link, 失败 ir_018 init。 引用: ir_010, ir_012, ir_014, ir_016, ir_018, edge_008, edge_010, edge_013

R04-F03 | 条件

按需引用 deployment-patterns 的决策树后，未关联且非 Git 仓库时直接 init；新站或首次发布与已有站点分别选择生产发布和预览发布。

归属: block / deployment decision tree | 原文: SKILL.md:L243-L247; references/deployment-patterns.md:L5-L19

运行	初审	理由与实际输出引用
1	部分保留	edge_009 保留非 Git 未关联时直接 init, ir_025.constraints 保留新站生产/已有站点预览并分支发布。不过该操作同时消费并非每条路径都产生的 link_result/init_result, 且引用资料整体被排除, 未记录新站直接生产与 Always Preview First 的冲突。 引用: ir_015, ir_025, ir_025.constraints, edge_009, edge_017, edge_018, diagnostics
2	部分保留	edge_017 保留未关联且非 Git 直接 init, edge_025/026 也区分新站生产与已有站点预览。但 ir_034 消费的新_site_created result_014 只在 init 路径产生, 已关联/关联成功路径没有该值; 同时没有保留 bundled reference 中新站直接生产与 Always Preview First 的来源冲突, 决策并非无歧义完整。 引用: ir_026, ir_034, edge_017, edge_025, edge_026, diagnostics
3	部分保留	edge_011 保留未关联非 Git 直接 init, ir_026.constraints 和 mode 分支保留新站生产/已有站点预览。但 site_creation_result result_010 只由 init 产生, 已有站点/关联成功路径没有该值; references 被整体当指导排除, 也没有记录首次生产与 Always Preview First 的冲突。 引用: ir_018, ir_026, ir_026.constraints, edge_011, edge_018, edge_019, diagnostics

R04-F04 | 条件

已有站点默认 `preview`；新站或明确生产请求采用 `--prod`；预览测试场景的参考流程要求预览获批后才生产发布，不能把两个 `deploy` 命令无条件串联。

归属: block / preview versus production | 原文: SKILL.md:L118-L136; SKILL.md:L243-L247;
references/deployment-patterns.md:L70-L90

运行	初审	理由与实际输出引用
1	部分保留	ir_025 按请求/站点选择 <code>preview</code> 或 <code>production</code>，两个命令互斥，未无条件串联；原文的新站或明确生产请求说明也在局部约束。但 <code>preview</code> 后只报告结果，没有测试场景的预览获批再生产路径，且未保留先预览与首次直接生产的来源冲突。 引用: ir_025, ir_027, ir_029, edge_017, edge_018, edge_019, diagnostics
2	部分保留	edge_025/026 保留新站或明确生产请求使用 <code>--prod</code>、已有站点默认 <code>preview</code>，两个 <code>deploy</code> 没有无条件串联。可是 <code>preview</code> 后只返回报告，没有参考测试场景的批准后转生产路径；诊断把 <code>references</code> 当非执行内容，也没有记录新站生产与先预览要求的冲突。 引用: ir_036, ir_038, ir_041, edge_025, edge_026, diagnostics
3	部分保留	ir_026 按请求和站点选择 <code>mode</code>，ir_028/030 互斥预览/生产，未无条件串联；默认/例外文字有保留。<code>preview</code> 直接报告返回，无测试场景的批准后生产分支，来源中先预览和首次直接生产的冲突没有任何诊断。 引用: ir_026, ir_028, ir_030, ir_033, edge_018, edge_019, diagnostics

R04-F05 | 数据流

发布前确保项目依赖已安装并根据包管理器选命令；发布读取构建设置，在本地构建并上传构建产物至 `Netlify`，取得并向用户报告部署 `URL`、适用的生产 `URL` 和日志链接。

归属: block / build deploy and report | 原文: SKILL.md:L106-L116; SKILL.md:L138-L150

运行	初审	理由与实际输出引用
1	部分保留	ir_017/019 检测包管理器，ir_021 安装依赖，ir_027/029/031 返回部署 <code>URL</code>、日志；ir_039/040 尝试报告给用户。但部署输入只有 <code>CLI</code>，未保留明确的配置读取、本地构建和构建产物上传 <code>I/O</code>；报告还同时消费互斥三种部署的结果而没有选择绑定，见 <code>additional finding</code>。 引用: ir_017, ir_019, ir_021, ir_027, ir_029, ir_031, ir_039, ir_040
2	部分保留	ir_028/ir_030 检测包管理器，ir_032 安装依赖；ir_036/038 调用部署并取得 <code>URL</code>/日志，ir_040 - 043 按生产与预览分别向用户返回结果。但部署输入仅 <code>CLI</code> 资源，未保留实际构建设置读取、本地构建及产物上传的来源/副作用说明，诊断直接称内部步骤不展开。允许黑盒边界不等于可省去其明确 <code>I/O</code>。 引用: ir_028, ir_030, ir_032, ir_036, ir_038, ir_040, ir_041, ir_042, ir_043, diagnostics
3	部分保留	ir_020/022 选择包管理器并安装依赖，ir_028/030 部署产出结果，ir_032/034 与 <code>return</code> 报告 <code>URL</code>/适用生产 <code>URL</code>和日志的意图。可是部署输入仅 <code>CLI</code>，配置读取只在 <code>constraints</code>，本地构建和构建产物上传未保留为黑盒 <code>I/O</code>；允许单调用边界并不省略明确数据来源/副作用。 引用: ir_020, ir_022, ir_028, ir_030, ir_032, ir_033, ir_034, ir_035

R04-F06 | 约束

已有 `netlify.toml` 由 CLI 自动使用；缺失时询问 `build command` 和 `publish directory`，可读 `package.json` 判断框架并建议值。参考 TOML 的 `command/publish/base` 属构建配置，不能把所有示例配置同时执行。

归属：operation / deployment configuration | 原文：SKILL.md:L152-L163；SKILL.md:L243-L247；
references/netlify-toml.md:L13-L33

运行	初审	理由与实际输出引用
1	部分保留	<code>netlify.toml</code> 存在自动使用/缺失提示的文字和框架默认值被放在 <code>Skill constraints</code> ， <code>ir_017/019</code> 也有 <code>package.json</code> 检测，没有执行所有示例配置。但配置文件或提示回复没有实际来源， <code>suggested_build_settings</code> 只给类型判定/报错，未绑定到部署调用，操作范围与实际数据链不足。 引用： <code>skill.constraints</code> ， <code>ir_017</code> ， <code>ir_019</code> ， <code>ir_025</code> ， <code>ir_027</code> ， <code>ir_029</code>
2	部分保留	<code>ir_028/ir_030</code> 读取 <code>package.json</code> 并建议框架配置，没有把 TOML 示例全部执行。但未表达 <code>netlify.toml</code> 存在时自动使用、缺失时询问 <code>build command/publish directory</code> 的两种配置来源，建议结果只进入报告而未输入部署。 引用： <code>ir_028</code> ， <code>ir_030</code> ， <code>block_017.constraints</code> ， <code>ir_036</code> ， <code>ir_038</code> ， <code>ir_040</code> ， <code>ir_042</code>
3	部分保留	<code>ir_020/024</code> 保留 <code>package.json</code> 框架识别及建议， <code>ir_028/030.constraints</code> 提到 CLI 从 <code>netlify.toml</code> 或用户提示检测配置，没有把所有 TOML 示例执行。但缺少文件存在/缺失条件及实际 <code>build command/publish directory</code> 来源，建议结果只进报告而未供部署。 引用： <code>ir_020</code> ， <code>ir_024</code> ， <code>ir_028.constraints</code> ， <code>ir_030.constraints</code> ， <code>ir_032</code> ， <code>ir_034</code>

R04-F07 | 条件

构建失败时检查构建命令、发布目录、依赖和日志；`publish directory` 不存在时验证构建已成功以及目录路径。保留这些特定故障条件。

归属：block / deployment failure handling | 原文：SKILL.md:L192-L209

运行	初审	理由与实际输出引用
1	部分保留	<code>ir_033 - 036</code> 保留检查构建命令、发布目录、依赖与日志以及验证构建成功的动作。但入口是所有 <code>other failure/重跑失败</code> ，未区分 <code>build failed</code> 与 <code>publish directory not found</code> 的特定条件，四项恢复全部顺序执行，扩大了各自适用范围。 引用： <code>ir_033</code> ， <code>ir_034</code> ， <code>ir_035</code> ， <code>ir_036</code> ， <code>edge_021</code> ， <code>edge_024</code> ， <code>edge_026</code>
2	遗漏	整个 CFG 未包含 <code>build failed</code> 或 <code>publish directory not found</code> 的恢复块、检查动作或对应条件； <code>diagnostics</code> 明确以 <code>conditional advice</code> 为由排除了故障恢复步骤，冻结事实要求的特定故障条件被遗漏。 引用： <code>diagnostics</code>
3	遗漏	CFG 没有 <code>build failed</code> 或 <code>publish directory not found</code> 对应的检查/恢复操作或分支； <code>diagnostics</code> 明确把 <code>Error Handling/Troubleshooting</code> 当非执行指导，冻结事实要求的特定恢复行为缺失。 引用： <code>diagnostics</code>

R04-F08 | 约束

网络超时、DNS 错误、连接重置或沙箱阻断出站请求时，上游文档要求以 `escalated permissions` 重跑部署；这是样例实际提出的有条件要求，静态分析应保留其范围，不在语料复核中执行。

归属: block / Escalated Network Access | 原文: SKILL.md:L213-L220

运行	初审	理由与实际输出引用
1	保留	preview/prod 的 network failure 分支 edge_020/023 才进入 ir_031 重跑，block_016.constraints 明确超时、DNS、连接重置及 escalated permissions；全局文字同时保留沙箱出站阻断的条件，没有将其改为所有部署默认升级。此处只复核语义，没有实际执行。 引用: ir_031, block_016.constraints, skill.constraints, edge_020, edge_023
2	部分保留	Skill constraints 保留沙箱阻断时以 escalated permissions 重跑的条件文字，但没有网络超时/DNS/连接重置对应范围或恢复调用路径；diagnostics 又排除 troubleshooting recovery。不是完全遗漏禁区文字，但局部条件范围和重跑行为未完整保留。本轮只静态复核该要求，没有执行权限请求。 引用: skill.constraints, diagnostics
3	部分保留	Skill constraints 仅保留沙箱阻断时 escalated 重跑的文字，没有超时、DNS、连接重置的完整条件范围及实际重跑路径；diagnostics 明确排除 troubleshooting，因而不能判完整保留。该权限文字始终仅作静态数据审查。 引用: skill.constraints, diagnostics

R04-F09 | 约束

“Never commit secrets to Git”约束整份部署流程；需要环境秘密时写入 Netlify 配置并由构建访问。CLI reference 中 `env:import .env` 只是能力条目，不能凭此制造每次自动导入。

归属: skill / secret handling | 原文: SKILL.md:L223-L229; SKILL.md:L243-L247; references/cli-commands.md:L82-L96

运行	初审	理由与实际输出引用
1	部分保留	Never commit secrets to Git 在 Skill constraints 保留，且没有 env:import 自动调用。写 Netlify 环境配置与构建 process.env 访问也保留文字，但整图未获取/设置所需环境值，实际秘密来源与 Netlify 配置去向仅被约束文字代替。 引用: skill.constraints
2	部分保留	Skill constraints 原样保留 Never commit secrets to Git，且没有自动 env:import .env。但需要环境秘密时写入 Netlify 配置并由构建访问的来源/去向未表达，diagnostics 明确排除环境变量 setup guidance，因此仅保留禁止的一部分。 引用: skill.constraints, diagnostics
3	部分保留	Skill constraints 保留 Never commit secrets to Git，没有自动 env:import；但环境秘密需要写入 Netlify 配置并在构建访问的行为/数据去向均未表达，仅剩全局禁止的一部分。 引用: skill.constraints

R04-F10 | 不得凭空增加

References 中的 Netlify 官网文档仅提供 URL，包内 Bundled References 才作为已提供文本；不得把未下载外链文档的隐含步骤补入本轮输入。

归属: block / reference boundary | 原文: SKILL.md:L238-L247

运行	初审	理由与实际输出引用
1	保留	没有读取外部 Netlify 文档 URL 或添加其未提供内容的步骤；CLI/Git/项目文件是明确来源。diagnostics 明确资料边界，误把 bundled 内容整体排除的问题在 F03/F04/F06 记录，不等同于借入包外资料。 引用: diagnostics, block_001, block_005, block_009
2	保留	全图没有读取包外 Netlify 文档 URL 或凭其生成下载内容；仅 CLI、Git 和项目输入进入行为。diagnostics 将非执行资料排除，虽然误排 bundled references 的问题已在其他事实记录，未见外链隐含步骤进入输入的证据。 引用: diagnostics, block_002, block_011, block_016
3	保留	实际 CFG 没有读取包外 Netlify 文档 URL 或借用未下载资料的隐含步骤。references 整体被当指导排除的损失另计，不把该错误与输入混入外链内容混淆。 引用: diagnostics, block_002, block_007, block_011

R05 | linear

R05 | 真实 Skill | 原始快照 | 保留验证集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	23	10	37
边数	30	10	45
操作数	62	22	78
三层约束条数	15	13	19
生成尝试	3	1	1
结构修复	2	0	0
HTTP 尝试	3	1	1
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

独立阅读冻结 Linear Skill/元数据及 DeepSeek 三轮终态 CFG，未复用旧模型判定。r1/r3 展开九种工作流但分支汇合输入不完整；r2 通用调用链较清楚且连接恢复较完整，却丢失被引用工作流细节和生产力提示。三轮均遗漏按故障类型的 Troubleshooting 恢复。

三轮均保留主要前提、工具目录与条件工作流的大体区分，仍存在事实层面的条件、参数及被采纳工作流细节缺失。仅评价本基线输出，不执行 Linear 操作，不据保留集改动 Prompt 或生产管线。外置事实的条件约束允许用明确作用域文字保留；实际结果流与恢复路径则检查 CFG 操作和边。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 R05 的三次明细。

事实表以外的发现

运行 3 | 多处 e.g. 示例被固定成真实参数：Sprint 的 Q1 Performance Sprint、文档查询 API auth、Release 的 v2.0 Release。它们属于示例而非所有该类任务的强制取值；这是额外的参数范围扩大，未修改冻结事实。

引用：ir_023, ir_031, ir_043

运行 1 | 初始连接检查可视为前提检查的合理抽象，但不能据此认定工作流中的每次 MCP 断连均已恢复；本次未实际运行这些工具。

引用：ir_001, edge_001

运行 1, 3 | 冻结开放问题指出 create cycle 在叙述中被要求、工具目录却仅列 list_cycles。本审没有因采用语义创建操作而直接扣分，也没有据此扩充工具目录。

引用：diagnostics

逐事实初审（参考事实已确认，结果判定待共同复核）

R05-F01 | 约束

Linear MCP 必须已连接且经 OAuth 可访问，并确认相关 workspace/team/project 权限；主流程步骤按顺序执行，不能跳步。

归属：skill / workflow prerequisites and ordering | 原文：SKILL.md:L14-L20

运行	初审	理由与实际输出引用
1	保留	Skill 约束明确保留 OAuth 可达、相关 workspace/team/project 访问权限以及按序且不跳步骤；连接检查后才进入用户目标澄清。 引用：skill.constraints, ir_001, edge_001, edge_002
2	保留	Skill 级保留 OAuth、workspace/team/project 访问与按序要求；目标澄清、选择工具、读取、写入和交付依次相连。 引用：skill.constraints, block_002, block_003, edge_003, edge_006
3	保留	Skill 与连接检查块保留 OAuth、相关资源访问和不可跳步约束，连接可用分支才进入用户请求读取。 引用：skill.constraints, block_001.constraints, ir_001, edge_002

R05-F02 | 条件

任一 MCP 调用因未连接失败时暂停并配置连接：添加服务器、启用远程客户端、OAuth 登录；成功后告知重启 Codex并结束本次回答，后续从 Step 1 恢复。

归属：block / MCP connection recovery | 原文：SKILL.md:L22-L33

运行	初审	理由与实际输出引用
1	部分保留	初始未连接分支确有 add server、启用客户端、OAuth 和重启/结束提示，但实际工作流 MCP 调用失败没有恢复边；登录操作没有成功值或成功条件便直接返回已完成提示，不能完整表达任一调用未连接时暂停以及成功登录后的停止条件。 引用：ir_001, ir_003, ir_004, ir_005, ir_006, edge_001, block_002.constraints
2	保留	读取和写入调用均返回错误并以 not connected 条件进入设置；添加、启用、OAuth 后由成功标志分支抵达重启 Codex、结束本次并下次从 Step 1 继续的返回块。 引用：ir_009, ir_013, ir_017, ir_018, ir_020, ir_022, edge_005, edge_008, edge_010
3	部分保留	初始连接检查与设置三步、登录成功分支及重启提示均保留，但其他 MCP 调用没有未连接恢复边；Windows 配置分支又无条件进入成功登录后的提示，未重新确认 OAuth 成功。设置块文字中的 any MCP call 不能补齐这些路径。 引用：block_002.constraints, ir_003, ir_004, ir_005, ir_009, edge_001, edge_003, edge_004, edge_005

R05-F03 | 不得凭空增加

Windows/WSL JSON 片段仅在 Windows 出现连接错误时作为尝试的配置分支；不应据此无条件启动 wsl 或建立第二个 MCP 会话。

归属: block / Windows connection recovery | 原文: SKILL.md:L35-L38

运行	初审	理由与实际输出引用
1	保留	Windows 出现连接错误才尝试 WSL 的条件保留在设置块约束，未把配置说明扩成无条件 wsl 执行或第二个 MCP 会话；配置的具体 JSON 未展开，按本事实的禁止额外推导范围判为保留。 引用: block_002.constraints, ir_003, edge_001
2	保留	设置块原样保留 Windows 连接错误时尝试 WSL 及 JSON 配置，主 CFG 未额外添加无条件 wsl 或第二个会话调用。 引用: block_008.constraints, ir_017, diagnostics
3	保留	WSL 配置 JSON 是配置操作的字面量，唯一进入该块的边明确要求 Windows connection errors；没有把 WSL 配置变成正常路径必做的新会话。该分支后续登录成功判定问题另记 F02。 引用: ir_007, block_003.constraints, edge_004

R05-F04 | 数据流

先明确用户目标和范围以及所需 team/project 等字段，再选择适合 workflow；调用前确认所选工具所需的标识（如 issue ID、project ID、team key），之后先 list/get/search 读取上下文，再创建或更新对象。

归属: block / Step 1 to Step 3 | 原文: SKILL.md:L40-L50

运行	初审	理由与实际输出引用
1	部分保留	目标/字段澄清、标识符确认及 workflow 选择均有实际数据流，多数 workflow 也先读后写；Release Planning 却从选择直接进入创建项目，没有前置 list/get/search，因此 Skill 级 Read first 文本未落实到所有写入路径。 引用: ir_009, ir_011, ir_012, ir_036, edge_009, skill.constraints
2	保留	澄清结果进入 workflow/工具选择与标识符确认，所选工具和标识符进入 MCP 读取；读取结果进入变更计划再交写入，先读后写与必要标识符的关系清楚。 引用: ir_003, ir_005, ir_006, ir_007, ir_009, ir_011, ir_013, edge_004, edge_006
3	部分保留	澄清与选择/确认先于工具调用，但 Release Planning 从选择直达 create_project，没有 read-first 操作；另外多数调用不消费已确认的标识符值，确认结果主要仅进入 sprint list，实际参数联系不完整。 引用: ir_012, ir_013, ir_015, ir_017, ir_019, ir_043, edge_012, block_007.constraints

R05-F05 | 约束

批量变更执行前解释分组逻辑；create/update 调用携带全部必填字段。这些要求作用于相应批处理和调用，不能据此凭空增加外部参数校验服务。

归属：operation / batched create/update | 原文：SKILL.md:L46-L50

运行	初审	理由与实际输出引用
1	保留	约束显式限定 create/update 带齐必需字段，并限定 bulk operations 在应用变更前解释分组；保留条件文字足以表明适用范围，没有新增外部参数校验器。 引用：skill.constraints, ir_017, ir_028, ir_036
2	保留	写入块直接携带 all required fields 与 bulk 分组解释须在变更前的局部约束，计划值进入写入；未增加原文不存在的外部校验调用。 引用：block_006.constraints, ir_011, ir_013
3	保留	Step 3 相关条件要求在 Skill 与选择块约束中显式限定为 create/update 和 bulk operations；没有把字段要求新增为外部校验器。具体调用参数联系不足另在 F04 记录。 引用：skill.constraints, block_007.constraints, ir_035, ir_043

R05-F06 | 不得凭空增加

Available Tools 是能力清单，Step 2 才根据目标选择工具；不能把 list_issues、create_issue、update_issue 等全部生成无条件调用。

归属：block / Available Tools | 原文：SKILL.md:L43-L44；SKILL.md:L55-L61

运行	初审	理由与实际输出引用
1	保留	先选工作流与所需工具，九类实际工作流由互斥条件进入；没有将 Available Tools 全表转为无条件调用序列。 引用：ir_012, edge_005, edge_007, edge_009
2	保留	工具目录仅保存在 identify_required_tools 的元数据，所选工具作为读取调用输入；流程没有无条件逐个调用整个工具目录。 引用：ir_005, ir_006, ir_009
3	保留	工作流选择和工具识别发生于分支前，后续按选定工作流进入不同路径，并非无条件执行工具目录。 引用：ir_015, ir_016, edge_008, edge_010, edge_012

R05-F07 | 行为

Step 2 明确采用的 Practical Workflows 具有实际行为：例如文档审计先 search documentation，再为缺口或过期章节开带 documentation 标签及详细修复建议的 issue。仅被目标选中的流程进入执行路径。

归属: block / selected Documentation Audit workflow | 原文: SKILL.md:L43-L44; SKILL.md:L63-L73

运行	初审	理由与实际输出引用
1	部分保留	文档审计被采纳为条件执行路径，包含搜索、找缺口和建 issue；建 issue 的输入、局部约束及元数据均未保留 documentation 标签或 detailed fixes， workflow 细节不全。 引用: edge_007, ir_025, ir_027, ir_028
2	部分保留	九种 Practical Workflows 只以名称作为选择元数据，再进入通用 read/plan/write。选定 workflow 确实驱动调用，但文档审计的搜索、缺口/过时章节、documentation 标签和 detailed fixes 没有具体保留。 引用: ir_005, ir_009, ir_011, ir_013, diagnostics
3	部分保留	文档审计分支保留 search → 找缺口 → create_issue，且 documentation 标签为实际输入；详细修复意见仍未进入 issue 内容或约束，搜索示例 API auth 还被固定为必传字面量。 引用: edge_010, ir_031, ir_033, ir_035, edge_023, edge_024

R05-F08 | 数据流

操作后总结结果，列出未解决缺口或阻碍并提出后续动作；大规模更新拆成较小批次以避免限流，频繁列举时可以缓存或复用筛选器。

归属: block / results and batching | 原文: SKILL.md:L52-L53; SKILL.md:L75-L80

运行	初审	理由与实际输出引用
1	部分保留	总结、缺口/阻碍、后续行动与小批次/缓存筛选提示均保留；但总结操作同时要求来自九个互斥 workflow 的所有结果，没有选中分支结果的汇合值，当前路径无法产生其完整输入集。 引用: block_023.constraints, ir_061, ir_062, skill.constraints, edge_015, edge_019, edge_022
2	部分保留	读取与写入结果实际进入总结并返回，交付约束包含缺口/阻碍/后续行动；大更新拆小批次以免限流、频繁列举缓存或复用筛选的提示被诊断说明作为非执行建议排除，CFG 及其约束中未保留。 引用: ir_015, ir_016, block_007.constraints, skill.constraints, diagnostics
3	部分保留	总结交付要求及小批次/缓存筛选提示均保留，但总结输入仍同时消费九个互斥分支产物；没有选择当前 workflow 产物的汇合值，结果到交付的数据流不完整。 引用: ir_077, ir_078, block_037.constraints, skill.constraints, edge_019, edge_025, edge_030

R05-F09 | 不得凭空增加

agents/openai.yaml 声明 Linear MCP 依赖、transport 和 URL，并给出 default_prompt；这是能力与接口元数据，不能单独制造一次远程调用或把 default_prompt 当成新用户任务。

归属: block / agent metadata | 原文: SKILL.md:L14-L16; agents/openai.yaml:L1-L14

运行	初审	理由与实际输出引用
1	保留	接口元数据未生成额外用户请求或独立 MCP 会话；外部资源用于已有连接检查及所选工作流，实际任务从 user_request 读取。 引用: ir_001, ir_007, ir_012, skill.constraints
2	保留	Linear 依赖声明作为既有 MCP 调用的资源/元数据，任务来自 user_request；未以 default_prompt 产生新任务，也未为依赖目录新增独立调用。 引用: ir_001, ir_009, ir_013, diagnostics
3	部分保留	未新增独立 MCP 会话，实际用户请求也单独读取；但 agents/openai.yaml 的 default_prompt 被诊断明确提升为 Skill 全局行为约束，要求本次任务 triage/update，未保持它仅为可选界面提示的元数据地位。 引用: skill.constraints, ir_010, diagnostics

R05-F10 | 条件

故障恢复按类型分别适用：认证故障时清理浏览器 cookies、重新 OAuth、核查 workspace 权限和 API 访问是否启用；工具调用错误时确认模型支持多次工具调用、补齐必填字段并拆分复杂请求；数据缺失时刷新 token、核查 workspace 访问、归档项目及 team 选择；性能问题时考虑 API 限流，采用批处理、特定筛选器或缓存频繁查询。这些恢复操作仅属于相应故障分支，不能拼成所有任务都执行的步骤。

归属: block / Troubleshooting branches by failure type | 原文: SKILL.md:L82-L87

运行	初审	理由与实际输出引用
1	遗漏	除初始未连接设置外，没有按 Authentication、Tool Calling Errors、Missing Data、Performance 分类的恢复条件或对应恢复动作。一般生产力提示中的批次/缓存不能替代这些分类恢复分支。 引用: skill.constraints, block_001, block_002, diagnostics
2	遗漏	只保留未连接触发设置；四类 Troubleshooting 被诊断归为非执行指导，CFG 未保留清 cookies/重 OAuth、工具调用能力及复杂请求拆分、缺失数据核查、性能问题筛选缓存等按故障类型生效的恢复行为。 引用: block_008, edge_005, edge_008, diagnostics
3	遗漏	诊断明确将 Troubleshooting 排除为 advisory；CFG 仅有初始连接和 Windows 配置分支，缺少四类故障的条件及完整恢复行为。Skill 中通用批次/缓存文本没有相应故障类型作用域。 引用: block_001, block_002, block_003, skill.constraints, diagnostics

R06 transcribe

R06 | 真实 Skill | 原始快照 | 保留验证集 | en

指标	第 1 次	第 2 次	第 3 次
结构状态	complete	complete	complete
块数	12	8	9
边数	11	8	9
操作数	22	15	17
三层约束条数	27	26	28
生成尝试	1	1	2
结构修复	0	0	1
HTTP 尝试	1	1	2
HTTP 重试	0	0	0
远端结果未知 HTTP	0	0	0

三次运行差异与初审意见

三轮均保留真实音频/参考来源、完整转写脚本黑盒和验证/调整主线，script_content 均与冻结原字符串逐字匹配。第 1 轮将调整固定成一次重跑后返回；第 2、3 轮有回边，但首轮就消费尚未产生的 targeted_change。三轮外层都把缺失密钥无条件停止，压掉 dry-run 无 key 例外。第 1、2 轮固定 output/transcribe 基目录，第 3 轮输出约定主要在参数约束；第 3 轮缺少用户要求说话人标签的独立输入来源。

完整黑盒源码为 speaker 文件读取/data URL、payload、兼容性终止、输出规则、超限仅告警及非 diarize speaker 警告后继续提供有效证据，不要求展开代码内部步骤。但外围控制流与参数绑定仍可改变这些能力的可达性；密钥门、输出范围和首轮回边结果问题不能仅靠保留源码视为通过。所有判断为实际来源对照初审，待共同复核；未运行转写或请求包外 API 验证未知结果。

已接受 CFG 的精确 JSON 有 3 种。编号、顺序、措辞变化均会造成差异，此统计不判定语义等价。全部节点、边与数据引用见后半册 R06 的三次明细。

事实表以外的发现

运行 1 | 第 1 轮初次输出先验证，需调整后只执行一次新 CLI 就直接返回，未复查调整后的质量、标签或分段。原文说必要时以单一针对性改动迭代，没有规定最多重跑一次；图将迭代固定为一次且跳过调整后的验证，记录为待共同复核的控制流粒度问题。

引用: ir_015, ir_017, ir_019, ir_022, edge_008, edge_010, edge_011

运行 2 | ir_009 首次执行即消费 targeted_change result_013，但该值只会第一次转录完成并验证失败后由 ir_013 产生。入口到 CLI 的 edge_004 没有初始值，诊断虽承认回边来源，没有补当前路径选择/默认绑定；结构接受不代表第一轮调用时结果可用，不要求固定 phi opcode。

引用: ir_009, ir_013, edge_004, edge_006, edge_008, diagnostics

运行 3 | 首次参数准备 ir_009 消费 targeted_change result_012，而该值只由转录验证后的 ir_015 产生；初次进入准备参数的路径尚未经过该操作。缺少首轮默认值或当前路径绑定，结构成功不能说明前向引用可执行。

引用: ir_009, ir_015, edge_004, edge_008, edge_009

逐事实初审（参考事实已确认，结果判定待共同复核）

R06-F01 | 数据流

收集音频文件路径、输出格式、可选语言提示和已知说话人参考，调用 bundled CLI；随后检查转录质量、说话人标签和分段边界，必要时做单一针对性调整；在此仓库工作时，将输出保存到 output/transcribe/。

归属: block / transcription workflow | 原文: SKILL.md:L11-L16

运行	初审	理由与实际输出引用
1	部分保留	ir_001/003 从用户请求收集音频、格式、语言、说话人参考并传入 ir_013; ir_015 校验质量、标签和边界，需调整才 ir_017/019 单次针对性重跑。但两个 CLI 无条件绑定 output/transcribe/，没有“在此仓库工作时”的条件；第 2 次输出直接返回而未再次校验，迭代粒度另列 finding。 引用: ir_001, ir_003, ir_013, ir_015, ir_017, ir_019, ir_022, edge_008, edge_009
2	部分保留	ir_001 收集真实输入，ir_009 CLI 产出转录及写出路径，ir_011 验证，失败 ir_013 调整并回边重跑。输出目录实际无条件固定为 output/transcribe，虽全局文字保留仓库条件；而首次调用还消费仅失败回边产生的 targeted_change，数据可用性另列 finding。 引用: ir_001, ir_009, ir_011, ir_013, ir_015, skill.constraints, edge_006, edge_007, edge_008
3	部分保留	ir_001 读取实际音频、格式、语言和参考；ir_009 准备参数，ir_011 调 CLI 产出转录/路径，ir_013 验证且需调整才 ir_015 回边。没有普通仓库运行时保存 output/transcribe/ 的条件及目的地参数，只有评估路径的约束；首轮参数又消费回边才产生的 targeted_change，另列 finding。 引用: ir_001, ir_009, ir_011, ir_013, ir_015, ir_017, ir_009.constraints, edge_007, edge_008, edge_009

R06-F02 | 约束

实时 API 调用前要求本地 OPENAI_API_KEY；缺失则让用户在本地设置，绝不要求把完整密钥粘贴到聊天。

归属: skill / API key handling | 原文: SKILL.md:L13-L14; SKILL.md:L39-L42

运行	初审	理由与实际输出引用
1	保留	Skill constraints 明确 live API 需本地 OPENAI_API_KEY、缺失时本地设置且不得粘贴聊天；ir_007 实际读取环境，missing 分支 ir_009 指导本地设置并返回。对于 live 调用的密钥要求和禁止要求保留；对 dry-run 扩大门槛另按 F07 记录。 引用: skill.constraints, ir_007, ir_009, edge_004, edge_005
2	保留	全局及 block_002 约束保留本地设置密钥和不得在聊天粘贴；ir_003 查本地 OPENAI_API_KEY，缺失进入 ir_005 指导并停止，只有已设置才去 CLI。live 语义保留，dry-run 例外被挡的问题单列 F07。 引用: skill.constraints, block_002.constraints, ir_003, ir_005, edge_002, edge_003
3	保留	Skill constraints 保留 live API 密钥前置条件与绝不索取聊天密钥；ir_003 实际读取环境，缺失去 ir_005 请求本地设置并 return，未把密钥当转录内容或输出。F07 单独检查此门对 dry-run 的不当扩大。 引用: skill.constraints, ir_003, ir_005, ir_005.constraints, edge_002, edge_003

R06-F03 | 条件

默认 gpt-4o-mini-transcribe 与 text；用户要求说话人标签或分离时选择 gpt-4o-transcribe-diarize 与 diarized_json，超过约 30 秒保持 auto 分块。

归属: operation / transcription model and parameters | 原文: SKILL.md:L18-L22

运行	初审	理由与实际输出引用
1	保留	ir_005 消费 requested_response_format 与明确的 diarization_requested，局部约束保留 mini/text 默认和要求标签时 diarize/diarized_json；选出的 model/format 实际进入 ir_013，auto 作为参数且有 >约30秒条件说明。冻结 acceptable_variants 允许模型选择用调用约束表达，不要求固定分支 opcode。 引用: ir_005, ir_005.constraints, ir_013, ir_013.constraints
2	保留	ir_007 从 wants_diarization 和请求格式选择 model/format，约束保留默认、标签触发规则及 >约30秒 auto；ir_009 消费选出结果和 auto 常量。依据冻结允许变体，这些参数条件可由调用约束保留，无需强求固定分支形式。 引用: ir_007, ir_007.constraints, ir_009
3	部分保留	ir_007.constraints 保留正确默认和标签触发的模型/格式规则，ir_009 也保留长音频 auto 说明。但入口只读取文件、格式、语言和参考，未读取用户是否要求标签/diarization；选择操作实际只消费格式和参考，单纯要求标签且没有参考时的触发信息来源缺失。问题是所需输入，非缺少固定条件 opcode。 引用: ir_001, ir_007, ir_007.constraints, ir_009.constraints

R06-F04 | 约束

diarize 模型不支持 prompt；脚本对该组合直接终止，并对非 diarize 模型请求 diarized_json 直接终止，不能静默忽略或改变请求参数。

归属: operation / CLI argument compatibility | 原文: SKILL.md:L22-L22; SKILL.md:L14-L14;
scripts/transcribe_diarize.py:L241-L244

运行	初审	理由与实际输出引用
1	保留	ir_013 和 ir_019 均携带逐字匹配的 CLI 原脚本: prompt 与 transcribe-diarize 组合直接 _die，非 diarize 的 diarized_json 也 _die；调用限制另明确禁 prompt。没有静默删 prompt、改模型或改格式的替代步骤。 引用: ir_013, ir_019, ir_005.constraints
2	保留	ir_009 的原始 script_content 保留两条参数不兼容的 _die，ir_007.constraints 也注明 diarize 禁 prompt。代码作为完整黑盒条件证据，未被外层改成忽略参数或静默转换。 引用: ir_009, ir_007.constraints
3	保留	ir_011 原代码保留 prompt+diarize 与非 diarize+diarized_json 两类直接终止，ir_009.constraints 保留 prompt 禁令；不要求将内部验证展开，图也没有替换不兼容请求参数的操作。 引用: ir_011, ir_009.constraints

R06-F05 | 数据流

已知说话人按 NAME=PATH 解析、读取参考音频并编码 data URL；姓名与参考列表一起传入 extra_body。参考最多 4 个，缺文件、格式不符或超量均终止。

归属: operation / known speaker references | 原文: SKILL.md:L61-L70; SKILL.md:L80-L81; references/api.md:L7-L7; scripts/transcribe_diarize.py:L74-L98; scripts/transcribe_diarize.py:L169-L174

运行	初审	理由与实际输出引用
1	保留	实际已收集的 known_speaker_refs result_005 进入 ir_013，调整后的 result_017 进入 ir_019；两个黑盒都保留原代码按 NAME=PATH 解析、读参考音频、编码 data URL、名字/参考写入 extra_body，以及格式/缺文件/>4 时终止，未使用示例 Alice/Bob。 引用: ir_003, ir_013, ir_019
2	保留	ir_001 读取实际参考为 result_004，ir_009 消费该结果；其完整原脚本保留参考文件 read_bytes、data URL、姓名/参考成对进入 extra_body 及最多4个和错误终止。黑盒实际 I/O 有代码证据，不只存在约束条目。 引用: ir_001, ir_009
3	保留	known_speaker_references 经 ir_001 的 result_004 进入 ir_009 参数，再以 cli_arguments 进入 ir_011。原代码明确 NAME=PATH、文件读取/data URL 编码、名字与参考列表传 extra_body 和三类终止条件，黑盒边界内的来源与去向保留。 引用: ir_001, ir_009, ir_011

R06-F06 | 数据流

CLI payload 始终含 model、response_format、chunking_strategy；language/prompt 仅有值才加入，speaker names 非空才加入 extra_body。音频以二进制文件对象连同 payload 传给 audio.transcriptions.create。

归属: operation / API request payload | 原文: SKILL.md:L14-L14; scripts/transcribe_diarize.py:L155-L186

运行	初审	理由与实际输出引用
1	保留	ir_013/019 绑定真实音频、选出的 model/format、language、speaker references 与 auto；逐字保留的 _build_payload 始终写三项基础参数，仅有值才写 language/prompt/extra_body，_run_one 以 rb 打开实际音频并将文件对象和 payload 交 audio.transcriptions.create。 引用: ir_013, ir_019
2	保留	ir_009 输入真实 audio_paths、selected_model/format、language_hint、speaker refs 与 auto，并保留 _build_payload/_run_one 原代码；基础字段、可选字段条件和二进制音频随 payload 传给 API 的行为都在同一黑盒中，没有只用约束替代。 引用: ir_009
3	保留	ir_009 将已读取的音频及各项参数组织成 cli_arguments，ir_011 消费该结果且原代码完整保留基础 payload、可选 language/prompt/speaker 条件和 rb 文件对象传 API。没有新增无关输入或把资源名当音频内容。 引用: ir_009, ir_011

R06-F07 | 条件

dry-run 在验证与 payload 组装后只打印 JSON 并返回，允许没有 API key；正常分支才创建客户端和调用 API。不能因源码出现 API 调用而为 dry-run 增加网络行为。

归属: block / dry-run branch | 原文: SKILL.md:L14-L14; scripts/transcribe_diarize.py:L33-L40; scripts/transcribe_diarize.py:L246-L264

运行	初审	理由与实际输出引用
1	冲突	ir_013/019 原代码仍保留 dry-run 在验证和 payload 之后打印 JSON 返回、不创建客户端的逻辑。但外层 ir_007 的 missing-key 分支无条件进入 ir_009/010 结束，既未读取 dry_run 也未提供无 key 的旁路，阻止原代码明确允许的无密钥 dry-run。记录可用路径冲突，不因源码存在 API 调用而宣称 dry-run 一定联网。 引用: ir_007, ir_009, ir_010, ir_013, edge_004, edge_005
2	冲突	ir_009 原脚本保留 dry-run 的验证/JSON 打印/早退及无 key 例外，然而 CLI 前的 ir_003/004 只按 key 是否存在分支，missing 时 ir_005/006 直接结束；没有 dry_run 条件读取或例外路径。外层门实际消除了源码允许的无 key dry-run。 引用: ir_003, ir_005, ir_006, ir_009, edge_002, edge_003
3	冲突	ir_011 黑盒代码仍有 dry-run 不创建客户端、无 key 仅警告的路径；图却在 ir_003 查 key 后把所有 missing 情况送 ir_005/006 结束，未提供 dry-run 分支或参数来源。对无密钥 dry-run 的前置拦截与冻结事实冲突；不推断其真实联网。 引用: ir_003, ir_005, ir_006, ir_011, edge_002, edge_003

R06-F08 | 约束

文档要求评估运行使用 `output/transcribe/<job-id>/`，多文件使用 `--out-dir`；`--out` 和 `--stdout` 仅支持单音频，`--stdout` 与 `--out/--out-dir` 不能同时使用。正常写文件分支优先采用 `--out`：若指向现有目录，则在其下按音频 `stem` 生成文件名；无后缀则补格式后缀，否则保留显式文件路径。未提供 `--out` 时，使用 `--out-dir` 或当前目录，文件名为 `<stem>.transcript.txt`（text）或 `<stem>.transcript.json`（其他格式），随后写出 UTF-8。`--out-dir` 本身不保证避免同 `stem` 输入覆盖同名输出；脚本未检测这类冲突，也未自动重命名。

归属：operation / transcript output | 原文：SKILL.md:L24-L26; SKILL.md:L14-L14;
scripts/transcribe_diarize.py:L234-L239; scripts/transcribe_diarize.py:L101-L123;
scripts/transcribe_diarize.py:L263-L272

运行	初审	理由与实际输出引用
1	部分保留	两个 CLI 原脚本完整保留 <code>--out/--stdout</code> 单文件与互斥限制、 <code>--out</code> 优先级、目录/后缀命名、UTF-8 写出及同 <code>stem</code> 无防覆盖逻辑。全局文字也保留评估 <code>job-id</code> /多文件 <code>--out-dir</code> ，但实际两个调用均硬传 <code>output/transcribe/</code> ，没有评估上下文/ <code>job-id</code> 或用户 <code>--out/--stdout</code> 参数来源；诊断以未给 <code>job-id</code> 为由退成基目录，外层调用未完整保留输出约定。 引用：ir_013, ir_019, skill.constraints, diagnostics
2	部分保留	ir_009 保留脚本完整输出路径优先级、格式后缀、单文件/互斥检查、UTF-8 与同 <code>stem</code> 不自动防覆盖；也显式产出 <code>written_output_paths</code> 。可是调用无条件用 <code>output/transcribe, job-id/--out-dir</code> 仅块约束，实际未收集评估和显式输出选项，诊断主动把 <code>job-id</code> 目录退成基目录。 引用：ir_009, block_005.constraints, diagnostics
3	部分保留	ir_011 原代码完整保留 native CLI 输出选项、命名优先级和同 <code>stem</code> 覆盖边界，图产出 <code>saved_output_path</code> 并 <code>return</code> ；ir_009 约束也写评估 <code>job-id</code> 与多文件 <code>--out-dir</code> 。但入口没有评估上下文、 <code>job-id</code> 或 <code>--out/--out-dir/--stdout</code> 选项来源，参数准备中未形成相应输出位置的实际绑定，指定输出约定仍主要停留在约束。 引用：ir_001, ir_009, ir_009.constraints, ir_011, ir_017

R06-F09 | 不得凭空增加

参考文档把单次请求最大文件大小写为 25 MB，但脚本超过 25*1024*1024 bytes 仅警告，不停止、不裁切；不能将限制说明自动扩展为音频压缩、转码或客户端强制拒绝。

归属：unresolved / document size limit versus script behavior | 原文：SKILL.md:L80-L81; references/api.md:L3-L6;
scripts/transcribe_diarize.py:L18-L19; scripts/transcribe_diarize.py:L145-L152

运行	初审	理由与实际输出引用
1	保留	ir_013/019 的局部约束保留文档的 25 MB/request，原脚本 <code>_validate_audio</code> 对 >25*1024*1024 只 <code>_warn</code> ，不 <code>_die</code> 或裁切。没有新增压缩、转码、客户端硬拒绝；文档限制与已知客户端告警分别存在，不推断包外 API 会接受或拒绝。 引用：ir_013, ir_013.constraints, ir_019, ir_019.constraints
2	保留	ir_009.constraints 保留单次请求 25 MB 文字，原代码保留确切字节阈值与仅警告继续；全图无压缩、转码或超限客户端强制拒绝。以文档限制和客户端行为两层证据记录，不用外部 API 假设补结论。 引用：ir_009, ir_009.constraints
3	保留	全局/ir_011.constraints 有文档 25 MB 限制，ir_011 的原脚本超过确切阈值只警告并继续；没有被改成客户端停止、压缩或转码。未把包外服务处理结果写成已知，保留文档与源码的不同边界。 引用：skill.constraints, ir_011, ir_011.constraints

R06-F10 | 不得凭空增加

Quick start 的 Alice/Bob 参考路径和 meeting.m4a 是示例实参，不是每次用户任务的固定输入；流程必须消费实际收集到的文件和参考。

归属: block / CLI quick start examples | 原文: SKILL.md:L12-L12; SKILL.md:L53-L78

运行	初审	理由与实际输出引用
1	保留	ir_001/003 从当前用户请求取得音频与说话人参考，ir_013/019 消费对应结果；未把 meeting.m4a、Alice/Bob 或 refs 示例路径作为固定参数。 引用: ir_001, ir_003, ir_013, ir_019
2	保留	ir_001 从 user_request 产出实际音频和参考，ir_009 直接消费这些结果，没有采用 quick start 的固定音频/参考示例实参。 引用: ir_001, ir_009
3	保留	ir_001 读取实际 audio_file_paths 和 known_speaker_references，经参数准备进入 ir_011；没有把 Alice/Bob、meeting.m4a 或示例参考路径写成每次任务输入。 引用: ir_001, ir_009, ir_011

R06-F11 | 条件

在前置验证通过且已知说话人姓名非空时，若 model 不含 transcribe-diarize，脚本发出仅支持 diarize 的警告，但仍将姓名和参考 data URLs 放入 extra_body，不会因该警告停止或删除参数。dry-run 只打印 payload 并返回；正常分支继续向 audio.transcriptions.create 传入该 payload。SDK/API 是否接受、拒绝或如何处理该组合属于包外未知，不能据此补写成功结果、服务端错误或自动回退。

归属: block / known-speaker warning and external API boundary | 原文: SKILL.md:L14-L14; scripts/transcribe_diarize.py:L169-L186; scripts/transcribe_diarize.py:L252-L264

运行	初审	理由与实际输出引用
1	保留	ir_013/019 保留完整原代码：前置验证后有 speaker names 且非 diarize 只 _warn，随后仍 _build_payload 写 names/reference data URLs，dry-run 打印返回、正常调用传入 payload。没有因警告删除参数或自动回退，也没有补写 SDK/API 对该组合必成功/必失败的结论。 引用: ir_013, ir_019
2	保留	ir_009 的原代码在 known_names 非空且 model 不含 transcribe-diarize 时仅警告，仍构建带 extra_body 的 payload，正常交 API、dry-run 打印后返回。图没有新增因此终止、参数删除或包外接受/拒绝结果；F07 的外层密钥问题不改写这项警告后的客户端行为。 引用: ir_009
3	保留	ir_011 保留原脚本警告后继续组装 speaker extra_body 的分支，dry-run/正常分支对 payload 的去向也在代码中。没有把 warning 升为终止、删参数或自动换模型，未宣称包外服务对该组合必然接受或拒绝。 引用: ir_011

附录：90 次提取的实际 CFG 与全部操作

按下列页码或 PDF 书签定位本册已展示的明细。正式 CFG 缺失时明确显示失败边界；不自动修饰候选或补造图。每条边的编号仅是本册按原边数组生成的可追溯索引。

N01 / 第 1 次 实际输出明细	173
N01 / 第 2 次 实际输出明细	178
N01 / 第 3 次 实际输出明细	182
N02 / 第 1 次 实际输出明细	183
N02 / 第 2 次 实际输出明细	187
N02 / 第 3 次 实际输出明细	192
N03 / 第 1 次 实际输出明细	196
N03 / 第 2 次 实际输出明细	200
N03 / 第 3 次 实际输出明细	205
N04 / 第 1 次 实际输出明细	210
N04 / 第 2 次 实际输出明细	215
N04 / 第 3 次 实际输出明细	218
N05 / 第 1 次 实际输出明细	223
N05 / 第 2 次 实际输出明细	227
N05 / 第 3 次 实际输出明细	231
N06 / 第 1 次 实际输出明细	235
N06 / 第 2 次 实际输出明细	239
N06 / 第 3 次 实际输出明细	243
Q01 / 第 1 次 实际输出明细	246
Q01 / 第 2 次 实际输出明细	250
Q01 / 第 3 次 实际输出明细	255
Q02 / 第 1 次 实际输出明细	264
Q02 / 第 2 次 实际输出明细	269
Q02 / 第 3 次 实际输出明细	274
Q03 / 第 1 次 实际输出明细	279

Q03 / 第 2 次 实际输出明细	284
Q03 / 第 3 次 实际输出明细	285
Q04 / 第 1 次 实际输出明细	288
Q04 / 第 2 次 实际输出明细	297
Q04 / 第 3 次 实际输出明细	304
Q05 / 第 1 次 实际输出明细	309
Q05 / 第 2 次 实际输出明细	314
Q05 / 第 3 次 实际输出明细	320
Q06 / 第 1 次 实际输出明细	325
Q06 / 第 2 次 实际输出明细	331
Q06 / 第 3 次 实际输出明细	340
F01 / 第 1 次 实际输出明细	347
F01 / 第 2 次 实际输出明细	355
F01 / 第 3 次 实际输出明细	364
F02 / 第 1 次 实际输出明细	370
F02 / 第 2 次 实际输出明细	377
F02 / 第 3 次 实际输出明细	385
F03 / 第 1 次 实际输出明细	386
F03 / 第 2 次 实际输出明细	387
F03 / 第 3 次 实际输出明细	388
F04 / 第 1 次 实际输出明细	394
F04 / 第 2 次 实际输出明细	400
F04 / 第 3 次 实际输出明细	407
F05 / 第 1 次 实际输出明细	415
F05 / 第 2 次 实际输出明细	420
F05 / 第 3 次 实际输出明细	428
F06 / 第 1 次 实际输出明细	435
F06 / 第 2 次 实际输出明细	442

F06 / 第 3 次 实际输出明细	450
D01 / 第 1 次 实际输出明细	458
D01 / 第 2 次 实际输出明细	462
D01 / 第 3 次 实际输出明细	467
D02 / 第 1 次 实际输出明细	472
D02 / 第 2 次 实际输出明细	477
D02 / 第 3 次 实际输出明细	481
D03 / 第 1 次 实际输出明细	486
D03 / 第 2 次 实际输出明细	491
D03 / 第 3 次 实际输出明细	496
D04 / 第 1 次 实际输出明细	501
D04 / 第 2 次 实际输出明细	506
D04 / 第 3 次 实际输出明细	511
D05 / 第 1 次 实际输出明细	516
D05 / 第 2 次 实际输出明细	520
D05 / 第 3 次 实际输出明细	524
D06 / 第 1 次 实际输出明细	528
D06 / 第 2 次 实际输出明细	533
D06 / 第 3 次 实际输出明细	538
R01 / 第 1 次 实际输出明细	543
R01 / 第 2 次 实际输出明细	556
R01 / 第 3 次 实际输出明细	566
R02 / 第 1 次 实际输出明细	578
R02 / 第 2 次 实际输出明细	585
R02 / 第 3 次 实际输出明细	593
R03 / 第 1 次 实际输出明细	600
R03 / 第 2 次 实际输出明细	608
R03 / 第 3 次 实际输出明细	617

R04 / 第 1 次 实际输出明细	631
R04 / 第 2 次 实际输出明细	644
R04 / 第 3 次 实际输出明细	657
R05 / 第 1 次 实际输出明细	667
R05 / 第 2 次 实际输出明细	683
R05 / 第 3 次 实际输出明细	690
R06 / 第 1 次 实际输出明细	711
R06 / 第 2 次 实际输出明细	719
R06 / 第 3 次 实际输出明细	725

N01 / 第 1 次 实际输出明细

条件通知与数据发送 · 编号步骤

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N01/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: 源文中的“处理条数”未明确是选中记录数还是读取记录总数；本候选按选中并调用 notify.send 的记录数解释。

candidate warning: notify.send 未说明返回值，因此按纯副作用处理。

candidate warning: events.json 按外部文件读取，未声明运行时 context key。

candidate warning: 包内没有代码块，因此未提供 metadata.script_content。

Skill 层约束与上下文声明

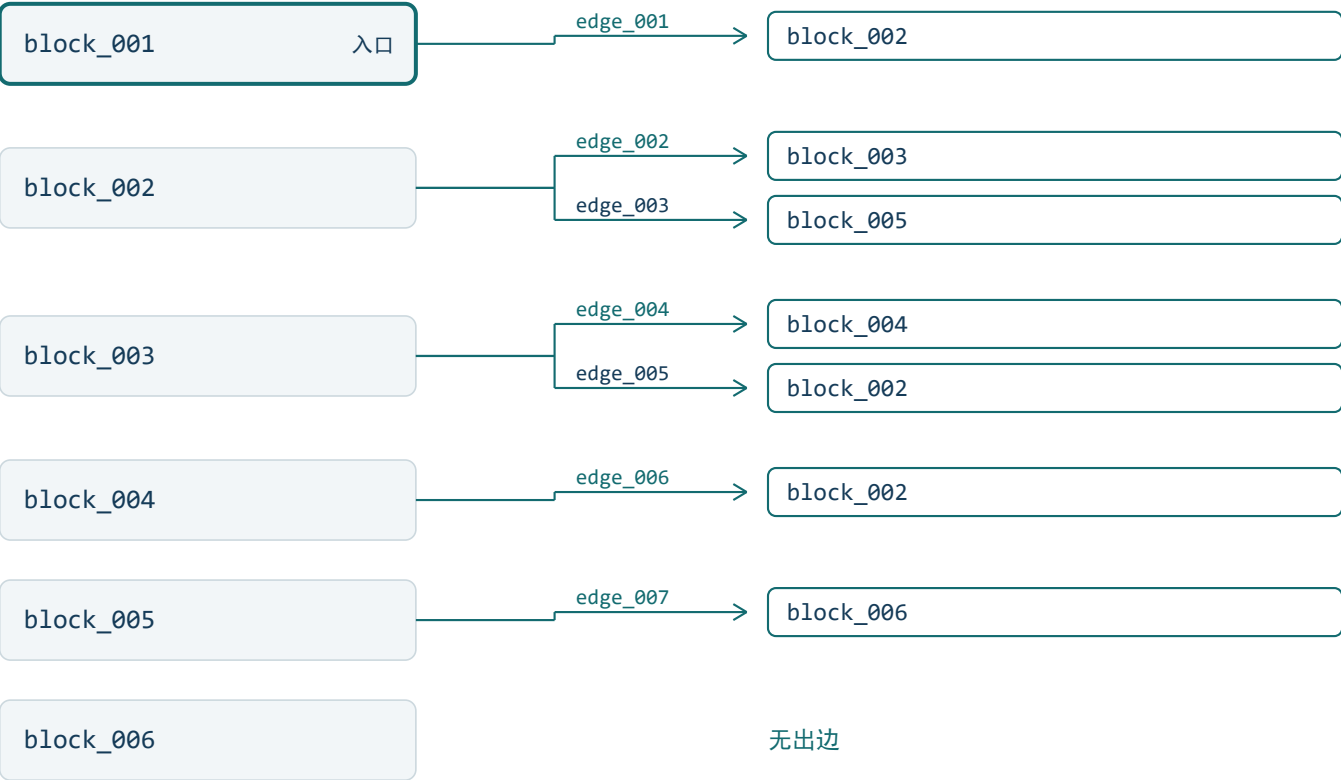
declared_context_keys = []

skill.constraints[0]: 整份流程禁止将 access_token 字段发送给任何接收对象。

skill.constraints[1]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"has more event records"
edge_003	block_002 -> block_005	"no more event records"
edge_004	block_003 -> block_004	"selected: opted_out is not true and (urgent is true or value >= 100)"
edge_005	block_003 -> block_002	"not selected: do not call notify.send"
edge_006	block_004 -> block_002	null
edge_007	block_005 -> block_006	null

全部操作、数据引用与约束落点

block_001 Read user-provided events.json

data_source_kind = "external"

block_001.constraints[0]: 读取用户提供的 events.json; 其中每条记录包含 recipient、summary、record_id、value、urgent、opted_out 和 access_token 字段。

ir_001 read_events_file

inputs[0] external_resource: events.json | events.json

outputs[0] result: result_001 | event_records

ir_001.constraints: []

draft_instruction_id = read_events

ir_002 dispatch

inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = go_iterate_events

block_002 Iterate over event records

data_source_kind = null

block_002.constraints: []

ir_003 get_next_event_record

inputs[0] result: result_001 | event_records

outputs[0] result: result_002 | current_event_record

outputs[1] result: result_003 | has_more_event_records

ir_003.constraints: []

draft_instruction_id = get_next_event

ir_004 dispatch

```
inputs[0] result: result_003 | has_more_event_records
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = dispatch_on_more
```

block_003 Extract fields and evaluate notification selection

```
data_source_kind = null
```

block_003.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

block_003.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

ir_005 extract_event_notification_fields

```
inputs[0] result: result_002 | current_event_record
```

```
outputs[0] result: result_004 | opted_out_flag
```

```
outputs[1] result: result_005 | urgent_flag
```

```
outputs[2] result: result_006 | record_value
```

```
outputs[3] result: result_007 | notification_recipient
```

```
outputs[4] result: result_008 | notification_body
```

ir_005.constraints[0]: 接收对象取该记录的 recipient 字段。

ir_005.constraints[1]: notify.send 的 body 参数直接取该记录的 summary 字段原值。

ir_005.constraints[2]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

```
draft_instruction_id = extract_event_fields
```

ir_006 check_notification_condition

```
inputs[0] result: result_004 | opted_out_flag
```

```
inputs[1] result: result_005 | urgent_flag
```

```
inputs[2] result: result_006 | record_value
```

```
inputs[3] literal = 100
```

```
outputs[0] result: result_009 | should_notify
```

ir_006.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

ir_006.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

```
draft_instruction_id = check_notification_condition
```

ir_007 dispatch

```
inputs[0] result: result_009 | should_notify
```

```
outputs = []
```

```
ir_007.constraints: []
```

```
draft_instruction_id = dispatch_on_selection
```

block_004 Send notify.send for selected record

data_source_kind = null

block_004.constraints[0]: 对每条选中记录调用一次 notify.send, 接收对象取该记录的 recipient 字段。

block_004.constraints[1]: notify.send 的 body 参数直接取该记录的 summary 字段原值。

block_004.constraints[2]: 未选中的记录不调用 notify.send。

block_004.constraints[3]: 整份流程禁止将 access_token 字段发送给任何接收对象。

ir_008 send_notification

inputs[0] external_resource: notify.send | notify.send

inputs[1] result: result_007 | notification_recipient

inputs[2] result: result_008 | notification_body

outputs = []

ir_008.constraints[0]: 对每条选中记录调用一次 notify.send。

ir_008.constraints[1]: 整份流程禁止将 access_token 字段发送给任何接收对象。

draft_instruction_id = send_notify

ir_009 dispatch

inputs = []

outputs = []

ir_009.constraints: []

draft_instruction_id = go_next_after_send

block_005 Count selected event records for processed total

data_source_kind = null

block_005.constraints: []

ir_010 count_selected_event_records

inputs[0] result: result_001 | event_records

inputs[1] literal = 100

outputs[0] result: result_010 | processed_count

ir_010.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

draft_instruction_id = count_selected

ir_011 dispatch

inputs = []

outputs = []

ir_011.constraints: []

draft_instruction_id = go_write_count

block_006 Write processed count to count.txt

data_source_kind = null

block_006.constraints[0]: 处理完成后, 将处理条数写入本地 count.txt。

ir_012 write_processed_count_to_file

inputs[0] external_resource: count.txt | count.txt

inputs[1] result: result_010 | processed_count


```
outputs = []

ir_012.constraints[0]: 处理完成后，将处理条数写入本地 count.txt。

draft_instruction_id = write_count_file

ir_013 return

inputs = []

outputs = []

ir_013.constraints: []

draft_instruction_id = return_after_write
```

N01 / 第 2 次 实际输出明细

条件通知与数据发送 · 编号步骤

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N01/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: “处理条数”未明确是全部读取记录还是选中记录；此处按选中记录数（即实际进入通知处理的记录数）建模。

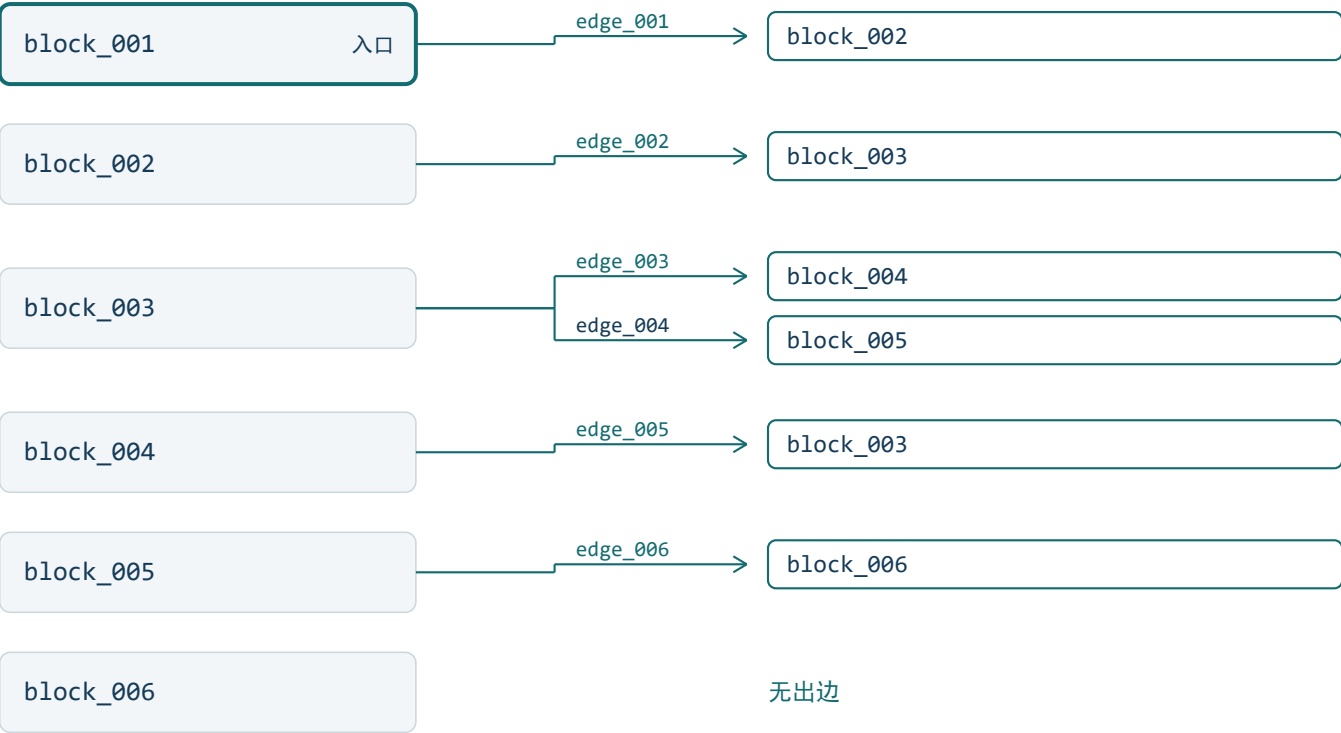
Skill 层约束与上下文声明

declared_context_keys = []

skill.constraints[0]: 整份流程禁止将 access_token 字段发送给任何接收对象。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"has_more_selected_records is true"
edge_004	block_003 -> block_005	"has_more_selected_records is false"
edge_005	block_004 -> block_003	null
edge_006	block_005 -> block_006	null

全部操作、数据引用与约束落点

block_001 读取用户提供的 events.json

```
data_source_kind = "external"

block_001.constraints: []

ir_001 read_events_json

inputs[0] external_resource: events.json | events.json
outputs[0] result: result_001 | event_records

ir_001.constraints: []

draft_instruction_id = read_events_json
```

ir_002 dispatch

```
inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = go_select
```

block_002 按订阅状态、紧急标记和数值筛选事件记录

```
data_source_kind = null

block_002.constraints: []

ir_003 select_event_records

inputs[0] result: result_001 | event_records
outputs[0] result: result_002 | selected_event_records

ir_003.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。
ir_003.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

draft_instruction_id = select_event_records
```

ir_004 dispatch

```
inputs = []

outputs = []

ir_004.constraints: []

draft_instruction_id = go_next_selected
```

block_003 获取下一条选中记录

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 get_next_selected_event_record

```
inputs[0] result: result_002 | selected_event_records
```

```
outputs[0] result: result_003 | current_event_record
```

```
outputs[1] result: result_004 | has_more_selected_records
```

```
ir_005.constraints: []
```

```
draft_instruction_id = next_selected_record
```

ir_006 dispatch

```
inputs[0] result: result_004 | has_more_selected_records
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = choose_has_more
```

block_004 提取收件人与摘要并发送通知

```
data_source_kind = null
```

```
block_004.constraints: []
```

ir_007 extract_recipient_and_summary

```
inputs[0] result: result_003 | current_event_record
```

```
outputs[0] result: result_005 | notification_recipient
```

```
outputs[1] result: result_006 | notification_summary
```

```
ir_007.constraints: []
```

```
draft_instruction_id = extract_notification_fields
```

ir_008 notify.send

```
inputs[0] external_resource: notify.send | notify.send
```

```
inputs[1] result: result_005 | notification_recipient
```

```
inputs[2] result: result_006 | notification_summary
```

```
outputs = []
```

ir_008.constraints[0]: 对每条选中记录调用一次 notify.send, 接收对象取该记录的 recipient 字段。

ir_008.constraints[1]: notify.send 的 body 参数直接取该记录的 summary 字段原值。

ir_008.constraints[2]: 未选中的记录不调用 notify.send。

ir_008.constraints[3]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

```
draft_instruction_id = send_notification
```

ir_009 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_009.constraints: []
```

```
draft_instruction_id = go_next_after_send
```

block_005 统计已处理条数

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_010 count_processed_event_records

```
inputs[0] result: result_002 | selected_event_records
```

```
outputs[0] result: result_007 | processed_count
```

```
ir_010.constraints: []
```

```
draft_instruction_id = count_processed_records
```

ir_011 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = go_write_count
```

block_006 将处理条数写入本地 count.txt

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_012 write_count_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_007 | processed_count
```

```
outputs = []
```

```
ir_012.constraints[0]: 处理完成后，将处理条数写入本地 count.txt。
```

```
draft_instruction_id = write_count_file
```

ir_013 return

```
inputs = []
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = return_done
```

N01 / 第 3 次 实际输出明细

条件通知与数据发送 · 编号步骤

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N01/deepseek-v4-flash-max/3

结构状态: error | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: 未记录 | HTTP 1 (200): deepseek-flash

诊断与失败边界

LLM SSE network failure: [WinError 5] 拒绝访问。: 'D:\\projects\\SkillFlow\\packages\\skill-ir\\experiments\\semantics_baseline\\runs\\baseline-deepseek-v4-flash-max-20260910\\trials\\N01\\deepseek-v4-flash-max\\3\\trace.json.2666c79218ae4348beb137508ec2adf0.tmp' -> 'D:\\projects\\SkillFlow\\packages\\skill-ir\\experiments\\semantics_baseline\\runs\\baseline-deepseek-v4-flash-max-20260910\\trials\\N01\\deepseek-v4-flash-max\\3\\trace.json'

流式请求的远端最终结果未知: 生成 1 / HTTP 尝试 1 / 已接收 HTTP 200

已收到成功响应头后，SSE 未达到完整结束条件；本地 error 不代表远端未生成结果，也不代表模型语义失败。未完成的流不能作为有效响应或有效 CFG，远端用量和最终处理结果以原始记录为准。

本次没有已接受的实际 CFG；不能补画或将候选当作有效图。原始响应 / 候选保存在同目录 trace.json、candidate.json 和 analysis.json（如生成）。

N02 / 第 1 次 实际输出明细

条件通知与数据发送 · 连贯段落

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N02/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: 处理条数 (processed count) is interpreted as the number of selected event records that triggered notify.send, because unselected records are skipped.

Skill 层约束与上下文声明

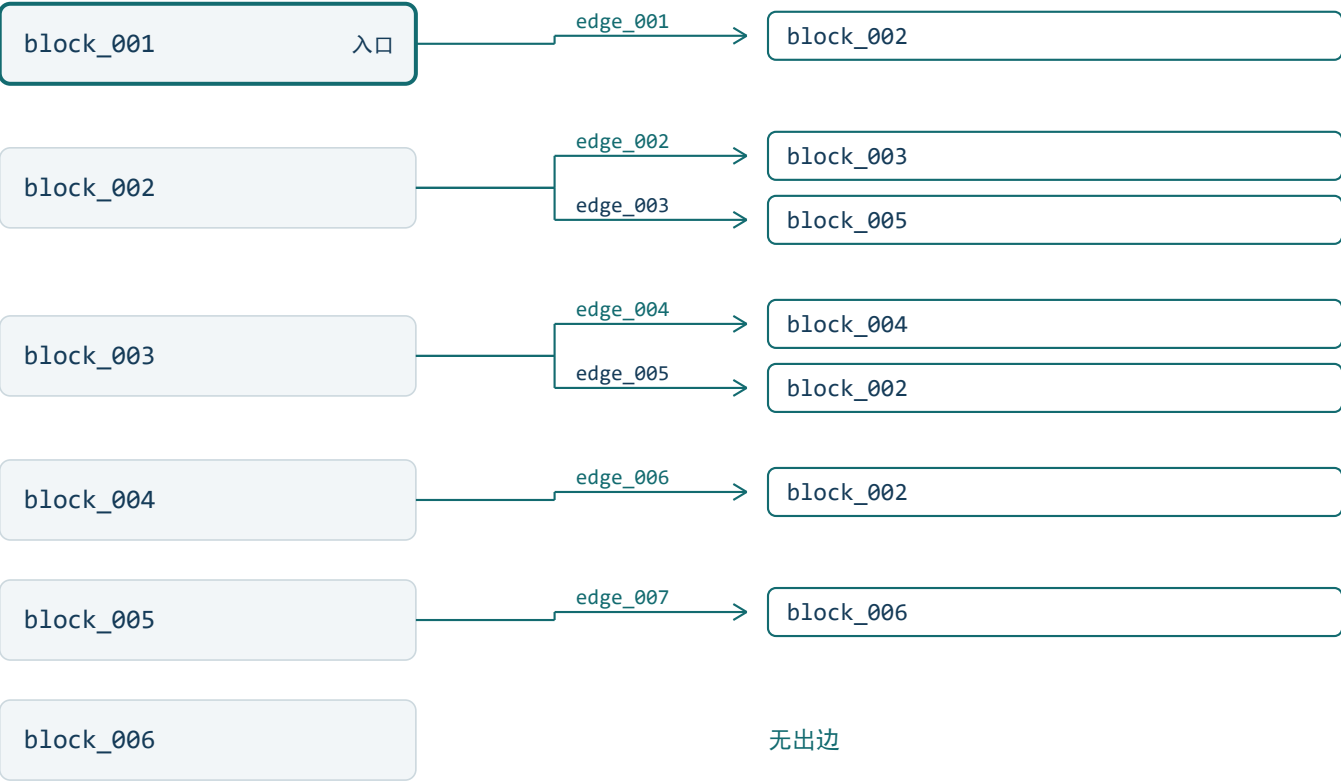
declared_context_keys = []

skill.constraints[0]: 整份流程禁止将 access_token 字段发送给任何接收对象。

skill.constraints[1]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"has more records"
edge_003	block_002 -> block_005	"no more records"
edge_004	block_003 -> block_004	"selected"
edge_005	block_003 -> block_002	"not selected"
edge_006	block_004 -> block_002	null
edge_007	block_005 -> block_006	null

全部操作、数据引用与约束落点

block_001 Read user-provided events.json

```
data_source_kind = "external"
block_001.constraints: []

ir_001 read_events_json_file

inputs[0] external_resource: events.json | events.json
outputs[0] result: result_001 | events_records
ir_001.constraints: []
draft_instruction_id = read_events
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_process_records
```

block_002 Get the next event record

```
data_source_kind = null
block_002.constraints: []

ir_003 get_next_event_record

inputs[0] result: result_001 | events_records
outputs[0] result: result_002 | current_record
outputs[1] result: result_003 | has_more_records
ir_003.constraints: []
draft_instruction_id = get_next_record
```


ir_004 dispatch

inputs[0] result: result_003 | has_more_records

outputs = []

ir_004.constraints: []

draft_instruction_id = check_more_records

block_003 Evaluate whether the record is selected

data_source_kind = null

block_003.constraints[0]: 未选中的记录不调用 notify.send。

ir_005 evaluate_event_record_selection

inputs[0] result: result_002 | current_record

outputs[0] result: result_004 | selected

ir_005.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

ir_005.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

draft_instruction_id = evaluate_selection

ir_006 dispatch

inputs[0] result: result_004 | selected

outputs = []

ir_006.constraints: []

draft_instruction_id = branch_on_selection

block_004 Send notification for the selected record

data_source_kind = null

block_004.constraints: []

ir_007 extract_notification_fields

inputs[0] result: result_002 | current_record

outputs[0] result: result_005 | recipient

outputs[1] result: result_006 | summary

ir_007.constraints: []

draft_instruction_id = extract_fields

ir_008 notify_send

inputs[0] external_resource: notify.send | notify.send

inputs[1] result: result_005 | recipient

inputs[2] result: result_006 | summary

outputs = []

ir_008.constraints[0]: 对每条选中记录调用一次 notify.send, 接收对象取该记录的 recipient 字段。

ir_008.constraints[1]: notify.send 的 body 参数直接取该记录的 summary 字段原值。

draft_instruction_id = send_notify

ir_009 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_009.constraints: []
```

```
draft_instruction_id = go_next_record
```

block_005 Count selected event records

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_010 count_selected_event_records

```
inputs[0] result: result_001 | events_records
```

```
outputs[0] result: result_007 | processed_count
```

ir_010.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

ir_010.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

```
draft_instruction_id = count_records
```

ir_011 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = go_write_count
```

block_006 Write processed count to local count.txt

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_012 write_count_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_007 | processed_count
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = write_count_file
```

ir_013 return

```
inputs = []
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = return
```

N02 / 第 2 次 实际输出明细

条件通知与数据发送 · 连贯段落

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N02/deepseek-v4-flash-max/2

结构状态: complete | 生成 2 | 修复 1 | HTTP 尝试 2 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

生成 2 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

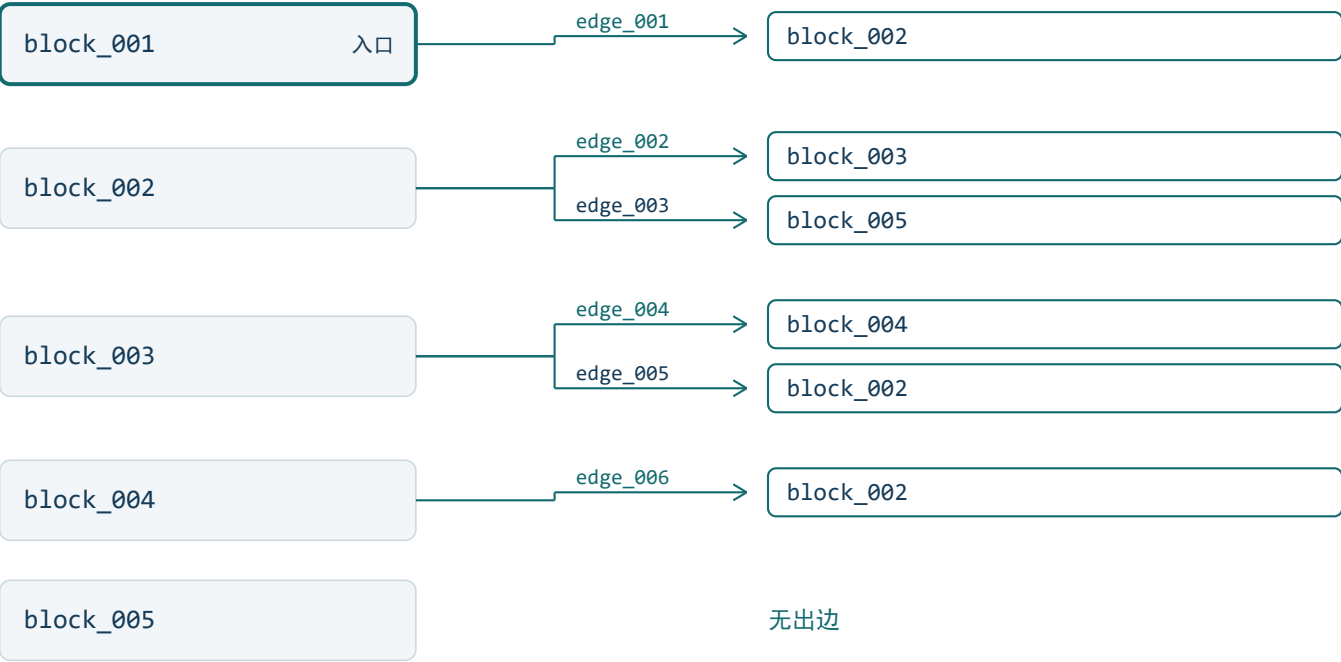
declared_context_keys = []

skill.constraints[0]: 整份流程禁止将 access_token 字段发送给任何接收对象。

skill.constraints[1]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"has_more_event_records"
edge_003	block_002 -> block_005	"no more event records"
edge_004	block_003 -> block_004	"record_selected"
edge_005	block_003 -> block_002	"not record_selected"
edge_006	block_004 -> block_002	null

全部操作、数据引用与约束落点

block_001 Read user-provided events.json

```
data_source_kind = "external"

block_001.constraints: []

ir_001 read_events_file

inputs[0] external_resource: events.json | events.json
outputs[0] result: result_001 | event_records

ir_001.constraints: []

draft_instruction_id = read_events_file

metadata = {"source_text": "读取用户提供的 events.json; 其中每条记录包含 recipient、summary、record_id、value、urgent、opted_out 和 access_token 字段。"}
```

ir_002 dispatch

```
inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = go_to_loop
```

block_002 Get the next event record

```
data_source_kind = null

block_002.constraints: []
```

ir_003 get_next_event_record

```
inputs[0] result: result_001 | event_records
outputs[0] result: result_002 | current_event_record
outputs[1] result: result_003 | has_more_event_records

ir_003.constraints: []

draft_instruction_id = get_next_event_record
```

ir_004 dispatch

inputs[0] result: result_003 | has_more_event_records

outputs = []

ir_004.constraints: []

draft_instruction_id = dispatch_on_more_records

block_003 Check if the event record is selected

data_source_kind = null

block_003.constraints: []

ir_005 extract_opted_out_from_record

inputs[0] result: result_002 | current_event_record

outputs[0] result: result_004 | record_opted_out

ir_005.constraints: []

draft_instruction_id = extract_opted_out

ir_006 extract_urgent_from_record

inputs[0] result: result_002 | current_event_record

outputs[0] result: result_005 | record_urgent

ir_006.constraints: []

draft_instruction_id = extract_urgent

ir_007 extract_value_from_record

inputs[0] result: result_002 | current_event_record

outputs[0] result: result_006 | record_value

ir_007.constraints: []

draft_instruction_id = extract_value

ir_008 check_record_eligibility

inputs[0] result: result_004 | record_opted_out

inputs[1] result: result_005 | record_urgent

inputs[2] result: result_006 | record_value

outputs[0] result: result_007 | record_selected

ir_008.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

ir_008.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

draft_instruction_id = check_record_eligibility

metadata = {"source_text": "仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。
urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。"}

ir_009 dispatch

inputs[0] result: result_007 | record_selected

outputs = []

ir_009.constraints: []

draft_instruction_id = dispatch_on_selected

block_004 Send notification for a selected record

data_source_kind = null

block_004.constraints: []

ir_010 extract_recipient_from_record

inputs[0] result: result_002 | current_event_record

outputs[0] result: result_008 | notification_recipient

ir_010.constraints: []

draft_instruction_id = extract_recipient

metadata = {"source_text": "接收对象取该记录的 recipient 字段。"}

ir_011 extract_summary_from_record

inputs[0] result: result_002 | current_event_record

outputs[0] result: result_009 | notification_body

ir_011.constraints: []

draft_instruction_id = extract_summary

metadata = {"source_text": "notify.send 的 body 参数直接取该记录的 summary 字段原值。"}

ir_012 notify.send

inputs[0] external_resource: notify.send | notify.send

inputs[1] result: result_008 | notification_recipient

inputs[2] result: result_009 | notification_body

outputs = []

ir_012.constraints[0]: 对每条选中记录调用一次 notify.send, 接收对象取该记录的 recipient 字段。

ir_012.constraints[1]: notify.send 的 body 参数直接取该记录的 summary 字段原值。

ir_012.constraints[2]: 未选中的记录不调用 notify.send。

draft_instruction_id = send_notify

metadata = {"source_text": "对每条选中记录调用一次 notify.send, 接收对象取该记录的 recipient 字段。 notify.send 的 body 参数直接取该记录的 summary 字段原值。 未选中的记录不调用 notify.send。"}

ir_013 dispatch

inputs = []

outputs = []

ir_013.constraints: []

draft_instruction_id = go_to_next_record

block_005 Count processed records and write count.txt

data_source_kind = null

block_005.constraints: []

ir_014 count_selected_records

inputs[0] result: result_001 | event_records

outputs[0] result: result_010 | processed_count

ir_014.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

ir_014.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

draft_instruction_id = count_processed_records

```
metadata = {"source_text": "处理完成后，将处理条数写入本地 count.txt。"}

---


```

ir_015 write_processed_count_to_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_010 | processed_count
```

```
outputs = []
```

```
ir_015.constraints[0]: 处理完成后，将处理条数写入本地 count.txt。
```

```
draft_instruction_id = write_count
```

```
metadata = {"source_text": "处理完成后，将处理条数写入本地 count.txt。"}

---


```

ir_016 return

```
inputs = []
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = return_done
```

N02 / 第 3 次 实际输出明细

条件通知与数据发送 · 连贯段落

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N02/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: The source lists record_id but does not use it in any behavior; it is omitted from operands.

candidate warning: The phrase '处理条数' is not explicitly defined; modeled as the number of selected event records that satisfy the notification criteria and are processed.

candidate warning: notify.send return value is not specified, so no result is produced for it.

candidate warning: events.json is treated as an external resource provided by the user.

Skill 层约束与上下文声明

declared_context_keys = []

skill.constraints[0]: 整份流程禁止将 access_token 字段发送给任何接收对象。

skill.constraints[1]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

skill.constraints[2]: notify.send 的 body 参数直接取该记录的 summary 字段原值。

skill.constraints[3]: 仅当 opted_out 不为 true，且 urgent 为 true 或 value 大于等于 100 时，选中该记录。

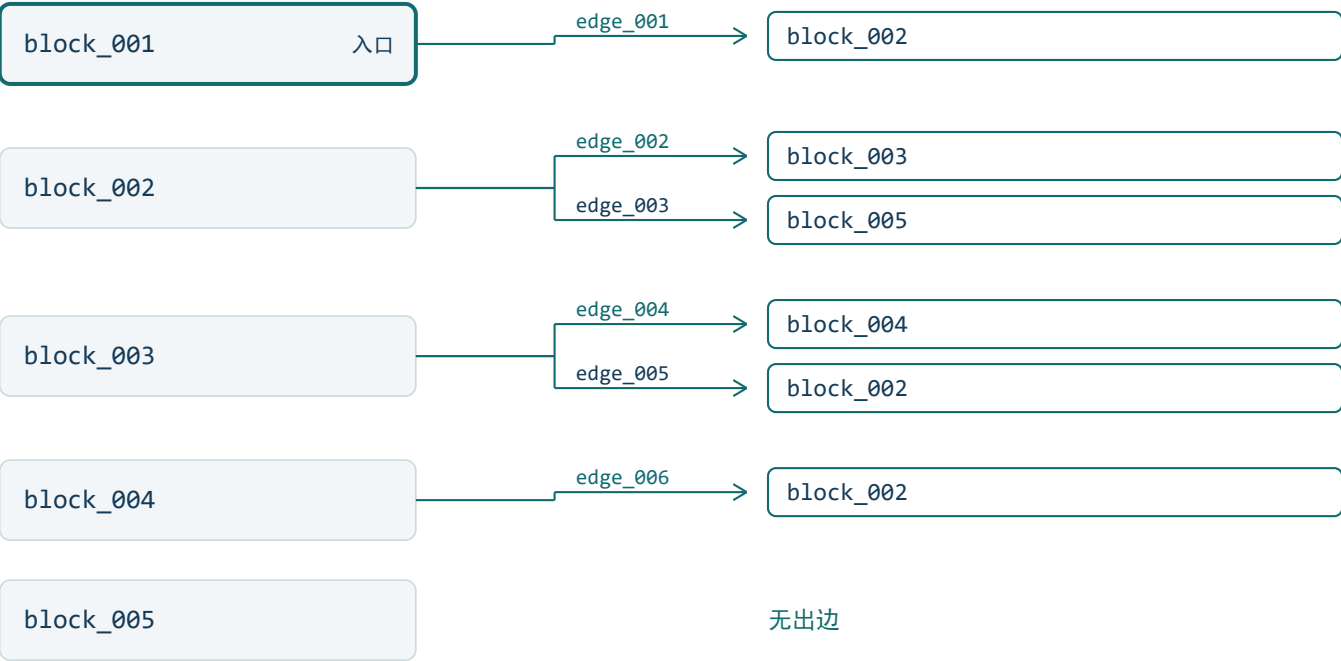
skill.constraints[4]: urgent 为 true 只豁免数值门槛，不豁免 opted_out 的限制。

skill.constraints[5]: 对每条选中记录调用一次 notify.send；未选中的记录不调用 notify.send。

skill.constraints[6]: 处理完成后，将处理条数写入本地 count.txt。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"has_more_records is true"
edge_003	block_002 -> block_005	"has_more_records is false"
edge_004	block_003 -> block_004	"record_selected is true"
edge_005	block_003 -> block_002	"record_selected is false"
edge_006	block_004 -> block_002	null

全部操作、数据引用与约束落点

block_001 Read events.json

```
data_source_kind = "external"
block_001.constraints: []
ir_001 read_events_file
inputs[0] external_resource: events.json | events.json
outputs[0] result: result_001 | events_records
ir_001.constraints: []
draft_instruction_id = read_events_json
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_to_iterate
```

block_002 Iterate over event records

```
data_source_kind = null
block_002.constraints: []
```

ir_003 get_next_event_record

```
inputs[0] result: result_001 | events_records
outputs[0] result: result_002 | current_event_record
outputs[1] result: result_003 | has_more_records
ir_003.constraints: []
draft_instruction_id = get_next_event_record
```

ir_004 dispatch

```
inputs[0] result: result_003 | has_more_records
outputs = []
ir_004.constraints: []
draft_instruction_id = branch_on_more_records
```

block_003 Evaluate event record selection criteria

data_source_kind = null

block_003.constraints[0]: 未选中的记录不调用 notify.send。

ir_005 extract_event_record_fields

inputs[0] result: result_002 | current_event_record

outputs[0] result: result_004 | event_recipient

outputs[1] result: result_005 | event_summary

outputs[2] result: result_006 | event_value

outputs[3] result: result_007 | event_urgent

outputs[4] result: result_008 | event_opted_out

ir_005.constraints: []

draft_instruction_id = extract_event_fields

ir_006 check_event_selection_criteria

inputs[0] result: result_008 | event_opted_out

inputs[1] result: result_007 | event_urgent

inputs[2] result: result_006 | event_value

inputs[3] literal = 100

outputs[0] result: result_009 | record_selected

ir_006.constraints[0]: Select only when opted_out is not true and (urgent is true or value >= 100).

ir_006.constraints[1]: urgent true only exempts the value threshold, not the opted_out restriction.

draft_instruction_id = check_event_selection

ir_007 dispatch

inputs[0] result: result_009 | record_selected

outputs = []

ir_007.constraints: []

draft_instruction_id = branch_on_selection

block_004 Send notification via notify.send

data_source_kind = null

block_004.constraints: []

ir_008 notify.send

inputs[0] external_resource: notify | notify

inputs[1] result: result_004 | event_recipient

inputs[2] result: result_005 | event_summary

outputs = []

ir_008.constraints[0]: Call notify.send once for each selected record.

ir_008.constraints[1]: The recipient is the record's recipient field.

ir_008.constraints[2]: The body parameter is the record's original summary field value.

ir_008.constraints[3]: Do not send the access_token field to any recipient.

ir_008.constraints[4]: Do not perform summarization, content rewriting, or extra formatting.

draft_instruction_id = send_notify

ir_009 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_009.constraints: []
```

```
draft_instruction_id = continue_after_send
```

block_005 Count processed records and write count.txt

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_010 count_processed_event_records

```
inputs[0] result: result_001 | events_records
```

```
outputs[0] result: result_010 | processed_record_count
```

```
ir_010.constraints[0]: Count selected event records that satisfy the notification criteria.
```

```
draft_instruction_id = count_processed_records
```

ir_011 write_count_to_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_010 | processed_record_count
```

```
outputs = []
```

```
ir_011.constraints[0]: Write the processed count to the local count.txt file.
```

```
draft_instruction_id = write_count_to_file
```

ir_012 return

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = return_after_count
```

N03 / 第 1 次 实际输出明细

条件通知与数据发送 · 表格

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N03/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: 源文未明确“处理条数”是统计全部输入记录还是仅统计选中记录；此处按选中并发送的记录数处理。

candidate warning: 源文未说明 notify.send 的返回值，因此将发送建模为无返回值的副作用调用。

Skill 层约束与上下文声明

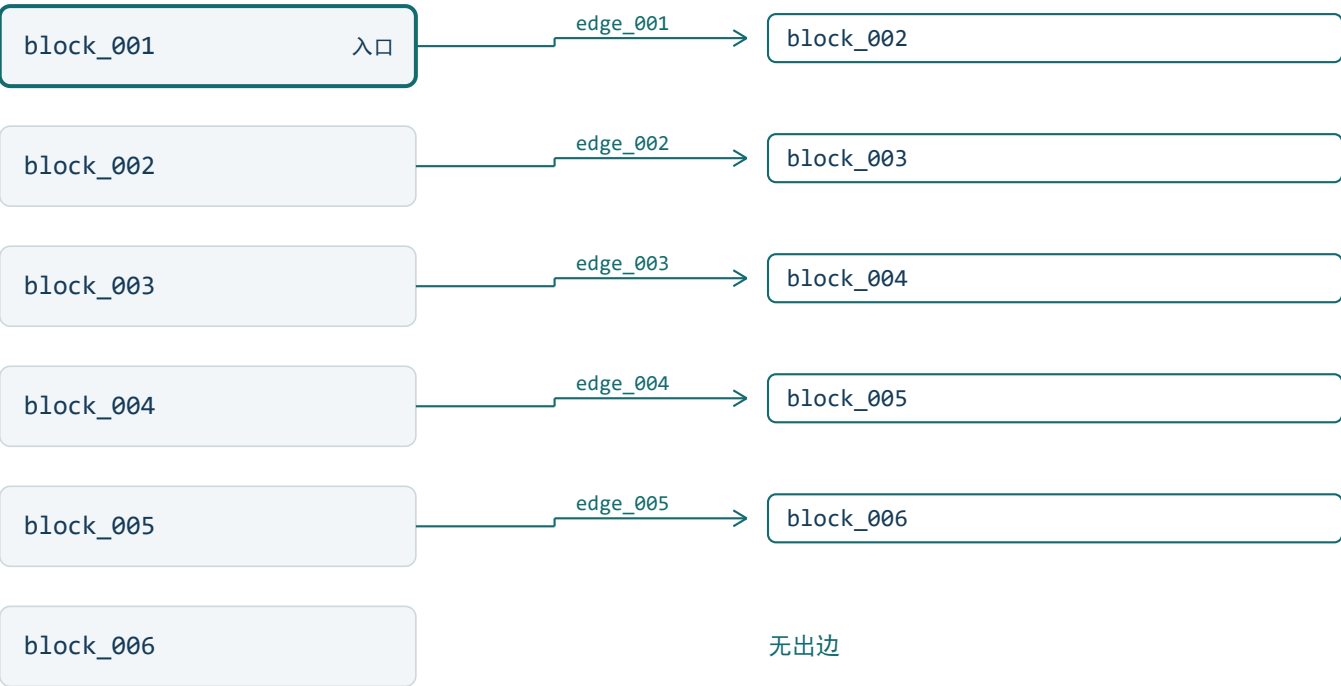
declared_context_keys = []

skill.constraints[0]: 整份流程禁止将 access_token 字段发送给任何接收对象。

skill.constraints[1]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null

全部操作、数据引用与约束落点

block_001 读取 events.json 事件记录

```
data_source_kind = "external"

block_001.constraints[0]: 其中每条记录包含 recipient、summary、record_id、value、urgent、opted_out 和 access_token 字段。

ir_001 read_events_json

inputs[0] external_resource: events.json | events.json
outputs[0] result: result_001 | events_records
ir_001.constraints: []

draft_instruction_id = read_events_file
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []

draft_instruction_id = go_select_events
```

block_002 按订阅状态、紧急标记和数值选择记录

```
data_source_kind = null

block_002.constraints: []

ir_003 select_eligible_event_records

inputs[0] result: result_001 | events_records
outputs[0] result: result_002 | eligible_records
ir_003.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。
ir_003.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

draft_instruction_id = select_eligible_events
```

ir_004 dispatch

```
inputs = []
outputs = []
ir_004.constraints: []

draft_instruction_id = go_prepare_notifications
```

block_003 提取选中记录的接收对象和摘要原值

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 extract_notification_recipient_and_body

```
inputs[0] result: result_002 | eligible_records
```

```
outputs[0] result: result_003 | notification_recipients
```

```
outputs[1] result: result_004 | notification_bodies
```

ir_005.constraints[0]: 接收对象取该记录的 recipient 字段。

ir_005.constraints[1]: notify.send 的 body 参数直接取该记录的 summary 字段原值。

```
draft_instruction_id = extract_notification_parameters
```

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = go_send_notifications
```

block_004 对每条选中记录调用 notify.send

```
data_source_kind = null
```

```
block_004.constraints: []
```

ir_007 notify.send

```
inputs[0] external_resource: notify.send | notify.send
```

```
inputs[1] result: result_003 | notification_recipients
```

```
inputs[2] result: result_004 | notification_bodies
```

```
outputs = []
```

ir_007.constraints[0]: 对每条选中记录调用一次 notify.send。

ir_007.constraints[1]: 未选中的记录不调用 notify.send。

```
draft_instruction_id = send_notifications
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_count_processed
```

block_005 统计已处理条数

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 count_selected_records

```
inputs[0] result: result_002 | eligible_records
```

```
outputs[0] result: result_005 | processed_count
```

```
ir_009.constraints: []
```

```
draft_instruction_id = count_processed_records
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_write_count
```

block_006 将处理条数写入本地 count.txt

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 write_count_to_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_005 | processed_count
```

```
outputs = []
```

```
ir_011.constraints[0]: 处理完成后, 将处理条数写入本地 count.txt。
```

```
draft_instruction_id = write_count
```

ir_012 return

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = return_count_write
```

N03 / 第 2 次 实际输出明细

条件通知与数据发送 · 表格

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N03/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: The source does not explicitly define whether '处理条数' counts only selected records that trigger notify.send or all records processed. This extraction interprets it as the number of selected records notified.

Skill 层约束与上下文声明

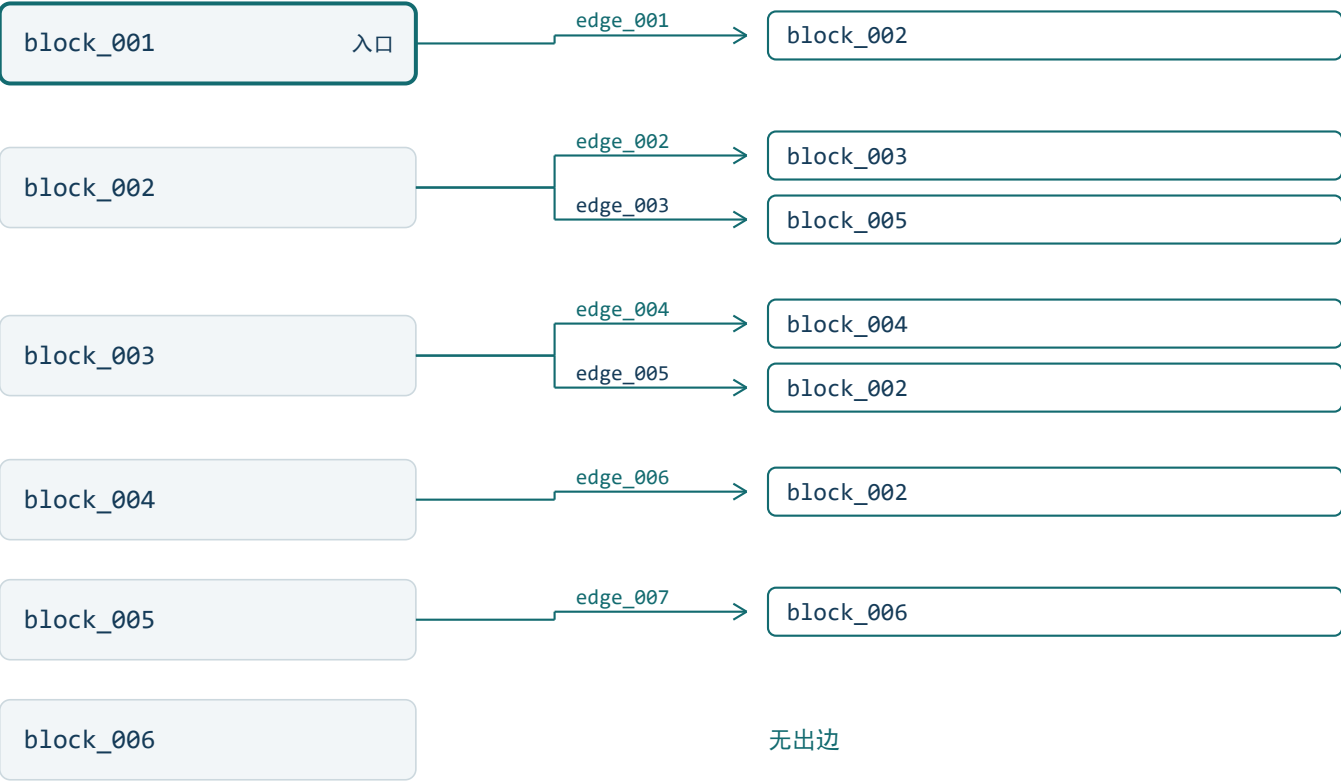
declared_context_keys = []

skill.constraints[0]: 整份流程禁止将 access_token 字段发送给任何接收对象。

skill.constraints[1]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"has_more_records is true"
edge_003	block_002 -> block_005	"has_more_records is false"
edge_004	block_003 -> block_004	"selected is true"
edge_005	block_003 -> block_002	"selected is false"
edge_006	block_004 -> block_002	null
edge_007	block_005 -> block_006	null

全部操作、数据引用与约束落点

block_001 Read the user-provided events. json

data_source_kind = "external"

block_001.constraints[0]: 读取用户提供的 events. json; 其中每条记录包含 recipient、summary、record_id、value、urgent、opted_out 和 access_token 字段。

ir_001 read_events_json

inputs[0] external_resource: events. json | events. json

outputs[0] result: result_001 | events. json records

ir_001.constraints: []

draft_instruction_id = read_events_json

ir_002 dispatch

inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = dispatch_to_next_record

block_002 Get the next event record

data_source_kind = null

block_002.constraints: []

ir_003 get_next_event_record

inputs[0] result: result_001 | events_records

outputs[0] result: result_002 | current event record

outputs[1] result: result_003 | whether more records remain

ir_003.constraints: []

draft_instruction_id = get_next_event_record

ir_004 dispatch

```
inputs[0] result: result_003 | has_more_records
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = dispatch_more_records
```

block_003 Evaluate whether the current record should trigger a notification

```
data_source_kind = null
```

block_003.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

block_003.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

ir_005 extract_event_record_fields

```
inputs[0] result: result_002 | current_record
```

```
outputs[0] result: result_004 | recipient
```

```
outputs[1] result: result_005 | summary
```

```
outputs[2] result: result_006 | opted_out
```

```
outputs[3] result: result_007 | urgent
```

```
outputs[4] result: result_008 | value
```

```
ir_005.constraints: []
```

```
draft_instruction_id = extract_event_fields
```

ir_006 evaluate_notification_condition

```
inputs[0] result: result_006 | record_opted_out
```

```
inputs[1] result: result_007 | record_urgent
```

```
inputs[2] result: result_008 | record_value
```

```
inputs[3] literal = true
```

```
inputs[4] literal = 100
```

```
outputs[0] result: result_009 | record selected for notification
```

ir_006.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

ir_006.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

```
draft_instruction_id = evaluate_notification_condition
```

ir_007 dispatch

```
inputs[0] result: result_009 | selected
```

```
outputs = []
```

```
ir_007.constraints: []
```

```
draft_instruction_id = dispatch_selected
```

block_004 Send a notification for the selected record

```
data_source_kind = null
```

block_004.constraints[0]: 对每条选中记录调用一次 `notify.send`, 接收对象取该记录的 `recipient` 字段。

block_004.constraints[1]: `notify.send` 的 `body` 参数直接取该记录的 `summary` 字段原值。

block_004.constraints[2]: 未选中的记录不调用 `notify.send`。

ir_008 notify.send

```
inputs[0] external_resource: notify.send | notify.send
```

```
inputs[1] result: result_004 | record_recipient
```

```
inputs[2] result: result_005 | record_summary
```

```
outputs = []
```

ir_008.constraints[0]: 对每条选中记录调用一次 `notify.send`, 接收对象取该记录的 `recipient` 字段。

ir_008.constraints[1]: `notify.send` 的 `body` 参数直接取该记录的 `summary` 字段原值。

```
draft_instruction_id = notify_send
```

ir_009 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_009.constraints: []
```

```
draft_instruction_id = dispatch_after_send
```

block_005 Count the processed records

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_010 count_selected_event_records

```
inputs[0] result: result_001 | events_records
```

```
outputs[0] result: result_010 | processed record count
```

```
ir_010.constraints: []
```

```
draft_instruction_id = count_selected_event_records
```

ir_011 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = dispatch_to_write_count
```

block_006 Write the processed count to count.txt

```
data_source_kind = null
```

block_006.constraints[0]: 处理完成后, 将处理条数写入本地 `count.txt`。

ir_012 write_processed_count_to_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_010 | processed_count
```

```
outputs = []
```

ir_012.constraints[0]: 处理完成后, 将处理条数写入本地 `count.txt`。

```
draft_instruction_id = write_processed_count
```

```
ir_013 return
```

```
inputs = []
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = return_done
```

N03 / 第 3 次 实际输出明细

条件通知与数据发送 · 表格

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N03/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: The source phrase '处理条数' is ambiguous; modeled as the count of all event records processed.

Skill 层约束与上下文声明

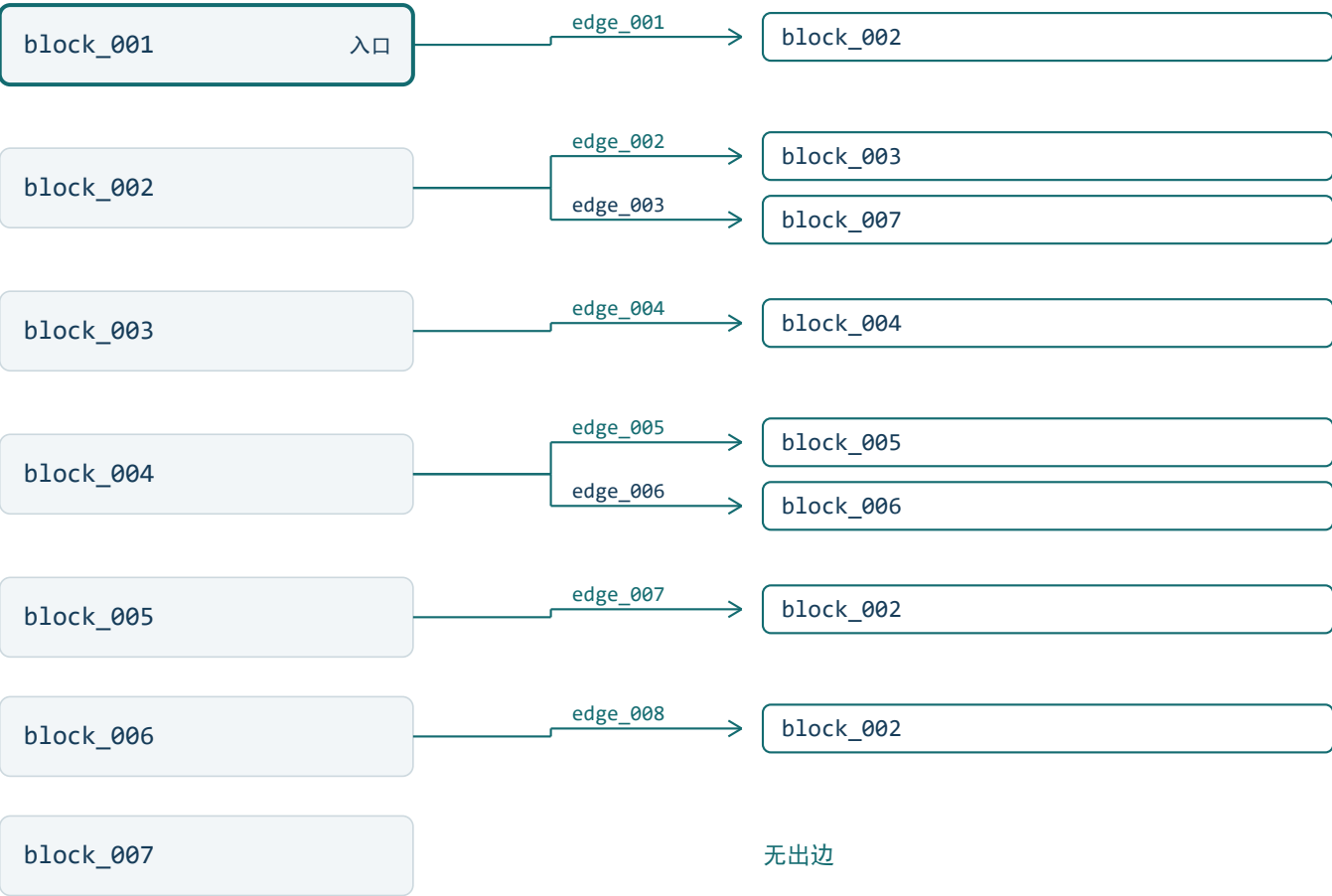
declared_context_keys = []

skill.constraints[0]: 整份流程禁止将 access_token 字段发送给任何接收对象。

skill.constraints[1]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"has_more_records is true"
edge_003	block_002 -> block_007	"no more records"
edge_004	block_003 -> block_004	null
edge_005	block_004 -> block_005	"selected: opted_out is not true and (urgent is true or value >= 100)"
edge_006	block_004 -> block_006	"not selected"
edge_007	block_005 -> block_002	null
edge_008	block_006 -> block_002	null

全部操作、数据引用与约束落点

block_001 Read events. json

```
data_source_kind = "external"
block_001.constraints: []

ir_001 read_events_file

inputs[0] external_resource: events.json | events.json
outputs[0] result: result_001 | event_records
ir_001.constraints: []
draft_instruction_id = read_events
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_loop
```

block_002 Get next event record

```
data_source_kind = null
block_002.constraints: []
```

ir_003 get_next_event_record

```
inputs[0] result: result_001 | event_records
outputs[0] result: result_002 | current_event_record
outputs[1] result: result_003 | has_more_records
ir_003.constraints: []
draft_instruction_id = next_event
```

ir_004 dispatch

inputs[0] result: result_003 | has_more_records

outputs = []

ir_004.constraints: []

draft_instruction_id = check_more

block_003 Extract fields from current event record

data_source_kind = null

block_003.constraints: []

ir_005 extract_event_fields

inputs[0] result: result_002 | current_event_record

outputs[0] result: result_004 | recipient

outputs[1] result: result_005 | summary

outputs[2] result: result_006 | opted_out

outputs[3] result: result_007 | urgent

outputs[4] result: result_008 | event_value

ir_005.constraints: []

draft_instruction_id = extract_fields

ir_006 dispatch

inputs = []

outputs = []

ir_006.constraints: []

draft_instruction_id = go_evaluate

block_004 Evaluate notification selection condition

data_source_kind = null

block_004.constraints: []

ir_007 evaluate_notification_selection

inputs[0] result: result_006 | opted_out

inputs[1] result: result_007 | urgent

inputs[2] result: result_008 | event_value

inputs[3] literal = 100

outputs[0] result: result_009 | should_notify

ir_007.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

ir_007.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

draft_instruction_id = evaluate_selection

ir_008 dispatch

inputs[0] result: result_009 | should_notify

outputs = []

ir_008.constraints: []

draft_instruction_id = branch_selection

block_005 Send notification for selected record

data_source_kind = null

block_005.constraints[0]: 对每条选中记录调用一次 notify.send。

ir_009 notify_send

inputs[0] external_resource: notify | notify

inputs[1] result: result_004 | recipient

inputs[2] result: result_005 | summary

outputs = []

ir_009.constraints[0]: notify.send 的接收对象取该记录的 recipient 字段。

ir_009.constraints[1]: notify.send 的 body 参数直接取该记录的 summary 字段原值。

draft_instruction_id = send_notify

ir_010 dispatch

inputs = []

outputs = []

ir_010.constraints: []

draft_instruction_id = go_next_from_send

block_006 Skip notification for unselected record

data_source_kind = null

block_006.constraints[0]: 未选中的记录不调用 notify.send。

ir_011 dispatch

inputs = []

outputs = []

ir_011.constraints: []

draft_instruction_id = go_next_from_skip

block_007 Write processed count to count.txt

data_source_kind = null

block_007.constraints: []

ir_012 count_event_records

inputs[0] result: result_001 | event_records

outputs[0] result: result_010 | processed_count

ir_012.constraints: []

draft_instruction_id = count_processed

ir_013 write_count_file

inputs[0] external_resource: count.txt | count.txt

inputs[1] result: result_010 | processed_count

outputs = []

ir_013.constraints[0]: 处理完成后, 将处理条数写入本地 count.txt。

draft_instruction_id = write_count

ir_014 return

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = return
```

N04 / 第 1 次 实际输出明细

条件通知与数据发送 · 跨文件混合

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N04/deepseek-v4-flash-max/1

结构状态: complete | 生成 2 | 修复 1 | HTTP 尝试 2 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

生成 2 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: ‘处理条数’ 未明确说明是总记录数还是选中记录数；本分析按选中并处理的记录数建模。

candidate warning: notify.send 未声明返回值，因此按纯外部副作用建模。

Skill 层约束与上下文声明

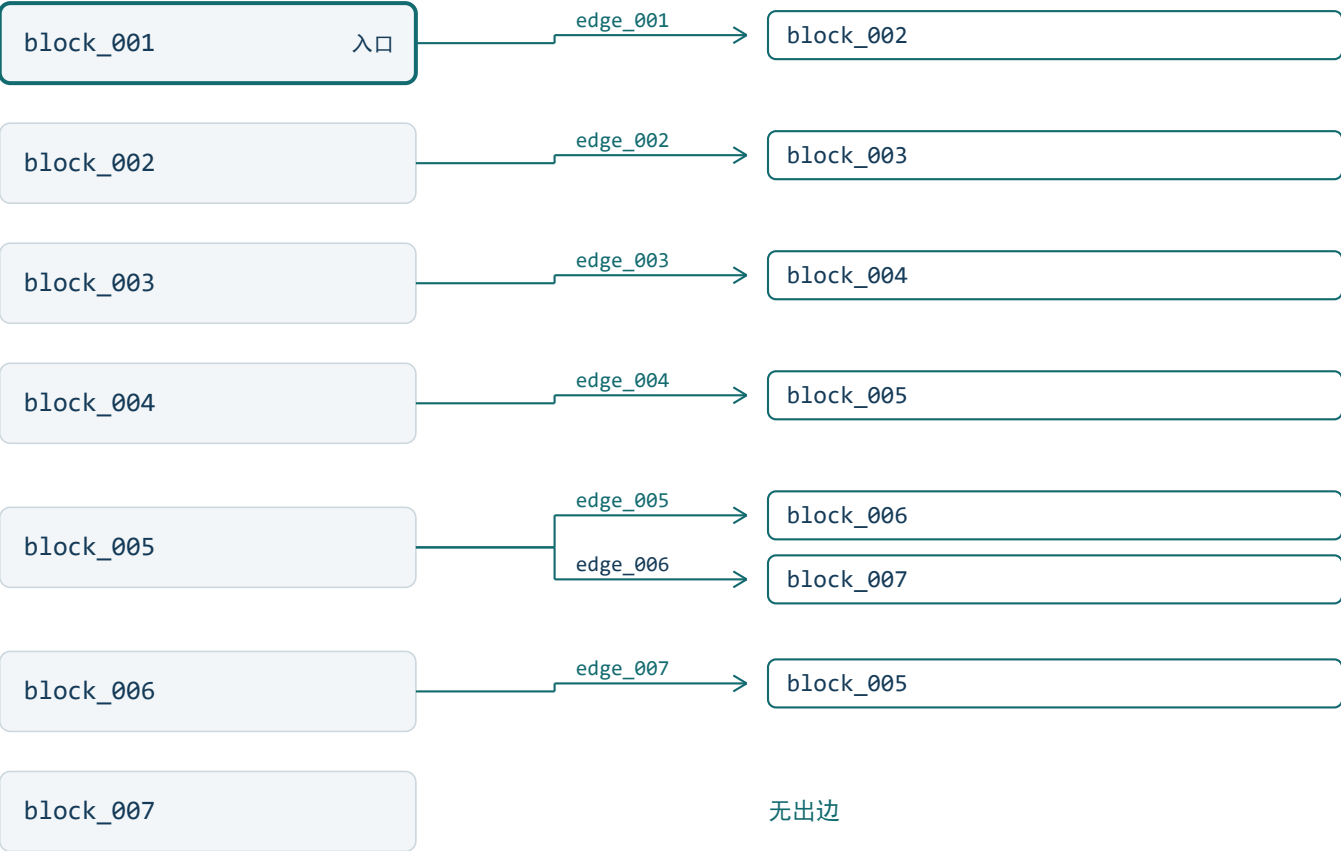
declared_context_keys = []

skill.constraints[0]: 整份流程禁止将 access_token 字段发送给任何接收对象。

skill.constraints[1]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	"has_more"
edge_006	block_005 -> block_007	"not has_more"
edge_007	block_006 -> block_005	null

全部操作、数据引用与约束落点

block_001 Read user-provided events.json

```
data_source_kind = "external"
block_001.constraints: []

ir_001 read_events_file

inputs[0] external_resource: events.json | user-provided events file
outputs[0] result: result_001 | event records
ir_001.constraints: []
draft_instruction_id = read_events_file
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_read_body_config
```

block_002 Read payload.json body_from configuration

```
data_source_kind = "external"
block_002.constraints: []

ir_003 read_body_from_config

inputs[0] external_resource: payload.json | sending configuration
outputs[0] result: result_002 | body_from field name
ir_003.constraints[0]: 正文配置同时由 payload.json 的 body_from 字段声明，按该字段直接取值。
draft_instruction_id = read_body_from_config
```

ir_004 dispatch

```
inputs = []
outputs = []
ir_004.constraints: []
draft_instruction_id = go_select_records
```

block_003 Select records requiring notification

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 select_notification_records

```
inputs[0] result: result_001 | events_records
```

```
inputs[1] literal = 100
```

```
inputs[2] literal = true
```

```
outputs[0] result: result_003 | selected event records
```

ir_005.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

ir_005.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

```
draft_instruction_id = select_notification_records
```

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = go_count_selected
```

block_004 Count selected records

```
data_source_kind = null
```

```
block_004.constraints: []
```

ir_007 count_selected_records

```
inputs[0] result: result_003 | selected_records
```

```
outputs[0] result: result_004 | processed record count
```

```
ir_007.constraints: []
```

```
draft_instruction_id = count_selected_records
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_iterate_selected
```

block_005 Iterate over selected records

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 iterate_selected_records

```
inputs[0] result: result_003 | selected_records
```

```
outputs[0] result: result_005 | current selected record
```

```
outputs[1] result: result_006 | more selected records flag
```

```
ir_009.constraints: []
```

```
draft_instruction_id = iterate_selected_records
```

ir_010 dispatch

inputs[0] result: result_006 | has_more

outputs = []

ir_010.constraints: []

draft_instruction_id = choose_record_or_done

block_006 Send notification for current selected record

data_source_kind = null

block_006.constraints: []

ir_011 extract_recipient

inputs[0] result: result_005 | current_record

outputs[0] result: result_007 | recipient field value

ir_011.constraints[0]: 接收对象取该记录的 recipient 字段。

draft_instruction_id = extract_recipient

ir_012 extract_configured_body_field

inputs[0] result: result_005 | current_record

inputs[1] result: result_002 | body_from_field

outputs[0] result: result_008 | notification body value

ir_012.constraints[0]: notify.send 的 body 参数直接取该记录的 summary 字段原值。

ir_012.constraints[1]: 正文配置同时由 payload.json 的 body_from 字段声明, 按该字段直接取值。

draft_instruction_id = extract_configured_body_field

ir_013 notify.send

inputs[0] external_resource: notify.send | notification service

inputs[1] result: result_007 | recipient

inputs[2] result: result_008 | notification_body

outputs = []

ir_013.constraints[0]: 对每条选中记录调用一次 notify.send。

ir_013.constraints[1]: 未选中的记录不调用 notify.send。

draft_instruction_id = send_notify

ir_014 dispatch

inputs = []

outputs = []

ir_014.constraints: []

draft_instruction_id = go_next_selected

block_007 Write processed count to count.txt

data_source_kind = null

block_007.constraints: []

ir_015 write_count_to_file

inputs[0] external_resource: count.txt | local count file

inputs[1] result: result_004 | processed_count

```
outputs = []  
  
ir_015.constraints[0]: 处理完成后，将处理条数写入本地 count.txt。  
draft_instruction_id = write_count_to_file  
  
ir_016 return  
  
inputs = []  
outputs = []  
  
ir_016.constraints: []  
draft_instruction_id = return_done
```

N04 / 第 2 次 实际输出明细

条件通知与数据发送 · 跨文件混合

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N04/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Interpreting payload.json body_from as a runtime configuration read that selects the record field used for notify.send body; the workflow also names summary directly.

Skill 层约束与上下文声明

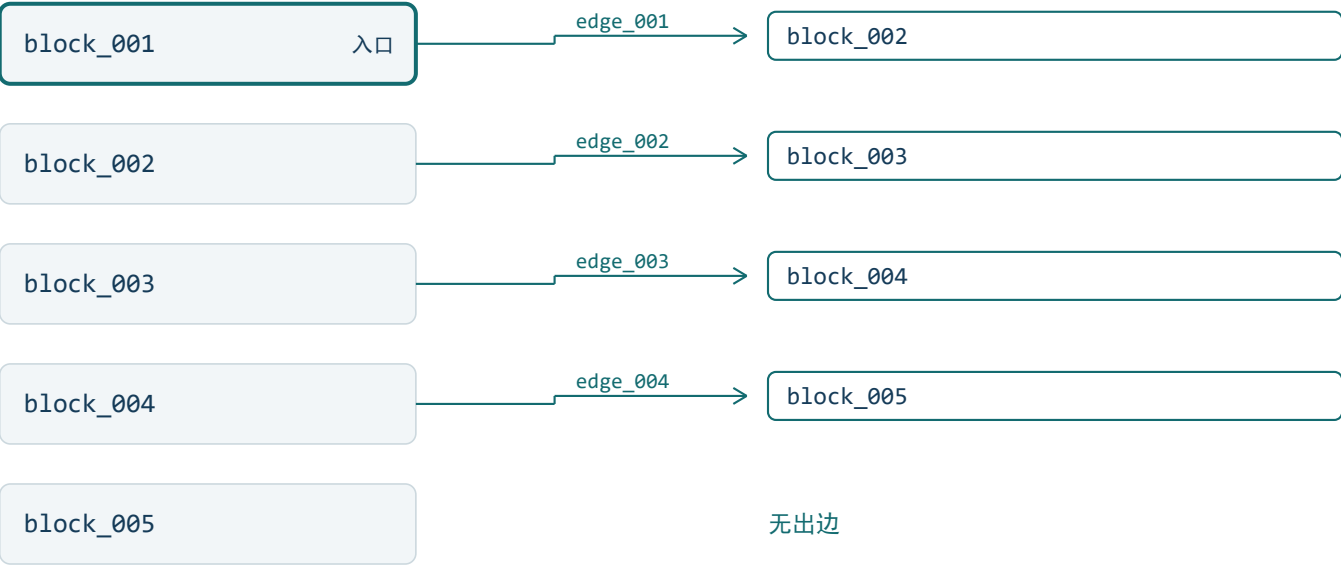
declared_context_keys = []

skill.constraints[0]: 必须执行 [流程说明] (references/workflow.md) 中的要求；该文件是当前流程的组成部分。

skill.constraints[1]: 整份流程禁止将 access_token 字段发送给任何接收对象。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null

全部操作、数据引用与约束落点

block_001 Read user-provided events.json

data_source_kind = "external"

block_001.constraints: []

ir_001 read_events_json

inputs[0] external_resource: events.json | events_file

outputs[0] result: result_001 | event_records

ir_001.constraints: []

draft_instruction_id = read_events

ir_002 dispatch

inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = go_filter

block_002 Select records that should receive notifications

data_source_kind = null

block_002.constraints: []

ir_003 filter_eligible_event_records

inputs[0] result: result_001 | events_records

outputs[0] result: result_002 | selected_event_records

ir_003.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

ir_003.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

draft_instruction_id = filter_events

ir_004 dispatch

inputs = []

outputs = []

ir_004.constraints: []

draft_instruction_id = go_read_payload

block_003 Read body_from from payload.json

data_source_kind = "external"

block_003.constraints: []

ir_005 read_body_from_payload_config

inputs[0] external_resource: payload.json | payload_config_file

outputs[0] result: result_003 | body_from_field_name

ir_005.constraints: []

draft_instruction_id = read_payload

ir_006 dispatch

inputs = []

outputs = []

ir_006.constraints: []

draft_instruction_id = go_send

block_004 Send one notification per selected record

data_source_kind = null

block_004.constraints: []

ir_007 notify_send

inputs[0] external_resource: notify.send | notify_service

inputs[1] result: result_002 | selected_records

inputs[2] result: result_003 | body_from_field

outputs[0] result: result_004 | processed_record_count

ir_007.constraints[0]: 对每条选中记录调用一次 notify.send, 接收对象取该记录的 recipient 字段。

ir_007.constraints[1]: notify.send 的 body 参数直接取该记录的 summary 字段原值。

ir_007.constraints[2]: 正文配置同时由 [发送配置](./payload.json) 的 body_from 字段声明, 按该字段直接取值。

ir_007.constraints[3]: 未选中的记录不调用 notify.send。

ir_007.constraints[4]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

draft_instruction_id = send_notifications

metadata = {"body_field_config_result": "body_from_field", "body_field_default": "summary", "recipient_field": "recipient"}

ir_008 dispatch

inputs = []

outputs = []

ir_008.constraints: []

draft_instruction_id = go_write_count

block_005 Write processed count to local count.txt

data_source_kind = null

block_005.constraints: []

ir_009 write_processed_count_to_file

inputs[0] external_resource: count.txt | count_file

inputs[1] result: result_004 | processed_count

outputs = []

ir_009.constraints[0]: 处理完成后, 将处理条数写入本地 count.txt。

draft_instruction_id = write_count

ir_010 return

inputs = []

outputs = []

ir_010.constraints: []

draft_instruction_id = return

N04 / 第 3 次 实际输出明细

条件通知与数据发送 · 跨文件混合

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N04/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: 将 payload.json 的 body_from 视为运行时配置读取。

candidate warning: 源未明确“处理条数”是选中记录数还是全部读取记录数；本候选按选中记录数处理。

Skill 层约束与上下文声明

declared_context_keys = []

skill.constraints[0]: 必须执行 references/workflow.md 中的要求；该文件是当前流程的组成部分。

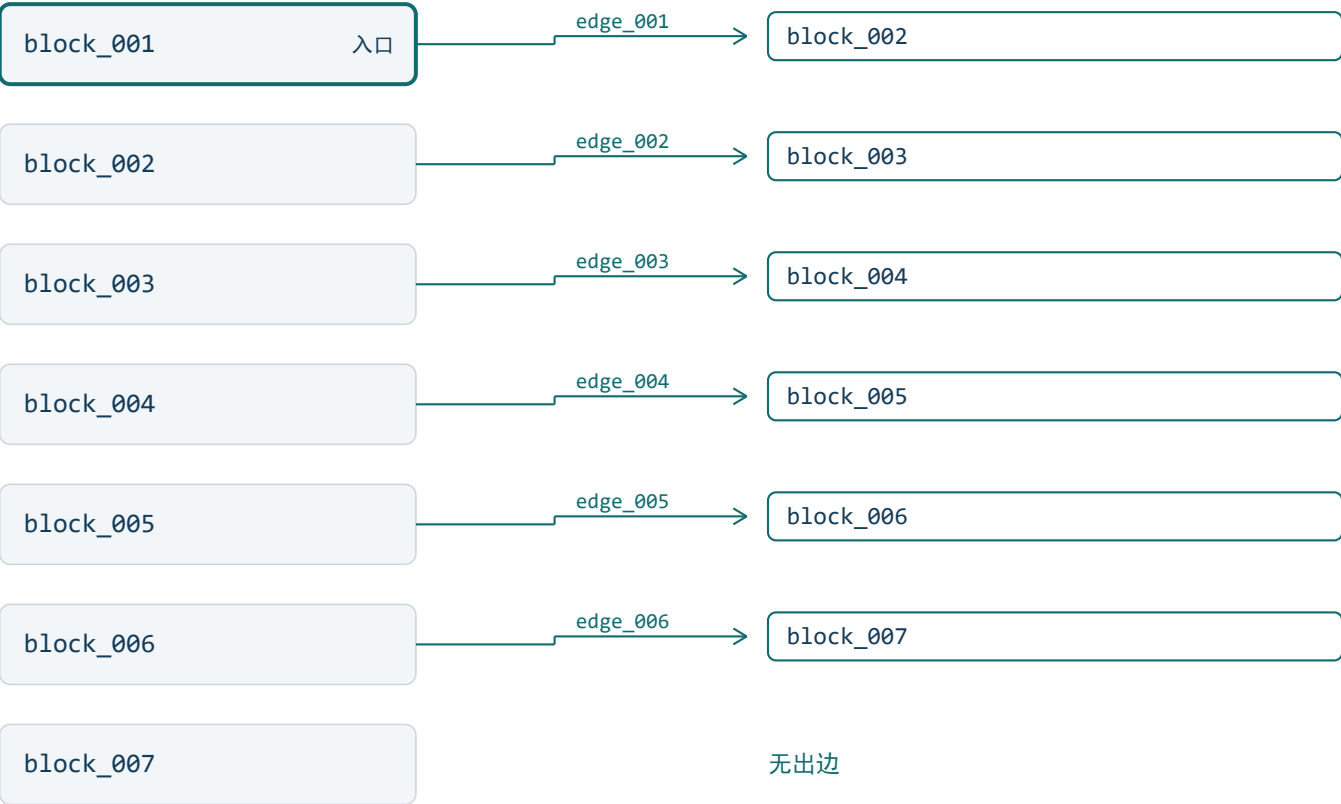
skill.constraints[1]: 整份流程禁止将 access_token 字段发送给任何接收对象。

skill.constraints[2]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

skill.constraints[3]: 处理完成后，将处理条数写入本地 count.txt。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null

全部操作、数据引用与约束落点

block_001 Read user-provided events.json

```
data_source_kind = "external"
```

block_001.constraints[0]: 读取用户提供的 events.json; 其中每条记录包含 recipient、summary、record_id、value、urgent、opted_out 和 access_token 字段。

ir_001 read_events_json

```
inputs[0] external_resource: events.json | events.json
outputs[0] result: result_001 | event_records
```

ir_001.constraints[0]: 读取用户提供的 events.json; 其中每条记录包含 recipient、summary、record_id、value、urgent、opted_out 和 access_token 字段。

draft_instruction_id = read_events_json

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_read_config
```

block_002 Read body_from from payload.json

```
data_source_kind = "external"
```

block_002.constraints[0]: 正文配置同时由 payload.json 的 body_from 字段声明，按该字段直接取值。

ir_003 read_payload_body_from

```
inputs[0] external_resource: payload.json | payload.json
outputs[0] result: result_002 | body_field_name
```

ir_003.constraints[0]: 正文配置同时由 payload.json 的 body_from 字段声明，按该字段直接取值。

draft_instruction_id = read_payload_body_from

ir_004 dispatch

inputs = []

outputs = []

ir_004.constraints: []

draft_instruction_id = go_filter

block_003 Filter eligible event records

data_source_kind = null

block_003.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

block_003.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

ir_005 filter_eligible_event_records

inputs[0] result: result_001 | event_records

inputs[1] literal = 100

outputs[0] result: result_003 | selected_records

ir_005.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

ir_005.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

draft_instruction_id = filter_eligible_event_records

ir_006 dispatch

inputs = []

outputs = []

ir_006.constraints: []

draft_instruction_id = go_extract

block_004 Extract recipient and body for selected records

data_source_kind = null

block_004.constraints[0]: 接收对象取该记录的 recipient 字段。

block_004.constraints[1]: notify.send 的 body 参数直接取该记录的 summary 字段原值。

block_004.constraints[2]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

ir_007 extract_notification_fields

inputs[0] result: result_003 | selected_records

inputs[1] result: result_002 | body_field_name

outputs[0] result: result_004 | notification_recipients

outputs[1] result: result_005 | notification_bodies

ir_007.constraints[0]: 接收对象取该记录的 recipient 字段。

ir_007.constraints[1]: notify.send 的 body 参数直接取该记录的 summary 字段原值。

ir_007.constraints[2]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

draft_instruction_id = extract_notification_fields

ir_008 dispatch

inputs = []

outputs = []

ir_008.constraints: []

draft_instruction_id = go_send

block_005 Send notify.send for each selected record

data_source_kind = null

block_005.constraints[0]: 对每条选中记录调用一次 notify.send。

block_005.constraints[1]: 未选中的记录不调用 notify.send。

block_005.constraints[2]: 整份流程禁止将 access_token 字段发送给任何接收对象。

ir_009 notify.send

inputs[0] external_resource: notify | notify

inputs[1] result: result_004 | notification_recipients

inputs[2] result: result_005 | notification_bodies

outputs = []

ir_009.constraints[0]: 对每条选中记录调用一次 notify.send。

ir_009.constraints[1]: 未选中的记录不调用 notify.send。

ir_009.constraints[2]: 整份流程禁止将 access_token 字段发送给任何接收对象。

draft_instruction_id = send_notify_per_selected_record

ir_010 dispatch

inputs = []

outputs = []

ir_010.constraints: []

draft_instruction_id = go_count

block_006 Count processed records

data_source_kind = null

block_006.constraints: []

ir_011 count_processed_records

inputs[0] result: result_003 | selected_records

outputs[0] result: result_006 | processed_count

ir_011.constraints: []

draft_instruction_id = count_processed_records

ir_012 dispatch

inputs = []

outputs = []

ir_012.constraints: []

draft_instruction_id = go_write

block_007 Write processed count to count.txt and finish

```
data_source_kind = null
```

block_007.constraints[0]: 处理完成后, 将处理条数写入本地 count.txt。

ir_013 write_count_to_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_006 | processed_count
```

```
outputs = []
```

ir_013.constraints[0]: 处理完成后, 将处理条数写入本地 count.txt。

```
draft_instruction_id = write_count_to_file
```

ir_014 return

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = return_done
```

N05 / 第 1 次 实际输出明细

条件通知与数据发送 · 用途最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N05/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: 步骤 9 被明确标记为历史示例且当前流程不执行，因此未提取写入 count.txt 的行为。

candidate warning: notify.send 未提及返回值，因此建模为无返回数据的纯副作用调用。

Skill 层约束与上下文声明

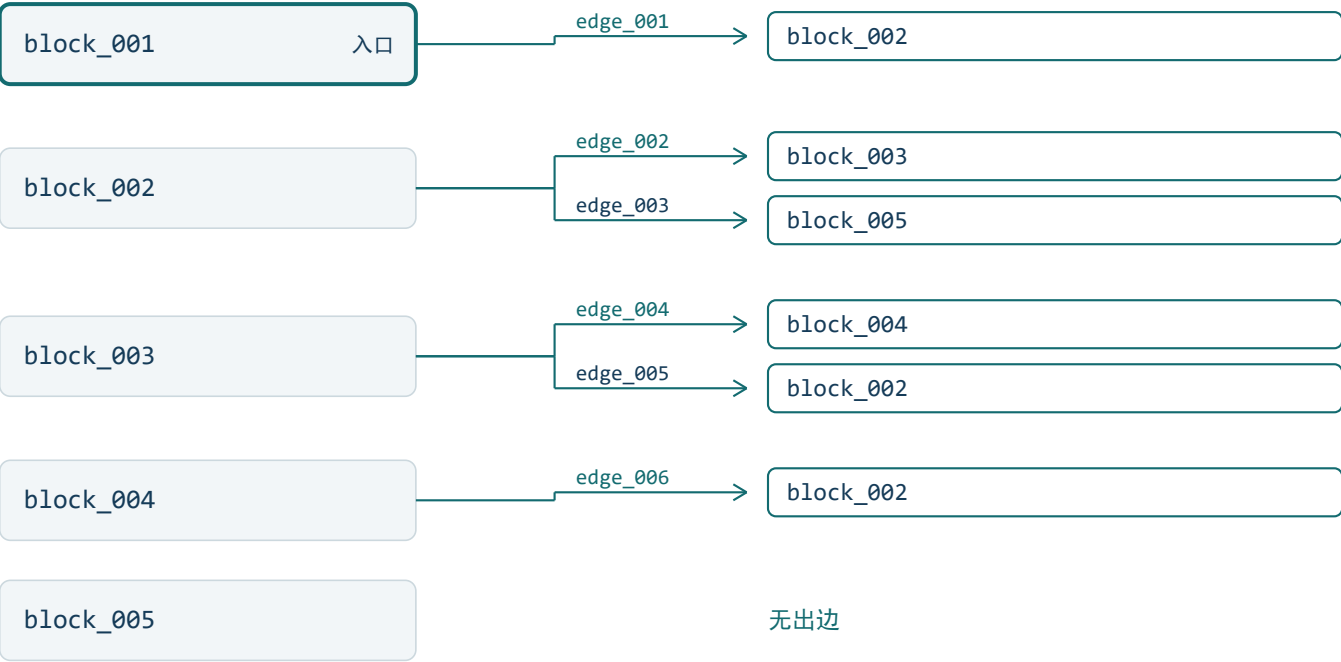
declared_context_keys = []

skill.constraints[0]: 整份流程禁止将 access_token 字段发送给任何接收对象。

skill.constraints[1]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"has_more_events 为 true"
edge_003	block_002 -> block_005	"has_more_events 为 false"
edge_004	block_003 -> block_004	"记录被选中"
edge_005	block_003 -> block_002	"记录未被选中"
edge_006	block_004 -> block_002	null

全部操作、数据引用与约束落点

```
block_001 Read the user-provided events.json

data_source_kind = "external"
block_001.constraints: []

ir_001 read_events_json

inputs[0] external_resource: events.json | events.json
outputs[0] result: result_001 | event_records
ir_001.constraints: []
draft_instruction_id = read_events_json

ir_002 dispatch

inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_to_next_event

block_002 Get next event record and check whether more records remain

data_source_kind = null
block_002.constraints: []

ir_003 get_next_event_record

inputs[0] result: result_001 | event_records
outputs[0] result: result_002 | current_event
outputs[1] result: result_003 | has_more_events
ir_003.constraints: []
draft_instruction_id = take_next_event

ir_004 dispatch

inputs[0] result: result_003 | has_more_events
outputs = []
ir_004.constraints: []
draft_instruction_id = dispatch_after_next_event
```


block_003 Evaluate whether the current event is selected

data_source_kind = null

block_003.constraints[0]: 未选中的记录不调用 notify.send。

ir_005 extract_event_selection_fields

inputs[0] result: result_002 | current_event

outputs[0] result: result_004 | event_value

outputs[1] result: result_005 | event_urgent

outputs[2] result: result_006 | event_opted_out

ir_005.constraints: []

draft_instruction_id = extract_selection_fields

ir_006 check_event_selection

inputs[0] result: result_006 | event_opted_out

inputs[1] result: result_005 | event_urgent

inputs[2] result: result_004 | event_value

inputs[3] literal = 100

outputs[0] result: result_007 | event_is_selected

ir_006.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

ir_006.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

draft_instruction_id = check_event_selection

ir_007 dispatch

inputs[0] result: result_007 | event_is_selected

outputs = []

ir_007.constraints: []

draft_instruction_id = dispatch_selection

block_004 Send notification for the selected event

data_source_kind = null

block_004.constraints: []

ir_008 extract_event_notification_fields

inputs[0] result: result_002 | current_event

outputs[0] result: result_008 | event_recipient

outputs[1] result: result_009 | event_summary

ir_008.constraints: []

draft_instruction_id = extract_notification_fields

ir_009 notify_send

inputs[0] external_resource: notify.send | notify.send

inputs[1] result: result_008 | event_recipient

inputs[2] result: result_009 | event_summary

outputs = []

ir_009.constraints[0]: 对每条选中记录调用一次 notify.send。

```
ir_009.constraints[1]: 接收对象取该记录的 recipient 字段。
ir_009.constraints[2]: notify.send 的 body 参数直接取该记录的 summary 字段原值。
draft_instruction_id = send_notify

ir_010 dispatch

inputs = []
outputs = []
ir_010.constraints: []
draft_instruction_id = dispatch_after_send

block_005 Finish after all event records are processed

data_source_kind = null
block_005.constraints: []

ir_011 return

inputs = []
outputs = []
ir_011.constraints: []
draft_instruction_id = return_done
```

N05 / 第 2 次 实际输出明细

条件通知与数据发送 · 用途最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N05/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: 第 9 条明确标注为历史示例且当前流程不执行，因此未生成 count.txt 写入行为。

candidate warning: events.json 作为外部资源读取；本包未声明 runtime context keys。

candidate warning: notify.send 未声明返回值，因此建模为无返回数据的副作用块。

Skill 层约束与上下文声明

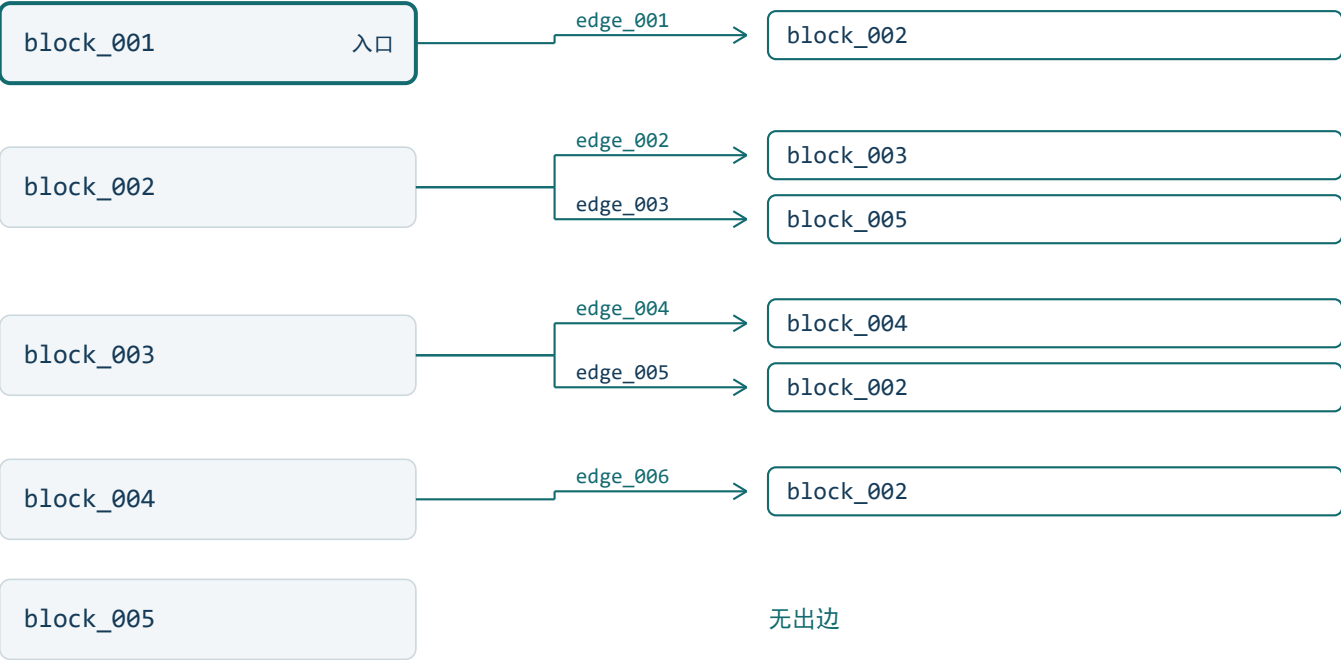
declared_context_keys = []

skill.constraints[0]: 整份流程禁止将 access_token 字段发送给任何接收对象。

skill.constraints[1]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"has_more_event_records is true"
edge_003	block_002 -> block_005	"has_more_event_records is false"
edge_004	block_003 -> block_004	"record_selected is true"
edge_005	block_003 -> block_002	"record_selected is false"
edge_006	block_004 -> block_002	null

全部操作、数据引用与约束落点

block_001 Read events.json

```
data_source_kind = "external"
block_001.constraints: []
ir_001 read_events_json
inputs[0] external_resource: events.json | user-provided events.json
outputs[0] result: result_001 | event_records
ir_001.constraints: []
draft_instruction_id = read_events_json
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_iterate
```

block_002 Iterate over event records

```
data_source_kind = null
block_002.constraints: []
ir_003 get_next_event_record
inputs[0] result: result_001 | event_records
outputs[0] result: result_002 | current_event_record
outputs[1] result: result_003 | has_more_event_records
ir_003.constraints: []
draft_instruction_id = get_next_event_record
```

ir_004 dispatch

```
inputs[0] result: result_003 | has_more_event_records
outputs = []
ir_004.constraints: []
draft_instruction_id = choose_more_or_end
```

block_003 Evaluate whether the current event record is selected

```
data_source_kind = null
```

```
block_003.constraints[0]: 未选中的记录不调用 notify.send。
```

ir_005 extract_event_record_fields

```
inputs[0] result: result_002 | current_event_record
```

```
outputs[0] result: result_004 | event_recipient
```

```
outputs[1] result: result_005 | event_summary
```

```
outputs[2] result: result_006 | event_value
```

```
outputs[3] result: result_007 | event_urgent
```

```
outputs[4] result: result_008 | event_opted_out
```

```
ir_005.constraints: []
```

```
draft_instruction_id = extract_event_record_fields
```

ir_006 check_event_record_selected

```
inputs[0] result: result_008 | event_opted_out
```

```
inputs[1] result: result_007 | event_urgent
```

```
inputs[2] result: result_006 | event_value
```

```
outputs[0] result: result_009 | record_selected
```

```
ir_006.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。
```

```
ir_006.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。
```

```
draft_instruction_id = check_event_record_selected
```

ir_007 dispatch

```
inputs[0] result: result_009 | record_selected
```

```
outputs = []
```

```
ir_007.constraints: []
```

```
draft_instruction_id = choose_send_or_skip
```

block_004 Send notification for the selected event record

```
data_source_kind = null
```

```
block_004.constraints: []
```

ir_008 notify.send

```
inputs[0] external_resource: notify.send | notify.send
```

```
inputs[1] result: result_004 | event_recipient
```

```
inputs[2] result: result_005 | event_summary
```

```
outputs = []
```

```
ir_008.constraints[0]: 对每条选中记录调用一次 notify.send, 接收对象取该记录的 recipient 字段。
```

```
ir_008.constraints[1]: notify.send 的 body 参数直接取该记录的 summary 字段原值。
```

```
draft_instruction_id = notify_send
```

ir_009 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_009.constraints: []
```

```
draft_instruction_id = go_next_record
```

block_005 Finish after processing all event records

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_010 return

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = return_done
```

N05 / 第 3 次 实际输出明细

条件通知与数据发送 · 用途最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N05/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: 历史示例（当前流程不执行）中的 count.txt 写入未建模为执行行为。

Skill 层约束与上下文声明

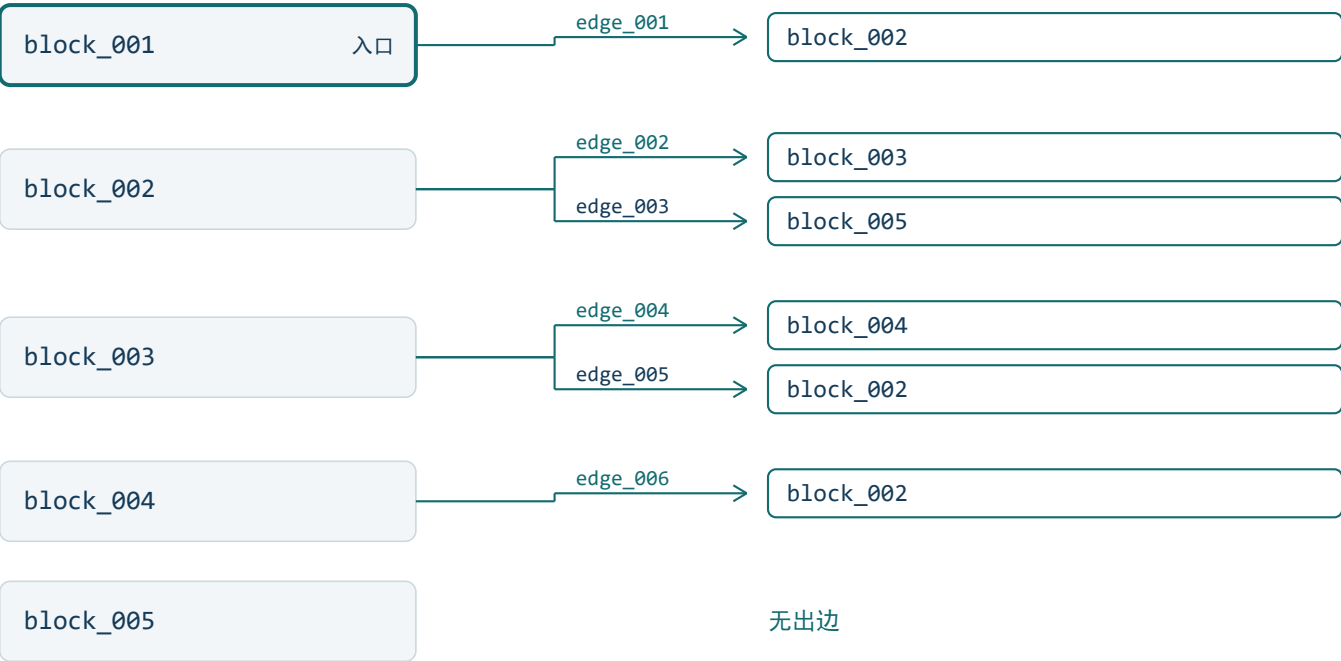
declared_context_keys = []

skill.constraints[0]: 整份流程禁止将 access_token 字段发送给任何接收对象。

skill.constraints[1]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"has_more_records is true"
edge_003	block_002 -> block_005	"has_more_records is false"
edge_004	block_003 -> block_004	"record_selected is true"
edge_005	block_003 -> block_002	"record_selected is false"
edge_006	block_004 -> block_002	null

全部操作、数据引用与约束落点

block_001 Read the user-provided events.json file

data_source_kind = "external"

block_001.constraints[0]: 读取用户提供的 events.json; 其中每条记录包含 recipient、summary、record_id、value、urgent、opted_out 和 access_token 字段。

ir_001 read_events_file

inputs[0] external_resource: events.json | events.json

outputs[0] result: result_001 | event_records

ir_001.constraints: []

draft_instruction_id = read_events_file

ir_002 dispatch

inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = read_events_dispatch

block_002 Get the next event record and check whether any remain

data_source_kind = null

block_002.constraints: []

ir_003 get_next_event_record

inputs[0] result: result_001 | event_records

outputs[0] result: result_002 | current_event_record

outputs[1] result: result_003 | has_more_records

ir_003.constraints: []

draft_instruction_id = get_next_event_record

ir_004 dispatch

inputs[0] result: result_003 | has_more_records

outputs = []

ir_004.constraints: []

draft_instruction_id = loop_control_dispatch

block_003 Select the current record if not opted out and urgent or high value

data_source_kind = null

block_003.constraints[0]: 未选中的记录不调用 notify.send。

ir_005 extract_event_fields

inputs[0] result: result_002 | current_event_record

outputs[0] result: result_004 | event_recipient

outputs[1] result: result_005 | event_summary

outputs[2] result: result_006 | event_value

outputs[3] result: result_007 | event_urgent

outputs[4] result: result_008 | event_opted_out

ir_005.constraints: []

draft_instruction_id = extract_event_fields

ir_006 evaluate_event_selection

inputs[0] result: result_008 | event_opted_out

inputs[1] result: result_007 | event_urgent

inputs[2] result: result_006 | event_value

outputs[0] result: result_009 | record_selected

ir_006.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

ir_006.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

draft_instruction_id = evaluate_event_selection

ir_007 dispatch

inputs[0] result: result_009 | record_selected

outputs = []

ir_007.constraints: []

draft_instruction_id = select_record_dispatch

block_004 Send one notification for the selected record

data_source_kind = null

block_004.constraints: []

ir_008 notify.send

inputs[0] external_resource: notify.send | service

inputs[1] result: result_004 | recipient

inputs[2] result: result_005 | body

outputs = []

ir_008.constraints[0]: 对每条选中记录调用一次 notify.send, 接收对象取该记录的 recipient 字段。

ir_008.constraints[1]: notify.send 的 body 参数直接取该记录的 summary 字段原值。

draft_instruction_id = send_notification

ir_009 dispatch

inputs = []

outputs = []

ir_009.constraints: []

draft_instruction_id = send_notification_dispatch

block_005 Finish after all event records are processed

data_source_kind = null

block_005.constraints: []

ir_010 return

inputs = []

outputs = []

ir_010.constraints: []

draft_instruction_id = finish_return

N06 / 第 1 次 实际输出明细

条件通知与数据发送 · 语义最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N06/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: 处理条数 was interpreted as the number of selected records processed/notified; if it means all read records, count_processed_records should consume events_records instead of selected_records.

Skill 层约束与上下文声明

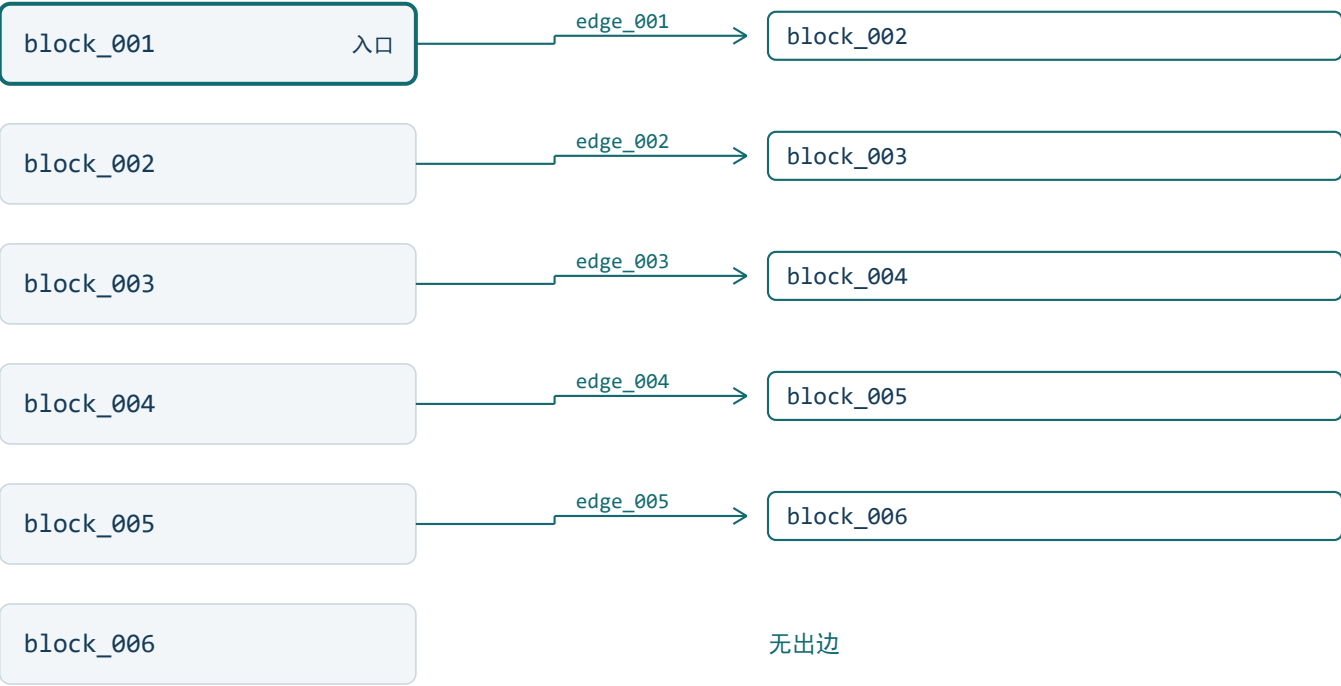
declared_context_keys = []

skill.constraints[0]: 整份流程禁止将 access_token 字段发送给任何接收对象。

skill.constraints[1]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null

全部操作、数据引用与约束落点

block_001 Read events.json

```
data_source_kind = "external"

block_001.constraints: []

ir_001 read_events_json

inputs[0] external_resource: events.json | events.json
outputs[0] result: result_001 | events_records
ir_001.constraints: []

draft_instruction_id = read_events_json
```

ir_002 dispatch

```
inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = go_select
```

block_002 Select records eligible for notification

```
data_source_kind = null

block_002.constraints: []

ir_003 select_notifiable_records

inputs[0] result: result_001 | events_records
outputs[0] result: result_002 | selected_records

ir_003.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。
ir_003.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

draft_instruction_id = select_notifiable_records
```

ir_004 dispatch

```
inputs = []

outputs = []

ir_004.constraints: []

draft_instruction_id = go_extract
```

block_003 Extract notification payloads

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 extract_notification_payloads

```
inputs[0] result: result_002 | selected_records
```

```
outputs[0] result: result_003 | notification_payloads
```

ir_005.constraints[0]: 接收对象取该记录的 recipient 字段。

ir_005.constraints[1]: notify.send 的 body 参数直接取该记录的 record_id 字段原值。

```
draft_instruction_id = extract_notification_payloads
```

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = go_send
```

block_004 Send notifications for selected records

```
data_source_kind = null
```

```
block_004.constraints: []
```

ir_007 notify_send_for_selected_records

```
inputs[0] external_resource: notify.send | notify.send
```

```
inputs[1] result: result_003 | notification_payloads
```

```
outputs = []
```

ir_007.constraints[0]: 对每条选中记录调用一次 notify.send。

ir_007.constraints[1]: 未选中的记录不调用 notify.send。

```
draft_instruction_id = send_notify
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_count
```

block_005 Count processed records

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 count_processed_records

```
inputs[0] result: result_002 | selected_records
```

```
outputs[0] result: result_004 | processed_count
```

```
ir_009.constraints: []
```

```
draft_instruction_id = count_processed_records
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_write
```

block_006 Write processed count to count.txt

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 write_count_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_004 | processed_count
```

```
outputs = []
```

```
ir_011.constraints[0]: 处理完成后，将处理条数写入本地 count.txt。
```

```
draft_instruction_id = write_count_file
```

ir_012 return

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = return_done
```

N06 / 第 2 次 实际输出明细

条件通知与数据发送 · 语义最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N06/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

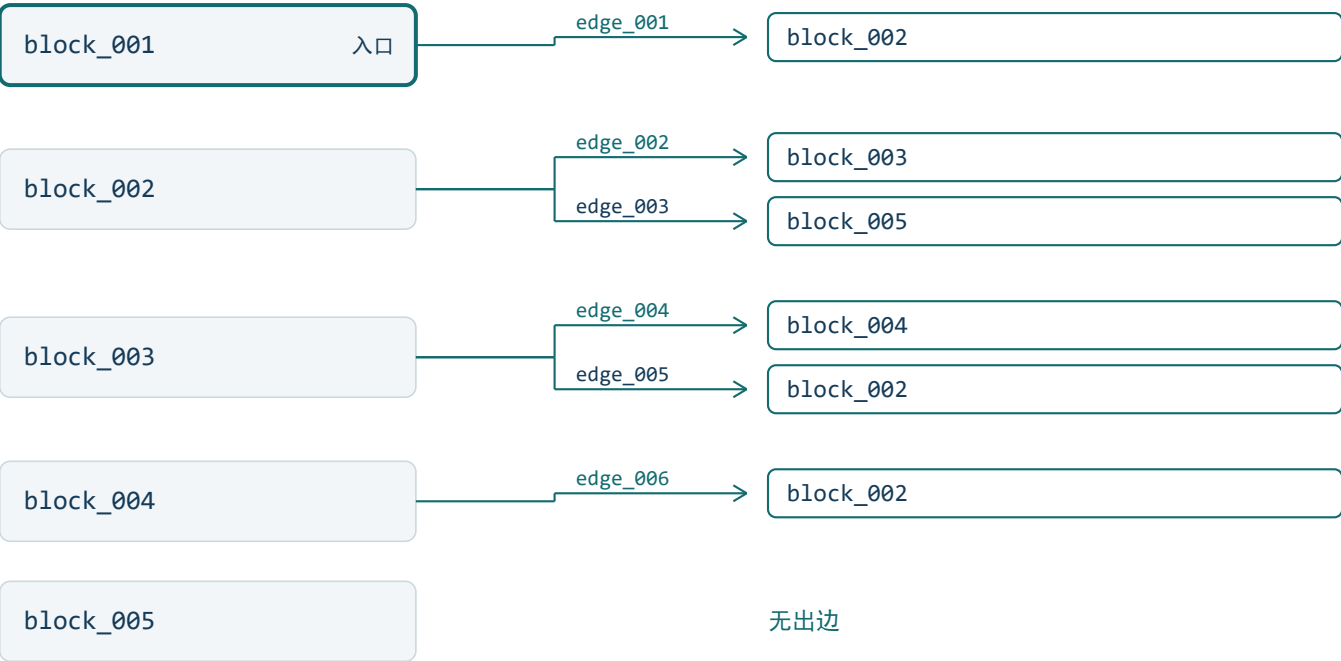
declared_context_keys = []

skill.constraints[0]: 整份流程禁止将 access_token 字段发送给任何接收对象。

skill.constraints[1]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"有更多记录"
edge_003	block_002 -> block_005	"没有更多记录"
edge_004	block_003 -> block_004	"选中该记录"
edge_005	block_003 -> block_002	"未选中该记录"
edge_006	block_004 -> block_002	null

全部操作、数据引用与约束落点

block_001 Read events.json

data_source_kind = "external"

block_001.constraints[0]: 读取用户提供的 events.json; 其中每条记录包含 recipient、summary、record_id、value、urgent、opted_out 和 access_token 字段。

ir_001 read_events_json

inputs[0] external_resource: events.json | events.json

outputs[0] result: result_001 | event_records

ir_001.constraints: []

draft_instruction_id = read_events_json

ir_002 dispatch

inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = go_to_process_next_record

block_002 Get next event record

data_source_kind = null

block_002.constraints: []

ir_003 get_next_event_record

inputs[0] result: result_001 | event_records

outputs[0] result: result_002 | current_event_record

outputs[1] result: result_003 | has_more_records

ir_003.constraints: []

draft_instruction_id = get_next_event_record

ir_004 dispatch

inputs[0] result: result_003 | has_more_records

outputs = []

ir_004.constraints: []

draft_instruction_id = dispatch_more_or_done

block_003 Check whether the current record is selected

data_source_kind = null

block_003.constraints[0]: 未选中的记录不调用 notify.send。

ir_005 check_record_selected

inputs[0] result: result_002 | current_event_record

outputs[0] result: result_004 | is_selected

ir_005.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

ir_005.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

draft_instruction_id = check_record_selected

ir_006 dispatch

inputs[0] result: result_004 | is_selected

outputs = []

ir_006.constraints: []

draft_instruction_id = dispatch_selected_or_not

block_004 Send notification for selected record

data_source_kind = null

block_004.constraints: []

ir_007 extract_event_recipient

inputs[0] result: result_002 | current_event_record

outputs[0] result: result_005 | recipient_value

ir_007.constraints: []

draft_instruction_id = extract_recipient

ir_008 extract_event_record_id

inputs[0] result: result_002 | current_event_record

outputs[0] result: result_006 | record_id_value

ir_008.constraints: []

draft_instruction_id = extract_record_id

ir_009 notify.send

inputs[0] external_resource: notify.send | notify.send

inputs[1] result: result_005 | recipient_value

inputs[2] result: result_006 | record_id_value

outputs = []

ir_009.constraints[0]: 对每条选中记录调用一次 notify.send, 接收对象取该记录的 recipient 字段。

ir_009.constraints[1]: notify.send 的 body 参数直接取该记录的 record_id 字段原值。

```
draft_instruction_id = notify_send
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_next_after_send
```

block_005 Write processed record count to count.txt

```
data_source_kind = null
```

```
block_005.constraints[0]: 处理完成后, 将处理条数写入本地 count.txt。
```

ir_011 count_processed_event_records

```
inputs[0] result: result_001 | event_records
```

```
outputs[0] result: result_007 | processed_count
```

```
ir_011.constraints: []
```

```
draft_instruction_id = count_processed_records
```

ir_012 write_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_007 | processed_count
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = write_count_file
```

ir_013 return

```
inputs = []
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = return_end
```

N06 / 第 3 次 实际输出明细

条件通知与数据发送 · 语义最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/N06/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

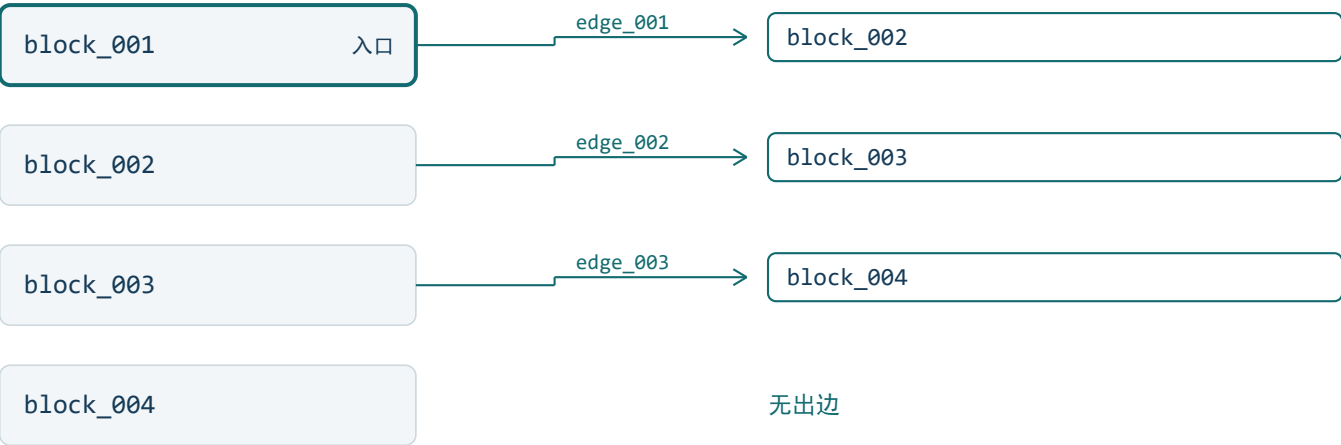
declared_context_keys = []

skill.constraints[0]: 整份流程禁止将 access_token 字段发送给任何接收对象。

skill.constraints[1]: 字段取值要求不包含摘要生成、内容改写或额外格式转换。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null

全部操作、数据引用与约束落点

block_001 Read event records from events.json

data_source_kind = "external"

block_001.constraints[0]: 每条记录包含 recipient、summary、record_id、value、urgent、opted_out 和 access_token 字段。

ir_001 read_events_json_file

inputs[0] external_resource: events.json | events.json

```
outputs[0] result: result_001 | event_records
```

```
ir_001.constraints: []
```

```
draft_instruction_id = read_events_json
```

ir_002 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_002.constraints: []
```

```
draft_instruction_id = go_select
```

block_002 Select eligible records and provide recipient and record_id for notification

```
data_source_kind = null
```

```
block_002.constraints: []
```

ir_003 select_eligible_notification_targets

```
inputs[0] result: result_001 | event_records
```

```
outputs[0] result: result_002 | selected_recipients
```

```
outputs[1] result: result_003 | selected_record_ids
```

ir_003.constraints[0]: 仅当 opted_out 不为 true, 且 urgent 为 true 或 value 大于等于 100 时, 选中该记录。

ir_003.constraints[1]: urgent 为 true 只豁免数值门槛, 不豁免 opted_out 的限制。

```
draft_instruction_id = select_targets
```

ir_004 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = go_send
```

block_003 Send notifications for selected records via notify.send

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 notify.send

```
inputs[0] external_resource: notify.send | notify.send
```

```
inputs[1] result: result_002 | selected_recipients
```

```
inputs[2] result: result_003 | selected_record_ids
```

```
outputs = []
```

ir_005.constraints[0]: 对每条选中记录调用一次 notify.send, 接收对象取该记录的 recipient 字段。

ir_005.constraints[1]: notify.send 的 body 参数直接取该记录的 record_id 字段原值。

ir_005.constraints[2]: 未选中的记录不调用 notify.send。

```
draft_instruction_id = send_notify
```

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = go_write_count
```

block_004 Count processed records and write to local count.txt

```
data_source_kind = null
```

```
block_004.constraints[0]: 处理完成后, 将处理条数写入本地 count.txt。
```

ir_007 count_processed_records

```
inputs[0] result: result_002 | selected_recipients
```

```
outputs[0] result: result_004 | processed_count
```

```
ir_007.constraints: []
```

```
draft_instruction_id = count_processed
```

ir_008 write_count_to_local_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_004 | processed_count
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = write_count_file
```

ir_009 return

```
inputs = []
```

```
outputs = []
```

```
ir_009.constraints: []
```

```
draft_instruction_id = return_done
```

Q01 / 第 1 次 实际输出明细

查询调用与参数使用 · 编号步骤

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q01/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

```
declared_context_keys = []

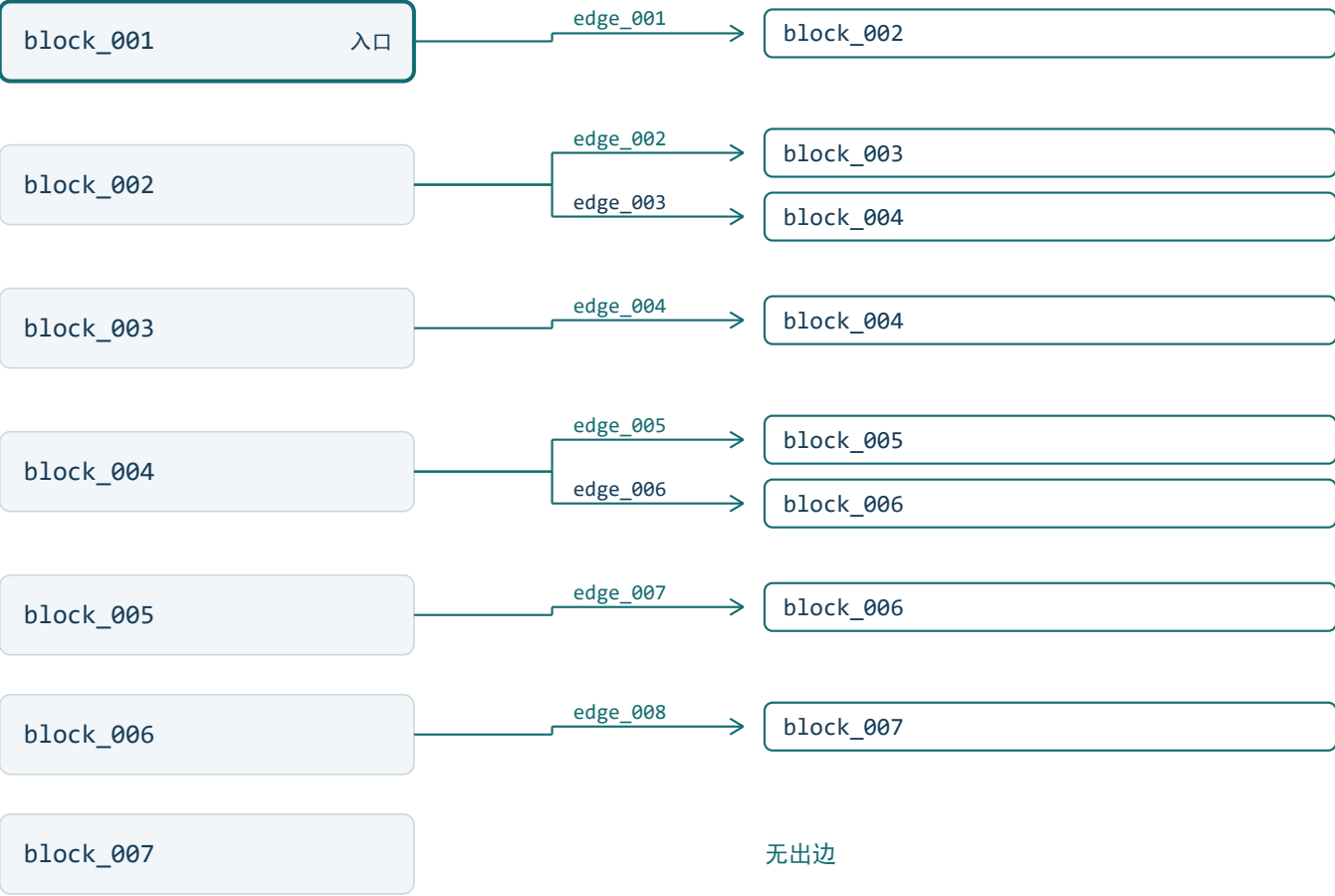
skill.constraints[0]: index.search supports fuzzy matching; this note does not request fuzzy expansion in this workflow.

skill.constraints[1]: Do not invoke index.delete anywhere in this workflow.

skill.constraints[2]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"from_date is present"
edge_003	block_002 -> block_004	"from_date is missing"
edge_004	block_003 -> block_004	null
edge_005	block_004 -> block_005	"limit is present"
edge_006	block_004 -> block_006	"limit is missing"
edge_007	block_005 -> block_006	null
edge_008	block_006 -> block_007	null

全部操作、数据引用与约束落点

block_001 Read the user-supplied request. json

```
data_source_kind = "external"
block_001.constraints: []

ir_001 read_request_json

inputs[0] external_resource: request.json | user-supplied request.json
outputs[0] result: result_001 | request.term
outputs[1] result: result_002 | request.json contents
outputs[2] result: result_003 | whether from_date is present
outputs[3] result: result_004 | whether limit is present
ir_001.constraints: []
draft_instruction_id = read_request_json
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_check_from_date
```

block_002 Check whether from_date is present

```
data_source_kind = null
block_002.constraints: []
```

ir_003 dispatch

```
inputs[0] result: result_003 | from_date_present
outputs = []
ir_003.constraints: []
draft_instruction_id = choose_from_date
```

block_003 Extract present from_date unchanged

data_source_kind = null

block_003.constraints: []

ir_004 extract_request_from_date

inputs[0] result: result_002 | request_json

outputs[0] result: result_005 | from_date value

ir_004.constraints: []

draft_instruction_id = extract_request_from_date

ir_005 dispatch

inputs = []

outputs = []

ir_005.constraints: []

draft_instruction_id = go_check_limit

block_004 Check whether limit is present

data_source_kind = null

block_004.constraints: []

ir_006 dispatch

inputs[0] result: result_004 | limit_present

outputs = []

ir_006.constraints: []

draft_instruction_id = choose_limit

block_005 Extract present limit unchanged

data_source_kind = null

block_005.constraints: []

ir_007 extract_request_limit

inputs[0] result: result_002 | request_json

outputs[0] result: result_006 | limit value

ir_007.constraints: []

draft_instruction_id = extract_request_limit

ir_008 dispatch

inputs = []

outputs = []

ir_008.constraints: []

draft_instruction_id = go_call_search

block_006 Call index.search exactly once with optional arguments

data_source_kind = "external"

block_006.constraints: []

ir_009 index_search

inputs[0] external_resource: index.search | index.search service


```
inputs[1] result: result_001 | request_term
inputs[2] result: result_005 | from_date_argument
inputs[3] result: result_006 | limit_argument
outputs[0] result: result_007 | response.total
outputs[1] result: result_008 | response.items
ir_009.constraints[0]: Call index.search exactly once.
ir_009.constraints[1]: Use request.term unchanged as its query argument.
ir_009.constraints[2]: When from_date is present, pass its value unchanged; the accepted format is YYYY-MM-DD.
ir_009.constraints[3]: When from_date is missing, omit the from_date argument.
ir_009.constraints[4]: When limit is present, pass its value unchanged as the limit argument.
ir_009.constraints[5]: When limit is missing, omit the limit argument.
draft_instruction_id = call_index_search
```

ir_010 dispatch

```
inputs = []
outputs = []
ir_010.constraints: []
draft_instruction_id = go_write_and_return
```

block_007 Write total to count.txt and return items

```
data_source_kind = null
block_007.constraints: []
```

ir_011 write_file

```
inputs[0] external_resource: count.txt | local count.txt
inputs[1] result: result_007 | search_total
outputs = []
ir_011.constraints: []
draft_instruction_id = write_count
```

ir_012 return

```
inputs[0] result: result_008 | search_items
outputs = []
ir_012.constraints[0]: Return the search response's items value unchanged.
draft_instruction_id = return_items
```

Q01 / 第 2 次 实际输出明细

查询调用与参数使用 · 编号步骤

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q01/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

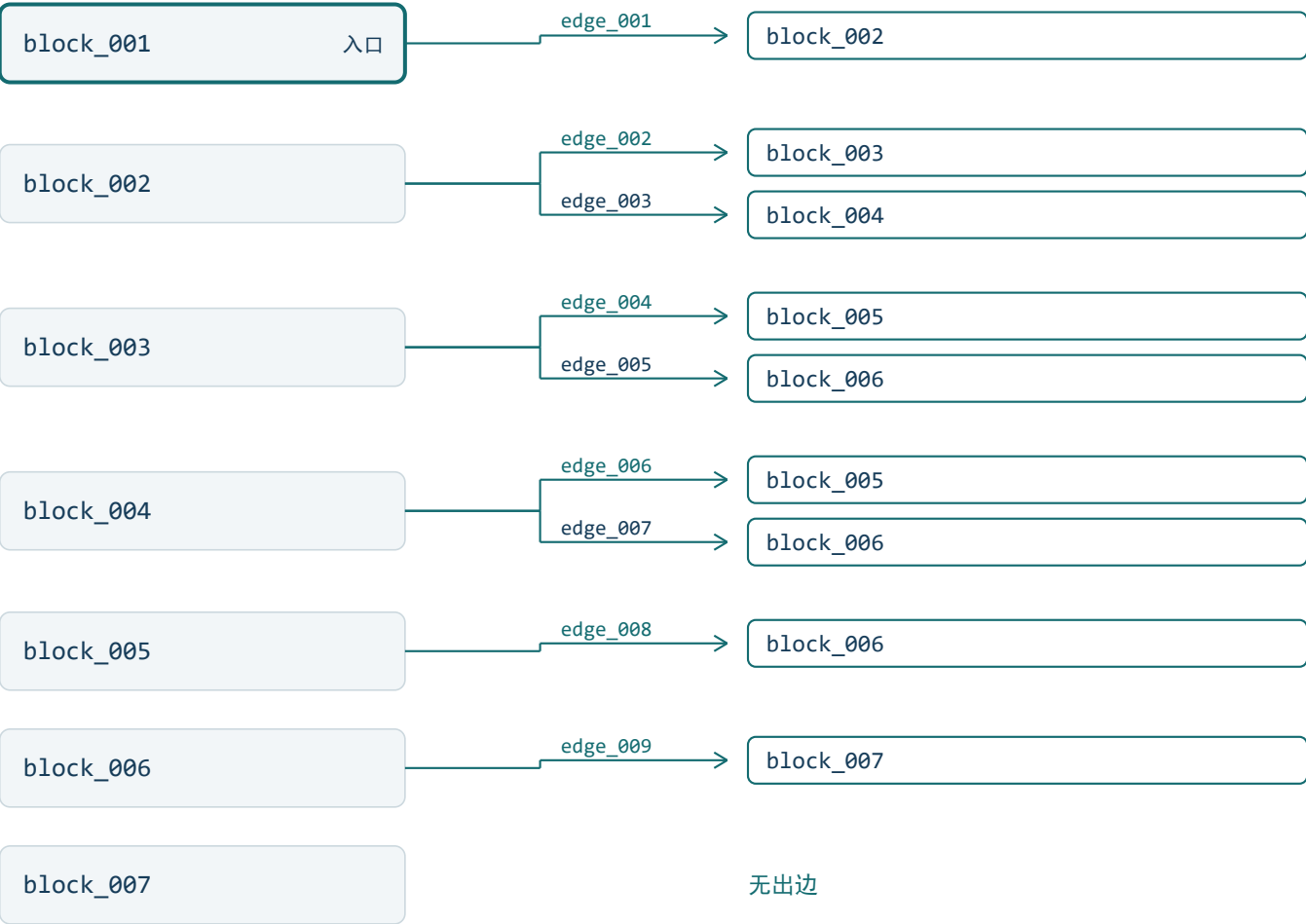
diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

```
declared_context_keys = []  
  
skill.constraints[0]: Call index.search exactly once.  
  
skill.constraints[1]: index.search supports fuzzy matching; this note does not request fuzzy expansion in this workflow.  
  
skill.constraints[2]: Do not invoke index.delete anywhere in this workflow.  
  
skill.constraints[3]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"from_date is present"
edge_003	block_002 -> block_004	"from_date is missing"
edge_004	block_003 -> block_005	"limit is present"
edge_005	block_003 -> block_006	"limit is missing"
edge_006	block_004 -> block_005	"limit is present"
edge_007	block_004 -> block_006	"limit is missing"
edge_008	block_005 -> block_006	null
edge_009	block_006 -> block_007	null

全部操作、数据引用与约束落点

```
block_001 Read the user-supplied request.json

data_source_kind = "external"
block_001.constraints: []

ir_001 read_request_json

inputs[0] external_resource: request.json | request.json
outputs[0] result: result_001 | request_data
ir_001.constraints: []
draft_instruction_id = read_request_json

ir_002 dispatch

inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_inspect

block_002 Extract term and check optional-parameter presence

data_source_kind = null
block_002.constraints: []

ir_003 extract_request_term

inputs[0] result: result_001 | request_data
outputs[0] result: result_002 | request_term
ir_003.constraints: []
draft_instruction_id = extract_term

ir_004 check_from_date_presence

inputs[0] result: result_001 | request_data
```

```
outputs[0] result: result_003 | from_date_present
ir_004.constraints: []
draft_instruction_id = check_from_date_presence
ir_005 check_limit_presence
inputs[0] result: result_001 | request_data
outputs[0] result: result_004 | limit_present
ir_005.constraints: []
draft_instruction_id = check_limit_presence
ir_006 dispatch
inputs[0] result: result_003 | from_date_present
outputs = []
ir_006.constraints: []
draft_instruction_id = go_from_date
block_003 Extract from_date when present
data_source_kind = null
block_003.constraints: []
ir_007 extract_request_from_date
inputs[0] result: result_001 | request_data
outputs[0] result: result_005 | request_from_date
ir_007.constraints: []
draft_instruction_id = extract_from_date_value
ir_008 dispatch
inputs[0] result: result_004 | limit_present
outputs = []
ir_008.constraints: []
draft_instruction_id = go_limit_after_from_date
block_004 Choose limit path when from_date is absent
data_source_kind = null
block_004.constraints: []
ir_009 dispatch
inputs[0] result: result_004 | limit_present
outputs = []
ir_009.constraints: []
draft_instruction_id = go_limit_without_from_date
block_005 Extract limit when present
data_source_kind = null
block_005.constraints: []
ir_010 extract_request_limit
inputs[0] result: result_001 | request_data
```

```
outputs[0] result: result_006 | request_limit
```

```
ir_010.constraints: []
```

```
draft_instruction_id = extract_limit_value
```

ir_011 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = go_search
```

block_006 Call index.search once with term and optional parameters

```
data_source_kind = "external"
```

```
block_006.constraints: []
```

ir_012 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_002 | request_term
```

```
inputs[2] result: result_003 | from_date_present
```

```
inputs[3] result: result_005 | request_from_date
```

```
inputs[4] result: result_004 | limit_present
```

```
inputs[5] result: result_006 | request_limit
```

```
outputs[0] result: result_007 | search_total
```

```
outputs[1] result: result_008 | search_items
```

```
ir_012.constraints[0]: Use request.term unchanged as the query argument.
```

```
ir_012.constraints[1]: When from_date is present, pass its value unchanged; the accepted format is YYYY-MM-DD.
```

```
ir_012.constraints[2]: When from_date is missing, omit the from_date argument.
```

```
ir_012.constraints[3]: When limit is present, pass its value unchanged as the limit argument.
```

```
ir_012.constraints[4]: When limit is missing, omit the limit argument.
```

```
draft_instruction_id = call_index_search
```

ir_013 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = go_finish
```

block_007 Write total to count.txt and return items

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_014 write_total_to_count_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_007 | search_total
```

```
outputs = []
```

```
ir_014.constraints[0]: Write the response's total value to local count.txt.
```

```
draft_instruction_id = write_total
```

ir_015 return

```
inputs[0] result: result_008 | search_items
```

```
outputs = []
```

```
ir_015.constraints[0]: Return the search response's items value unchanged.
```

```
draft_instruction_id = return_items
```

Q01 / 第 3 次 实际输出明细

查询调用与参数使用 · 编号步骤

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q01/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: The capability note about index.search fuzzy matching is non-executable and does not request fuzzy expansion in this workflow.

candidate warning: The workflow branches on optional from_date and limit presence, resulting in multiple index.search instructions on mutually exclusive paths to preserve optional-argument omission semantics.

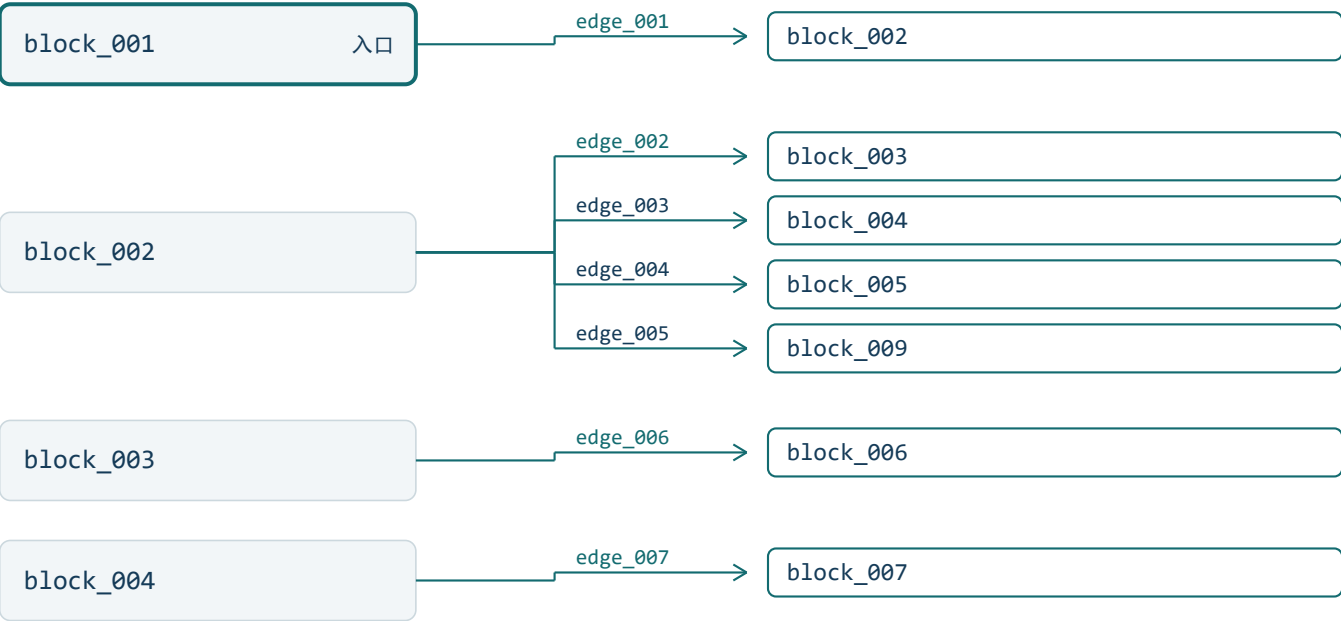
Skill 层约束与上下文声明

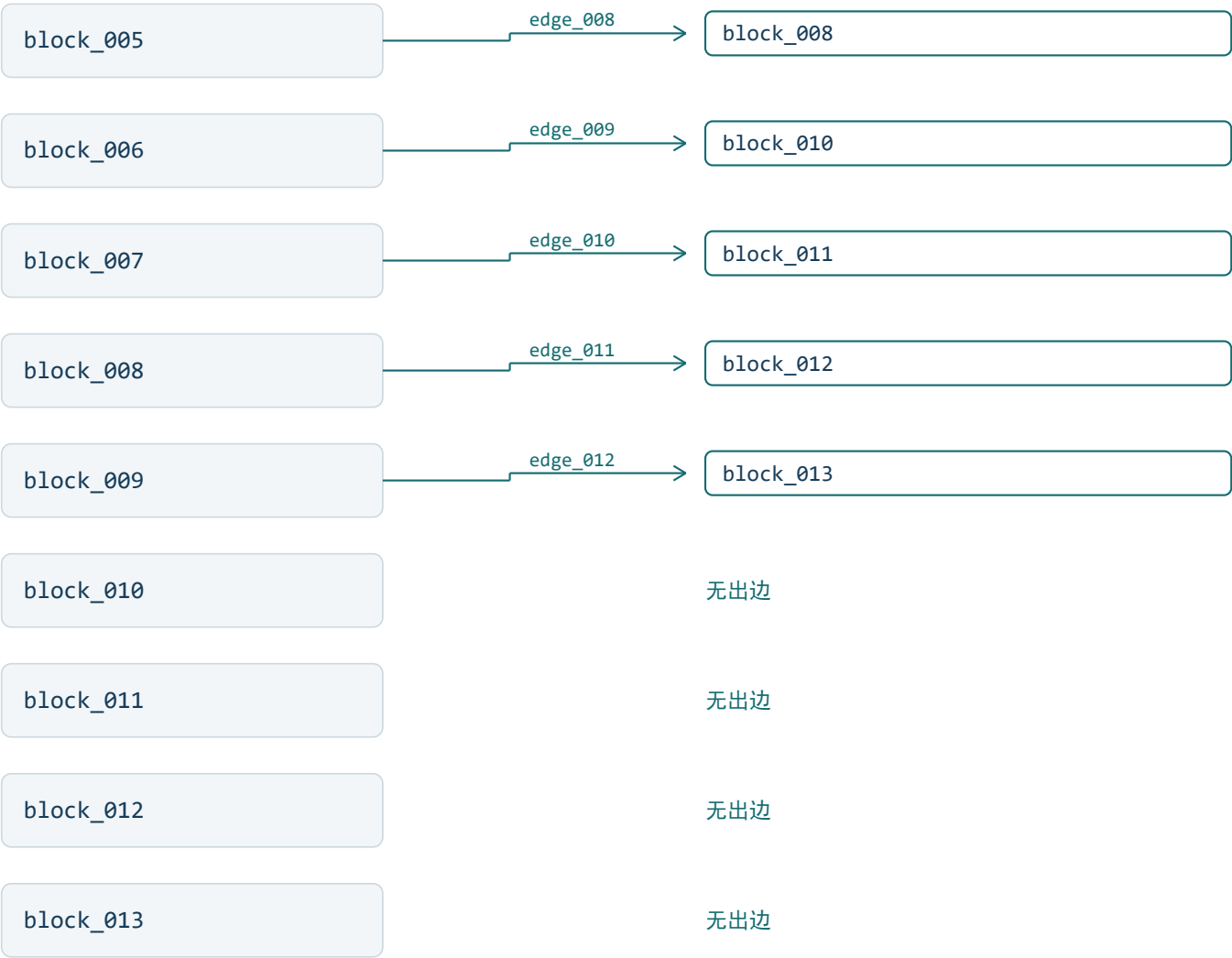
```
declared_context_keys = []

skill.constraints[0]: Call index.search exactly once, using request.term unchanged as its query argument.
skill.constraints[1]: When from_date is present, pass its value unchanged; the accepted format is YYYY-MM-DD.
skill.constraints[2]: When from_date is missing, omit the from_date argument.
skill.constraints[3]: When limit is present, pass its value unchanged as the limit argument.
skill.constraints[4]: When limit is missing, omit the limit argument.
skill.constraints[5]: Do not invoke index.delete anywhere in this workflow.
skill.constraints[6]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"from_date present and limit present"
edge_003	block_002 -> block_004	"from_date present and limit missing"
edge_004	block_002 -> block_005	"from_date missing and limit present"
edge_005	block_002 -> block_009	"from_date missing and limit missing"
edge_006	block_003 -> block_006	null
edge_007	block_004 -> block_007	null
edge_008	block_005 -> block_008	null
edge_009	block_006 -> block_010	null
edge_010	block_007 -> block_011	null
edge_011	block_008 -> block_012	null
edge_012	block_009 -> block_013	null

全部操作、数据引用与约束落点

```
block_001 Read the user-supplied request.json

data_source_kind = "external"
block_001.constraints: []

ir_001 read_request_json

inputs[0] external_resource: request.json | request.json
outputs[0] result: result_001 | request_data
ir_001.constraints: []
draft_instruction_id = read_request_json_instr

ir_002 dispatch

inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_inspect_request

block_002 Inspect request fields and optional parameter presence

data_source_kind = null
block_002.constraints: []

ir_003 extract_request_term

inputs[0] result: result_001 | request_data
outputs[0] result: result_002 | request_term
```

```
ir_003.constraints: []

draft_instruction_id = extract_request_term_instr

ir_004 check_from_date_presence

inputs[0] result: result_001 | request_data
outputs[0] result: result_003 | from_date_present
ir_004.constraints: []

draft_instruction_id = check_from_date_presence_instr

ir_005 check_limit_presence

inputs[0] result: result_001 | request_data
outputs[0] result: result_004 | limit_present
ir_005.constraints: []

draft_instruction_id = check_limit_presence_instr

ir_006 dispatch

inputs[0] result: result_003 | from_date_present
inputs[1] result: result_004 | limit_present
outputs = []

ir_006.constraints: []

draft_instruction_id = dispatch_on_optional_presence

block_003 Extract from_date and limit values

data_source_kind = null

block_003.constraints: []

ir_007 extract_from_date_value

inputs[0] result: result_001 | request_data
outputs[0] result: result_005 | from_date_value_both_present
ir_007.constraints: []

draft_instruction_id = extract_from_date_both

ir_008 extract_limit_value

inputs[0] result: result_001 | request_data
outputs[0] result: result_006 | limit_value_both_present
ir_008.constraints: []

draft_instruction_id = extract_limit_both

ir_009 dispatch

inputs = []
outputs = []

ir_009.constraints: []

draft_instruction_id = go_call_both
```

block_004 Extract from_date value

data_source_kind = null

block_004.constraints: []

ir_010 extract_from_date_value

inputs[0] result: result_001 | request_data

outputs[0] result: result_007 | from_date_value_from_only

ir_010.constraints: []

draft_instruction_id = extract_from_date_from_only

ir_011 dispatch

inputs = []

outputs = []

ir_011.constraints: []

draft_instruction_id = go_call_from_only

block_005 Extract limit value

data_source_kind = null

block_005.constraints: []

ir_012 extract_limit_value

inputs[0] result: result_001 | request_data

outputs[0] result: result_008 | limit_value_limit_only

ir_012.constraints: []

draft_instruction_id = extract_limit_limit_only

ir_013 dispatch

inputs = []

outputs = []

ir_013.constraints: []

draft_instruction_id = go_call_limit_only

block_006 Call index.search with from_date and limit

data_source_kind = "external"

block_006.constraints: []

ir_014 index.search

inputs[0] external_resource: index.search | index.search

inputs[1] result: result_002 | request_term

inputs[2] result: result_005 | from_date_value_both_present

inputs[3] result: result_006 | limit_value_both_present

outputs[0] result: result_009 | search_response_both_present

ir_014.constraints: []

draft_instruction_id = call_index_search_both

ir_015 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_015.constraints: []
```

```
draft_instruction_id = go_post_both
```

block_007 Call index.search with from_date only

```
data_source_kind = "external"
```

```
block_007.constraints: []
```

ir_016 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_002 | request_term
```

```
inputs[2] result: result_007 | from_date_value_from_only
```

```
outputs[0] result: result_010 | search_response_from_only
```

```
ir_016.constraints: []
```

```
draft_instruction_id = call_index_search_from_only
```

ir_017 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_017.constraints: []
```

```
draft_instruction_id = go_post_from_only
```

block_008 Call index.search with limit only

```
data_source_kind = "external"
```

```
block_008.constraints: []
```

ir_018 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_002 | request_term
```

```
inputs[2] result: result_008 | limit_value_limit_only
```

```
outputs[0] result: result_011 | search_response_limit_only
```

```
ir_018.constraints: []
```

```
draft_instruction_id = call_index_search_limit_only
```

ir_019 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_019.constraints: []
```

```
draft_instruction_id = go_post_limit_only
```

block_009 Call index.search with neither optional argument

```
data_source_kind = "external"
```

```
block_009.constraints: []
```

ir_020 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_002 | request_term
```

```
outputs[0] result: result_012 | search_response_neither_present
```

```
ir_020.constraints: []
```

```
draft_instruction_id = call_index_search_neither
```

ir_021 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_021.constraints: []
```

```
draft_instruction_id = go_post_neither
```

block_010 Write count and return items

```
data_source_kind = null
```

```
block_010.constraints: []
```

ir_022 extract_response_total

```
inputs[0] result: result_009 | search_response_both_present
```

```
outputs[0] result: result_013 | total_both_present
```

```
ir_022.constraints: []
```

```
draft_instruction_id = extract_total_both
```

ir_023 write_total_to_count_file

```
inputs[0] result: result_013 | total_both_present
```

```
inputs[1] external_resource: count.txt | count.txt
```

```
outputs = []
```

```
ir_023.constraints: []
```

```
draft_instruction_id = write_count_both
```

ir_024 extract_response_items

```
inputs[0] result: result_009 | search_response_both_present
```

```
outputs[0] result: result_014 | items_both_present
```

```
ir_024.constraints: []
```

```
draft_instruction_id = extract_items_both
```

ir_025 return

```
inputs[0] result: result_014 | items_both_present
```

```
outputs = []
```

```
ir_025.constraints: []
```

```
draft_instruction_id = return_items_both
```

block_011 Write count and return items

data_source_kind = null

block_011.constraints: []

ir_026 extract_response_total

inputs[0] result: result_010 | search_response_from_only

outputs[0] result: result_015 | total_from_only

ir_026.constraints: []

draft_instruction_id = extract_total_from_only

ir_027 write_total_to_count_file

inputs[0] result: result_015 | total_from_only

inputs[1] external_resource: count.txt | count.txt

outputs = []

ir_027.constraints: []

draft_instruction_id = write_count_from_only

ir_028 extract_response_items

inputs[0] result: result_010 | search_response_from_only

outputs[0] result: result_016 | items_from_only

ir_028.constraints: []

draft_instruction_id = extract_items_from_only

ir_029 return

inputs[0] result: result_016 | items_from_only

outputs = []

ir_029.constraints: []

draft_instruction_id = return_items_from_only

block_012 Write count and return items

data_source_kind = null

block_012.constraints: []

ir_030 extract_response_total

inputs[0] result: result_011 | search_response_limit_only

outputs[0] result: result_017 | total_limit_only

ir_030.constraints: []

draft_instruction_id = extract_total_limit_only

ir_031 write_total_to_count_file

inputs[0] result: result_017 | total_limit_only

inputs[1] external_resource: count.txt | count.txt

outputs = []

ir_031.constraints: []

draft_instruction_id = write_count_limit_only

ir_032 extract_response_items

```
inputs[0] result: result_011 | search_response_limit_only
```

```
outputs[0] result: result_018 | items_limit_only
```

```
ir_032.constraints: []
```

```
draft_instruction_id = extract_items_limit_only
```

ir_033 return

```
inputs[0] result: result_018 | items_limit_only
```

```
outputs = []
```

```
ir_033.constraints: []
```

```
draft_instruction_id = return_items_limit_only
```

block_013 Write count and return items

```
data_source_kind = null
```

```
block_013.constraints: []
```

ir_034 extract_response_total

```
inputs[0] result: result_012 | search_response_neither_present
```

```
outputs[0] result: result_019 | total_neither_present
```

```
ir_034.constraints: []
```

```
draft_instruction_id = extract_total_neither
```

ir_035 write_total_to_count_file

```
inputs[0] result: result_019 | total_neither_present
```

```
inputs[1] external_resource: count.txt | count.txt
```

```
outputs = []
```

```
ir_035.constraints: []
```

```
draft_instruction_id = write_count_neither
```

ir_036 extract_response_items

```
inputs[0] result: result_012 | search_response_neither_present
```

```
outputs[0] result: result_020 | items_neither_present
```

```
ir_036.constraints: []
```

```
draft_instruction_id = extract_items_neither
```

ir_037 return

```
inputs[0] result: result_020 | items_neither_present
```

```
outputs = []
```

```
ir_037.constraints: []
```

```
draft_instruction_id = return_items_neither
```

Q02 / 第 1 次 实际输出明细

查询调用与参数使用 · 连贯段落

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q02/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

declared_context_keys = []

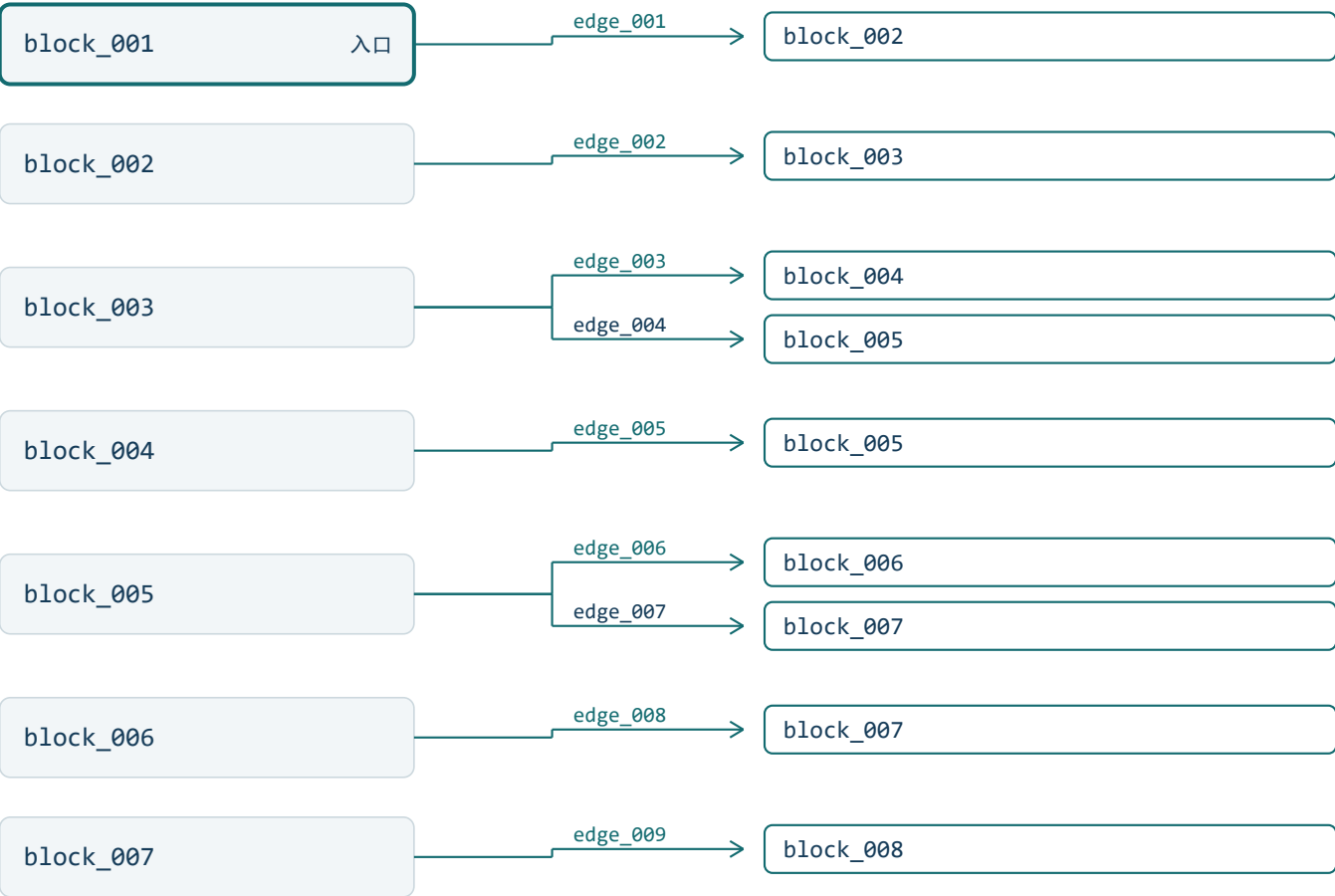
skill.constraints[0]: Do not invoke index.delete anywhere in this workflow.

skill.constraints[1]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.

skill.constraints[2]: index.search supports fuzzy matching; this note does not request fuzzy expansion in this workflow.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



block_008

无出边

完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"from_date present"
edge_004	block_003 -> block_005	"from_date missing"
edge_005	block_004 -> block_005	null
edge_006	block_005 -> block_006	"limit present"
edge_007	block_005 -> block_007	"limit missing"
edge_008	block_006 -> block_007	null
edge_009	block_007 -> block_008	null

全部操作、数据引用与约束落点

```
block_001 Read the user-supplied request.json

data_source_kind = "external"
block_001.constraints: []

ir_001 read_request_json

inputs[0] external_resource: request.json | request.json
outputs[0] result: result_001 | raw_request_json
ir_001.constraints: []
draft_instruction_id = read_request_json

ir_002 dispatch

inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_parse_request

block_002 Parse request fields and presence flags

data_source_kind = null
block_002.constraints: []

ir_003 parse_request_json

inputs[0] result: result_001 | raw_request_json
outputs[0] result: result_002 | request_term
outputs[1] result: result_003 | has_from_date
```

```
outputs[2] result: result_004 | has_limit
ir_003.constraints: []
draft_instruction_id = parse_request_json

ir_004 dispatch

inputs = []
outputs = []
ir_004.constraints: []
draft_instruction_id = go_decide_from_date

block_003 Check whether from_date is present

data_source_kind = null
block_003.constraints: []

ir_005 dispatch

inputs[0] result: result_003 | has_from_date
outputs = []
ir_005.constraints: []
draft_instruction_id = choose_from_date_path

block_004 Extract the from_date value

data_source_kind = null
block_004.constraints: []

ir_006 extract_request_from_date

inputs[0] result: result_001 | raw_request_json
outputs[0] result: result_005 | request_from_date
ir_006.constraints: []
draft_instruction_id = extract_request_from_date

ir_007 dispatch

inputs = []
outputs = []
ir_007.constraints: []
draft_instruction_id = go_decide_limit_from_date

block_005 Check whether limit is present

data_source_kind = null
block_005.constraints: []

ir_008 dispatch

inputs[0] result: result_004 | has_limit
outputs = []
ir_008.constraints: []
draft_instruction_id = choose_limit_path
```

block_006 Extract the limit value

data_source_kind = null

block_006.constraints: []

ir_009 extract_request_limit

inputs[0] result: result_001 | raw_request_json

outputs[0] result: result_006 | request_limit

ir_009.constraints: []

draft_instruction_id = extract_request_limit

ir_010 dispatch

inputs = []

outputs = []

ir_010.constraints: []

draft_instruction_id = go_search_from_limit

block_007 Call index.search with the request parameters

data_source_kind = "external"

block_007.constraints: []

ir_011 index.search

inputs[0] external_resource: index.search | index.search

inputs[1] result: result_002 | request_term

inputs[2] result: result_005 | request_from_date

inputs[3] result: result_006 | request_limit

outputs[0] result: result_007 | search_response

ir_011.constraints[0]: Call index.search exactly once.

ir_011.constraints[1]: Use request.term unchanged as its query argument.

ir_011.constraints[2]: When from_date is present, pass its value unchanged; the accepted format is YYYY-MM-DD.

ir_011.constraints[3]: When from_date is missing, omit the from_date argument.

ir_011.constraints[4]: When limit is present, pass its value unchanged as the limit argument.

ir_011.constraints[5]: When limit is missing, omit the limit argument.

draft_instruction_id = call_index_search

ir_012 dispatch

inputs = []

outputs = []

ir_012.constraints: []

draft_instruction_id = go_postprocess

block_008 Write total to count.txt and return items

data_source_kind = null

block_008.constraints: []

ir_013 extract_response_total

inputs[0] result: result_007 | search_response

```
outputs[0] result: result_008 | response_total
ir_013.constraints: []
draft_instruction_id = extract_response_total

ir_014 extract_response_items

inputs[0] result: result_007 | search_response
outputs[0] result: result_009 | response_items
ir_014.constraints: []
draft_instruction_id = extract_response_items

ir_015 write_count_file

inputs[0] external_resource: count.txt | count.txt
inputs[1] result: result_008 | response_total
outputs = []
ir_015.constraints: []
draft_instruction_id = write_count_file

ir_016 return

inputs[0] result: result_009 | response_items
outputs = []
ir_016.constraints: []
draft_instruction_id = return_items
```

Q02 / 第 2 次 实际输出明细

查询调用与参数使用 · 连贯段落

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q02/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

```
declared_context_keys = []

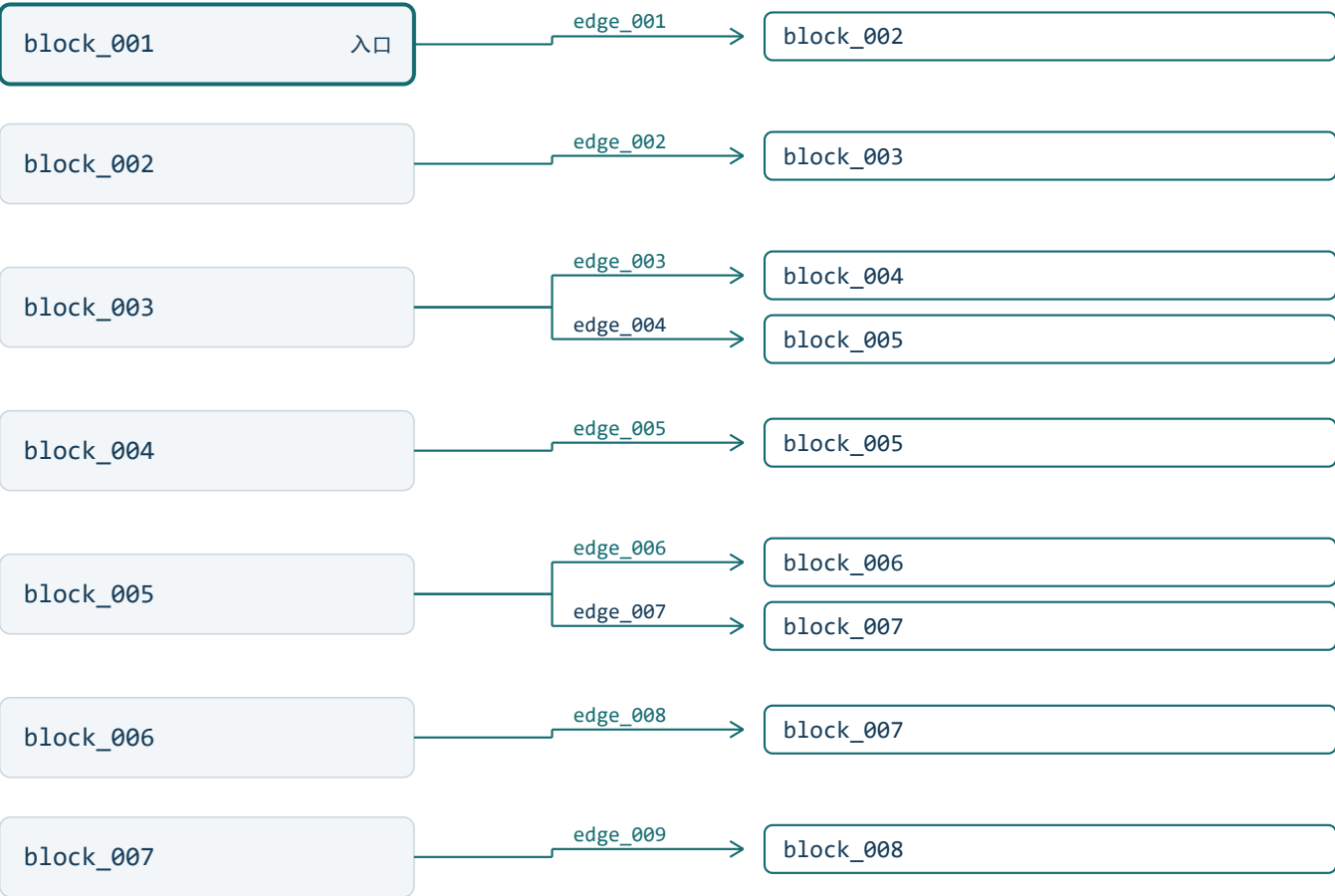
skill.constraints[0]: Capability note: index.search supports fuzzy matching; this note does not request fuzzy expansion in this workflow.

skill.constraints[1]: Do not invoke index.delete anywhere in this workflow.

skill.constraints[2]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



block_008

无出边

完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"from_date is present"
edge_004	block_003 -> block_005	"from_date is missing"
edge_005	block_004 -> block_005	null
edge_006	block_005 -> block_006	"limit is present"
edge_007	block_005 -> block_007	"limit is missing"
edge_008	block_006 -> block_007	null
edge_009	block_007 -> block_008	null

全部操作、数据引用与约束落点

block_001 Read request.json

```
data_source_kind = "external"

block_001.constraints[0]: Read the user-supplied request.json, which contains term and may contain from_date and limit.
```

ir_001 read_request_json

```
inputs[0] external_resource: request.json | request.json
outputs[0] result: result_001 | request_data
ir_001.constraints: []
draft_instruction_id = read_request_json
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = dispatch_to_extract_term
```

block_002 Extract request term

```
data_source_kind = null

block_002.constraints: []
```

ir_003 extract_request_term

```
inputs[0] result: result_001 | request_data
outputs[0] result: result_002 | request_term
ir_003.constraints: []
```

```
draft_instruction_id = extract_request_term

ir_004 dispatch

inputs = []
outputs = []
ir_004.constraints: []
draft_instruction_id = dispatch_to_check_from_date

block_003 Check whether from_date is present

data_source_kind = null
block_003.constraints: []

ir_005 check_from_date_presence

inputs[0] result: result_001 | request_data
outputs[0] result: result_003 | from_date_present
ir_005.constraints: []
draft_instruction_id = check_from_date_presence

ir_006 dispatch

inputs[0] result: result_003 | from_date_present
outputs = []
ir_006.constraints: []
draft_instruction_id = dispatch_from_date

block_004 Extract from_date value

data_source_kind = null
block_004.constraints: []

ir_007 extract_from_date_value

inputs[0] result: result_001 | request_data
outputs[0] result: result_004 | from_date_value
ir_007.constraints: []
draft_instruction_id = extract_from_date_value

ir_008 dispatch

inputs = []
outputs = []
ir_008.constraints: []
draft_instruction_id = dispatch_to_check_limit

block_005 Check whether limit is present

data_source_kind = null
block_005.constraints: []

ir_009 check_limit_presence

inputs[0] result: result_001 | request_data
outputs[0] result: result_005 | limit_present
ir_009.constraints: []
```

```
draft_instruction_id = check_limit_presence
```

ir_010 dispatch

```
inputs[0] result: result_005 | limit_present
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_limit
```

block_006 Extract limit value

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 extract_limit_value

```
inputs[0] result: result_001 | request_data
```

```
outputs[0] result: result_006 | limit_value
```

```
ir_011.constraints: []
```

```
draft_instruction_id = extract_limit_value
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_to_index_search
```

block_007 Call index.search with optional parameters

```
data_source_kind = "external"
```

```
block_007.constraints: []
```

ir_013 index_search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_002 | request_term
```

```
inputs[2] result: result_003 | from_date_present
```

```
inputs[3] result: result_004 | from_date_value
```

```
inputs[4] result: result_005 | limit_present
```

```
inputs[5] result: result_006 | limit_value
```

```
outputs[0] result: result_007 | search_response
```

```
ir_013.constraints[0]: Call index.search exactly once, using request.term unchanged as its query argument.
```

```
ir_013.constraints[1]: When from_date is present, pass its value unchanged; the accepted format is YYYY-MM-DD.
```

```
ir_013.constraints[2]: When from_date is missing, omit the from_date argument.
```

```
ir_013.constraints[3]: When limit is present, pass its value unchanged as the limit argument.
```

```
ir_013.constraints[4]: When limit is missing, omit the limit argument.
```

```
draft_instruction_id = index_search
```


ir_014 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = dispatch_to_write_and_return
```

block_008 Write total to count.txt and return items

```
data_source_kind = null
```

```
block_008.constraints[0]: After the search, write the response's total value to local count.txt.
```

```
block_008.constraints[1]: Return the search response's items value unchanged.
```

ir_015 extract_search_total

```
inputs[0] result: result_007 | search_response
```

```
outputs[0] result: result_008 | search_total
```

```
ir_015.constraints: []
```

```
draft_instruction_id = extract_search_total
```

ir_016 extract_search_items

```
inputs[0] result: result_007 | search_response
```

```
outputs[0] result: result_009 | search_items
```

```
ir_016.constraints: []
```

```
draft_instruction_id = extract_search_items
```

ir_017 write_count_to_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_008 | search_total
```

```
outputs = []
```

```
ir_017.constraints: []
```

```
draft_instruction_id = write_count_to_file
```

ir_018 return

```
inputs[0] result: result_009 | search_items
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = return_items
```

Q02 / 第 3 次 实际输出明细

查询调用与参数使用 · 连贯段落

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q02/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: request.json is read as an external file resource; the package declares no runtime context keys.

candidate warning: Presence checks for from_date and limit implement the explicit pass-when-present / omit-when-missing argument semantics; no pre-call format validation or normalization is modeled.

Skill 层约束与上下文声明

```
declared_context_keys = []

skill.constraints[0]: Capability note: index.search supports fuzzy matching; this note does not request fuzzy expansion in this workflow.

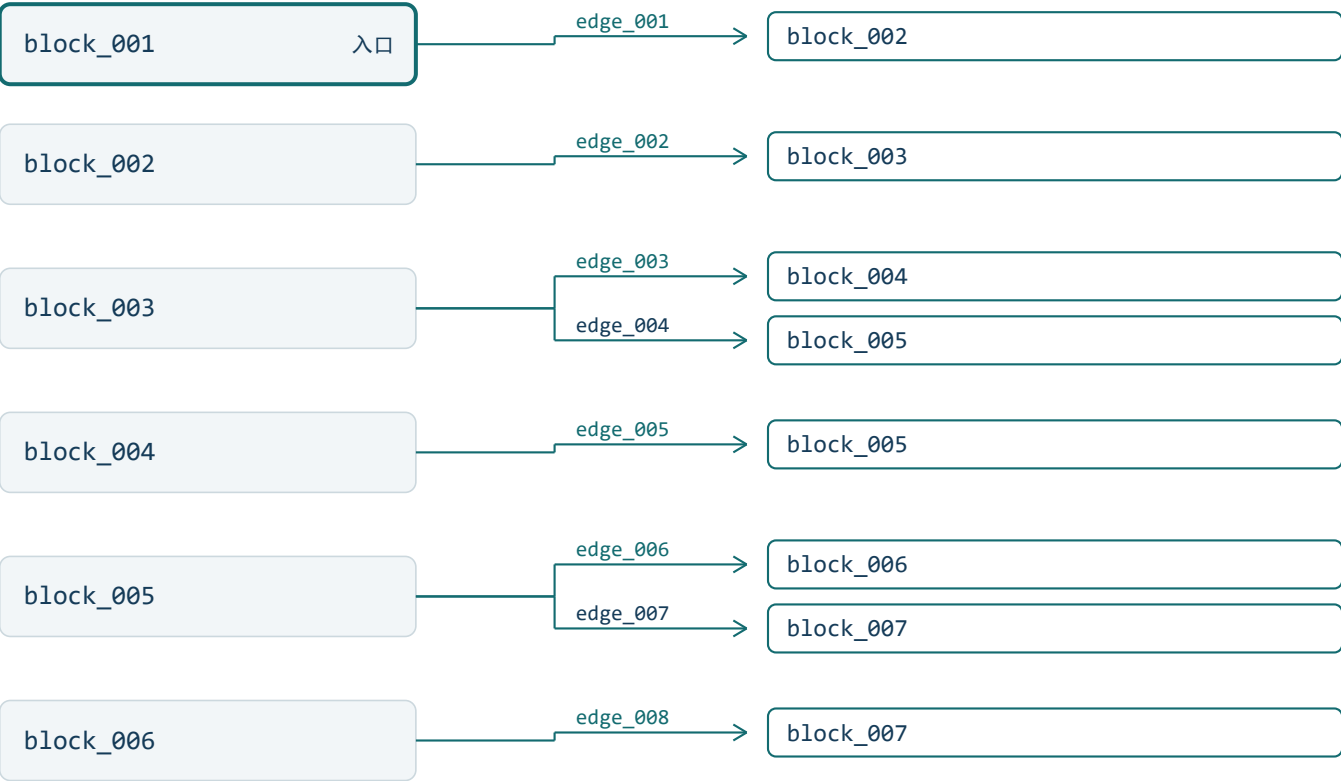
skill.constraints[1]: Call index.search exactly once.

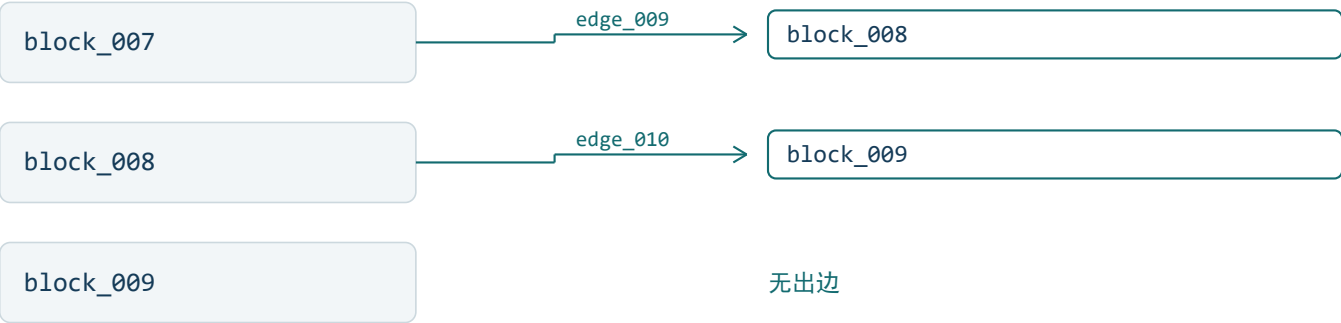
skill.constraints[2]: Do not invoke index.delete anywhere in this workflow.

skill.constraints[3]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"from_date is present"
edge_004	block_003 -> block_005	"from_date is missing"
edge_005	block_004 -> block_005	null
edge_006	block_005 -> block_006	"limit is present"
edge_007	block_005 -> block_007	"limit is missing"
edge_008	block_006 -> block_007	null
edge_009	block_007 -> block_008	null
edge_010	block_008 -> block_009	null

全部操作、数据引用与约束落点

block_001 Read the user-supplied request.json

```
data_source_kind = "external"

block_001.constraints: []

ir_001 read_request_json

inputs[0] external_resource: request.json | request.json
outputs[0] result: result_001 | request

ir_001.constraints: []

draft_instruction_id = read_request_json

ir_002 dispatch

inputs = []
outputs = []

ir_002.constraints: []

draft_instruction_id = go_extract_term
```

block_002 Extract the search term from the request

data_source_kind = null

block_002.constraints: []

ir_003 extract_request_term

inputs[0] result: result_001 | request

outputs[0] result: result_002 | request_term

ir_003.constraints: []

draft_instruction_id = extract_term

ir_004 dispatch

inputs = []

outputs = []

ir_004.constraints: []

draft_instruction_id = go_check_from_date

block_003 Check whether from_date is present in the request

data_source_kind = null

block_003.constraints: []

ir_005 check_from_date_present

inputs[0] result: result_001 | request

outputs[0] result: result_003 | from_date_present

ir_005.constraints: []

draft_instruction_id = check_from_date

ir_006 dispatch

inputs[0] result: result_003 | from_date_present

outputs = []

ir_006.constraints: []

draft_instruction_id = go_after_from_date

block_004 Extract the from_date value from the request

data_source_kind = null

block_004.constraints: []

ir_007 extract_request_from_date

inputs[0] result: result_001 | request

outputs[0] result: result_004 | from_date_value

ir_007.constraints: []

draft_instruction_id = extract_from_date

ir_008 dispatch

inputs = []

outputs = []

ir_008.constraints: []

draft_instruction_id = go_after_from_date_extraction

block_005 Check whether limit is present in the request

data_source_kind = null

block_005.constraints: []

ir_009 check_limit_present

inputs[0] result: result_001 | request

outputs[0] result: result_005 | limit_present

ir_009.constraints: []

draft_instruction_id = check_limit

ir_010 dispatch

inputs[0] result: result_005 | limit_present

outputs = []

ir_010.constraints: []

draft_instruction_id = go_after_limit

block_006 Extract the limit value from the request

data_source_kind = null

block_006.constraints: []

ir_011 extract_request_limit

inputs[0] result: result_001 | request

outputs[0] result: result_006 | limit_value

ir_011.constraints: []

draft_instruction_id = extract_limit

ir_012 dispatch

inputs = []

outputs = []

ir_012.constraints: []

draft_instruction_id = go_after_limit_extraction

block_007 Call index.search with the request term and optional from_date and limit

data_source_kind = "external"

block_007.constraints: []

ir_013 index_search

inputs[0] external_resource: index.search | index.search

inputs[1] result: result_002 | request_term

inputs[2] result: result_004 | from_date_value

inputs[3] result: result_006 | limit_value

outputs[0] result: result_007 | search_response

ir_013.constraints[0]: Use request.term unchanged as its query argument.

ir_013.constraints[1]: When from_date is present, pass its value unchanged; the accepted format is YYYY-MM-DD.

ir_013.constraints[2]: When from_date is missing, omit the from_date argument.

ir_013.constraints[3]: When limit is present, pass its value unchanged as the limit argument.

```
ir_013.constraints[4]: When limit is missing, omit the limit argument.
```

```
draft_instruction_id = call_index_search
```

ir_014 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = go_write_count
```

block_008 Write the response total to count.txt

```
data_source_kind = null
```

```
block_008.constraints: []
```

ir_015 extract_response_total

```
inputs[0] result: result_007 | search_response
```

```
outputs[0] result: result_008 | response_total
```

```
ir_015.constraints: []
```

```
draft_instruction_id = extract_total
```

ir_016 write_total_to_count_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_008 | response_total
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = write_count
```

ir_017 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_017.constraints: []
```

```
draft_instruction_id = go_return_items
```

block_009 Return the response items

```
data_source_kind = null
```

```
block_009.constraints: []
```

ir_018 extract_response_items

```
inputs[0] result: result_007 | search_response
```

```
outputs[0] result: result_009 | response_items
```

```
ir_018.constraints: []
```

```
draft_instruction_id = extract_items
```

ir_019 return

```
inputs[0] result: result_009 | response_items
```

```
outputs = []
```

```
ir_019.constraints: []
```

```
draft_instruction_id = return_search_items
```

Q03 / 第 1 次 实际输出明细

查询调用与参数使用 · 表格

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q03/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

declared_context_keys = []

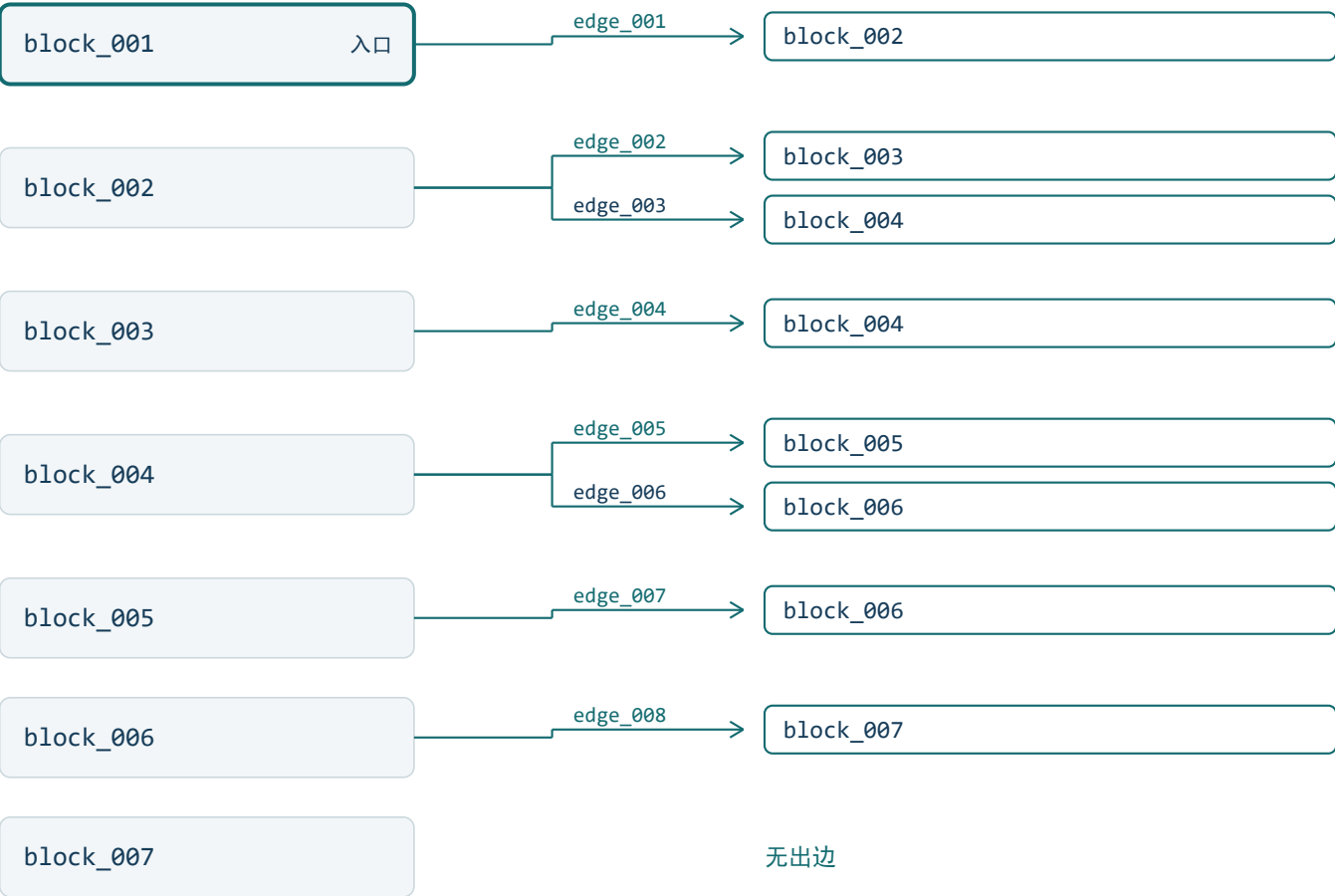
skill.constraints[0]: Do not invoke index.delete anywhere in this workflow.

skill.constraints[1]: index.search supports fuzzy matching; this note does not request fuzzy expansion in this workflow.

skill.constraints[2]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"from_date present"
edge_003	block_002 -> block_004	"from_date missing"
edge_004	block_003 -> block_004	null
edge_005	block_004 -> block_005	"limit present"
edge_006	block_004 -> block_006	"limit missing"
edge_007	block_005 -> block_006	null
edge_008	block_006 -> block_007	null

全部操作、数据引用与约束落点

```
block_001 Read request.json

data_source_kind = "external"
block_001.constraints: []

ir_001 read_file

inputs[0] external_resource: request.json | request.json
outputs[0] result: result_001 | request_data
ir_001.constraints: []
draft_instruction_id = read_request_file

ir_002 dispatch

inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_parse_request

block_002 Extract term and check from_date presence

data_source_kind = null
block_002.constraints: []

ir_003 extract_request_term

inputs[0] result: result_001 | request_data
outputs[0] result: result_002 | term
ir_003.constraints: []
draft_instruction_id = extract_term

ir_004 check_request_from_date_presence

inputs[0] result: result_001 | request_data
outputs[0] result: result_003 | from_date_present
```



```
ir_004.constraints: []
draft_instruction_id = check_from_date_present

ir_005 dispatch

inputs[0] result: result_003 | from_date_present
outputs = []
ir_005.constraints: []
draft_instruction_id = branch_on_from_date

block_003 Extract from_date value when present

data_source_kind = null
block_003.constraints: []

ir_006 extract_request_from_date

inputs[0] result: result_001 | request_data
outputs[0] result: result_004 | from_date
ir_006.constraints: []
draft_instruction_id = extract_from_date_value

ir_007 dispatch

inputs = []
outputs = []
ir_007.constraints: []
draft_instruction_id = go_check_limit

block_004 Check limit presence

data_source_kind = null
block_004.constraints: []

ir_008 check_request_limit_presence

inputs[0] result: result_001 | request_data
outputs[0] result: result_005 | limit_present
ir_008.constraints: []
draft_instruction_id = check_limit_present

ir_009 dispatch

inputs[0] result: result_005 | limit_present
outputs = []
ir_009.constraints: []
draft_instruction_id = branch_on_limit

block_005 Extract limit value when present

data_source_kind = null
block_005.constraints: []

ir_010 extract_request_limit

inputs[0] result: result_001 | request_data
outputs[0] result: result_006 | limit
```

```
ir_010.constraints: []
```

```
draft_instruction_id = extract_limit_value
```

ir_011 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = go_search
```

block_006 Call index.search with required and optional arguments

```
data_source_kind = "external"
```

```
block_006.constraints: []
```

ir_012 index.search

```
inputs[0] external_resource: index | index
```

```
inputs[1] result: result_002 | term
```

```
inputs[2] result: result_004 | from_date
```

```
inputs[3] result: result_006 | limit
```

```
outputs[0] result: result_007 | search_response
```

```
ir_012.constraints[0]: Call index.search exactly once.
```

```
ir_012.constraints[1]: Use request.term unchanged as the query argument.
```

```
ir_012.constraints[2]: When from_date is present, pass its value unchanged; the accepted format is YYYY-MM-DD.  
When from_date is missing, omit the from_date argument.
```

```
ir_012.constraints[3]: When limit is present, pass its value unchanged as the limit argument. When limit is  
missing, omit the limit argument.
```

```
draft_instruction_id = call_index_search
```

ir_013 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = go_write_return
```

block_007 Write total to count.txt and return items

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_014 extract_response_total

```
inputs[0] result: result_007 | search_response
```

```
outputs[0] result: result_008 | search_total
```

```
ir_014.constraints: []
```

```
draft_instruction_id = extract_total
```

ir_015 extract_response_items

```
inputs[0] result: result_007 | search_response
```

```
outputs[0] result: result_009 | search_items
```

```
ir_015.constraints: []
```

```
draft_instruction_id = extract_items
```

ir_016 write_total_to_count_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_008 | search_total
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = write_count
```

ir_017 return

```
inputs[0] result: result_009 | search_items
```

```
outputs = []
```

```
ir_017.constraints: []
```

```
draft_instruction_id = return_items
```

Q03 / 第 2 次 实际输出明细

查询调用与参数使用 · 表格

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q03/deepseek-v4-flash-max/2

结构状态: error | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: 未记录 | HTTP 1 (200): deepseek-flash

诊断与失败边界

IncompleteRead(0 bytes read)

流式请求的远端最终结果未知: 生成 1 / HTTP 尝试 1 / 已接收 HTTP 200

已收到成功响应头后，SSE 未达到完整结束条件；本地 error 不代表远端未生成结果，也不代表模型语义失败。未完成的流不能作为有效响应或有效 CFG，远端用量和最终处理结果以原始记录为准。

本次没有已接受的实际 CFG；不能补画或将候选当作有效图。原始响应 / 候选保存在同目录 trace.json、candidate.json 和 analysis.json（如生成）。

Q03 / 第 3 次 实际输出明细

查询调用与参数使用 · 表格

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q03/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

```
declared_context_keys = []

skill.constraints[0]: Call index.search exactly once, using request.term unchanged as its query argument.

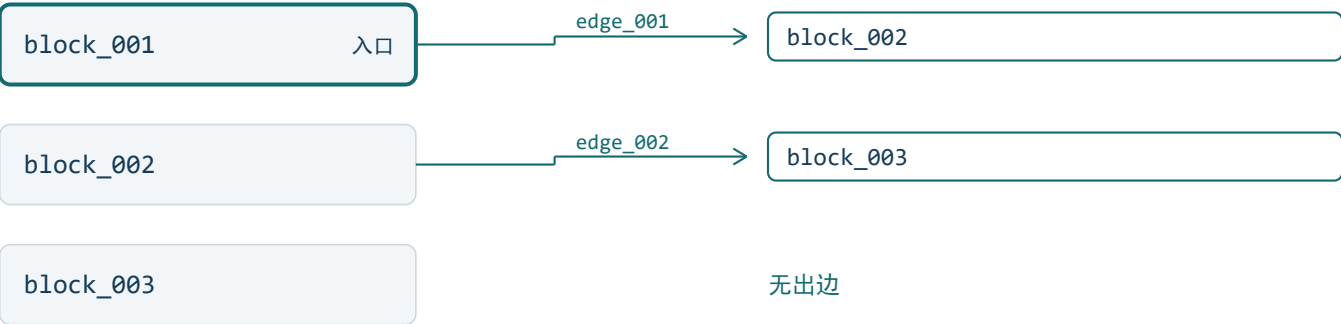
skill.constraints[1]: Do not invoke index.delete anywhere in this workflow.

skill.constraints[2]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.

skill.constraints[3]: Capability note: index.search supports fuzzy matching; this note does not request fuzzy expansion in this workflow.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null

全部操作、数据引用与约束落点

```
block_001 Read the user-supplied request.json

data_source_kind = "external"

block_001.constraints: []

ir_001 read_request_json

inputs[0] external_resource: request.json | request.json
```

```
outputs[0] result: result_001 | request_term
outputs[1] result: result_002 | request_from_date
outputs[2] result: result_003 | request_limit
ir_001.constraints: []
draft_instruction_id = read_request_json
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_to_search
```

block_002 Call index.search with the request parameters

```
data_source_kind = "external"
block_002.constraints: []
```

ir_003 index.search

```
inputs[0] external_resource: index | index
inputs[1] result: result_001 | request_term
inputs[2] result: result_002 | request_from_date
inputs[3] result: result_003 | request_limit
outputs[0] result: result_004 | search_response_total
outputs[1] result: result_005 | search_response_items
ir_003.constraints[0]: Use request.term unchanged as the query argument.
ir_003.constraints[1]: When from_date is present, pass its value unchanged; the accepted format is YYYY-MM-DD.
When from_date is missing, omit the from_date argument.
ir_003.constraints[2]: When limit is present, pass its value unchanged as the limit argument. When limit is
missing, omit the limit argument.
draft_instruction_id = search_index
```

ir_004 dispatch

```
inputs = []
outputs = []
ir_004.constraints: []
draft_instruction_id = go_to_write
```

block_003 Write the response total to count.txt and return the items

```
data_source_kind = null
block_003.constraints: []
```

ir_005 write_total_to_count_file

```
inputs[0] result: result_004 | search_response_total
inputs[1] external_resource: count.txt | count.txt
outputs = []
ir_005.constraints: []
draft_instruction_id = write_count
```

ir_006 return

```
inputs[0] result: result_005 | search_response_items
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = return_items
```

Q04 / 第 1 次 实际输出明细

查询调用与参数使用 · 跨文件混合

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q04/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Optional from_date and limit parameters were represented as four mutually exclusive branches to preserve the omit-when-missing semantics.

candidate warning: query.yaml's missing_arguments settings were applied as constraints on the index.search calls.

candidate warning: The fuzzy-matching capability note was treated as a non-operational constraint and did not introduce fuzzy expansion.

Skill 层约束与上下文声明

```
declared_context_keys = []

skill.constraints[0]: Call index.search exactly once.

skill.constraints[1]: Do not invoke index.delete anywhere in this workflow.

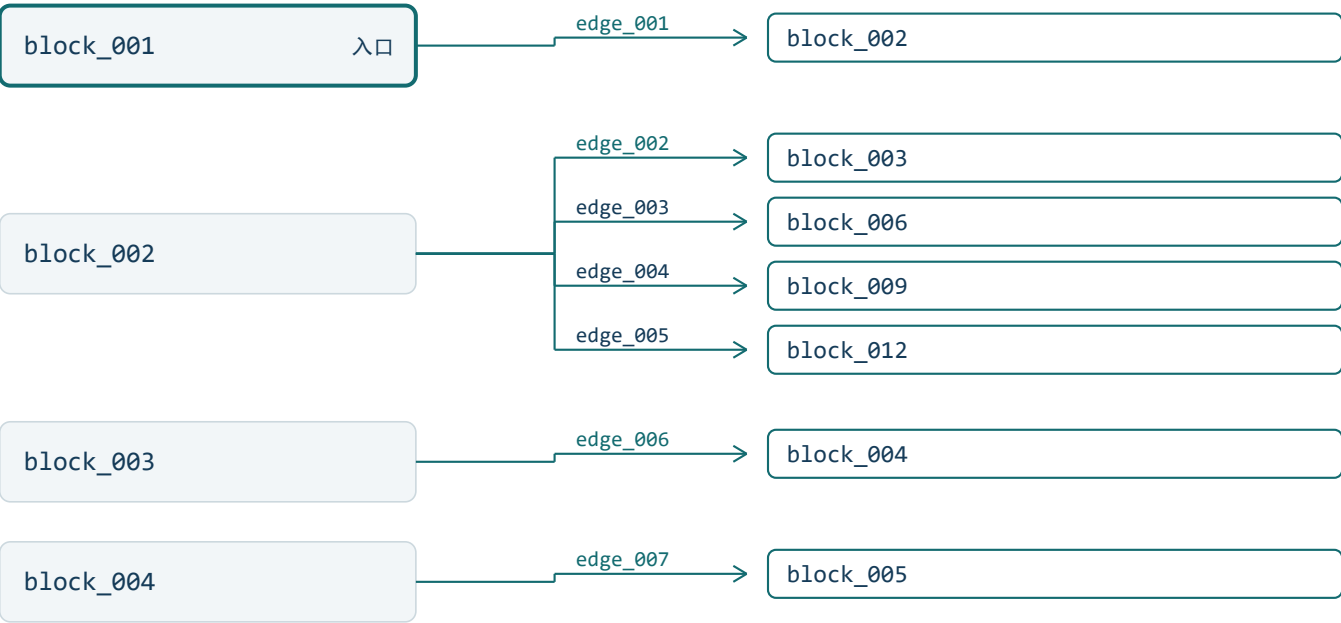
skill.constraints[2]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.

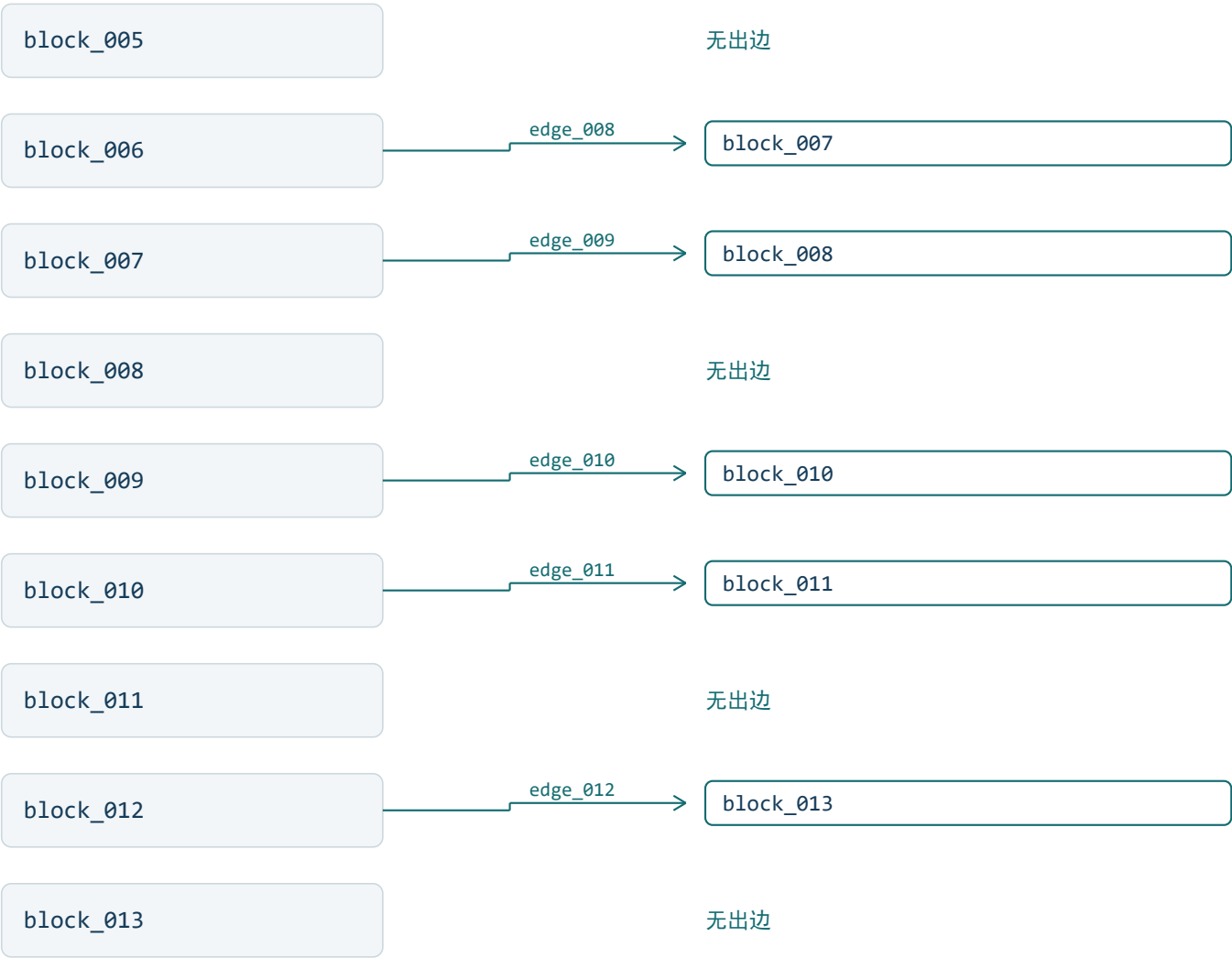
skill.constraints[3]: Capability note: index.search supports fuzzy matching; this note does not request fuzzy expansion in this workflow.

skill.constraints[4]: Apply the missing-argument settings in query.yaml to the index.search call: from_date: omit; limit: omit.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"from_date present and limit present"
edge_003	block_002 -> block_006	"from_date present and limit missing"
edge_004	block_002 -> block_009	"from_date missing and limit present"
edge_005	block_002 -> block_012	"from_date missing and limit missing"
edge_006	block_003 -> block_004	null
edge_007	block_004 -> block_005	null
edge_008	block_006 -> block_007	null
edge_009	block_007 -> block_008	null
edge_010	block_009 -> block_010	null
edge_011	block_010 -> block_011	null
edge_012	block_012 -> block_013	null

全部操作、数据引用与约束落点

block_001 Read the user-supplied request.json

```
data_source_kind = "external"

block_001.constraints[0]: Read the user-supplied request.json, which contains term and may contain from_date and limit.

ir_001 read_request_json

inputs[0] external_resource: request.json | request.json
outputs[0] result: result_001 | request_document

ir_001.constraints: []

draft_instruction_id = read_request_json
```

ir_002 dispatch

```
inputs = []
outputs = []

ir_002.constraints: []

draft_instruction_id = go_inspect_request
```

block_002 Extract term and check optional parameter presence

```
data_source_kind = null

block_002.constraints: []
```

ir_003 extract_request_term

```
inputs[0] result: result_001 | request_document
```

```
outputs[0] result: result_002 | request_term
ir_003.constraints: []
draft_instruction_id = extract_request_term

ir_004 check_from_date_present

inputs[0] result: result_001 | request_document
outputs[0] result: result_003 | from_date_present
ir_004.constraints: []
draft_instruction_id = check_from_date_present

ir_005 check_limit_present

inputs[0] result: result_001 | request_document
outputs[0] result: result_004 | limit_present
ir_005.constraints: []
draft_instruction_id = check_limit_present

ir_006 dispatch

inputs[0] result: result_003 | from_date_present
inputs[1] result: result_004 | limit_present
outputs = []
ir_006.constraints: []
draft_instruction_id = dispatch_optional_params

block_003 Extract present from_date and limit values

data_source_kind = null
block_003.constraints: []

ir_007 extract_from_date_value

inputs[0] result: result_001 | request_document
outputs[0] result: result_005 | from_date_value_both
ir_007.constraints: []
draft_instruction_id = extract_from_date_value_both

ir_008 extract_limit_value

inputs[0] result: result_001 | request_document
outputs[0] result: result_006 | limit_value_both
ir_008.constraints: []
draft_instruction_id = extract_limit_value_both

ir_009 dispatch

inputs = []
outputs = []
ir_009.constraints: []
draft_instruction_id = dispatch_to_call_both
```

block_004 Call index.search with term, from_date, and limit

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_010 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_002 | request_term
```

```
inputs[2] result: result_005 | from_date_value_both
```

```
inputs[3] result: result_006 | limit_value_both
```

```
outputs[0] result: result_007 | search_total_both
```

```
outputs[1] result: result_008 | search_items_both
```

```
ir_010.constraints[0]: Call index.search exactly once.
```

```
ir_010.constraints[1]: Use request.term unchanged as its query argument.
```

```
ir_010.constraints[2]: When from_date is present, pass its value unchanged; the accepted format is YYYY-MM-DD.
```

```
ir_010.constraints[3]: When limit is present, pass its value unchanged as the limit argument.
```

```
ir_010.constraints[4]: Apply the missing-argument settings in query.yaml to this same call: from_date: omit;  
limit: omit.
```

```
draft_instruction_id = index_search_both
```

ir_011 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = go_write_return_both
```

block_005 Write total to count.txt and return items

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_012 write_count_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_007 | search_total_both
```

```
outputs = []
```

```
ir_012.constraints[0]: After the search, write the response's total value to local count.txt.
```

```
draft_instruction_id = write_count_both
```

ir_013 return

```
inputs[0] result: result_008 | search_items_both
```

```
outputs = []
```

```
ir_013.constraints[0]: Return the search response's items value unchanged.
```

```
draft_instruction_id = return_items_both
```

block_006 Extract present from_date value

data_source_kind = null

block_006.constraints: []

ir_014 extract_from_date_value

inputs[0] result: result_001 | request_document

outputs[0] result: result_009 | from_date_value_only

ir_014.constraints: []

draft_instruction_id = extract_from_date_value_only

ir_015 dispatch

inputs = []

outputs = []

ir_015.constraints: []

draft_instruction_id = dispatch_to_call_from_date_only

block_007 Call index.search with term and from_date

data_source_kind = "external"

block_007.constraints: []

ir_016 index.search

inputs[0] external_resource: index.search | index.search

inputs[1] result: result_002 | request_term

inputs[2] result: result_009 | from_date_value_only

outputs[0] result: result_010 | search_total_from_date_only

outputs[1] result: result_011 | search_items_from_date_only

ir_016.constraints[0]: Call index.search exactly once.

ir_016.constraints[1]: Use request.term unchanged as its query argument.

ir_016.constraints[2]: When from_date is present, pass its value unchanged; the accepted format is YYYY-MM-DD.

ir_016.constraints[3]: When limit is missing, omit the limit argument.

ir_016.constraints[4]: Apply the missing-argument settings in query.yaml to this same call: from_date: omit;
limit: omit.

draft_instruction_id = index_search_from_date_only

ir_017 dispatch

inputs = []

outputs = []

ir_017.constraints: []

draft_instruction_id = go_write_return_from_date_only

block_008 Write total to count.txt and return items

data_source_kind = null

block_008.constraints: []

ir_018 write_count_file

inputs[0] external_resource: count.txt | count.txt

```
inputs[1] result: result_010 | search_total_from_date_only
outputs = []
ir_018.constraints[0]: After the search, write the response's total value to local count.txt.
draft_instruction_id = write_count_from_date_only
```

ir_019 return

```
inputs[0] result: result_011 | search_items_from_date_only
outputs = []
ir_019.constraints[0]: Return the search response's items value unchanged.
draft_instruction_id = return_items_from_date_only
```

block_009 Extract present limit value

```
data_source_kind = null
block_009.constraints: []
```

ir_020 extract_limit_value

```
inputs[0] result: result_001 | request_document
outputs[0] result: result_012 | limit_value_only
ir_020.constraints: []
draft_instruction_id = extract_limit_value_only
```

ir_021 dispatch

```
inputs = []
outputs = []
ir_021.constraints: []
draft_instruction_id = dispatch_to_call_limit_only
```

block_010 Call index.search with term and limit

```
data_source_kind = "external"
block_010.constraints: []
```

ir_022 index.search

```
inputs[0] external_resource: index.search | index.search
inputs[1] result: result_002 | request_term
inputs[2] result: result_012 | limit_value_only
outputs[0] result: result_013 | search_total_limit_only
outputs[1] result: result_014 | search_items_limit_only
ir_022.constraints[0]: Call index.search exactly once.
ir_022.constraints[1]: Use request.term unchanged as its query argument.
ir_022.constraints[2]: When from_date is missing, omit the from_date argument.
ir_022.constraints[3]: When limit is present, pass its value unchanged as the limit argument.
ir_022.constraints[4]: Apply the missing-argument settings in query.yaml to this same call: from_date: omit;
limit: omit.
draft_instruction_id = index_search_limit_only
```

ir_023 dispatch

```
inputs = []
outputs = []
ir_023.constraints: []
draft_instruction_id = go_write_return_limit_only
```

block_011 Write total to count.txt and return items

```
data_source_kind = null
block_011.constraints: []
```

ir_024 write_count_file

```
inputs[0] external_resource: count.txt | count.txt
inputs[1] result: result_013 | search_total_limit_only
outputs = []
ir_024.constraints[0]: After the search, write the response's total value to local count.txt.
draft_instruction_id = write_count_limit_only
```

ir_025 return

```
inputs[0] result: result_014 | search_items_limit_only
outputs = []
ir_025.constraints[0]: Return the search response's items value unchanged.
draft_instruction_id = return_items_limit_only
```

block_012 Call index.search with term only

```
data_source_kind = "external"
block_012.constraints: []
```

ir_026 index.search

```
inputs[0] external_resource: index.search | index.search
inputs[1] result: result_002 | request_term
outputs[0] result: result_015 | search_total_neither
outputs[1] result: result_016 | search_items_neither
ir_026.constraints[0]: Call index.search exactly once.
ir_026.constraints[1]: Use request.term unchanged as its query argument.
ir_026.constraints[2]: When from_date is missing, omit the from_date argument.
ir_026.constraints[3]: When limit is missing, omit the limit argument.
ir_026.constraints[4]: Apply the missing-argument settings in query.yaml to this same call: from_date: omit;
limit: omit.
draft_instruction_id = index_search_neither
```

ir_027 dispatch

```
inputs = []
outputs = []
ir_027.constraints: []
draft_instruction_id = go_write_return_neither
```

block_013 Write total to count.txt and return items

```
data_source_kind = null
```

```
block_013.constraints: []
```

ir_028 write_count_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_015 | search_total_neither
```

```
outputs = []
```

```
ir_028.constraints[0]: After the search, write the response's total value to local count.txt.
```

```
draft_instruction_id = write_count_neither
```

ir_029 return

```
inputs[0] result: result_016 | search_items_neither
```

```
outputs = []
```

```
ir_029.constraints[0]: Return the search response's items value unchanged.
```

```
draft_instruction_id = return_items_neither
```


Q04 / 第 2 次 实际输出明细

查询调用与参数使用 · 跨文件混合

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q04/deepseek-v4-flash-max/2

结构状态: complete | 生成 2 | 修复 1 | HTTP 尝试 2 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

生成 2 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Optional from_date and limit presence is modeled with explicit branches; request_from_date and request_limit are produced by extraction but only consumed on paths where the corresponding presence flag is true.

Skill 层约束与上下文声明

```
declared_context_keys = []

skill.constraints[0]: Call index.search exactly once, using request.term unchanged as its query argument.

skill.constraints[1]: Do not invoke index.delete anywhere in this workflow.

skill.constraints[2]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.

skill.constraints[3]: Capability note: index.search supports fuzzy matching; this note does not request fuzzy expansion in this workflow.

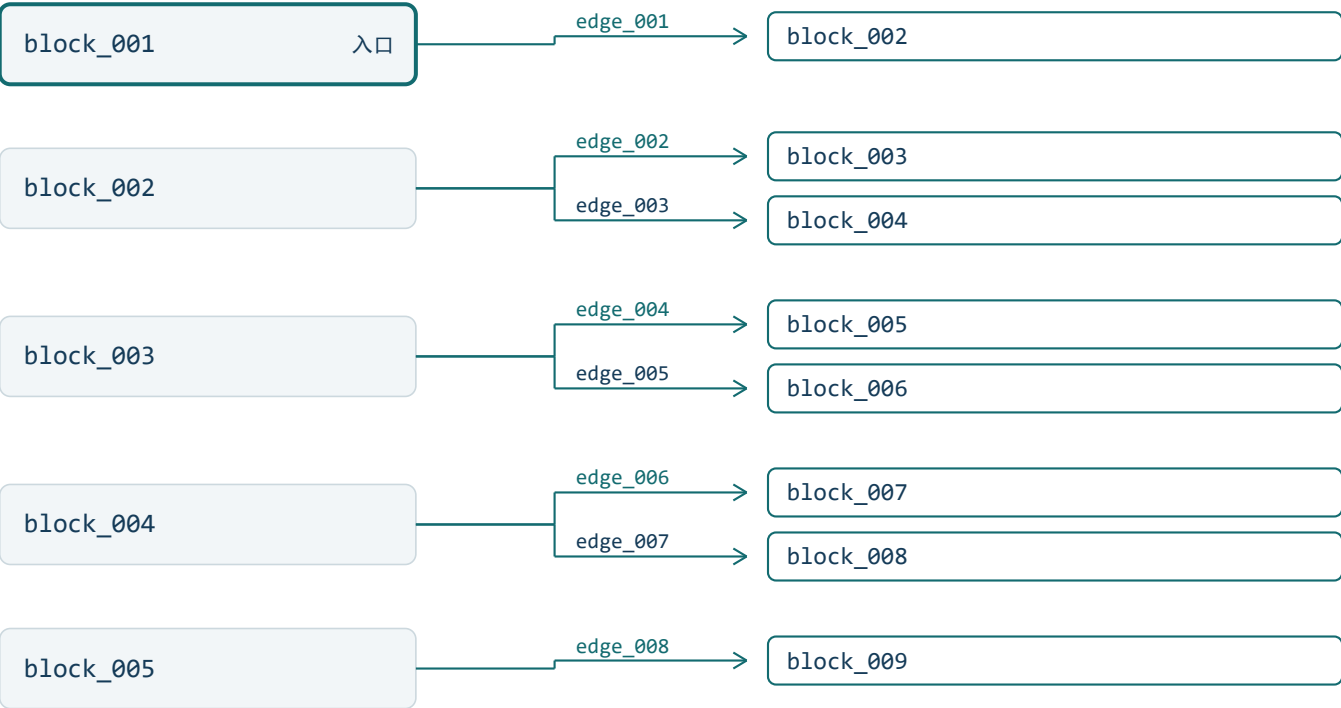
skill.constraints[4]: Apply the missing-argument settings in query configuration to this same call.

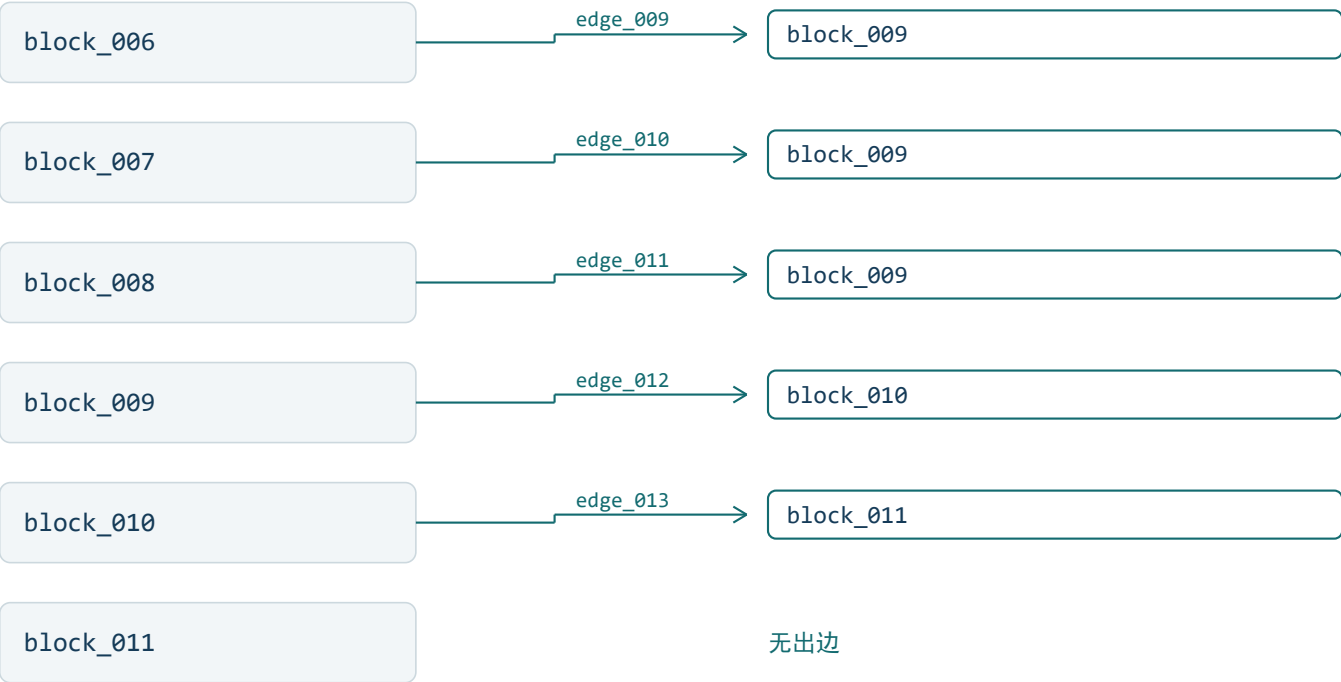
skill.constraints[5]: After the search, write the response's total value to local count.txt.

skill.constraints[6]: Return the search response's items value unchanged.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"from_date present"
edge_003	block_002 -> block_004	"from_date missing"
edge_004	block_003 -> block_005	"limit present"
edge_005	block_003 -> block_006	"limit missing"
edge_006	block_004 -> block_007	"limit present"
edge_007	block_004 -> block_008	"limit missing"
edge_008	block_005 -> block_009	null
edge_009	block_006 -> block_009	null
edge_010	block_007 -> block_009	null
edge_011	block_008 -> block_009	null
edge_012	block_009 -> block_010	null
edge_013	block_010 -> block_011	null

全部操作、数据引用与约束落点

block_001 Read the user-supplied request.json

data_source_kind = "external"

block_001.constraints: []

ir_001 read_request_json

inputs[0] external_resource: request.json | request.json

outputs[0] result: result_001 | request_document

ir_001.constraints: []

draft_instruction_id = read_request_json

ir_002 dispatch

inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = go_extract_request_fields

block_002 Extract request fields and check from_date presence

data_source_kind = null

block_002.constraints: []

ir_003 extract_request_fields

inputs[0] result: result_001 | request_document

outputs[0] result: result_002 | request_term

outputs[1] result: result_003 | request_from_date

outputs[2] result: result_004 | request_limit

outputs[3] result: result_005 | from_date_present

outputs[4] result: result_006 | limit_present

ir_003.constraints: []

draft_instruction_id = extract_fields

ir_004 dispatch

inputs[0] result: result_005 | from_date_present

outputs = []

ir_004.constraints: []

draft_instruction_id = decide_from_date_presence

block_003 Check limit presence when from_date is present

data_source_kind = null

block_003.constraints: []

ir_005 dispatch

inputs[0] result: result_006 | limit_present

outputs = []

ir_005.constraints: []

draft_instruction_id = decide_limit_when_from_date_present

block_004 Check limit presence when from_date is missing

```
data_source_kind = null
```

```
block_004.constraints: []
```

ir_006 dispatch

```
inputs[0] result: result_006 | limit_present
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = decide_limit_when_from_date_missing
```

block_005 Build search arguments with from_date and limit

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_007 build_index_search_arguments

```
inputs[0] result: result_002 | request_term
```

```
inputs[1] result: result_003 | request_from_date
```

```
inputs[2] result: result_004 | request_limit
```

```
outputs[0] result: result_007 | search_args_both
```

```
ir_007.constraints[0]: Pass from_date value unchanged; accepted format is YYYY-MM-DD.
```

```
ir_007.constraints[1]: Pass limit value unchanged.
```

```
draft_instruction_id = build_args_with_both
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_select_args_from_both
```

block_006 Build search arguments with from_date only

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_009 build_index_search_arguments

```
inputs[0] result: result_002 | request_term
```

```
inputs[1] result: result_003 | request_from_date
```

```
outputs[0] result: result_008 | search_args_from_date_only
```

```
ir_009.constraints[0]: Pass from_date value unchanged; accepted format is YYYY-MM-DD.
```

```
ir_009.constraints[1]: Omit the limit argument.
```

```
draft_instruction_id = build_args_with_from_date_only
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_select_args_from_date_only
```

block_007 Build search arguments with limit only

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_011 build_index_search_arguments

```
inputs[0] result: result_002 | request_term
```

```
inputs[1] result: result_004 | request_limit
```

```
outputs[0] result: result_009 | search_args_limit_only
```

```
ir_011.constraints[0]: Omit the from_date argument.
```

```
ir_011.constraints[1]: Pass limit value unchanged.
```

```
draft_instruction_id = build_args_with_limit_only
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_select_args_from_limit_only
```

block_008 Build search arguments with no optional parameters

```
data_source_kind = null
```

```
block_008.constraints: []
```

ir_013 build_index_search_arguments

```
inputs[0] result: result_002 | request_term
```

```
outputs[0] result: result_010 | search_args_none
```

```
ir_013.constraints[0]: Omit the from_date argument.
```

```
ir_013.constraints[1]: Omit the limit argument.
```

```
draft_instruction_id = build_args_with_no_optionals
```

ir_014 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = go_select_args_from_none
```

block_009 Select the search arguments for the current path

```
data_source_kind = null
```

```
block_009.constraints: []
```

ir_015 select_index_search_arguments

```
inputs[0] result: result_007 | search_args_both
```

```
inputs[1] result: result_008 | search_args_from_date_only
```

```
inputs[2] result: result_009 | search_args_limit_only
```

```
inputs[3] result: result_010 | search_args_none
```

```
outputs[0] result: result_011 | search_arguments
```

```
ir_015.constraints: []
```

```
draft_instruction_id = select_args
```

ir_016 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = go_call_index_search
```

block_010 Call index.search once with the selected arguments

```
data_source_kind = "external"
```

```
block_010.constraints: []
```

ir_017 index_search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_011 | search_arguments
```

```
outputs[0] result: result_012 | search_response
```

```
ir_017.constraints[0]: Call index.search exactly once.
```

```
ir_017.constraints[1]: Use request.term unchanged as the query argument.
```

```
ir_017.constraints[2]: When from_date is present, pass its value unchanged; the accepted format is YYYY-MM-DD.
```

```
ir_017.constraints[3]: When from_date is missing, omit the from_date argument.
```

```
ir_017.constraints[4]: When limit is present, pass its value unchanged as the limit argument.
```

```
ir_017.constraints[5]: When limit is missing, omit the limit argument.
```

```
ir_017.constraints[6]: Apply the missing-argument settings in query.yaml: from_date omit, limit omit.
```

```
ir_017.constraints[7]: Do not request fuzzy expansion even though index.search supports fuzzy matching.
```

```
draft_instruction_id = invoke_index_search
```

ir_018 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = go_process_response
```

block_011 Write total to count.txt and return items

```
data_source_kind = null
```

```
block_011.constraints[0]: After the search, write the response's total value to local count.txt.
```

```
block_011.constraints[1]: Return the search response's items value unchanged.
```

ir_019 extract_search_response_total

```
inputs[0] result: result_012 | search_response
```

```
outputs[0] result: result_013 | search_total
```

```
ir_019.constraints: []
```

```
draft_instruction_id = extract_total
```

ir_020 write_total_to_count_file

```
inputs[0] result: result_013 | search_total
```

```
inputs[1] external_resource: count.txt | count.txt
```

```
outputs = []

ir_020.constraints: []

draft_instruction_id = write_count

ir_021 extract_search_response_items

inputs[0] result: result_012 | search_response
outputs[0] result: result_014 | search_items

ir_021.constraints: []

draft_instruction_id = extract_items

ir_022 return

inputs[0] result: result_014 | search_items

outputs = []

ir_022.constraints: []

draft_instruction_id = return_items
```

Q04 / 第 3 次 实际输出明细

查询调用与参数使用 · 跨文件混合

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q04/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: query.yaml is treated as a static configuration for missing-argument semantics; no separate runtime read block was created because the workflow does not explicitly instruct reading it.

Skill 层约束与上下文声明

```
declared_context_keys = []  
  
skill.constraints[0]: Call index.search exactly once.  
  
skill.constraints[1]: Do not invoke index.delete anywhere in this workflow.  
  
skill.constraints[2]: Do not perform fuzzy expansion in this workflow.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"from_date present"
edge_003	block_002 -> block_004	"from_date missing"
edge_004	block_003 -> block_004	null
edge_005	block_004 -> block_005	"limit present"
edge_006	block_004 -> block_006	"limit missing"
edge_007	block_005 -> block_006	null
edge_008	block_006 -> block_007	null

全部操作、数据引用与约束落点

block_001 Read the user-supplied request.json

data_source_kind = "external"

block_001.constraints: []

ir_001 read_request_json

inputs[0] external_resource: request.json | request.json

outputs[0] result: result_001 | request_term

outputs[1] result: result_002 | request_from_date

outputs[2] result: result_003 | request_limit

ir_001.constraints: []

draft_instruction_id = read_request_json

ir_002 dispatch

inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = go_check_from_date

block_002 Check whether from_date is present

data_source_kind = null

block_002.constraints: []

ir_003 check_from_date_presence

inputs[0] result: result_002 | request_from_date

outputs[0] result: result_004 | from_date_present

ir_003.constraints: []

draft_instruction_id = check_from_date_presence

ir_004 dispatch

```
inputs[0] result: result_004 | from_date_present
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = dispatch_from_date
```

block_003 Pass from_date value unchanged as from_date argument

```
data_source_kind = null
```

```
block_003.constraints[0]: Pass the from_date value unchanged.
```

ir_005 pass_from_date_argument

```
inputs[0] result: result_002 | request_from_date
```

```
outputs[0] result: result_005 | from_date_arg
```

```
ir_005.constraints: []
```

```
draft_instruction_id = pass_from_date_argument
```

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = go_check_limit_from_date
```

block_004 Check whether limit is present

```
data_source_kind = null
```

```
block_004.constraints: []
```

ir_007 check_limit_presence

```
inputs[0] result: result_003 | request_limit
```

```
outputs[0] result: result_006 | limit_present
```

```
ir_007.constraints: []
```

```
draft_instruction_id = check_limit_presence
```

ir_008 dispatch

```
inputs[0] result: result_006 | limit_present
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_limit
```

block_005 Pass limit value unchanged as limit argument

```
data_source_kind = null
```

```
block_005.constraints[0]: Pass the limit value unchanged as the limit argument.
```

ir_009 pass_limit_argument

```
inputs[0] result: result_003 | request_limit
```

```
outputs[0] result: result_007 | limit_arg
```

```
ir_009.constraints: []
```

```
draft_instruction_id = pass_limit_argument
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_call_search
```

block_006 Call index.search once with the request term and optional arguments

```
data_source_kind = "external"
```

```
block_006.constraints[0]: Apply the missing-argument settings in query.yaml to this same call.
```

ir_011 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_001 | request_term
```

```
inputs[2] result: result_005 | from_date_arg
```

```
inputs[3] result: result_007 | limit_arg
```

```
outputs[0] result: result_008 | search_total
```

```
outputs[1] result: result_009 | search_items
```

```
ir_011.constraints[0]: Use request.term unchanged as the query argument.
```

```
ir_011.constraints[1]: When from_date is present, pass its value unchanged; accepted format is YYYY-MM-DD.
```

```
ir_011.constraints[2]: When from_date is missing, omit the from_date argument.
```

```
ir_011.constraints[3]: When limit is present, pass its value unchanged as the limit argument.
```

```
ir_011.constraints[4]: When limit is missing, omit the limit argument.
```

```
ir_011.constraints[5]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.
```

```
draft_instruction_id = search_index
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_write_count
```

block_007 Write total to count.txt and return items

```
data_source_kind = null
```

```
block_007.constraints[0]: After the search, write the response's total value to local count.txt.
```

```
block_007.constraints[1]: Return the search response's items value unchanged.
```

ir_013 write_count_txt

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_008 | search_total
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = write_count_txt
```

ir_014 return

```
inputs[0] result: result_009 | search_items
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = return_items
```

Q05 / 第 1 次 实际输出明细

查询调用与参数使用 · 用途最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q05/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Capability note about index.search supporting fuzzy matching is informational and does not request fuzzy expansion; it is not modeled as executable behavior.

candidate warning: Historical example about writing the response's total value to local count.txt is explicitly not executed and is not modeled.

candidate warning: Optional parameter presence checks are modeled as control-flow branches so that missing from_date or limit arguments are omitted.

Skill 层约束与上下文声明

declared_context_keys = []

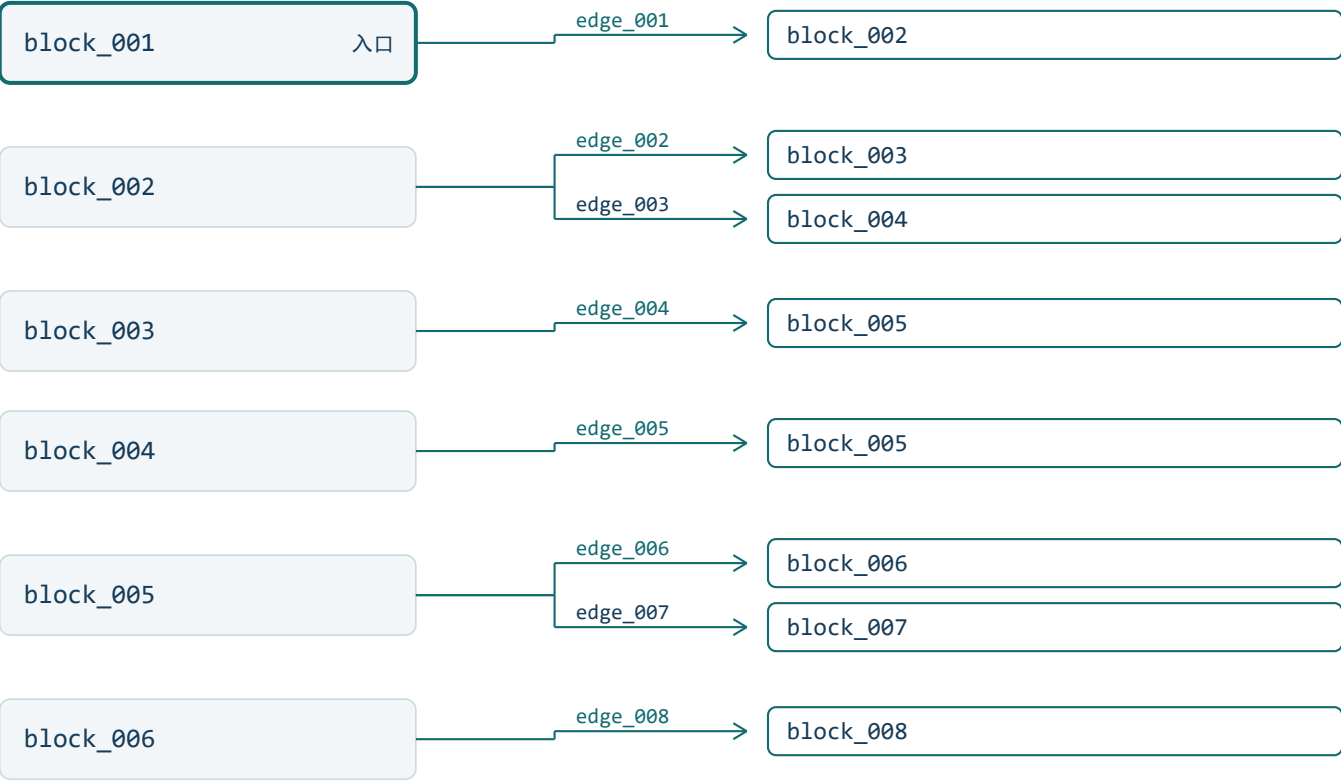
skill.constraints[0]: Call index.search exactly once.

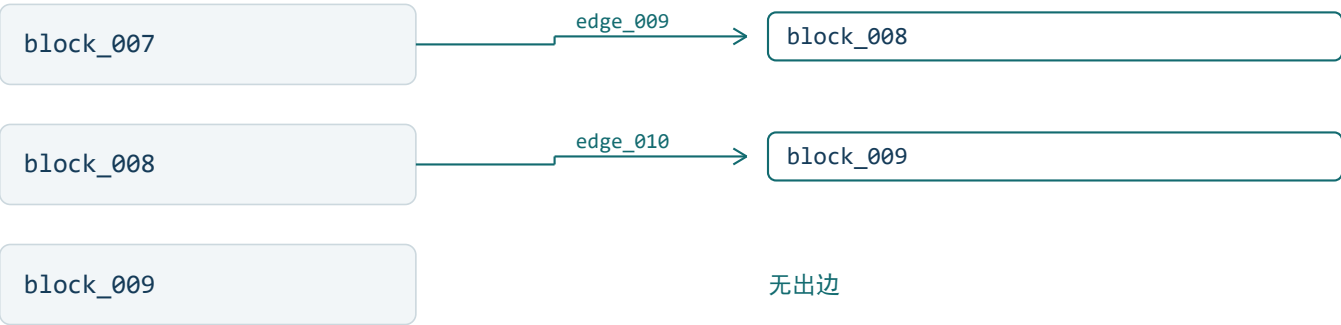
skill.constraints[1]: Do not invoke index.delete anywhere in this workflow.

skill.constraints[2]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"from_date present"
edge_003	block_002 -> block_004	"from_date missing"
edge_004	block_003 -> block_005	null
edge_005	block_004 -> block_005	null
edge_006	block_005 -> block_006	"limit present"
edge_007	block_005 -> block_007	"limit missing"
edge_008	block_006 -> block_008	null
edge_009	block_007 -> block_008	null
edge_010	block_008 -> block_009	null

全部操作、数据引用与约束落点

block_001 Read the user-supplied request.json

```
data_source_kind = "external"

block_001.constraints: []

ir_001 read_request_file

inputs[0] external_resource: request.json | request.json
outputs[0] result: result_001 | request_payload

ir_001.constraints: []

draft_instruction_id = read_request

ir_002 dispatch

inputs = []
outputs = []

ir_002.constraints: []

draft_instruction_id = dispatch_to_inspect
```

block_002 Inspect request fields and optional-parameter presence

```
data_source_kind = null
```

```
block_002.constraints: []
```

ir_003 extract_request_term

```
inputs[0] result: result_001 | request_payload
```

```
outputs[0] result: result_002 | request_term
```

```
ir_003.constraints: []
```

```
draft_instruction_id = extract_term
```

ir_004 check_from_date_present

```
inputs[0] result: result_001 | request_payload
```

```
outputs[0] result: result_003 | from_date_present
```

```
ir_004.constraints: []
```

```
draft_instruction_id = check_from_date_presence
```

ir_005 check_limit_present

```
inputs[0] result: result_001 | request_payload
```

```
outputs[0] result: result_004 | limit_present
```

```
ir_005.constraints: []
```

```
draft_instruction_id = check_limit_presence
```

ir_006 dispatch

```
inputs[0] result: result_003 | from_date_present
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = dispatch_on_from_date
```

block_003 Extract from_date because it is present

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_007 extract_from_date_value

```
inputs[0] result: result_001 | request_payload
```

```
outputs[0] result: result_005 | from_date_value
```

```
ir_007.constraints: []
```

```
draft_instruction_id = extract_from_date
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_to_limit_check
```

block_004 Omit from_date argument because it is missing

```
data_source_kind = null
```

```
block_004.constraints: []
```

ir_009 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_009.constraints: []
```

```
draft_instruction_id = dispatch_to_limit_check_missing_from_date
```

block_005 Check whether limit is present

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_010 dispatch

```
inputs[0] result: result_004 | limit_present
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_on_limit
```

block_006 Extract limit because it is present

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 extract_limit_value

```
inputs[0] result: result_001 | request_payload
```

```
outputs[0] result: result_006 | limit_value
```

```
ir_011.constraints: []
```

```
draft_instruction_id = extract_limit
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_to_search
```

block_007 Omit limit argument because it is missing

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = dispatch_to_search_missing_limit
```


block_008 Call index.search with the request term and optional parameters

```
data_source_kind = "external"
```

```
block_008.constraints: []
```

ir_014 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_002 | request_term
```

```
inputs[2] result: result_005 | from_date_value
```

```
inputs[3] result: result_006 | limit_value
```

```
outputs[0] result: result_007 | search_response
```

```
ir_014.constraints[0]: Use request.term unchanged as its query argument.
```

```
ir_014.constraints[1]: When from_date is present, pass its value unchanged; the accepted format is YYYY-MM-DD.
```

```
ir_014.constraints[2]: When from_date is missing, omit the from_date argument.
```

```
ir_014.constraints[3]: When limit is present, pass its value unchanged as the limit argument.
```

```
ir_014.constraints[4]: When limit is missing, omit the limit argument.
```

```
draft_instruction_id = call_index_search_once
```

ir_015 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_015.constraints: []
```

```
draft_instruction_id = dispatch_to_return
```

block_009 Return the search response's items value

```
data_source_kind = null
```

```
block_009.constraints: []
```

ir_016 extract_search_response_items

```
inputs[0] result: result_007 | search_response
```

```
outputs[0] result: result_008 | search_items
```

```
ir_016.constraints: []
```

```
draft_instruction_id = extract_search_items
```

ir_017 return

```
inputs[0] result: result_008 | search_items
```

```
outputs = []
```

```
ir_017.constraints[0]: Return the search response's items value unchanged.
```

```
draft_instruction_id = return_items
```

Q05 / 第 2 次 实际输出明细

查询调用与参数使用 · 用途最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q05/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Historical example about writing the search response's total value to local count.txt is not executed in this workflow and was excluded.

Skill 层约束与上下文声明

```
declared_context_keys = []

skill.constraints[0]: index.search supports fuzzy matching; this note does not request fuzzy expansion in this workflow.

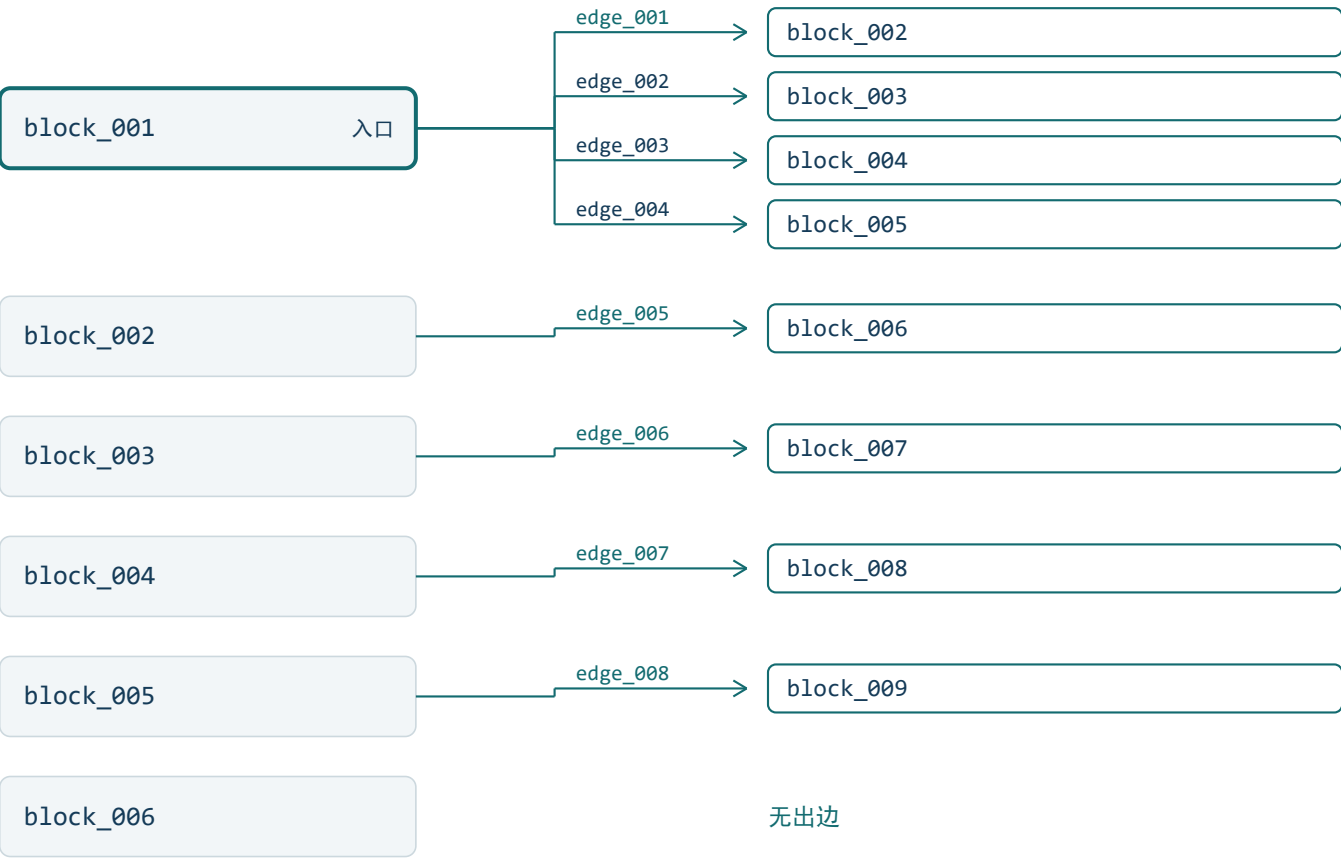
skill.constraints[1]: Call index.search exactly once, using request.term unchanged as its query argument.

skill.constraints[2]: Do not invoke index.delete anywhere in this workflow.

skill.constraints[3]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



block_007	无出边
block_008	无出边
block_009	无出边

完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	"from_date present and limit present"
edge_002	block_001 -> block_003	"from_date present and limit missing"
edge_003	block_001 -> block_004	"from_date missing and limit present"
edge_004	block_001 -> block_005	"from_date missing and limit missing"
edge_005	block_002 -> block_006	null
edge_006	block_003 -> block_007	null
edge_007	block_004 -> block_008	null
edge_008	block_005 -> block_009	null

全部操作、数据引用与约束落点

block_001 Read request.json and check optional fields

```
data_source_kind = "external"
block_001.constraints: []

ir_001 read_request_json

inputs[0] external_resource: request.json | request.json
outputs[0] result: result_001 | request.term
outputs[1] result: result_002 | request.from_date
outputs[2] result: result_003 | from_date present
outputs[3] result: result_004 | request.limit
outputs[4] result: result_005 | limit present
ir_001.constraints: []
draft_instruction_id = read_request_json
```

ir_002 dispatch

```
inputs[0] result: result_003 | request_from_date_present
```

```
inputs[1] result: result_005 | request_limit_present
```

```
outputs = []
```

```
ir_002.constraints: []
```

```
draft_instruction_id = dispatch_optional
```

block_002 Call index.search with from_date and limit

```
data_source_kind = "external"
```

```
block_002.constraints: []
```

ir_003 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_001 | request_term
```

```
inputs[2] result: result_002 | request_from_date
```

```
inputs[3] result: result_004 | request_limit
```

```
outputs[0] result: result_006 | search response
```

```
ir_003.constraints[0]: When from_date is present, pass its value unchanged; the accepted format is YYYY-MM-DD.
```

```
ir_003.constraints[1]: When limit is present, pass its value unchanged as the limit argument.
```

```
draft_instruction_id = call_search_both
```

ir_004 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = dispatch_to_extract_both
```

block_003 Call index.search with from_date only

```
data_source_kind = "external"
```

```
block_003.constraints: []
```

ir_005 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_001 | request_term
```

```
inputs[2] result: result_002 | request_from_date
```

```
outputs[0] result: result_007 | search response
```

```
ir_005.constraints[0]: When from_date is present, pass its value unchanged; the accepted format is YYYY-MM-DD.
```

```
ir_005.constraints[1]: When limit is missing, omit the limit argument.
```

```
draft_instruction_id = call_search_from_date_only
```

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = dispatch_to_extract_from_date_only
```

block_004 Call index.search with limit only

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_007 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_001 | request_term
```

```
inputs[2] result: result_004 | request_limit
```

```
outputs[0] result: result_008 | search response
```

```
ir_007.constraints[0]: When from_date is missing, omit the from_date argument.
```

```
ir_007.constraints[1]: When limit is present, pass its value unchanged as the limit argument.
```

```
draft_instruction_id = call_search_limit_only
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_to_extract_limit_only
```

block_005 Call index.search without optional arguments

```
data_source_kind = "external"
```

```
block_005.constraints: []
```

ir_009 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_001 | request_term
```

```
outputs[0] result: result_009 | search response
```

```
ir_009.constraints[0]: When from_date is missing, omit the from_date argument.
```

```
ir_009.constraints[1]: When limit is missing, omit the limit argument.
```

```
draft_instruction_id = call_search_neither
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_to_extract_neither
```

block_006 Extract items and return

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 extract_search_items

```
inputs[0] result: result_006 | search_response_both
```

```
outputs[0] result: result_010 | search response items
```

```
ir_011.constraints: []
```

```
draft_instruction_id = extract_items_both
```

ir_012 return

inputs[0] result: result_010 | search_items_both

outputs = []

ir_012.constraints[0]: Return the search response's items value unchanged.

draft_instruction_id = return_items_both

block_007 Extract items and return

data_source_kind = null

block_007.constraints: []

ir_013 extract_search_items

inputs[0] result: result_007 | search_response_from_date_only

outputs[0] result: result_011 | search response items

ir_013.constraints: []

draft_instruction_id = extract_items_from_date_only

ir_014 return

inputs[0] result: result_011 | search_items_from_date_only

outputs = []

ir_014.constraints[0]: Return the search response's items value unchanged.

draft_instruction_id = return_items_from_date_only

block_008 Extract items and return

data_source_kind = null

block_008.constraints: []

ir_015 extract_search_items

inputs[0] result: result_008 | search_response_limit_only

outputs[0] result: result_012 | search response items

ir_015.constraints: []

draft_instruction_id = extract_items_limit_only

ir_016 return

inputs[0] result: result_012 | search_items_limit_only

outputs = []

ir_016.constraints[0]: Return the search response's items value unchanged.

draft_instruction_id = return_items_limit_only

block_009 Extract items and return

data_source_kind = null

block_009.constraints: []

ir_017 extract_search_items

inputs[0] result: result_009 | search_response_neither

outputs[0] result: result_013 | search response items

ir_017.constraints: []

draft_instruction_id = extract_items_neither

ir_018 return

```
inputs[0] result: result_013 | search_items_neither
```

```
outputs = []
```

```
ir_018.constraints[0]: Return the search response's items value unchanged.
```

```
draft_instruction_id = return_items_neither
```

Q05 / 第 3 次 实际输出明细

查询调用与参数使用 · 用途最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q05/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Historical example about writing response.total to local count.txt is explicitly not executed in this workflow and was not modeled.

candidate warning: Capability note indicates index.search supports fuzzy matching, but this workflow does not request fuzzy expansion.

candidate warning: Optional from_date and limit are modeled with presence branches; on missing paths the corresponding pass-through result is not produced, representing argument omission.

candidate warning: No runtime context keys are declared because request.json is read as an external resource.

Skill 层约束与上下文声明

declared_context_keys = []

skill.constraints[0]: Call index.search exactly once, using request.term unchanged as its query argument.

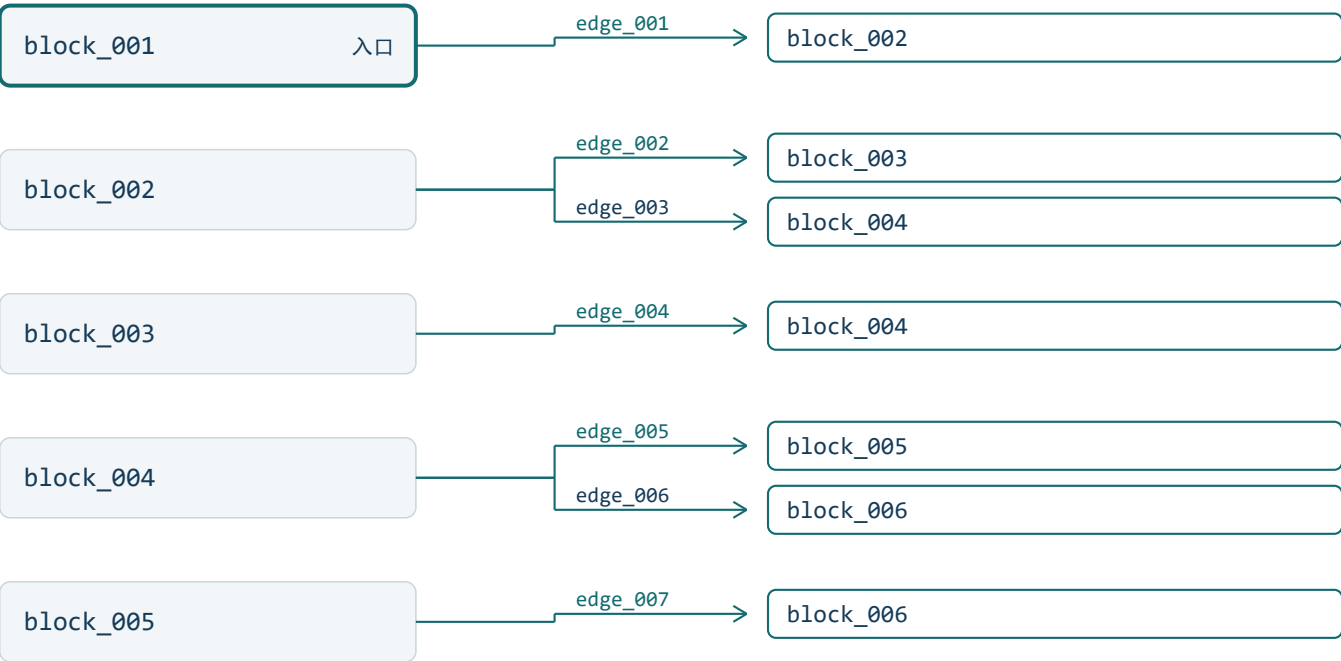
skill.constraints[1]: Do not invoke index.delete anywhere in this workflow.

skill.constraints[2]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.

skill.constraints[3]: index.search supports fuzzy matching; this note does not request fuzzy expansion in this workflow.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"from_date is present"
edge_003	block_002 -> block_004	"from_date is missing; omit from_date argument"
edge_004	block_003 -> block_004	null
edge_005	block_004 -> block_005	"limit is present"
edge_006	block_004 -> block_006	"limit is missing; omit limit argument"
edge_007	block_005 -> block_006	null
edge_008	block_006 -> block_007	null

全部操作、数据引用与约束落点

block_001 Read request.json

```
data_source_kind = "external"
block_001.constraints: []

ir_001 read_request_json

inputs[0] external_resource: request.json | request.json
outputs[0] result: result_001 | request_object
outputs[1] result: result_002 | request_term
outputs[2] result: result_003 | request_from_date
outputs[3] result: result_004 | request_limit
ir_001.constraints: []
draft_instruction_id = read_request_json
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = dispatch_read_request
```

block_002 Check whether from_date is present

data_source_kind = null

block_002.constraints: []

ir_003 check_from_date_present

inputs[0] result: result_001 | request_object

outputs[0] result: result_005 | from_date_present

ir_003.constraints: []

draft_instruction_id = check_from_date_present

ir_004 dispatch

inputs[0] result: result_005 | from_date_present

outputs = []

ir_004.constraints: []

draft_instruction_id = dispatch_from_date

block_003 Pass from_date unchanged when present

data_source_kind = null

block_003.constraints: []

ir_005 pass_from_date_unchanged

inputs[0] result: result_003 | request_from_date

outputs[0] result: result_006 | from_date_argument

ir_005.constraints[0]: Pass from_date value unchanged; the accepted format is YYYY-MM-DD.

draft_instruction_id = pass_from_date_unchanged

ir_006 dispatch

inputs = []

outputs = []

ir_006.constraints: []

draft_instruction_id = dispatch_pass_from_date

block_004 Check whether limit is present

data_source_kind = null

block_004.constraints: []

ir_007 check_limit_present

inputs[0] result: result_001 | request_object

outputs[0] result: result_007 | limit_present

ir_007.constraints: []

draft_instruction_id = check_limit_present

ir_008 dispatch

inputs[0] result: result_007 | limit_present

outputs = []

ir_008.constraints: []

draft_instruction_id = dispatch_limit

block_005 Pass limit unchanged when present

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 pass_limit_unchanged

```
inputs[0] result: result_004 | request_limit
```

```
outputs[0] result: result_008 | limit_argument
```

```
ir_009.constraints[0]: Pass limit value unchanged as the limit argument.
```

```
draft_instruction_id = pass_limit_unchanged
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_pass_limit
```

block_006 Call index.search once with the request arguments

```
data_source_kind = "external"
```

```
block_006.constraints: []
```

ir_011 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_002 | request_term
```

```
inputs[2] result: result_006 | from_date_argument
```

```
inputs[3] result: result_008 | limit_argument
```

```
outputs[0] result: result_009 | search_response
```

```
ir_011.constraints[0]: When from_date is present, pass its value unchanged; the accepted format is YYYY-MM-DD.
```

```
ir_011.constraints[1]: When from_date is missing, omit the from_date argument.
```

```
ir_011.constraints[2]: When limit is present, pass its value unchanged as the limit argument.
```

```
ir_011.constraints[3]: When limit is missing, omit the limit argument.
```

```
draft_instruction_id = index_search
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_search
```

block_007 Extract and return the search response items

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 extract_search_response_items

```
inputs[0] result: result_009 | search_response
```

```
outputs[0] result: result_010 | search_items
```

```
ir_013.constraints: []
```

```
draft_instruction_id = extract_items
```

```
ir_014 return
```

```
inputs[0] result: result_010 | search_items
```

```
outputs = []
```

```
ir_014.constraints[0]: Return the search response's items value unchanged.
```

```
draft_instruction_id = return_items
```

Q06 / 第 1 次 实际输出明细

查询调用与参数使用 · 语义最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q06/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

```
declared_context_keys = []

skill.constraints[0]: index.search supports fuzzy matching; this note does not request fuzzy expansion in this workflow.

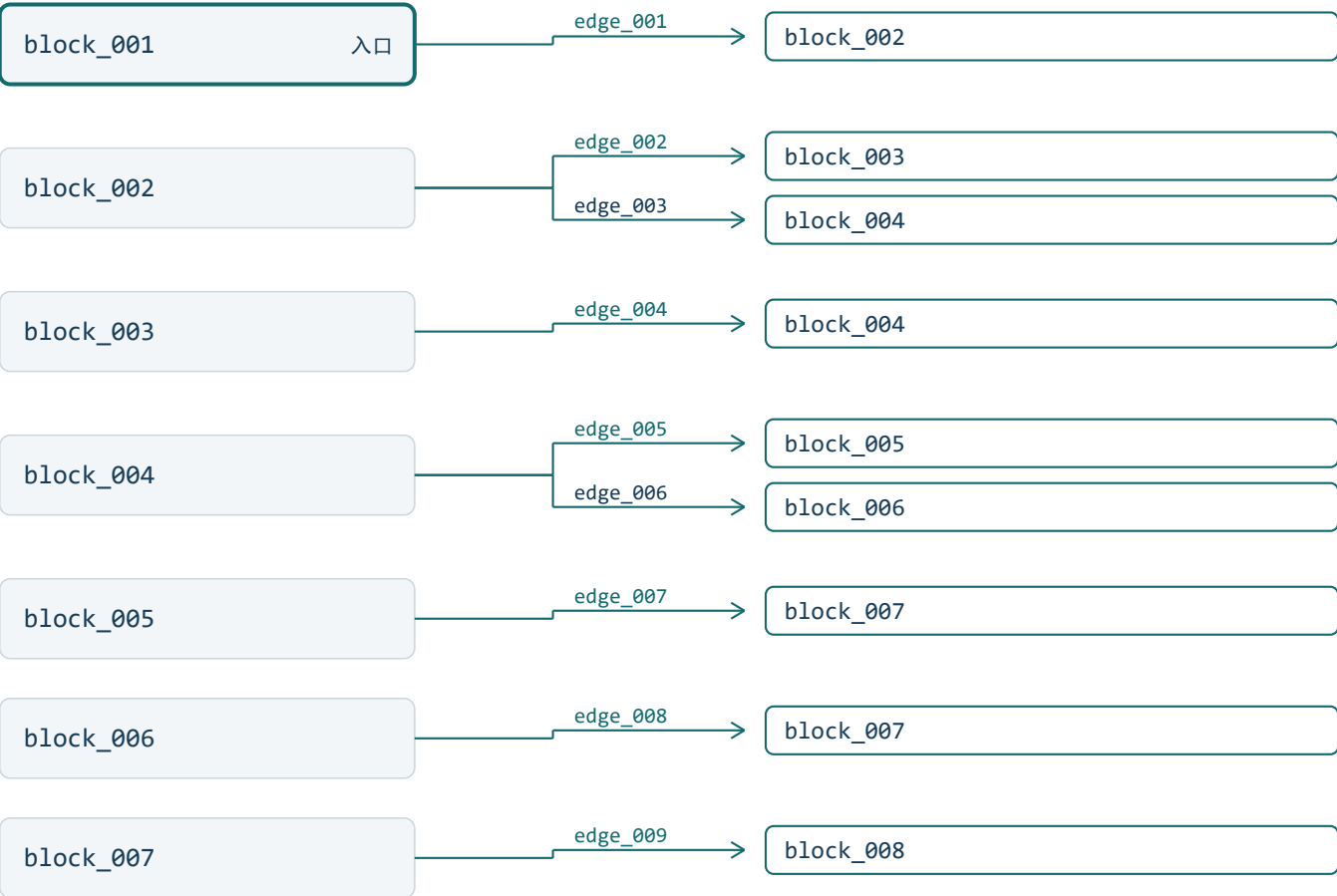
skill.constraints[1]: Call index.search exactly once.

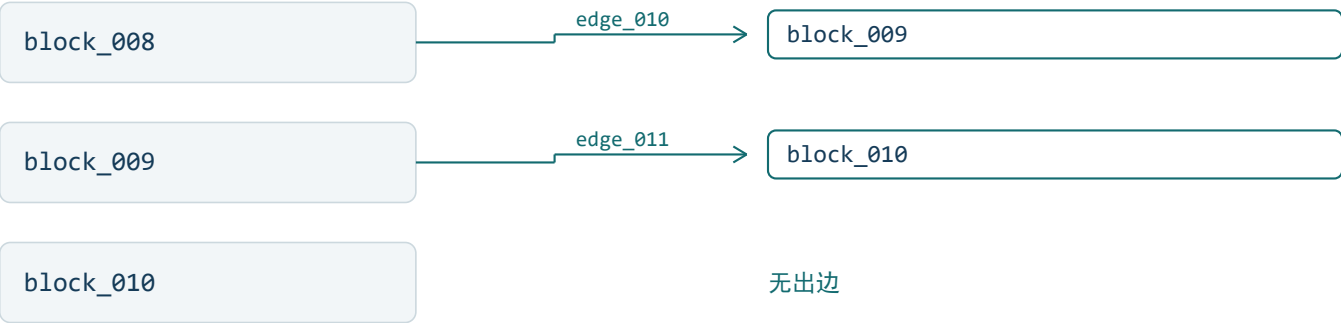
skill.constraints[2]: Do not invoke index.delete anywhere in this workflow.

skill.constraints[3]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"from_date is present"
edge_003	block_002 -> block_004	"from_date is missing"
edge_004	block_003 -> block_004	null
edge_005	block_004 -> block_005	"limit is present"
edge_006	block_004 -> block_006	"limit is missing"
edge_007	block_005 -> block_007	null
edge_008	block_006 -> block_007	null
edge_009	block_007 -> block_008	null
edge_010	block_008 -> block_009	null
edge_011	block_009 -> block_010	null

全部操作、数据引用与约束落点

block_001 Read the user-supplied request.json

```
data_source_kind = "external"
block_001.constraints: []

ir_001 read_request_json

inputs[0] external_resource: request.json | request.json
outputs[0] result: result_001 | request_data
ir_001.constraints: []
draft_instruction_id = read_request_file

ir_002 dispatch

inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_check_from_date
```

block_002 Extract the request term and check whether from_date is present

data_source_kind = null

block_002.constraints: []

ir_003 extract_request_term

inputs[0] result: result_001 | request_data

outputs[0] result: result_002 | request_term

ir_003.constraints: []

draft_instruction_id = extract_request_term

ir_004 check_from_date_presence

inputs[0] result: result_001 | request_data

outputs[0] result: result_003 | from_date_present

ir_004.constraints: []

draft_instruction_id = check_from_date_presence

ir_005 dispatch

inputs[0] result: result_003 | from_date_present

outputs = []

ir_005.constraints: []

draft_instruction_id = choose_from_date

block_003 Extract the from_date value when it is present

data_source_kind = null

block_003.constraints: []

ir_006 extract_from_date_value

inputs[0] result: result_001 | request_data

outputs[0] result: result_004 | from_date_argument

ir_006.constraints: []

draft_instruction_id = extract_from_date_value

ir_007 dispatch

inputs = []

outputs = []

ir_007.constraints: []

draft_instruction_id = go_check_limit_from_date

block_004 Check whether the limit parameter is present

data_source_kind = null

block_004.constraints: []

ir_008 check_limit_presence

inputs[0] result: result_001 | request_data

outputs[0] result: result_005 | limit_present

ir_008.constraints: []

draft_instruction_id = check_limit_presence

ir_009 dispatch

```
inputs[0] result: result_005 | limit_present
```

```
outputs = []
```

```
ir_009.constraints: []
```

```
draft_instruction_id = choose_limit
```

block_005 Use the request's limit value unchanged

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_010 extract_request_limit

```
inputs[0] result: result_001 | request_data
```

```
outputs[0] result: result_006 | request_limit
```

```
ir_010.constraints: []
```

```
draft_instruction_id = extract_request_limit
```

ir_011 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = go_resolve_from_request_limit
```

block_006 Use the default limit value 10

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_012 use_default_limit

```
inputs[0] literal = 10
```

```
outputs[0] result: result_007 | default_limit
```

```
ir_012.constraints: []
```

```
draft_instruction_id = use_default_limit_value
```

ir_013 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = go_resolve_from_default_limit
```

block_007 Resolve the final limit argument

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_014 resolve_limit_argument

```
inputs[0] result: result_006 | request_limit
```

```
inputs[1] result: result_007 | default_limit
```

```
inputs[2] result: result_005 | limit_present
```

```
outputs[0] result: result_008 | limit_argument
```



```
ir_014.constraints: []
```

```
draft_instruction_id = resolve_limit_argument
```

ir_015 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_015.constraints: []
```

```
draft_instruction_id = go_search
```

block_008 Call index.search exactly once with the resolved arguments

```
data_source_kind = "external"
```

```
block_008.constraints: []
```

ir_016 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_002 | request_term
```

```
inputs[2] result: result_004 | from_date_argument
```

```
inputs[3] result: result_008 | limit_argument
```

```
outputs[0] result: result_009 | search_response
```

```
ir_016.constraints[0]: Call index.search exactly once.
```

```
ir_016.constraints[1]: Use request.term unchanged as its query argument.
```

```
ir_016.constraints[2]: When from_date is present, pass its value unchanged; the accepted format is YYYY-MM-DD.
```

```
ir_016.constraints[3]: When from_date is missing, omit the from_date argument.
```

```
ir_016.constraints[4]: When limit is present, pass its value unchanged as the limit argument.
```

```
ir_016.constraints[5]: When limit is missing, pass 10 as the limit argument.
```

```
draft_instruction_id = index_search
```

ir_017 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_017.constraints: []
```

```
draft_instruction_id = go_write_count
```

block_009 Write the search response total to count.txt

```
data_source_kind = null
```

```
block_009.constraints: []
```

ir_018 extract_search_total

```
inputs[0] result: result_009 | search_response
```

```
outputs[0] result: result_010 | search_total
```

```
ir_018.constraints: []
```

```
draft_instruction_id = extract_search_total
```

ir_019 write_count_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_010 | search_total
```

```
outputs = []

ir_019.constraints: []

draft_instruction_id = write_count_file

ir_020 dispatch

inputs = []

outputs = []

ir_020.constraints: []

draft_instruction_id = go_return_items

block_010 Return the search response items unchanged

data_source_kind = null

block_010.constraints: []

ir_021 extract_search_items

inputs[0] result: result_009 | search_response

outputs[0] result: result_011 | search_items

ir_021.constraints: []

draft_instruction_id = extract_search_items

ir_022 return

inputs[0] result: result_011 | search_items

outputs = []

ir_022.constraints: []

draft_instruction_id = return_items
```

Q06 / 第 2 次 实际输出明细

查询调用与参数使用 · 语义最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q06/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

```
declared_context_keys = []

skill.constraints[0]: Call index.search exactly once.

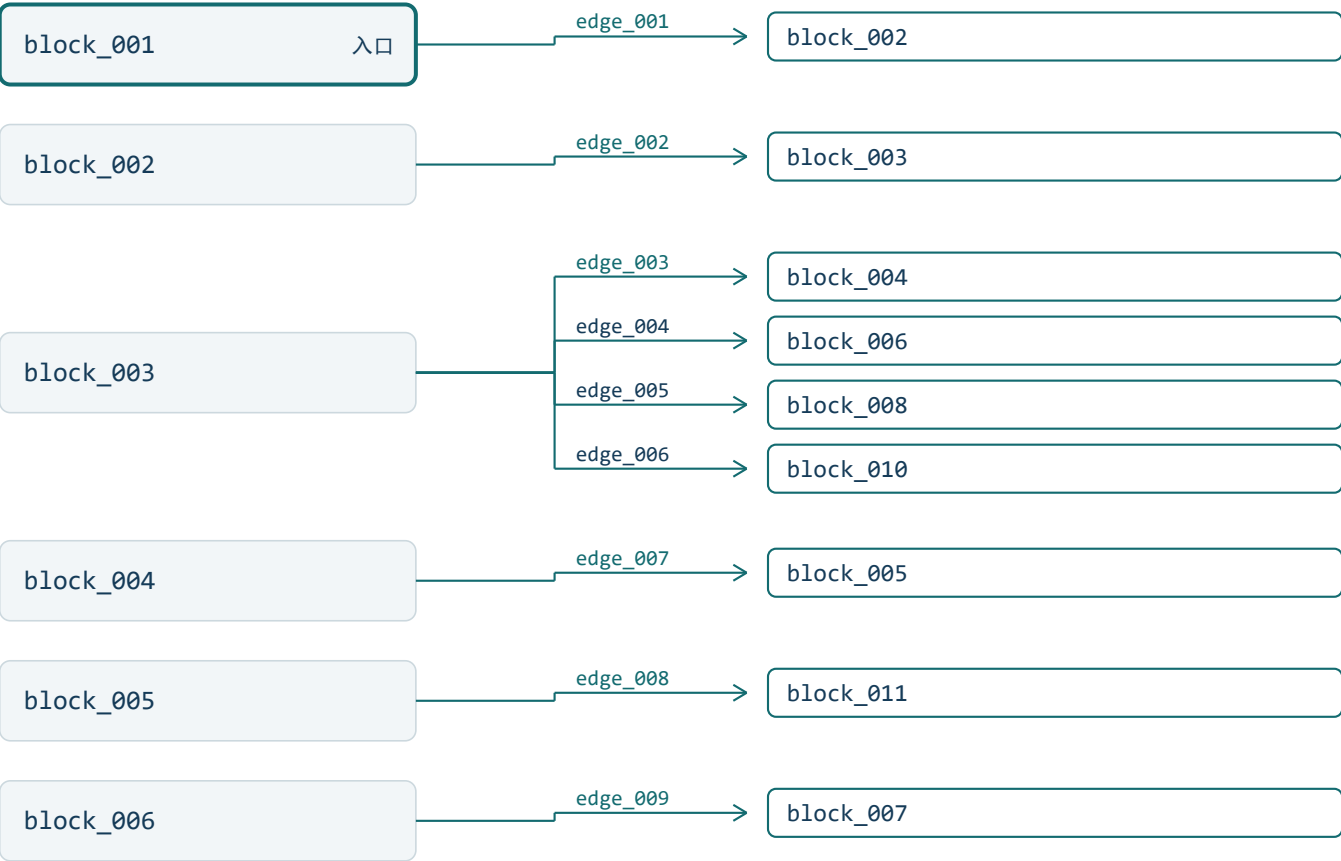
skill.constraints[1]: Do not invoke index.delete anywhere in this workflow.

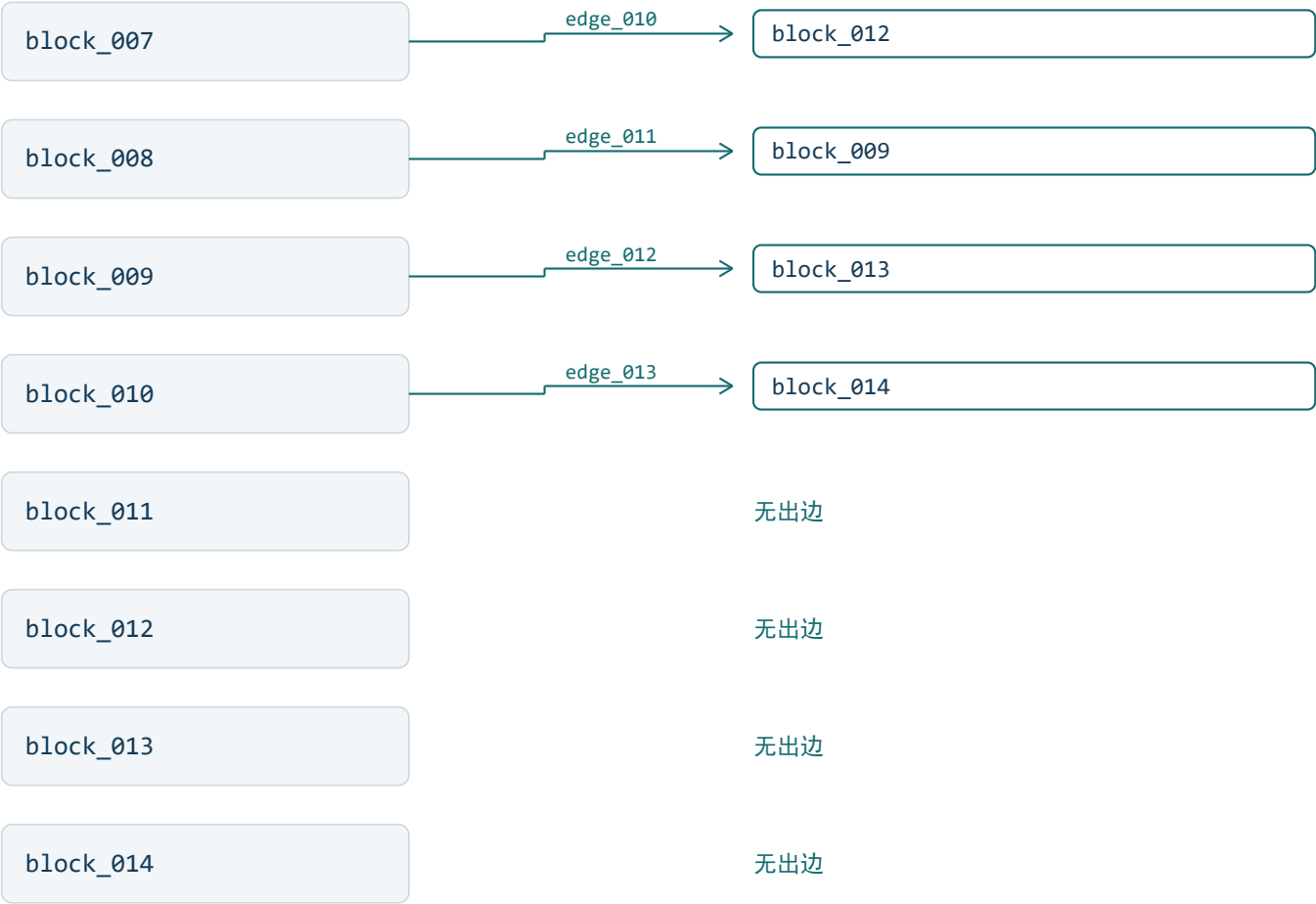
skill.constraints[2]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.

skill.constraints[3]: Capability note: index.search supports fuzzy matching; this note does not request fuzzy expansion in this workflow.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"from_date is present and limit is present"
edge_004	block_003 -> block_006	"from_date is present and limit is missing"
edge_005	block_003 -> block_008	"from_date is missing and limit is present"
edge_006	block_003 -> block_010	"from_date is missing and limit is missing"
edge_007	block_004 -> block_005	null
edge_008	block_005 -> block_011	null
edge_009	block_006 -> block_007	null
edge_010	block_007 -> block_012	null
edge_011	block_008 -> block_009	null
edge_012	block_009 -> block_013	null
edge_013	block_010 -> block_014	null

全部操作、数据引用与约束落点

block_001 Read the user-supplied request.json

data_source_kind = "external"

block_001.constraints: []

ir_001 read_file

inputs[0] external_resource: request.json | request.json

outputs[0] result: result_001 | request_data

ir_001.constraints: []

draft_instruction_id = read_request_file

ir_002 dispatch

inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = go_extract_term

block_002 Extract request.term

data_source_kind = null

block_002.constraints: []

ir_003 extract_request_term

inputs[0] result: result_001 | request_data

outputs[0] result: result_002 | request_term

ir_003.constraints: []

draft_instruction_id = extract_request_term

ir_004 dispatch

inputs = []

outputs = []

ir_004.constraints: []

draft_instruction_id = go_check_optional

block_003 Check presence of from_date and limit

data_source_kind = null

block_003.constraints: []

ir_005 check_from_date_present

inputs[0] result: result_001 | request_data

outputs[0] result: result_003 | from_date_present

ir_005.constraints: []

draft_instruction_id = check_from_date_present

ir_006 check_limit_present

inputs[0] result: result_001 | request_data

outputs[0] result: result_004 | limit_present

```
ir_006.constraints: []
draft_instruction_id = check_limit_present

ir_007 dispatch

inputs[0] result: result_003 | from_date_present
inputs[1] result: result_004 | limit_present
outputs = []

ir_007.constraints: []
draft_instruction_id = choose_optional_args

block_004 Extract from_date and limit

data_source_kind = null
block_004.constraints: []

ir_008 extract_from_date

inputs[0] result: result_001 | request_data
outputs[0] result: result_005 | request_from_date_both
ir_008.constraints: []
draft_instruction_id = extract_from_date_both

ir_009 extract_limit

inputs[0] result: result_001 | request_data
outputs[0] result: result_006 | request_limit_both
ir_009.constraints: []
draft_instruction_id = extract_limit_both

ir_010 dispatch

inputs = []
outputs = []
ir_010.constraints: []
draft_instruction_id = go_call_both

block_005 Call index.search with from_date and limit

data_source_kind = "external"
block_005.constraints: []

ir_011 index.search

inputs[0] external_resource: index.search | index.search
inputs[1] result: result_002 | request_term
inputs[2] result: result_005 | request_from_date_both
inputs[3] result: result_006 | request_limit_both
outputs[0] result: result_007 | search_response_both
ir_011.constraints[0]: use request.term unchanged as query argument
ir_011.constraints[1]: pass from_date value unchanged; accepted format is YYYY-MM-DD
ir_011.constraints[2]: pass limit value unchanged as limit argument
draft_instruction_id = call_index_search_both
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_write_both
```

block_006 Extract from_date when limit is missing

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_013 extract_from_date

```
inputs[0] result: result_001 | request_data
```

```
outputs[0] result: result_008 | request_from_date_fd_lm
```

```
ir_013.constraints: []
```

```
draft_instruction_id = extract_from_date_fd_lm
```

ir_014 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = go_call_fd_lm
```

block_007 Call index.search with from_date and default limit

```
data_source_kind = "external"
```

```
block_007.constraints: []
```

ir_015 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_002 | request_term
```

```
inputs[2] result: result_008 | request_from_date_fd_lm
```

```
inputs[3] literal = 10
```

```
outputs[0] result: result_009 | search_response_fd_lm
```

```
ir_015.constraints[0]: use request.term unchanged as query argument
```

```
ir_015.constraints[1]: pass from_date value unchanged; accepted format is YYYY-MM-DD
```

```
ir_015.constraints[2]: when limit is missing, pass 10 as the limit argument
```

```
draft_instruction_id = call_index_search_fd_lm
```

ir_016 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = go_write_fd_lm
```

block_008 Extract limit when from_date is missing

```
data_source_kind = null
```

```
block_008.constraints: []
```

ir_017 extract_limit

```
inputs[0] result: result_001 | request_data
```

```
outputs[0] result: result_010 | request_limit_fd_m
```

```
ir_017.constraints: []
```

```
draft_instruction_id = extract_limit_fd_m
```

ir_018 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = go_call_fd_m
```

block_009 Call index.search without from_date

```
data_source_kind = "external"
```

```
block_009.constraints: []
```

ir_019 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_002 | request_term
```

```
inputs[2] result: result_010 | request_limit_fd_m
```

```
outputs[0] result: result_011 | search_response_fd_m
```

```
ir_019.constraints[0]: use request.term unchanged as query argument
```

```
ir_019.constraints[1]: when from_date is missing, omit the from_date argument
```

```
ir_019.constraints[2]: pass limit value unchanged as limit argument
```

```
draft_instruction_id = call_index_search_fd_m
```

ir_020 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = go_write_fd_m
```

block_010 Call index.search with default limit and no from_date

```
data_source_kind = "external"
```

```
block_010.constraints: []
```

ir_021 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_002 | request_term
```

```
inputs[2] literal = 10
```

```
outputs[0] result: result_012 | search_response_both_missing
```

```
ir_021.constraints[0]: use request.term unchanged as query argument
```


ir_021.constraints[1]: when from_date is missing, omit the from_date argument

ir_021.constraints[2]: when limit is missing, pass 10 as the limit argument

draft_instruction_id = call_index_search_both_missing

ir_022 dispatch

inputs = []

outputs = []

ir_022.constraints: []

draft_instruction_id = go_write_both_missing

block_011 Write total and return items

data_source_kind = null

block_011.constraints: []

ir_023 extract_response_total

inputs[0] result: result_007 | search_response_both

outputs[0] result: result_013 | response_total_both

ir_023.constraints: []

draft_instruction_id = extract_total_both

ir_024 write_file

inputs[0] external_resource: count.txt | count.txt

inputs[1] result: result_013 | response_total_both

outputs = []

ir_024.constraints[0]: write the response's total value to local count.txt

draft_instruction_id = write_count_both

ir_025 extract_response_items

inputs[0] result: result_007 | search_response_both

outputs[0] result: result_014 | response_items_both

ir_025.constraints: []

draft_instruction_id = extract_items_both

ir_026 return

inputs[0] result: result_014 | response_items_both

outputs = []

ir_026.constraints[0]: Return the search response's items value unchanged.

draft_instruction_id = return_items_both

block_012 Write total and return items (from_date present, limit missing)

data_source_kind = null

block_012.constraints: []

ir_027 extract_response_total

inputs[0] result: result_009 | search_response_fd_lm

outputs[0] result: result_015 | response_total_fd_lm

ir_027.constraints: []

```
draft_instruction_id = extract_total_fd_lm

ir_028 write_file

inputs[0] external_resource: count.txt | count.txt
inputs[1] result: result_015 | response_total_fd_lm
outputs = []

ir_028.constraints[0]: write the response's total value to local count.txt
draft_instruction_id = write_count_fd_lm

ir_029 extract_response_items

inputs[0] result: result_009 | search_response_fd_lm
outputs[0] result: result_016 | response_items_fd_lm
ir_029.constraints: []

draft_instruction_id = extract_items_fd_lm

ir_030 return

inputs[0] result: result_016 | response_items_fd_lm
outputs = []

ir_030.constraints[0]: Return the search response's items value unchanged.
draft_instruction_id = return_items_fd_lm

block_013 Write total and return items (from_date missing, limit present)

data_source_kind = null
block_013.constraints: []

ir_031 extract_response_total

inputs[0] result: result_011 | search_response_fd_m
outputs[0] result: result_017 | response_total_fd_m
ir_031.constraints: []

draft_instruction_id = extract_total_fd_m

ir_032 write_file

inputs[0] external_resource: count.txt | count.txt
inputs[1] result: result_017 | response_total_fd_m
outputs = []

ir_032.constraints[0]: write the response's total value to local count.txt
draft_instruction_id = write_count_fd_m

ir_033 extract_response_items

inputs[0] result: result_011 | search_response_fd_m
outputs[0] result: result_018 | response_items_fd_m
ir_033.constraints: []

draft_instruction_id = extract_items_fd_m
```

ir_034 return

```
inputs[0] result: result_018 | response_items_fd_m
```

```
outputs = []
```

```
ir_034.constraints[0]: Return the search response's items value unchanged.
```

```
draft_instruction_id = return_items_fd_m
```

block_014 Write total and return items (both optional missing)

```
data_source_kind = null
```

```
block_014.constraints: []
```

ir_035 extract_response_total

```
inputs[0] result: result_012 | search_response_both_missing
```

```
outputs[0] result: result_019 | response_total_both_missing
```

```
ir_035.constraints: []
```

```
draft_instruction_id = extract_total_both_missing
```

ir_036 write_file

```
inputs[0] external_resource: count.txt | count.txt
```

```
inputs[1] result: result_019 | response_total_both_missing
```

```
outputs = []
```

```
ir_036.constraints[0]: write the response's total value to local count.txt
```

```
draft_instruction_id = write_count_both_missing
```

ir_037 extract_response_items

```
inputs[0] result: result_012 | search_response_both_missing
```

```
outputs[0] result: result_020 | response_items_both_missing
```

```
ir_037.constraints: []
```

```
draft_instruction_id = extract_items_both_missing
```

ir_038 return

```
inputs[0] result: result_020 | response_items_both_missing
```

```
outputs = []
```

```
ir_038.constraints[0]: Return the search response's items value unchanged.
```

```
draft_instruction_id = return_items_both_missing
```

Q06 / 第 3 次 实际输出明细

查询调用与参数使用 · 语义最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/Q06/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

declared_context_keys = []

skill.constraints[0]: Call index.search exactly once in this workflow.

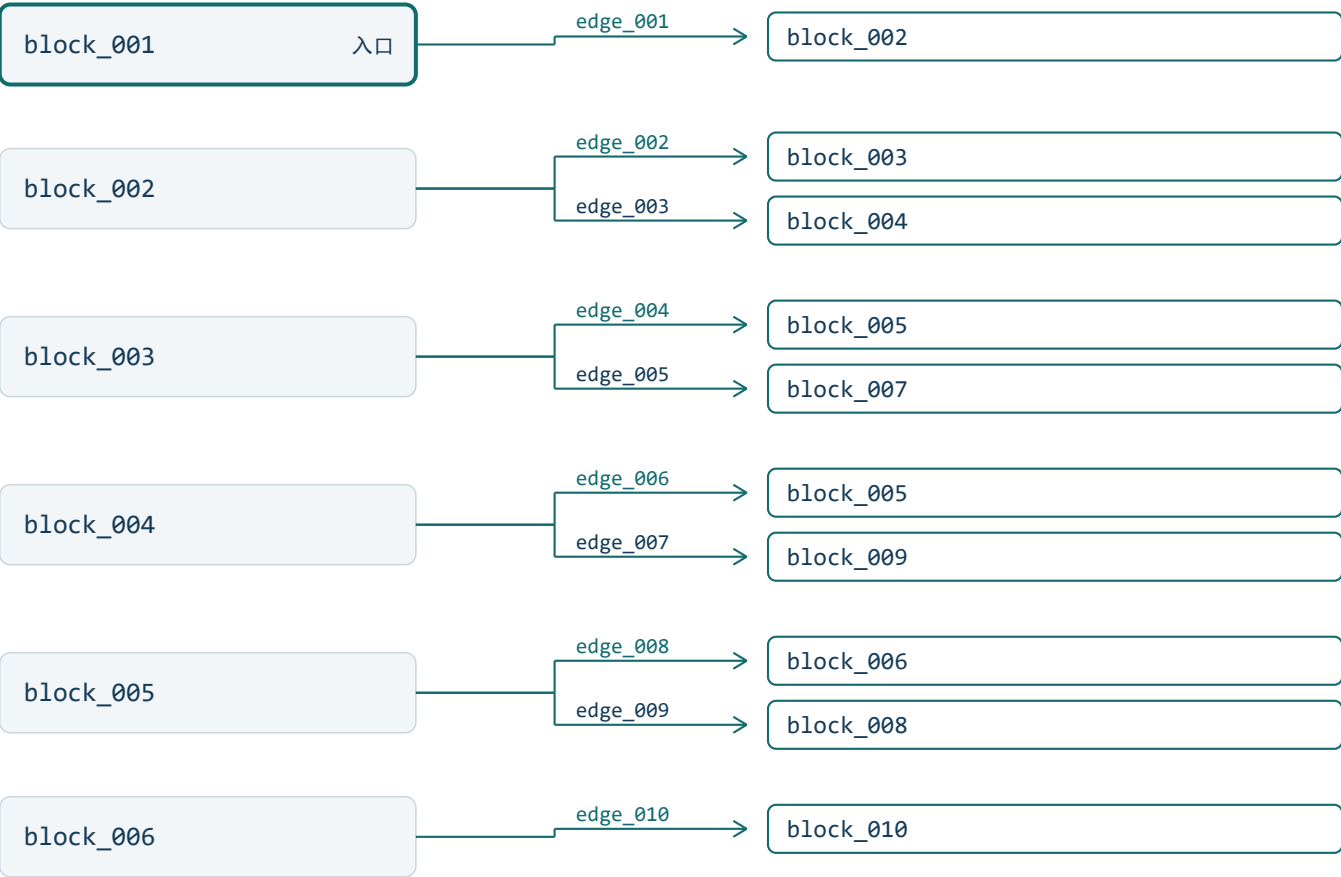
skill.constraints[1]: Do not invoke index.delete anywhere in this workflow.

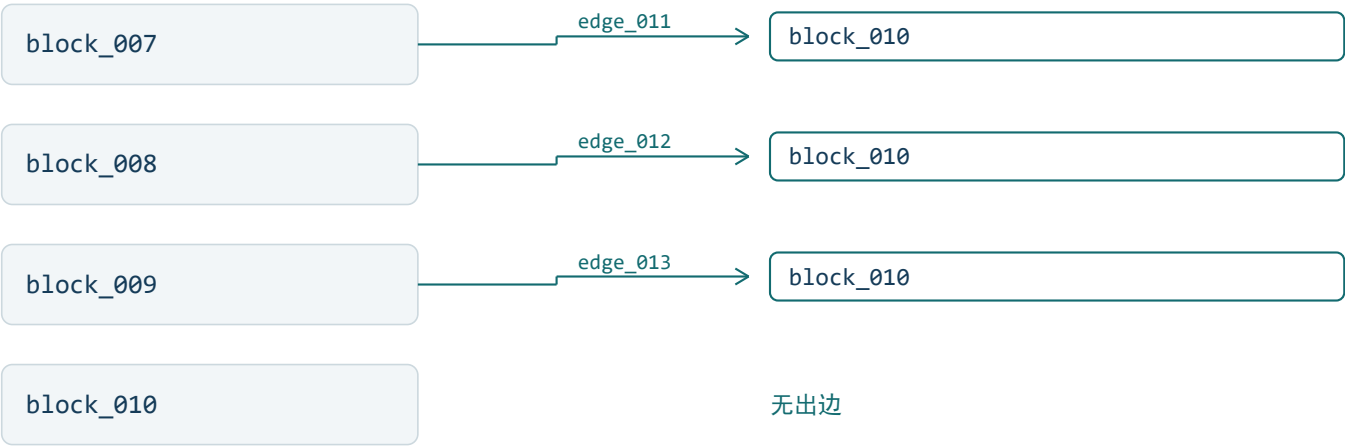
skill.constraints[2]: index.search supports fuzzy matching; this note does not request fuzzy expansion in this workflow.

skill.constraints[3]: Parameter descriptions are call requirements and do not instruct pre-call validation or normalization.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"from_date is present"
edge_003	block_002 -> block_004	"from_date is missing"
edge_004	block_003 -> block_005	"limit is present"
edge_005	block_003 -> block_007	"limit is missing"
edge_006	block_004 -> block_005	"limit is present"
edge_007	block_004 -> block_009	"limit is missing"
edge_008	block_005 -> block_006	"from_date is present"
edge_009	block_005 -> block_008	"from_date is missing"
edge_010	block_006 -> block_010	null
edge_011	block_007 -> block_010	null
edge_012	block_008 -> block_010	null
edge_013	block_009 -> block_010	null

全部操作、数据引用与约束落点

```
block_001 Read request.json

data_source_kind = "external"
block_001.constraints: []

ir_001 read_request_json

inputs[0] external_resource: request.json | request.json
outputs[0] result: result_001 | request_document

ir_001.constraints: []

draft_instruction_id = read_request_json
```

ir_002 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_002.constraints: []
```

```
draft_instruction_id = go_inspect
```

block_002 Inspect request parameters

```
data_source_kind = null
```

```
block_002.constraints: []
```

ir_003 inspect_request_parameters

```
inputs[0] result: result_001 | request_document
```

```
outputs[0] result: result_002 | request_term
```

```
outputs[1] result: result_003 | from_date_present
```

```
outputs[2] result: result_004 | limit_present
```

```
ir_003.constraints: []
```

```
draft_instruction_id = inspect_parameters
```

ir_004 dispatch

```
inputs[0] result: result_003 | from_date_present
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = dispatch_from_date
```

block_003 Extract from_date and check limit presence

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 extract_from_date_value

```
inputs[0] result: result_001 | request_document
```

```
outputs[0] result: result_005 | request_from_date
```

```
ir_005.constraints: []
```

```
draft_instruction_id = extract_from_date
```

ir_006 dispatch

```
inputs[0] result: result_004 | limit_present
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = dispatch_limit
```

block_004 from_date is missing; check limit presence

```
data_source_kind = null
```

```
block_004.constraints: []
```

ir_007 dispatch

```
inputs[0] result: result_004 | limit_present
```

```
outputs = []
```

```
ir_007.constraints: []
```

```
draft_instruction_id = dispatch_limit_from_missing_date
```

block_005 Extract limit value

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_008 extract_limit_value

```
inputs[0] result: result_001 | request_document
```

```
outputs[0] result: result_006 | request_limit
```

```
ir_008.constraints: []
```

```
draft_instruction_id = extract_limit
```

ir_009 dispatch

```
inputs[0] result: result_003 | from_date_present
```

```
outputs = []
```

```
ir_009.constraints: []
```

```
draft_instruction_id = dispatch_from_date_after_limit
```

block_006 Call index.search with from_date and limit

```
data_source_kind = "external"
```

```
block_006.constraints: []
```

ir_010 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_002 | request_term
```

```
inputs[2] result: result_005 | request_from_date
```

```
inputs[3] result: result_006 | request_limit
```

```
outputs[0] result: result_007 | search_response_pp
```

```
outputs[1] result: result_008 | search_total_pp
```

```
outputs[2] result: result_009 | search_items_pp
```

```
ir_010.constraints[0]: Use request.term unchanged as query argument.
```

```
ir_010.constraints[1]: When from_date is present, pass its value unchanged; accepted format is YYYY-MM-DD.
```

```
ir_010.constraints[2]: When limit is present, pass its value unchanged as the limit argument.
```

```
draft_instruction_id = search_pp
```

ir_011 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = go_finish_pp
```

block_007 Call index.search with from_date and default limit

```
data_source_kind = "external"
```

```
block_007.constraints: []
```

ir_012 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_002 | request_term
```

```
inputs[2] result: result_005 | request_from_date
```

```
inputs[3] literal = 10
```

```
outputs[0] result: result_010 | search_response_pm
```

```
outputs[1] result: result_011 | search_total_pm
```

```
outputs[2] result: result_012 | search_items_pm
```

```
ir_012.constraints[0]: Use request.term unchanged as query argument.
```

```
ir_012.constraints[1]: When from_date is present, pass its value unchanged; accepted format is YYYY-MM-DD.
```

```
ir_012.constraints[2]: When limit is missing, pass 10 as the limit argument.
```

```
draft_instruction_id = search_pm
```

ir_013 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = go_finish_pm
```

block_008 Call index.search without from_date but with limit

```
data_source_kind = "external"
```

```
block_008.constraints: []
```

ir_014 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_002 | request_term
```

```
inputs[2] result: result_006 | request_limit
```

```
outputs[0] result: result_013 | search_response_mp
```

```
outputs[1] result: result_014 | search_total_mp
```

```
outputs[2] result: result_015 | search_items_mp
```

```
ir_014.constraints[0]: Use request.term unchanged as query argument.
```

```
ir_014.constraints[1]: When from_date is missing, omit the from_date argument.
```

```
ir_014.constraints[2]: When limit is present, pass its value unchanged as the limit argument.
```

```
draft_instruction_id = search_mp
```


ir_015 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_015.constraints: []
```

```
draft_instruction_id = go_finish_mp
```

block_009 Call index.search without from_date and with default limit

```
data_source_kind = "external"
```

```
block_009.constraints: []
```

ir_016 index.search

```
inputs[0] external_resource: index.search | index.search
```

```
inputs[1] result: result_002 | request_term
```

```
inputs[2] literal = 10
```

```
outputs[0] result: result_016 | search_response_mm
```

```
outputs[1] result: result_017 | search_total_mm
```

```
outputs[2] result: result_018 | search_items_mm
```

```
ir_016.constraints[0]: Use request.term unchanged as query argument.
```

```
ir_016.constraints[1]: When from_date is missing, omit the from_date argument.
```

```
ir_016.constraints[2]: When limit is missing, pass 10 as the limit argument.
```

```
draft_instruction_id = search_mm
```

ir_017 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_017.constraints: []
```

```
draft_instruction_id = go_finish_mm
```

block_010 Write total and return items

```
data_source_kind = null
```

```
block_010.constraints: []
```

ir_018 write_search_total_to_count_txt

```
inputs[0] result: result_008 | search_total_pp
```

```
inputs[1] result: result_011 | search_total_pm
```

```
inputs[2] result: result_014 | search_total_mp
```

```
inputs[3] result: result_017 | search_total_mm
```

```
inputs[4] external_resource: count.txt | count.txt
```

```
outputs = []
```

```
ir_018.constraints[0]: Write the search response's total value to local count.txt.
```

```
draft_instruction_id = write_total
```

ir_019 return

```
inputs[0] result: result_009 | search_items_pp
```

```
inputs[1] result: result_012 | search_items_pm
```

```
inputs[2] result: result_015 | search_items_mp
inputs[3] result: result_018 | search_items_mm
outputs = []
ir_019.constraints[0]: Return the search response's items value unchanged.
draft_instruction_id = return_items
```

F01 / 第 1 次 实际输出明细

工具选择与失败回退 · 编号步骤

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F01/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: The exact final status text appended to status.txt is not specified; 'success' and 'failure' literals are used to represent the final status.

Skill 层约束与上下文声明

```
declared_context_keys = ["source_id", "FAST_KEY"]

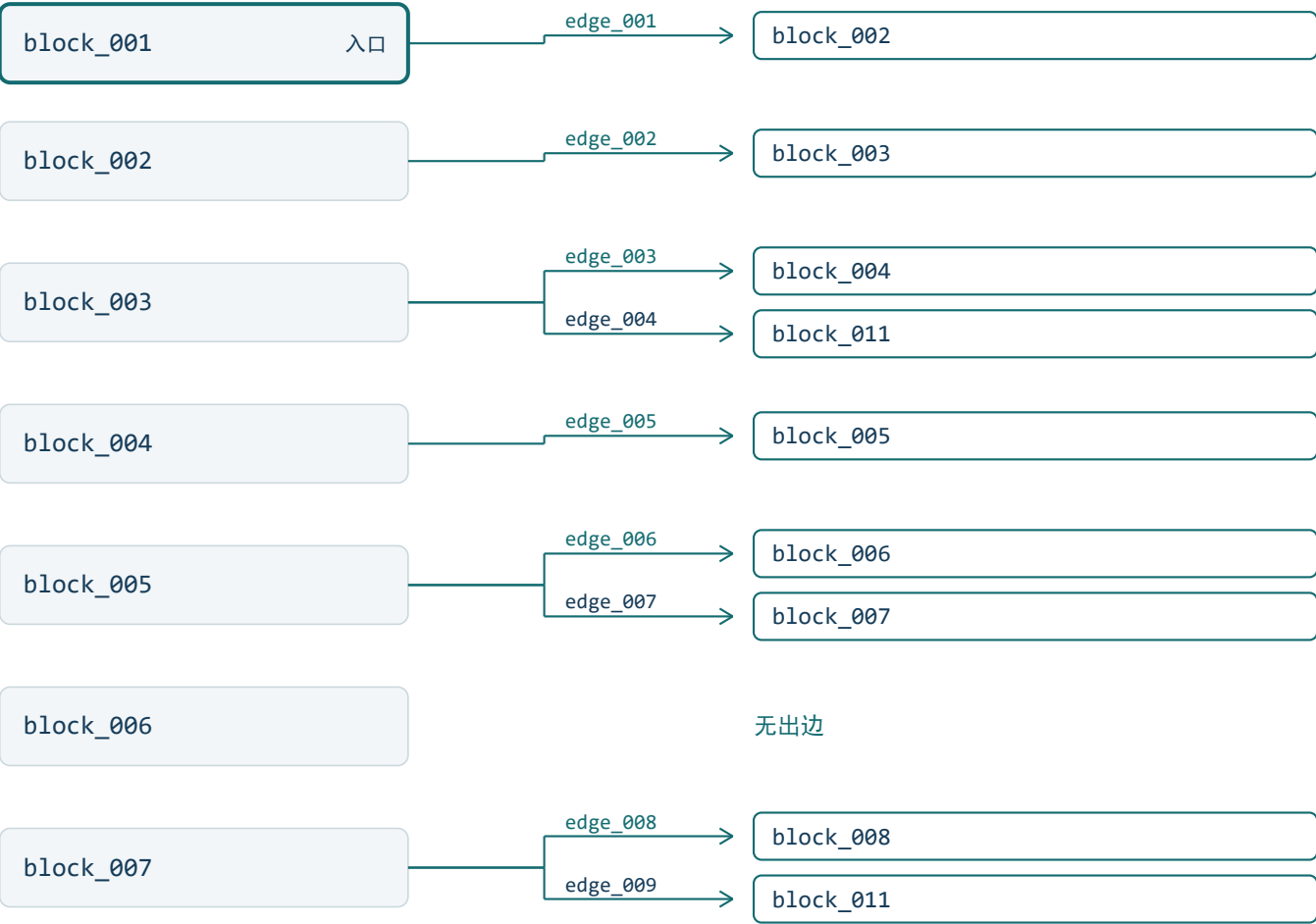
skill.constraints[0]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

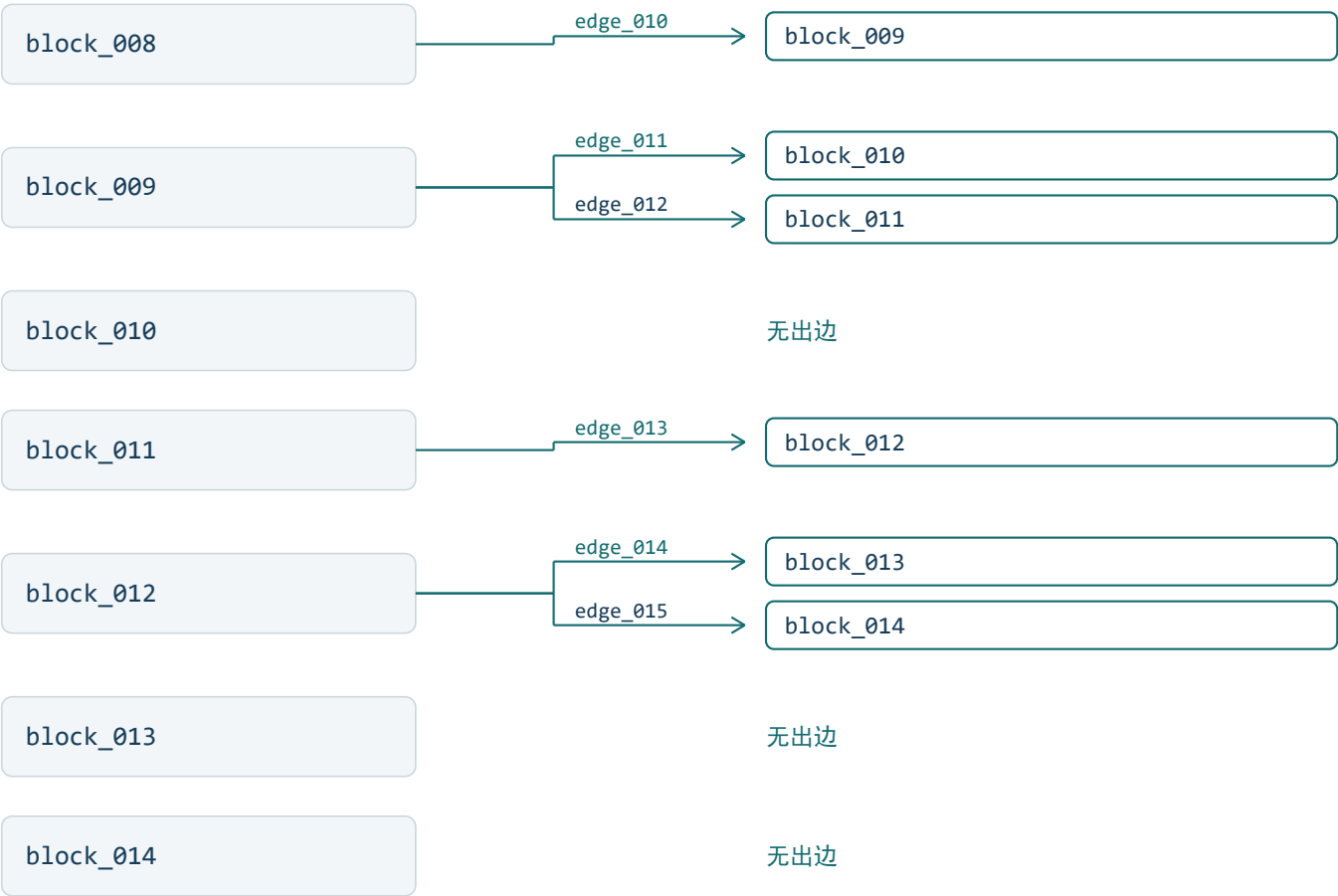
skill.constraints[1]: Before returning from every success or failure path, append the final status to local status.txt.

skill.constraints[2]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"FAST_KEY is present"
edge_004	block_003 -> block_011	"FAST_KEY is absent"
edge_005	block_004 -> block_005	null
edge_006	block_005 -> block_006	"first fast.fetch attempt succeeded"
edge_007	block_005 -> block_007	"first fast.fetch attempt failed"
edge_008	block_007 -> block_008	"first fast.fetch failure was transient"
edge_009	block_007 -> block_011	"first fast.fetch failure was non-transient"
edge_010	block_008 -> block_009	null
edge_011	block_009 -> block_010	"fast.fetch retry succeeded"
edge_012	block_009 -> block_011	"fast.fetch retry failed"
edge_013	block_011 -> block_012	null
edge_014	block_012 -> block_013	"archive.fetch succeeded"
edge_015	block_012 -> block_014	"archive.fetch failed"

全部操作、数据引用与约束落点

block_001 Read source_id from the user's request

```
data_source_kind = "context"
block_001.constraints: []

ir_001 read_source_id_from_request

inputs[0] context_key: source_id | source_id
outputs[0] result: result_001 | source_id_value
ir_001.constraints: []
draft_instruction_id = read_source_id

ir_002 dispatch

inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_read_fast_key
```

block_002 Read FAST_KEY from the environment

```
data_source_kind = "context"
```

```
block_002.constraints: []
```

ir_003 read_fast_key_from_environment

```
inputs[0] context_key: FAST_KEY | FAST_KEY
```

```
outputs[0] result: result_002 | fast_key
```

```
ir_003.constraints: []
```

```
draft_instruction_id = read_fast_key
```

ir_004 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = go_check_fast_key
```

block_003 Check whether FAST_KEY is present

```
data_source_kind = null
```

```
block_003.constraints[0]: If FAST_KEY is present, try fast.fetch first with source_id and FAST_KEY; if it is absent, go directly to archive.fetch without calling fast.fetch.
```

ir_005 check_fast_key_presence

```
inputs[0] result: result_002 | fast_key
```

```
outputs[0] result: result_003 | fast_key_present
```

```
ir_005.constraints: []
```

```
draft_instruction_id = check_fast_key
```

ir_006 dispatch

```
inputs[0] result: result_003 | fast_key_present
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = go_branch_on_fast_key
```

block_004 Try fast.fetch first with source_id and FAST_KEY

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_007 call_fast_fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id_value
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_004 | fast_fetch_response_body
```

```
outputs[1] result: result_005 | fast_fetch_error
```

```
ir_007.constraints: []
```

```
draft_instruction_id = call_fast_fetch_first
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_check_fast_fetch_first_success
```

block_005 Check whether the first fast.fetch attempt succeeded

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 check_fast_fetch_success

```
inputs[0] result: result_005 | fast_fetch_error
```

```
outputs[0] result: result_006 | fast_fetch_succeeded
```

```
ir_009.constraints: []
```

```
draft_instruction_id = check_fast_fetch_first_success
```

ir_010 dispatch

```
inputs[0] result: result_006 | fast_fetch_succeeded
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_branch_on_fast_fetch_first_success
```

block_006 Append success status and return the first fast.fetch response body

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 append_final_status_to_status_file

```
inputs[0] external_resource: status.txt | local status.txt
```

```
inputs[1] literal = "success"
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = append_success_status_first_fast_fetch
```

ir_012 return

```
inputs[0] result: result_004 | fast_fetch_response_body
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = return_first_fast_fetch_body
```

block_007 Check whether the first fast.fetch failure was transient

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 check_transient_error

```
inputs[0] result: result_005 | fast_fetch_error
```

```
outputs[0] result: result_007 | fast_fetch_error_transient
```

```
ir_013.constraints: []
```

```
draft_instruction_id = check_fast_fetch_first_transient
```

ir_014 dispatch

```
inputs[0] result: result_007 | fast_fetch_error_transient
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = go_branch_on_fast_fetch_first_transient
```

block_008 Retry fast.fetch once

```
data_source_kind = "external"
```

```
block_008.constraints: []
```

ir_015 retry_fast_fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id_value
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_008 | retry_fast_fetch_response_body
```

```
outputs[1] result: result_009 | retry_fast_fetch_error
```

```
ir_015.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do not retry a non-transient first failure.
```

```
draft_instruction_id = call_fast_fetch_retry
```

ir_016 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = go_check_retry_fast_fetch_success
```

block_009 Check whether the fast.fetch retry succeeded

```
data_source_kind = null
```

```
block_009.constraints: []
```

ir_017 check_fast_fetch_success

```
inputs[0] result: result_009 | retry_fast_fetch_error
```

```
outputs[0] result: result_010 | retry_fast_fetch_succeeded
```

```
ir_017.constraints: []
```

```
draft_instruction_id = check_retry_fast_fetch_success
```

ir_018 dispatch

```
inputs[0] result: result_010 | retry_fast_fetch_succeeded
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = go_branch_on_retry_fast_fetch_success
```


block_010 Append success status and return the fast.fetch retry response body

```
data_source_kind = null
```

```
block_010.constraints: []
```

ir_019 append_final_status_to_status_file

```
inputs[0] external_resource: status.txt | local status.txt
```

```
inputs[1] literal = "success"
```

```
outputs = []
```

```
ir_019.constraints: []
```

```
draft_instruction_id = append_success_status_retry_fast_fetch
```

ir_020 return

```
inputs[0] result: result_008 | retry_fast_fetch_response_body
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = return_retry_fast_fetch_body
```

block_011 Call archive.fetch once with source_id

```
data_source_kind = "external"
```

```
block_011.constraints: []
```

ir_021 call_archive_fetch

```
inputs[0] external_resource: archive.fetch | archive.fetch
```

```
inputs[1] result: result_001 | source_id_value
```

```
outputs[0] result: result_011 | archive_fetch_response_body
```

```
outputs[1] result: result_012 | archive_fetch_error
```

```
ir_021.constraints[0]: After a non-transient first failure or any failed retry of fast.fetch, call archive.fetch with source_id.
```

```
ir_021.constraints[1]: Call archive.fetch at most once and pass source_id as its only argument.
```

```
ir_021.constraints[2]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

```
draft_instruction_id = call_archive_fetch
```

ir_022 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_022.constraints: []
```

```
draft_instruction_id = go_check_archive_fetch_success
```

block_012 Check whether archive.fetch succeeded

```
data_source_kind = null
```

```
block_012.constraints: []
```

ir_023 check_archive_fetch_success

```
inputs[0] result: result_012 | archive_fetch_error
```

```
outputs[0] result: result_013 | archive_fetch_succeeded
```

```
ir_023.constraints: []
```

```
draft_instruction_id = check_archive_fetch_success
```

ir_024 dispatch

```
inputs[0] result: result_013 | archive_fetch_succeeded
```

```
outputs = []
```

```
ir_024.constraints: []
```

```
draft_instruction_id = go_branch_on_archive_fetch_success
```

block_013 Append success status and return the archive.fetch response body

```
data_source_kind = null
```

```
block_013.constraints: []
```

ir_025 append_final_status_to_status_file

```
inputs[0] external_resource: status.txt | local status.txt
```

```
inputs[1] literal = "success"
```

```
outputs = []
```

```
ir_025.constraints: []
```

```
draft_instruction_id = append_success_status_archive_fetch
```

ir_026 return

```
inputs[0] result: result_011 | archive_fetch_response_body
```

```
outputs = []
```

```
ir_026.constraints: []
```

```
draft_instruction_id = return_archive_fetch_body
```

block_014 Append failure status and return the archive.fetch error

```
data_source_kind = null
```

```
block_014.constraints: []
```

ir_027 append_final_status_to_status_file

```
inputs[0] external_resource: status.txt | local status.txt
```

```
inputs[1] literal = "failure"
```

```
outputs = []
```

```
ir_027.constraints: []
```

```
draft_instruction_id = append_failure_status_archive_fetch
```

ir_028 return

```
inputs[0] result: result_012 | archive_fetch_error
```

```
outputs = []
```

```
ir_028.constraints: []
```

```
draft_instruction_id = return_archive_fetch_error
```

F01 / 第 2 次 实际输出明细

工具选择与失败回退 · 编号步骤

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F01/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

```
declared_context_keys = ["source_id", "FAST_KEY"]

skill.constraints[0]: If FAST_KEY is present, try fast.fetch first with source_id and FAST_KEY; if it is absent,
go directly to archive.fetch without calling fast.fetch.

skill.constraints[1]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do
not retry a non-transient first failure.

skill.constraints[2]: After a non-transient first failure or any failed retry of fast.fetch, call archive.fetch
with source_id.

skill.constraints[3]: Call archive.fetch at most once and pass source_id as its only argument.

skill.constraints[4]: On either tool's success, return that successful response's body value unchanged and make no
further fetch calls.

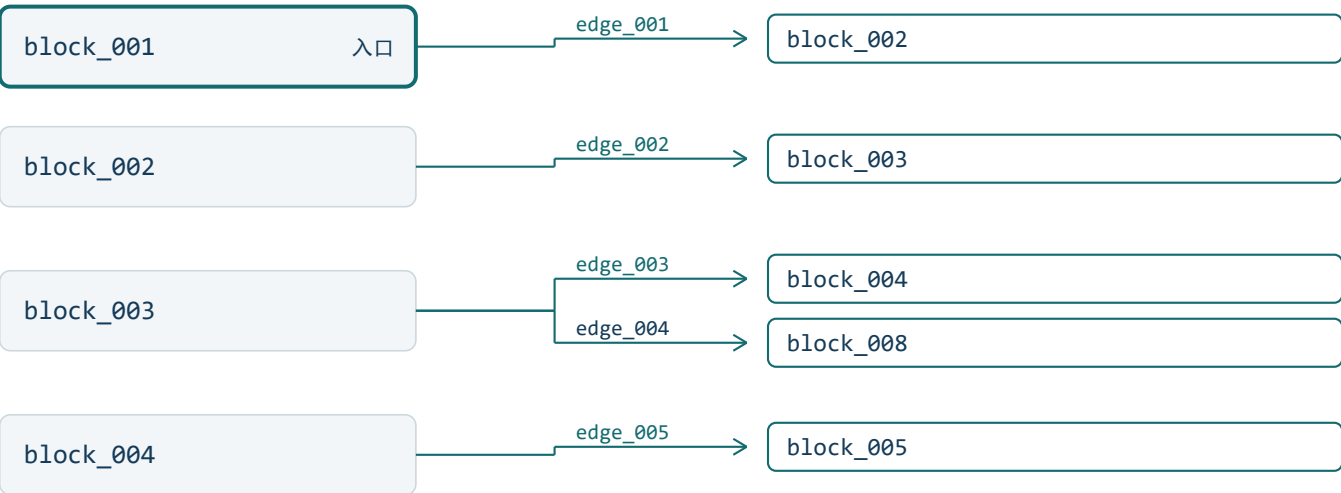
skill.constraints[5]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.

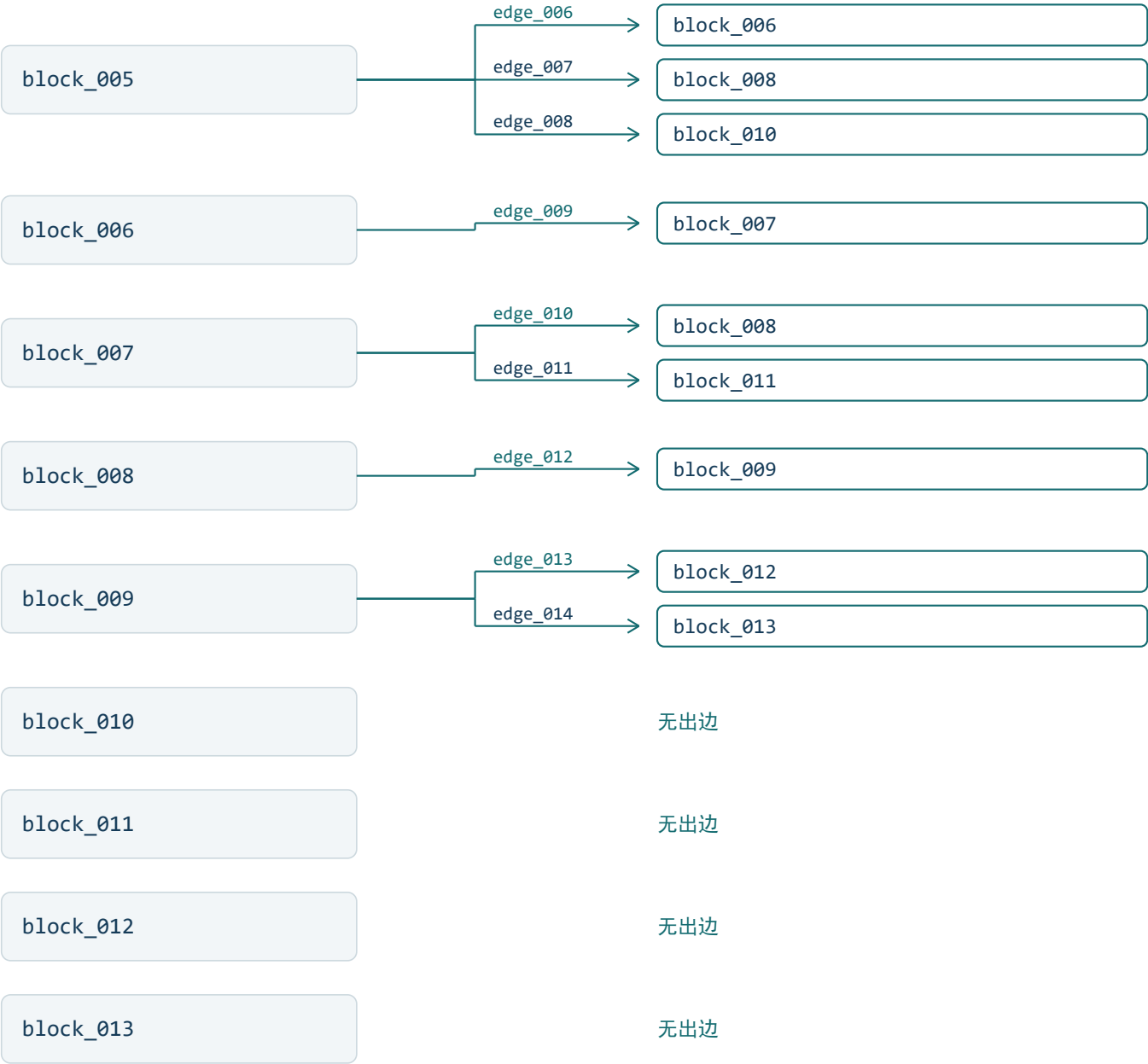
skill.constraints[6]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

skill.constraints[7]: Before returning from every success or failure path, append the final status to local
status.txt.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"FAST_KEY present"
edge_004	block_003 -> block_008	"FAST_KEY absent"
edge_005	block_004 -> block_005	null
edge_006	block_005 -> block_006	"transient error"
edge_007	block_005 -> block_008	"non-transient error"
edge_008	block_005 -> block_010	"success"
edge_009	block_006 -> block_007	null
edge_010	block_007 -> block_008	"retry failed"
edge_011	block_007 -> block_011	"retry success"
edge_012	block_008 -> block_009	null
edge_013	block_009 -> block_012	"success"
edge_014	block_009 -> block_013	"failure"

全部操作、数据引用与约束落点

block_001 Read source_id from the user's request

```
data_source_kind = "context"
block_001.constraints: []

ir_001 read_source_id_from_request

inputs[0] context_key: source_id | source_id
outputs[0] result: result_001 | source_id
ir_001.constraints: []

draft_instruction_id = read_source_id_from_request
```

ir_002 dispatch

```
inputs = []
outputs = []

ir_002.constraints: []

draft_instruction_id = go_read_fast_key
```

block_002 Read FAST_KEY from the environment

data_source_kind = "context"

block_002.constraints: []

ir_003 read_fast_key_from_environment

inputs[0] context_key: FAST_KEY | FAST_KEY

outputs[0] result: result_002 | fast_key

ir_003.constraints: []

draft_instruction_id = read_fast_key_from_environment

ir_004 dispatch

inputs = []

outputs = []

ir_004.constraints: []

draft_instruction_id = go_check_fast_key_presence

block_003 Check whether FAST_KEY is present

data_source_kind = null

block_003.constraints[0]: If FAST_KEY is present, try fast.fetch first with source_id and FAST_KEY; if it is absent, go directly to archive.fetch without calling fast.fetch.

ir_005 check_fast_key_presence

inputs[0] result: result_002 | fast_key

outputs[0] result: result_003 | fast_key_present

ir_005.constraints: []

draft_instruction_id = check_fast_key_present

ir_006 dispatch

inputs[0] result: result_003 | fast_key_present

outputs = []

ir_006.constraints: []

draft_instruction_id = dispatch_on_fast_key_presence

block_004 Try fast.fetch first with source_id and FAST_KEY

data_source_kind = "external"

block_004.constraints[0]: Try fast.fetch first with source_id and FAST_KEY only when FAST_KEY is present.

ir_007 fast_fetch

inputs[0] external_resource: fast.fetch | fast.fetch

inputs[1] result: result_001 | source_id

inputs[2] result: result_002 | fast_key

outputs[0] result: result_004 | fast_fetch_response

outputs[1] result: result_005 | fast_fetch_body

ir_007.constraints[0]: Try fast.fetch first with source_id and FAST_KEY only when FAST_KEY is present.

draft_instruction_id = fast_fetch_first_attempt

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_evaluate_fast_fetch_first
```

block_005 Evaluate the first fast.fetch attempt

```
data_source_kind = null
```

```
block_005.constraints[0]: After a non-transient first failure of fast.fetch, call archive.fetch with source_id.
```

```
block_005.constraints[1]: Retry fast.fetch exactly once only when its first attempt fails with a transient error;  
do not retry a non-transient first failure.
```

ir_009 evaluate_fast_fetch_outcome

```
inputs[0] result: result_004 | fast_fetch_response
```

```
outputs[0] result: result_006 | fast_fetch_first_outcome
```

```
ir_009.constraints: []
```

```
draft_instruction_id = evaluate_fast_fetch_first_outcome
```

ir_010 dispatch

```
inputs[0] result: result_006 | fast_fetch_first_outcome
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_on_fast_fetch_first_outcome
```

block_006 Retry fast.fetch once after a transient first-attempt error

```
data_source_kind = "external"
```

```
block_006.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error;  
do not retry a non-transient first failure.
```

ir_011 fast_fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_007 | fast_fetch_retry_response
```

```
outputs[1] result: result_008 | fast_fetch_retry_body
```

```
ir_011.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do  
not retry a non-transient first failure.
```

```
draft_instruction_id = fast_fetch_retry_attempt
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_evaluate_fast_fetch_retry
```

block_007 Evaluate the fast.fetch retry

```
data_source_kind = null
```

```
block_007.constraints[0]: After any failed retry of fast.fetch, call archive.fetch with source_id.
```

ir_013 evaluate_fast_fetch_retry_outcome

```
inputs[0] result: result_007 | fast_fetch_retry_response
```

```
outputs[0] result: result_009 | fast_fetch_retry_outcome
```

```
ir_013.constraints: []
```

```
draft_instruction_id = evaluate_fast_fetch_retry_outcome
```

ir_014 dispatch

```
inputs[0] result: result_009 | fast_fetch_retry_outcome
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = dispatch_on_fast_fetch_retry_outcome
```

block_008 Call archive.fetch with source_id

```
data_source_kind = "external"
```

```
block_008.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.
```

```
block_008.constraints[1]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.
```

ir_015 archive_fetch

```
inputs[0] external_resource: archive.fetch | archive.fetch
```

```
inputs[1] result: result_001 | source_id
```

```
outputs[0] result: result_010 | archive_response
```

```
outputs[1] result: result_011 | archive_body
```

```
outputs[2] result: result_012 | archive_error
```

```
ir_015.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.
```

```
ir_015.constraints[1]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.
```

```
draft_instruction_id = archive_fetch_call
```

ir_016 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = go_evaluate_archive_fetch
```

block_009 Evaluate the archive.fetch result

```
data_source_kind = null
```

```
block_009.constraints[0]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

ir_017 evaluate_archive_fetch_outcome

```
inputs[0] result: result_010 | archive_response
```

```
outputs[0] result: result_013 | archive_outcome
```

```
ir_017.constraints: []
```

```
draft_instruction_id = evaluate_archive_fetch_outcome
```


ir_018 dispatch

inputs[0] result: result_013 | archive_outcome

outputs = []

ir_018.constraints: []

draft_instruction_id = dispatch_on_archive_fetch_outcome

block_010 Append status and return the successful fast.fetch body

data_source_kind = null

block_010.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

block_010.constraints[1]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

ir_019 append_final_status_to_status_file

inputs[0] external_resource: local status.txt | local status.txt

inputs[1] literal = "success"

outputs = []

ir_019.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

draft_instruction_id = append_status_fast_fetch_first_success

ir_020 return

inputs[0] result: result_005 | fast_fetch_body

outputs = []

ir_020.constraints[0]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

draft_instruction_id = return_fast_fetch_first_body

block_011 Append status and return the successful fast.fetch retry body

data_source_kind = null

block_011.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

block_011.constraints[1]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

ir_021 append_final_status_to_status_file

inputs[0] external_resource: local status.txt | local status.txt

inputs[1] literal = "success"

outputs = []

ir_021.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

draft_instruction_id = append_status_fast_fetch_retry_success

ir_022 return

```
inputs[0] result: result_008 | fast_fetch_retry_body
```

```
outputs = []
```

ir_022.constraints[0]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

```
draft_instruction_id = return_fast_fetch_retry_body
```

block_012 Append status and return the successful archive.fetch body

```
data_source_kind = null
```

block_012.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

block_012.constraints[1]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

ir_023 append_final_status_to_status_file

```
inputs[0] external_resource: local status.txt | local status.txt
```

```
inputs[1] literal = "success"
```

```
outputs = []
```

ir_023.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

```
draft_instruction_id = append_status_archive_success
```

ir_024 return

```
inputs[0] result: result_011 | archive_body
```

```
outputs = []
```

ir_024.constraints[0]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

```
draft_instruction_id = return_archive_body
```

block_013 Append status and return the archive.fetch error

```
data_source_kind = null
```

block_013.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

block_013.constraints[1]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.

ir_025 append_final_status_to_status_file

```
inputs[0] external_resource: local status.txt | local status.txt
```

```
inputs[1] literal = "failure"
```

```
outputs = []
```

ir_025.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

```
draft_instruction_id = append_status_archive_failure
```

ir_026 return

```
inputs[0] result: result_012 | archive_error
```

```
outputs = []
```

```
ir_026.constraints[0]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

```
draft_instruction_id = return_archive_error
```

F01 / 第 3 次 实际输出明细

工具选择与失败回退 · 编号步骤

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F01/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

```
declared_context_keys = ["user_request.source_id", "environment.FAST_KEY"]

skill.constraints[0]: If FAST_KEY is present, try fast.fetch first with source_id and FAST_KEY; if it is absent, go directly to archive.fetch without calling fast.fetch.

skill.constraints[1]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do not retry a non-transient first failure.

skill.constraints[2]: After a non-transient first failure or any failed retry of fast.fetch, call archive.fetch with source_id.

skill.constraints[3]: Call archive.fetch at most once and pass source_id as its only argument.

skill.constraints[4]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

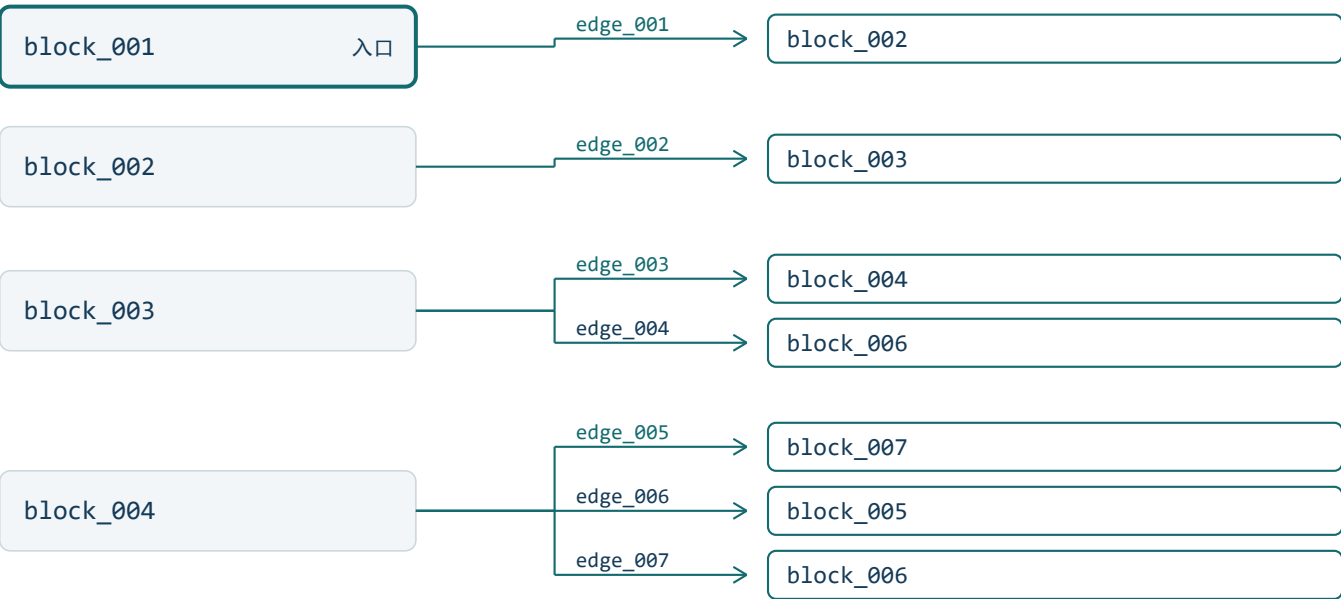
skill.constraints[5]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.

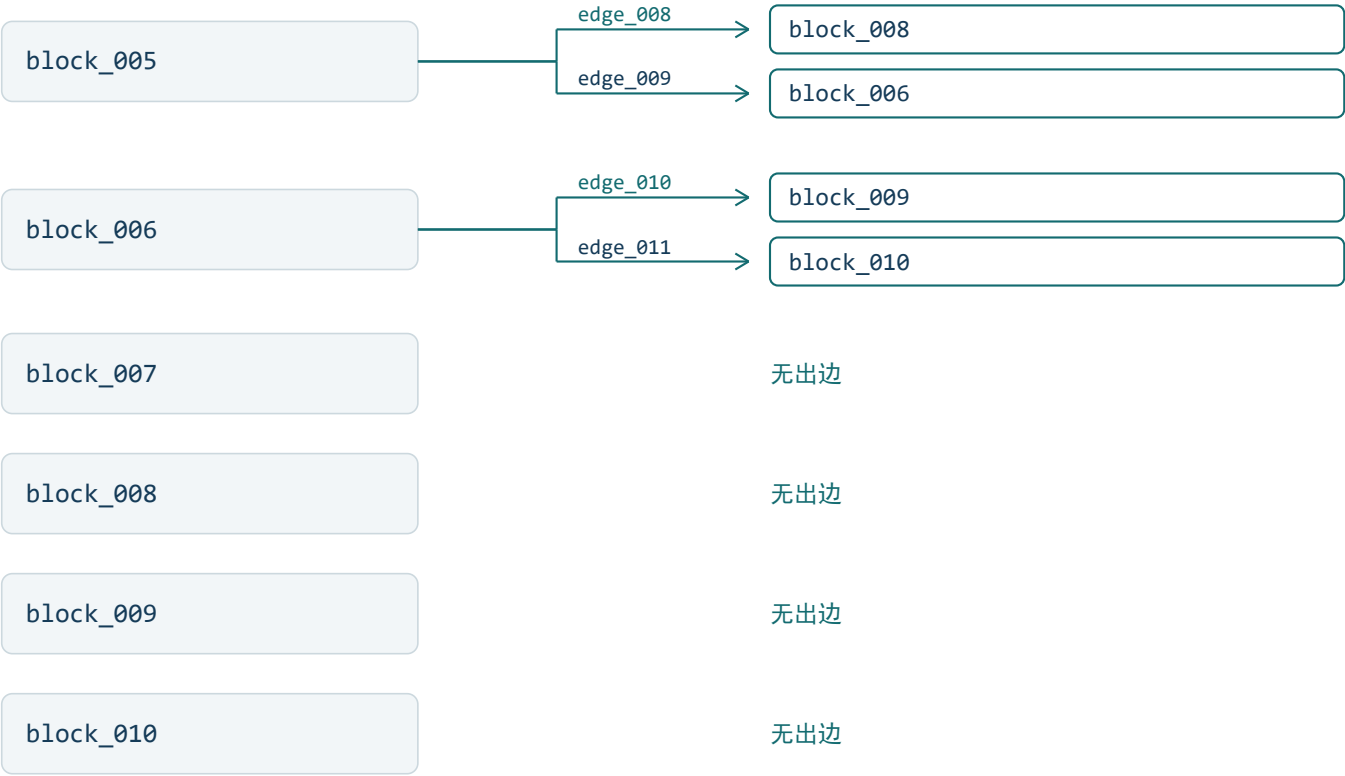
skill.constraints[6]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

skill.constraints[7]: Before returning from every success or failure path, append the final status to local status.txt.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"FAST_KEY present"
edge_004	block_003 -> block_006	"FAST_KEY absent"
edge_005	block_004 -> block_007	"fast.fetch first attempt succeeds"
edge_006	block_004 -> block_005	"fast.fetch first attempt fails with a transient error"
edge_007	block_004 -> block_006	"fast.fetch first attempt fails with a non-transient error"
edge_008	block_005 -> block_008	"fast.fetch retry succeeds"
edge_009	block_005 -> block_006	"fast.fetch retry fails"
edge_010	block_006 -> block_009	"archive.fetch succeeds"
edge_011	block_006 -> block_010	"archive.fetch fails"

全部操作、数据引用与约束落点

block_001 Read `source_id` from the user's request

`data_source_kind = "context"`

`block_001.constraints: []`

ir_001 `read_source_id_from_request`

`inputs[0] context_key: user_request.source_id | user_request.source_id`

`outputs[0] result: result_001 | source_id`

`ir_001.constraints: []`

`draft_instruction_id = read_source_id_from_request`

ir_002 `dispatch`

`inputs = []`

`outputs = []`

`ir_002.constraints: []`

`draft_instruction_id = go_read_fast_key`

block_002 Read `FAST_KEY` from the environment

`data_source_kind = "context"`

`block_002.constraints: []`

ir_003 `read_fast_key_from_environment`

`inputs[0] context_key: environment.FAST_KEY | environment.FAST_KEY`

`outputs[0] result: result_002 | fast_key`

`ir_003.constraints: []`

`draft_instruction_id = read_fast_key_from_environment`

ir_004 `dispatch`

`inputs = []`

`outputs = []`

`ir_004.constraints: []`

`draft_instruction_id = go_check_fast_key_present`

block_003 Check whether `FAST_KEY` is present

`data_source_kind = null`

`block_003.constraints: []`

ir_005 `check_fast_key_present`

`inputs[0] result: result_002 | fast_key`

`outputs[0] result: result_003 | fast_key_present`

`ir_005.constraints: []`

`draft_instruction_id = check_fast_key_present`

ir_006 dispatch

```
inputs[0] result: result_003 | fast_key_present
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = dispatch_fast_key_present
```

block_004 Call fast.fetch first with source_id and FAST_KEY

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_007 fast.fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_004 | fast_first_body
```

```
outputs[1] result: result_005 | fast_first_status
```

```
ir_007.constraints: []
```

```
draft_instruction_id = call_fast_fetch_first
```

ir_008 dispatch

```
inputs[0] result: result_005 | fast_first_status
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_fast_first_outcome
```

block_005 Retry fast.fetch exactly once

```
data_source_kind = "external"
```

```
block_005.constraints: []
```

ir_009 fast.fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_006 | fast_retry_body
```

```
outputs[1] result: result_007 | fast_retry_status
```

```
ir_009.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do not retry a non-transient first failure.
```

```
draft_instruction_id = call_fast_fetch_retry
```

ir_010 dispatch

```
inputs[0] result: result_007 | fast_retry_status
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_fast_retry_outcome
```

block_006 Call archive.fetch with source_id only

```
data_source_kind = "external"
```

```
block_006.constraints: []
```

ir_011 archive.fetch

```
inputs[0] external_resource: archive.fetch | archive.fetch
```

```
inputs[1] result: result_001 | source_id
```

```
outputs[0] result: result_008 | archive_body
```

```
outputs[1] result: result_009 | archive_error
```

```
outputs[2] result: result_010 | archive_status
```

```
ir_011.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.
```

```
ir_011.constraints[1]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.
```

```
draft_instruction_id = call_archive_fetch
```

ir_012 dispatch

```
inputs[0] result: result_010 | archive_status
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_archive_outcome
```

block_007 Append status and return fast.fetch first success body

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 append_final_status_to_status_file

```
inputs[0] external_resource: status.txt | status.txt
```

```
inputs[1] result: result_005 | fast_first_status
```

```
outputs = []
```

```
ir_013.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

```
draft_instruction_id = append_status_after_fast_first_success
```

ir_014 return

```
inputs[0] result: result_004 | fast_first_body
```

```
outputs = []
```

```
ir_014.constraints[0]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.
```

```
draft_instruction_id = return_fast_first_body
```

block_008 Append status and return fast.fetch retry success body

```
data_source_kind = null
```

```
block_008.constraints: []
```

ir_015 append_final_status_to_status_file

```
inputs[0] external_resource: status.txt | status.txt
```

```
inputs[1] result: result_007 | fast_retry_status
```

```
outputs = []
```


ir_015.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

draft_instruction_id = append_status_after_fast_retry_success

ir_016 return

inputs[0] result: result_006 | fast_retry_body

outputs = []

ir_016.constraints[0]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

draft_instruction_id = return_fast_retry_body

block_009 Append status and return archive.fetch success body

data_source_kind = null

block_009.constraints: []

ir_017 append_final_status_to_status_file

inputs[0] external_resource: status.txt | status.txt

inputs[1] result: result_010 | archive_status

outputs = []

ir_017.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

draft_instruction_id = append_status_after_archive_success

ir_018 return

inputs[0] result: result_008 | archive_body

outputs = []

ir_018.constraints[0]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

draft_instruction_id = return_archive_body

block_010 Append status and return archive.fetch error

data_source_kind = null

block_010.constraints: []

ir_019 append_final_status_to_status_file

inputs[0] external_resource: status.txt | status.txt

inputs[1] result: result_010 | archive_status

outputs = []

ir_019.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

draft_instruction_id = append_status_after_archive_failure

ir_020 return

inputs[0] result: result_009 | archive_error

outputs = []

ir_020.constraints[0]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.

draft_instruction_id = return_archive_error

F02 / 第 1 次 实际输出明细

工具选择与失败回退 · 连贯段落

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F02/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

```
declared_context_keys = ["source_id", "FAST_KEY"]

skill.constraints[0]: If FAST_KEY is present, try fast.fetch first with source_id and FAST_KEY; if it is absent,
go directly to archive.fetch without calling fast.fetch.

skill.constraints[1]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do
not retry a non-transient first failure.

skill.constraints[2]: After a non-transient first failure or any failed retry of fast.fetch, call archive.fetch
with source_id.

skill.constraints[3]: Call archive.fetch at most once and pass source_id as its only argument.

skill.constraints[4]: On either tool's success, return that successful response's body value unchanged and make no
further fetch calls.

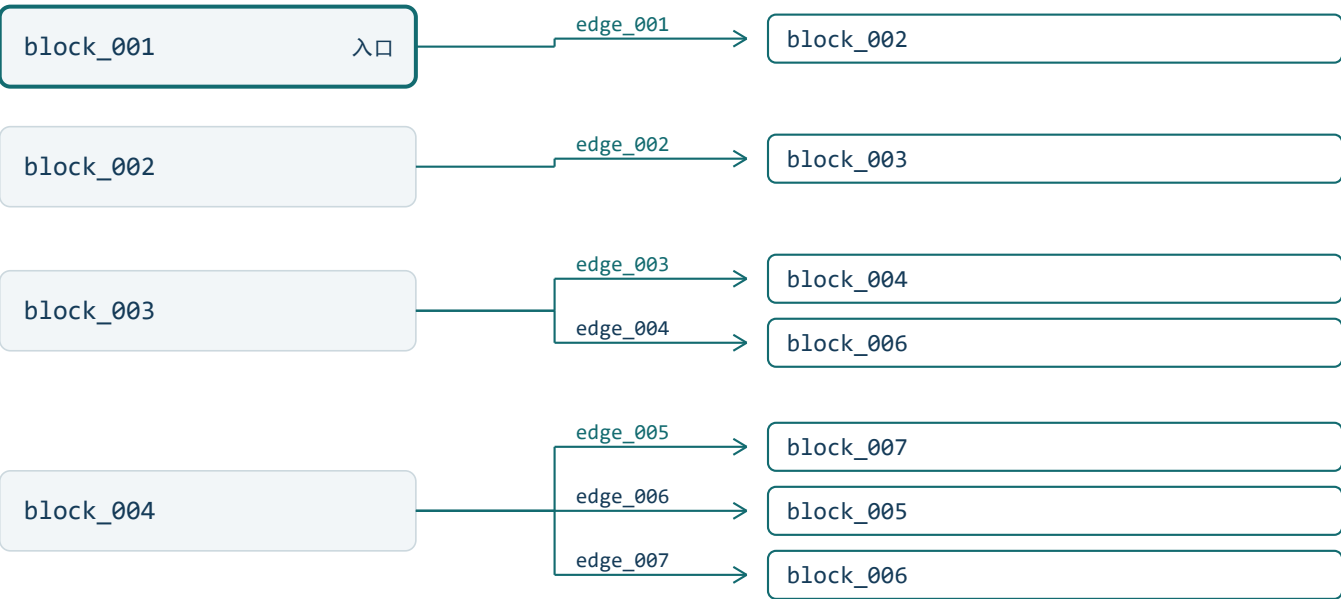
skill.constraints[5]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.

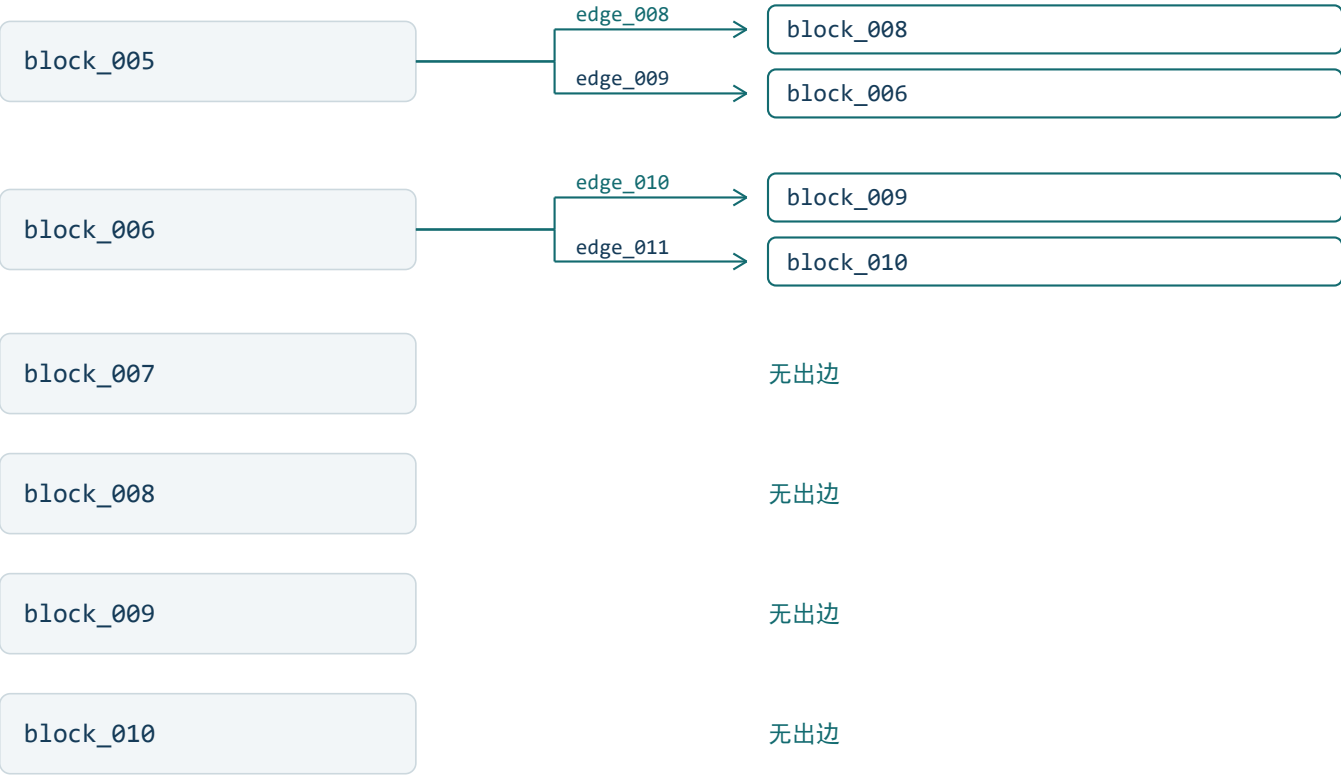
skill.constraints[6]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

skill.constraints[7]: Before returning from every success or failure path, append the final status to local
status.txt.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"FAST_KEY is present"
edge_004	block_003 -> block_006	"FAST_KEY is absent"
edge_005	block_004 -> block_007	"fast.fetch succeeds"
edge_006	block_004 -> block_005	"first fast.fetch attempt fails with a transient error"
edge_007	block_004 -> block_006	"first fast.fetch attempt fails with a non-transient error"
edge_008	block_005 -> block_008	"fast.fetch retry succeeds"
edge_009	block_005 -> block_006	"fast.fetch retry fails"
edge_010	block_006 -> block_009	"archive.fetch succeeds"
edge_011	block_006 -> block_010	"archive.fetch fails"

全部操作、数据引用与约束落点

block_001 Read `source_id` from the user's request

`data_source_kind = "context"`

`block_001.constraints: []`

ir_001 `read_source_id_from_request`

`inputs[0] context_key: source_id | source_id`

`outputs[0] result: result_001 | source_id`

`ir_001.constraints: []`

`draft_instruction_id = read_source_id_from_request`

ir_002 `dispatch`

`inputs = []`

`outputs = []`

`ir_002.constraints: []`

`draft_instruction_id = go_read_fast_key`

block_002 Read `FAST_KEY` from the environment

`data_source_kind = "context"`

`block_002.constraints: []`

ir_003 `read_fast_key_from_environment`

`inputs[0] context_key: FAST_KEY | FAST_KEY`

`outputs[0] result: result_002 | fast_key`

`ir_003.constraints: []`

`draft_instruction_id = read_fast_key_from_environment`

ir_004 `dispatch`

`inputs = []`

`outputs = []`

`ir_004.constraints: []`

`draft_instruction_id = go_check_fast_key_presence`

block_003 Check whether `FAST_KEY` is present

`data_source_kind = null`

`block_003.constraints: []`

ir_005 `check_fast_key_presence`

`inputs[0] result: result_002 | fast_key`

`outputs[0] result: result_003 | fast_key_present`

`ir_005.constraints: []`

`draft_instruction_id = check_fast_key_presence`

ir_006 dispatch

```
inputs[0] result: result_003 | fast_key_present
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = choose_fetch_path
```

block_004 Try fast.fetch with source_id and FAST_KEY

```
data_source_kind = "external"
```

```
block_004.constraints[0]: Try fast.fetch first with source_id and FAST_KEY when FAST_KEY is present.
```

ir_007 fast.fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_004 | fast_fetch_status
```

```
outputs[1] result: result_005 | fast_fetch_response_body
```

```
ir_007.constraints[0]: Try fast.fetch first with source_id and FAST_KEY when FAST_KEY is present.
```

```
draft_instruction_id = try_fast_fetch
```

ir_008 dispatch

```
inputs[0] result: result_004 | fast_fetch_status
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_fast_fetch_outcome
```

block_005 Retry fast.fetch once after a transient first failure

```
data_source_kind = "external"
```

```
block_005.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error.
```

ir_009 fast.fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_006 | fast_fetch_retry_status
```

```
outputs[1] result: result_007 | fast_fetch_retry_response_body
```

```
ir_009.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error.
```

```
draft_instruction_id = retry_fast_fetch
```

ir_010 dispatch

```
inputs[0] result: result_006 | fast_fetch_retry_status
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_fast_fetch_retry_outcome
```

block_006 Call archive.fetch with source_id

data_source_kind = "external"

block_006.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.

block_006.constraints[1]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

ir_011 archive.fetch

inputs[0] external_resource: archive.fetch | archive.fetch

inputs[1] result: result_001 | source_id

outputs[0] result: result_008 | archive_fetch_status

outputs[1] result: result_009 | archive_fetch_response_body

outputs[2] result: result_010 | archive_fetch_error

ir_011.constraints[0]: Pass source_id as its only argument.

ir_011.constraints[1]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

draft_instruction_id = call_archive_fetch

ir_012 dispatch

inputs[0] result: result_008 | archive_fetch_status

outputs = []

ir_012.constraints: []

draft_instruction_id = dispatch_archive_fetch_outcome

block_007 Append fast.fetch success status and return its response body

data_source_kind = null

block_007.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

ir_013 append_final_status_to_status_file

inputs[0] external_resource: local status.txt | local status.txt

inputs[1] result: result_004 | fast_fetch_status

outputs = []

ir_013.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

draft_instruction_id = append_fast_success_status

ir_014 return

inputs[0] result: result_005 | fast_fetch_response_body

outputs = []

ir_014.constraints: []

draft_instruction_id = return_fast_success_body

block_008 Append fast.fetch retry success status and return its response body

data_source_kind = null

block_008.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

ir_015 append_final_status_to_status_file

inputs[0] external_resource: local status.txt | local status.txt

```
inputs[1] result: result_006 | fast_fetch_retry_status
```

```
outputs = []
```

```
ir_015.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

```
draft_instruction_id = append_fast_retry_success_status
```

ir_016 return

```
inputs[0] result: result_007 | fast_fetch_retry_response_body
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = return_fast_retry_success_body
```

block_009 Append archive.fetch success status and return its response body

```
data_source_kind = null
```

```
block_009.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

ir_017 append_final_status_to_status_file

```
inputs[0] external_resource: local status.txt | local status.txt
```

```
inputs[1] result: result_008 | archive_fetch_status
```

```
outputs = []
```

```
ir_017.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

```
draft_instruction_id = append_archive_success_status
```

ir_018 return

```
inputs[0] result: result_009 | archive_fetch_response_body
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = return_archive_success_body
```

block_010 Append archive.fetch failure status and return its error

```
data_source_kind = null
```

```
block_010.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

```
block_010.constraints[1]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

ir_019 append_final_status_to_status_file

```
inputs[0] external_resource: local status.txt | local status.txt
```

```
inputs[1] result: result_008 | archive_fetch_status
```

```
outputs = []
```

```
ir_019.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

```
draft_instruction_id = append_archive_failure_status
```

ir_020 return

```
inputs[0] result: result_010 | archive_fetch_error
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = return_archive_error
```


F02 / 第 2 次 实际输出明细

工具选择与失败回退 · 连贯段落

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F02/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: The exact text of the final status appended to local status.txt is not specified; literals ' success' and 'failure' represent the path outcome.

candidate warning: Response body and error values are modeled as extracted from tool responses by the check_* instructions.

Skill 层约束与上下文声明

```
declared_context_keys = ["source_id", "FAST_KEY"]

skill.constraints[0]: Read source_id from the user's request and read FAST_KEY from the environment.

skill.constraints[1]: If FAST_KEY is present, try fast.fetch first with source_id and FAST_KEY; if it is absent, go directly to archive.fetch without calling fast.fetch.

skill.constraints[2]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do not retry a non-transient first failure.

skill.constraints[3]: After a non-transient first failure or any failed retry of fast.fetch, call archive.fetch with source_id.

skill.constraints[4]: Call archive.fetch at most once and pass source_id as its only argument.

skill.constraints[5]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

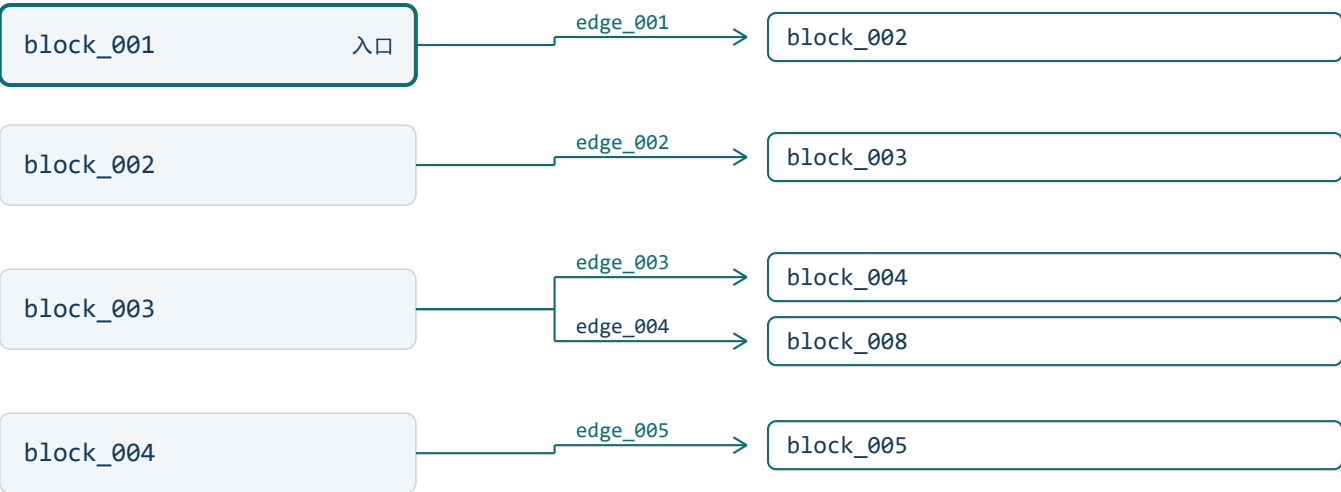
skill.constraints[6]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.

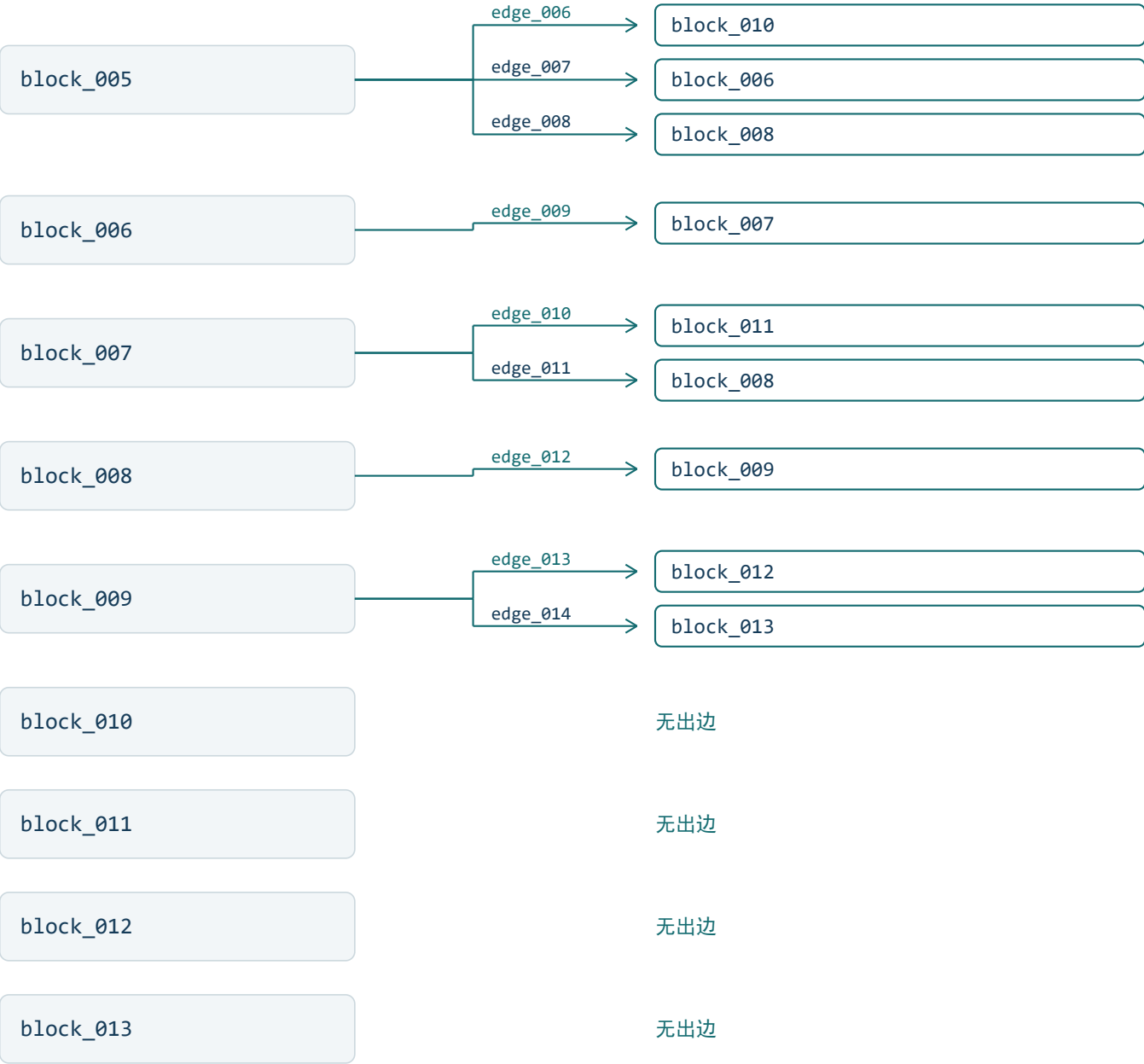
skill.constraints[7]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

skill.constraints[8]: Before returning from every success or failure path, append the final status to local status.txt.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"FAST_KEY is present"
edge_004	block_003 -> block_008	"FAST_KEY is absent"
edge_005	block_004 -> block_005	null
edge_006	block_005 -> block_010	"fast.fetch first attempt succeeded"
edge_007	block_005 -> block_006	"fast.fetch first attempt failed with a transient error"
edge_008	block_005 -> block_008	"fast.fetch first attempt failed with a non-transient error"
edge_009	block_006 -> block_007	null
edge_010	block_007 -> block_011	"fast.fetch retry succeeded"
edge_011	block_007 -> block_008	"fast.fetch retry failed"
edge_012	block_008 -> block_009	null
edge_013	block_009 -> block_012	"archive.fetch succeeded"
edge_014	block_009 -> block_013	"archive.fetch failed"

全部操作、数据引用与约束落点

block_001 Read source_id from the user's request

data_source_kind = "context"

block_001.constraints: []

ir_001 read_source_id_from_request

inputs[0] context_key: source_id | source_id

outputs[0] result: result_001 | source_id_from_request

ir_001.constraints: []

draft_instruction_id = read_source_id

ir_002 dispatch

inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = go_read_fast_key

block_002 Read FAST_KEY from the environment

```
data_source_kind = "context"
```

```
block_002.constraints: []
```

ir_003 read_fast_key_from_environment

```
inputs[0] context_key: FAST_KEY | FAST_KEY
```

```
outputs[0] result: result_002 | fast_key
```

```
ir_003.constraints: []
```

```
draft_instruction_id = read_fast_key
```

ir_004 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = go_check_fast_key_presence
```

block_003 Check whether FAST_KEY is present

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 check_fast_key_presence

```
inputs[0] result: result_002 | fast_key
```

```
outputs[0] result: result_003 | fast_key_present
```

```
ir_005.constraints: []
```

```
draft_instruction_id = check_fast_key_presence
```

ir_006 dispatch

```
inputs[0] result: result_003 | fast_key_present
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = choose_fetch_path
```

block_004 Try fast.fetch first with source_id and FAST_KEY

```
data_source_kind = "external"
```

```
block_004.constraints[0]: If FAST_KEY is present, try fast.fetch first with source_id and FAST_KEY.
```

ir_007 fast.fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id_from_request
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_004 | fast_fetch_first_response
```

```
ir_007.constraints: []
```

```
draft_instruction_id = call_fast_fetch_first
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_evaluate_fast_fetch_first
```

block_005 Check the first fast.fetch result for success and transient failure

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 check_fast_fetch_result

```
inputs[0] result: result_004 | fast_fetch_first_response
```

```
outputs[0] result: result_005 | fast_fetch_first_succeeded
```

```
outputs[1] result: result_006 | fast_fetch_first_failure_is_transient
```

```
outputs[2] result: result_007 | fast_fetch_first_body
```

```
ir_009.constraints: []
```

```
draft_instruction_id = check_fast_fetch_first
```

ir_010 dispatch

```
inputs[0] result: result_005 | fast_fetch_first_succeeded
```

```
inputs[1] result: result_006 | fast_fetch_first_failure_is_transient
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = choose_after_fast_fetch_first
```

block_006 Retry fast.fetch once after a transient first failure

```
data_source_kind = "external"
```

```
block_006.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error;  
do not retry a non-transient first failure.
```

ir_011 fast.fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id_from_request
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_008 | fast_fetch_retry_response
```

```
ir_011.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error.
```

```
draft_instruction_id = call_fast_fetch_retry
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_evaluate_fast_fetch_retry
```

block_007 Check whether the fast.fetch retry succeeded

data_source_kind = null

block_007.constraints: []

ir_013 check_fast_fetch_result

inputs[0] result: result_008 | fast_fetch_retry_response

outputs[0] result: result_009 | fast_fetch_retry_succeeded

outputs[1] result: result_010 | fast_fetch_retry_body

ir_013.constraints: []

draft_instruction_id = check_fast_fetch_retry

ir_014 dispatch

inputs[0] result: result_009 | fast_fetch_retry_succeeded

outputs = []

ir_014.constraints: []

draft_instruction_id = choose_after_fast_fetch_retry

block_008 Call archive.fetch with source_id

data_source_kind = "external"

block_008.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.

block_008.constraints[1]: Do not retry archive.fetch.

block_008.constraints[2]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

ir_015 archive.fetch

inputs[0] external_resource: archive.fetch | archive.fetch

inputs[1] result: result_001 | source_id_from_request

outputs[0] result: result_011 | archive_fetch_response

ir_015.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.

ir_015.constraints[1]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

draft_instruction_id = call_archive_fetch

ir_016 dispatch

inputs = []

outputs = []

ir_016.constraints: []

draft_instruction_id = go_evaluate_archive_fetch

block_009 Check the archive.fetch result

data_source_kind = null

block_009.constraints: []

ir_017 check_archive_fetch_result

inputs[0] result: result_011 | archive_fetch_response

outputs[0] result: result_012 | archive_fetch_succeeded

outputs[1] result: result_013 | archive_fetch_body

outputs[2] result: result_014 | archive_fetch_error

```
ir_017.constraints: []
```

```
draft_instruction_id = check_archive_fetch
```

ir_018 dispatch

```
inputs[0] result: result_012 | archive_fetch_succeeded
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = choose_after_archive_fetch
```

block_010 Append success status and return the first fast.fetch response body

```
data_source_kind = null
```

block_010.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

block_010.constraints[1]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

ir_019 append_final_status

```
inputs[0] external_resource: status.txt | status.txt
```

```
inputs[1] literal = "success"
```

```
outputs = []
```

```
ir_019.constraints[0]: Must not include FAST_KEY.
```

```
draft_instruction_id = append_status_fast_first_success
```

ir_020 return

```
inputs[0] result: result_007 | fast_fetch_first_body
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = return_fast_first_success
```

block_011 Append success status and return the fast.fetch retry response body

```
data_source_kind = null
```

block_011.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

block_011.constraints[1]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

ir_021 append_final_status

```
inputs[0] external_resource: status.txt | status.txt
```

```
inputs[1] literal = "success"
```

```
outputs = []
```

```
ir_021.constraints[0]: Must not include FAST_KEY.
```

```
draft_instruction_id = append_status_fast_retry_success
```

ir_022 return

```
inputs[0] result: result_010 | fast_fetch_retry_body
```

```
outputs = []
```

```
ir_022.constraints: []
```

```
draft_instruction_id = return_fast_retry_success
```

block_012 Append success status and return the archive.fetch response body

```
data_source_kind = null
```

```
block_012.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

```
block_012.constraints[1]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.
```

ir_023 append_final_status

```
inputs[0] external_resource: status.txt | status.txt
```

```
inputs[1] literal = "success"
```

```
outputs = []
```

```
ir_023.constraints[0]: Must not include FAST_KEY.
```

```
draft_instruction_id = append_status_archive_success
```

ir_024 return

```
inputs[0] result: result_013 | archive_fetch_body
```

```
outputs = []
```

```
ir_024.constraints: []
```

```
draft_instruction_id = return_archive_success
```

block_013 Append failure status and return the archive.fetch error

```
data_source_kind = null
```

```
block_013.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

```
block_013.constraints[1]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

ir_025 append_final_status

```
inputs[0] external_resource: status.txt | status.txt
```

```
inputs[1] literal = "failure"
```

```
outputs = []
```

```
ir_025.constraints[0]: Must not include FAST_KEY.
```

```
draft_instruction_id = append_status_archive_failure
```

ir_026 return

```
inputs[0] result: result_014 | archive_fetch_error
```

```
outputs = []
```

```
ir_026.constraints: []
```

```
draft_instruction_id = return_archive_failure
```


F02 / 第 3 次 实际输出明细

工具选择与失败回退 · 连贯段落

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F02/deepseek-v4-flash-max/3

结构状态: error | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: 未记录 | HTTP 1 (200): deepseek-flash

诊断与失败边界

IncompleteRead(0 bytes read)

流式请求的远端最终结果未知: 生成 1 / HTTP 尝试 1 / 已接收 HTTP 200

已收到成功响应头后，SSE 未达到完整结束条件；本地 error 不代表远端未生成结果，也不代表模型语义失败。未完成的流不能作为有效响应或有效 CFG，远端用量和最终处理结果以原始记录为准。

本次没有已接受的实际 CFG；不能补画或将候选当作有效图。原始响应 / 候选保存在同目录 trace.json、candidate.json 和 analysis.json（如生成）。

F03 / 第 1 次 实际输出明细

工具选择与失败回退 · 表格

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F03/deepseek-v4-flash-max/1

结构状态: error | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: 未记录 | HTTP 1 (200): deepseek-flash

诊断与失败边界

IncompleteRead(0 bytes read)

流式请求的远端最终结果未知: 生成 1 / HTTP 尝试 1 / 已接收 HTTP 200

已收到成功响应头后，SSE 未达到完整结束条件；本地 error 不代表远端未生成结果，也不代表模型语义失败。未完成的流不能作为有效响应或有效 CFG，远端用量和最终处理结果以原始记录为准。

本次没有已接受的实际 CFG；不能补画或将候选当作有效图。原始响应 / 候选保存在同目录 trace.json、candidate.json 和 analysis.json（如生成）。

F03 / 第 2 次 实际输出明细

工具选择与失败回退 · 表格

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F03/deepseek-v4-flash-max/2

结构状态: error | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: 未记录 | HTTP 1 (200): deepseek-flash

诊断与失败边界

IncompleteRead(0 bytes read)

流式请求的远端最终结果未知: 生成 1 / HTTP 尝试 1 / 已接收 HTTP 200

已收到成功响应头后，SSE 未达到完整结束条件；本地 error 不代表远端未生成结果，也不代表模型语义失败。未完成的流不能作为有效响应或有效 CFG，远端用量和最终处理结果以原始记录为准。

本次没有已接受的实际 CFG；不能补画或将候选当作有效图。原始响应 / 候选保存在同目录 trace.json、candidate.json 和 analysis.json（如生成）。

F03 / 第 3 次 实际输出明细

工具选择与失败回退 · 表格

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F03/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Exact final status text is not specified in the source; 'success' and 'failure' are used as literal status values for status.txt append operations.

candidate warning: Transient error classification is modeled as a computation over the returned fast.fetch error.

candidate warning: The source does not specify the internal structure of tool responses; body and error are modeled as separate outputs of each external fetch.

Skill 层约束与上下文声明

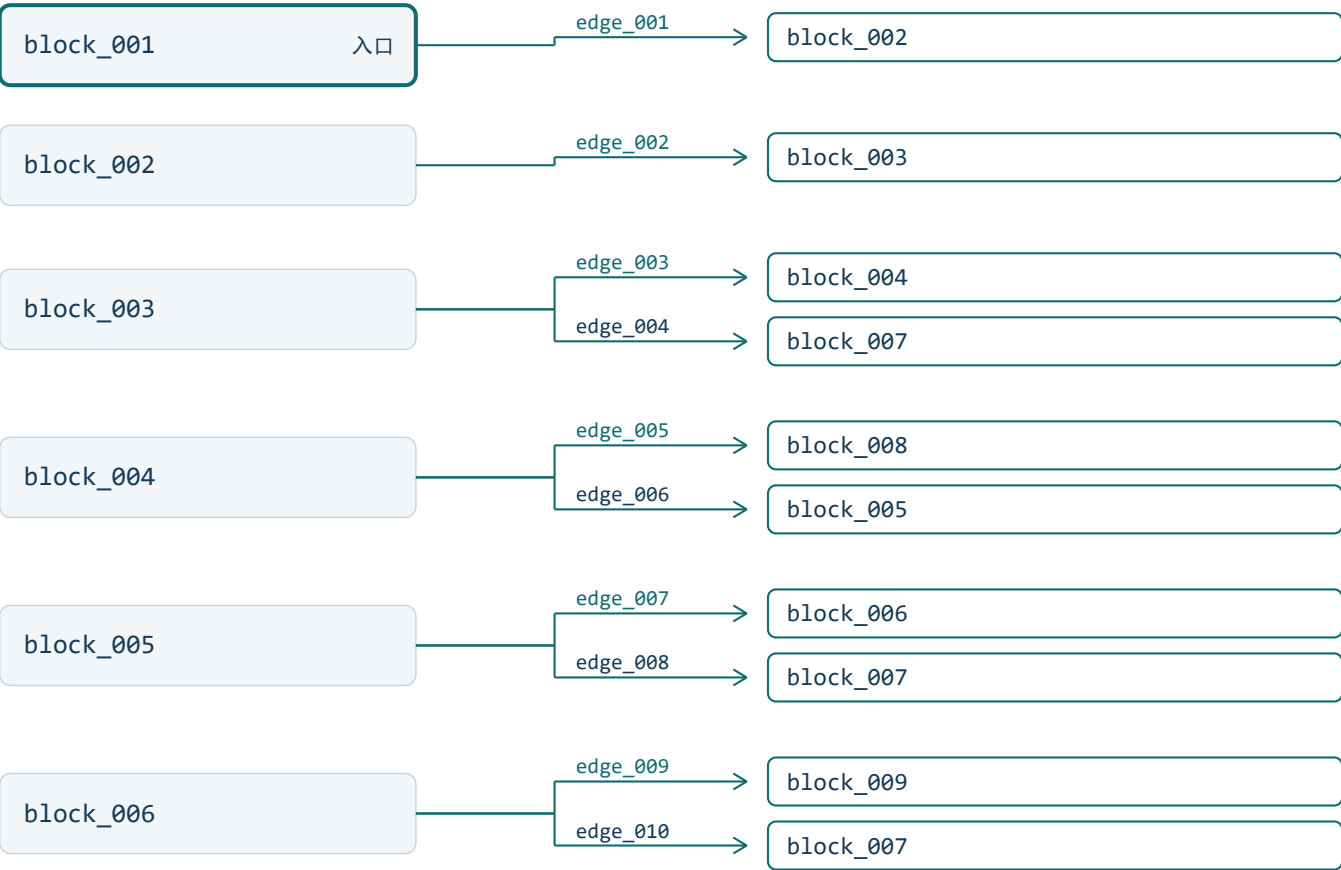
```
declared_context_keys = ["source_id", "FAST_KEY"]
```

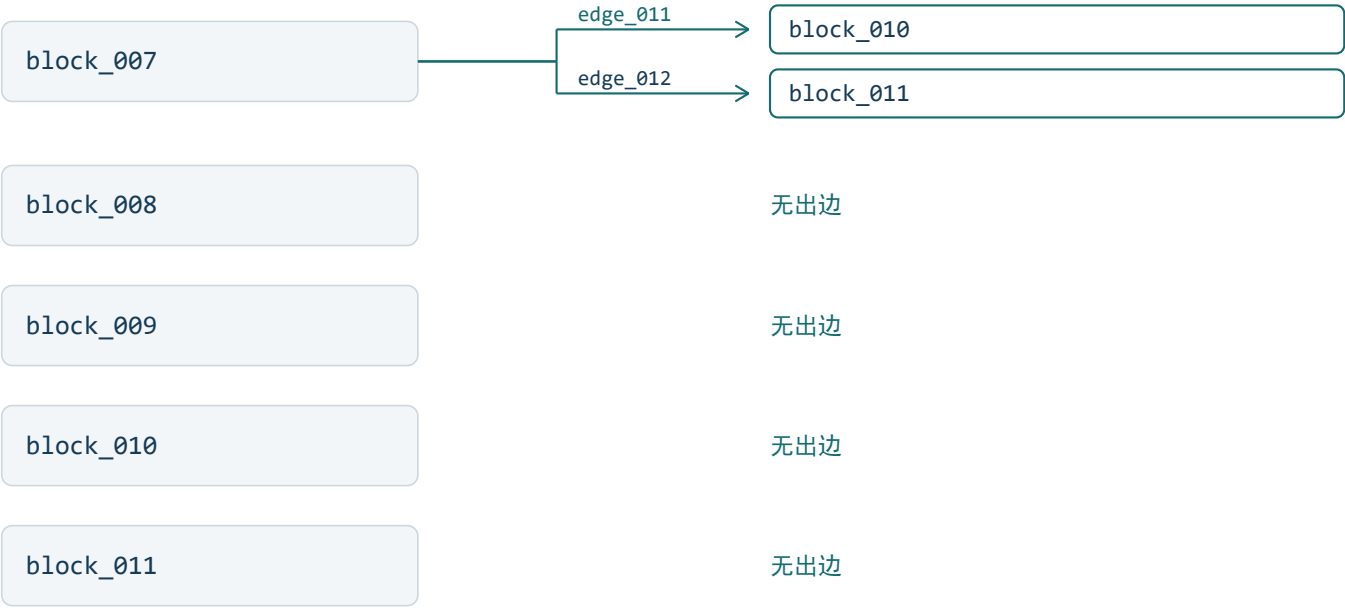
skill.constraints[0]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

skill.constraints[1]: Before returning from every success or failure path, append the final status to local status.txt.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"FAST_KEY is present"
edge_004	block_003 -> block_007	"FAST_KEY is absent"
edge_005	block_004 -> block_008	"fast.fetch first attempt succeeds"
edge_006	block_004 -> block_005	"fast.fetch first attempt fails"
edge_007	block_005 -> block_006	"first failure is transient"
edge_008	block_005 -> block_007	"first failure is non-transient"
edge_009	block_006 -> block_009	"fast.fetch retry succeeds"
edge_010	block_006 -> block_007	"fast.fetch retry fails"
edge_011	block_007 -> block_010	"archive.fetch succeeds"
edge_012	block_007 -> block_011	"archive.fetch fails"

全部操作、数据引用与约束落点

```
block_001 Read source_id from the user's request

data_source_kind = "context"

block_001.constraints: []

ir_001 read_source_id_from_user_request

inputs[0] context_key: source_id | source_id
outputs[0] result: result_001 | source_id_value
```

```
ir_001.constraints: []  
draft_instruction_id = read_source_id_inst  
  
ir_002 dispatch  
  
inputs = []  
outputs = []  
ir_002.constraints: []  
draft_instruction_id = go_read_fast_key  
  
block_002 Read FAST_KEY from the environment  
  
data_source_kind = "context"  
block_002.constraints: []  
  
ir_003 read_fast_key_from_environment  
  
inputs[0] context_key: FAST_KEY | FAST_KEY  
outputs[0] result: result_002 | fast_key  
ir_003.constraints: []  
draft_instruction_id = read_fast_key_inst  
  
ir_004 dispatch  
  
inputs = []  
outputs = []  
ir_004.constraints: []  
draft_instruction_id = go_decide_fast_key_present  
  
block_003 Check whether FAST_KEY is present  
  
data_source_kind = null  
block_003.constraints: []  
  
ir_005 check_fast_key_present  
  
inputs[0] result: result_002 | fast_key  
outputs[0] result: result_003 | fast_key_present  
ir_005.constraints: []  
draft_instruction_id = check_fast_key_present_inst  
  
ir_006 dispatch  
  
inputs[0] result: result_003 | fast_key_present  
outputs = []  
ir_006.constraints: []  
draft_instruction_id = dispatch_on_fast_key_present  
  
block_004 Try fast.fetch with source_id and FAST_KEY  
  
data_source_kind = "external"  
block_004.constraints: []  
  
ir_007 fast.fetch  
  
inputs[0] external_resource: fast.fetch | fast.fetch  
inputs[1] result: result_001 | source_id_value
```

```
inputs[2] result: result_002 | fast_key
outputs[0] result: result_004 | fast_fetch_first_body
outputs[1] result: result_005 | fast_fetch_first_error
ir_007.constraints: []
draft_instruction_id = fast_fetch_first_inst
```

ir_008 dispatch

```
inputs[0] result: result_005 | fast_fetch_first_error
outputs = []
ir_008.constraints: []
draft_instruction_id = dispatch_on_first_error
```

block_005 Classify first fast.fetch failure as transient or non-transient

```
data_source_kind = null
block_005.constraints: []
```

ir_009 check_transient_error

```
inputs[0] result: result_005 | fast_fetch_first_error
outputs[0] result: result_006 | fast_fetch_first_failure_is_transient
ir_009.constraints: []
draft_instruction_id = classify_first_error_inst
```

ir_010 dispatch

```
inputs[0] result: result_006 | fast_fetch_first_failure_is_transient
outputs = []
ir_010.constraints: []
draft_instruction_id = dispatch_on_transient
```

block_006 Retry fast.fetch once with source_id and FAST_KEY

```
data_source_kind = "external"
block_006.constraints: []
```

ir_011 fast.fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
inputs[1] result: result_001 | source_id_value
inputs[2] result: result_002 | fast_key
outputs[0] result: result_007 | fast_fetch_retry_body
outputs[1] result: result_008 | fast_fetch_retry_error
```

ir_011.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do not retry a non-transient first failure.

```
draft_instruction_id = fast_fetch_retry_inst
```

ir_012 dispatch

```
inputs[0] result: result_008 | fast_fetch_retry_error
outputs = []
ir_012.constraints: []
draft_instruction_id = dispatch_on_retry_error
```

block_007 Call archive.fetch with source_id

data_source_kind = "external"

block_007.constraints: []

ir_013 archive.fetch

inputs[0] external_resource: archive.fetch | archive.fetch

inputs[1] result: result_001 | source_id_value

outputs[0] result: result_009 | archive_fetch_body

outputs[1] result: result_010 | archive_fetch_error

ir_013.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.

draft_instruction_id = archive_fetch_inst

ir_014 dispatch

inputs[0] result: result_010 | archive_fetch_error

outputs = []

ir_014.constraints: []

draft_instruction_id = dispatch_on_archive_error

block_008 Return fast.fetch first successful response body and append final status

data_source_kind = null

block_008.constraints[0]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

ir_015 append_final_status_to_file

inputs[0] external_resource: status.txt | status.txt

inputs[1] literal = "success"

outputs = []

ir_015.constraints: []

draft_instruction_id = append_status_first_success

ir_016 return

inputs[0] result: result_004 | fast_fetch_first_body

outputs = []

ir_016.constraints: []

draft_instruction_id = return_first_success

block_009 Return fast.fetch retry successful response body and append final status

data_source_kind = null

block_009.constraints[0]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

ir_017 append_final_status_to_file

inputs[0] external_resource: status.txt | status.txt

inputs[1] literal = "success"

outputs = []

ir_017.constraints: []

draft_instruction_id = append_status_retry_success

ir_018 return

```
inputs[0] result: result_007 | fast_fetch_retry_body
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = return_retry_success
```

block_010 Return archive.fetch successful response body and append final status

```
data_source_kind = null
```

```
block_010.constraints[0]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.
```

ir_019 append_final_status_to_file

```
inputs[0] external_resource: status.txt | status.txt
```

```
inputs[1] literal = "success"
```

```
outputs = []
```

```
ir_019.constraints: []
```

```
draft_instruction_id = append_status_archive_success
```

ir_020 return

```
inputs[0] result: result_009 | archive_fetch_body
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = return_archive_success
```

block_011 Return archive.fetch error and append final status

```
data_source_kind = null
```

```
block_011.constraints[0]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

ir_021 append_final_status_to_file

```
inputs[0] external_resource: status.txt | status.txt
```

```
inputs[1] literal = "failure"
```

```
outputs = []
```

```
ir_021.constraints: []
```

```
draft_instruction_id = append_status_archive_failure
```

ir_022 return

```
inputs[0] result: result_010 | archive_fetch_error
```

```
outputs = []
```

```
ir_022.constraints: []
```

```
draft_instruction_id = return_archive_failure
```

F04 / 第 1 次 实际输出明细

工具选择与失败回退 · 跨文件混合

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F04/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

```
declared_context_keys = ["user_request.source_id", "environment.FAST_KEY"]

skill.constraints[0]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

skill.constraints[1]: Before returning from every success or failure path, append the final status to local status.txt.

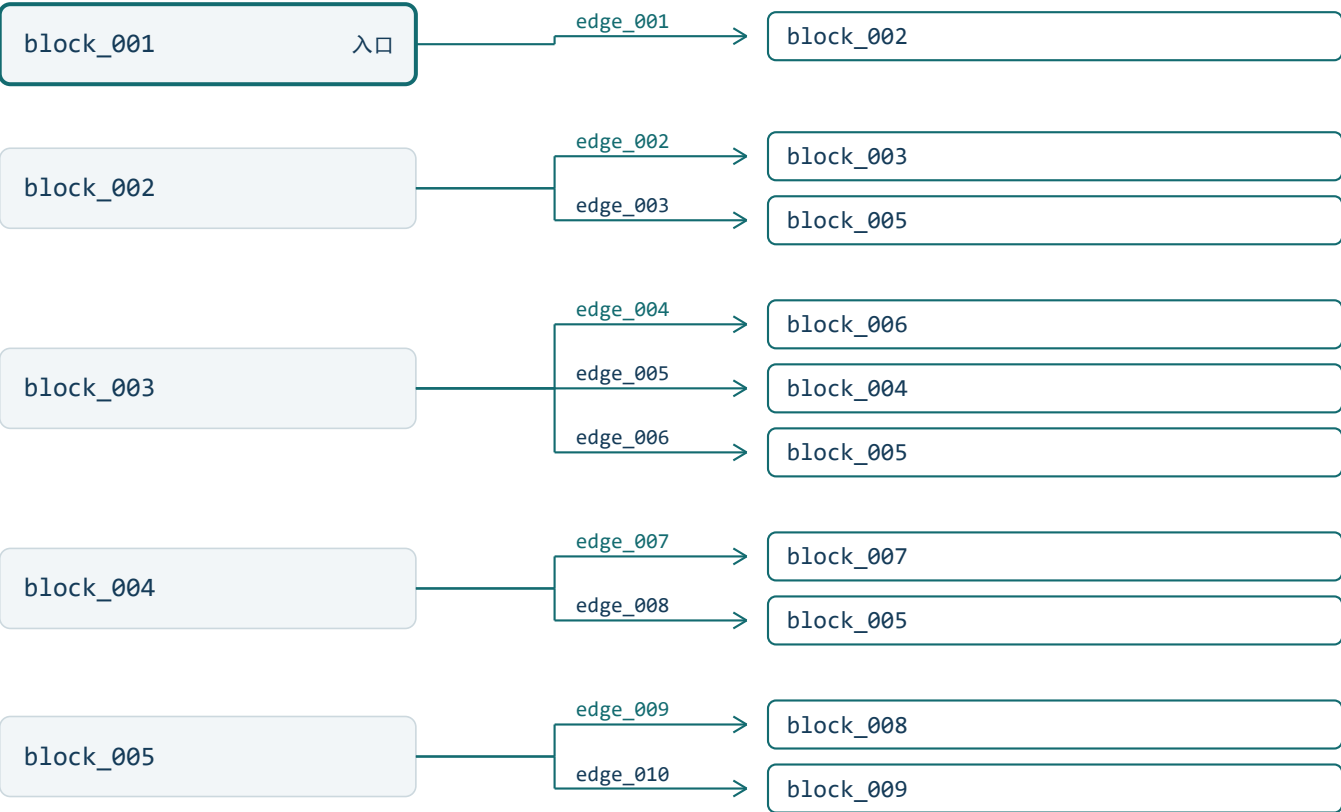
skill.constraints[2]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

skill.constraints[3]: After a non-transient first failure or any failed retry of fast.fetch, call archive.fetch with source_id.

skill.constraints[4]: The same retry policy is specified by retry configuration; apply it to fast.fetch only.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



block_006	无出边
block_007	无出边
block_008	无出边
block_009	无出边

完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"FAST_KEY is present"
edge_003	block_002 -> block_005	"FAST_KEY is absent"
edge_004	block_003 -> block_006	"fast.fetch first attempt succeeds"
edge_005	block_003 -> block_004	"fast.fetch first attempt fails with a transient error"
edge_006	block_003 -> block_005	"fast.fetch first attempt fails with a non-transient error"
edge_007	block_004 -> block_007	"fast.fetch retry succeeds"
edge_008	block_004 -> block_005	"fast.fetch retry fails"
edge_009	block_005 -> block_008	"archive.fetch succeeds"
edge_010	block_005 -> block_009	"archive.fetch fails"

全部操作、数据引用与约束落点

block_001 Read source_id from the user request

```
data_source_kind = "context"
block_001.constraints: []

ir_001 read_source_id_from_user_request

inputs[0] context_key: user_request.source_id | user_request.source_id
outputs[0] result: result_001 | source_id
ir_001.constraints: []
draft_instruction_id = read_source_id
```

ir_002 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_002.constraints: []
```

```
draft_instruction_id = dispatch_to_read_fast_key
```

block_002 Read FAST_KEY from the environment and decide whether fast.fetch can run

```
data_source_kind = "context"
```

```
block_002.constraints: []
```

ir_003 read_fast_key_from_environment

```
inputs[0] context_key: environment.FAST_KEY | environment.FAST_KEY
```

```
outputs[0] result: result_002 | fast_key
```

```
ir_003.constraints: []
```

```
draft_instruction_id = read_fast_key
```

ir_004 dispatch

```
inputs[0] result: result_002 | fast_key
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = dispatch_on_fast_key
```

block_003 Call fast.fetch first with source_id and FAST_KEY

```
data_source_kind = "external"
```

```
block_003.constraints: []
```

ir_005 fast.fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_003 | first_fast_fetch_body
```

```
outputs[1] result: result_004 | first_fast_fetch_status
```

```
ir_005.constraints: []
```

```
draft_instruction_id = call_fast_fetch_first
```

ir_006 dispatch

```
inputs[0] result: result_004 | first_fast_fetch_status
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = dispatch_on_first_fast_fetch
```

block_004 Retry fast.fetch once with source_id and FAST_KEY

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_007 fast.fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id
inputs[2] result: result_002 | fast_key
outputs[0] result: result_005 | retry_fast_fetch_body
outputs[1] result: result_006 | retry_fast_fetch_status
ir_007.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do
not retry a non-transient first failure.
ir_007.constraints[1]: fast_fetch: max_attempts: 2, retry_first_error: transient_only.
draft_instruction_id = call_fast_fetch_retry

ir_008 dispatch

inputs[0] result: result_006 | retry_fast_fetch_status
outputs = []
ir_008.constraints: []
draft_instruction_id = dispatch_on_retry_fast_fetch

block_005 Call archive.fetch with source_id

data_source_kind = "external"
block_005.constraints: []

ir_009 archive.fetch

inputs[0] external_resource: archive.fetch | archive.fetch
inputs[1] result: result_001 | source_id
outputs[0] result: result_007 | archive_fetch_body
outputs[1] result: result_008 | archive_fetch_error
outputs[2] result: result_009 | archive_fetch_status
ir_009.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.
ir_009.constraints[1]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
ir_009.constraints[2]: archive_fetch: max_attempts: 1.
draft_instruction_id = call_archive_fetch

ir_010 dispatch

inputs[0] result: result_009 | archive_fetch_status
outputs = []
ir_010.constraints: []
draft_instruction_id = dispatch_on_archive_fetch

block_006 Append final status and return fast.fetch first response body

data_source_kind = null
block_006.constraints: []

ir_011 append_final_status

inputs[0] result: result_004 | first_fast_fetch_status
inputs[1] external_resource: local status.txt | local status.txt
outputs = []
ir_011.constraints: []
draft_instruction_id = append_first_fast_fetch_success_status
```

ir_012 return

```
inputs[0] result: result_003 | first_fast_fetch_body
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = return_first_fast_fetch_body
```

block_007 Append final status and return fast.fetch retry response body

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 append_final_status

```
inputs[0] result: result_006 | retry_fast_fetch_status
```

```
inputs[1] external_resource: local status.txt | local status.txt
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = append_retry_fast_fetch_success_status
```

ir_014 return

```
inputs[0] result: result_005 | retry_fast_fetch_body
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = return_retry_fast_fetch_body
```

block_008 Append final status and return archive.fetch response body

```
data_source_kind = null
```

```
block_008.constraints: []
```

ir_015 append_final_status

```
inputs[0] result: result_009 | archive_fetch_status
```

```
inputs[1] external_resource: local status.txt | local status.txt
```

```
outputs = []
```

```
ir_015.constraints: []
```

```
draft_instruction_id = append_archive_success_status
```

ir_016 return

```
inputs[0] result: result_007 | archive_fetch_body
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = return_archive_fetch_body
```

block_009 Append final status and return archive.fetch error

```
data_source_kind = null
```

```
block_009.constraints: []
```

ir_017 append_final_status

```
inputs[0] result: result_009 | archive_fetch_status
```

```
inputs[1] external_resource: local status.txt | local status.txt
```

```
outputs = []  
ir_017.constraints: []  
draft_instruction_id = append_archive_failure_status  
ir_018 return  
inputs[0] result: result_008 | archive_fetch_error  
outputs = []  
ir_018.constraints: []  
draft_instruction_id = return_archive_fetch_error
```

F04 / 第 2 次 实际输出明细

工具选择与失败回退 · 跨文件混合

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F04/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: SKILL.md points to references/workflow.md; executable steps were taken from that workflow.

candidate warning: retry.yaml is referenced as the retry-policy specification; no explicit runtime read of retry.yaml is stated, so its values are captured as constraints rather than as an acquisition block.

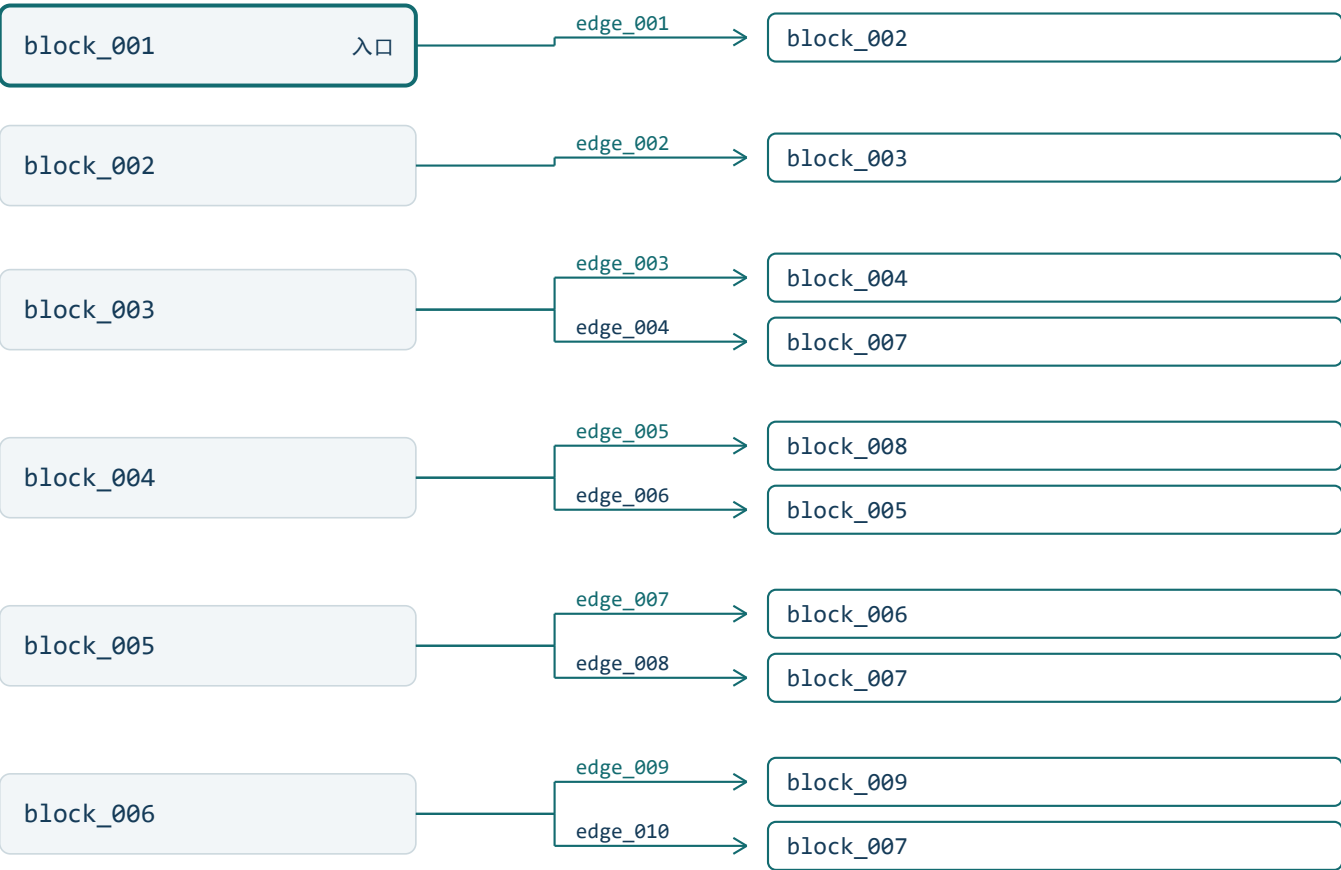
Skill 层约束与上下文声明

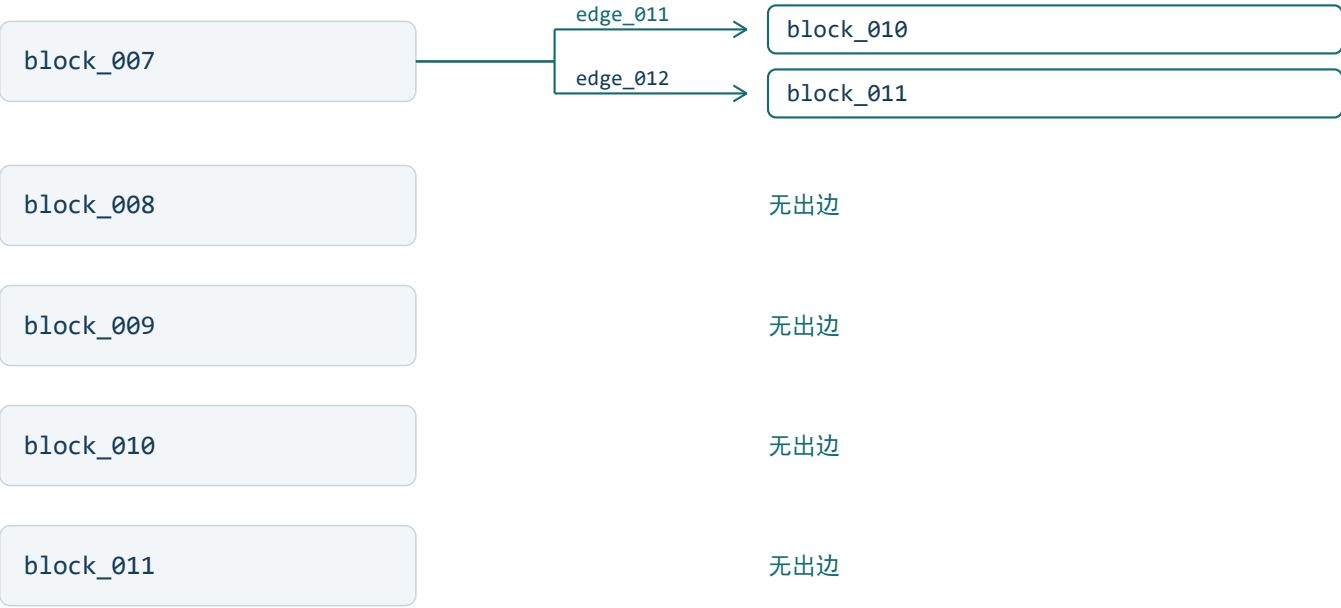
```
declared_context_keys = ["user_request", "FAST_KEY"]

skill.constraints[0]: Fetch one source using a credential-gated preferred tool and a bounded archive fallback.
skill.constraints[1]: The same retry policy is specified by retry configuration; apply it to fast.fetch only.
skill.constraints[2]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.
skill.constraints[3]: Before returning from every success or failure path, append the final status to local
status.txt.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"FAST_KEY is present"
edge_004	block_003 -> block_007	"FAST_KEY is absent"
edge_005	block_004 -> block_008	"fast.fetch first attempt succeeds"
edge_006	block_004 -> block_005	"fast.fetch first attempt fails"
edge_007	block_005 -> block_006	"first error is transient"
edge_008	block_005 -> block_007	"first error is non-transient"
edge_009	block_006 -> block_009	"retry succeeds"
edge_010	block_006 -> block_007	"retry fails"
edge_011	block_007 -> block_010	"archive.fetch succeeds"
edge_012	block_007 -> block_011	"archive.fetch fails"

全部操作、数据引用与约束落点

```
block_001 Read source_id from the user's request

data_source_kind = "context"

block_001.constraints: []

ir_001 read_source_id_from_request

inputs[0] context_key: user_request | user_request
outputs[0] result: result_001 | source_id
```

```
ir_001.constraints: []

draft_instruction_id = read_source_id_from_request

ir_002 dispatch

inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = go_read_fast_key

block_002 Read FAST_KEY from the environment

data_source_kind = "context"

block_002.constraints: []

ir_003 read_fast_key_from_environment

inputs[0] context_key: FAST_KEY | FAST_KEY

outputs[0] result: result_002 | FAST_KEY

ir_003.constraints: []

draft_instruction_id = read_fast_key_from_environment

ir_004 dispatch

inputs = []

outputs = []

ir_004.constraints: []

draft_instruction_id = go_check_fast_key

block_003 Check whether FAST_KEY is present

data_source_kind = null

block_003.constraints: []

ir_005 check_fast_key_presence

inputs[0] result: result_002 | fast_key

outputs[0] result: result_003 | FAST_KEY presence

ir_005.constraints: []

draft_instruction_id = check_fast_key_presence

ir_006 dispatch

inputs[0] result: result_003 | fast_key_present

outputs = []

ir_006.constraints: []

draft_instruction_id = choose_fetch_path

block_004 Try fast.fetch first with source_id and FAST_KEY

data_source_kind = "external"

block_004.constraints: []

ir_007 fast.fetch

inputs[0] external_resource: fast.fetch | fast.fetch

inputs[1] result: result_001 | source_id
```

```
inputs[2] result: result_002 | fast_key
outputs[0] result: result_004 | fast.fetch first response body
outputs[1] result: result_005 | fast.fetch first error
ir_007.constraints[0]: fast_fetch: max_attempts: 2, retry_first_error: transient_only
draft_instruction_id = fast_fetch_first
```

ir_008 dispatch

```
inputs[0] result: result_005 | fast_fetch_first_error
outputs = []
ir_008.constraints: []
draft_instruction_id = choose_after_first_fast_fetch
```

block_005 Classify the first fast.fetch error as transient or non-transient

```
data_source_kind = null
block_005.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error;
do not retry a non-transient first failure.
```

ir_009 check_fast_fetch_error_transience

```
inputs[0] result: result_005 | fast_fetch_first_error
outputs[0] result: result_006 | first fast.fetch error transience
ir_009.constraints: []
draft_instruction_id = check_fast_fetch_first_error_transience
```

ir_010 dispatch

```
inputs[0] result: result_006 | fast_fetch_first_error_is_transient
outputs = []
ir_010.constraints: []
draft_instruction_id = choose_after_classify_first_fast_fetch
```

block_006 Retry fast.fetch once after a transient first error

```
data_source_kind = "external"
block_006.constraints: []
```

ir_011 fast.fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
inputs[1] result: result_001 | source_id
inputs[2] result: result_002 | fast_key
outputs[0] result: result_007 | fast.fetch retry response body
outputs[1] result: result_008 | fast.fetch retry error
ir_011.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do
not retry a non-transient first failure.
ir_011.constraints[1]: After any failed retry of fast.fetch, call archive.fetch with source_id.
draft_instruction_id = fast_fetch_retry
```

ir_012 dispatch

inputs[0] result: result_008 | fast_fetch_retry_error

outputs = []

ir_012.constraints: []

draft_instruction_id = choose_after_retry_fast_fetch

block_007 Call archive.fetch with source_id

data_source_kind = "external"

block_007.constraints[0]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.

ir_013 archive.fetch

inputs[0] external_resource: archive.fetch | archive.fetch

inputs[1] result: result_001 | source_id

outputs[0] result: result_009 | archive.fetch response body

outputs[1] result: result_010 | archive.fetch error

ir_013.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.

ir_013.constraints[1]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

ir_013.constraints[2]: archive_fetch: max_attempts: 1

draft_instruction_id = archive_fetch

ir_014 dispatch

inputs[0] result: result_010 | archive_fetch_error

outputs = []

ir_014.constraints: []

draft_instruction_id = choose_after_archive_fetch

block_008 Append success status and return the first fast.fetch body

data_source_kind = null

block_008.constraints[0]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

block_008.constraints[1]: Before returning from every success or failure path, append the final status to local status.txt.

ir_015 append_final_status_to_local_file

inputs[0] external_resource: local status.txt | local status.txt

inputs[1] literal = "success"

outputs = []

ir_015.constraints[0]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

draft_instruction_id = append_success_status_first_fast_fetch

ir_016 return

inputs[0] result: result_004 | fast_fetch_first_body

outputs = []

ir_016.constraints: []

draft_instruction_id = return_first_fast_fetch_body

block_009 Append success status and return the retried fast.fetch body

```
data_source_kind = null
```

```
block_009.constraints[0]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.
```

```
block_009.constraints[1]: Before returning from every success or failure path, append the final status to local status.txt.
```

ir_017 append_final_status_to_local_file

```
inputs[0] external_resource: local status.txt | local status.txt
```

```
inputs[1] literal = "success"
```

```
outputs = []
```

```
ir_017.constraints[0]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.
```

```
draft_instruction_id = append_success_status_retry_fast_fetch
```

ir_018 return

```
inputs[0] result: result_007 | fast_fetch_retry_body
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = return_retry_fast_fetch_body
```

block_010 Append success status and return the archive.fetch body

```
data_source_kind = null
```

```
block_010.constraints[0]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.
```

```
block_010.constraints[1]: Before returning from every success or failure path, append the final status to local status.txt.
```

ir_019 append_final_status_to_local_file

```
inputs[0] external_resource: local status.txt | local status.txt
```

```
inputs[1] literal = "success"
```

```
outputs = []
```

```
ir_019.constraints[0]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.
```

```
draft_instruction_id = append_success_status_archive_fetch
```

ir_020 return

```
inputs[0] result: result_009 | archive_fetch_body
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = return_archive_fetch_body
```

block_011 Append failure status and return the archive.fetch error

```
data_source_kind = null
```

```
block_011.constraints[0]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

```
block_011.constraints[1]: Before returning from every success or failure path, append the final status to local status.txt.
```

ir_021 append_final_status_to_local_file

```
inputs[0] external_resource: local status.txt | local status.txt
```

```
inputs[1] literal = "failure"

outputs = []

ir_021.constraints[0]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.
draft_instruction_id = append_failure_status_archive_fetch

ir_022 return

inputs[0] result: result_010 | archive_fetch_error

outputs = []

ir_022.constraints: []

draft_instruction_id = return_archive_fetch_error
```

F04 / 第 3 次 实际输出明细

工具选择与失败回退 · 跨文件混合

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F04/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: retry.yaml is referenced as retry configuration; its contents match the workflow retry policy (fast_fetch max_attempts 2, retry_first_error transient_only; archive_fetch max_attempts 1). It is not modeled as a runtime acquisition because the workflow states the policy directly and does not instruct reading the file.

candidate warning: The final status appended to local status.txt is represented by the outcome status produced by each fetch attempt classification; the source does not define a separate status computation.

Skill 层约束与上下文声明

declared_context_keys = ["source_id", "FAST_KEY"]

skill.constraints[0]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

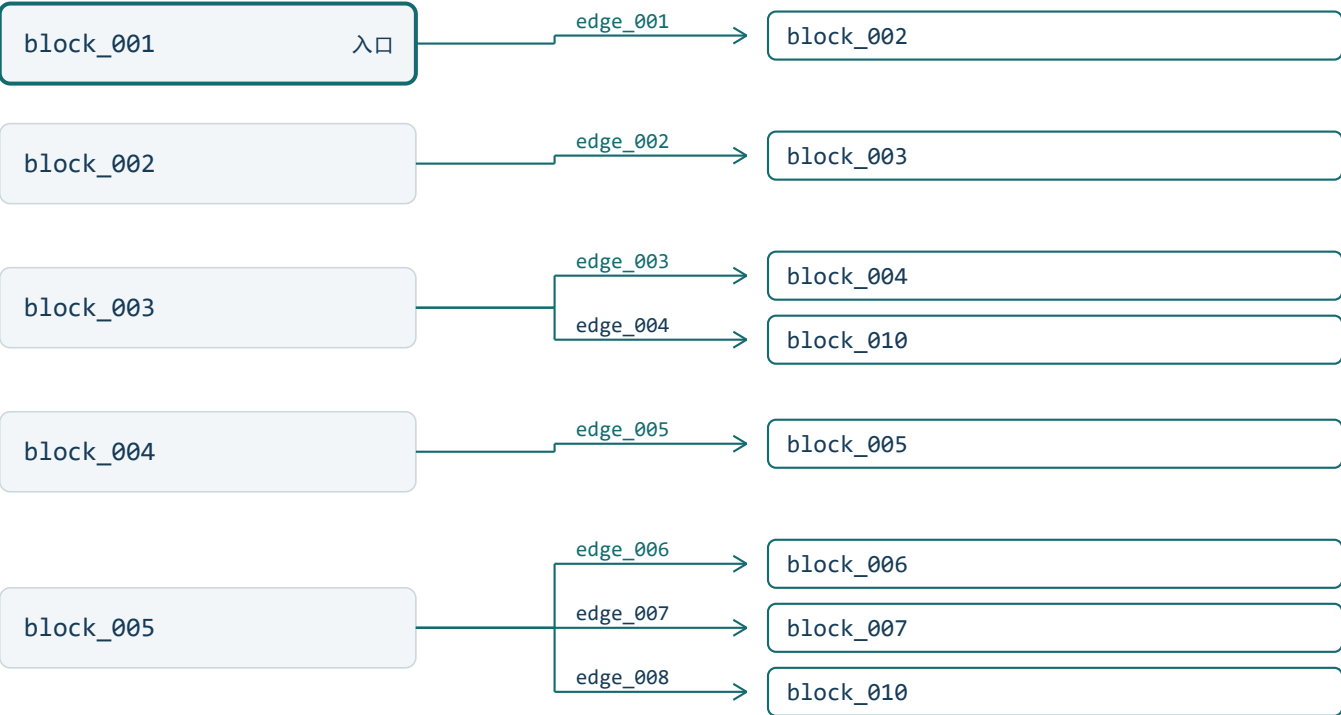
skill.constraints[1]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

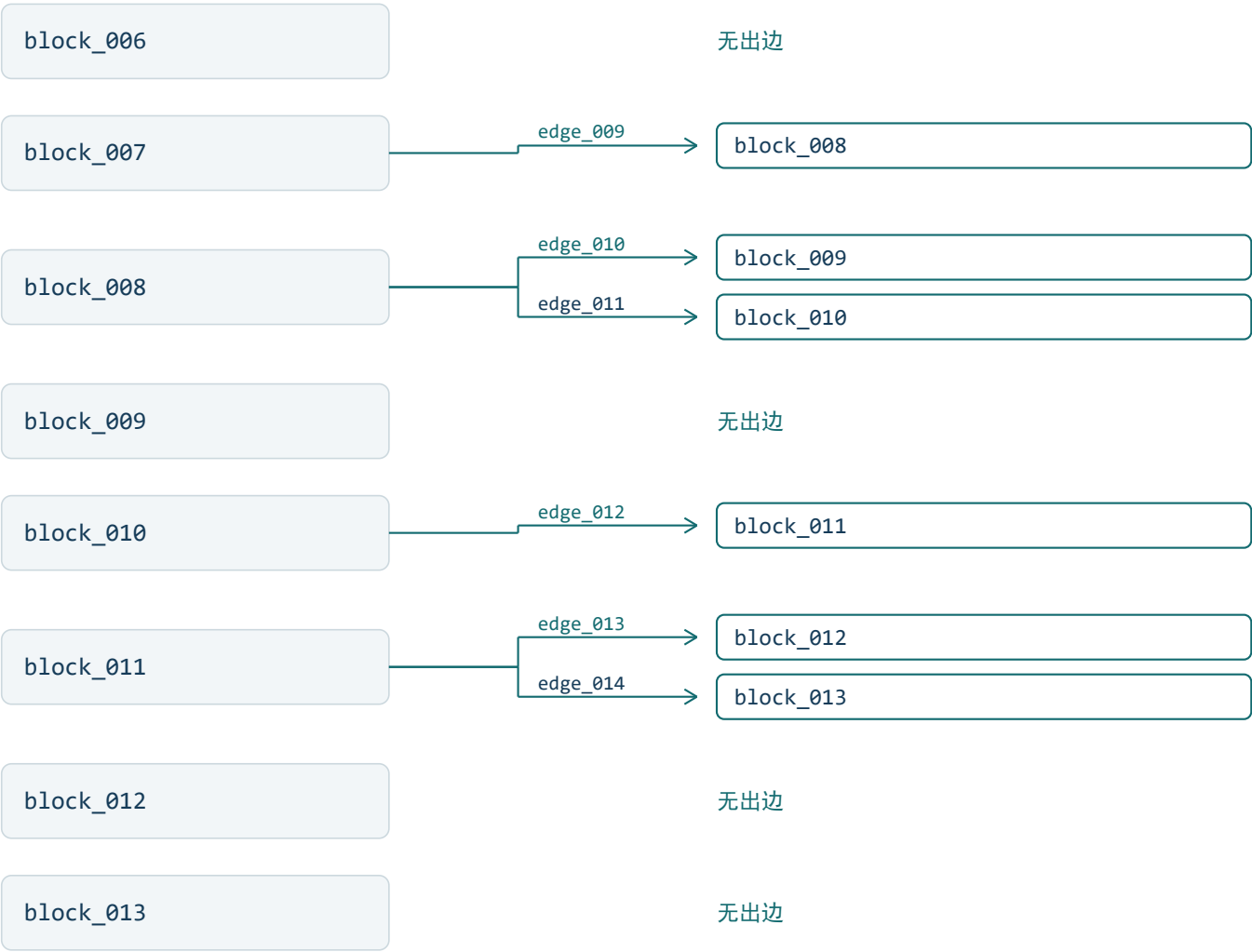
skill.constraints[2]: Before returning from every success or failure path, append the final status to local status.txt.

skill.constraints[3]: The same retry policy is specified by retry configuration; apply it to fast.fetch only.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"FAST_KEY is present"
edge_004	block_003 -> block_010	"FAST_KEY is absent"
edge_005	block_004 -> block_005	null
edge_006	block_005 -> block_006	"first fast.fetch attempt succeeds"
edge_007	block_005 -> block_007	"first fast.fetch attempt fails with a transient error"
edge_008	block_005 -> block_010	"first fast.fetch attempt fails with a non-transient error"
edge_009	block_007 -> block_008	null
edge_010	block_008 -> block_009	"fast.fetch retry succeeds"
edge_011	block_008 -> block_010	"fast.fetch retry fails"
edge_012	block_010 -> block_011	null
edge_013	block_011 -> block_012	"archive.fetch succeeds"
edge_014	block_011 -> block_013	"archive.fetch fails"

全部操作、数据引用与约束落点

block_001 Read source_id from the user's request

```
data_source_kind = "context"
block_001.constraints: []
ir_001 read_source_id_from_request
inputs[0] context_key: source_id | source_id
outputs[0] result: result_001 | source_id_value
ir_001.constraints: []
draft_instruction_id = read_source_id_from_request_1
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = dispatch_to_read_fast_key
```

block_002 Read FAST_KEY from the environment

```
data_source_kind = "context"
```

```
block_002.constraints: []
```

ir_003 read_fast_key_from_environment

```
inputs[0] context_key: FAST_KEY | FAST_KEY
```

```
outputs[0] result: result_002 | fast_key_value
```

```
ir_003.constraints: []
```

```
draft_instruction_id = read_fast_key_from_environment_1
```

ir_004 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = dispatch_to_decide_fast_key
```

block_003 Check whether FAST_KEY is present

```
data_source_kind = null
```

```
block_003.constraints[0]: If FAST_KEY is present, try fast.fetch first with source_id and FAST_KEY; if it is absent, go directly to archive.fetch without calling fast.fetch.
```

ir_005 check_fast_key_present

```
inputs[0] result: result_002 | fast_key_value
```

```
outputs[0] result: result_003 | fast_key_present
```

```
ir_005.constraints: []
```

```
draft_instruction_id = check_fast_key_present_1
```

ir_006 dispatch

```
inputs[0] result: result_003 | fast_key_present
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = dispatch_on_fast_key_presence
```

block_004 Try fast.fetch first with source_id and FAST_KEY

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_007 fast_fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id_value
```

```
inputs[2] result: result_002 | fast_key_value
```

```
outputs[0] result: result_004 | fast_fetch_first_body
```

```
outputs[1] result: result_005 | fast_fetch_first_error
```

```
ir_007.constraints: []
```

```
draft_instruction_id = call_fast_fetch_first
```

ir_008 dispatch

```
inputs = []
outputs = []
ir_008.constraints: []
draft_instruction_id = dispatch_to_classify_first_attempt
```

block_005 Determine the first fast.fetch attempt outcome

```
data_source_kind = null
block_005.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error;
do not retry a non-transient first failure.
block_005.constraints[1]: After a non-transient first failure or any failed retry of fast.fetch, call
archive.fetch with source_id.
```

ir_009 classify_fast_fetch_first_attempt

```
inputs[0] result: result_004 | fast_fetch_first_body
inputs[1] result: result_005 | fast_fetch_first_error
outputs[0] result: result_006 | first_attempt_status
ir_009.constraints: []
draft_instruction_id = classify_fast_fetch_first_attempt_1
```

ir_010 dispatch

```
inputs[0] result: result_006 | first_attempt_status
outputs = []
ir_010.constraints: []
draft_instruction_id = dispatch_on_first_attempt_status
```

block_006 Append final status and return the first fast.fetch body

```
data_source_kind = null
block_006.constraints: []
```

ir_011 append_final_status_to_local_status_file

```
inputs[0] external_resource: local_status_file | local_status.txt
inputs[1] result: result_006 | first_attempt_status
outputs = []
ir_011.constraints: []
draft_instruction_id = append_status_success_first
```

ir_012 return

```
inputs[0] result: result_004 | fast_fetch_first_body
outputs = []
ir_012.constraints: []
draft_instruction_id = return_first_fast_fetch_body
```

block_007 Retry fast.fetch exactly once with source_id and FAST_KEY

```
data_source_kind = "external"
```

```
block_007.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error.
```

ir_013 fast_fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id_value
```

```
inputs[2] result: result_002 | fast_key_value
```

```
outputs[0] result: result_007 | fast_fetch_retry_body
```

```
outputs[1] result: result_008 | fast_fetch_retry_error
```

```
ir_013.constraints: []
```

```
draft_instruction_id = call_fast_fetch_retry
```

ir_014 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = dispatch_to_classify_retry_attempt
```

block_008 Determine whether the fast.fetch retry succeeded or failed

```
data_source_kind = null
```

```
block_008.constraints[0]: After any failed retry of fast.fetch, call archive.fetch with source_id.
```

ir_015 classify_fast_fetch_retry_attempt

```
inputs[0] result: result_007 | fast_fetch_retry_body
```

```
inputs[1] result: result_008 | fast_fetch_retry_error
```

```
outputs[0] result: result_009 | retry_attempt_status
```

```
ir_015.constraints: []
```

```
draft_instruction_id = classify_fast_fetch_retry_attempt_1
```

ir_016 dispatch

```
inputs[0] result: result_009 | retry_attempt_status
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = dispatch_on_retry_attempt_status
```

block_009 Append final status and return the retry fast.fetch body

```
data_source_kind = null
```

```
block_009.constraints: []
```

ir_017 append_final_status_to_local_status_file

```
inputs[0] external_resource: local_status_file | local_status.txt
```

```
inputs[1] result: result_009 | retry_attempt_status
```

```
outputs = []
```

```
ir_017.constraints: []
```

```
draft_instruction_id = append_status_success_retry
```

ir_018 return

```
inputs[0] result: result_007 | fast_fetch_retry_body
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = return_retry_fast_fetch_body
```

block_010 Call archive.fetch with source_id only

```
data_source_kind = "external"
```

```
block_010.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.
```

ir_019 archive_fetch

```
inputs[0] external_resource: archive.fetch | archive.fetch
```

```
inputs[1] result: result_001 | source_id_value
```

```
outputs[0] result: result_010 | archive_fetch_body
```

```
outputs[1] result: result_011 | archive_fetch_error
```

```
ir_019.constraints: []
```

```
draft_instruction_id = call_archive_fetch
```

ir_020 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = dispatch_to_classify_archive_attempt
```

block_011 Determine whether archive.fetch succeeded or failed

```
data_source_kind = null
```

```
block_011.constraints[0]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

ir_021 classify_archive_fetch_attempt

```
inputs[0] result: result_010 | archive_fetch_body
```

```
inputs[1] result: result_011 | archive_fetch_error
```

```
outputs[0] result: result_012 | archive_attempt_status
```

```
ir_021.constraints: []
```

```
draft_instruction_id = classify_archive_fetch_attempt_1
```

ir_022 dispatch

```
inputs[0] result: result_012 | archive_attempt_status
```

```
outputs = []
```

```
ir_022.constraints: []
```

```
draft_instruction_id = dispatch_on_archive_attempt_status
```

block_012 Append final status and return the archive.fetch body

```
data_source_kind = null
```

```
block_012.constraints: []
```

ir_023 append_final_status_to_local_status_file

```
inputs[0] external_resource: local_status_file | local_status.txt
```

```
inputs[1] result: result_012 | archive_attempt_status
outputs = []
ir_023.constraints: []
draft_instruction_id = append_status_success_archive

ir_024 return

inputs[0] result: result_010 | archive_fetch_body
outputs = []
ir_024.constraints: []
draft_instruction_id = return_archive_fetch_body

block_013 Append final status and return the archive.fetch error

data_source_kind = null
block_013.constraints: []

ir_025 append_final_status_to_local_status_file

inputs[0] external_resource: local_status_file | local status.txt
inputs[1] result: result_012 | archive_attempt_status
outputs = []
ir_025.constraints: []
draft_instruction_id = append_status_failure_archive

ir_026 return

inputs[0] result: result_011 | archive_fetch_error
outputs = []
ir_026.constraints: []
draft_instruction_id = return_archive_fetch_error
```

F05 / 第 1 次 实际输出明细

工具选择与失败回退 · 用途最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F05/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Step 9 is explicitly marked as a historical example, not executed in this workflow; no status.txt append behavior was modeled.

Skill 层约束与上下文声明

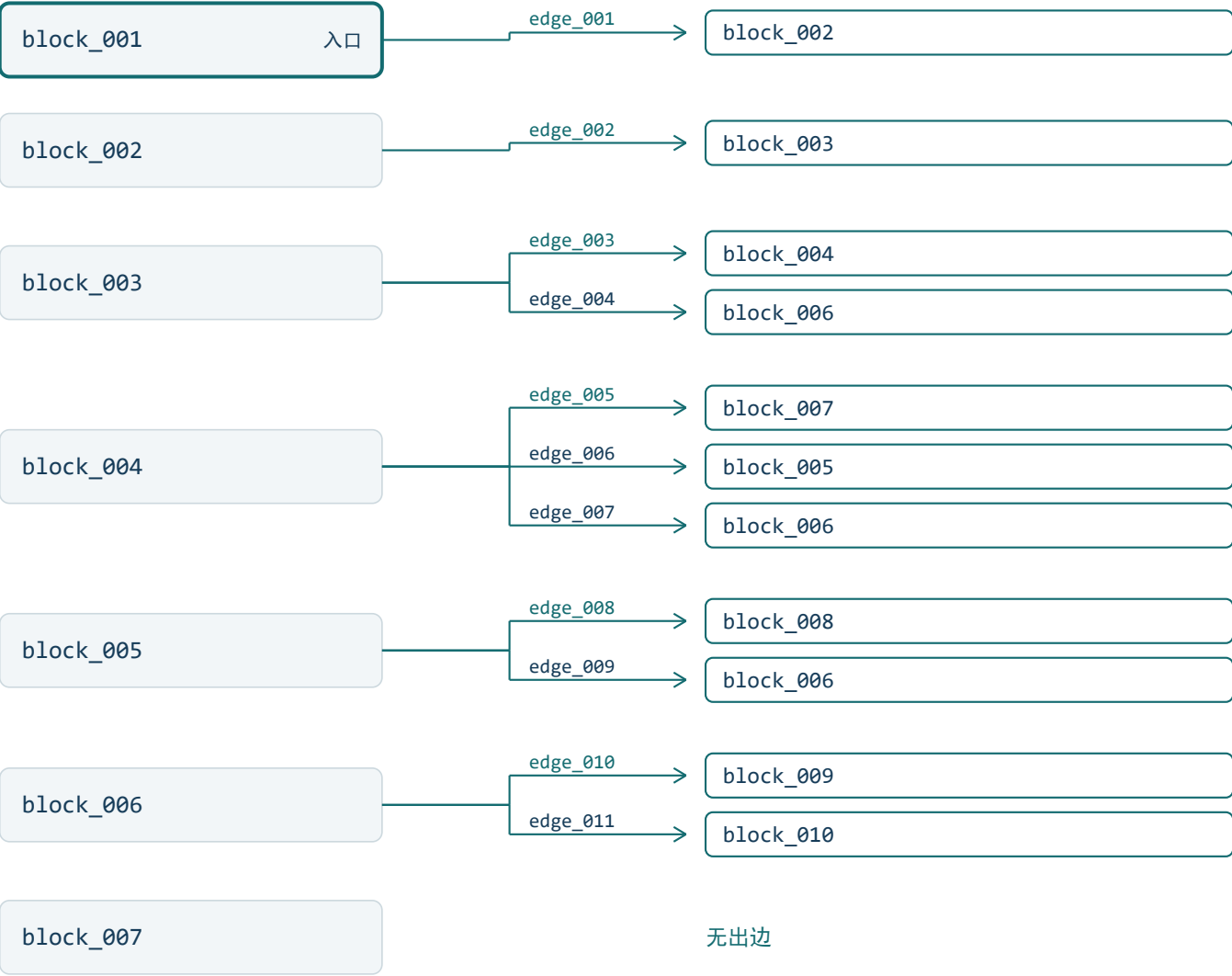
```
declared_context_keys = ["source_id", "FAST_KEY"]

skill.constraints[0]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

skill.constraints[1]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



block_008	无出边
block_009	无出边
block_010	无出边

完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"FAST_KEY is present"
edge_004	block_003 -> block_006	"FAST_KEY is absent"
edge_005	block_004 -> block_007	"success"
edge_006	block_004 -> block_005	"transient error"
edge_007	block_004 -> block_006	"non-transient error"
edge_008	block_005 -> block_008	"success"
edge_009	block_005 -> block_006	"failure"
edge_010	block_006 -> block_009	"success"
edge_011	block_006 -> block_010	"failure"

全部操作、数据引用与约束落点

block_001 Read source_id from the user's request

```
data_source_kind = "context"
block_001.constraints: []

ir_001 read_source_id_from_request
inputs[0] context_key: source_id | source_id
outputs[0] result: result_001 | source_id
ir_001.constraints: []
draft_instruction_id = read_source_id_from_request

ir_002 dispatch
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_read_fast_key
```


block_002 Read FAST_KEY from the environment

```
data_source_kind = "context"
```

```
block_002.constraints: []
```

ir_003 read_fast_key_from_environment

```
inputs[0] context_key: FAST_KEY | FAST_KEY
```

```
outputs[0] result: result_002 | fast_key
```

```
ir_003.constraints: []
```

```
draft_instruction_id = read_fast_key_from_environment
```

ir_004 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = go_check_fast_key_present
```

block_003 Check whether FAST_KEY is present

```
data_source_kind = null
```

```
block_003.constraints[0]: If FAST_KEY is present, try fast.fetch first with source_id and FAST_KEY; if it is absent, go directly to archive.fetch without calling fast.fetch.
```

ir_005 check_fast_key_present

```
inputs[0] result: result_002 | fast_key
```

```
outputs[0] result: result_003 | fast_key_present
```

```
ir_005.constraints: []
```

```
draft_instruction_id = check_fast_key_present
```

ir_006 dispatch

```
inputs[0] result: result_003 | fast_key_present
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = branch_on_fast_key_presence
```

block_004 Try fast.fetch first with source_id and FAST_KEY

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_007 fast.fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_004 | fast_fetch_attempt_1_outcome
```

```
outputs[1] result: result_005 | fast_fetch_attempt_1_body
```

```
ir_007.constraints: []
```

```
draft_instruction_id = try_fast_fetch_first
```

ir_008 dispatch

inputs[0] result: result_004 | fast_fetch_attempt_1_outcome

outputs = []

ir_008.constraints: []

draft_instruction_id = branch_on_first_fast_fetch_outcome

block_005 Retry fast.fetch once after a transient first failure

data_source_kind = "external"

block_005.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do not retry a non-transient first failure.

ir_009 fast.fetch

inputs[0] external_resource: fast.fetch | fast.fetch

inputs[1] result: result_001 | source_id

inputs[2] result: result_002 | fast_key

outputs[0] result: result_006 | fast_fetch_attempt_2_outcome

outputs[1] result: result_007 | fast_fetch_attempt_2_body

ir_009.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do not retry a non-transient first failure.

draft_instruction_id = retry_fast_fetch_once

ir_010 dispatch

inputs[0] result: result_006 | fast_fetch_attempt_2_outcome

outputs = []

ir_010.constraints: []

draft_instruction_id = branch_on_retry_fast_fetch_outcome

block_006 Call archive.fetch with source_id

data_source_kind = "external"

block_006.constraints[0]: After a non-transient first failure or any failed retry of fast.fetch, call archive.fetch with source_id.

block_006.constraints[1]: Call archive.fetch at most once and pass source_id as its only argument.

ir_011 archive.fetch

inputs[0] external_resource: archive.fetch | archive.fetch

inputs[1] result: result_001 | source_id

outputs[0] result: result_008 | archive_outcome

outputs[1] result: result_009 | archive_body

outputs[2] result: result_010 | archive_error

ir_011.constraints[0]: After a non-transient first failure or any failed retry of fast.fetch, call archive.fetch with source_id.

ir_011.constraints[1]: Call archive.fetch at most once and pass source_id as its only argument.

draft_instruction_id = call_archive_fetch

ir_012 dispatch

```
inputs[0] result: result_008 | archive_outcome
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = branch_on_archive_fetch_outcome
```

block_007 Return the successful fast.fetch response body from the first attempt

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 return

```
inputs[0] result: result_005 | fast_fetch_attempt_1_body
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = return_fast_body_1
```

block_008 Return the successful fast.fetch response body from the retry

```
data_source_kind = null
```

```
block_008.constraints: []
```

ir_014 return

```
inputs[0] result: result_007 | fast_fetch_attempt_2_body
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = return_fast_body_2
```

block_009 Return the successful archive.fetch response body

```
data_source_kind = null
```

```
block_009.constraints: []
```

ir_015 return

```
inputs[0] result: result_009 | archive_body
```

```
outputs = []
```

```
ir_015.constraints: []
```

```
draft_instruction_id = return_archive_body
```

block_010 Return the archive.fetch error

```
data_source_kind = null
```

```
block_010.constraints[0]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

ir_016 return

```
inputs[0] result: result_010 | archive_error
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = return_archive_error
```

F05 / 第 2 次 实际输出明细

工具选择与失败回退 · 用途最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F05/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Step 9 is explicitly a historical example not executed in this workflow, so it was not modeled as runtime behavior.

Skill 层约束与上下文声明

```
declared_context_keys = ["source_id", "FAST_KEY"]

skill.constraints[0]: If FAST_KEY is present, try fast.fetch first with source_id and FAST_KEY; if it is absent, go directly to archive.fetch without calling fast.fetch.

skill.constraints[1]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do not retry a non-transient first failure.

skill.constraints[2]: After a non-transient first failure or any failed retry of fast.fetch, call archive.fetch with source_id.

skill.constraints[3]: Call archive.fetch at most once and pass source_id as its only argument.

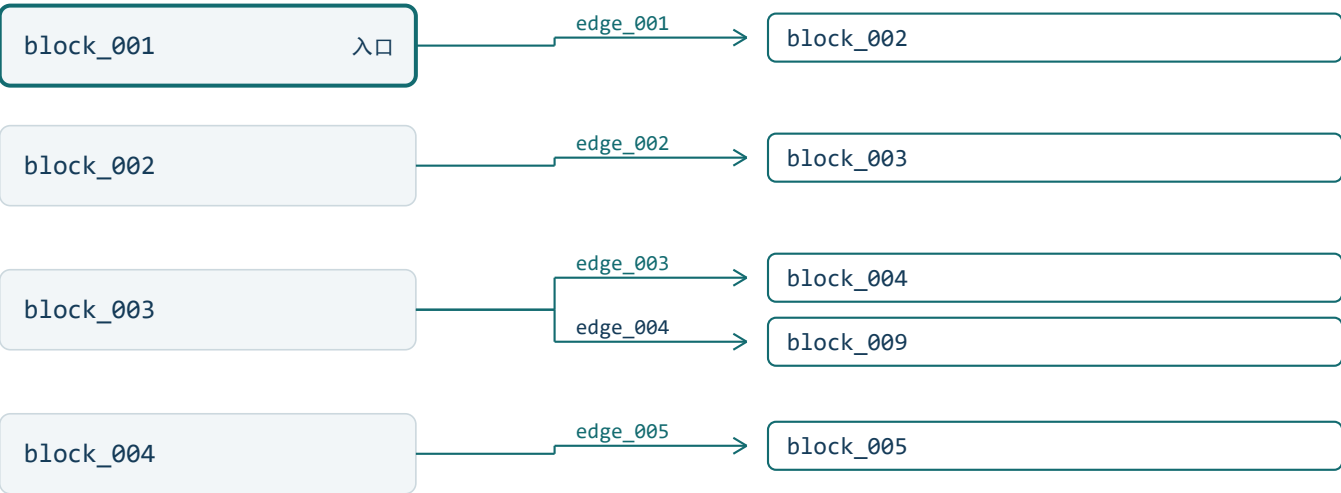
skill.constraints[4]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

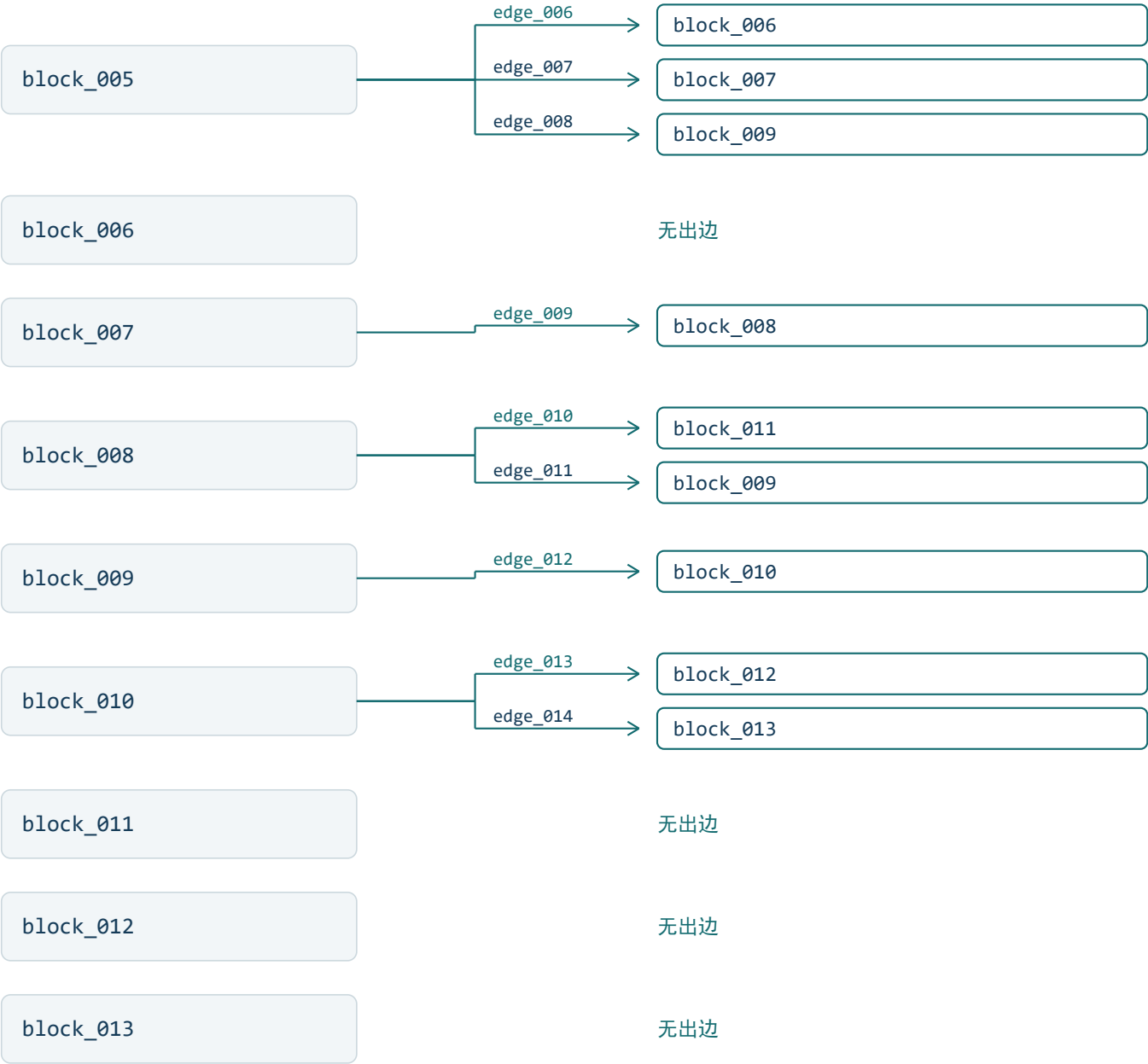
skill.constraints[5]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.

skill.constraints[6]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"FAST_KEY is present"
edge_004	block_003 -> block_009	"FAST_KEY is absent"
edge_005	block_004 -> block_005	null
edge_006	block_005 -> block_006	"fast.fetch first attempt succeeded"
edge_007	block_005 -> block_007	"fast.fetch first attempt failed with a transient error"
edge_008	block_005 -> block_009	"fast.fetch first attempt failed with a non-transient error"
edge_009	block_007 -> block_008	null
edge_010	block_008 -> block_011	"fast.fetch retry succeeded"
edge_011	block_008 -> block_009	"fast.fetch retry failed"
edge_012	block_009 -> block_010	null
edge_013	block_010 -> block_012	"archive.fetch succeeded"
edge_014	block_010 -> block_013	"archive.fetch failed"

全部操作、数据引用与约束落点

block_001 Read source_id from the user's request

data_source_kind = "context"

block_001.constraints: []

ir_001 read_source_id_from_user_request

inputs[0] context_key: source_id | source_id

outputs[0] result: result_001 | source_id

ir_001.constraints: []

draft_instruction_id = read_source_id_from_user_request

ir_002 dispatch

inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = go_read_fast_key

block_002 Read FAST_KEY from the environment

```
data_source_kind = "context"
```

```
block_002.constraints: []
```

ir_003 read_fast_key_from_environment

```
inputs[0] context_key: FAST_KEY | FAST_KEY
```

```
outputs[0] result: result_002 | fast_key
```

```
ir_003.constraints: []
```

```
draft_instruction_id = read_fast_key_from_environment
```

ir_004 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = go_check_fast_key_presence
```

block_003 Check whether FAST_KEY is present

```
data_source_kind = null
```

```
block_003.constraints[0]: If FAST_KEY is present, try fast.fetch first with source_id and FAST_KEY; if it is absent, go directly to archive.fetch without calling fast.fetch.
```

ir_005 check_fast_key_presence

```
inputs[0] result: result_002 | fast_key
```

```
outputs[0] result: result_003 | fast_key_present
```

```
ir_005.constraints: []
```

```
draft_instruction_id = check_fast_key_presence
```

ir_006 dispatch

```
inputs[0] result: result_003 | fast_key_present
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = choose_fast_path
```

block_004 Try fast.fetch first with source_id and FAST_KEY

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_007 fast_fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_004 | fast_fetch_first_result
```

```
ir_007.constraints: []
```

```
draft_instruction_id = call_fast_fetch_first
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_classify_first
```

block_005 Classify fast.fetch first attempt outcome

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 classify_fast_fetch_first_outcome

```
inputs[0] result: result_004 | fast_fetch_first_result
```

```
outputs[0] result: result_005 | fast_first_succeeded
```

```
outputs[1] result: result_006 | fast_first_transient_failure
```

```
outputs[2] result: result_007 | fast_first_nontransient_failure
```

```
outputs[3] result: result_008 | fast_first_body
```

```
outputs[4] result: result_009 | fast_first_error
```

```
ir_009.constraints: []
```

```
draft_instruction_id = classify_fast_fetch_first_outcome
```

ir_010 dispatch

```
inputs[0] result: result_005 | fast_first_succeeded
```

```
inputs[1] result: result_006 | fast_first_transient_failure
```

```
inputs[2] result: result_007 | fast_first_nontransient_failure
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = choose_first_outcome
```

block_006 Return fast.fetch first success body

```
data_source_kind = null
```

```
block_006.constraints[0]: On success, return the successful response's body value unchanged and make no further fetch calls.
```

ir_011 return

```
inputs[0] result: result_008 | fast_first_body
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = return_fast_first_body
```

block_007 Retry fast.fetch once after transient first failure

```
data_source_kind = "external"
```

```
block_007.constraints: []
```

ir_012 fast_fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id
```



```
inputs[2] result: result_002 | fast_key
outputs[0] result: result_010 | fast_fetch_retry_result
ir_012.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do
not retry a non-transient first failure.
draft_instruction_id = call_fast_fetch_retry

ir_013 dispatch

inputs = []
outputs = []
ir_013.constraints: []
draft_instruction_id = go_classify_retry

block_008 Classify fast.fetch retry outcome

data_source_kind = null
block_008.constraints: []

ir_014 classify_fast_fetch_retry_outcome

inputs[0] result: result_010 | fast_fetch_retry_result
outputs[0] result: result_011 | fast_retry_succeeded
outputs[1] result: result_012 | fast_retry_body
outputs[2] result: result_013 | fast_retry_error
ir_014.constraints: []
draft_instruction_id = classify_fast_fetch_retry_outcome

ir_015 dispatch

inputs[0] result: result_011 | fast_retry_succeeded
outputs = []
ir_015.constraints: []
draft_instruction_id = choose_retry_outcome

block_009 Call archive.fetch with source_id

data_source_kind = "external"
block_009.constraints: []

ir_016 archive_fetch

inputs[0] external_resource: archive.fetch | archive.fetch
inputs[1] result: result_001 | source_id
outputs[0] result: result_014 | archive_result
ir_016.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.
ir_016.constraints[1]: Never pass FAST_KEY to archive.fetch.
draft_instruction_id = call_archive_fetch

ir_017 dispatch

inputs = []
outputs = []
ir_017.constraints: []
draft_instruction_id = go_classify_archive
```

block_010 Classify archive.fetch outcome

data_source_kind = null

block_010.constraints: []

ir_018 classify_archive_fetch_outcome

inputs[0] result: result_014 | archive_result

outputs[0] result: result_015 | archive_succeeded

outputs[1] result: result_016 | archive_body

outputs[2] result: result_017 | archive_error

ir_018.constraints: []

draft_instruction_id = classify_archive_fetch_outcome

ir_019 dispatch

inputs[0] result: result_015 | archive_succeeded

outputs = []

ir_019.constraints: []

draft_instruction_id = choose_archive_outcome

block_011 Return fast.fetch retry success body

data_source_kind = null

block_011.constraints[0]: On success, return the successful response's body value unchanged and make no further fetch calls.

ir_020 return

inputs[0] result: result_012 | fast_retry_body

outputs = []

ir_020.constraints: []

draft_instruction_id = return_fast_retry_body

block_012 Return archive.fetch success body

data_source_kind = null

block_012.constraints[0]: On success, return the successful response's body value unchanged and make no further fetch calls.

ir_021 return

inputs[0] result: result_016 | archive_body

outputs = []

ir_021.constraints: []

draft_instruction_id = return_archive_body

block_013 Return archive.fetch error and stop

```
data_source_kind = null
```

```
block_013.constraints[0]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

ir_022 return

```
inputs[0] result: result_017 | archive_error
```

```
outputs = []
```

```
ir_022.constraints: []
```

```
draft_instruction_id = return_archive_error
```

F05 / 第 3 次 实际输出明细

工具选择与失败回退 · 用途最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F05/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Step 9 is a historical example and is not executed; no runtime block was created for appending status to local status.txt.

Skill 层约束与上下文声明

```
declared_context_keys = ["user_request", "FAST_KEY"]

skill.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do not retry a non-transient first failure.

skill.constraints[1]: After a non-transient first failure or any failed retry of fast.fetch, call archive.fetch with source_id.

skill.constraints[2]: Call archive.fetch at most once and pass source_id as its only argument.

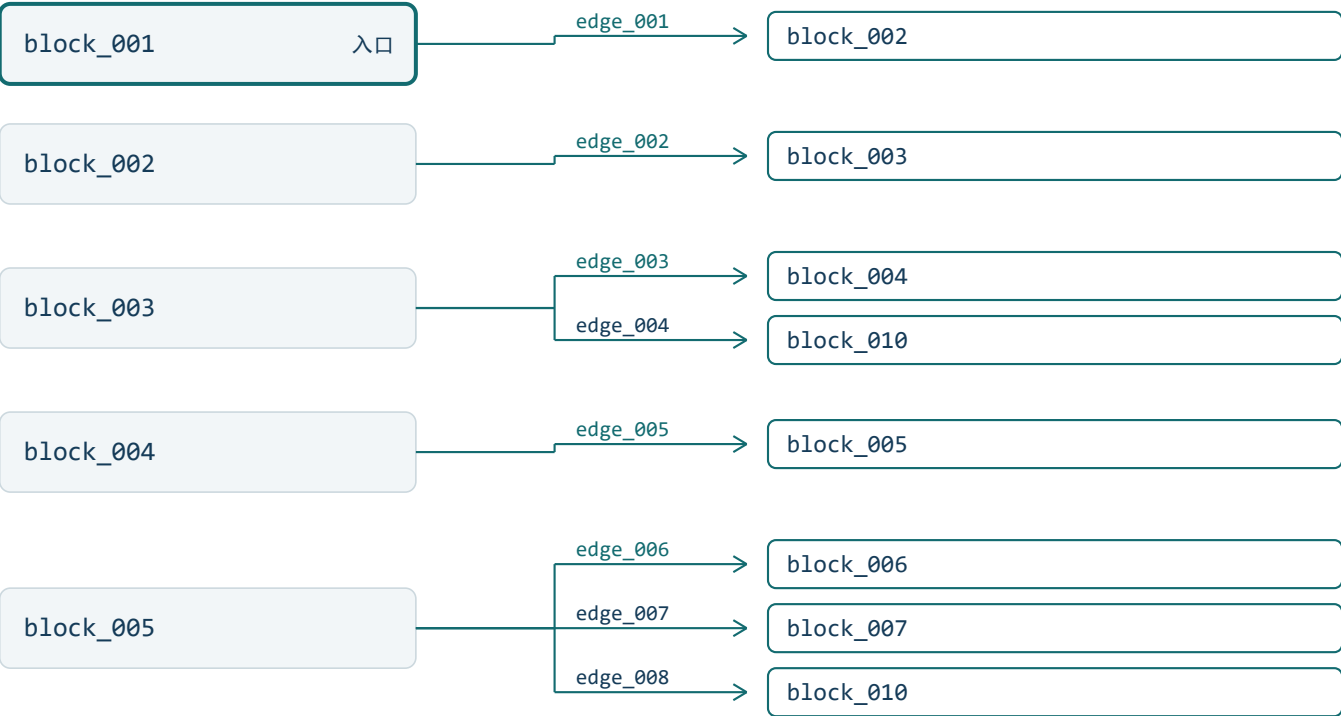
skill.constraints[3]: On either tool's success, return that successful response's body value unchanged and make no further fetch calls.

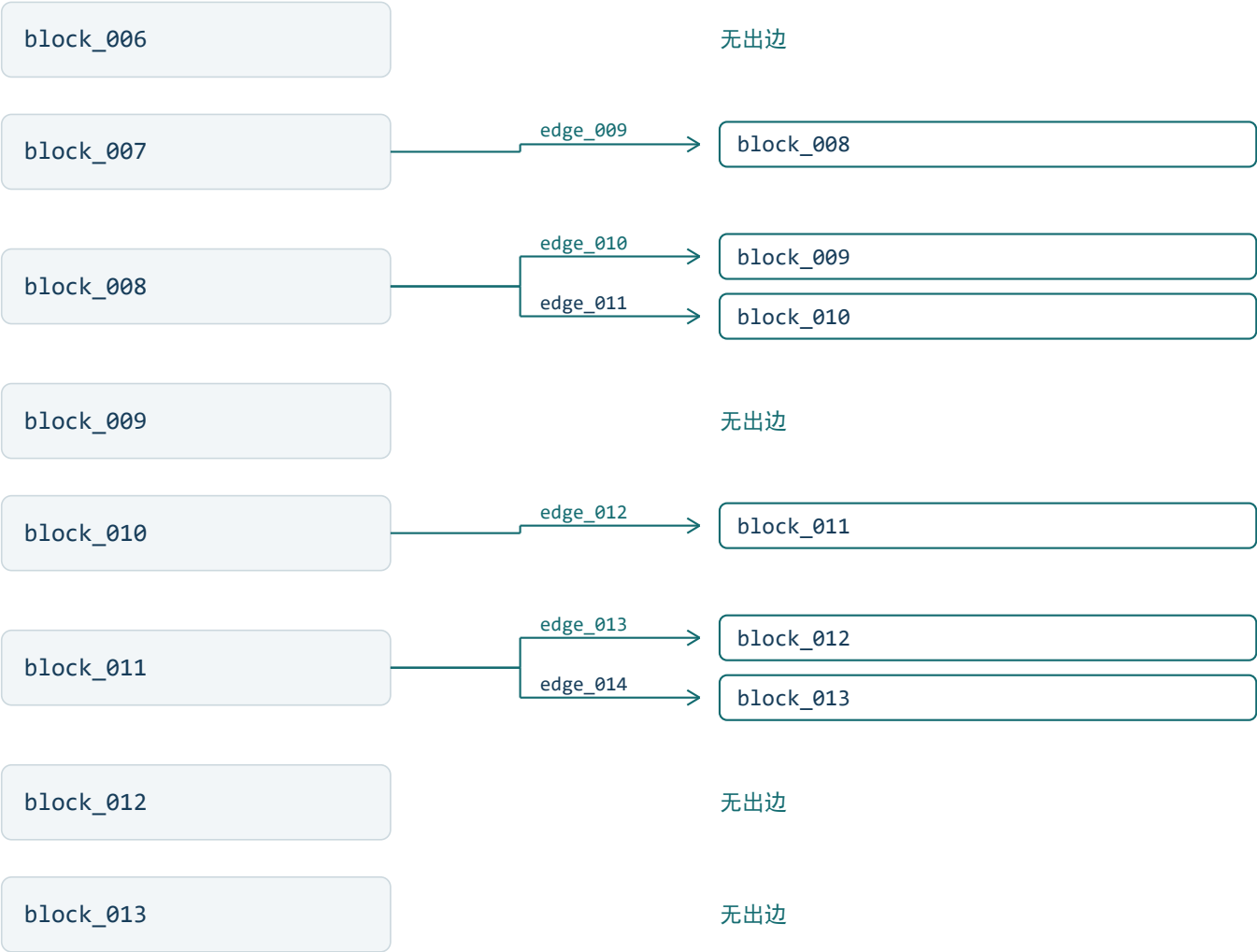
skill.constraints[4]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.

skill.constraints[5]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"FAST_KEY present"
edge_004	block_003 -> block_010	"FAST_KEY absent"
edge_005	block_004 -> block_005	null
edge_006	block_005 -> block_006	"fast.fetch attempt 1 succeeds"
edge_007	block_005 -> block_007	"fast.fetch attempt 1 fails with a transient error"
edge_008	block_005 -> block_010	"fast.fetch attempt 1 fails with a non-transient error"
edge_009	block_007 -> block_008	null
edge_010	block_008 -> block_009	"fast.fetch retry succeeds"
edge_011	block_008 -> block_010	"fast.fetch retry fails"
edge_012	block_010 -> block_011	null
edge_013	block_011 -> block_012	"archive.fetch succeeds"
edge_014	block_011 -> block_013	"archive.fetch fails"

全部操作、数据引用与约束落点

block_001 Read source_id from the user's request

```
data_source_kind = "context"
```

```
block_001.constraints: []
```

ir_001 read_source_id_from_user_request

```
inputs[0] context_key: user_request | user_request
```

```
outputs[0] result: result_001 | source_id
```

```
ir_001.constraints: []
```

```
draft_instruction_id = read_source_id_from_user_request
```

ir_002 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_002.constraints: []
```

```
draft_instruction_id = go_read_fast_key
```

block_002 Read FAST_KEY from the environment

```
data_source_kind = "context"
```

```
block_002.constraints: []
```

ir_003 read_fast_key_from_environment

```
inputs[0] context_key: FAST_KEY | FAST_KEY
```

```
outputs[0] result: result_002 | fast_key
```

```
ir_003.constraints: []
```

```
draft_instruction_id = read_fast_key_from_environment
```

ir_004 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = go_decide_fast_key_present
```

block_003 Check whether FAST_KEY is present

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 check_fast_key_present

```
inputs[0] result: result_002 | fast_key
```

```
outputs[0] result: result_003 | fast_key_present
```

```
ir_005.constraints: []
```

```
draft_instruction_id = check_fast_key_present
```

ir_006 dispatch

```
inputs[0] result: result_003 | fast_key_present
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = choose_fetch_path
```

block_004 Try fast.fetch with source_id and FAST_KEY

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_007 fast.fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_004 | fast_fetch_attempt1_response
```

```
outputs[1] result: result_005 | fast_fetch_attempt1_body
```

```
ir_007.constraints: []
```

```
draft_instruction_id = fast_fetch_attempt1_call
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_classify_fast_fetch_attempt1
```

block_005 Classify the first fast.fetch attempt

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 classify_fast_fetch_attempt1_outcome

```
inputs[0] result: result_004 | fast_fetch_attempt1_response
```

```
outputs[0] result: result_006 | fast_fetch_attempt1_status
```

```
ir_009.constraints: []
```

```
draft_instruction_id = classify_fast_fetch_attempt1_outcome
```

ir_010 dispatch

```
inputs[0] result: result_006 | fast_fetch_attempt1_status
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = choose_after_fast_fetch_attempt1
```

block_006 Return the first fast.fetch response body

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 return

```
inputs[0] result: result_005 | fast_fetch_attempt1_body
```

```
outputs = []
```

```
ir_011.constraints[0]: Return the successful response's body value unchanged and make no further fetch calls.
```

```
draft_instruction_id = return_fast_fetch_attempt1_body
```

block_007 Retry fast.fetch once after a transient failure

```
data_source_kind = "external"
```

```
block_007.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error;  
do not retry a non-transient first failure.
```

ir_012 fast.fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_007 | fast_fetch_attempt2_response
```

```
outputs[1] result: result_008 | fast_fetch_attempt2_body
```

```
ir_012.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error.
```

```
draft_instruction_id = fast_fetch_attempt2_call
```


ir_013 dispatch

```
inputs = []
outputs = []
ir_013.constraints: []
draft_instruction_id = go_classify_fast_fetch_attempt2
```

block_008 Classify the retry of fast.fetch

```
data_source_kind = null
block_008.constraints: []
```

ir_014 classify_fast_fetch_attempt2_outcome

```
inputs[0] result: result_007 | fast_fetch_attempt2_response
outputs[0] result: result_009 | fast_fetch_attempt2_status
ir_014.constraints: []
draft_instruction_id = classify_fast_fetch_attempt2_outcome
```

ir_015 dispatch

```
inputs[0] result: result_009 | fast_fetch_attempt2_status
outputs = []
ir_015.constraints: []
draft_instruction_id = choose_after_fast_fetch_attempt2
```

block_009 Return the fast.fetch retry response body

```
data_source_kind = null
block_009.constraints: []
```

ir_016 return

```
inputs[0] result: result_008 | fast_fetch_attempt2_body
outputs = []
ir_016.constraints[0]: Return the successful response's body value unchanged and make no further fetch calls.
draft_instruction_id = return_fast_fetch_attempt2_body
```

block_010 Call archive.fetch with source_id

```
data_source_kind = "external"
block_010.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.
block_010.constraints[1]: Never pass FAST_KEY to archive.fetch.
```

ir_017 archive.fetch

```
inputs[0] external_resource: archive.fetch | archive.fetch
inputs[1] result: result_001 | source_id
outputs[0] result: result_010 | archive_fetch_response
outputs[1] result: result_011 | archive_fetch_body
outputs[2] result: result_012 | archive_fetch_error
ir_017.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.
ir_017.constraints[1]: Never pass FAST_KEY to archive.fetch.
draft_instruction_id = archive_fetch_call
```

ir_018 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = go_classify_archive_fetch
```

block_011 Classify the archive.fetch result

```
data_source_kind = null
```

```
block_011.constraints: []
```

ir_019 classify_archive_fetch_outcome

```
inputs[0] result: result_010 | archive_fetch_response
```

```
outputs[0] result: result_013 | archive_fetch_status
```

```
ir_019.constraints: []
```

```
draft_instruction_id = classify_archive_fetch_outcome
```

ir_020 dispatch

```
inputs[0] result: result_013 | archive_fetch_status
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = choose_after_archive_fetch
```

block_012 Return the archive.fetch response body

```
data_source_kind = null
```

```
block_012.constraints: []
```

ir_021 return

```
inputs[0] result: result_011 | archive_fetch_body
```

```
outputs = []
```

```
ir_021.constraints[0]: Return the successful response's body value unchanged and make no further fetch calls.
```

```
draft_instruction_id = return_archive_fetch_body
```

block_013 Return the archive.fetch error

```
data_source_kind = null
```

```
block_013.constraints[0]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

ir_022 return

```
inputs[0] result: result_012 | archive_fetch_error
```

```
outputs = []
```

```
ir_022.constraints[0]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

```
draft_instruction_id = return_archive_fetch_error
```

F06 / 第 1 次 实际输出明细

工具选择与失败回退 · 语义最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F06/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

```
declared_context_keys = ["user_request", "FAST_KEY"]

skill.constraints[0]: If FAST_KEY is present, try fast.fetch first with source_id and FAST_KEY; if it is absent, go directly to archive.fetch without calling fast.fetch.

skill.constraints[1]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do not retry a non-transient first failure.

skill.constraints[2]: After a non-transient first failure or any failed retry of fast.fetch, call archive.fetch with source_id.

skill.constraints[3]: Call archive.fetch at most once and pass source_id as its only argument.

skill.constraints[4]: On either tool's success, return that successful response's summary value unchanged and make no further fetch calls.

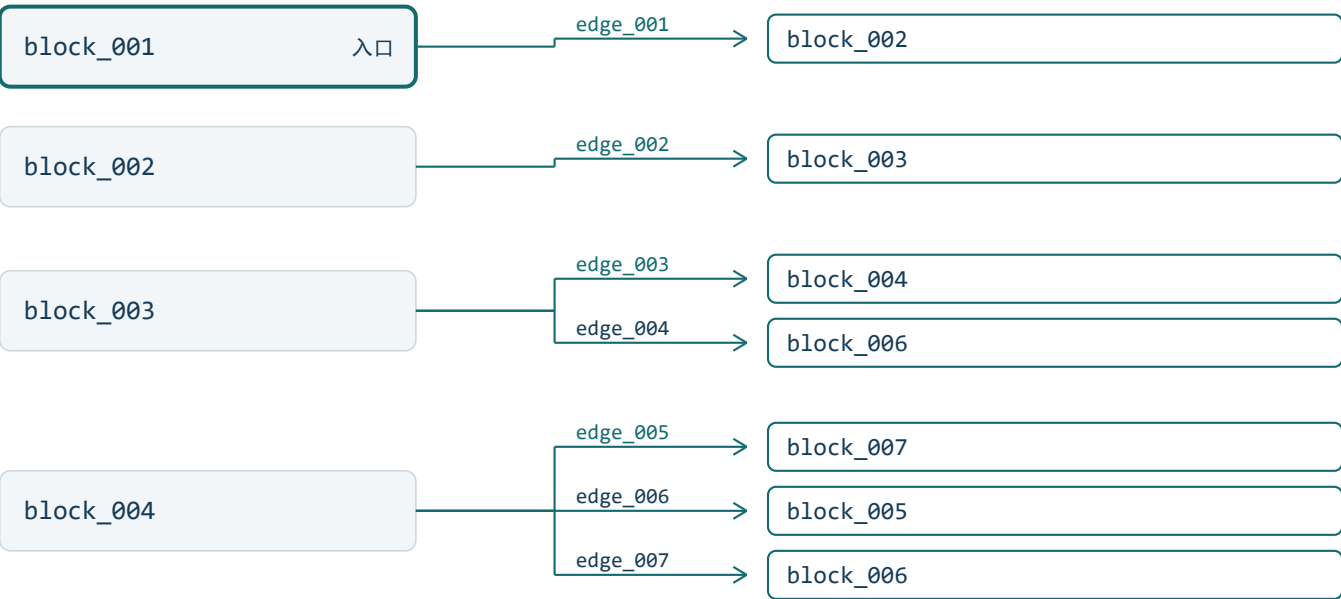
skill.constraints[5]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.

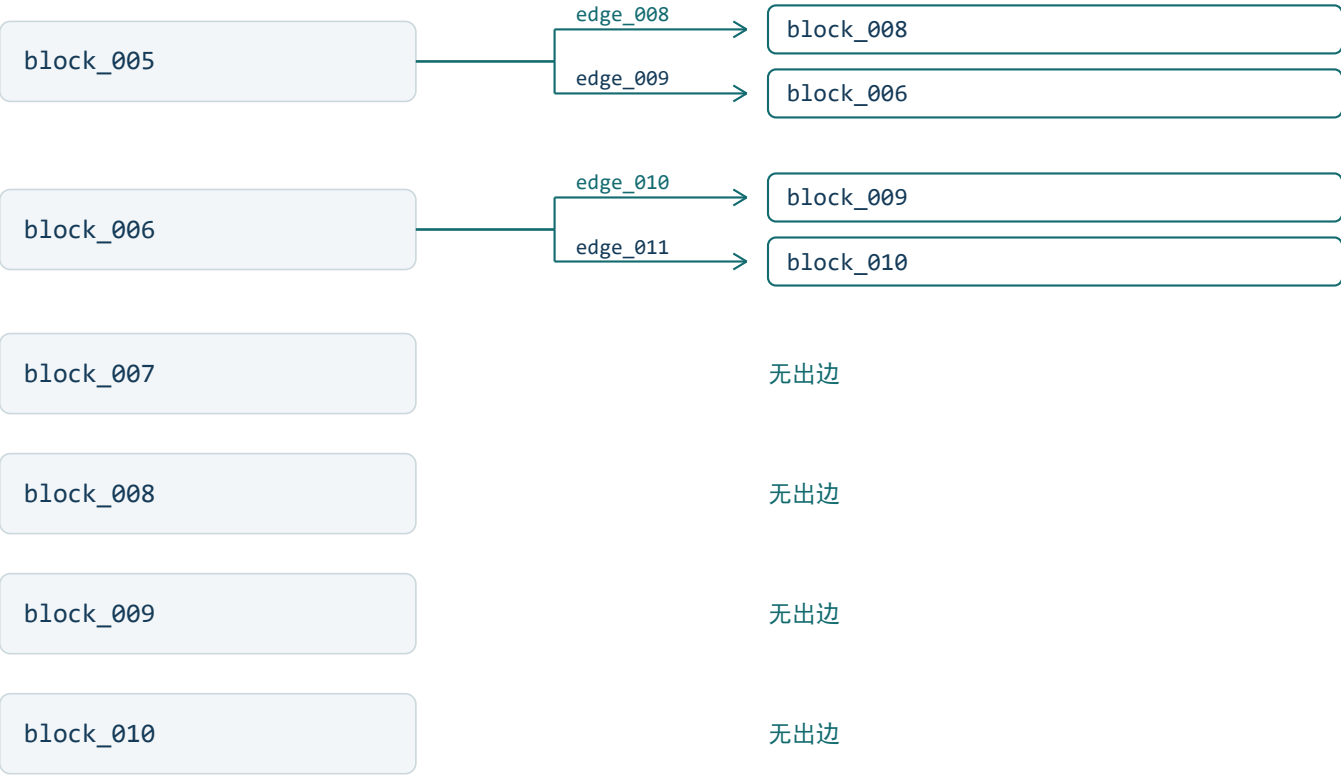
skill.constraints[6]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

skill.constraints[7]: Before returning from every success or failure path, append the final status to local status.txt.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"FAST_KEY present"
edge_004	block_003 -> block_006	"FAST_KEY absent"
edge_005	block_004 -> block_007	"first fast.fetch succeeds"
edge_006	block_004 -> block_005	"first fast.fetch fails with a transient error"
edge_007	block_004 -> block_006	"first fast.fetch fails with a non-transient error"
edge_008	block_005 -> block_008	"retry fast.fetch succeeds"
edge_009	block_005 -> block_006	"retry fast.fetch fails"
edge_010	block_006 -> block_009	"archive.fetch succeeds"
edge_011	block_006 -> block_010	"archive.fetch fails"

全部操作、数据引用与约束落点

block_001 Read source_id from the user's request

data_source_kind = "context"

block_001.constraints: []

ir_001 read_source_id_from_user_request

inputs[0] context_key: user_request | user_request

outputs[0] result: result_001 | source_id

ir_001.constraints: []

draft_instruction_id = read_source_id_from_request

ir_002 dispatch

inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = go_read_fast_key

block_002 Read FAST_KEY from the environment

data_source_kind = "context"

block_002.constraints: []

ir_003 read_fast_key_from_environment

inputs[0] context_key: FAST_KEY | FAST_KEY

outputs[0] result: result_002 | fast_key

ir_003.constraints: []

draft_instruction_id = read_fast_key_from_environment

ir_004 dispatch

inputs = []

outputs = []

ir_004.constraints: []

draft_instruction_id = go_check_fast_key

block_003 Check whether FAST_KEY is present

data_source_kind = null

block_003.constraints: []

ir_005 check_fast_key_present

inputs[0] result: result_002 | fast_key

outputs[0] result: result_003 | fast_key_present

ir_005.constraints: []

draft_instruction_id = check_fast_key_presence

ir_006 dispatch

```
inputs[0] result: result_003 | fast_key_present
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = choose_fast_or_archive
```

block_004 Try fast.fetch with source_id and FAST_KEY

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_007 fast.fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_004 | first_fast_fetch_outcome
```

```
outputs[1] result: result_005 | first_fast_fetch_summary
```

```
outputs[2] result: result_006 | first_fast_fetch_error
```

```
ir_007.constraints[0]: If FAST_KEY is present, try fast.fetch first with source_id and FAST_KEY.
```

```
draft_instruction_id = fast_fetch_first_attempt
```

ir_008 dispatch

```
inputs[0] result: result_004 | first_fast_fetch_outcome
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_fast_fetch_result
```

block_005 Retry fast.fetch once after a transient first failure

```
data_source_kind = "external"
```

```
block_005.constraints: []
```

ir_009 fast.fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_007 | retry_fast_fetch_outcome
```

```
outputs[1] result: result_008 | retry_fast_fetch_summary
```

```
outputs[2] result: result_009 | retry_fast_fetch_error
```

```
ir_009.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do not retry a non-transient first failure.
```

```
draft_instruction_id = fast_fetch_retry
```

ir_010 dispatch

```
inputs[0] result: result_007 | retry_fast_fetch_outcome
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_retry_fast_fetch_result
```

block_006 Call archive.fetch with source_id

data_source_kind = "external"

block_006.constraints: []

ir_011 archive.fetch

inputs[0] external_resource: archive.fetch | archive.fetch

inputs[1] result: result_001 | source_id

outputs[0] result: result_010 | archive_fetch_outcome

outputs[1] result: result_011 | archive_fetch_summary

outputs[2] result: result_012 | archive_fetch_error

ir_011.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.

ir_011.constraints[1]: Never pass FAST_KEY to archive.fetch.

draft_instruction_id = archive_fetch_once

ir_012 dispatch

inputs[0] result: result_010 | archive_fetch_outcome

outputs = []

ir_012.constraints: []

draft_instruction_id = dispatch_archive_fetch_result

block_007 Append final status and return fast.fetch summary

data_source_kind = null

block_007.constraints: []

ir_013 append_final_status_to_local_status_txt

inputs[0] external_resource: status.txt | status.txt

inputs[1] literal = "success"

outputs = []

ir_013.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

ir_013.constraints[1]: Never pass FAST_KEY to diagnostic output anywhere in this workflow.

draft_instruction_id = append_status_after_first_fast_fetch_success

ir_014 return

inputs[0] result: result_005 | first_fast_fetch_summary

outputs = []

ir_014.constraints[0]: On either tool's success, return that successful response's summary value unchanged and make no further fetch calls.

draft_instruction_id = return_first_fast_fetch_summary

block_008 Append final status and return retry fast.fetch summary

data_source_kind = null

block_008.constraints: []

ir_015 append_final_status_to_local_status_txt

inputs[0] external_resource: status.txt | status.txt

inputs[1] literal = "success"

```
outputs = []

ir_015.constraints[0]: Before returning from every success or failure path, append the final status to local
status.txt.

ir_015.constraints[1]: Never pass FAST_KEY to diagnostic output anywhere in this workflow.

draft_instruction_id = append_status_after_retry_fast_fetch_success

ir_016 return

inputs[0] result: result_008 | retry_fast_fetch_summary

outputs = []

ir_016.constraints[0]: On either tool's success, return that successful response's summary value unchanged and
make no further fetch calls.

draft_instruction_id = return_retry_fast_fetch_summary

block_009 Append final status and return archive.fetch summary

data_source_kind = null

block_009.constraints: []

ir_017 append_final_status_to_local_status_txt

inputs[0] external_resource: status.txt | status.txt

inputs[1] literal = "success"

outputs = []

ir_017.constraints[0]: Before returning from every success or failure path, append the final status to local
status.txt.

ir_017.constraints[1]: Never pass FAST_KEY to diagnostic output anywhere in this workflow.

draft_instruction_id = append_status_after_archive_success

ir_018 return

inputs[0] result: result_011 | archive_fetch_summary

outputs = []

ir_018.constraints[0]: On either tool's success, return that successful response's summary value unchanged and
make no further fetch calls.

draft_instruction_id = return_archive_fetch_summary

block_010 Append final status and return archive.fetch error

data_source_kind = null

block_010.constraints: []

ir_019 append_final_status_to_local_status_txt

inputs[0] external_resource: status.txt | status.txt

inputs[1] literal = "failure"

outputs = []

ir_019.constraints[0]: Before returning from every success or failure path, append the final status to local
status.txt.

ir_019.constraints[1]: Never pass FAST_KEY to diagnostic output anywhere in this workflow.

draft_instruction_id = append_status_after_archive_failure
```


ir_020 return

```
inputs[0] result: result_012 | archive_fetch_error
```

```
outputs = []
```

```
ir_020.constraints[0]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

```
draft_instruction_id = return_archive_fetch_error
```

F06 / 第 2 次 实际输出明细

工具选择与失败回退 · 语义最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F06/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

```
declared_context_keys = ["source_id", "FAST_KEY"]

skill.constraints[0]: If FAST_KEY is present, try fast.fetch first with source_id and FAST_KEY; if it is absent, go directly to archive.fetch without calling fast.fetch.

skill.constraints[1]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do not retry a non-transient first failure.

skill.constraints[2]: After a non-transient first failure or any failed retry of fast.fetch, call archive.fetch with source_id.

skill.constraints[3]: Call archive.fetch at most once and pass source_id as its only argument.

skill.constraints[4]: On either tool's success, return that successful response's summary value unchanged and make no further fetch calls.

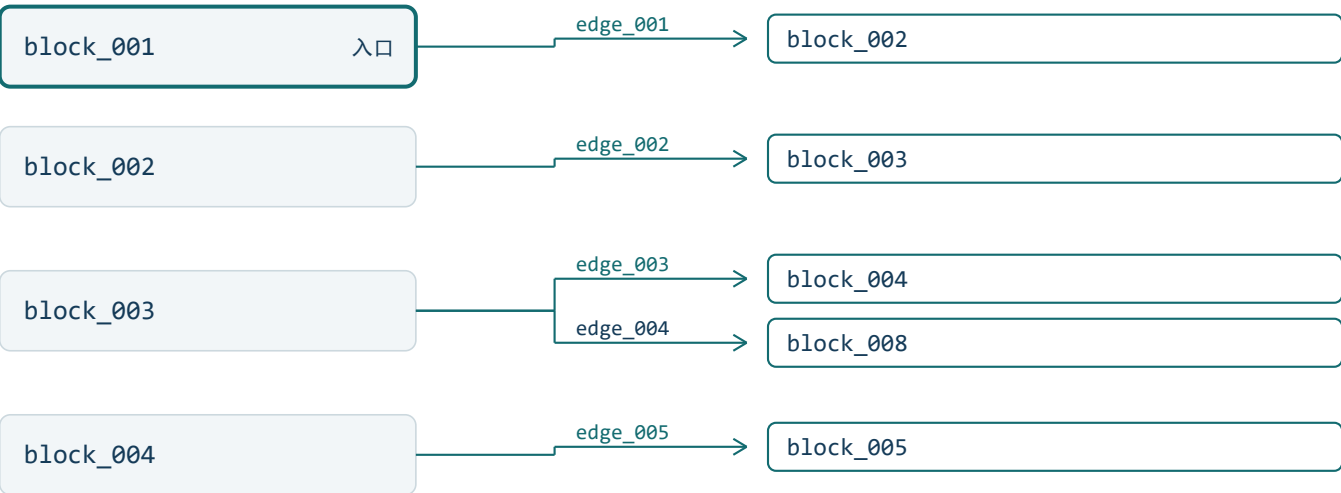
skill.constraints[5]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.

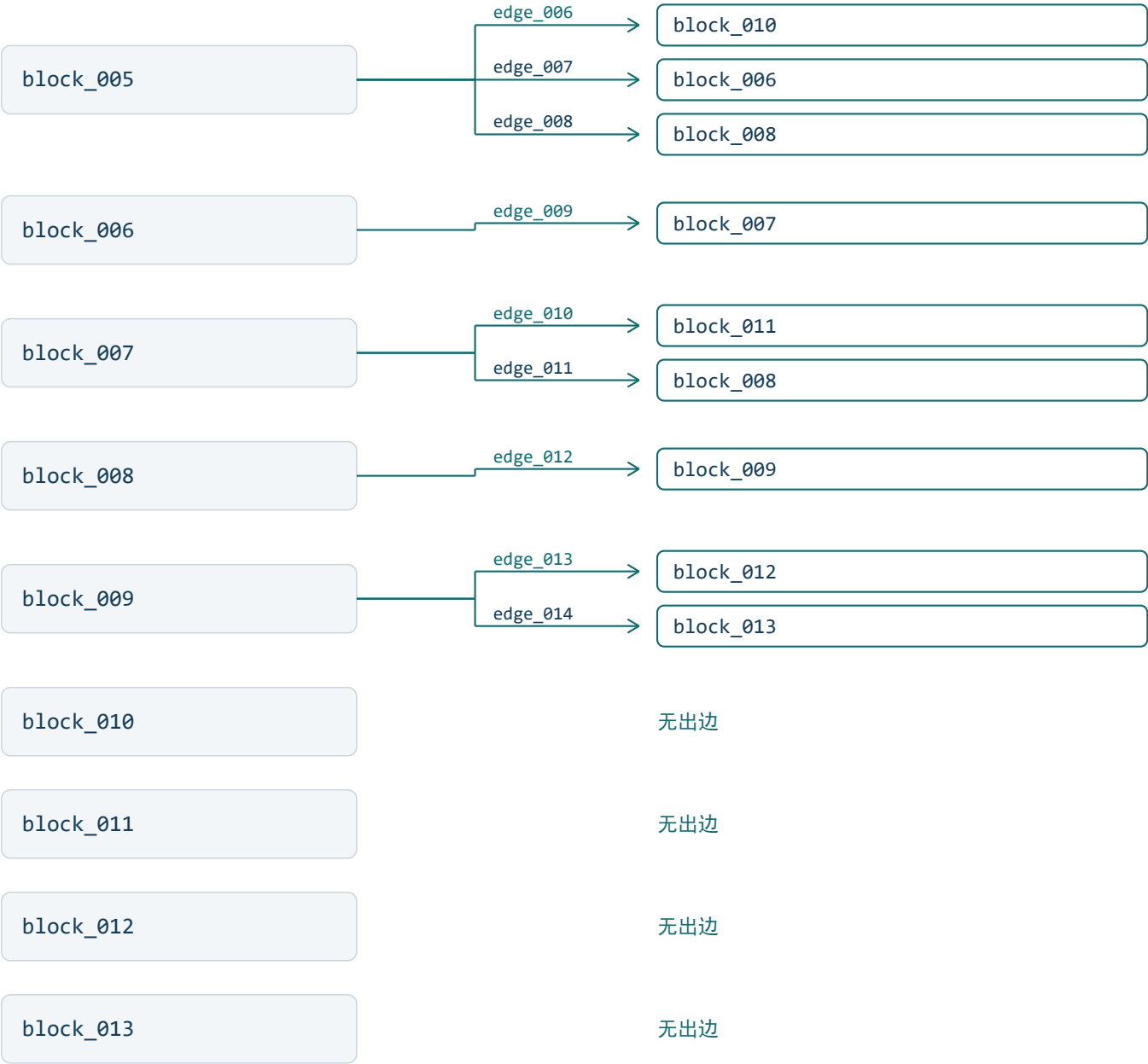
skill.constraints[6]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.

skill.constraints[7]: Before returning from every success or failure path, append the final status to local status.txt.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"FAST_KEY present"
edge_004	block_003 -> block_008	"FAST_KEY absent"
edge_005	block_004 -> block_005	null
edge_006	block_005 -> block_010	"success"
edge_007	block_005 -> block_006	"transient failure"
edge_008	block_005 -> block_008	"non-transient failure"
edge_009	block_006 -> block_007	null
edge_010	block_007 -> block_011	"success"
edge_011	block_007 -> block_008	"failure"
edge_012	block_008 -> block_009	null
edge_013	block_009 -> block_012	"success"
edge_014	block_009 -> block_013	"failure"

全部操作、数据引用与约束落点

block_001 Read source_id from the user's request

```
data_source_kind = "context"
block_001.constraints: []

ir_001 read_source_id_from_request

inputs[0] context_key: source_id | source_id
outputs[0] result: result_001 | source_id_value
ir_001.constraints: []
draft_instruction_id = read_source_id_from_request
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = dispatch_after_source_id
```

block_002 Read FAST_KEY from the environment

```
data_source_kind = "context"
```

```
block_002.constraints: []
```

ir_003 read_fast_key_from_environment

```
inputs[0] context_key: FAST_KEY | FAST_KEY
```

```
outputs[0] result: result_002 | fast_key_value
```

```
ir_003.constraints: []
```

```
draft_instruction_id = read_fast_key_from_environment
```

ir_004 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = dispatch_after_fast_key
```

block_003 Decide whether FAST_KEY is present

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 check_fast_key_present

```
inputs[0] result: result_002 | fast_key_value
```

```
outputs[0] result: result_003 | fast_key_present
```

```
ir_005.constraints: []
```

```
draft_instruction_id = check_fast_key_present
```

ir_006 dispatch

```
inputs[0] result: result_003 | fast_key_present
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = dispatch_on_fast_key_presence
```

block_004 Try fast.fetch first with source_id and FAST_KEY

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_007 fast_fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id_value
```

```
inputs[2] result: result_002 | fast_key_value
```

```
outputs[0] result: result_004 | first_fast_fetch_result
```

```
ir_007.constraints: []
```

```
draft_instruction_id = fast_fetch_first_call
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_after_first_fast_fetch
```

block_005 Classify the first fast.fetch outcome

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 classify_fast_fetch_outcome

```
inputs[0] result: result_004 | first_fast_fetch_result
```

```
outputs[0] result: result_005 | first_fast_fetch_status
```

```
outputs[1] result: result_006 | first_fast_fetch_summary
```

```
ir_009.constraints: []
```

```
draft_instruction_id = classify_first_fast_fetch_outcome
```

ir_010 dispatch

```
inputs[0] result: result_005 | first_fast_fetch_status
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_on_first_fast_fetch_status
```

block_006 Retry fast.fetch once with source_id and FAST_KEY

```
data_source_kind = "external"
```

```
block_006.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do not retry a non-transient first failure.
```

ir_011 fast_fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | source_id_value
```

```
inputs[2] result: result_002 | fast_key_value
```

```
outputs[0] result: result_007 | retry_fast_fetch_result
```

```
ir_011.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do not retry a non-transient first failure.
```

```
draft_instruction_id = fast_fetch_retry_call
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_after_retry_fast_fetch
```

block_007 Classify the fast.fetch retry outcome

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 classify_fast_fetch_outcome

```
inputs[0] result: result_007 | retry_fast_fetch_result
```

```
outputs[0] result: result_008 | retry_fast_fetch_status
```

```
outputs[1] result: result_009 | retry_fast_fetch_summary
```

```
ir_013.constraints: []
```

```
draft_instruction_id = classify_retry_fast_fetch_outcome
```

ir_014 dispatch

```
inputs[0] result: result_008 | retry_fast_fetch_status
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = dispatch_on_retry_fast_fetch_status
```

block_008 Call archive.fetch with source_id

```
data_source_kind = "external"
```

```
block_008.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.
```

```
block_008.constraints[1]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.
```

```
block_008.constraints[2]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

ir_015 archive_fetch

```
inputs[0] external_resource: archive.fetch | archive.fetch
```

```
inputs[1] result: result_001 | source_id_value
```

```
outputs[0] result: result_010 | archive_fetch_result
```

```
ir_015.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.
```

```
ir_015.constraints[1]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.
```

```
ir_015.constraints[2]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

```
draft_instruction_id = archive_fetch_call
```

ir_016 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = dispatch_after_archive_fetch
```

block_009 Classify the archive.fetch outcome

```
data_source_kind = null
```

```
block_009.constraints: []
```

ir_017 classify_archive_fetch_outcome

```
inputs[0] result: result_010 | archive_fetch_result
```

```
outputs[0] result: result_011 | archive_fetch_status
```

```
outputs[1] result: result_012 | archive_fetch_summary
```

```
outputs[2] result: result_013 | archive_fetch_error
```

```
ir_017.constraints: []
```

```
draft_instruction_id = classify_archive_fetch_outcome
```

ir_018 dispatch

```
inputs[0] result: result_011 | archive_fetch_status
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = dispatch_on_archive_fetch_status
```

block_010 Append status and return the first fast.fetch summary

```
data_source_kind = null
```

```
block_010.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

ir_019 append_final_status_to_status_file

```
inputs[0] external_resource: status.txt | status.txt
```

```
inputs[1] result: result_005 | first_fast_fetch_status
```

```
outputs = []
```

```
ir_019.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

```
draft_instruction_id = append_first_fast_fetch_status
```

ir_020 return

```
inputs[0] result: result_006 | first_fast_fetch_summary
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = return_first_fast_fetch_summary
```

block_011 Append status and return the retry fast.fetch summary

```
data_source_kind = null
```

```
block_011.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

ir_021 append_final_status_to_status_file

```
inputs[0] external_resource: status.txt | status.txt
```

```
inputs[1] result: result_008 | retry_fast_fetch_status
```

```
outputs = []
```

```
ir_021.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

```
draft_instruction_id = append_retry_fast_fetch_status
```

ir_022 return

```
inputs[0] result: result_009 | retry_fast_fetch_summary
```

```
outputs = []
```

```
ir_022.constraints: []
```

```
draft_instruction_id = return_retry_fast_fetch_summary
```


block_012 Append status and return the archive.fetch summary

```
data_source_kind = null
```

```
block_012.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

ir_023 append_final_status_to_status_file

```
inputs[0] external_resource: status.txt | status.txt
```

```
inputs[1] result: result_011 | archive_fetch_status
```

```
outputs = []
```

```
ir_023.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

```
draft_instruction_id = append_archive_fetch_status
```

ir_024 return

```
inputs[0] result: result_012 | archive_fetch_summary
```

```
outputs = []
```

```
ir_024.constraints: []
```

```
draft_instruction_id = return_archive_fetch_summary
```

block_013 Append status and return the archive.fetch error

```
data_source_kind = null
```

```
block_013.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

ir_025 append_final_status_to_status_file

```
inputs[0] external_resource: status.txt | status.txt
```

```
inputs[1] result: result_011 | archive_fetch_status
```

```
outputs = []
```

```
ir_025.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

```
draft_instruction_id = append_archive_fetch_failure_status
```

ir_026 return

```
inputs[0] result: result_013 | archive_fetch_error
```

```
outputs = []
```

```
ir_026.constraints: []
```

```
draft_instruction_id = return_archive_fetch_error
```

F06 / 第 3 次 实际输出明细

工具选择与失败回退 · 语义最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/F06/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

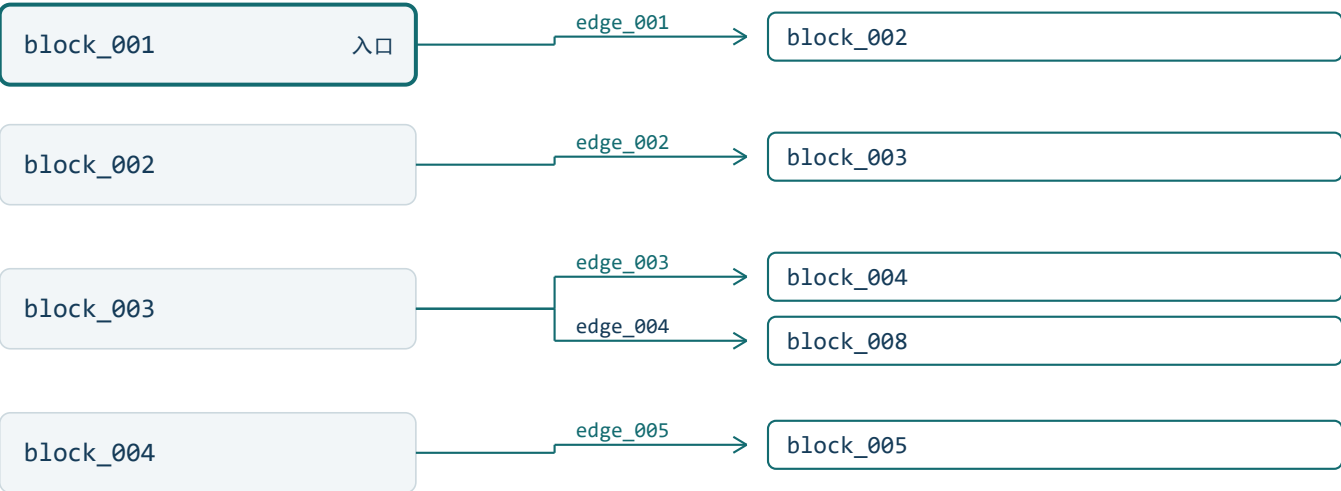
Skill 层约束与上下文声明

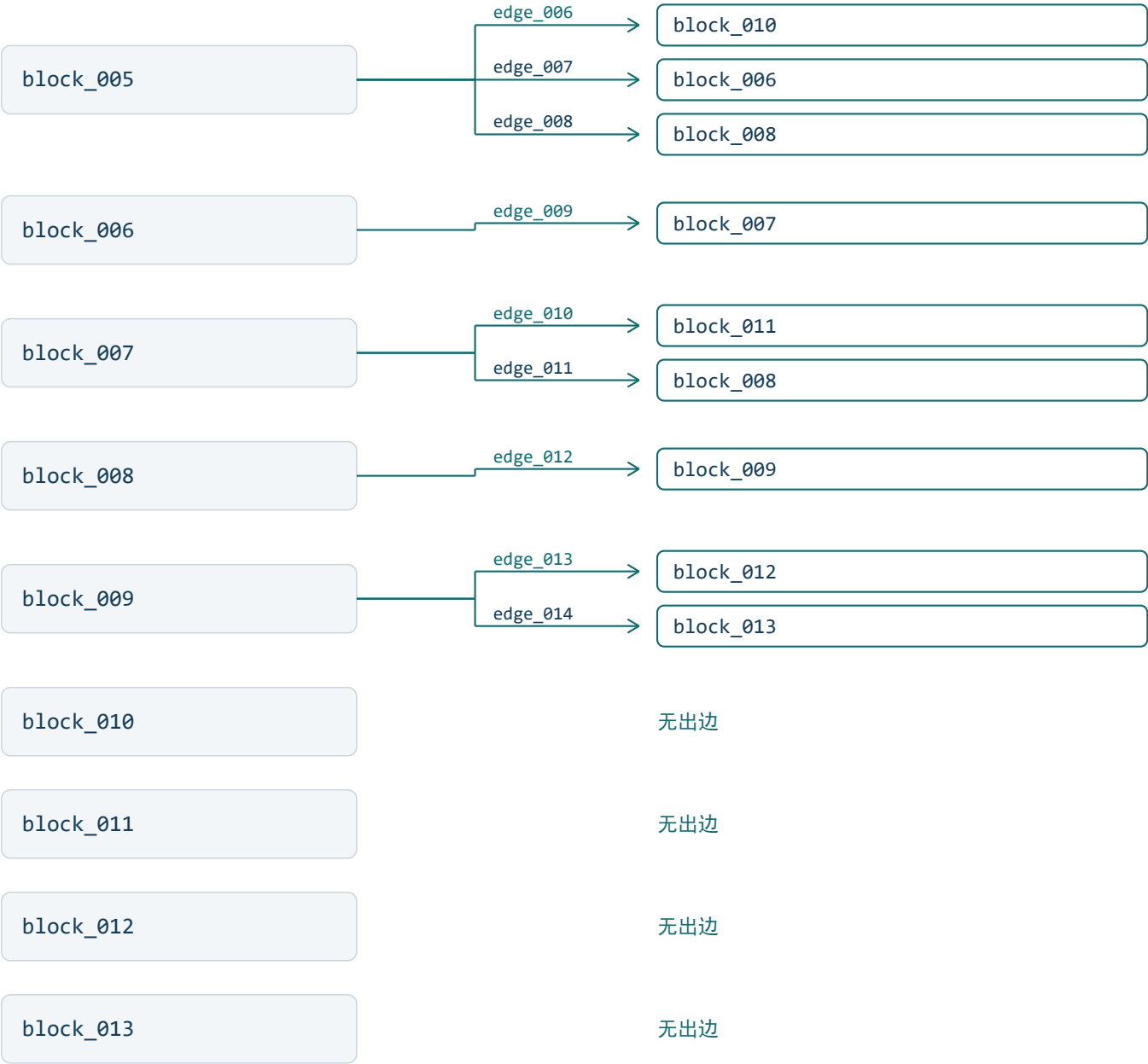
```
declared_context_keys = ["source_id", "FAST_KEY"]

skill.constraints[0]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.
skill.constraints[1]: Call archive.fetch at most once and pass source_id as its only argument.
skill.constraints[2]: On either tool's success, return that successful response's summary value unchanged and make no further fetch calls.
skill.constraints[3]: Before returning from every success or failure path, append the final status to local status.txt.
skill.constraints[4]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
skill.constraints[5]: Retry fast.fetch exactly once only when its first attempt fails with a transient error; do not retry a non-transient first failure.
skill.constraints[6]: After a non-transient first failure or any failed retry of fast.fetch, call archive.fetch with source_id.
skill.constraints[7]: If FAST_KEY is present, try fast.fetch first with source_id and FAST_KEY; if it is absent, go directly to archive.fetch without calling fast.fetch.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"FAST_KEY is present"
edge_004	block_003 -> block_008	"FAST_KEY is absent"
edge_005	block_004 -> block_005	null
edge_006	block_005 -> block_010	"first attempt succeeded"
edge_007	block_005 -> block_006	"first attempt failed with a transient error"
edge_008	block_005 -> block_008	"first attempt failed with a non-transient error"
edge_009	block_006 -> block_007	null
edge_010	block_007 -> block_011	"retry succeeded"
edge_011	block_007 -> block_008	"retry failed"
edge_012	block_008 -> block_009	null
edge_013	block_009 -> block_012	"archive.fetch succeeded"
edge_014	block_009 -> block_013	"archive.fetch failed"

全部操作、数据引用与约束落点

block_001 Read source_id from the user's request

```
data_source_kind = "context"
block_001.constraints: []

ir_001 read_source_id_from_request

inputs[0] context_key: source_id | source_id
outputs[0] result: result_001 | requested_source_id
ir_001.constraints: []

draft_instruction_id = read_source_id
```

ir_002 dispatch

```
inputs = []
outputs = []

ir_002.constraints: []

draft_instruction_id = go_read_fast_key
```

block_002 Read FAST_KEY from the environment

```
data_source_kind = "context"
```

```
block_002.constraints: []
```

ir_003 read_fast_key_from_environment

```
inputs[0] context_key: FAST_KEY | FAST_KEY
```

```
outputs[0] result: result_002 | fast_key
```

```
ir_003.constraints: []
```

```
draft_instruction_id = read_fast_key
```

ir_004 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = go_check_fast_key_presence
```

block_003 Check whether FAST_KEY is present

```
data_source_kind = null
```

```
block_003.constraints[0]: If FAST_KEY is present, try fast.fetch first with source_id and FAST_KEY; if it is absent, go directly to archive.fetch without calling fast.fetch.
```

ir_005 check_fast_key_presence

```
inputs[0] result: result_002 | fast_key
```

```
outputs[0] result: result_003 | fast_key_present
```

```
ir_005.constraints: []
```

```
draft_instruction_id = check_fast_key_presence
```

ir_006 dispatch

```
inputs[0] result: result_003 | fast_key_present
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = dispatch_key_presence
```

block_004 Try fast.fetch first with source_id and FAST_KEY

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_007 fast_fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | requested_source_id
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_004 | fast_fetch_first_summary
```

```
outputs[1] result: result_005 | fast_fetch_first_error
```

```
ir_007.constraints: []
```

```
draft_instruction_id = fast_fetch_first
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_classify_first
```

block_005 Classify the first fast.fetch attempt outcome

```
data_source_kind = null
```

```
block_005.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error;  
do not retry a non-transient first failure.
```

```
block_005.constraints[1]: After a non-transient first failure or any failed retry of fast.fetch, call  
archive.fetch with source_id.
```

ir_009 classify_fast_fetch_outcome

```
inputs[0] result: result_004 | fast_fetch_first_summary
```

```
inputs[1] result: result_005 | fast_fetch_first_error
```

```
outputs[0] result: result_006 | fast_fetch_first_status
```

```
ir_009.constraints: []
```

```
draft_instruction_id = classify_first
```

ir_010 dispatch

```
inputs[0] result: result_006 | fast_fetch_first_status
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_first_outcome
```

block_006 Retry fast.fetch exactly once after a transient first failure

```
data_source_kind = "external"
```

```
block_006.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error.
```

ir_011 fast_fetch

```
inputs[0] external_resource: fast.fetch | fast.fetch
```

```
inputs[1] result: result_001 | requested_source_id
```

```
inputs[2] result: result_002 | fast_key
```

```
outputs[0] result: result_007 | fast_fetch_retry_summary
```

```
outputs[1] result: result_008 | fast_fetch_retry_error
```

```
ir_011.constraints[0]: Retry fast.fetch exactly once only when its first attempt fails with a transient error.
```

```
draft_instruction_id = fast_fetch_retry
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_classify_retry
```

block_007 Classify the fast.fetch retry outcome

```
data_source_kind = null
```

```
block_007.constraints[0]: After a non-transient first failure or any failed retry of fast.fetch, call archive.fetch with source_id.
```

ir_013 classify_fast_fetch_retry_outcome

```
inputs[0] result: result_007 | fast_fetch_retry_summary
```

```
inputs[1] result: result_008 | fast_fetch_retry_error
```

```
outputs[0] result: result_009 | fast_fetch_retry_status
```

```
ir_013.constraints: []
```

```
draft_instruction_id = classify_retry
```

ir_014 dispatch

```
inputs[0] result: result_009 | fast_fetch_retry_status
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = dispatch_retry_outcome
```

block_008 Call archive.fetch with source_id as the only argument

```
data_source_kind = "external"
```

```
block_008.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.
```

```
block_008.constraints[1]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.
```

ir_015 archive_fetch

```
inputs[0] external_resource: archive.fetch | archive.fetch
```

```
inputs[1] result: result_001 | requested_source_id
```

```
outputs[0] result: result_010 | archive_fetch_summary
```

```
outputs[1] result: result_011 | archive_fetch_error
```

```
ir_015.constraints[0]: Call archive.fetch at most once and pass source_id as its only argument.
```

```
ir_015.constraints[1]: Never pass FAST_KEY to archive.fetch or diagnostic output anywhere in this workflow.
```

```
draft_instruction_id = archive_fetch
```

ir_016 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = go_classify_archive
```

block_009 Classify the archive.fetch outcome

```
data_source_kind = null
```

```
block_009.constraints[0]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

ir_017 classify_archive_fetch_outcome

```
inputs[0] result: result_010 | archive_fetch_summary
```

```
inputs[1] result: result_011 | archive_fetch_error
```

```
outputs[0] result: result_012 | archive_fetch_status
```

```
ir_017.constraints: []
```

```
draft_instruction_id = classify_archive
```

ir_018 dispatch

```
inputs[0] result: result_012 | archive_fetch_status
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = dispatch_archive_outcome
```

block_010 Append success status and return the first fast.fetch response's summary

```
data_source_kind = null
```

block_010.constraints[0]: On either tool's success, return that successful response's summary value unchanged and make no further fetch calls.

block_010.constraints[1]: Before returning from every success or failure path, append the final status to local status.txt.

ir_019 append_final_status_to_status_file

```
inputs[0] external_resource: local status.txt | local status.txt
```

```
inputs[1] result: result_006 | fast_fetch_first_status
```

```
outputs = []
```

ir_019.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

```
draft_instruction_id = append_first_fast_status
```

ir_020 return

```
inputs[0] result: result_004 | fast_fetch_first_summary
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = return_first_fast_summary
```

block_011 Append success status and return the retry fast.fetch response's summary

```
data_source_kind = null
```

block_011.constraints[0]: On either tool's success, return that successful response's summary value unchanged and make no further fetch calls.

block_011.constraints[1]: Before returning from every success or failure path, append the final status to local status.txt.

ir_021 append_final_status_to_status_file

```
inputs[0] external_resource: local status.txt | local status.txt
```

```
inputs[1] result: result_009 | fast_fetch_retry_status
```

```
outputs = []
```

ir_021.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.

```
draft_instruction_id = append_retry_fast_status
```


ir_022 return

```
inputs[0] result: result_007 | fast_fetch_retry_summary
```

```
outputs = []
```

```
ir_022.constraints: []
```

```
draft_instruction_id = return_retry_fast_summary
```

block_012 Append success status and return the archive.fetch response's summary

```
data_source_kind = null
```

```
block_012.constraints[0]: On either tool's success, return that successful response's summary value unchanged and make no further fetch calls.
```

```
block_012.constraints[1]: Before returning from every success or failure path, append the final status to local status.txt.
```

ir_023 append_final_status_to_status_file

```
inputs[0] external_resource: local status.txt | local status.txt
```

```
inputs[1] result: result_012 | archive_fetch_status
```

```
outputs = []
```

```
ir_023.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

```
draft_instruction_id = append_archive_success_status
```

ir_024 return

```
inputs[0] result: result_010 | archive_fetch_summary
```

```
outputs = []
```

```
ir_024.constraints: []
```

```
draft_instruction_id = return_archive_summary
```

block_013 Append failure status and return the archive.fetch error

```
data_source_kind = null
```

```
block_013.constraints[0]: If archive.fetch fails, stop and return its error; do not retry archive.fetch.
```

```
block_013.constraints[1]: Before returning from every success or failure path, append the final status to local status.txt.
```

ir_025 append_final_status_to_status_file

```
inputs[0] external_resource: local status.txt | local status.txt
```

```
inputs[1] result: result_012 | archive_fetch_status
```

```
outputs = []
```

```
ir_025.constraints[0]: Before returning from every success or failure path, append the final status to local status.txt.
```

```
draft_instruction_id = append_archive_failure_status
```

ir_026 return

```
inputs[0] result: result_011 | archive_fetch_error
```

```
outputs = []
```

```
ir_026.constraints: []
```

```
draft_instruction_id = return_archive_error
```

D01 / 第 1 次 实际输出明细

多文档处理与交付 · 编号步骤

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D01/deepseek-v4-flash-max/1

结构状态: complete | 生成 2 | 修复 1 | HTTP 尝试 2 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

生成 2 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

declared_context_keys = ["output_dir", "delivery_path"]

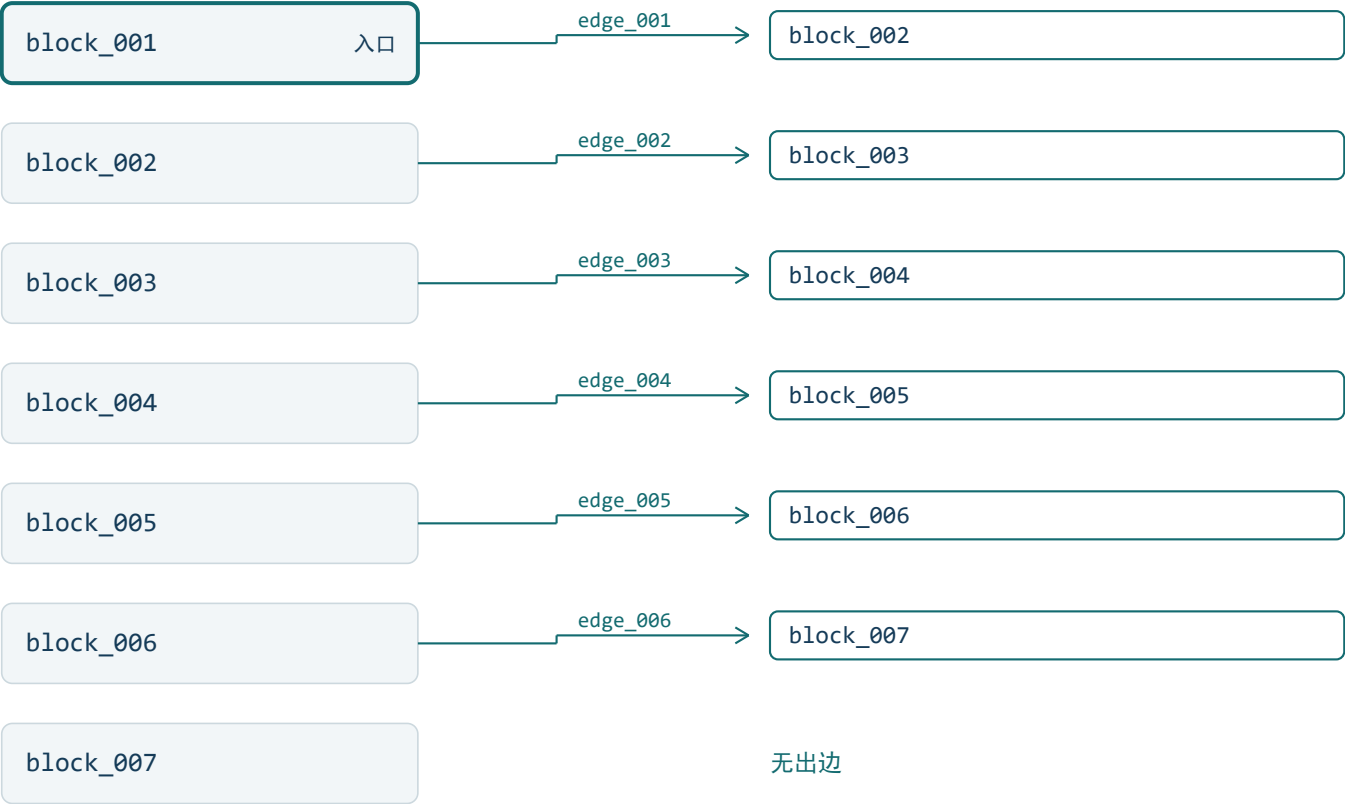
skill.constraints[0]: 整份流程不得修改输入原文文件。

skill.constraints[1]: 整份流程禁止把文档上传到外部服务。

skill.constraints[2]: 必要时保留原顺序。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null

全部操作、数据引用与约束落点

block_001 Read the user-provided manifest

```
data_source_kind = "external"

block_001.constraints: []

ir_001 read_manifest_json

inputs[0] external_resource: manifest.json | user-provided manifest
outputs[0] result: result_001 | manifest contents

ir_001.constraints: []

draft_instruction_id = read_manifest_file
```

ir_002 dispatch

```
inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = dispatch_to_parse_manifest
```

block_002 Extract document paths from the manifest

```
data_source_kind = null

block_002.constraints: []

ir_003 extract_document_paths_from_manifest

inputs[0] result: result_001 | manifest_data
outputs[0] result: result_002 | manifest paths list

ir_003.constraints: []

draft_instruction_id = extract_document_paths
```

ir_004 dispatch

```
inputs = []

outputs = []

ir_004.constraints: []

draft_instruction_id = dispatch_to_read_request_settings
```

block_003 Read output_dir and delivery_path from the request

```
data_source_kind = "context"
```

```
block_003.constraints: []
```

ir_005 read_request_output_locations

```
inputs[0] context_key: output_dir | output_dir
```

```
inputs[1] context_key: delivery_path | delivery_path
```

```
outputs[0] result: result_003 | requested output directory
```

```
outputs[1] result: result_004 | requested delivery path
```

```
ir_005.constraints: []
```

```
draft_instruction_id = read_request_locations
```

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = dispatch_to_convert_documents
```

block_004 Convert each document with scripts/convert.py

```
data_source_kind = "external"
```

block_004.constraints[0]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。

ir_007 convert_document_paths_with_script

```
inputs[0] external_resource: scripts/convert.py | convert script
```

```
inputs[1] result: result_002 | document_paths
```

```
inputs[2] result: result_003 | request_output_dir
```

```
outputs[0] result: result_005 | conversion output paths
```

ir_007.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。

```
draft_instruction_id = convert_documents_with_script
```

metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。

```
metadata = {"command": "python scripts/convert.py <文档路径> <output_dir>"}
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_to_package_bundle
```

block_005 Package converted artifacts with scripts/package.py

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 package_converted_artifacts_with_script

```
inputs[0] external_resource: scripts/package.py | package script
```

```
inputs[1] result: result_005 | converted_paths
```

```
inputs[2] literal = "bundle.zip"
```

```
outputs[0] result: result_006 | generated bundle.zip path
```

ir_009.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 `--paths`。

```
draft_instruction_id = package_converted_artifacts
```

metadata.script_content: 457 字符; SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

```
metadata = {"command": "python scripts/package.py --paths <转换产物路径列表> --output bundle.zip"}
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_to_deliver_bundle
```

block_006 Deliver bundle.zip to the requested delivery_path

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 deliver_bundle_to_path

```
inputs[0] result: result_006 | bundle_zip_path
```

```
inputs[1] result: result_004 | request_delivery_path
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = deliver_bundle
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_to_write_receipt
```

block_007 Write converted paths to receipt.txt

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 write_converted_paths_to_receipt

```
inputs[0] external_resource: receipt.txt | local receipt file
```

```
inputs[1] result: result_005 | converted_paths
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = write_receipt
```

ir_014 return

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = return_from_skill
```

D01 / 第 2 次 实际输出明细

多文档处理与交付 · 编号步骤

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D01/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

declared_context_keys = ["output_dir", "delivery_path"]

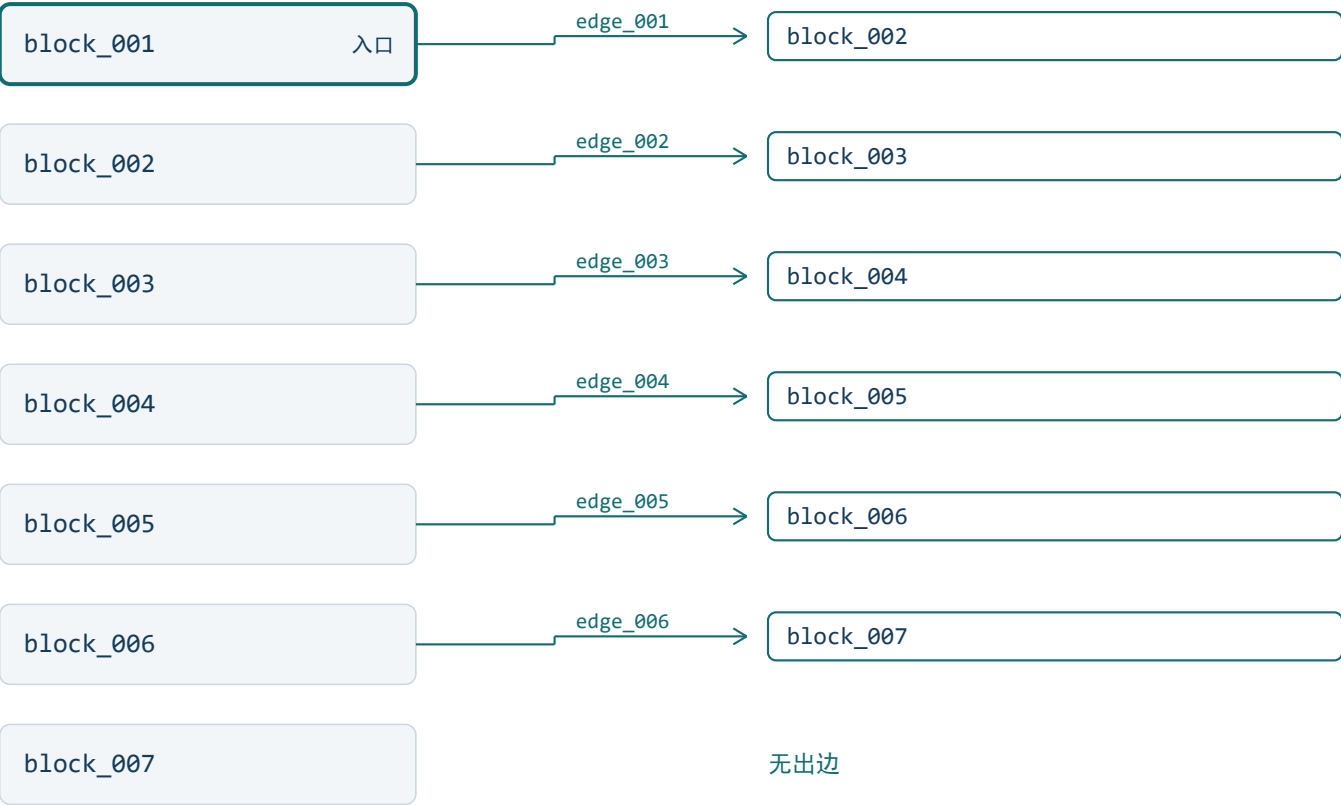
skill.constraints[0]: 整份流程不得修改输入原文文件。

skill.constraints[1]: 整份流程禁止把文档上传到外部服务。

skill.constraints[2]: 必要时保留原顺序。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null

全部操作、数据引用与约束落点

```
block_001 Read the user-provided manifest.json

data_source_kind = "external"
block_001.constraints: []

ir_001 read_manifest_file

inputs[0] external_resource: manifest.json | manifest.json
outputs[0] result: result_001 | manifest_content
ir_001.constraints: []
draft_instruction_id = read_manifest_file

ir_002 dispatch

inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_read_request

block_002 Read output_dir and delivery_path from the request

data_source_kind = "context"
block_002.constraints: []

ir_003 read_output_dir_and_delivery_path_from_request

inputs[0] context_key: output_dir | output_dir
inputs[1] context_key: delivery_path | delivery_path
outputs[0] result: result_002 | resolved_output_dir
outputs[1] result: result_003 | resolved_delivery_path
ir_003.constraints: []
draft_instruction_id = read_request_paths
```

ir_004 dispatch

inputs = []

outputs = []

ir_004.constraints: []

draft_instruction_id = go_parse_manifest

block_003 Extract document paths from the manifest

data_source_kind = null

block_003.constraints: []

ir_005 extract_paths_from_manifest

inputs[0] result: result_001 | manifest_content

outputs[0] result: result_004 | document_paths

ir_005.constraints: []

draft_instruction_id = extract_manifest_paths

ir_006 dispatch

inputs = []

outputs = []

ir_006.constraints: []

draft_instruction_id = go_convert_documents

block_004 Convert each document with scripts/convert.py

data_source_kind = "external"

block_004.constraints: []

ir_007 run_convert_script_for_each_document

inputs[0] external_resource: scripts/convert.py | scripts/convert.py

inputs[1] result: result_004 | document_paths

inputs[2] result: result_002 | resolved_output_dir

outputs[0] result: result_005 | converted_artifact_paths

ir_007.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。

ir_007.constraints[1]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。

draft_instruction_id = run_convert_script_for_each_document

metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。

ir_008 dispatch

inputs = []

outputs = []

ir_008.constraints: []

draft_instruction_id = go_package_bundle

block_005 Package converted artifacts into bundle.zip

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 run_package_script

```
inputs[0] external_resource: scripts/package.py | scripts/package.py
```

```
inputs[1] result: result_005 | converted_artifact_paths
```

```
inputs[2] literal = "bundle.zip"
```

```
outputs[0] result: result_006 | bundle_zip_path
```

ir_009.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 `--paths`。

```
draft_instruction_id = run_package_script
```

metadata.script_content: 457 字符; SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_deliver_bundle
```

block_006 Deliver bundle.zip to the requested delivery_path

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 deliver_bundle_zip_to_delivery_path

```
inputs[0] result: result_006 | bundle_zip_path
```

```
inputs[1] result: result_003 | resolved_delivery_path
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = deliver_bundle_zip
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_write_receipt
```

block_007 Write converted artifact paths to local receipt.txt

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 write_converted_paths_to_receipt

```
inputs[0] external_resource: receipt.txt | receipt.txt
```

```
inputs[1] result: result_005 | converted_artifact_paths
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = write_receipt_file  
  
ir_014 return  
  
inputs = []  
outputs = []  
  
ir_014.constraints: []  
  
draft_instruction_id = return_done
```

D01 / 第 3 次 实际输出明细

多文档处理与交付 · 编号步骤

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D01/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: 备注“必要时保留原顺序”的作用域不明确，未提升为 Skill-wide 约束。

Skill 层约束与上下文声明

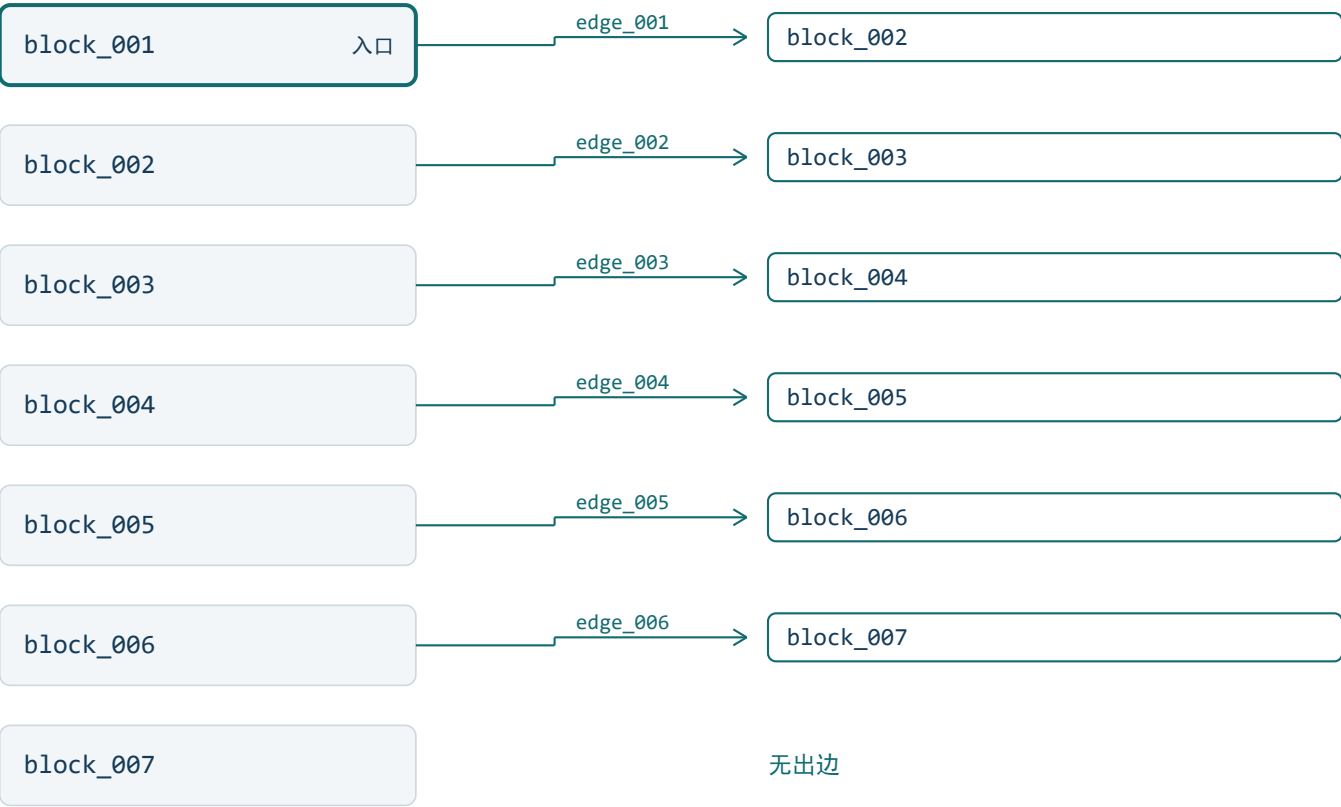
declared_context_keys = ["output_dir", "delivery_path"]

skill.constraints[0]: 整份流程不得修改输入原文文件。

skill.constraints[1]: 整份流程禁止把文档上传到外部服务。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null

全部操作、数据引用与约束落点

```
block_001 Read the user-provided manifest.json

data_source_kind = "external"
block_001.constraints: []

ir_001 read_manifest_json

inputs[0] external_resource: manifest.json | manifest.json
outputs[0] result: result_001 | manifest_json
ir_001.constraints: []
draft_instruction_id = read_manifest_json

ir_002 dispatch

inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_parse_manifest

block_002 Extract document paths from the manifest paths list

data_source_kind = null
block_002.constraints: []

ir_003 extract_document_paths_from_manifest

inputs[0] result: result_001 | manifest_json
outputs[0] result: result_002 | document_paths
ir_003.constraints: []
draft_instruction_id = extract_document_paths

ir_004 dispatch

inputs = []
outputs = []
ir_004.constraints: []
draft_instruction_id = go_read_request
```

block_003 Read output_dir and delivery_path from the request

```
data_source_kind = "context"
```

```
block_003.constraints: []
```

ir_005 read_output_dir_and_delivery_path_from_request

```
inputs[0] context_key: output_dir | output_dir
```

```
inputs[1] context_key: delivery_path | delivery_path
```

```
outputs[0] result: result_003 | requested_output_dir
```

```
outputs[1] result: result_004 | requested_delivery_path
```

```
ir_005.constraints: []
```

```
draft_instruction_id = read_request_settings
```

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = go_convert
```

block_004 Convert each document via scripts/convert.py and collect stdout paths

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_007 execute_convert_script_for_each_document

```
inputs[0] result: result_002 | document_paths
```

```
inputs[1] result: result_003 | requested_output_dir
```

```
inputs[2] external_resource: scripts/convert.py | scripts/convert.py
```

```
outputs[0] result: result_005 | converted_artifact_paths
```

```
ir_007.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。
```

```
ir_007.constraints[1]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。
```

```
draft_instruction_id = convert_documents
```

```
metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。
```

```
metadata = {"command": "python scripts/convert.py <文档路径> <output_dir>"}
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_package
```

block_005 Package converted artifact paths into bundle.zip using scripts/package.py

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 package_converted_documents_with_package_script

```
inputs[0] result: result_005 | converted_artifact_paths
```

```
inputs[1] literal = "bundle.zip"
inputs[2] external_resource: scripts/package.py | scripts/package.py
outputs[0] result: result_006 | bundle_zip
ir_009.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 --paths。
draft_instruction_id = package_bundle
metadata.script_content: 457 字符；SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。
metadata = {"reference_command": "python scripts/package.py --paths <转换产物路径列表> --output bundle.zip"}
```

ir_010 dispatch

```
inputs = []
outputs = []
ir_010.constraints: []
draft_instruction_id = go_deliver
```

block_006 Deliver bundle.zip to the requested delivery_path

```
data_source_kind = null
block_006.constraints: []
```

ir_011 deliver_bundle_to_delivery_path

```
inputs[0] result: result_006 | bundle_zip
inputs[1] result: result_004 | requested_delivery_path
outputs = []
ir_011.constraints: []
draft_instruction_id = deliver_bundle
```

ir_012 dispatch

```
inputs = []
outputs = []
ir_012.constraints: []
draft_instruction_id = go_write_receipt
```

block_007 Write converted artifact paths to local receipt.txt

```
data_source_kind = null
block_007.constraints: []
```

ir_013 write_converted_artifact_paths_to_receipt

```
inputs[0] result: result_005 | converted_artifact_paths
inputs[1] external_resource: receipt.txt | receipt.txt
outputs = []
ir_013.constraints: []
draft_instruction_id = write_receipt
```

ir_014 return

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = return
```

D02 / 第 1 次 实际输出明细

多文档处理与交付 · 连贯段落

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D02/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: The note '必要时保留原顺序' does not specify which operation or list must preserve order, so it is not promoted to a skill-wide constraint.

candidate warning: The per-document conversion loop is represented as a single batch operation because the package does not define loop state or iteration variables.

candidate warning: The final return has no inputs because the package does not specify a return value for the skill.

Skill 层约束与上下文声明

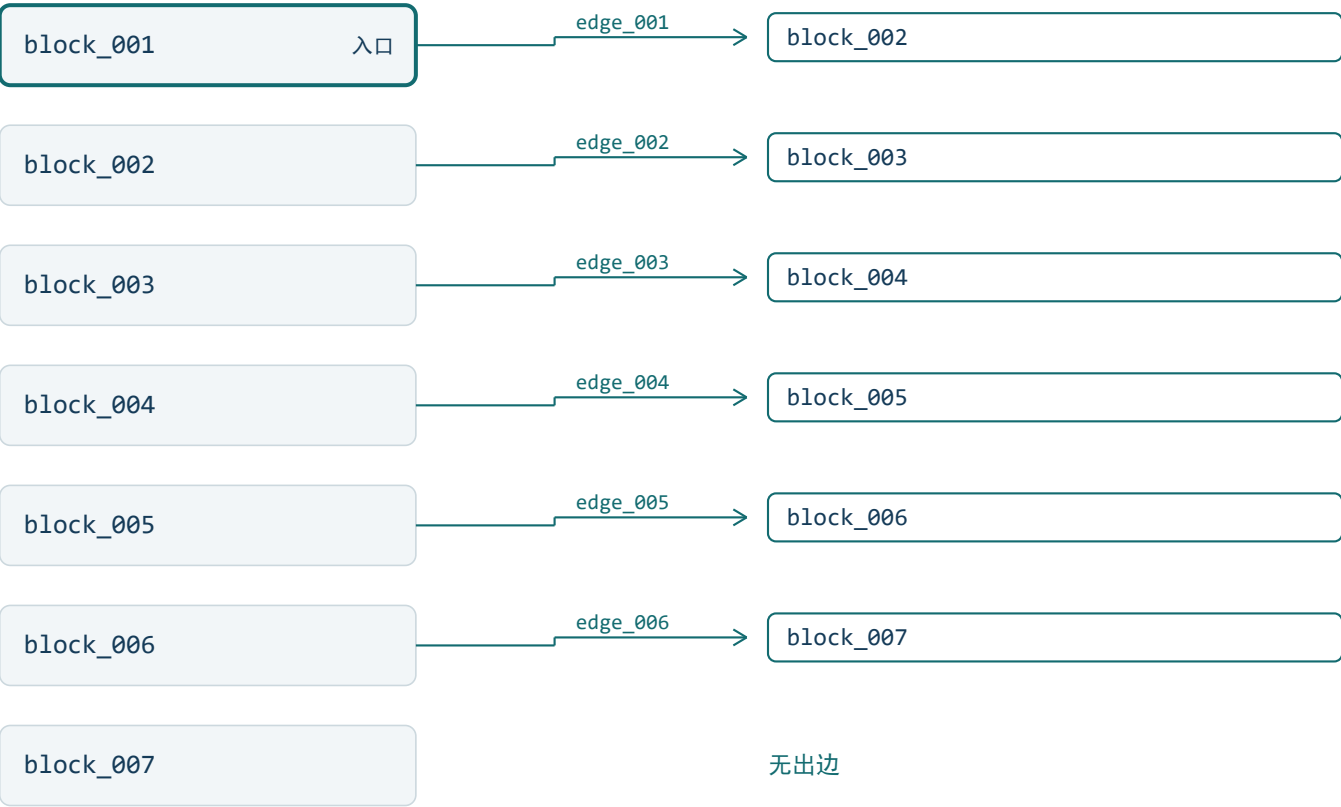
declared_context_keys = ["output_dir", "delivery_path"]

skill.constraints[0]: 整份流程不得修改输入原文文件。

skill.constraints[1]: 整份流程禁止把文档上传到外部服务。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null

全部操作、数据引用与约束落点

```
block_001 Read the user-provided manifest.json

data_source_kind = "external"
block_001.constraints: []

ir_001 read_manifest_file

inputs[0] external_resource: manifest.json | manifest.json
outputs[0] result: result_001 | manifest_contents
ir_001.constraints: []
draft_instruction_id = read_manifest_file

ir_002 dispatch

inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_extract_paths

block_002 Extract document paths from the manifest paths list

data_source_kind = null
block_002.constraints: []

ir_003 extract_document_paths_from_manifest

inputs[0] result: result_001 | manifest_contents
outputs[0] result: result_002 | document_paths
ir_003.constraints: []
draft_instruction_id = extract_paths_from_manifest

ir_004 dispatch

inputs = []
outputs = []
ir_004.constraints: []
draft_instruction_id = go_read_request_paths
```

block_003 Read output_dir and delivery_path from the request

data_source_kind = "context"

block_003.constraints: []

ir_005 read_output_dir_and_delivery_path_from_request

inputs[0] context_key: output_dir | output_dir

inputs[1] context_key: delivery_path | delivery_path

outputs[0] result: result_003 | request_output_dir

outputs[1] result: result_004 | request_delivery_path

ir_005.constraints: []

draft_instruction_id = read_output_dir_and_delivery_path

ir_006 dispatch

inputs = []

outputs = []

ir_006.constraints: []

draft_instruction_id = go_convert_documents

block_004 Convert each manifest document with scripts/convert.py

data_source_kind = "external"

block_004.constraints: []

ir_007 run_convert_script_for_each_document

inputs[0] external_resource: scripts/convert.py | scripts/convert.py

inputs[1] result: result_002 | document_paths

inputs[2] result: result_003 | request_output_dir

outputs[0] result: result_005 | converted_product_paths

ir_007.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。

ir_007.constraints[1]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。

draft_instruction_id = run_convert_script_for_each_document

metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。

metadata = {"command": "python scripts/convert.py <文档路径> <output_dir>", "script_path": "scripts/convert.py"}

ir_008 dispatch

inputs = []

outputs = []

ir_008.constraints: []

draft_instruction_id = go_package_documents

block_005 Package converted documents into bundle.zip with scripts/package.py

data_source_kind = null

block_005.constraints: []

ir_009 run_package_script

inputs[0] external_resource: scripts/package.py | scripts/package.py

```
inputs[1] result: result_005 | converted_product_paths
```

```
inputs[2] external_resource: bundle.zip | bundle.zip
```

```
outputs = []
```

ir_009.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 `--paths`。

```
draft_instruction_id = run_package_script
```

metadata.script_content: 457 字符; SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

```
metadata = {"command": "python scripts/package.py --paths <转换产物路径列表> --output bundle.zip", "script_path": "scripts/package.py"}
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_deliver_bundle
```

block_006 Deliver bundle.zip to the requested delivery_path

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 deliver_bundle_to_delivery_path

```
inputs[0] external_resource: bundle.zip | bundle.zip
```

```
inputs[1] result: result_004 | request_delivery_path
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = deliver_bundle_to_delivery_path
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_write_receipt
```

block_007 Write converted product paths to local receipt.txt

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 write_converted_paths_to_receipt

```
inputs[0] result: result_005 | converted_product_paths
```

```
inputs[1] external_resource: receipt.txt | receipt.txt
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = write_receipt_file
```

ir_014 return

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = return
```

D02 / 第 2 次 实际输出明细

多文档处理与交付 · 连贯段落

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D02/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

declared_context_keys = ["output_dir", "delivery_path"]

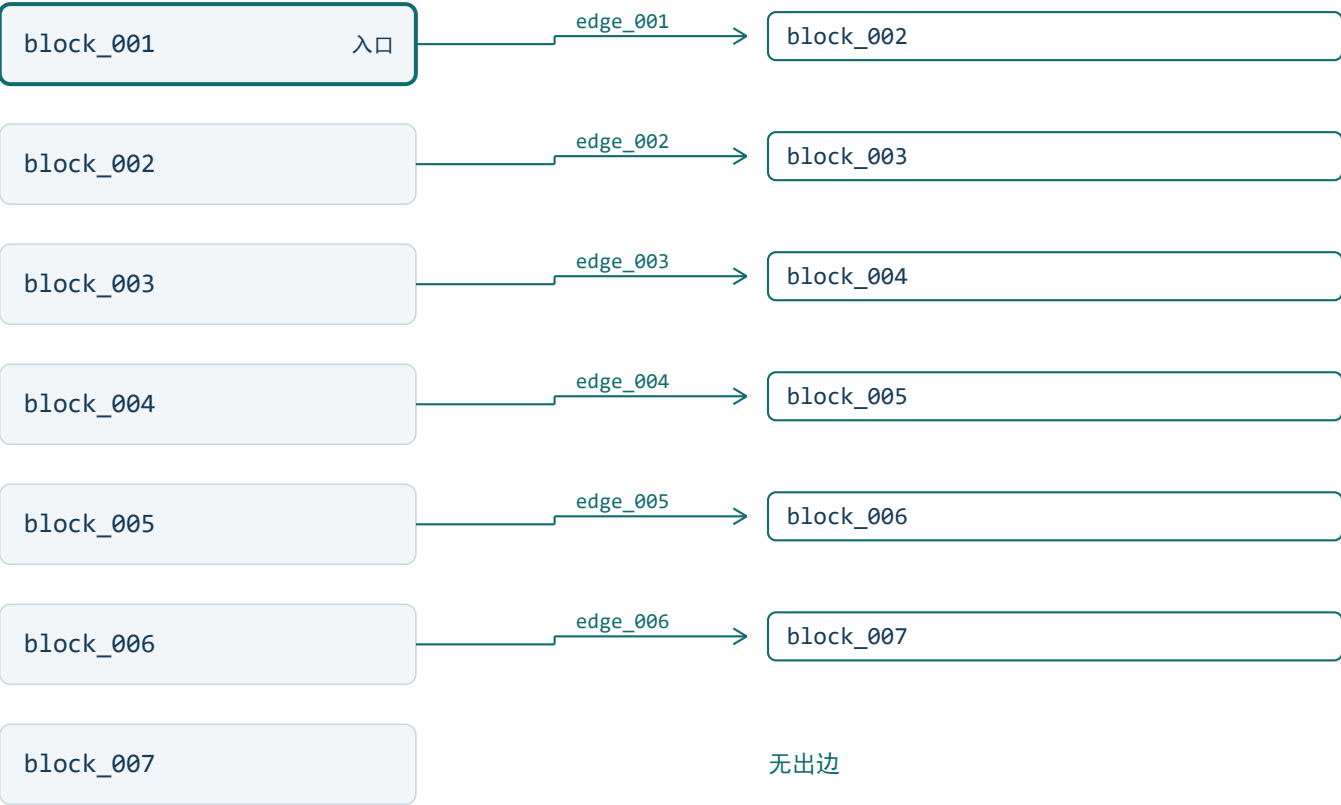
skill.constraints[0]: 整份流程不得修改输入原文文件。

skill.constraints[1]: 整份流程禁止把文档上传到外部服务。

skill.constraints[2]: 备注: 必要时保留原顺序。

实际 CFG: 分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null

全部操作、数据引用与约束落点

block_001 Read user-provided manifest.json

```
data_source_kind = "external"
block_001.constraints: []
ir_001 read_manifest_json
inputs[0] external_resource: manifest.json | manifest.json
outputs[0] result: result_001 | manifest_data
ir_001.constraints: []
draft_instruction_id = read_manifest_file
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_extract_paths
```

block_002 Extract document paths from manifest

```
data_source_kind = null
block_002.constraints: []
ir_003 extract_paths_from_manifest
inputs[0] result: result_001 | manifest_data
outputs[0] result: result_002 | document_paths
ir_003.constraints: []
draft_instruction_id = extract_document_paths
```

ir_004 dispatch

```
inputs = []
outputs = []
ir_004.constraints: []
draft_instruction_id = go_read_request
```

block_003 Read output_dir and delivery_path from request

```
data_source_kind = "context"
```

```
block_003.constraints: []
```

ir_005 read_output_dir_and_delivery_path

```
inputs[0] context_key: output_dir | output_dir
```

```
inputs[1] context_key: delivery_path | delivery_path
```

```
outputs[0] result: result_003 | output_dir
```

```
outputs[1] result: result_004 | delivery_path
```

```
ir_005.constraints: []
```

```
draft_instruction_id = read_request_paths
```

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = go_convert_documents
```

block_004 Convert each document with scripts/convert.py

```
data_source_kind = "external"
```

block_004.constraints[0]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。

ir_007 convert_documents_in_order

```
inputs[0] external_resource: scripts/convert.py | scripts/convert.py
```

```
inputs[1] result: result_002 | document_paths
```

```
inputs[2] result: result_003 | output_dir
```

```
outputs[0] result: result_005 | conversion_artifact_paths
```

ir_007.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。

```
draft_instruction_id = run_convert_script_per_document
```

metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。

```
metadata = {"invocation_command": "python scripts/convert.py <文档路径> <output_dir>"}
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_package_bundle
```

block_005 Package conversion artifacts into bundle.zip

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 run_package_script

```
inputs[0] external_resource: scripts/package.py | scripts/package.py
```

```
inputs[1] result: result_005 | conversion_artifact_paths
```

```
inputs[2] literal = "bundle.zip"
```

```
outputs[0] result: result_006 | bundle_zip_path
```

ir_009.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 `--paths`。

```
draft_instruction_id = run_package_script
```

metadata.script_content: 457 字符; SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

```
metadata = {"reference_command": "python scripts/package.py --paths <转换产物路径列表> --output bundle.zip"}
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_deliver_bundle
```

block_006 Deliver bundle.zip to requested delivery_path

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 deliver_bundle

```
inputs[0] result: result_006 | bundle_zip_path
```

```
inputs[1] result: result_004 | delivery_path
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = deliver_bundle_file
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_write_receipt
```

block_007 Write conversion artifact paths to receipt.txt

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 write_receipt_file

```
inputs[0] result: result_005 | conversion_artifact_paths
```

```
inputs[1] external_resource: receipt.txt | receipt.txt
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = write_receipt_file
```

ir_014 return

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = return_success
```


D02 / 第 3 次 实际输出明细

多文档处理与交付 · 连贯段落

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D02/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

declared_context_keys = ["output_dir", "delivery_path"]

skill.constraints[0]: 整份流程不得修改输入原文文件。

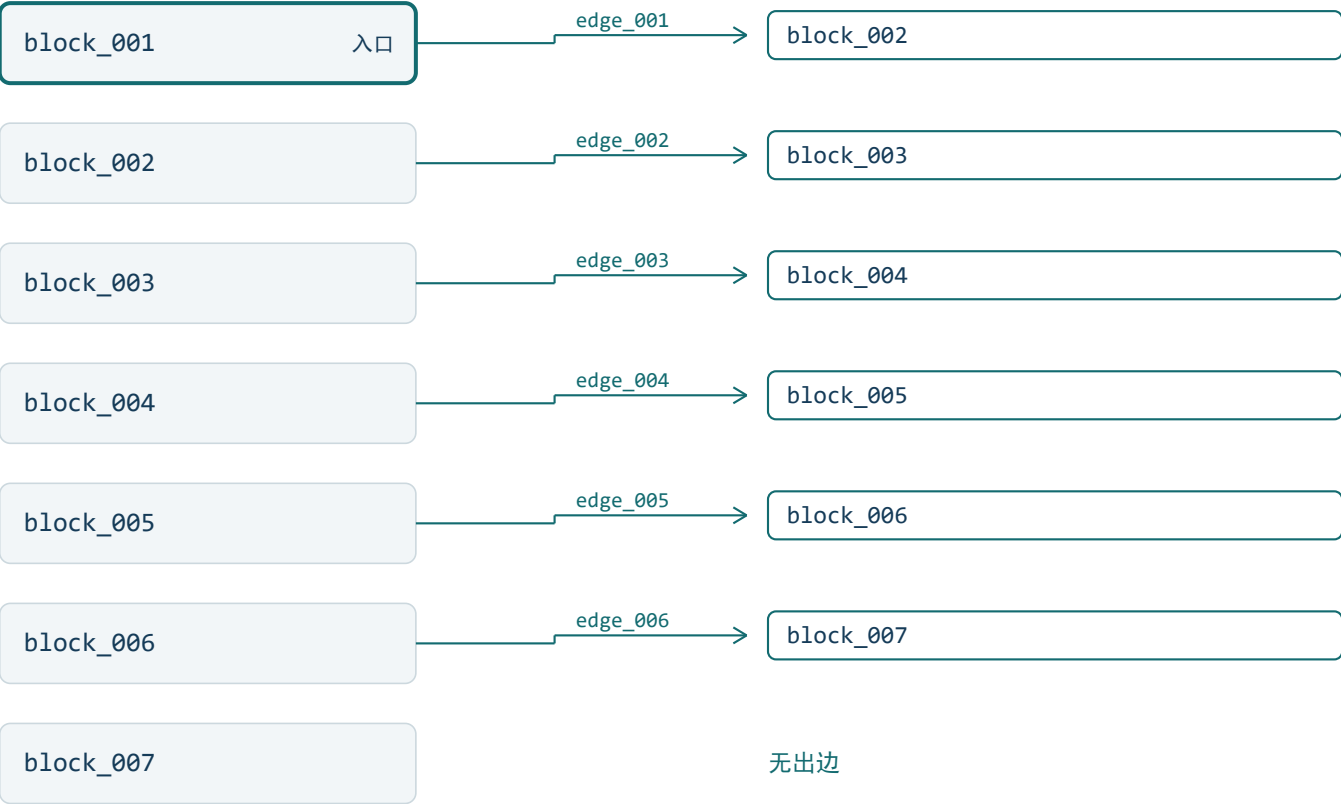
skill.constraints[1]: 整份流程禁止把文档上传到外部服务。

skill.constraints[2]: 备注: 必要时保留原顺序。

skill.constraints[3]: 转换通过 scripts/convert.py 完成; 主流程只使用脚本输出路径, 不重复执行脚本内部的处理。

实际 CFG: 分页邻接图

每个左框是实际 block ID; 箭头连接到右框标明的目标 ID。目标可位于其他行 / 页, 回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识; 完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null

全部操作、数据引用与约束落点

```
block_001 Read the user-provided manifest.json

data_source_kind = "external"
block_001.constraints: []

ir_001 read_manifest_file

inputs[0] external_resource: manifest.json | manifest.json
outputs[0] result: result_001 | manifest_content
ir_001.constraints: []
draft_instruction_id = read_manifest_file

ir_002 dispatch

inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = dispatch_to_extract_paths

block_002 Extract document paths from manifest paths list

data_source_kind = null
block_002.constraints: []

ir_003 extract_manifest_paths

inputs[0] result: result_001 | manifest_content
outputs[0] result: result_002 | document_paths
ir_003.constraints: []
draft_instruction_id = extract_manifest_paths

ir_004 dispatch

inputs = []
outputs = []
ir_004.constraints: []
draft_instruction_id = dispatch_to_read_request
```

block_003 Read output_dir and delivery_path from the request

data_source_kind = "context"

block_003.constraints: []

ir_005 read_output_dir_and_delivery_path_from_request

inputs[0] context_key: output_dir | output_dir

inputs[1] context_key: delivery_path | delivery_path

outputs[0] result: result_003 | output_dir_path

outputs[1] result: result_004 | delivery_target_path

ir_005.constraints: []

draft_instruction_id = read_output_dir_and_delivery_path

ir_006 dispatch

inputs = []

outputs = []

ir_006.constraints: []

draft_instruction_id = dispatch_to_convert

block_004 Convert each document using scripts/convert.py

data_source_kind = "external"

block_004.constraints: []

ir_007 run_convert_script_for_each_path

inputs[0] external_resource: scripts/convert.py | scripts/convert.py

inputs[1] result: result_002 | document_paths

inputs[2] result: result_003 | output_dir_path

outputs[0] result: result_005 | converted_artifact_paths

ir_007.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。

draft_instruction_id = run_convert_script_for_each_path

metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。

metadata = {"command": "python scripts/convert.py <文档路径> <output_dir>", "side_effects": ["writes each converted document to output_dir", "prints the output path for each converted document"]}

ir_008 dispatch

inputs = []

outputs = []

ir_008.constraints: []

draft_instruction_id = dispatch_to_package

block_005 Package converted artifacts into bundle.zip

data_source_kind = null

block_005.constraints: []

ir_009 run_package_script

inputs[0] external_resource: scripts/package.py | scripts/package.py

inputs[1] result: result_005 | converted_artifact_paths

```
inputs[2] literal = "bundle.zip"
```

```
outputs = []
```

ir_009.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 `--paths`。

```
draft_instruction_id = run_package_script
```

metadata.script_content: 457 字符; SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

```
metadata = {"command": "python scripts/package.py --paths <转换产物路径列表> --output bundle.zip", "side_effects": ["writes bundle.zip containing the converted artifact paths"]}
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_to_deliver
```

block_006 Deliver bundle.zip to requested delivery_path

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 deliver_bundle_to_delivery_path

```
inputs[0] literal = "bundle.zip"
```

```
inputs[1] result: result_004 | delivery_target_path
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = deliver_bundle_to_delivery_path
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_to_write_receipt
```

block_007 Write converted artifact path list to local receipt.txt

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 write_receipt_file

```
inputs[0] result: result_005 | converted_artifact_paths
```

```
inputs[1] external_resource: receipt.txt | receipt.txt
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = write_receipt_file
```

ir_014 return

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = return
```

D03 / 第 1 次 实际输出明细

多文档处理与交付 · 表格

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D03/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Row 10 note '备注：必要时保留原顺序。' has unclear scope; not modeled as a constraint.

Skill 层约束与上下文声明

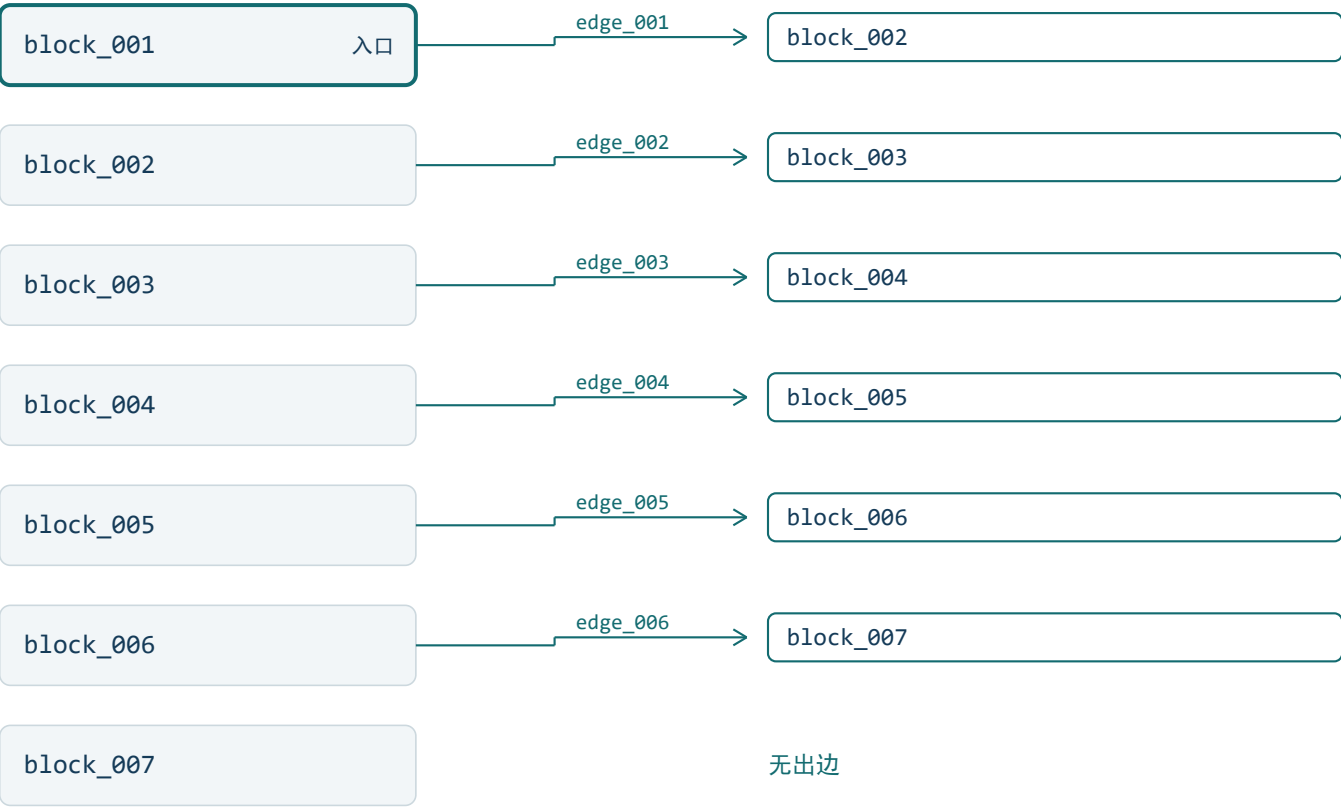
declared_context_keys = ["output_dir", "delivery_path"]

skill.constraints[0]: 整份流程不得修改输入原文文件。

skill.constraints[1]: 整份流程禁止把文档上传到外部服务。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null

全部操作、数据引用与约束落点

```
block_001 Read the user-provided manifest.json

data_source_kind = "external"
block_001.constraints: []

ir_001 read_manifest_json

inputs[0] external_resource: manifest.json | manifest.json
outputs[0] result: result_001 | manifest_contents
ir_001.constraints: []
draft_instruction_id = read_manifest_file

ir_002 dispatch

inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_extract_paths

block_002 Extract document paths from the manifest

data_source_kind = null
block_002.constraints: []

ir_003 extract_manifest_document_paths

inputs[0] result: result_001 | manifest_contents
outputs[0] result: result_002 | document_paths
ir_003.constraints: []
draft_instruction_id = extract_manifest_paths

ir_004 dispatch

inputs = []
outputs = []
ir_004.constraints: []
draft_instruction_id = go_read_request
```

block_003 Read output_dir and delivery_path from the request

```
data_source_kind = "context"
```

```
block_003.constraints: []
```

ir_005 read_output_dir_and_delivery_path

```
inputs[0] context_key: output_dir | output_dir
```

```
inputs[1] context_key: delivery_path | delivery_path
```

```
outputs[0] result: result_003 | requested_output_dir
```

```
outputs[1] result: result_004 | requested_delivery_path
```

```
ir_005.constraints: []
```

```
draft_instruction_id = read_delivery_request
```

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = go_convert
```

block_004 Convert each document with scripts/convert.py

```
data_source_kind = "external"
```

block_004.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。

block_004.constraints[1]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。

ir_007 convert_documents_with_script

```
inputs[0] external_resource: scripts/convert.py | scripts/convert.py
```

```
inputs[1] result: result_002 | document_paths
```

```
inputs[2] result: result_003 | requested_output_dir
```

```
outputs[0] result: result_005 | converted_artifact_paths
```

```
ir_007.constraints: []
```

```
draft_instruction_id = run_convert_script
```

metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。

```
metadata = {"reference_command": "python scripts/convert.py <文档路径> <output_dir>"}
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_package
```

block_005 Package converted artifacts into bundle.zip

```
data_source_kind = null
```

block_005.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 --paths。

ir_009 package_converted_artifacts

```
inputs[0] external_resource: scripts/package.py | scripts/package.py
```



```
inputs[1] result: result_005 | converted_artifact_paths
```

```
inputs[2] literal = "bundle.zip"
```

```
outputs = []
```

```
ir_009.constraints: []
```

```
draft_instruction_id = run_package_script
```

metadata.script_content: 457 字符; SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

```
metadata = {"reference_command": "python scripts/package.py --paths <转换产物路径列表> --output bundle.zip"}
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_deliver
```

block_006 Deliver bundle.zip to the requested delivery_path

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 deliver_bundle

```
inputs[0] external_resource: bundle.zip | bundle.zip
```

```
inputs[1] result: result_004 | requested_delivery_path
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = deliver_generated_bundle
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_receipt
```

block_007 Write converted artifact paths to local receipt.txt

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 write_converted_paths_to_receipt

```
inputs[0] result: result_005 | converted_artifact_paths
```

```
inputs[1] external_resource: receipt.txt | receipt.txt
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = write_receipt_file
```

ir_014 return

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = return_done
```

D03 / 第 2 次 实际输出明细

多文档处理与交付 · 表格

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D03/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Note in SKILL.md says '必要时保留原顺序'; scope unclear, not promoted to constraint.

Skill 层约束与上下文声明

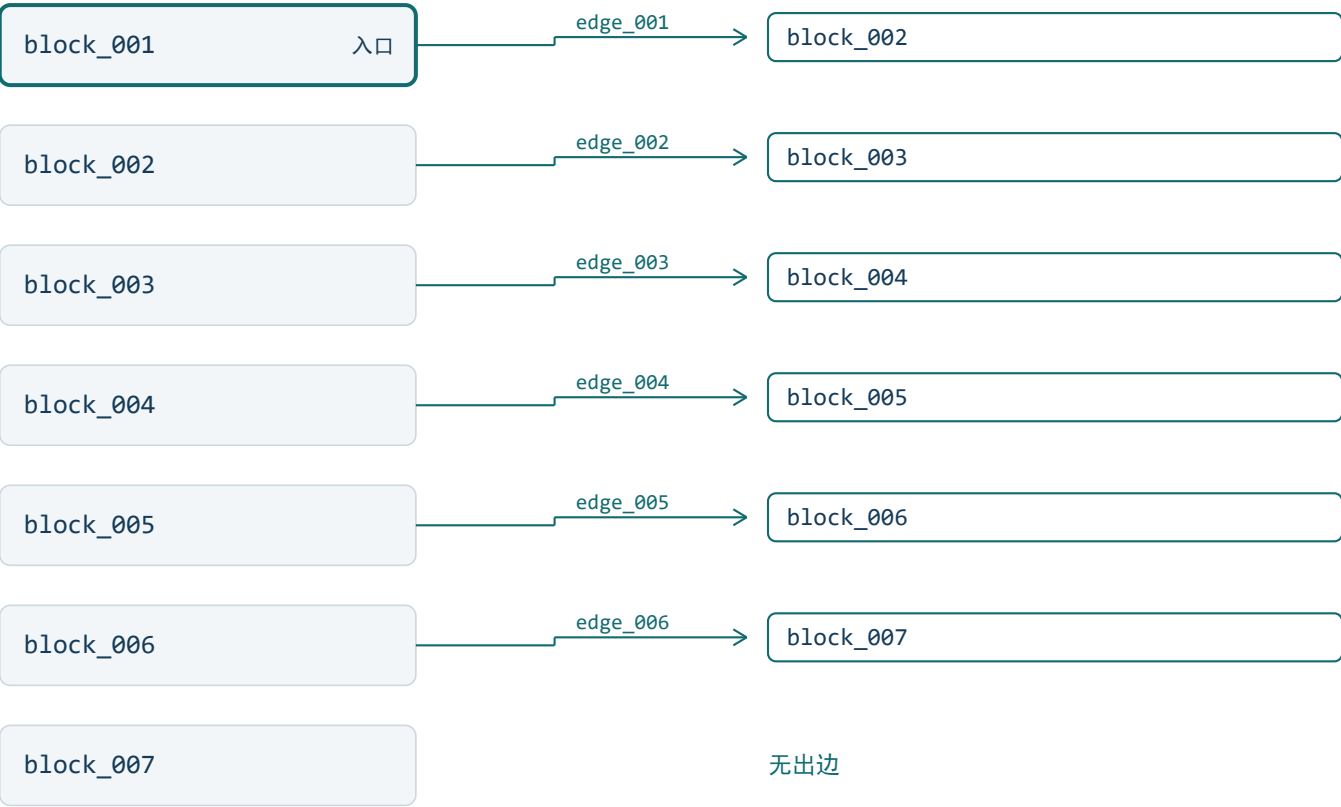
declared_context_keys = ["output_dir", "delivery_path"]

skill.constraints[0]: 整份流程不得修改输入原文文件。

skill.constraints[1]: 整份流程禁止把文档上传到外部服务。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null

全部操作、数据引用与约束落点

```
block_001 Read the user-provided manifest.json

data_source_kind = "external"
block_001.constraints: []

ir_001 read_manifest_json

inputs[0] external_resource: manifest.json | manifest.json
outputs[0] result: result_001 | manifest_data
ir_001.constraints: []
draft_instruction_id = read_manifest_json

ir_002 dispatch

inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = dispatch_after_manifest

block_002 Extract document paths from the manifest

data_source_kind = null
block_002.constraints: []

ir_003 extract_document_paths

inputs[0] result: result_001 | manifest_data
outputs[0] result: result_002 | document_paths
ir_003.constraints: []
draft_instruction_id = extract_document_paths

ir_004 dispatch

inputs = []
outputs = []
ir_004.constraints: []
draft_instruction_id = dispatch_after_extract
```

block_003 Read output_dir and delivery_path from the request

data_source_kind = "context"

block_003.constraints: []

ir_005 read_output_dir_and_delivery_path_from_request

inputs[0] context_key: output_dir | output_dir

inputs[1] context_key: delivery_path | delivery_path

outputs[0] result: result_003 | requested_output_dir

outputs[1] result: result_004 | requested_delivery_path

ir_005.constraints: []

draft_instruction_id = read_request_params

ir_006 dispatch

inputs = []

outputs = []

ir_006.constraints: []

draft_instruction_id = dispatch_after_request

block_004 Convert each document and collect output paths

data_source_kind = "external"

block_004.constraints: []

ir_007 convert_documents

inputs[0] external_resource: scripts/convert.py | scripts/convert.py

inputs[1] result: result_002 | document_paths

inputs[2] result: result_003 | requested_output_dir

outputs[0] result: result_005 | converted_artifact_paths

ir_007.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。

ir_007.constraints[1]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。

draft_instruction_id = run_convert_script

metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。

metadata = {"command": "python scripts/convert.py <文档路径> <output_dir>"}

ir_008 dispatch

inputs = []

outputs = []

ir_008.constraints: []

draft_instruction_id = dispatch_after_convert

block_005 Package converted artifacts into bundle.zip

data_source_kind = null

block_005.constraints: []

ir_009 package_artifacts_into_zip

inputs[0] external_resource: scripts/package.py | scripts/package.py

```
inputs[1] result: result_005 | converted_artifact_paths
```

```
inputs[2] literal = "bundle.zip"
```

```
outputs[0] result: result_006 | bundle_zip_path
```

ir_009.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 `--paths`。

```
draft_instruction_id = run_package_script
```

metadata.script_content: 457 字符; SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

```
metadata = {"command": "python scripts/package.py --paths <转换产物路径列表> --output bundle.zip"}
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_after_package
```

block_006 Deliver bundle.zip to the requested delivery path

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 deliver_bundle_zip

```
inputs[0] result: result_006 | bundle_zip_path
```

```
inputs[1] result: result_004 | requested_delivery_path
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = deliver_bundle
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_after_deliver
```

block_007 Write converted artifact paths to local receipt.txt

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 write_receipt_txt

```
inputs[0] result: result_005 | converted_artifact_paths
```

```
inputs[1] external_resource: receipt.txt | receipt.txt
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = write_receipt
```

ir_014 return

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = return_after_receipt
```

D03 / 第 3 次 实际输出明细

多文档处理与交付 · 表格

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D03/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: 第 10 行备注“必要时保留原顺序”的范围不明确，未提升为 Skill 级约束。

Skill 层约束与上下文声明

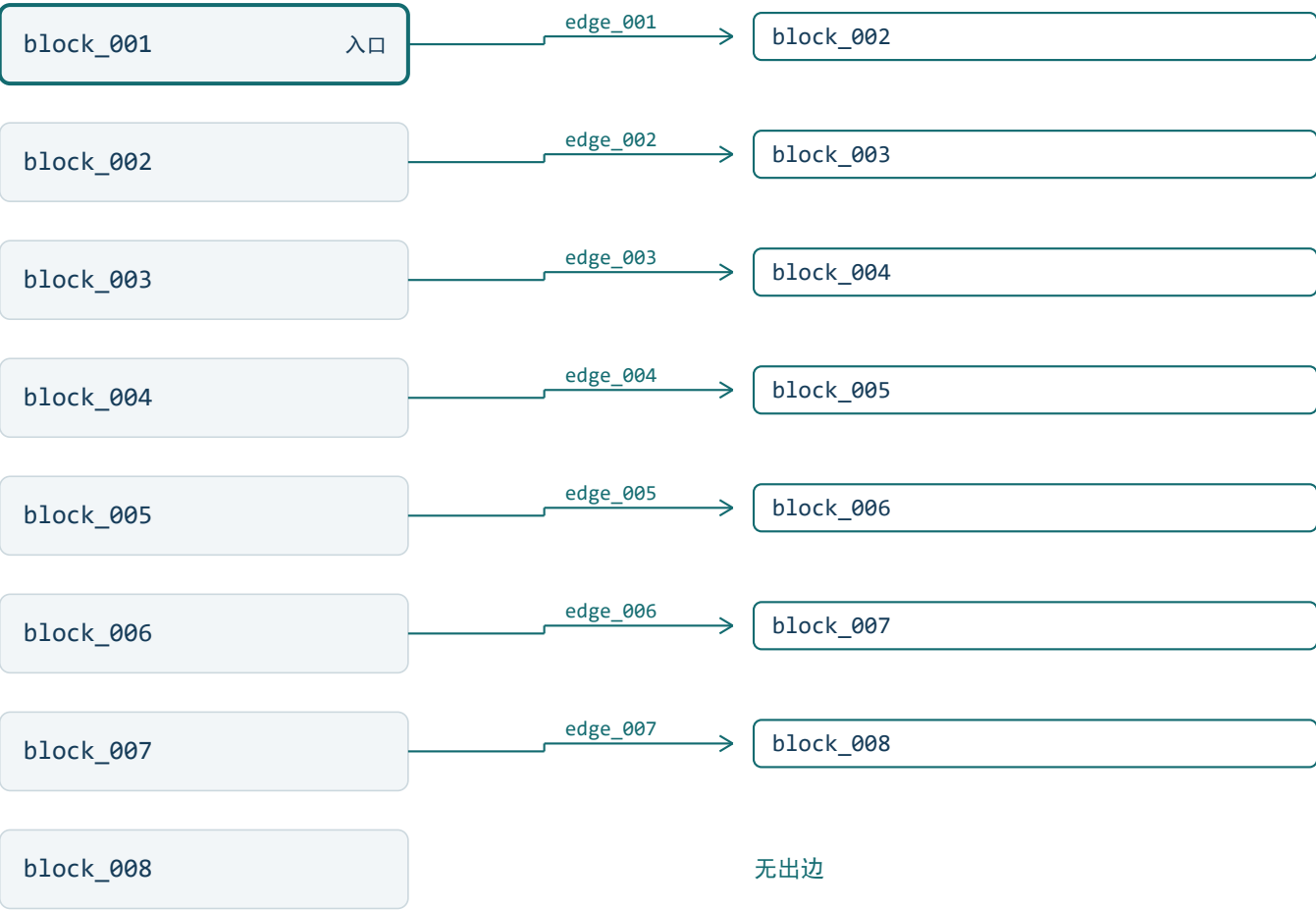
declared_context_keys = ["output_dir", "delivery_path"]

skill.constraints[0]: 整份流程不得修改输入原文文件。

skill.constraints[1]: 整份流程禁止把文档上传到外部服务。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null
edge_007	block_007 -> block_008	null

全部操作、数据引用与约束落点

block_001 Read user-provided manifest file

```
data_source_kind = "external"
block_001.constraints: []

ir_001 read_manifest_file

inputs[0] external_resource: manifest.json | manifest.json
outputs[0] result: result_001 | manifest_contents
ir_001.constraints: []
draft_instruction_id = read_manifest
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = dispatch_to_read_output_dir
```

block_002 Read output_dir from request

```
data_source_kind = "context"
block_002.constraints: []

ir_003 read_output_dir_from_request

inputs[0] context_key: output_dir | output_dir
outputs[0] result: result_002 | output_dir_path
ir_003.constraints: []
draft_instruction_id = read_output_dir
```

ir_004 dispatch

```
inputs = []
outputs = []
ir_004.constraints: []
draft_instruction_id = dispatch_to_read_delivery_path
```

block_003 Read delivery_path from request

data_source_kind = "context"

block_003.constraints: []

ir_005 read_delivery_path_from_request

inputs[0] context_key: delivery_path | delivery_path

outputs[0] result: result_003 | delivery_path_value

ir_005.constraints: []

draft_instruction_id = read_delivery_path

ir_006 dispatch

inputs = []

outputs = []

ir_006.constraints: []

draft_instruction_id = dispatch_to_extract_paths

block_004 Extract document paths from manifest

data_source_kind = null

block_004.constraints: []

ir_007 extract_paths_from_manifest

inputs[0] result: result_001 | manifest_contents

outputs[0] result: result_004 | document_paths

ir_007.constraints: []

draft_instruction_id = extract_paths

ir_008 dispatch

inputs = []

outputs = []

ir_008.constraints: []

draft_instruction_id = dispatch_to_convert_documents

block_005 Convert each document with scripts/convert.py

data_source_kind = "external"

block_005.constraints: []

ir_009 run_convert_script_for_each_document

inputs[0] external_resource: scripts/convert.py | scripts/convert.py

inputs[1] result: result_004 | document_paths

inputs[2] result: result_002 | output_dir_path

outputs[0] result: result_005 | converted_artifact_paths

ir_009.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。

ir_009.constraints[1]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。

draft_instruction_id = convert_documents

metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。

```
metadata = {"command_template": "python scripts/convert.py <文档路径> <output_dir>"}
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_to_package_documents
```

block_006 Package converted documents with scripts/package.py

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 package_converted_documents

```
inputs[0] external_resource: scripts/package.py | scripts/package.py
```

```
inputs[1] result: result_005 | converted_artifact_paths
```

```
inputs[2] external_resource: bundle.zip | bundle.zip
```

```
outputs = []
```

ir_011.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 `--paths`。

```
draft_instruction_id = package_documents
```

metadata.script_content: 457 字符; SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

```
metadata = {"command": "python scripts/package.py --paths <转换产物路径列表> --output bundle.zip"}
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_to_deliver_bundle
```

block_007 Deliver bundle.zip to delivery_path

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 deliver_bundle

```
inputs[0] external_resource: bundle.zip | bundle.zip
```

```
inputs[1] result: result_003 | delivery_path_value
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = deliver_bundle
```

ir_014 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = dispatch_to_write_receipt
```

block_008 Write converted artifact paths to receipt.txt

```
data_source_kind = null
```

```
block_008.constraints: []
```

ir_015 write_receipt_file

```
inputs[0] result: result_005 | converted_artifact_paths
```

```
inputs[1] external_resource: receipt.txt | receipt.txt
```

```
outputs = []
```

```
ir_015.constraints: []
```

```
draft_instruction_id = write_receipt
```

ir_016 return

```
inputs = []
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = return_receipt
```

D04 / 第 1 次 实际输出明细

多文档处理与交付 · 跨文件混合

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D04/deepseek-v4-flash-max/1

结构状态: complete | 生成 2 | 修复 1 | HTTP 尝试 2 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

生成 2 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: manifest.json is modeled as an external file resource because the workflow names it directly and does not declare a context key for it.

candidate warning: The per-document conversion loop is represented by one instruction that runs scripts/convert.py for each document path and collects the stdout paths.

candidate warning: The workflow does not specify the relative order of delivery and receipt writing; they are modeled sequentially after packaging, with receipt writing after delivery.

candidate warning: scripts/convert.py and scripts/package.py are treated as black-box external script operations; their original code bodies are preserved in instruction metadata.

Skill 层约束与上下文声明

declared_context_keys = ["output_dir", "delivery_path"]

skill.constraints[0]: 必须执行 references/workflow.md 中的要求；该文件是当前流程的组成部分。

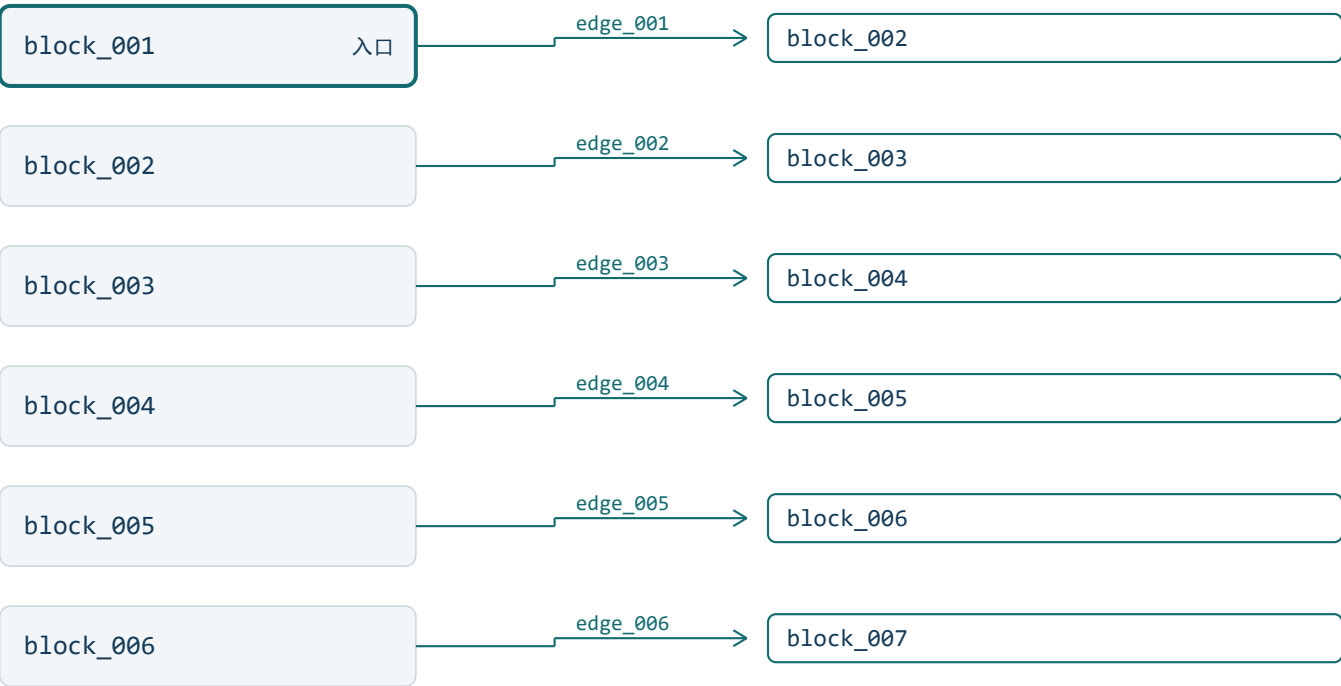
skill.constraints[1]: 整份流程不得修改输入原文文件。

skill.constraints[2]: 整份流程禁止把文档上传到外部服务。

skill.constraints[3]: 必要时保留原顺序。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



block_007

无出边

完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null

全部操作、数据引用与约束落点

```
block_001 Read the user-provided manifest.json
data_source_kind = "external"
block_001.constraints: []

ir_001 read_manifest_json
inputs[0] external_resource: manifest.json | manifest.json
outputs[0] result: result_001 | manifest contents
ir_001.constraints: []
draft_instruction_id = read_manifest_file

ir_002 dispatch
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_read_request_config

block_002 Read output_dir and delivery_path from the request
data_source_kind = "context"
block_002.constraints: []

ir_003 read_output_dir_and_delivery_path_from_request
inputs[0] context_key: output_dir | output_dir
inputs[1] context_key: delivery_path | delivery_path
outputs[0] result: result_002 | output_dir
outputs[1] result: result_003 | delivery_path
ir_003.constraints: []
draft_instruction_id = read_delivery_config
```

ir_004 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = go_extract_document_paths
```

block_003 Extract document paths from the manifest paths list

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 extract_manifest_document_paths

```
inputs[0] result: result_001 | manifest_contents
```

```
outputs[0] result: result_004 | document paths
```

```
ir_005.constraints: []
```

```
draft_instruction_id = extract_manifest_paths
```

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = go_convert_documents
```

block_004 Convert each document with scripts/convert.py

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_007 run_convert_script_for_each_document

```
inputs[0] external_resource: scripts/convert.py | scripts/convert.py
```

```
inputs[1] result: result_004 | document_paths
```

```
inputs[2] result: result_002 | requested_output_dir
```

```
outputs[0] result: result_005 | 转换产物路径列表
```

ir_007.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。

ir_007.constraints[1]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。

```
draft_instruction_id = run_convert_script_for_each_document
```

metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。

```
metadata = {"command": "python scripts/convert.py <文档路径> <output_dir>", "script_path": "scripts/convert.py", "side_effects": ["creates output_dir if needed", "writes a converted UTF-8 .txt file", "prints the output path"], "stdout": "每次标准输出作为对应的转换产物路径"}
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_package_bundle
```

block_005 Package converted documents into bundle.zip

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 run_package_script_to_build_bundle

```
inputs[0] external_resource: scripts/package.py | scripts/package.py
```

```
inputs[1] result: result_005 | conversion_artifact_paths
```

```
inputs[2] literal = "bundle.zip"
```

```
outputs[0] result: result_006 | bundle.zip
```

ir_009.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 `--paths`。

```
draft_instruction_id = run_package_script
```

metadata.script_content: 457 字符; SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

```
metadata = {"command": "python scripts/package.py --paths <转换产物路径列表> --output bundle.zip", "script_path": "scripts/package.py", "side_effects": ["creates bundle.zip ZIP archive"]}
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_deliver_bundle
```

block_006 Deliver bundle.zip to requested delivery_path

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 deliver_bundle_zip_to_delivery_path

```
inputs[0] result: result_006 | bundle_zip_path
```

```
inputs[1] result: result_003 | requested_delivery_path
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = deliver_bundle_zip
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_write_receipt
```

block_007 Write conversion artifact paths to local receipt.txt

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 write_conversion_paths_to_receipt

```
inputs[0] result: result_005 | conversion_artifact_paths
```

```
inputs[1] external_resource: receipt.txt | receipt.txt
```

```
outputs = []
```



```
ir_013.constraints: []  
draft_instruction_id = write_receipt_file  
  
ir_014 return  
  
inputs = []  
outputs = []  
  
ir_014.constraints: []  
draft_instruction_id = return_after_receipt
```

D04 / 第 2 次 实际输出明细

多文档处理与交付 · 跨文件混合

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D04/deepseek-v4-flash-max/2

结构状态: complete | 生成 2 | 修复 1 | HTTP 尝试 2 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

生成 2 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

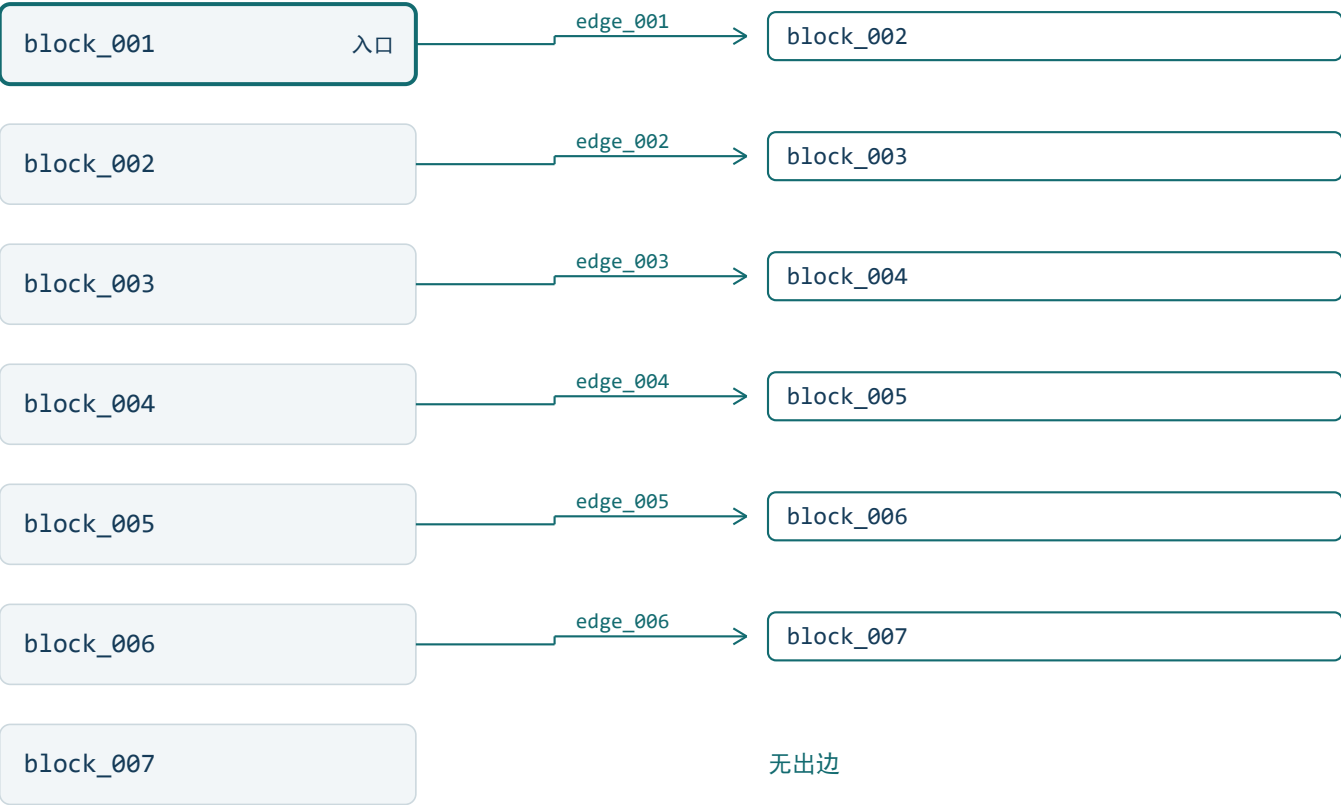
declared_context_keys = ["output_dir", "delivery_path"]

skill.constraints[0]: 整份流程不得修改输入原文文件。

skill.constraints[1]: 整份流程禁止把文档上传到外部服务。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null

全部操作、数据引用与约束落点

```
block_001 Read the user-provided manifest.json

data_source_kind = "external"
block_001.constraints: []

ir_001 read_manifest_file

inputs[0] external_resource: manifest.json | manifest.json
outputs[0] result: result_001 | manifest_contents
ir_001.constraints: []
draft_instruction_id = read_manifest_file

ir_002 dispatch

inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_extract_paths

block_002 Extract document paths from the manifest paths list

data_source_kind = null
block_002.constraints: []

ir_003 extract_manifest_document_paths

inputs[0] result: result_001 | manifest_contents
outputs[0] result: result_002 | document_paths
ir_003.constraints: []
draft_instruction_id = extract_paths

ir_004 dispatch

inputs = []
outputs = []
ir_004.constraints: []
draft_instruction_id = go_read_request
```

block_003 Read output_dir and delivery_path from the request

```
data_source_kind = "context"
```

```
block_003.constraints: []
```

ir_005 read_output_and_delivery_paths_from_request

```
inputs[0] context_key: output_dir | output_dir
```

```
inputs[1] context_key: delivery_path | delivery_path
```

```
outputs[0] result: result_003 | output_dir_path
```

```
outputs[1] result: result_004 | delivery_target_path
```

```
ir_005.constraints: []
```

```
draft_instruction_id = read_request_paths
```

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = go_convert
```

block_004 Convert each document with scripts/convert.py and collect output paths

```
data_source_kind = "external"
```

```
block_004.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。
```

```
block_004.constraints[1]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。
```

```
block_004.constraints[2]: 必要时保留原顺序。
```

ir_007 convert_documents_via_convert_script

```
inputs[0] external_resource: scripts/convert.py | scripts/convert.py
```

```
inputs[1] result: result_002 | document_paths
```

```
inputs[2] result: result_003 | output_dir_path
```

```
outputs[0] result: result_005 | converted_artifact_paths
```

```
ir_007.constraints: []
```

```
draft_instruction_id = convert_documents
```

```
metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。
```

```
metadata = {"command": "python scripts/convert.py <文档路径> <output_dir>"}
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_package
```

block_005 Package converted artifact paths into bundle.zip with scripts/package.py

```
data_source_kind = null
```

block_005.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 `--paths`。

ir_009 execute_package_script

```
inputs[0] external_resource: scripts/package.py | scripts/package.py
```

```
inputs[1] result: result_005 | converted_artifact_paths
```

```
inputs[2] literal = "bundle.zip"
```

```
outputs[0] result: result_006 | bundle_zip_path
```

```
ir_009.constraints: []
```

```
draft_instruction_id = package_documents
```

metadata.script_content: 457 字符; SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

```
metadata = {"command": "python scripts/package.py --paths <转换产物路径列表> --output bundle.zip"}
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_deliver
```

block_006 Deliver bundle.zip to the requested delivery_path

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 deliver_bundle_to_path

```
inputs[0] result: result_006 | bundle_zip_path
```

```
inputs[1] result: result_004 | delivery_target_path
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = deliver_bundle
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_write_receipt
```

block_007 Write converted artifact path list to receipt.txt and return

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 write_receipt_file

```
inputs[0] result: result_005 | converted_artifact_paths
```

```
inputs[1] external_resource: receipt.txt | receipt.txt
```

```
outputs = []
```

```
ir_013.constraints: []  
draft_instruction_id = write_receipt  
  
ir_014 return  
  
inputs = []  
outputs = []  
  
ir_014.constraints: []  
draft_instruction_id = return
```

D04 / 第 3 次 实际输出明细

多文档处理与交付 · 跨文件混合

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D04/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

diagnostics: []。无结构诊断并不表示语义通过。

Skill 层约束与上下文声明

declared_context_keys = ["output_dir", "delivery_path"]

skill.constraints[0]: 必须执行 [流程说明] (references/workflow.md) 中的要求；该文件是当前流程的组成部分。

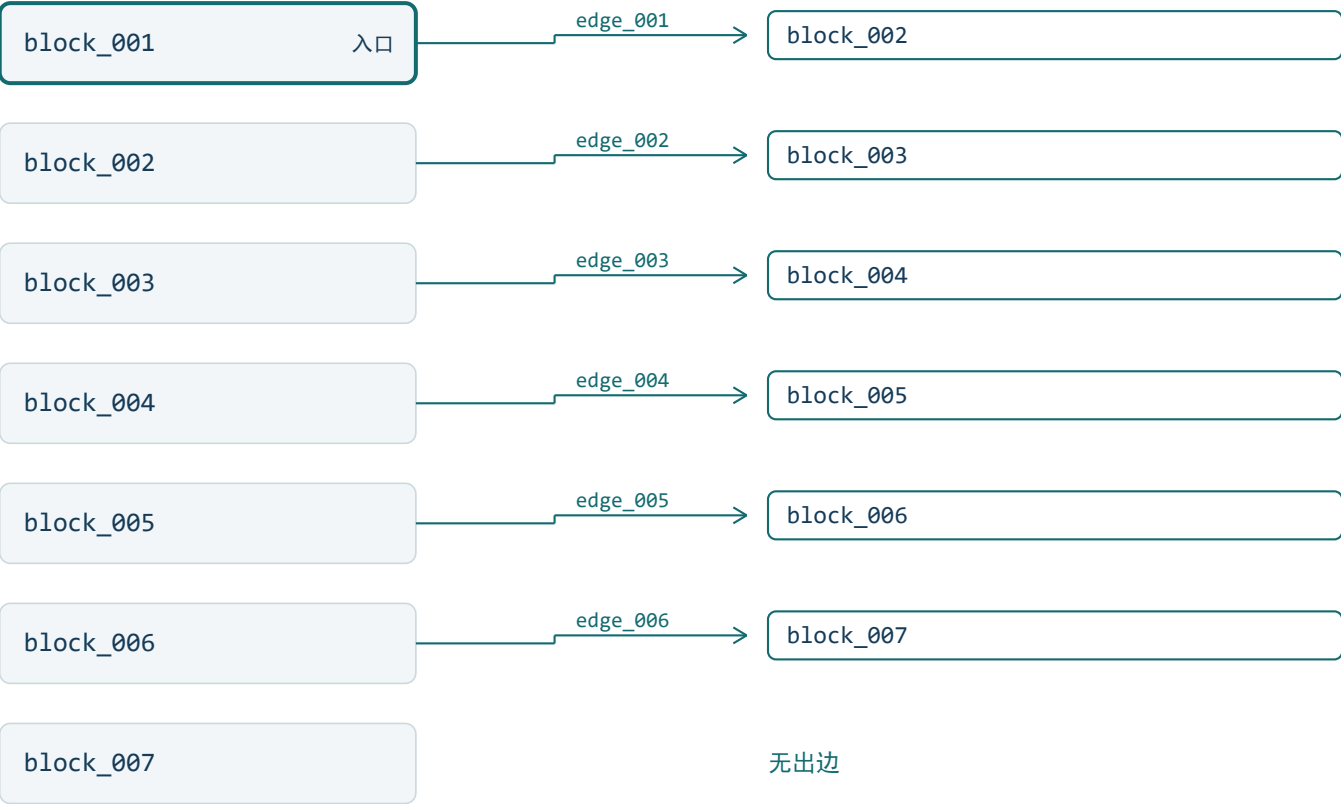
skill.constraints[1]: 整份流程不得修改输入原文文件。

skill.constraints[2]: 整份流程禁止把文档上传到外部服务。

skill.constraints[3]: 备注：必要时保留原顺序。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null

全部操作、数据引用与约束落点

```
block_001 Read the user-provided manifest.json

data_source_kind = "external"
block_001.constraints: []

ir_001 read_manifest_file

inputs[0] external_resource: manifest.json | manifest.json
outputs[0] result: result_001 | manifest_contents
ir_001.constraints: []
draft_instruction_id = read_manifest

ir_002 dispatch

inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = dispatch_to_extract_paths

block_002 Extract document paths from the manifest paths list

data_source_kind = null
block_002.constraints: []

ir_003 extract_document_paths_from_manifest

inputs[0] result: result_001 | manifest_contents
outputs[0] result: result_002 | document_paths
ir_003.constraints: []
draft_instruction_id = extract_paths

ir_004 dispatch

inputs = []
outputs = []
ir_004.constraints: []
draft_instruction_id = dispatch_to_read_request
```


block_003 Read output_dir and delivery_path from the request

data_source_kind = "context"

block_003.constraints: []

ir_005 read_output_dir_and_delivery_path_from_request

inputs[0] context_key: output_dir | output_dir

inputs[1] context_key: delivery_path | delivery_path

outputs[0] result: result_003 | requested_output_dir

outputs[1] result: result_004 | requested_delivery_path

ir_005.constraints: []

draft_instruction_id = read_request_locations

ir_006 dispatch

inputs = []

outputs = []

ir_006.constraints: []

draft_instruction_id = dispatch_to_convert

block_004 Convert each document path in order

data_source_kind = "external"

block_004.constraints: []

ir_007 convert_each_document_with_script

inputs[0] external_resource: scripts/convert.py | scripts/convert.py

inputs[1] result: result_002 | document_paths

inputs[2] result: result_003 | requested_output_dir

outputs[0] result: result_005 | converted_output_paths

ir_007.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。

ir_007.constraints[1]: 逐个执行 python scripts/convert.py <文档路径> <output_dir>，把每次标准输出作为对应的转换产物路径。

ir_007.constraints[2]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。

draft_instruction_id = convert_each_document

metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。

ir_008 dispatch

inputs = []

outputs = []

ir_008.constraints: []

draft_instruction_id = dispatch_to_package

block_005 Package converted documents into bundle.zip

data_source_kind = null

block_005.constraints: []

ir_009 package_converted_documents_with_script

inputs[0] external_resource: scripts/package.py | scripts/package.py

```
inputs[1] result: result_005 | converted_output_paths
```

```
inputs[2] literal = "bundle.zip"
```

```
outputs[0] result: result_006 | bundle_zip_path
```

ir_009.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 `--paths`。

ir_009.constraints[1]: `python scripts/package.py --paths <转换产物路径列表> --output bundle.zip`

```
draft_instruction_id = package_bundle
```

metadata.script_content: 457 字符; SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_to_deliver
```

block_006 Deliver bundle.zip to the requested delivery_path

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 deliver_bundle_to_requested_path

```
inputs[0] result: result_006 | bundle_zip_path
```

```
inputs[1] result: result_004 | requested_delivery_path
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = deliver_bundle_to_path
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_to_receipt
```

block_007 Write converted output paths to receipt.txt

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 write_receipt_file

```
inputs[0] external_resource: receipt.txt | receipt.txt
```

```
inputs[1] result: result_005 | converted_output_paths
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = write_receipt_file
```

ir_014 return

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = return_complete
```

D05 / 第 1 次 实际输出明细

多文档处理与交付 · 用途最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D05/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: 历史示例（当前流程不执行）：另外，将转换产物路径列表写入本地 receipt.txt。该步骤未被建模为执行指令。

candidate warning: 备注“必要时保留原顺序”的范围不明确，未提升为 Skill-wide 约束或执行步骤。

candidate warning: scripts/convert.py 和 scripts/package.py 按黑盒操作建模；其内部文件读写和 ZIP 循环未拆分为独立块。

candidate warning: manifest.json 按外部文件资源建模；output_dir 和 delivery_path 按请求上下文字段建模。

candidate warning: 步骤 2 的逐文档转换被视为一个黑盒操作 run_convert_script_for_each_document，未拆分逐文档迭代。

Skill 层约束与上下文声明

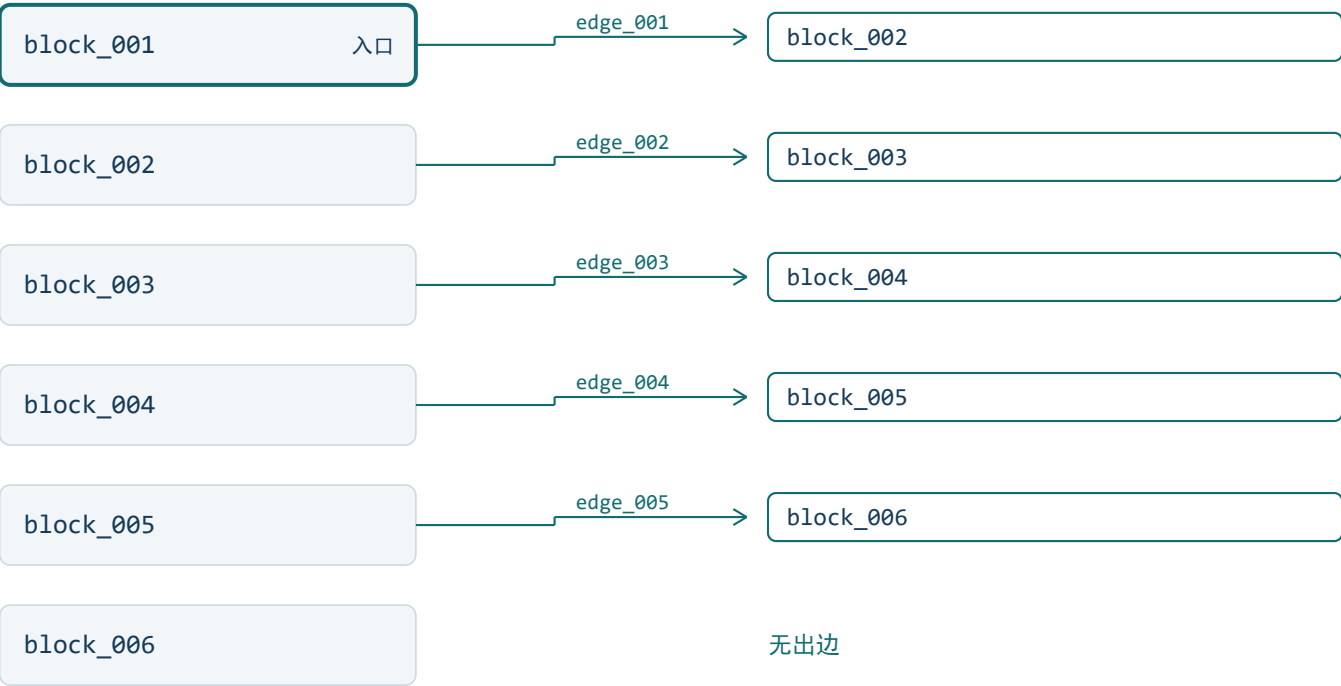
declared_context_keys = ["output_dir", "delivery_path"]

skill.constraints[0]：整份流程不得修改输入原文文件。

skill.constraints[1]：整份流程禁止把文档上传到外部服务。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null

全部操作、数据引用与约束落点

block_001 Read the user-provided manifest.json

```
data_source_kind = "external"
block_001.constraints: []

ir_001 read_manifest_json

inputs[0] external_resource: manifest.json | user-provided manifest.json
outputs[0] result: result_001 | manifest contents
ir_001.constraints: []

draft_instruction_id = read_manifest_file
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []

draft_instruction_id = go_parse_manifest
```

block_002 Extract document paths from the manifest

```
data_source_kind = null
block_002.constraints: []

ir_003 extract_paths_from_manifest

inputs[0] result: result_001 | manifest_data
outputs[0] result: result_002 | document paths
ir_003.constraints: []

draft_instruction_id = extract_document_paths
```

ir_004 dispatch

```
inputs = []
outputs = []
ir_004.constraints: []

draft_instruction_id = go_read_request
```

block_003 Read output_dir and delivery_path from the request

```
data_source_kind = "context"
```

```
block_003.constraints: []
```

ir_005 read_output_dir_and_delivery_path_from_request

```
inputs[0] context_key: output_dir | output_dir
```

```
inputs[1] context_key: delivery_path | delivery_path
```

```
outputs[0] result: result_003 | requested output_dir
```

```
outputs[1] result: result_004 | requested delivery_path
```

```
ir_005.constraints: []
```

```
draft_instruction_id = read_request_settings
```

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = go_convert_documents
```

block_004 Convert each document with scripts/convert.py and collect stdout paths

```
data_source_kind = "external"
```

```
block_004.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。
```

```
block_004.constraints[1]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。
```

ir_007 run_convert_script_for_each_document

```
inputs[0] external_resource: scripts/convert.py | scripts/convert.py
```

```
inputs[1] result: result_002 | document_paths
```

```
inputs[2] result: result_003 | requested_output_dir
```

```
outputs[0] result: result_005 | converted artifact paths from stdout
```

```
ir_007.constraints: []
```

```
draft_instruction_id = run_convert_script_for_each_document
```

```
metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。
```

```
metadata = {"script_path": "scripts/convert.py", "side_effects": ["creates output_dir if needed", "writes the converted document to output_dir/<source_stem>.txt", "prints the converted output path to stdout"]}
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_package_documents
```

block_005 Package converted artifacts with scripts/package.py

```
data_source_kind = null
```

```
block_005.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 --paths。
```

ir_009 run_package_script

```
inputs[0] external_resource: scripts/package.py | scripts/package.py
```

```
inputs[1] result: result_005 | converted_artifact_paths
inputs[2] literal = "bundle.zip"
outputs[0] result: result_006 | generated bundle.zip
ir_009.constraints: []
draft_instruction_id = run_package_script
metadata.script_content: 457 字符; SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。
metadata = {"script_path": "scripts/package.py", "side_effects": ["writes the ZIP archive to bundle.zip"]}

ir_010 dispatch

inputs = []
outputs = []
ir_010.constraints: []
draft_instruction_id = go_deliver_bundle

block_006 Deliver bundle.zip to the requested delivery_path

data_source_kind = null
block_006.constraints: []

ir_011 deliver_bundle_zip_to_path

inputs[0] result: result_006 | bundle_zip_path
inputs[1] result: result_004 | requested_delivery_path
outputs = []
ir_011.constraints: []
draft_instruction_id = deliver_bundle_zip

ir_012 return

inputs = []
outputs = []
ir_012.constraints: []
draft_instruction_id = return_delivery
```

D05 / 第 2 次 实际输出明细

多文档处理与交付 · 用途最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D05/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: 历史示例（当前流程不执行）：将转换产物路径列表写入本地 receipt.txt 未被提取为执行步骤，因为源文本说明当前流程不执行。

candidate warning: 备注：必要时保留原顺序。范围不明确，未提升为 Skill-wide 约束。

Skill 层约束与上下文声明

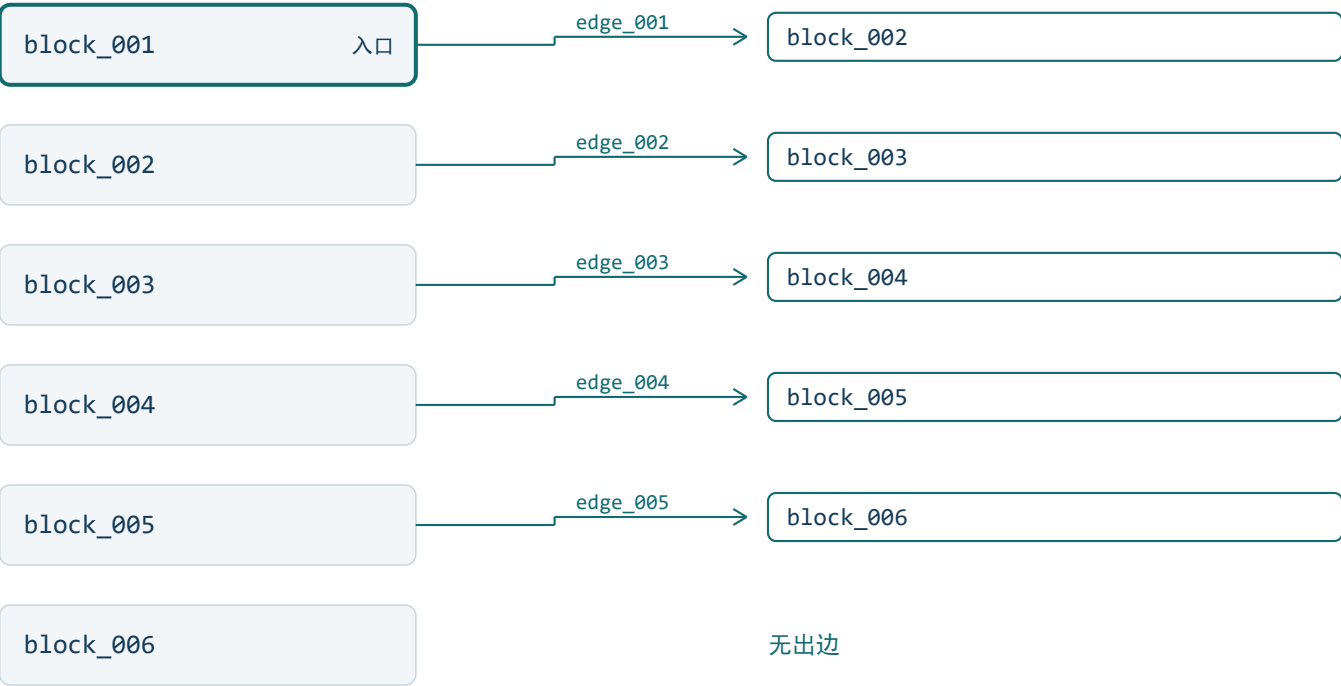
declared_context_keys = ["output_dir", "delivery_path"]

skill.constraints[0]：整份流程不得修改输入原文文件。

skill.constraints[1]：整份流程禁止把文档上传到外部服务。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null

全部操作、数据引用与约束落点

```
block_001 Read the user-provided manifest.json
data_source_kind = "external"
block_001.constraints: []

ir_001 read_manifest_file
inputs[0] external_resource: manifest.json | manifest.json
outputs[0] result: result_001 | manifest_data
ir_001.constraints: []
draft_instruction_id = read_manifest_file

ir_002 dispatch
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_parse_manifest

block_002 Extract document paths from the manifest paths list
data_source_kind = null
block_002.constraints: []

ir_003 extract_document_paths
inputs[0] result: result_001 | manifest_data
outputs[0] result: result_002 | document_paths
ir_003.constraints: []
draft_instruction_id = extract_document_paths

ir_004 dispatch
inputs = []
outputs = []
ir_004.constraints: []
draft_instruction_id = go_read_request_config
```

block_003 Read output_dir and delivery_path from the request

data_source_kind = "context"

block_003.constraints: []

ir_005 read_request_delivery_config

inputs[0] context_key: output_dir | output_dir

inputs[1] context_key: delivery_path | delivery_path

outputs[0] result: result_003 | request_output_dir

outputs[1] result: result_004 | request_delivery_path

ir_005.constraints: []

draft_instruction_id = read_request_delivery_config

ir_006 dispatch

inputs = []

outputs = []

ir_006.constraints: []

draft_instruction_id = go_convert_documents

block_004 Run scripts/convert.py for each document path

data_source_kind = "external"

block_004.constraints[0]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。

ir_007 run_convert_script_per_document

inputs[0] external_resource: scripts/convert.py | scripts/convert.py

inputs[1] result: result_002 | document_paths

inputs[2] result: result_003 | request_output_dir

outputs[0] result: result_005 | converted_artifact_paths

ir_007.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。

draft_instruction_id = run_convert_script_per_document

metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。

metadata = {"command": "python scripts/convert.py <文档路径> <output_dir>"}

ir_008 dispatch

inputs = []

outputs = []

ir_008.constraints: []

draft_instruction_id = go_package_bundle

block_005 Package converted artifacts with scripts/package.py

data_source_kind = null

block_005.constraints: []

ir_009 run_package_script

inputs[0] external_resource: scripts/package.py | scripts/package.py

inputs[1] result: result_005 | converted_artifact_paths

```
inputs[2] literal = "bundle.zip"
```

```
outputs[0] result: result_006 | bundle_zip_path
```

ir_009.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 `--paths`。

```
draft_instruction_id = run_package_script
```

metadata.script_content: 457 字符; SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

```
metadata = {"command": "python scripts/package.py --paths <转换产物路径列表> --output bundle.zip"}
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_deliver_bundle
```

block_006 Deliver bundle.zip to the requested delivery_path

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 deliver_bundle_to_path

```
inputs[0] result: result_006 | bundle_zip_path
```

```
inputs[1] result: result_004 | request_delivery_path
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = deliver_bundle_to_path
```

ir_012 return

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = return
```

D05 / 第 3 次 实际输出明细

多文档处理与交付 · 用途最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D05/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Step 7 describes a historical example (writing converted paths to receipt.txt) that is explicitly marked as not executed in the current flow; no instruction was created.

candidate warning: The note '必要时保留原顺序' has unclear scope and was not promoted to a Skill-wide constraint.

Skill 层约束与上下文声明

declared_context_keys = ["output_dir", "delivery_path"]

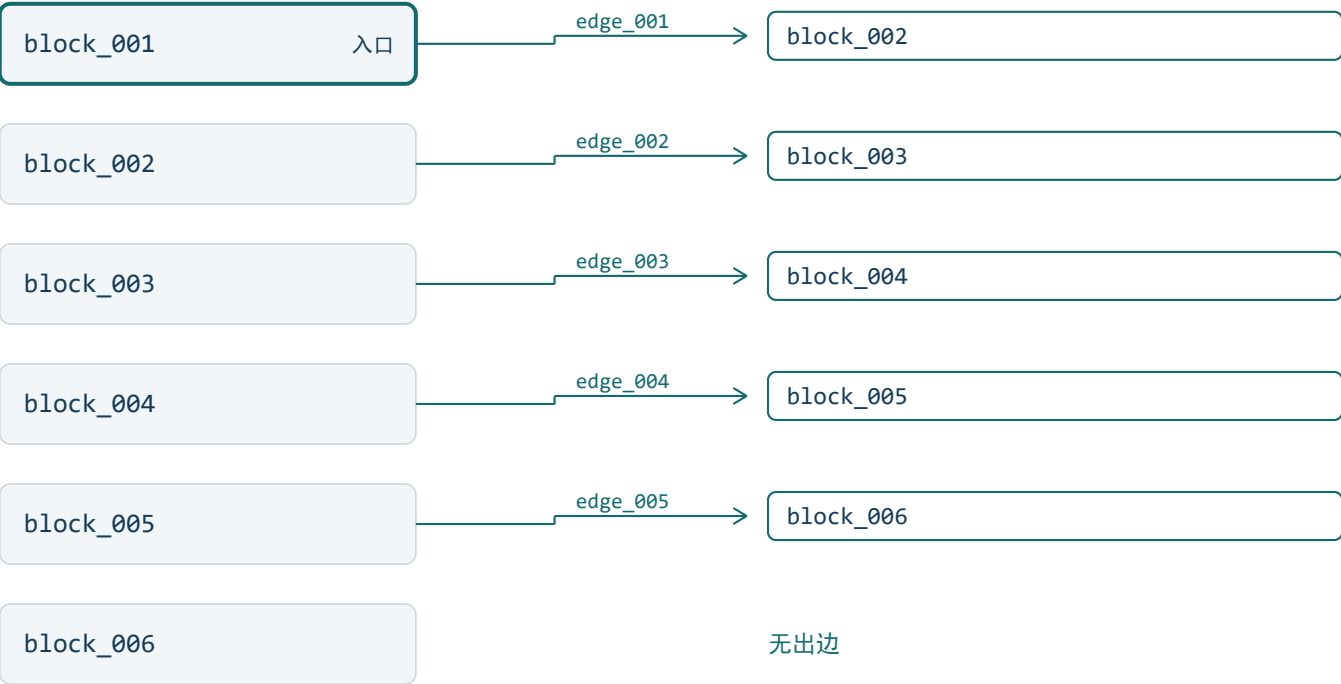
skill.constraints[0]: 整份流程不得修改输入原文文件。

skill.constraints[1]: 整份流程禁止把文档上传到外部服务。

skill.constraints[2]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null

全部操作、数据引用与约束落点

block_001 Read the user-provided manifest.json

```
data_source_kind = "external"
block_001.constraints: []

ir_001 read_manifest_file

inputs[0] external_resource: manifest.json | manifest.json
outputs[0] result: result_001 | manifest_content
ir_001.constraints: []
draft_instruction_id = read_manifest_file
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_extract_paths
```

block_002 Extract document paths from manifest

```
data_source_kind = null
block_002.constraints: []

ir_003 extract_paths_from_manifest

inputs[0] result: result_001 | manifest_content
outputs[0] result: result_002 | document_paths
ir_003.constraints: []
draft_instruction_id = extract_paths_from_manifest
```

ir_004 dispatch

```
inputs = []
outputs = []
ir_004.constraints: []
draft_instruction_id = go_read_request
```

block_003 Read output_dir and delivery_path from request

data_source_kind = "context"

block_003.constraints: []

ir_005 read_output_dir_and_delivery_path_from_request

inputs[0] context_key: output_dir | output_dir

inputs[1] context_key: delivery_path | delivery_path

outputs[0] result: result_003 | output_dir_value

outputs[1] result: result_004 | delivery_path_value

ir_005.constraints: []

draft_instruction_id = read_output_dir_and_delivery_path

ir_006 dispatch

inputs = []

outputs = []

ir_006.constraints: []

draft_instruction_id = go_convert

block_004 Convert each document using scripts/convert.py

data_source_kind = "external"

block_004.constraints: []

ir_007 run_convert_script_for_each_document

inputs[0] external_resource: scripts/convert.py | scripts/convert.py

inputs[1] result: result_002 | document_paths

inputs[2] result: result_003 | output_dir_value

outputs[0] result: result_005 | converted_paths

ir_007.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。

draft_instruction_id = run_convert_script_for_each_document

metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。

metadata = {"command": "python scripts/convert.py <文档路径> <output_dir>"}

ir_008 dispatch

inputs = []

outputs = []

ir_008.constraints: []

draft_instruction_id = go_package

block_005 Package converted documents into bundle.zip using scripts/package.py

data_source_kind = null

block_005.constraints: []

ir_009 run_package_script

inputs[0] external_resource: scripts/package.py | scripts/package.py

inputs[1] result: result_005 | converted_paths

```
inputs[2] literal = "bundle.zip"
```

```
outputs[0] result: result_006 | bundle_zip
```

ir_009.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 `--paths`。

```
draft_instruction_id = run_package_script
```

metadata.script_content: 457 字符; SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

```
metadata = {"command": "python scripts/package.py --paths <转换产物路径列表> --output bundle.zip"}
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_deliver
```

block_006 Deliver bundle.zip to the requested delivery_path

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 deliver_bundle_to_delivery_path

```
inputs[0] result: result_006 | bundle_zip
```

```
inputs[1] result: result_004 | delivery_path_value
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = deliver_bundle_to_delivery_path
```

ir_012 return

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = return_success
```

D06 / 第 1 次 实际输出明细

多文档处理与交付 · 语义最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D06/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: SKILL.md step 1 reads delivery_path from the request, but step 6 delivers to archive_path; delivery_path is read but not used in the extracted flow.

candidate warning: archive_path is not listed among the initial request fields in step 1, so it is acquired immediately before delivery.

candidate warning: The per-document conversion loop is represented as a single external acquisition instruction execute_convert_script_for_each_document to avoid splitting the repeated call and the internal script behavior.

Skill 层约束与上下文声明

declared_context_keys = ["output_dir", "delivery_path", "archive_path"]

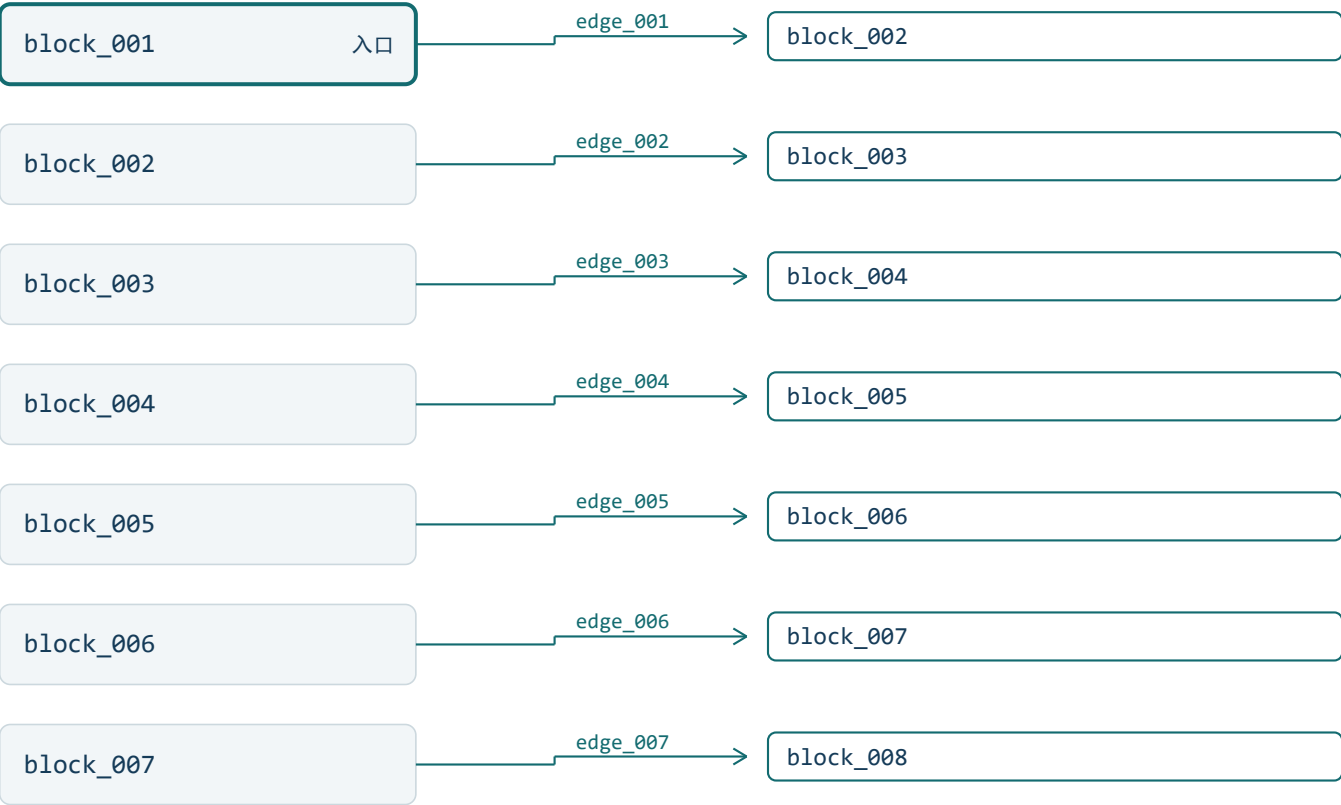
skill.constraints[0]: 整份流程不得修改输入原文文件。

skill.constraints[1]: 整份流程禁止把文档上传到外部服务。

skill.constraints[2]: 备注：必要时保留原顺序。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



block_008

无出边

完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null
edge_007	block_007 -> block_008	null

全部操作、数据引用与约束落点

```
block_001 Read the user-provided manifest.json
data_source_kind = "external"
block_001.constraints: []

ir_001 read_manifest_file
inputs[0] external_resource: manifest.json | manifest.json
outputs[0] result: result_001 | manifest_content
ir_001.constraints: []
draft_instruction_id = read_manifest_instruction

ir_002 dispatch
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = dispatch_to_request_settings

block_002 Read output_dir and delivery_path from the request
data_source_kind = "context"
block_002.constraints: []

ir_003 read_output_dir_and_delivery_path_from_request
inputs[0] context_key: output_dir | output_dir
inputs[1] context_key: delivery_path | delivery_path
outputs[0] result: result_002 | output_dir
outputs[1] result: result_003 | delivery_path
ir_003.constraints: []
draft_instruction_id = read_request_settings_instruction
```

ir_004 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = dispatch_to_extract_paths
```

block_003 Extract document paths from the manifest paths list

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 extract_document_paths_from_manifest

```
inputs[0] result: result_001 | manifest_content
```

```
outputs[0] result: result_004 | document_paths
```

```
ir_005.constraints: []
```

```
draft_instruction_id = extract_paths_instruction
```

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = dispatch_to_convert
```

block_004 Convert each document and collect its output path

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_007 execute_convert_script_for_each_document

```
inputs[0] external_resource: scripts/convert.py | scripts/convert.py
```

```
inputs[1] result: result_004 | document_paths
```

```
inputs[2] result: result_002 | output_dir_value
```

```
outputs[0] result: result_005 | converted_artifact_paths
```

ir_007.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。

ir_007.constraints[1]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。

```
draft_instruction_id = convert_each_document_instruction
```

metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_to_package
```

block_005 Package converted artifacts into bundle.zip

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 execute_package_script

```
inputs[0] external_resource: scripts/package.py | scripts/package.py
```

```
inputs[1] result: result_005 | converted_artifact_paths
```

```
inputs[2] literal = "bundle.zip"
```

```
outputs = []
```

ir_009.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 `--paths`。

ir_009.constraints[1]: `python scripts/package.py --paths <转换产物路径列表> --output bundle.zip`

```
draft_instruction_id = package_bundle_instruction
```

metadata.script_content: 457 字符; SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_to_read_archive
```

block_006 Read archive_path from the request

```
data_source_kind = "context"
```

```
block_006.constraints: []
```

ir_011 read_archive_path_from_request

```
inputs[0] context_key: archive_path | archive_path
```

```
outputs[0] result: result_006 | archive_path
```

```
ir_011.constraints: []
```

```
draft_instruction_id = read_archive_path_instruction
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_to_deliver
```

block_007 Deliver bundle.zip to the requested archive_path

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 deliver_bundle_zip

```
inputs[0] literal = "bundle.zip"
```

```
inputs[1] result: result_006 | archive_path_value
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = deliver_bundle_instruction
```

ir_014 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = dispatch_to_write_receipt
```

block_008 Write converted artifact paths to receipt.txt

```
data_source_kind = null
```

```
block_008.constraints: []
```

ir_015 write_converted_paths_to_receipt

```
inputs[0] result: result_005 | converted_artifact_paths
```

```
inputs[1] literal = "receipt.txt"
```

```
outputs = []
```

```
ir_015.constraints: []
```

```
draft_instruction_id = write_receipt_instruction
```

ir_016 return

```
inputs = []
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = return_instruction
```

D06 / 第 2 次 实际输出明细

多文档处理与交付 · 语义最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D06/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Step 1 reads delivery_path from the request, but step 6 says deliver to the requested archive_path; delivery_path is modeled as an acquisition but is not consumed by the modeled flow.

candidate warning: The note '备注：必要时保留原顺序' has unclear scope and is not promoted to a Skill-wide constraint.

Skill 层约束与上下文声明

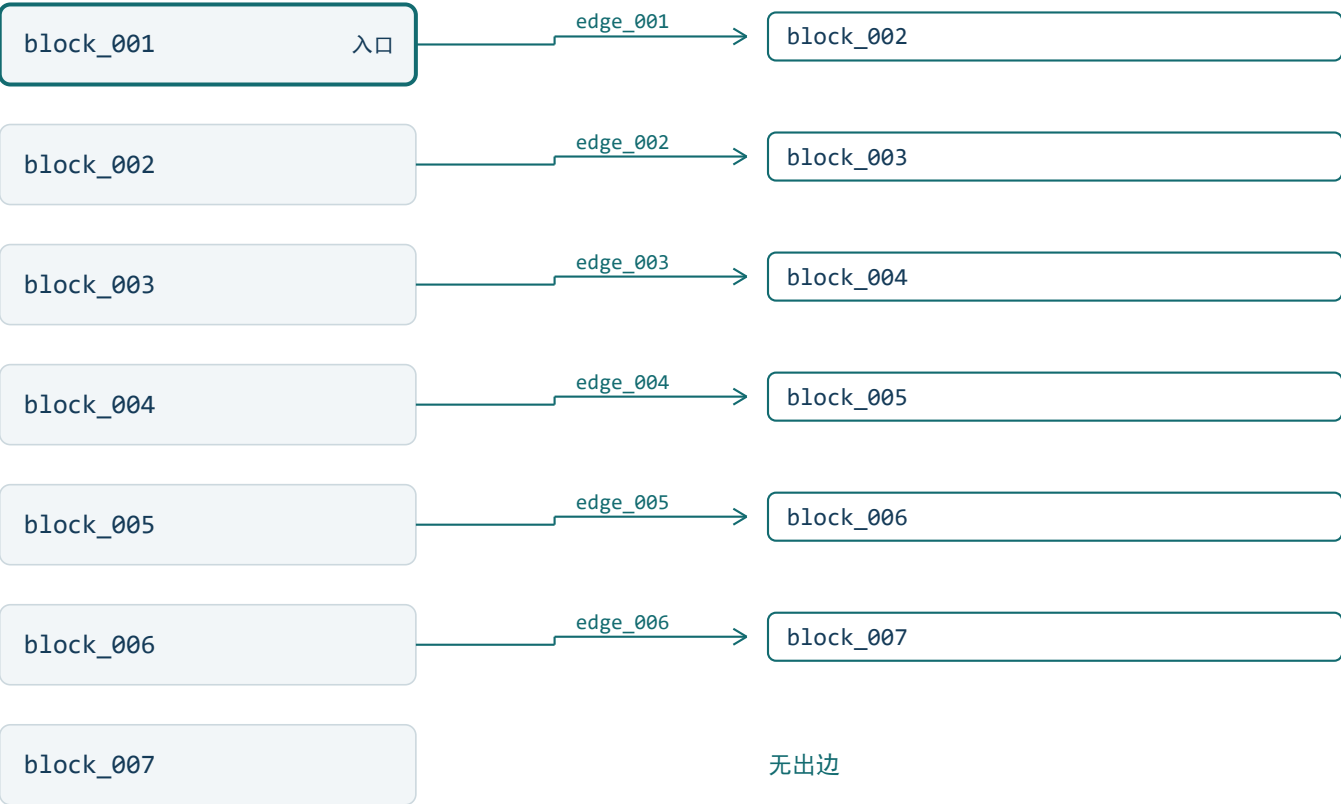
declared_context_keys = ["output_dir", "delivery_path", "archive_path"]

skill.constraints[0]: 整份流程不得修改输入原文文件。

skill.constraints[1]: 整份流程禁止把文档上传到外部服务。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null

全部操作、数据引用与约束落点

block_001 Read document paths from the user-provided manifest

```
data_source_kind = "external"
block_001.constraints: []
ir_001 read_document_paths_from_manifest
inputs[0] external_resource: manifest.json | manifest.json
outputs[0] result: result_001 | document_paths
ir_001.constraints: []
draft_instruction_id = read_manifest_paths
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_read_request_params
```

block_002 Read output directory and delivery path from the request

```
data_source_kind = "context"
block_002.constraints: []
ir_003 read_output_dir_and_delivery_path_from_request
inputs[0] context_key: output_dir | output_dir
inputs[1] context_key: delivery_path | delivery_path
outputs[0] result: result_002 | requested_output_dir
outputs[1] result: result_003 | requested_delivery_path
ir_003.constraints: []
draft_instruction_id = read_output_dir_and_delivery_path
```

ir_004 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = go_convert_documents
```

block_003 Convert each manifest document with scripts/convert.py and collect stdout paths

```
data_source_kind = "external"
```

```
block_003.constraints: []
```

ir_005 run_convert_script_for_each_document

```
inputs[0] result: result_001 | document_paths
```

```
inputs[1] result: result_002 | requested_output_dir
```

```
inputs[2] external_resource: scripts/convert.py | scripts/convert.py
```

```
outputs[0] result: result_004 | converted_artifact_paths
```

ir_005.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。

ir_005.constraints[1]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。

```
draft_instruction_id = convert_all_documents
```

metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。

```
metadata = {"script_path": "scripts/convert.py"}
```

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = go_package_documents
```

block_004 Build bundle.zip from converted artifact paths with scripts/package.py

```
data_source_kind = null
```

```
block_004.constraints: []
```

ir_007 run_package_script

```
inputs[0] result: result_004 | converted_artifact_paths
```

```
inputs[1] literal = "bundle.zip"
```

```
inputs[2] external_resource: scripts/package.py | scripts/package.py
```

```
outputs = []
```

ir_007.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 --paths。

```
draft_instruction_id = package_converted_documents
```

metadata.script_content: 457 字符；SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。

```
metadata = {"reference_command": "python scripts/package.py --paths <转换产物路径列表> --output bundle.zip", "script_path": "scripts/package.py"}
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_read_archive_path
```

block_005 Read the requested archive path from the request

```
data_source_kind = "context"
```

```
block_005.constraints: []
```

ir_009 read_archive_path_from_request

```
inputs[0] context_key: archive_path | archive_path
```

```
outputs[0] result: result_005 | requested_archive_path
```

```
ir_009.constraints: []
```

```
draft_instruction_id = read_archive_path
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_deliver_bundle
```

block_006 Deliver bundle.zip to the requested archive path

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 deliver_bundle_zip_to_archive_path

```
inputs[0] literal = "bundle.zip"
```

```
inputs[1] result: result_005 | requested_archive_path
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = deliver_bundle_zip
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_write_receipt
```

block_007 Write converted artifact paths to receipt.txt and finish

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 write_converted_artifact_paths_to_receipt

```
inputs[0] result: result_004 | converted_artifact_paths
```

```
inputs[1] external_resource: receipt.txt | receipt.txt
```

```
outputs = []
```



```
ir_013.constraints: []  
draft_instruction_id = write_receipt_file  
  
ir_014 return  
  
inputs = []  
outputs = []  
  
ir_014.constraints: []  
draft_instruction_id = return
```

D06 / 第 3 次 实际输出明细

多文档处理与交付 · 语义最小差异

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/D06/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: SKILL.md step 1 says to read delivery_path from the request, while step 6 says to deliver to the requested archive_path. Both context keys are declared and read; the delivery instruction uses archive_path because step 6 explicitly names it. delivery_path is read but not otherwise consumed.

candidate warning: The note '必要时保留原顺序' has unclear scope and is not promoted to a Skill-wide constraint.

Skill 层约束与上下文声明

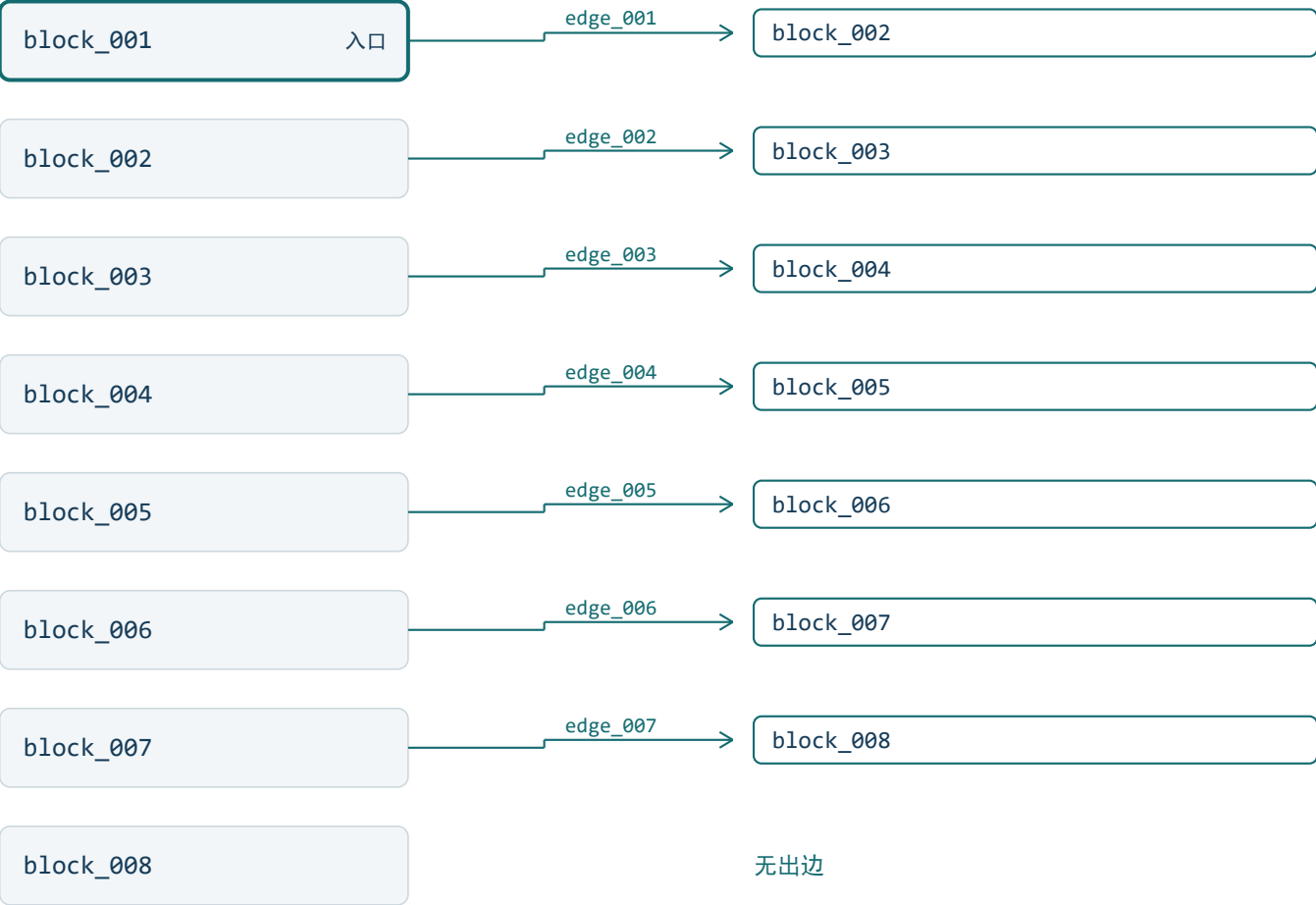
declared_context_keys = ["output_dir", "delivery_path", "archive_path"]

skill.constraints[0]: 整份流程不得修改输入原文文件。

skill.constraints[1]: 整份流程禁止把文档上传到外部服务。

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null
edge_007	block_007 -> block_008	null

全部操作、数据引用与约束落点

```
block_001 Read the user-provided manifest.json

data_source_kind = "external"
block_001.constraints: []

ir_001 read_manifest_file

inputs[0] external_resource: manifest.json | manifest.json
outputs[0] result: result_001 | manifest_contents
ir_001.constraints: []
draft_instruction_id = read_manifest_file

ir_002 dispatch

inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_parse_manifest

block_002 Extract document paths from the manifest paths list

data_source_kind = null
block_002.constraints: []

ir_003 extract_document_paths

inputs[0] result: result_001 | manifest_contents
outputs[0] result: result_002 | document_paths
ir_003.constraints: []
draft_instruction_id = extract_document_paths

ir_004 dispatch

inputs = []
outputs = []
ir_004.constraints: []
draft_instruction_id = go_read_request_paths
```

block_003 Read output_dir and delivery_path from the request

data_source_kind = "context"

block_003.constraints: []

ir_005 read_output_dir_and_delivery_path

inputs[0] context_key: output_dir | output_dir

inputs[1] context_key: delivery_path | delivery_path

outputs[0] result: result_003 | requested_output_dir

outputs[1] result: result_004 | requested_delivery_path

ir_005.constraints: []

draft_instruction_id = read_output_dir_and_delivery_path

ir_006 dispatch

inputs = []

outputs = []

ir_006.constraints: []

draft_instruction_id = go_convert_documents

block_004 Convert each document with scripts/convert.py and collect stdout paths

data_source_kind = "external"

block_004.constraints: []

ir_007 run_convert_script_for_each_document

inputs[0] external_resource: scripts/convert.py | scripts/convert.py

inputs[1] result: result_002 | document_paths

inputs[2] result: result_003 | requested_output_dir

outputs[0] result: result_005 | converted_product_paths

ir_007.constraints[0]: 转换期间保留原文标题；该要求仅限转换，不限制汇总标题。

ir_007.constraints[1]: 转换通过 scripts/convert.py 完成；主流程只使用脚本输出路径，不重复执行脚本内部的处理。

draft_instruction_id = run_convert_script_for_each_document

metadata.script_content: 337 字符；SHA256 60363d7a6cc5874be4cc9aa46ce415688ae9a5aee6bda6e09158ad15cae64808；计数与摘要依据：原文 UTF-8。完整值保留在本次 analysis.json；此处保留黑盒接口及全部数据引用。

metadata = {"script_path": "scripts/convert.py", "side_effects": ["reads each source UTF-8 text document", "creates output_dir if needed", "writes each converted text file", "returns each converted output path via stdout"]}

ir_008 dispatch

inputs = []

outputs = []

ir_008.constraints: []

draft_instruction_id = go_package_bundle

block_005 Package converted product paths into bundle.zip using scripts/package.py

data_source_kind = null

block_005.constraints: []

ir_009 run_package_script

inputs[0] external_resource: scripts/package.py | scripts/package.py

```
inputs[1] result: result_005 | converted_product_paths
```

```
inputs[2] literal = "bundle.zip"
```

```
outputs[0] result: result_006 | bundle_archive
```

ir_009.constraints[0]: 打包时必须采用下方“参考步骤”的命令，并将转换产物路径列表传给 `--paths`。

```
draft_instruction_id = run_package_script
```

metadata.script_content: 457 字符; SHA256 e991b2266ce205179dc2f57e311e151da812965f33722f448c8f7d5c52b5ce6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

```
metadata = {"reference_command": "python scripts/package.py --paths <转换产物路径列表> --output bundle.zip", "
script_path": "scripts/package.py", "side_effects": ["creates the requested ZIP archive at bundle.zip", "reads the
converted files listed in --paths"]}
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_read_archive_path
```

block_006 Read the requested archive_path from the request

```
data_source_kind = "context"
```

```
block_006.constraints: []
```

ir_011 read_requested_archive_path

```
inputs[0] context_key: archive_path | archive_path
```

```
outputs[0] result: result_007 | requested_archive_path
```

```
ir_011.constraints: []
```

```
draft_instruction_id = read_requested_archive_path
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_deliver_bundle
```

block_007 Deliver the generated bundle archive to the requested archive_path

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 deliver_bundle_to_archive_path

```
inputs[0] result: result_006 | bundle_archive
```

```
inputs[1] result: result_007 | requested_archive_path
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = deliver_bundle_to_archive_path
```

```
metadata = {"side_effects": ["delivers the generated bundle archive to requested archive_path"]}
```

ir_014 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = go_write_receipt
```

block_008 Write converted product paths to local receipt.txt and finish

```
data_source_kind = null
```

```
block_008.constraints: []
```

ir_015 write_converted_paths_receipt

```
inputs[0] result: result_005 | converted_product_paths
```

```
inputs[1] external_resource: receipt.txt | receipt.txt
```

```
outputs = []
```

```
ir_015.constraints: []
```

```
draft_instruction_id = write_converted_paths_receipt
```

```
metadata = {"side_effects": ["writes the converted product path list to local receipt.txt"]}
```

ir_016 return

```
inputs = []
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = return_done
```

R01 / 第 1 次 实际输出明细

pdf

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R01/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: The incoming PDF task type is inferred as context key 'pdf_task_type' from the 'When to use' section; the source does not explicitly name a runtime input key.

candidate warning: agents/openai.yaml contains interface metadata; its default_prompt informed the final summarize behavior but is not extracted as a separate executable block.

candidate warning: The dependency installation blocks form an independent component with its own starting block 'check_uv'.

Skill 层约束与上下文声明

```
declared_context_keys = ["pdf_task_type", "pdftoppm_available", "os_type", "uv_available"]
```

skill.constraints[0]: Use tmp/pdfs/ for intermediate files; delete when done.

skill.constraints[1]: Write final artifacts under output/pdf/ when working in this repo.

skill.constraints[2]: Keep filenames stable and descriptive.

skill.constraints[3]: Maintain polished visual design: consistent typography, spacing, margins, and section hierarchy.

skill.constraints[4]: Avoid rendering issues: clipped text, overlapping elements, broken tables, black squares, or unreadable glyphs.

skill.constraints[5]: Charts, tables, and images must be sharp, aligned, and clearly labeled.

skill.constraints[6]: Use ASCII hyphens only. Avoid U+2011 (non-breaking hyphen) and other Unicode dashes.

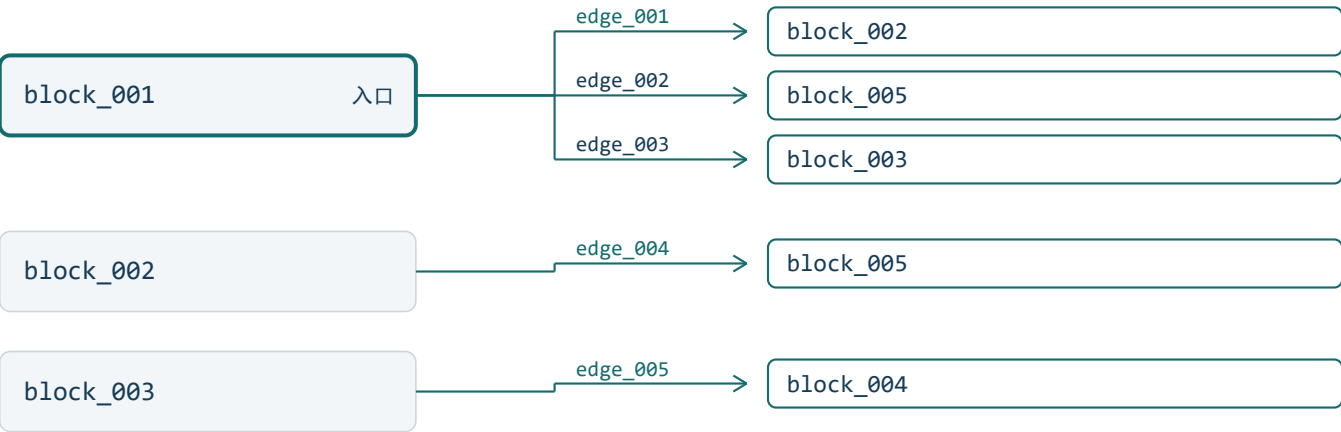
skill.constraints[7]: Citations and references must be human-readable; never leave tool tokens or placeholder strings.

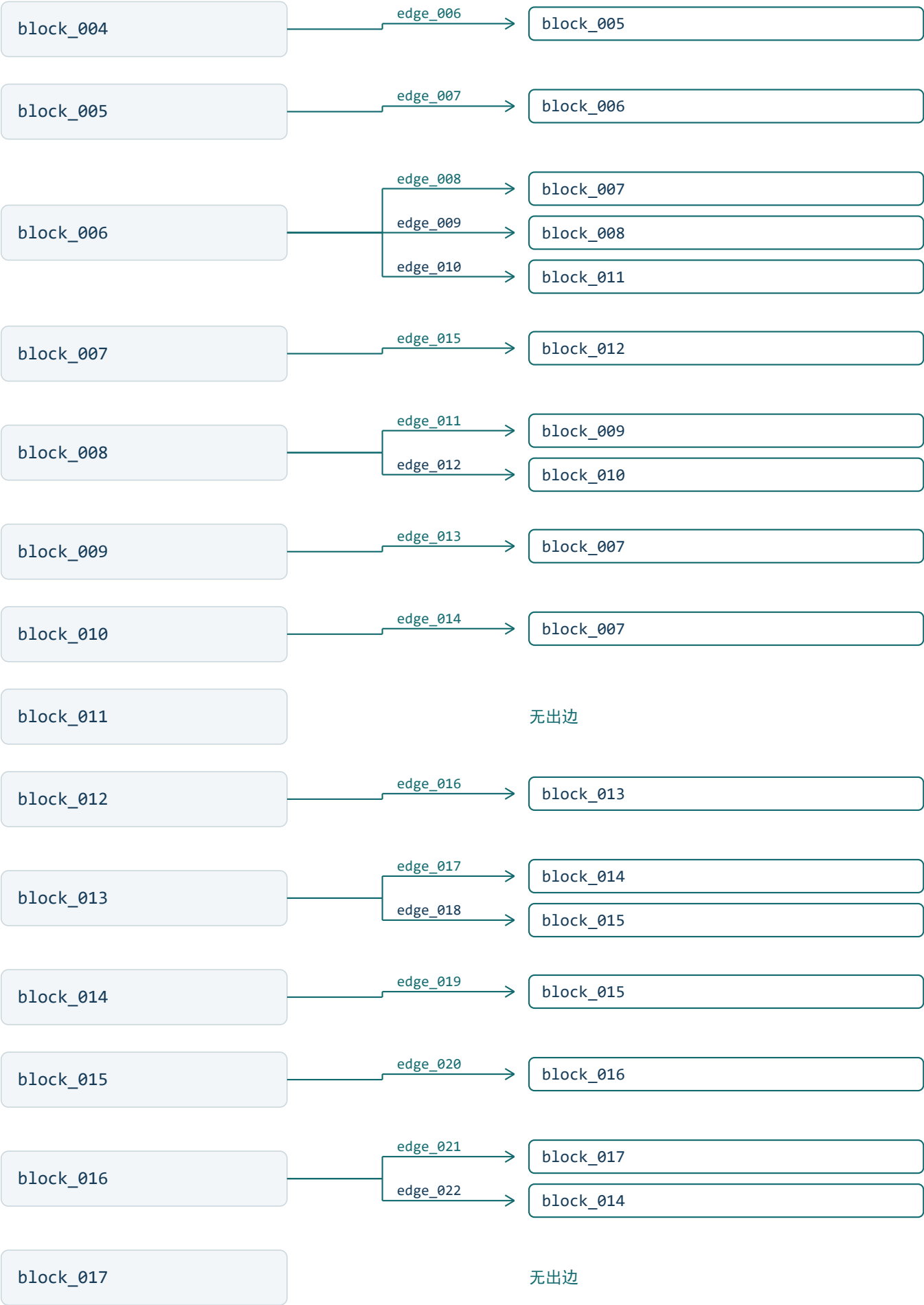
skill.constraints[8]: No required environment variables.

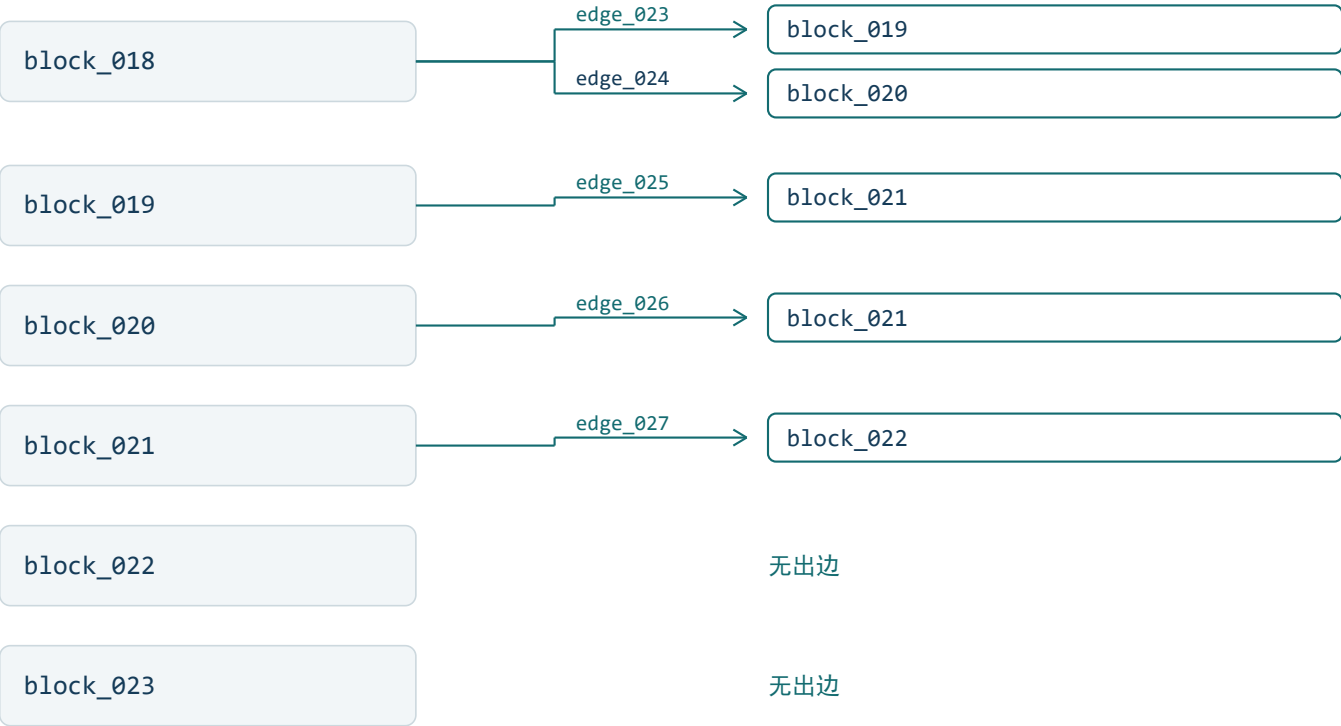
skill.constraints[9]: Keep intermediate files organized or remove them after final approval.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。







完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	"task is create"
edge_002	block_001 -> block_005	"task is read or review"
edge_003	block_001 -> block_003	"task is extract text"
edge_004	block_002 -> block_005	null
edge_005	block_003 -> block_004	null
edge_006	block_004 -> block_005	null
edge_007	block_005 -> block_006	null
edge_008	block_006 -> block_007	"pdftoppm available"
edge_009	block_006 -> block_008	"pdftoppm unavailable, install Poppler"
edge_010	block_006 -> block_011	"pdftoppm unavailable, ask user to review locally"
edge_011	block_008 -> block_009	"macOS"
edge_012	block_008 -> block_010	"Ubuntu or Debian"
edge_013	block_009 -> block_007	null
edge_014	block_010 -> block_007	null
edge_015	block_007 -> block_012	null
edge_016	block_012 -> block_013	null
edge_017	block_013 -> block_014	"visual defects present"
edge_018	block_013 -> block_015	"no visual defects"
edge_019	block_014 -> block_015	null
edge_020	block_015 -> block_016	null
edge_021	block_016 -> block_017	"final checks passed"
edge_022	block_016 -> block_014	"final checks not passed"
edge_023	block_018 -> block_019	"uv available"
edge_024	block_018 -> block_020	"uv unavailable"
edge_025	block_019 -> block_021	null
edge_026	block_020 -> block_021	null
edge_027	block_021 -> block_022	null

全部操作、数据引用与约束落点

block_001 Read the incoming PDF task type

data_source_kind = "context"

block_001.constraints: []

ir_001 read_pdf_task_type

inputs[0] context_key: pdf_task_type | pdf_task_type

outputs[0] result: result_001 | pdf_task_type_result

ir_001.constraints: []

draft_instruction_id = read_task

ir_002 dispatch

inputs[0] result: result_001 | pdf_task_type_result

outputs = []

ir_002.constraints: []

draft_instruction_id = dispatch_task

block_002 Generate a PDF with reportlab

data_source_kind = null

block_002.constraints[0]: Use reportlab to generate PDFs when creating new documents.

ir_003 generate_pdf_with_reportlab

inputs[0] external_resource: reportlab | reportlab

outputs[0] result: result_002 | generated_pdf

ir_003.constraints: []

draft_instruction_id = generate

ir_004 dispatch

inputs = []

outputs = []

ir_004.constraints: []

draft_instruction_id = dispatch_generate

block_003 Extract text from the PDF

data_source_kind = "external"

block_003.constraints[0]: Use pdfplumber (or pypdf) for text extraction and quick checks; do not rely on it for layout fidelity.

ir_005 extract_pdf_text

inputs[0] external_resource: pdfplumber | pdfplumber

inputs[1] external_resource: pypdf | pypdf

inputs[2] external_resource: input_pdf | input_pdf

outputs[0] result: result_003 | extracted_text

ir_005.constraints: []

draft_instruction_id = extract

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = dispatch_extract
```

block_004 Perform quick checks on the extracted text

```
data_source_kind = null
```

```
block_004.constraints: []
```

ir_007 perform_quick_text_checks

```
inputs[0] result: result_003 | extracted_text
```

```
outputs[0] result: result_004 | text_checks
```

```
ir_007.constraints: []
```

```
draft_instruction_id = quick_checks
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_quick_checks
```

block_005 Resolve the PDF to render

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 resolve_pdf_to_render

```
inputs[0] external_resource: input_pdf | input_pdf
```

```
inputs[1] result: result_002 | generated_pdf
```

```
outputs[0] result: result_005 | pdf_to_render
```

```
ir_009.constraints: []
```

```
draft_instruction_id = resolve
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_resolve
```

block_006 Check whether pdftoppm is available

```
data_source_kind = "context"
```

```
block_006.constraints[0]: Use pdftoppm if available.
```

ir_011 check_pdftoppm_availability

```
inputs[0] context_key: pdftoppm_available | pdftoppm_available
```

```
outputs[0] result: result_006 | pdftoppm_available_result
```

```
ir_011.constraints: []
```

```
draft_instruction_id = check_pdftoppm
```

ir_012 dispatch

```
inputs[0] result: result_006 | pdftoppm_available_result
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_pdftoppm
```

block_007 Render PDF pages to PNGs with pdftoppm

```
data_source_kind = "external"
```

```
block_007.constraints[0]: Prefer visual review: render PDF pages to PNGs and inspect them.
```

ir_013 render_pdf_pages_to_png

```
inputs[0] external_resource: pdftoppm | pdftoppm
```

```
inputs[1] result: result_005 | pdf_to_render
```

```
inputs[2] external_resource: output_prefix | output_prefix
```

```
outputs[0] result: result_007 | rendered_pngs
```

```
ir_013.constraints: []
```

```
draft_instruction_id = render
```

metadata.script_content: 39 字符; SHA256 79c02a9cfe88a2e47306257700712f22384532250bab1373082137d4bcf07d7c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_014 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = dispatch_render
```

block_008 Read the operating system for Poppler installation

```
data_source_kind = "context"
```

```
block_008.constraints: []
```

ir_015 read_operating_system

```
inputs[0] context_key: os_type | os_type
```

```
outputs[0] result: result_008 | os_type_result
```

```
ir_015.constraints: []
```

```
draft_instruction_id = read_os
```

ir_016 dispatch

```
inputs[0] result: result_008 | os_type_result
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = dispatch_os
```

block_009 Install Poppler with Homebrew

```
data_source_kind = null
```

```
block_009.constraints: []
```

ir_017 install_poppler_with_brew

```
inputs[0] external_resource: brew | brew
```

```
inputs[1] literal = "poppler"
```

```
outputs[0] result: result_009 | poppler_installed_macos
```

```
ir_017.constraints: []
```

```
draft_instruction_id = install_brew
```

metadata.script_content: 20 字符; SHA256 2c3a9cecc41fed843cad7f475da0b3a22a1f694f0631e585ce09402d19cfeb7b; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_018 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = dispatch_brew
```

block_010 Install Poppler with apt-get

```
data_source_kind = null
```

```
block_010.constraints: []
```

ir_019 install_poppler_with_apt

```
inputs[0] external_resource: apt-get | apt-get
```

```
inputs[1] literal = "poppler-utils"
```

```
outputs[0] result: result_010 | poppler_installed_linux
```

```
ir_019.constraints: []
```

```
draft_instruction_id = install_apt
```

metadata.script_content: 37 字符; SHA256 c62abd7e97e757bd3ff203cd92c2c01db1b1d14bad23af0531aaadbfaf972e90; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_020 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = dispatch_apt
```

block_011 Ask the user to review the output locally

```
data_source_kind = null
```

```
block_011.constraints: []
```

ir_021 ask_user_to_review_output_locally

```
inputs = []
```

```
outputs[0] result: result_011 | user_review_request
```

```
ir_021.constraints: []
```

```
draft_instruction_id = ask_user
```

ir_022 return

```
inputs[0] result: result_011 | user_review_request
```

```
outputs = []
```

```
ir_022.constraints: []
```

```
draft_instruction_id = return_ask
```

block_012 Inspect the rendered PNGs

```
data_source_kind = null
```

```
block_012.constraints: []
```

ir_023 inspect_rendered_pngs

```
inputs[0] result: result_007 | rendered_pngs
```

```
outputs[0] result: result_012 | inspection_findings
```

```
ir_023.constraints: []
```

```
draft_instruction_id = inspect
```

ir_024 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_024.constraints: []
```

```
draft_instruction_id = dispatch_inspect
```

block_013 Verify alignment, spacing, and legibility

```
data_source_kind = null
```

```
block_013.constraints[0]: After each meaningful update, re-render pages and verify alignment, spacing, and legibility.
```

ir_025 verify_visual_quality

```
inputs[0] result: result_012 | inspection_findings
```

```
outputs[0] result: result_013 | visual_defects_present
```

```
ir_025.constraints: []
```

```
draft_instruction_id = verify
```

ir_026 dispatch

```
inputs[0] result: result_013 | visual_defects_present
```

```
outputs = []
```

```
ir_026.constraints: []
```

```
draft_instruction_id = dispatch_verify
```

block_014 Apply update and re-render until visual quality is clean

```
data_source_kind = null
```

```
block_014.constraints[0]: After each meaningful update, re-render pages and verify alignment, spacing, and legibility.
```

ir_027 apply_meaningful_update_and_re_render

```
inputs[0] result: result_013 | visual_defects_present
```

```
inputs[1] result: result_005 | pdf_to_render
```

```
outputs[0] result: result_014 | final_inspection_findings
```

```
ir_027.constraints: []
```

```
draft_instruction_id = update_and_re_render
```

ir_028 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_028.constraints: []
```

```
draft_instruction_id = dispatch_update
```

block_015 Merge inspection findings from initial and updated renders

```
data_source_kind = null
```

```
block_015.constraints: []
```

ir_029 merge_inspection_findings

```
inputs[0] result: result_012 | inspection_findings
```

```
inputs[1] result: result_014 | final_inspection_findings
```

```
outputs[0] result: result_015 | latest_inspection_findings
```

```
ir_029.constraints: []
```

```
draft_instruction_id = merge
```

ir_030 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_030.constraints: []
```

```
draft_instruction_id = dispatch_merge
```

block_016 Confirm headers, footers, page numbering, and section transitions

```
data_source_kind = null
```

```
block_016.constraints[0]: Do not deliver until the latest PNG inspection shows zero visual or formatting defects.
```

```
block_016.constraints[1]: Confirm headers/footers, page numbering, and section transitions look polished.
```

ir_031 confirm_pdf_final_layout

```
inputs[0] result: result_015 | latest_inspection_findings
```

```
outputs[0] result: result_016 | final_checks_passed
```

```
ir_031.constraints: []
```

```
draft_instruction_id = confirm_layout
```

ir_032 dispatch

```
inputs[0] result: result_016 | final_checks_passed
```

```
outputs = []
```

```
ir_032.constraints: []
```

```
draft_instruction_id = dispatch_final_check
```


block_017 Deliver the final PDF and summarize the key output or changes

```
data_source_kind = null
```

```
block_017.constraints[0]: Keep intermediate files organized or remove them after final approval.
```

ir_033 summarize_key_output_or_changes

```
inputs[0] result: result_016 | final_checks_passed
```

```
inputs[1] result: result_015 | latest_inspection_findings
```

```
inputs[2] result: result_003 | extracted_text
```

```
inputs[3] result: result_004 | text_checks
```

```
outputs[0] result: result_017 | delivery_summary
```

```
ir_033.constraints: []
```

```
draft_instruction_id = summarize
```

ir_034 return

```
inputs[0] result: result_017 | delivery_summary
```

```
outputs = []
```

```
ir_034.constraints: []
```

```
draft_instruction_id = return_delivery
```

block_018 Check whether uv is available

```
data_source_kind = "context"
```

```
block_018.constraints[0]: Prefer uv for dependency management.
```

ir_035 check_uv_availability

```
inputs[0] context_key: uv_available | uv_available
```

```
outputs[0] result: result_018 | uv_available_result
```

```
ir_035.constraints: []
```

```
draft_instruction_id = check_uv
```

ir_036 dispatch

```
inputs[0] result: result_018 | uv_available_result
```

```
outputs = []
```

```
ir_036.constraints: []
```

```
draft_instruction_id = dispatch_uv
```

block_019 Install Python PDF packages with uv

```
data_source_kind = null
```

```
block_019.constraints: []
```

ir_037 install_python_packages_with_uv

```
inputs[0] external_resource: uv | uv
```

```
inputs[1] literal = "reportlab"
```

```
inputs[2] literal = "pdfplumber"
```

```
inputs[3] literal = "pypdf"
```

```
outputs[0] result: result_019 | python_packages_installed_with_uv
```

```
ir_037.constraints: []
```

```
draft_instruction_id = install_uv
```

metadata.script_content: 41 字符; SHA256 1a802bb05838a89ee97d7e99f50cc38fce36fbfeb2df71a4057d025837f5c8ad; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_038 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_038.constraints: []
```

```
draft_instruction_id = dispatch_install_uv
```

block_020 Install Python PDF packages with pip

```
data_source_kind = null
```

```
block_020.constraints: []
```

ir_039 install_python_packages_with_pip

```
inputs[0] external_resource: python3 | python3
```

```
inputs[1] literal = "reportlab"
```

```
inputs[2] literal = "pdfplumber"
```

```
inputs[3] literal = "pypdf"
```

```
outputs[0] result: result_020 | python_packages_installed_with_pip
```

```
ir_039.constraints: []
```

```
draft_instruction_id = install_pip
```

metadata.script_content: 49 字符; SHA256 aefc6265747e177c49719903453cfb18bd54baaba17cf07a2c618084b5ff4cc8; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_040 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_040.constraints: []
```

```
draft_instruction_id = dispatch_install_pip
```

block_021 Merge Python package installation results

```
data_source_kind = null
```

```
block_021.constraints: []
```

ir_041 merge_python_packages_installed

```
inputs[0] result: result_019 | python_packages_installed_with_uv
```

```
inputs[1] result: result_020 | python_packages_installed_with_pip
```

```
outputs[0] result: result_021 | python_packages_installed
```

```
ir_041.constraints: []
```

```
draft_instruction_id = merge_packages
```

ir_042 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_042.constraints: []
```

```
draft_instruction_id = dispatch_merge_packages
```

block_022 Return after installing Python packages

```
data_source_kind = null
```

```
block_022.constraints: []
```

ir_043 return

```
inputs[0] result: result_021 | python_packages_installed
```

```
outputs = []
```

```
ir_043.constraints: []
```

```
draft_instruction_id = return_packages
```

block_023 Tell the user which dependency is missing and how to install it locally

```
data_source_kind = null
```

```
block_023.constraints[0]: If installation isn't possible in this environment, tell the user which dependency is missing and how to install it locally.
```

ir_044 tell_user_missing_dependency_and_install_instructions

```
inputs[0] literal = "missing dependency"
```

```
outputs[0] result: result_022 | user_notified
```

```
ir_044.constraints: []
```

```
draft_instruction_id = tell_missing
```

ir_045 return

```
inputs[0] result: result_022 | user_notified
```

```
outputs = []
```

```
ir_045.constraints: []
```

```
draft_instruction_id = return_notified
```

R01 / 第 2 次 实际输出明细

pdf

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R01/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: LICENSE.txt contains license terms, not runtime behavior; ignored.

candidate warning: agents/openai.yaml contains interface metadata, not runtime behavior; ignored.

candidate warning: assets/pdf.png is binary and omitted; not analyzed.

candidate warning: SKILL.md states 'No required environment variables' but the rendering command references \$INPUT_PDF and \$OUTPUT_PREFIX; these are modeled as context keys for the rendering paths.

candidate warning: The dependency installation commands are represented as a single black-box block with script content preserved in metadata.

Skill 层约束与上下文声明

declared_context_keys = ["pdf_task", "INPUT_PDF", "OUTPUT_PREFIX"]

skill.constraints[0]: No required environment variables.

skill.constraints[1]: Use tmp/pdfs/ for intermediate files; delete when done.

skill.constraints[2]: Write final artifacts under output/pdf/ when working in this repo.

skill.constraints[3]: Keep filenames stable and descriptive.

skill.constraints[4]: Maintain polished visual design: consistent typography, spacing, margins, and section hierarchy.

skill.constraints[5]: Avoid rendering issues: clipped text, overlapping elements, broken tables, black squares, or unreadable glyphs.

skill.constraints[6]: Charts, tables, and images must be sharp, aligned, and clearly labeled.

skill.constraints[7]: Use ASCII hyphens only. Avoid U+2011 (non-breaking hyphen) and other Unicode dashes.

skill.constraints[8]: Citations and references must be human-readable; never leave tool tokens or placeholder strings.

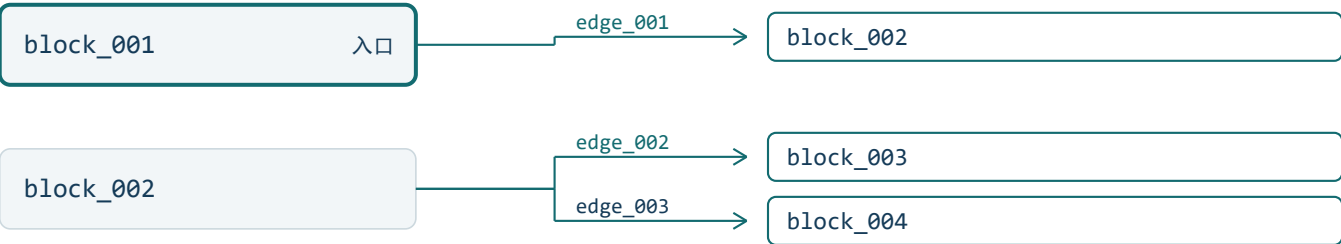
skill.constraints[9]: Do not deliver until the latest PNG inspection shows zero visual or formatting defects.

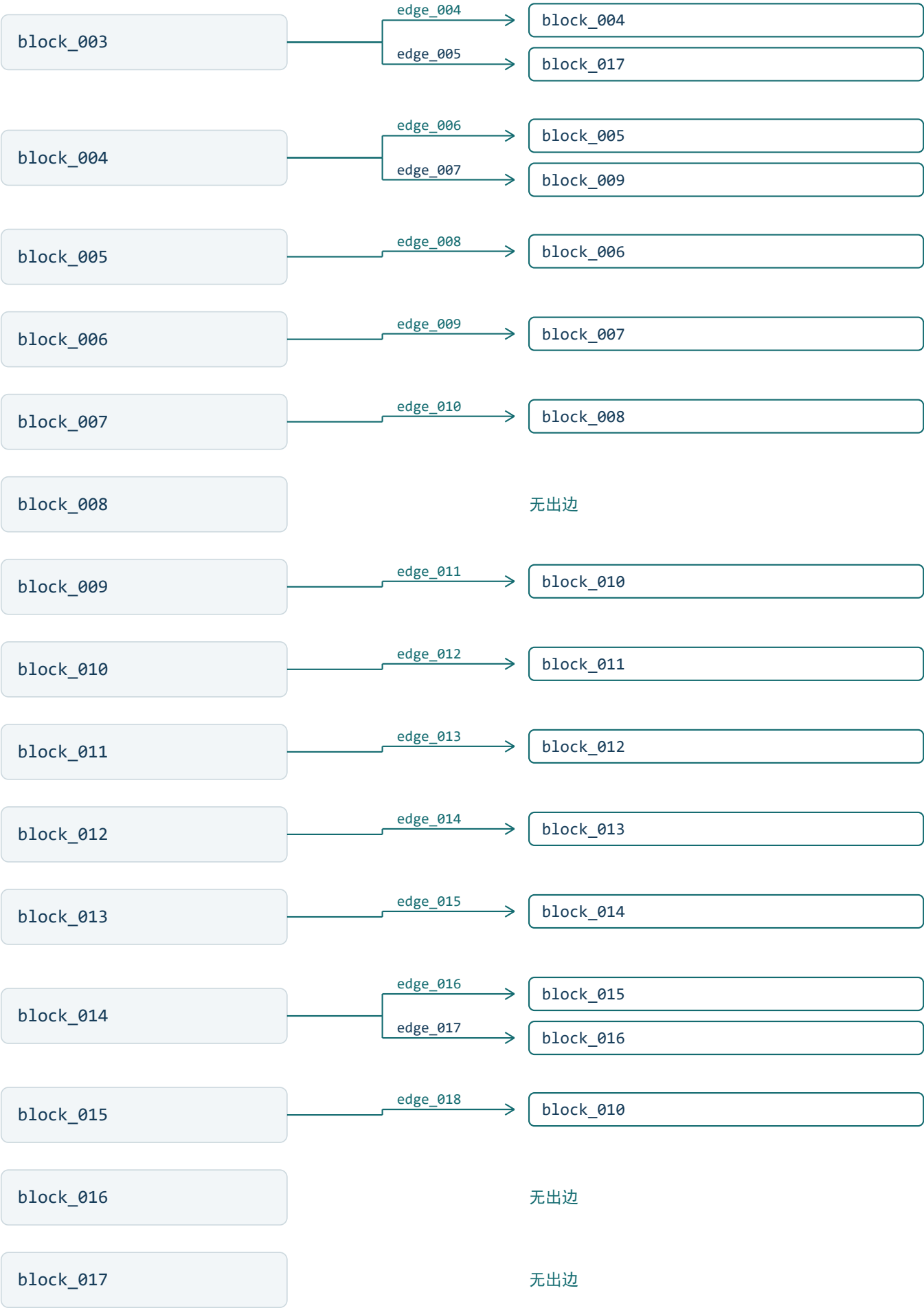
skill.constraints[10]: Confirm headers/footers, page numbering, and section transitions look polished.

skill.constraints[11]: Keep intermediate files organized or remove them after final approval.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"dependencies missing"
edge_003	block_002 -> block_004	"all dependencies available"
edge_004	block_003 -> block_004	"installation succeeded"
edge_005	block_003 -> block_017	"installation failed"
edge_006	block_004 -> block_005	"task involves reading or reviewing PDF content"
edge_007	block_004 -> block_009	"task involves creating or editing a PDF"
edge_008	block_005 -> block_006	null
edge_009	block_006 -> block_007	null
edge_010	block_007 -> block_008	null
edge_011	block_009 -> block_010	null
edge_012	block_010 -> block_011	null
edge_013	block_011 -> block_012	null
edge_014	block_012 -> block_013	null
edge_015	block_013 -> block_014	null
edge_016	block_014 -> block_015	"defects found"
edge_017	block_014 -> block_016	"no defects found"
edge_018	block_015 -> block_010	null

全部操作、数据引用与约束落点

block_001 Read the PDF task and rendering paths

data_source_kind = "context"

block_001.constraints: []

ir_001 read_pdf_task_and_render_paths

inputs[0] context_key: pdf_task | pdf_task

inputs[1] context_key: INPUT_PDF | INPUT_PDF

inputs[2] context_key: OUTPUT_PREFIX | OUTPUT_PREFIX

outputs[0] result: result_001 | pdf_task

outputs[1] result: result_002 | input_pdf_path

outputs[2] result: result_003 | output_prefix

ir_001.constraints: []

```
draft_instruction_id = read_pdf_task_context
```

ir_002 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_002.constraints: []
```

```
draft_instruction_id = go_check_dependencies
```

block_002 Check PDF dependency availability

```
data_source_kind = "external"
```

```
block_002.constraints: []
```

ir_003 check_pdf_dependencies_availability

```
inputs[0] external_resource: system | system
```

```
outputs[0] result: result_004 | dependencies_missing
```

```
outputs[1] result: result_005 | uv_available
```

```
ir_003.constraints: []
```

```
draft_instruction_id = check_pdf_deps
```

ir_004 dispatch

```
inputs[0] result: result_004 | dependencies_missing
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = dispatch_deps
```

block_003 Install missing PDF dependencies

```
data_source_kind = "external"
```

```
block_003.constraints: []
```

ir_005 install_missing_pdf_dependencies

```
inputs[0] external_resource: package_manager | package_manager
```

```
inputs[1] result: result_004 | dependencies_missing
```

```
inputs[2] result: result_005 | uv_available
```

```
outputs[0] result: result_006 | installation_result
```

```
ir_005.constraints: []
```

```
draft_instruction_id = install_deps
```

metadata.script_content: 150 字符; SHA256 bdb40149559834fd346dbc731ca397d18192f389a4e75a838fd97e568dce578e; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_006 dispatch

```
inputs[0] result: result_006 | installation_result
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = dispatch_install
```

block_004 Determine the PDF task type

```
data_source_kind = null
```

```
block_004.constraints: []
```

ir_007 determine_pdf_task_type

```
inputs[0] result: result_001 | pdf_task
```

```
outputs[0] result: result_007 | task_type
```

```
ir_007.constraints: []
```

```
draft_instruction_id = determine_task
```

ir_008 dispatch

```
inputs[0] result: result_007 | task_type
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_task
```

block_005 Render PDF pages to PNGs for review

```
data_source_kind = "external"
```

```
block_005.constraints: []
```

ir_009 render_pdf_pages_to_png

```
inputs[0] external_resource: pdftoppm | pdftoppm
```

```
inputs[1] result: result_002 | input_pdf_path
```

```
inputs[2] result: result_003 | output_prefix
```

```
inputs[3] literal = "-png"
```

```
outputs[0] result: result_008 | rendered_pngs_review
```

```
ir_009.constraints: []
```

```
draft_instruction_id = render_review_pdf
```

```
metadata.script_content: 39 字符; SHA256 79c02a9cfe88a2e47306257700712f22384532250bab1373082137d4bcf07d7c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_inspect_review
```

block_006 Inspect rendered PNGs for visual review

```
data_source_kind = "external"
```

```
block_006.constraints: []
```

ir_011 inspect_rendered_pngs

```
inputs[0] result: result_008 | rendered_pngs_review
```

```
outputs[0] result: result_009 | visual_inspection_review
```

```
ir_011.constraints: []
```

```
draft_instruction_id = inspect_review_pngs
```


ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_extract_review
```

block_007 Extract PDF text for quick checks

```
data_source_kind = "external"
```

```
block_007.constraints: []
```

ir_013 extract_pdf_text_with_pdfplumber_or_pypdf

```
inputs[0] external_resource: pdfplumber_or_pypdf | pdfplumber_or_pypdf
```

```
inputs[1] result: result_002 | input_pdf_path
```

```
outputs[0] result: result_010 | extracted_text_review
```

```
ir_013.constraints: []
```

```
draft_instruction_id = extract_review_text
```

ir_014 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = go_final_review
```

block_008 Perform final checks and return review result

```
data_source_kind = null
```

```
block_008.constraints: []
```

ir_015 perform_final_pdf_checks

```
inputs[0] result: result_009 | visual_inspection_review
```

```
inputs[1] result: result_010 | extracted_text_review
```

```
outputs[0] result: result_011 | final_validation_review
```

```
ir_015.constraints: []
```

```
draft_instruction_id = final_checks_review
```

ir_016 return

```
inputs[0] result: result_011 | final_validation_review
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = return_review
```

block_009 Generate a new PDF with reportlab

```
data_source_kind = "external"
```

```
block_009.constraints: []
```

ir_017 generate_pdf_with_reportlab

```
inputs[0] external_resource: reportlab | reportlab
```

```
inputs[1] result: result_001 | pdf_task
```

```
outputs[0] result: result_012 | generated_pdf_path
```

```
ir_017.constraints: []
```

```
draft_instruction_id = generate_pdf_reportlab
```

ir_018 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = go_select_pdf
```

block_010 Select the current PDF version for validation

```
data_source_kind = null
```

```
block_010.constraints: []
```

ir_019 select_current_pdf_version

```
inputs[0] result: result_012 | generated_pdf_path
```

```
inputs[1] result: result_018 | updated_pdf_path
```

```
outputs[0] result: result_013 | current_pdf_path
```

```
ir_019.constraints: []
```

```
draft_instruction_id = select_current_pdf
```

ir_020 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = go_extract_current
```

block_011 Extract text from the current PDF for quick checks

```
data_source_kind = "external"
```

```
block_011.constraints: []
```

ir_021 extract_pdf_text_with_pdfplumber_or_pypdf

```
inputs[0] external_resource: pdfplumber_or_pypdf | pdfplumber_or_pypdf
```

```
inputs[1] result: result_013 | current_pdf_path
```

```
outputs[0] result: result_014 | extracted_text_current
```

```
ir_021.constraints: []
```

```
draft_instruction_id = extract_current_text
```

ir_022 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_022.constraints: []
```

```
draft_instruction_id = go_render_current
```

block_012 Render the current PDF pages to PNGs

```
data_source_kind = "external"
```

```
block_012.constraints: []
```

ir_023 render_pdf_pages_to_png

```
inputs[0] external_resource: pdftoppm | pdftoppm
```

```
inputs[1] result: result_013 | current_pdf_path
```

```
inputs[2] result: result_003 | output_prefix
```

```
inputs[3] literal = "-png"
```

```
outputs[0] result: result_015 | rendered_pngs_current
```

```
ir_023.constraints: []
```

```
draft_instruction_id = render_current_pdf
```

metadata.script_content: 39 字符; SHA256 79c02a9cfe88a2e47306257700712f22384532250bab1373082137d4bcf07d7c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_024 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_024.constraints: []
```

```
draft_instruction_id = go_inspect_current
```

block_013 Inspect rendered PNGs of the current PDF

```
data_source_kind = "external"
```

```
block_013.constraints: []
```

ir_025 inspect_rendered_pngs

```
inputs[0] result: result_015 | rendered_pngs_current
```

```
outputs[0] result: result_016 | visual_inspection_current
```

```
ir_025.constraints: []
```

```
draft_instruction_id = inspect_current_pngs
```

ir_026 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_026.constraints: []
```

```
draft_instruction_id = go_check_defects
```

block_014 Check for visual and formatting defects

```
data_source_kind = null
```

```
block_014.constraints: []
```

ir_027 check_visual_and_formatting_defects

```
inputs[0] result: result_016 | visual_inspection_current
```

```
outputs[0] result: result_017 | defects_found
```

```
ir_027.constraints: []
```

```
draft_instruction_id = check_defects
```

ir_028 dispatch

```
inputs[0] result: result_017 | defects_found
```

```
outputs = []
```

```
ir_028.constraints: []
```

```
draft_instruction_id = dispatch_defects
```

block_015 Update the PDF after defects are found

```
data_source_kind = null
```

```
block_015.constraints: []
```

ir_029 update_pdf_after_defects

```
inputs[0] result: result_013 | current_pdf_path
```

```
inputs[1] result: result_017 | defects_found
```

```
outputs[0] result: result_018 | updated_pdf_path
```

```
ir_029.constraints: []
```

```
draft_instruction_id = update_pdf
```

ir_030 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_030.constraints: []
```

```
draft_instruction_id = go_select_again
```

block_016 Perform final checks and return generated PDF

```
data_source_kind = null
```

```
block_016.constraints: []
```

ir_031 perform_final_pdf_checks

```
inputs[0] result: result_016 | visual_inspection_current
```

```
inputs[1] result: result_014 | extracted_text_current
```

```
outputs[0] result: result_019 | final_validation_create
```

```
ir_031.constraints: []
```

```
draft_instruction_id = final_checks_create
```

ir_032 return

```
inputs[0] result: result_019 | final_validation_create
```

```
outputs = []
```

```
ir_032.constraints: []
```

```
draft_instruction_id = return_create
```

block_017 Ask the user to review the PDF locally

```
data_source_kind = null
```

```
block_017.constraints: []
```

ir_033 ask_user_to_review_pdf_locally

```
inputs[0] result: result_002 | input_pdf_path
```

```
outputs = []
```

```
ir_033.constraints: []  
draft_instruction_id = ask_user_review  
  
ir_034 return  
  
inputs = []  
outputs = []  
  
ir_034.constraints: []  
draft_instruction_id = return_user_review
```

R01 / 第 3 次 实际输出明细

pdf

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R01/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: The context key 'task_type' is inferred from the skill description's task categories (reading, creating, reviewing, validating); the source does not explicitly name a runtime context key.

candidate warning: The input PDF is modeled as an external resource 'input_pdf' because the source does not declare it as a context key.

candidate warning: The loop backedge references 'updated_rendered_pngs' produced by 're_render_pdf_pages'; this is intentional and allowed for loop-carried values.

candidate warning: The exact message for missing dependency notification is not specified in the source; a placeholder literal is used.

candidate warning: The YAML file contains only interface metadata and no executable behavior.

candidate warning: The LICENSE and binary asset are non-executable and ignored.

Skill 层约束与上下文声明

```
declared_context_keys = ["task_type"]
```

skill.constraints[0]: Prefer uv for dependency management.

skill.constraints[1]: Use tmp/pdfs/ for intermediate files; delete when done.

skill.constraints[2]: Write final artifacts under output/pdf/ when working in this repo.

skill.constraints[3]: Keep filenames stable and descriptive.

skill.constraints[4]: No required environment variables.

skill.constraints[5]: Maintain polished visual design: consistent typography, spacing, margins, and section hierarchy.

skill.constraints[6]: Avoid rendering issues: clipped text, overlapping elements, broken tables, black squares, or unreadable glyphs.

skill.constraints[7]: Charts, tables, and images must be sharp, aligned, and clearly labeled.

skill.constraints[8]: Use ASCII hyphens only. Avoid U+2011 (non-breaking hyphen) and other Unicode dashes.

skill.constraints[9]: Citations and references must be human-readable; never leave tool tokens or placeholder strings.

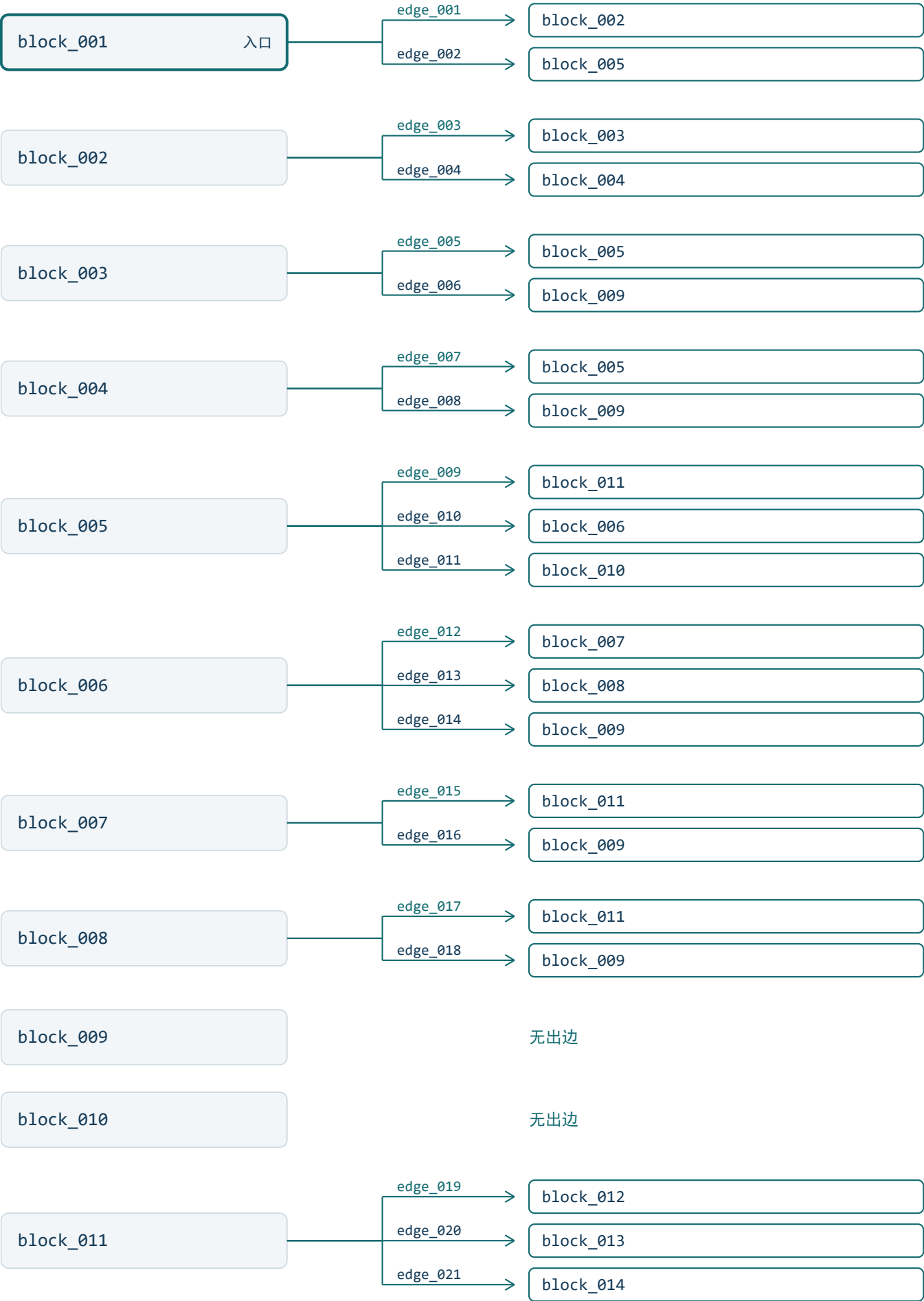
skill.constraints[10]: Do not deliver until the latest PNG inspection shows zero visual or formatting defects.

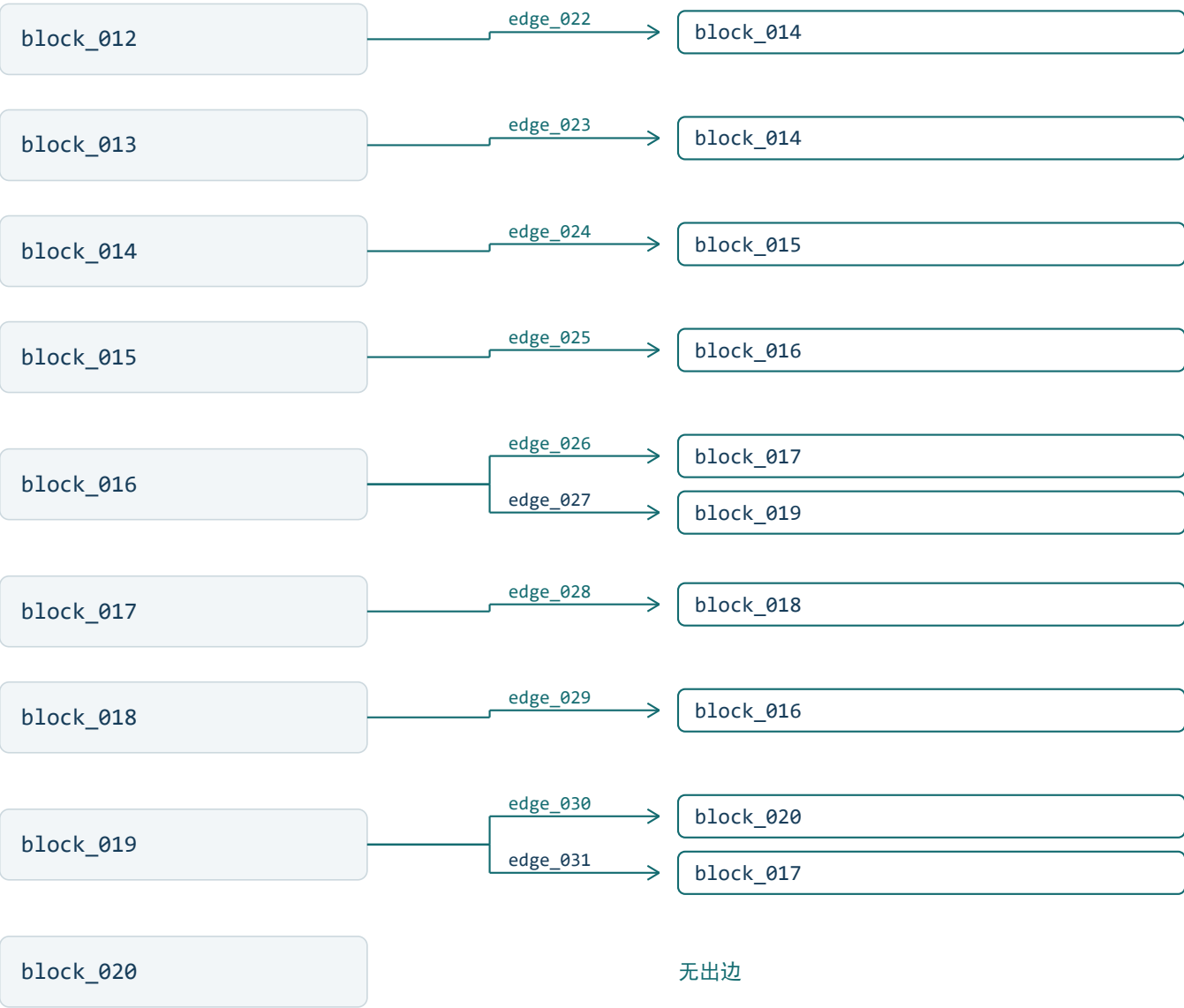
skill.constraints[11]: Confirm headers/footers, page numbering, and section transitions look polished.

skill.constraints[12]: Keep intermediate files organized or remove them after final approval.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	"missing_python_packages is non-empty"
edge_002	block_001 -> block_005	"missing_python_packages is empty"
edge_003	block_002 -> block_003	"uv_available is true"
edge_004	block_002 -> block_004	"uv_available is false"
edge_005	block_003 -> block_005	"python_deps_installed_uv is true"
edge_006	block_003 -> block_009	"python_deps_installed_uv is false"
edge_007	block_004 -> block_005	"python_deps_installed_pip is true"
edge_008	block_004 -> block_009	"python_deps_installed_pip is false"
edge_009	block_005 -> block_011	"pdftoppm_available is true"
edge_010	block_005 -> block_006	"pdftoppm_available is false; install Poppler"
edge_011	block_005 -> block_010	"pdftoppm_available is false; ask user to review locally"
edge_012	block_006 -> block_007	"os_type is macOS"
edge_013	block_006 -> block_008	"os_type is Ubuntu/Debian"
edge_014	block_006 -> block_009	"os_type is unsupported"
edge_015	block_007 -> block_011	"poppler_installed_macos is true"
edge_016	block_007 -> block_009	"poppler_installed_macos is false"
edge_017	block_008 -> block_011	"poppler_installed_ubuntu is true"
edge_018	block_008 -> block_009	"poppler_installed_ubuntu is false"
edge_019	block_011 -> block_012	"task_type == 'create'"
edge_020	block_011 -> block_013	"task_type == 'extract'"
edge_021	block_011 -> block_014	"task_type == 'review' or 'validate'"
edge_022	block_012 -> block_014	null
edge_023	block_013 -> block_014	null
edge_024	block_014 -> block_015	null
edge_025	block_015 -> block_016	null
edge_026	block_016 -> block_017	"defects_found is true"
edge_027	block_016 -> block_019	"defects_found is false"

边标识	源块 -> 目标块	condition_text
edge_028	block_017 -> block_018	null
edge_029	block_018 -> block_016	null
edge_030	block_019 -> block_020	"final_checks_passed is true"
edge_031	block_019 -> block_017	"final_checks_passed is false"

全部操作、数据引用与约束落点

block_001 Check whether required Python packages are installed

```
data_source_kind = "external"

block_001.constraints: []

ir_001 check_python_packages_installed

inputs[0] external_resource: python | python
inputs[1] literal = "reportlab"
inputs[2] literal = "pdfplumber"
inputs[3] literal = "pypdf"

outputs[0] result: result_001 | missing_python_packages

ir_001.constraints: []

draft_instruction_id = check_python_deps_inst
```

ir_002 dispatch

```
inputs[0] result: result_001 | missing_python_packages

outputs = []

ir_002.constraints: []

draft_instruction_id = dispatch_check_python_deps
```

block_002 Check if uv is available

```
data_source_kind = "external"

block_002.constraints: []

ir_003 check_uv_available

inputs[0] external_resource: uv | uv

outputs[0] result: result_002 | uv_available

ir_003.constraints: []

draft_instruction_id = check_uv_inst
```

ir_004 dispatch

```
inputs[0] result: result_002 | uv_available

outputs = []

ir_004.constraints: []

draft_instruction_id = dispatch_check_uv
```

block_003 Install Python packages using uv

```
data_source_kind = null
```

```
block_003.constraints[0]: Prefer uv for dependency management.
```

ir_005 uv_pip_install

```
inputs[0] external_resource: uv | uv
```

```
inputs[1] literal = "reportlab"
```

```
inputs[2] literal = "pdfplumber"
```

```
inputs[3] literal = "pypdf"
```

```
outputs[0] result: result_003 | python_deps_installed_uv
```

```
ir_005.constraints: []
```

```
draft_instruction_id = install_python_uv_inst
```

metadata.script_content: 41 字符; SHA256 1a802bb05838a89ee97d7e99f50cc38fce36fbfeb2df71a4057d025837f5c8ad; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_006 dispatch

```
inputs[0] result: result_003 | python_deps_installed_uv
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = dispatch_install_python_uv
```

block_004 Install Python packages using python3 -m pip

```
data_source_kind = null
```

```
block_004.constraints[0]: Use python3 -m pip if uv is unavailable.
```

ir_007 python3_m_pip_install

```
inputs[0] external_resource: python3 | python3
```

```
inputs[1] literal = "reportlab"
```

```
inputs[2] literal = "pdfplumber"
```

```
inputs[3] literal = "pypdf"
```

```
outputs[0] result: result_004 | python_deps_installed_pip
```

```
ir_007.constraints: []
```

```
draft_instruction_id = install_python_pip_inst
```

metadata.script_content: 49 字符; SHA256 aefc6265747e177c49719903453cfb18bd54baaba17cf07a2c618084b5ff4cc8; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_008 dispatch

```
inputs[0] result: result_004 | python_deps_installed_pip
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_install_python_pip
```

block_005 Check if pdftoppm is available

```
data_source_kind = "external"
```

```
block_005.constraints: []
```

ir_009 check_pdftoppm_available

```
inputs[0] external_resource: pdftoppm | pdftoppm
```

```
outputs[0] result: result_005 | pdftoppm_available
```

```
ir_009.constraints: []
```

```
draft_instruction_id = check_pdftoppm_inst
```

ir_010 dispatch

```
inputs[0] result: result_005 | pdftoppm_available
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_check_pdftoppm
```

block_006 Determine operating system for Poppler installation

```
data_source_kind = "external"
```

```
block_006.constraints: []
```

ir_011 check_operating_system

```
inputs[0] external_resource: os | os
```

```
outputs[0] result: result_006 | os_type
```

```
ir_011.constraints: []
```

```
draft_instruction_id = check_os_inst
```

ir_012 dispatch

```
inputs[0] result: result_006 | os_type
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_check_os
```

block_007 Install Poppler on macOS using Homebrew

```
data_source_kind = null
```

```
block_007.constraints[0]: On macOS, use Homebrew to install Poppler.
```

ir_013 brew_install_poppler

```
inputs[0] external_resource: brew | brew
```

```
outputs[0] result: result_007 | poppler_installed_macos
```

```
ir_013.constraints: []
```

```
draft_instruction_id = install_poppler_macos_inst
```

```
metadata.script_content: 39 字符; SHA256 26efaa1a95bb08c538bad0d276eba5bac9dbb68e348a8a0e11398c632e52f83a; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。
```

ir_014 dispatch

inputs[0] result: result_007 | poppler_installed_macos

outputs = []

ir_014.constraints: []

draft_instruction_id = dispatch_install_poppler_macos

block_008 Install Poppler on Ubuntu/Debian using apt-get

data_source_kind = null

block_008.constraints[0]: On Ubuntu/Debian, use apt-get to install poppler-utils.

ir_015 apt_get_install_poppler_utils

inputs[0] external_resource: apt-get | apt-get

outputs[0] result: result_008 | poppler_installed_ubuntu

ir_015.constraints: []

draft_instruction_id = install_poppler_ubuntu_inst

metadata.script_content: 53 字符; SHA256 a66364a67b042361093b99a6c633c6244f4898cc77301c35217ee8580502d8dd; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_016 dispatch

inputs[0] result: result_008 | poppler_installed_ubuntu

outputs = []

ir_016.constraints: []

draft_instruction_id = dispatch_install_poppler_ubuntu

block_009 Tell the user which dependency is missing and how to install it

data_source_kind = null

block_009.constraints[0]: If installation isn't possible in this environment, tell the user which dependency is missing and how to install it locally.

ir_017 tell_user_missing_dependency

inputs[0] literal = "Missing dependency. Please install it locally."

outputs = []

ir_017.constraints: []

draft_instruction_id = notify_missing_dependency_inst

ir_018 return

inputs = []

outputs = []

ir_018.constraints: []

draft_instruction_id = return_notify_missing_dependency

block_010 Ask the user to review the output locally

data_source_kind = null

block_010.constraints[0]: If pdftoppm is unavailable, ask the user to review the output locally.

ir_019 ask_user_to_review_locally

inputs[0] literal = "Please review the PDF output locally."

outputs = []

```
ir_019.constraints: []  
draft_instruction_id = ask_user_to_review_inst  
ir_020 return  
  
inputs = []  
outputs = []  
ir_020.constraints: []  
draft_instruction_id = return_ask_user  
block_011 Read the PDF task request type  
  
data_source_kind = "context"  
block_011.constraints: []  
ir_021 read_task_type  
  
inputs[0] context_key: task_type | task_type  
outputs[0] result: result_009 | task_type  
ir_021.constraints: []  
draft_instruction_id = read_task_type_inst  
ir_022 dispatch  
  
inputs[0] result: result_009 | task_type  
outputs = []  
ir_022.constraints: []  
draft_instruction_id = dispatch_read_task_type  
block_012 Generate a PDF using reportlab  
  
data_source_kind = null  
block_012.constraints[0]: Use reportlab to generate PDFs when creating new documents.  
ir_023 generate_pdf_with_reportlab  
  
inputs[0] external_resource: reportlab | reportlab  
outputs[0] result: result_010 | generated_pdf  
ir_023.constraints: []  
draft_instruction_id = generate_pdf_inst  
ir_024 dispatch  
  
inputs = []  
outputs = []  
ir_024.constraints: []  
draft_instruction_id = dispatch_generate_pdf  
block_013 Extract text from the PDF using pdfplumber or pypdf  
  
data_source_kind = "external"  
block_013.constraints[0]: Use pdfplumber or pypdf for text extraction and quick checks.  
block_013.constraints[1]: Do not rely on pdfplumber or pypdf for layout fidelity.  
ir_025 extract_pdf_text  
  
inputs[0] external_resource: pdfplumber | pdfplumber
```

```
inputs[1] external_resource: input_pdf | input_pdf
```

```
outputs[0] result: result_011 | extracted_text
```

```
ir_025.constraints: []
```

```
draft_instruction_id = extract_pdf_text_inst
```

ir_026 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_026.constraints: []
```

```
draft_instruction_id = dispatch_extract_text
```

block_014 Select the PDF to render for visual review

```
data_source_kind = null
```

```
block_014.constraints: []
```

ir_027 select_pdf_for_rendering

```
inputs[0] result: result_010 | generated_pdf
```

```
inputs[1] external_resource: input_pdf | input_pdf
```

```
outputs[0] result: result_012 | pdf_to_render
```

```
ir_027.constraints: []
```

```
draft_instruction_id = select_pdf_inst
```

ir_028 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_028.constraints: []
```

```
draft_instruction_id = dispatch_select_pdf
```

block_015 Render the PDF pages to PNGs

```
data_source_kind = "external"
```

```
block_015.constraints[0]: Use pdftoppm if available.
```

ir_029 pdftoppm_render_pdf

```
inputs[0] external_resource: pdftoppm | pdftoppm
```

```
inputs[1] result: result_012 | pdf_to_render
```

```
outputs[0] result: result_013 | rendered_pngs
```

```
ir_029.constraints: []
```

```
draft_instruction_id = render_pdf_pages_inst
```

metadata.script_content: 39 字符; SHA256 79c02a9cfe88a2e47306257700712f22384532250bab1373082137d4bcf07d7c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_030 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_030.constraints: []
```

```
draft_instruction_id = dispatch_render_pdf
```

block_016 Inspect rendered PNGs for visual and formatting defects

```
data_source_kind = null
```

```
block_016.constraints[0]: After each meaningful update, re-render pages and verify alignment, spacing, and legibility.
```

ir_031 inspect_pdf_pages

```
inputs[0] result: result_013 | rendered_pngs
```

```
inputs[1] result: result_016 | updated_rendered_pngs
```

```
outputs[0] result: result_014 | defects_found
```

```
ir_031.constraints: []
```

```
draft_instruction_id = inspect_pdf_pages_inst
```

ir_032 dispatch

```
inputs[0] result: result_014 | defects_found
```

```
outputs = []
```

```
ir_032.constraints: []
```

```
draft_instruction_id = dispatch_inspect_pdf
```

block_017 Update the PDF to fix visual or formatting defects

```
data_source_kind = null
```

```
block_017.constraints: []
```

ir_033 update_pdf_content

```
inputs[0] result: result_012 | pdf_to_render
```

```
inputs[1] result: result_014 | defects_found
```

```
outputs[0] result: result_015 | updated_pdf
```

```
ir_033.constraints: []
```

```
draft_instruction_id = update_pdf_inst
```

ir_034 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_034.constraints: []
```

```
draft_instruction_id = dispatch_update_pdf
```

block_018 Re-render the updated PDF pages to PNGs

```
data_source_kind = "external"
```

```
block_018.constraints: []
```

ir_035 pdftoppm_render_pdf

```
inputs[0] external_resource: pdftoppm | pdftoppm
```

```
inputs[1] result: result_015 | updated_pdf
```

```
outputs[0] result: result_016 | updated_rendered_pngs
```

```
ir_035.constraints: []
```

```
draft_instruction_id = re_render_pdf_pages_inst
```

metadata.script_content: 39 字符; SHA256 79c02a9cfe88a2e47306257700712f22384532250bab1373082137d4bcf07d7c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_036 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_036.constraints: []
```

```
draft_instruction_id = dispatch_re_render_pdf
```

block_019 Perform final checks on headers, footers, and page transitions

```
data_source_kind = null
```

```
block_019.constraints[0]: Do not deliver until the latest PNG inspection shows zero visual or formatting defects.
```

```
block_019.constraints[1]: Confirm headers/footers, page numbering, and section transitions look polished.
```

ir_037 final_pdf_checks

```
inputs[0] result: result_013 | rendered_pngs
```

```
inputs[1] result: result_016 | updated_rendered_pngs
```

```
outputs[0] result: result_017 | final_checks_passed
```

```
ir_037.constraints: []
```

```
draft_instruction_id = final_checks_inst
```

ir_038 dispatch

```
inputs[0] result: result_017 | final_checks_passed
```

```
outputs = []
```

```
ir_038.constraints: []
```

```
draft_instruction_id = dispatch_final_checks
```

block_020 Delete intermediate files under tmp/pdfs/

```
data_source_kind = null
```

```
block_020.constraints[0]: Use tmp/pdfs/ for intermediate files; delete when done.
```

```
block_020.constraints[1]: Keep intermediate files organized or remove them after final approval.
```

ir_039 cleanup_intermediate_files

```
inputs[0] literal = "tmp/pdfs/"
```

```
outputs = []
```

```
ir_039.constraints: []
```

```
draft_instruction_id = cleanup_inst
```

ir_040 return

```
inputs = []
```

```
outputs = []
```

```
ir_040.constraints: []
```

```
draft_instruction_id = return_cleanup
```

R02 / 第 1 次 实际输出明细

playwright

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R02/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: LICENSE.txt and NOTICE.txt are non-executable legal/notice files and were not modeled as runtime behavior.

candidate warning: agents/openai.yaml contains interface metadata only and was not modeled as runtime behavior.

candidate warning: references/cli.md and references/workflows.md are reference documentation and examples; their commands were not extracted as independent runtime instructions, except where they constrain the core workflow.

candidate warning: The wrapper script is modeled as a black-box component with its original script_content preserved; its internal prerequisite check and session-flag loop are not split into separate control-flow blocks.

candidate warning: The target page URL is modeled as a runtime context input to the skill, since the core workflow refers to 'the page' without specifying its source.

Skill 层约束与上下文声明

declared_context_keys = ["PATH", "CODEX_HOME", "HOME", "target_url", "command_line_args", "PLAYWRIGHT_CLI_SESSION"]

skill.constraints[0]: Drive a real browser from the terminal using `playwright-cli`. Prefer the bundled wrapper script so the CLI works even when it is not globally installed.

skill.constraints[1]: Treat this skill as CLI-first automation. Do not pivot to `@playwright/test` unless the user explicitly asks for test files.

skill.constraints[2]: Once `npx` is present, proceed with the wrapper script. A global install of `playwright-cli` is optional.

skill.constraints[3]: User-scoped skills install under `\$CODEX\_HOME/skills` (default: `~/codex/skills`).

skill.constraints[4]: Always snapshot before referencing element ids like `e12`.

skill.constraints[5]: Re-snapshot when refs seem stale.

skill.constraints[6]: Prefer explicit commands over `eval` and `run-code` unless needed.

skill.constraints[7]: When you do not have a fresh snapshot, use placeholder refs like `eX` and say why; do not bypass refs with `run-code`.

skill.constraints[8]: Use `--headed` when a visual check will help.

skill.constraints[9]: When capturing artifacts in this repo, use `output/playwright/` and avoid introducing new top-level artifact folders.

skill.constraints[10]: Default to CLI commands and workflows, not Playwright test specs.

skill.constraints[11]: Use the wrapper script unless the CLI is already installed globally.

skill.constraints[12]: Assume `PWCLI` is set and `pwcli` is an alias for `"\$PWCLI"`.

skill.constraints[13]: In this repo, run commands from `output/playwright/<label>/` to keep artifacts contained.

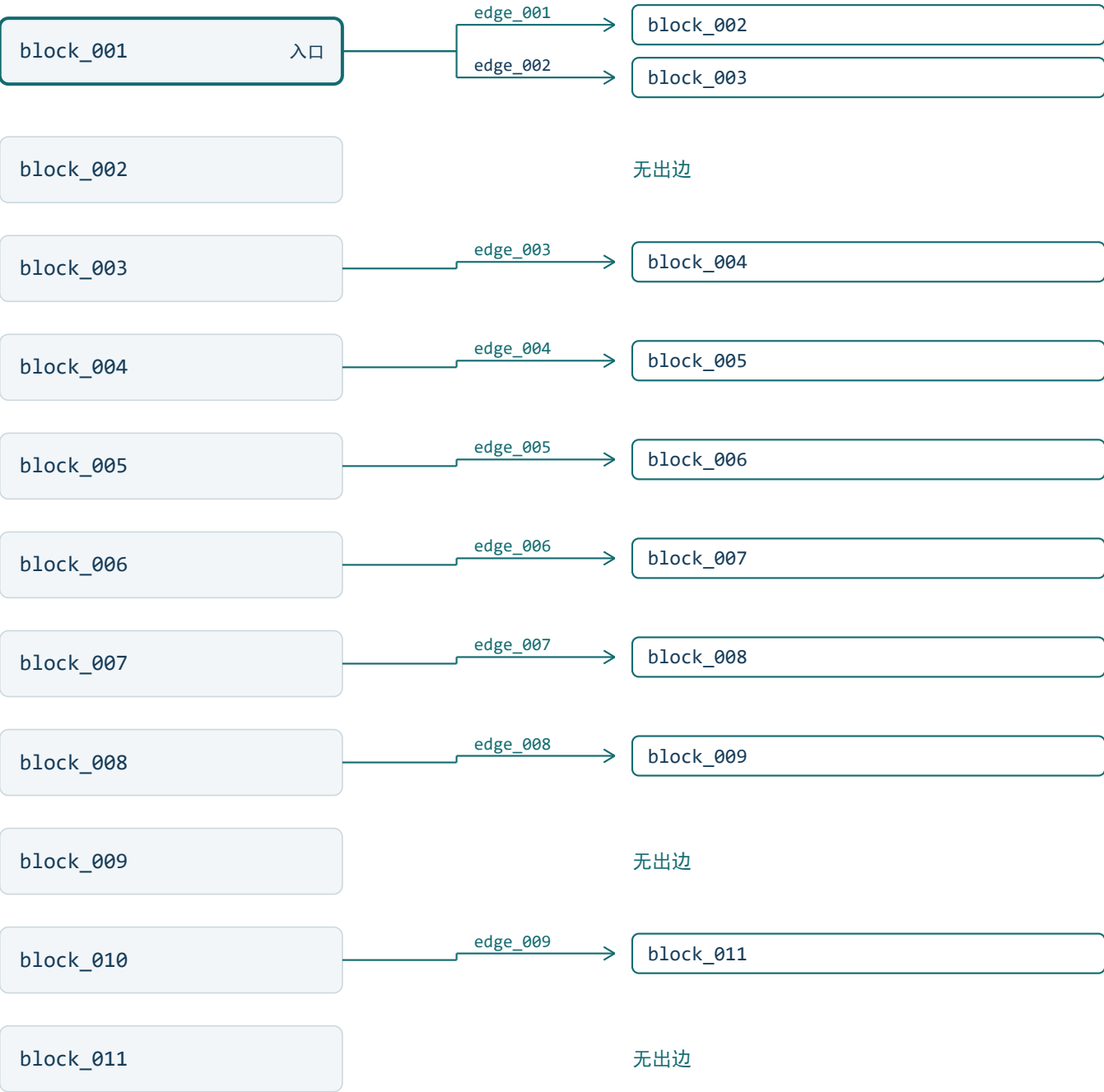
skill.constraints[14]: If an element ref fails, run `pwcli snapshot` again and retry.

skill.constraints[15]: If the page looks wrong, re-open with `--headed` and resize the window.

skill.constraints[16]: If a flow depends on prior state, use a named `--session`.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	"npx not available"
edge_002	block_001 -> block_003	"npx available"
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null
edge_007	block_007 -> block_008	null
edge_008	block_008 -> block_009	null
edge_009	block_010 -> block_011	null

全部操作、数据引用与约束落点

block_001 Check whether npx is available

```
data_source_kind = "context"
block_001.constraints: []

ir_001 check_npx_available

inputs[0] context_key: PATH | PATH
outputs[0] result: result_001 | npx_available

ir_001.constraints[0]: Before proposing commands, check whether `npx` is available (the wrapper depends on it).
draft_instruction_id = check_npx
```

ir_002 dispatch

```
inputs[0] result: result_001 | npx_available
outputs = []

ir_002.constraints: []

draft_instruction_id = branch_on_npx
```

block_002 Ask the user to install Node.js/npm

```
data_source_kind = null
block_002.constraints: []

ir_003 ask_user_to_install_node

inputs[0] literal = "node --version\nnpm --version\n\n# If missing, install Node.js/npm, then:\nnpm install -g\n@playwright/cli@latest\nplaywright-cli --help"
outputs = []

ir_003.constraints[0]: If it is not available, pause and ask the user to install Node.js/npm (which provides `npx`). Provide these steps verbatim.
draft_instruction_id = ask_install_node
```

ir_004 return

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = return_install_request
```

block_003 Resolve the bundled Playwright CLI wrapper path

```
data_source_kind = "context"
```

```
block_003.constraints: []
```

ir_005 resolve_playwright_cli_wrapper_path

```
inputs[0] context_key: CODEX_HOME | CODEX_HOME
```

```
inputs[1] context_key: HOME | HOME
```

```
outputs[0] result: result_002 | pwcli_path
```

```
ir_005.constraints[0]: Skill path (set once). User-scoped skills install under `$_CODEX_HOME/skills` (default: `~/codex/skills`).
```

```
draft_instruction_id = resolve_wrapper
```

metadata.script_content: 118 字符; SHA256 033d9a26a0c33ec79b782cddf9862e43fed962a400247835828692fd4b46f278; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = go_read_target
```

block_004 Read the target page URL from the task

```
data_source_kind = "context"
```

```
block_004.constraints: []
```

ir_007 read_target_url_from_task

```
inputs[0] context_key: target_url | target_url
```

```
outputs[0] result: result_003 | target_page_url
```

```
ir_007.constraints: []
```

```
draft_instruction_id = read_target_url
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_open
```

block_005 Open the target page in the browser

```
data_source_kind = null
```

```
block_005.constraints[0]: Use the wrapper script unless the CLI is already installed globally.
```

```
block_005.constraints[1]: Use `--headed` when a visual check will help.
```

ir_009 open_page

```
inputs[0] external_resource: playwright_cli_wrapper | playwright_cli_wrapper
```

```
inputs[1] result: result_002 | pwcli_path
```

```
inputs[2] result: result_003 | target_page_url
```

```
outputs = []
```

```
ir_009.constraints: []
```

```
draft_instruction_id = open_page
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_snapshot
```

block_006 Snapshot the page to obtain stable element refs

```
data_source_kind = "external"
```

```
block_006.constraints[0]: Always snapshot before referencing element ids like `e12`.
```

```
block_006.constraints[1]: Snapshot to get stable element refs.
```

ir_011 take_page_snapshot

```
inputs[0] external_resource: playwright_cli_wrapper | playwright_cli_wrapper
```

```
inputs[1] result: result_002 | pwcli_path
```

```
outputs[0] result: result_004 | element_refs
```

```
ir_011.constraints: []
```

```
draft_instruction_id = take_snapshot
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_interact
```

block_007 Interact with the page using refs from the latest snapshot

```
data_source_kind = null
```

```
block_007.constraints[0]: Interact using refs from the latest snapshot.
```

ir_013 interact_with_element_refs

```
inputs[0] external_resource: playwright_cli_wrapper | playwright_cli_wrapper
```

```
inputs[1] result: result_002 | pwcli_path
```

```
inputs[2] result: result_004 | element_refs
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = interact_with_refs
```

ir_014 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = go_resnapshot
```

block_008 Re-snapshot after navigation or significant DOM changes

```
data_source_kind = "external"
```

```
block_008.constraints[0]: Re-snapshot after navigation or significant DOM changes.
```

```
block_008.constraints[1]: Snapshot again after: navigation, clicking elements that change the UI substantially, opening/closing modals or menus, tab switches.
```

ir_015 take_page_snapshot

```
inputs[0] external_resource: playwright_cli_wrapper | playwright_cli_wrapper
```

```
inputs[1] result: result_002 | pwcli_path
```

```
outputs[0] result: result_005 | updated_element_refs
```

```
ir_015.constraints: []
```

```
draft_instruction_id = resnapshot
```

ir_016 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = go_capture
```

block_009 Capture browser artifacts when useful

```
data_source_kind = null
```

```
block_009.constraints[0]: Capture artifacts (screenshot, pdf, traces) when useful.
```

```
block_009.constraints[1]: When capturing artifacts in this repo, use `output/playwright/` and avoid introducing new top-level artifact folders.
```

ir_017 capture_browser_artifacts

```
inputs[0] external_resource: playwright_cli_wrapper | playwright_cli_wrapper
```

```
inputs[1] result: result_002 | pwcli_path
```

```
outputs = []
```

```
ir_017.constraints: []
```

```
draft_instruction_id = capture_artifacts
```

ir_018 return

```
inputs = []
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = return_success
```

block_010 Read wrapper arguments and session environment

```
data_source_kind = "context"
```

```
block_010.constraints: []
```

ir_019 read_wrapper_arguments_and_session

```
inputs[0] context_key: command_line_args | command_line_args
```

```
inputs[1] context_key: PLAYWRIGHT_CLI_SESSION | PLAYWRIGHT_CLI_SESSION
```

```
outputs[0] result: result_006 | cli_args
```

```
outputs[1] result: result_007 | session_name
```

```
ir_019.constraints: []
```

```
draft_instruction_id = read_wrapper_inputs
```

ir_020 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = go_exec_wrapper
```

block_011 Execute playwright-cli through npx

```
data_source_kind = "external"
```

```
block_011.constraints[0]: A global install of `playwright-cli` is optional.
```

ir_021 run_playwright_cli_wrapper

```
inputs[0] external_resource: npx | npx
```

```
inputs[1] external_resource: playwright-cli | playwright-cli
```

```
inputs[2] result: result_006 | cli_args
```

```
inputs[3] result: result_007 | session_name
```

```
outputs[0] result: result_008 | cli_output
```

```
ir_021.constraints[0]: The wrapper script uses `npx --package @playwright/cli playwright-cli` so the CLI can run without a global install.
```

```
draft_instruction_id = exec_playwright_cli
```

```
metadata.script_content: 526 字符; SHA256 aa3fdff5d0e4556177f4dfd5f04117e772aa54f94b6a2e34b6c0edf629c6b9b5; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。
```

ir_022 return

```
inputs[0] result: result_008 | cli_output
```

```
outputs = []
```

```
ir_022.constraints: []
```

```
draft_instruction_id = return_wrapper
```


R02 / 第 2 次 实际输出明细

playwright

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R02/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: The skill does not define a concrete return value; the return block is modeled with the last artifact result.

candidate warning: The wrapper script is treated as a black-box external tool; its internal npx check and session flag scan are not expanded into control flow.

candidate warning: Reference guides and example command blocks are not modeled as execution instructions; they are documentation.

candidate warning: The global-install alternative is modeled as a constraint rather than a separate execution path.

Skill 层约束与上下文声明

declared_context_keys = ["browser_automation_task", "CODEX_HOME", "HOME", "PLAYWRIGHT_CLI_SESSION"]

skill.constraints[0]: Prefer the bundled wrapper script so the CLI works even when it is not globally installed.

skill.constraints[1]: Treat this skill as CLI-first automation. Do not pivot to @playwright/test unless the user explicitly asks for test files.

skill.constraints[2]: Default to CLI commands and workflows, not Playwright test specs.

skill.constraints[3]: Prefer explicit commands over eval and run-code unless needed.

skill.constraints[4]: Always snapshot before referencing element ids like e12.

skill.constraints[5]: Re-snapshot when refs seem stale.

skill.constraints[6]: When you do not have a fresh snapshot, use placeholder refs like eX and say why; do not bypass refs with run-code.

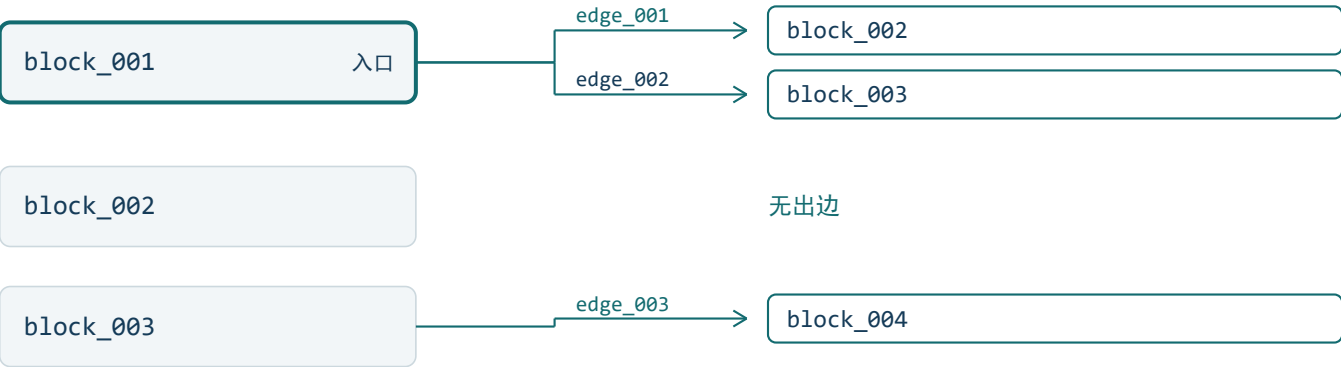
skill.constraints[7]: Use --headed when a visual check will help.

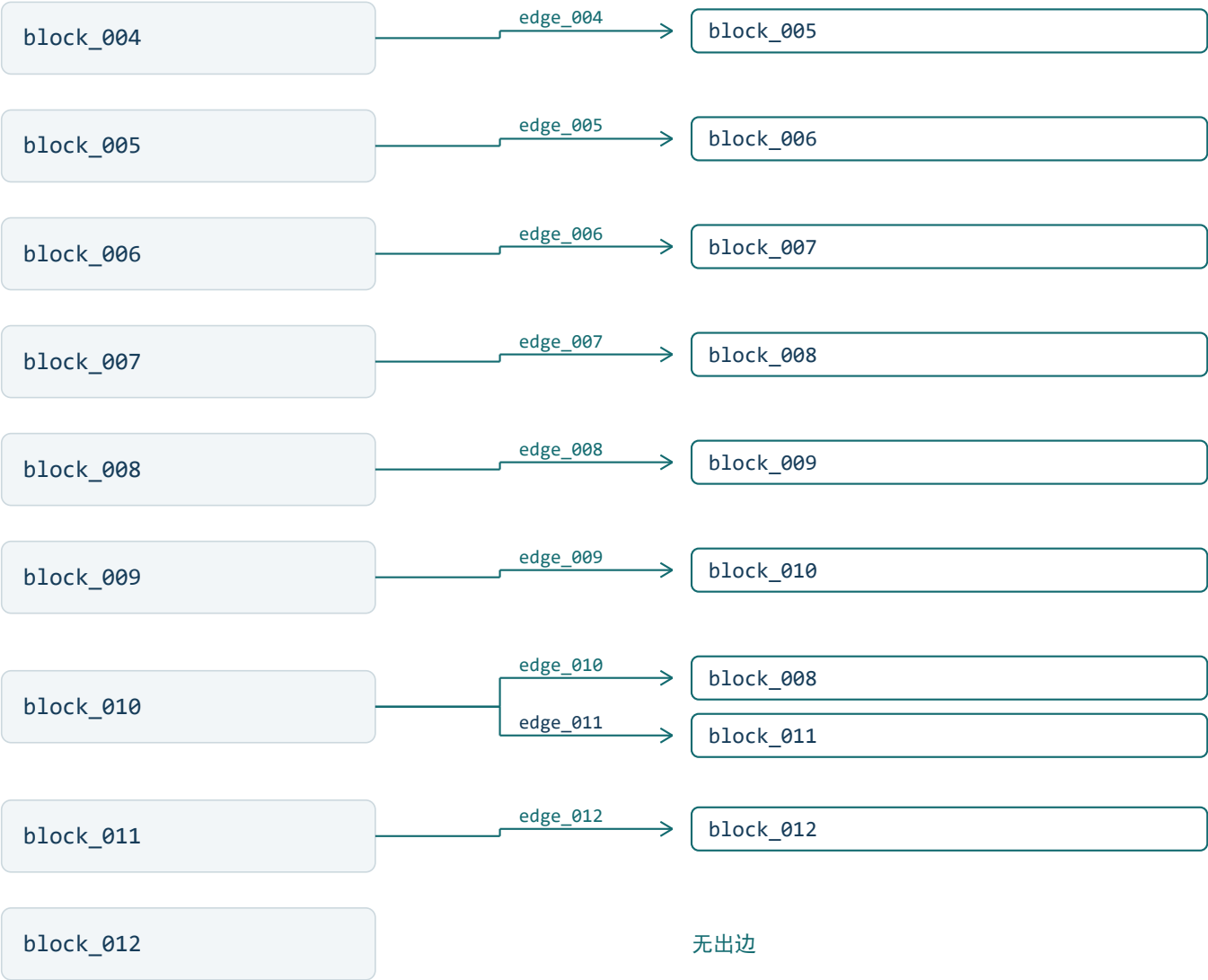
skill.constraints[8]: When capturing artifacts in this repo, use output/playwright/ and avoid introducing new top-level artifact folders.

skill.constraints[9]: A global install of playwright-cli is optional.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	"npx is not available"
edge_002	block_001 -> block_003	"npx is available"
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null
edge_007	block_007 -> block_008	null
edge_008	block_008 -> block_009	null
edge_009	block_009 -> block_010	null
edge_010	block_010 -> block_008	"re-snapshot needed (navigation, significant DOM change, modal/menu open/close, tab switch, or stale ref failure)"
edge_011	block_010 -> block_011	"no re-snapshot needed"
edge_012	block_011 -> block_012	null

全部操作、数据引用与约束落点

block_001 Check whether npx is available

```
data_source_kind = "external"

block_001.constraints: []

ir_001 check_npx_availability

inputs[0] external_resource: npx | npx
outputs[0] result: result_001 | npx_available

ir_001.constraints[0]: Before proposing commands, check whether npx is available (the wrapper depends on it).

draft_instruction_id = check_npx_availability
```

ir_002 dispatch

```
inputs[0] result: result_001 | npx_available
outputs = []

ir_002.constraints: []

draft_instruction_id = dispatch_npx_check
```

block_002 Ask the user to install Node.js/npm

```
data_source_kind = null
```

```
block_002.constraints: []
```

ir_003 ask_user_to_install_node_npm

```
inputs[0] literal = "node --version\nnpm --version\n\n# If missing, install Node.js/npm, then:\nnpm install -g\n@playwright/cli@latest\nplaywright-cli --help"
```

```
outputs = []
```

```
ir_003.constraints[0]: If npx is not available, pause and ask the user to install Node.js/npm (which provides npx).
```

```
ir_003.constraints[1]: Provide these steps verbatim.
```

```
draft_instruction_id = ask_install_node_npm
```

ir_004 return

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = return_missing_npx
```

block_003 Read CODEX_HOME and HOME environment variables

```
data_source_kind = "context"
```

```
block_003.constraints: []
```

ir_005 read_codex_home_and_home

```
inputs[0] context_key: CODEX_HOME | CODEX_HOME
```

```
inputs[1] context_key: HOME | HOME
```

```
outputs[0] result: result_002 | codex_home_env
```

```
outputs[1] result: result_003 | home_dir
```

```
ir_005.constraints: []
```

```
draft_instruction_id = read_codex_home_and_home
```

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = dispatch_read_env
```

block_004 Compute CODEX_HOME and PWCLI wrapper path

```
data_source_kind = null
```

```
block_004.constraints: []
```

ir_007 compute_codex_home_and_pwcli_path

```
inputs[0] result: result_002 | codex_home_env
```

```
inputs[1] result: result_003 | home_dir
```

```
outputs[0] result: result_004 | codex_home
```

```
outputs[1] result: result_005 | pwcli_path
```

```
ir_007.constraints[0]: export CODEX_HOME="${CODEX_HOME:-$HOME/.codex}"
```

```
ir_007.constraints[1]: export PWCLI="$CODEX_HOME/skills/playwright/scripts/playwright_cli.sh"
```

```
draft_instruction_id = compute_wrapper_path
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_set_wrapper
```

block_005 Read PLAYWRIGHT_CLI_SESSION environment variable

```
data_source_kind = "context"
```

```
block_005.constraints: []
```

ir_009 read_playwright_cli_session

```
inputs[0] context_key: PLAYWRIGHT_CLI_SESSION | PLAYWRIGHT_CLI_SESSION
```

```
outputs[0] result: result_006 | playwright_cli_session
```

```
ir_009.constraints: []
```

```
draft_instruction_id = read_playwright_cli_session
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_read_session
```

block_006 Read the browser automation task

```
data_source_kind = "context"
```

```
block_006.constraints: []
```

ir_011 read_browser_automation_task

```
inputs[0] context_key: browser_automation_task | browser_automation_task
```

```
outputs[0] result: result_007 | browser_task
```

```
ir_011.constraints: []
```

```
draft_instruction_id = read_browser_automation_task
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_read_task
```

block_007 Open the target page

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 run_playwright_cli_open

```
inputs[0] result: result_005 | pwcli_path
```

```
inputs[1] result: result_006 | playwright_cli_session
```

```
inputs[2] result: result_007 | browser_task
outputs = []

ir_013.constraints[0]: Use the wrapper script unless the repository already standardizes on a global install.
ir_013.constraints[1]: Use --headed when a visual check will help.

draft_instruction_id = open_page_via_wrapper

metadata.script_content: 526 字符; SHA256 aa3fdff5d0e4556177f4dfd5f04117e772aa54f94b6a2e34b6c0edf629c6b9b5; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_014 dispatch

inputs = []
outputs = []

ir_014.constraints: []

draft_instruction_id = dispatch_open

block_008 Snapshot the page to get stable element refs

data_source_kind = "external"

block_008.constraints: []

ir_015 run_playwright_cli_snapshot

inputs[0] result: result_005 | pwcli_path
inputs[1] result: result_006 | playwright_cli_session
outputs[0] result: result_008 | snapshot_refs
ir_015.constraints: []

draft_instruction_id = take_snapshot

ir_016 dispatch

inputs = []
outputs = []

ir_016.constraints: []

draft_instruction_id = dispatch_snapshot

block_009 Interact using refs from the latest snapshot

data_source_kind = "external"

block_009.constraints: []

ir_017 run_playwright_cli_interaction

inputs[0] result: result_005 | pwcli_path
inputs[1] result: result_006 | playwright_cli_session
inputs[2] result: result_008 | snapshot_refs
inputs[3] result: result_007 | browser_task
outputs[0] result: result_009 | interaction_result

ir_017.constraints[0]: Always snapshot before referencing element ids like e12.
ir_017.constraints[1]: When you do not have a fresh snapshot, use placeholder refs like eX and say why; do not bypass refs with run-code.
ir_017.constraints[2]: Prefer explicit commands over eval and run-code unless needed.

draft_instruction_id = interact_with_element
```

ir_018 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = dispatch_interact
```

block_010 Decide whether to re-snapshot

```
data_source_kind = null
```

```
block_010.constraints: []
```

ir_019 should_resnapshot

```
inputs[0] result: result_009 | interaction_result
```

```
outputs[0] result: result_010 | resnapshot_needed
```

```
ir_019.constraints[0]: Re-snapshot after navigation.
```

```
ir_019.constraints[1]: Re-snapshot after clicking elements that change the UI substantially.
```

```
ir_019.constraints[2]: Re-snapshot after opening/closing modals or menus.
```

```
ir_019.constraints[3]: Re-snapshot after tab switches.
```

```
ir_019.constraints[4]: Refs can go stale. When a command fails due to a missing ref, snapshot again.
```

```
draft_instruction_id = check_resnapshot_needed
```

ir_020 dispatch

```
inputs[0] result: result_010 | resnapshot_needed
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = dispatch_resnapshot
```

block_011 Capture artifacts when useful

```
data_source_kind = "external"
```

```
block_011.constraints: []
```

ir_021 run_playwright_cli_capture_artifacts_when_useful

```
inputs[0] result: result_005 | pwcli_path
```

```
inputs[1] result: result_006 | playwright_cli_session
```

```
inputs[2] result: result_007 | browser_task
```

```
outputs[0] result: result_011 | artifact_path
```

```
ir_021.constraints[0]: Capture artifacts (screenshot, pdf, traces) when useful.
```

```
ir_021.constraints[1]: When capturing artifacts in this repo, use output/playwright/ and avoid introducing new top-level artifact folders.
```

```
draft_instruction_id = capture_artifact
```

ir_022 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_022.constraints: []
```

```
draft_instruction_id = dispatch_capture
```

block_012 Return the browser automation result

```
data_source_kind = null
```

```
block_012.constraints: []
```

ir_023 return

```
inputs[0] result: result_011 | artifact_path
```

```
outputs = []
```

```
ir_023.constraints: []
```

```
draft_instruction_id = return_artifact
```


R02 / 第 3 次 实际输出明细

playwright

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R02/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Binary asset assets/playwright.png was omitted from analysis; no executable behavior inferred from it.

candidate warning: LICENSE.txt and NOTICE.txt contain legal text only and were not analyzed as executable behavior.

candidate warning: Command examples in SKILL.md, references/cli.md, and references/workflows.md were treated as documentation examples, not as fixed runtime steps.

Skill 层约束与上下文声明

declared_context_keys = ["browser_task", "codex_home", "home_directory", "playwright_cli_session"]

skill.constraints[0]: Prefer the bundled wrapper script so the CLI works even when it is not globally installed.

skill.constraints[1]: Treat this skill as CLI-first automation. Do not pivot to @playwright/test unless the user explicitly asks for test files.

skill.constraints[2]: Before proposing commands, check whether npx is available (the wrapper depends on it).

skill.constraints[3]: If npx is not available, pause and ask the user to install Node.js/npm (which provides npx). Provide these steps verbatim.

skill.constraints[4]: Once npx is present, proceed with the wrapper script. A global install of playwright-cli is optional.

skill.constraints[5]: User-scoped skills install under \$CODEX_HOME/skills (default: ~/.codex/skills).

skill.constraints[6]: Always snapshot before referencing element ids like e12.

skill.constraints[7]: Re-snapshot when refs seem stale.

skill.constraints[8]: Prefer explicit commands over eval and run-code unless needed.

skill.constraints[9]: When you do not have a fresh snapshot, use placeholder refs like eX and say why; do not bypass refs with run-code.

skill.constraints[10]: Use --headed when a visual check will help.

skill.constraints[11]: When capturing artifacts in this repo, use output/playwright/ and avoid introducing new top-level artifact folders.

skill.constraints[12]: Default to CLI commands and workflows, not Playwright test specs.

skill.constraints[13]: Open only what you need: CLI command reference: references/cli.md; Practical workflows and troubleshooting: references/workflows.md.

skill.constraints[14]: Use the wrapper script unless the CLI is already installed globally.

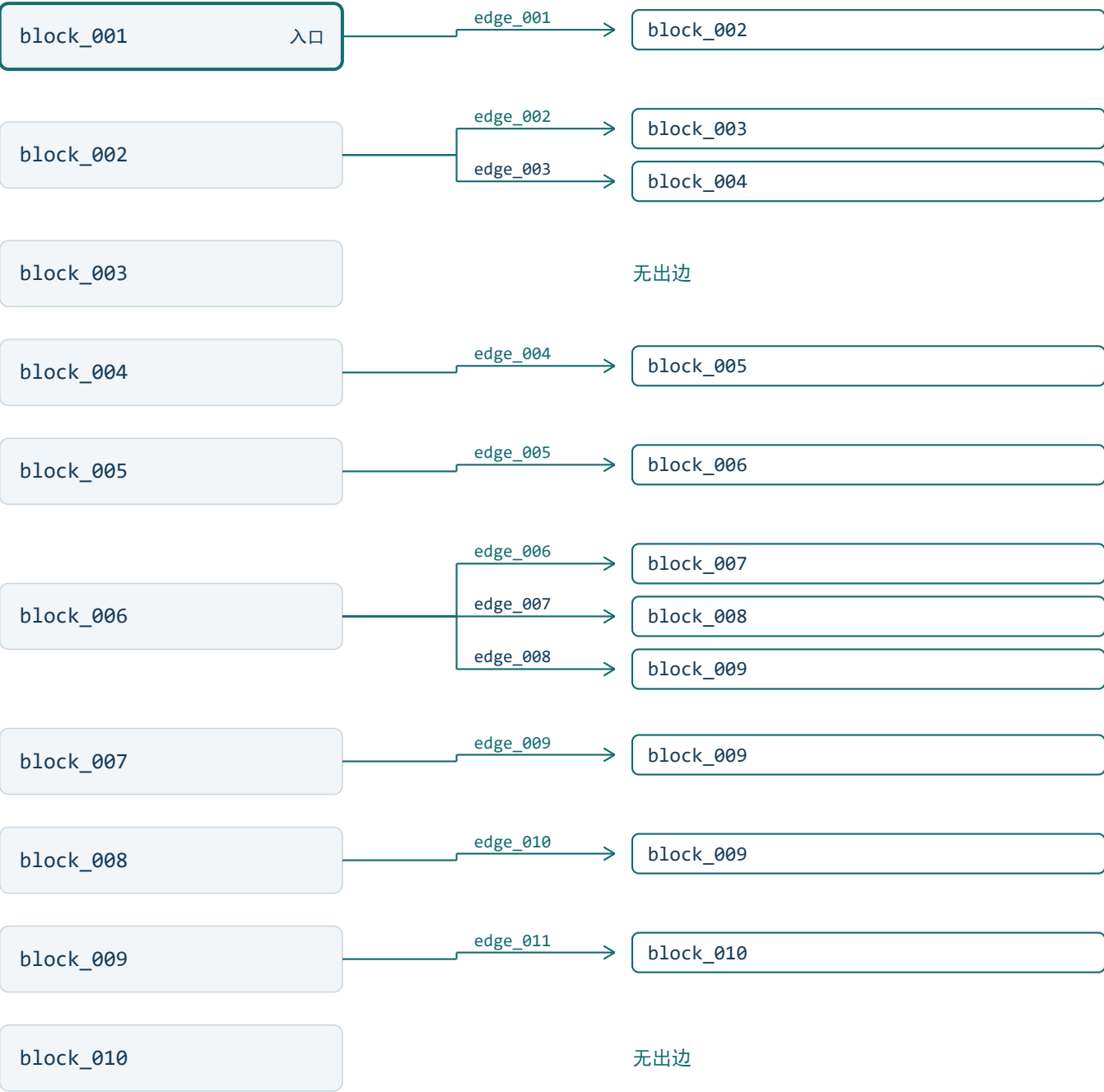
skill.constraints[15]: Run commands from output/playwright/<label>/ to keep artifacts contained.

skill.constraints[16]: By default, the CLI reads playwright-cli.json from the current directory. Use --config to point at a specific file.

skill.constraints[17]: Use a named session to isolate work.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。



完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"npx is not available"
edge_003	block_002 -> block_004	"npx is available"
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	"CLI command reference needed"
edge_007	block_006 -> block_008	"workflows or troubleshooting reference needed"
edge_008	block_006 -> block_009	"no reference needed"
edge_009	block_007 -> block_009	null
edge_010	block_008 -> block_009	null
edge_011	block_009 -> block_010	null

全部操作、数据引用与约束落点

block_001 Read the browser automation task

```
data_source_kind = "context"
block_001.constraints: []

ir_001 read_browser_automation_task

inputs[0] context_key: browser_task | browser_task
outputs[0] result: result_001 | browser_task_request
ir_001.constraints: []

draft_instruction_id = read_task_inst
```

ir_002 dispatch

```
inputs = []
outputs = []

ir_002.constraints: []

draft_instruction_id = dispatch_task
```

block_002 Check whether npx is available

```
data_source_kind = "external"

block_002.constraints[0]: Before proposing commands, check whether npx is available (the wrapper depends on it).

ir_003 check_npx_available

inputs[0] external_resource: npx | npx
outputs[0] result: result_002 | npx_available
ir_003.constraints: []
```

```
draft_instruction_id = check_npx_inst
```

ir_004 dispatch

```
inputs[0] result: result_002 | npx_available
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = dispatch_npx
```

block_003 Ask the user to install Node.js/npm

```
data_source_kind = null
```

```
block_003.constraints[0]: If npx is not available, pause and ask the user to install Node.js/npm (which provides npx). Provide these steps verbatim.
```

ir_005 request_node_npm_install

```
inputs[0] literal = ["# Verify Node/npm are installed", "node --version", "npm --version", "", "# If missing, install Node.js/npm, then:", "npm install -g @playwright/cli@latest", "playwright-cli --help"]
```

```
outputs[0] result: result_003 | install_request_sent
```

```
ir_005.constraints: []
```

```
draft_instruction_id = request_install_inst
```

ir_006 return

```
inputs[0] result: result_003 | install_request_sent
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = return_install
```

block_004 Read Codex home and home directory

```
data_source_kind = "context"
```

```
block_004.constraints[0]: User-scoped skills install under $CODEX_HOME/skills (default: ~/.codex/skills).
```

ir_007 read_codex_home_and_home_directory

```
inputs[0] context_key: codex_home | codex_home
```

```
inputs[1] context_key: home_directory | home_directory
```

```
outputs[0] result: result_004 | codex_home_setting
```

```
outputs[1] result: result_005 | home_directory_setting
```

```
ir_007.constraints: []
```

```
draft_instruction_id = read_codex_env_inst
```

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_codex_env
```

block_005 Compute the wrapper script path

```
data_source_kind = null
```

```
block_005.constraints[0]: Set PWCLI to $CODEX_HOME/skills/playwright/scripts/playwright_cli.sh.
```

ir_009 compute_wrapper_path

```
inputs[0] result: result_004 | codex_home_setting
```

```
inputs[1] result: result_005 | home_directory_setting
```

```
inputs[2] literal = "skills/playwright/scripts/playwright_cli.sh"
```

```
outputs[0] result: result_006 | pwcli_path
```

```
ir_009.constraints: []
```

```
draft_instruction_id = compute_wrapper_path_inst
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_compute
```

block_006 Decide which reference documents to open

```
data_source_kind = null
```

```
block_006.constraints[0]: Open only what you need: CLI command reference: references/cli.md; Practical workflows and troubleshooting: references/workflows.md.
```

ir_011 assess_reference_need

```
inputs[0] result: result_001 | browser_task_request
```

```
outputs[0] result: result_007 | needs_cli_reference
```

```
outputs[1] result: result_008 | needs_workflows_reference
```

```
ir_011.constraints: []
```

```
draft_instruction_id = assess_references_inst
```

ir_012 dispatch

```
inputs[0] result: result_007 | needs_cli_reference
```

```
inputs[1] result: result_008 | needs_workflows_reference
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_assess
```

block_007 Read the CLI command reference

```
data_source_kind = "external"
```

```
block_007.constraints: []
```

ir_013 read_file

```
inputs[0] external_resource: references/cli.md | references/cli.md
```

```
outputs[0] result: result_009 | cli_reference_content
```

```
ir_013.constraints: []
```

```
draft_instruction_id = read_cli_reference_inst
```

ir_014 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = dispatch_cli_ref
```

block_008 Read the workflows and troubleshooting reference

```
data_source_kind = "external"
```

```
block_008.constraints: []
```

ir_015 read_file

```
inputs[0] external_resource: references/workflows.md | references/workflows.md
```

```
outputs[0] result: result_010 | workflows_reference_content
```

```
ir_015.constraints: []
```

```
draft_instruction_id = read_workflows_reference_inst
```

ir_016 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = dispatch_workflows_ref
```

block_009 Read the Playwright CLI session

```
data_source_kind = "context"
```

```
block_009.constraints[0]: Use a named session to isolate work.
```

ir_017 read_playwright_cli_session

```
inputs[0] context_key: playwright_cli_session | playwright_cli_session
```

```
outputs[0] result: result_011 | session_name
```

```
ir_017.constraints: []
```

```
draft_instruction_id = read_session_env_inst
```

ir_018 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = dispatch_session
```

block_010 Run the requested Playwright CLI browser automation

```
data_source_kind = "external"
```

block_010.constraints[0]: Core workflow: 1. Open the page. 2. Snapshot to get stable element refs. 3. Interact using refs from the latest snapshot. 4. Re-snapshot after navigation or significant DOM changes. 5. Capture artifacts (screenshot, pdf, traces) when useful.

block_010.constraints[1]: Always snapshot before referencing element ids like e12.

block_010.constraints[2]: Re-snapshot when refs seem stale.

block_010.constraints[3]: Prefer explicit commands over eval and run-code unless needed.

block_010.constraints[4]: When you do not have a fresh snapshot, use placeholder refs like eX and say why; do not bypass refs with run-code.

block_010.constraints[5]: Use --headed when a visual check will help.

block_010.constraints[6]: When capturing artifacts in this repo, use output/playwright/ and avoid introducing new top-level artifact folders.

block_010.constraints[7]: Default to CLI commands and workflows, not Playwright test specs.

ir_019 run_playwright_cli_wrapper

```
inputs[0] result: result_006 | pwcli_path
```

```
inputs[1] result: result_001 | browser_task_request
```

```
inputs[2] result: result_011 | session_name
```

```
outputs[0] result: result_012 | browser_automation_result
```

```
ir_019.constraints: []
```

```
draft_instruction_id = run_browser_automation_inst
```

metadata.script_content: 526 字符; SHA256 aa3fdff5d0e4556177f4dfd5f04117e772aa54f94b6a2e34b6c0edf629c6b9b5; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_020 return

```
inputs[0] result: result_012 | browser_automation_result
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = return_browser_automation
```

R03 / 第 1 次 实际输出明细

gh-fix-ci

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R03/deepseek-v4-flash-max/1

结构状态: complete | 生成 2 | 修复 1 | HTTP 尝试 2 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

生成 2 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: LICENSE.txt and assets were not treated as executable behavior.

candidate warning: agents/openai.yaml provides interface metadata only and was not treated as runtime behavior.

candidate warning: The manual fallback is represented as a single external acquisition block because its internal per-check loop and conditionals are implementation details.

Skill 层约束与上下文声明

```
declared_context_keys = ["repo", "pr", "user_approval"]
```

skill.constraints[0]: Use gh to locate failing PR checks, fetch GitHub Actions logs for actionable failures, summarize the failure snippet, then propose a fix plan and implement after explicit approval.

skill.constraints[1]: If a plan-oriented skill (for example `create-plan`) is available, use it; otherwise draft a concise plan inline and request approval before implementing.

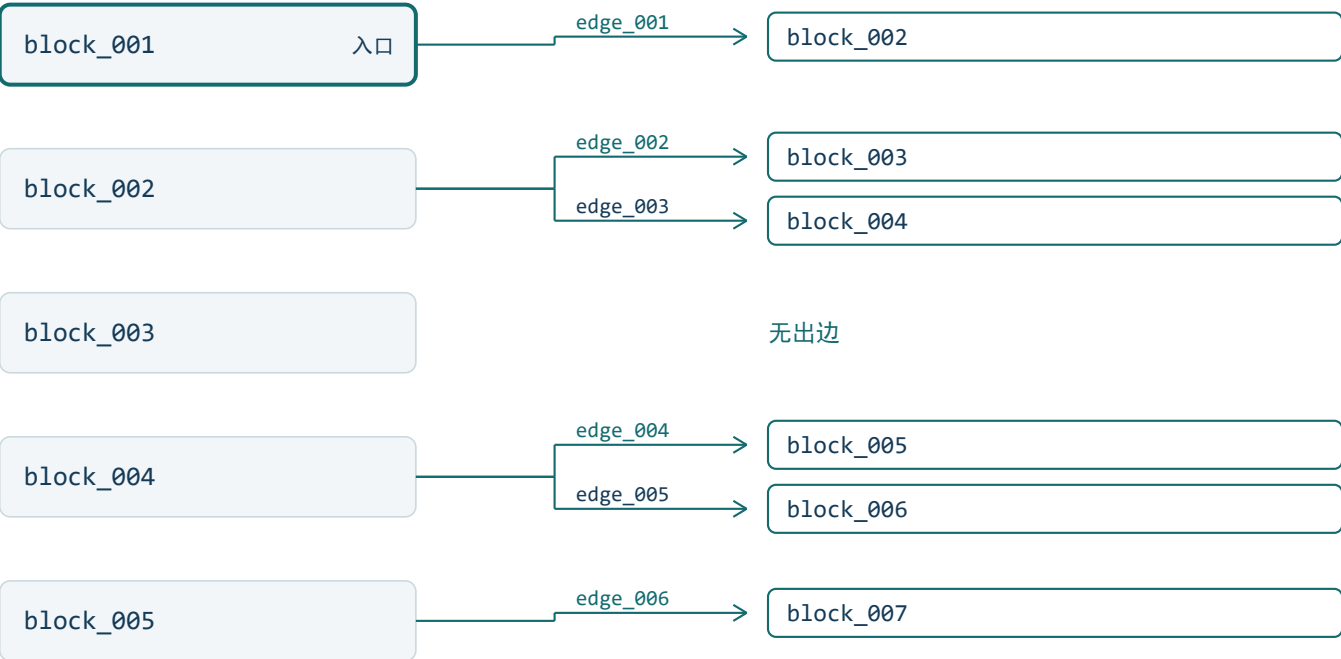
skill.constraints[2]: Prereq: authenticate with the standard GitHub CLI once (for example, run `gh auth login`), then confirm with `gh auth status` (repo + workflow scopes are typically required).

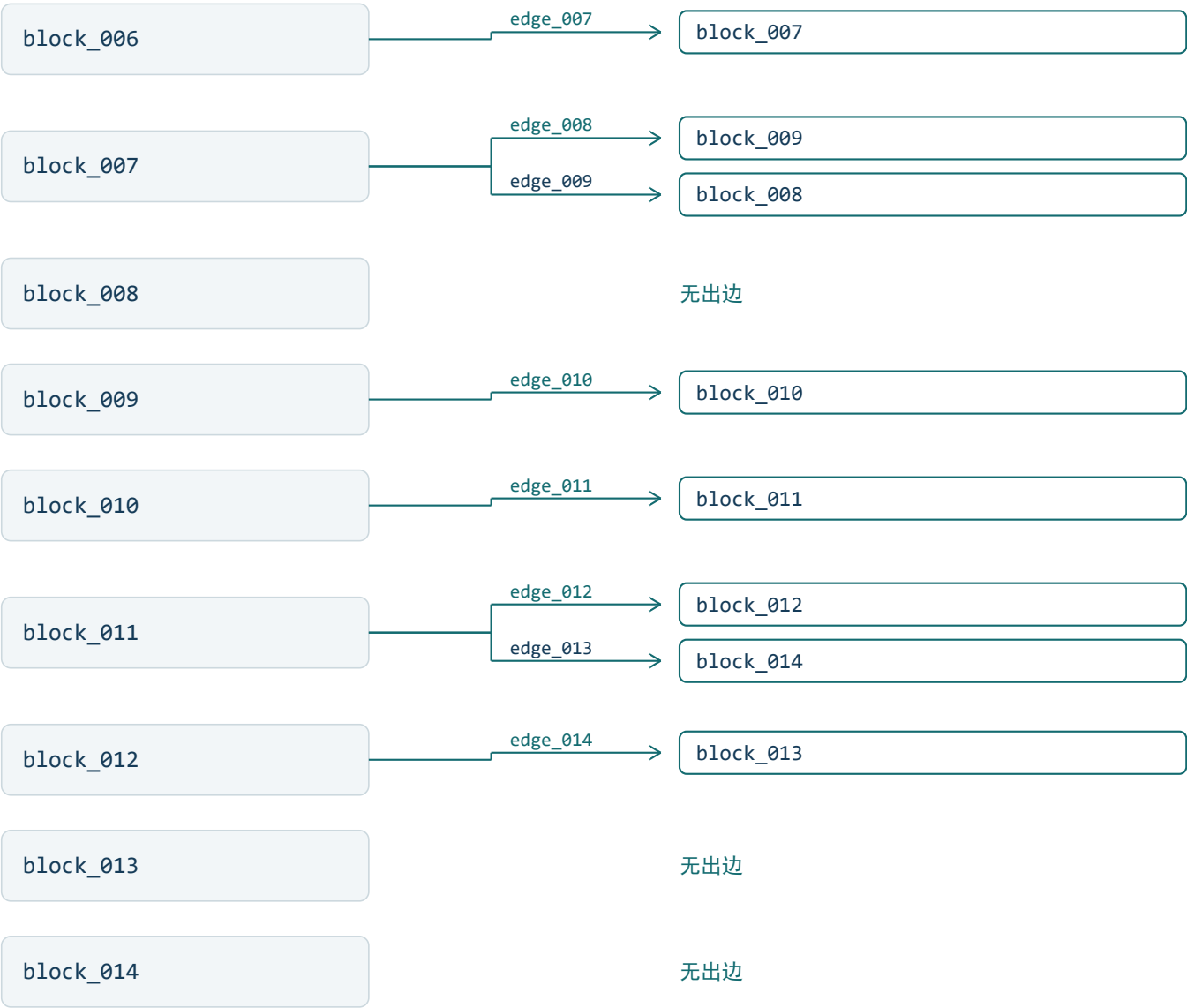
skill.constraints[3]: Treat external providers (for example Buildkite) as out of scope and report only the details URL.

skill.constraints[4]: Do not attempt Buildkite or other providers; keep the workflow lean.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"not authenticated"
edge_003	block_002 -> block_004	"authenticated"
edge_004	block_004 -> block_005	"bundled script chosen"
edge_005	block_004 -> block_006	"manual fallback chosen"
edge_006	block_005 -> block_007	null
edge_007	block_006 -> block_007	null
edge_008	block_007 -> block_009	"failures found"
edge_009	block_007 -> block_008	"no failures found"
edge_010	block_009 -> block_010	null
edge_011	block_010 -> block_011	null
edge_012	block_011 -> block_012	"approved"
edge_013	block_011 -> block_014	"not approved"
edge_014	block_012 -> block_013	null

全部操作、数据引用与约束落点

block_001 Read skill inputs

```
data_source_kind = "context"
block_001.constraints: []

ir_001 read_skill_inputs

inputs[0] context_key: repo | repo
inputs[1] context_key: pr | pr
outputs[0] result: result_001 | repo_path
outputs[1] result: result_002 | pr_input
ir_001.constraints: []
draft_instruction_id = read_inputs_instr
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = dispatch_to_verify
```

block_002 Verify gh authentication

```
data_source_kind = "external"
```

```
block_002.constraints[0]: Repo + workflow scopes are typically required.
```

ir_003 check_gh_auth_status

```
inputs[0] external_resource: gh | gh
```

```
inputs[1] result: result_001 | repo_path
```

```
outputs[0] result: result_003 | gh_auth_status
```

```
ir_003.constraints: []
```

```
draft_instruction_id = check_gh_auth
```

ir_004 dispatch

```
inputs[0] result: result_003 | gh_auth_status
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = dispatch_auth
```

block_003 Ask user to authenticate

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 ask_user_to_run_gh_auth_login

```
inputs[0] literal = "gh auth login"
```

```
inputs[1] literal = "repo + workflow scopes"
```

```
outputs = []
```

```
ir_005.constraints: []
```

```
draft_instruction_id = ask_login
```

ir_006 return

```
inputs[0] literal = "Please run `gh auth login` with repo and workflow scopes, then retry."
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = return_auth_prompt
```

block_004 Choose inspection method

```
data_source_kind = null
```

```
block_004.constraints[0]: Preferred method is the bundled script; manual fallback is available if needed.
```

ir_007 select_inspection_method

```
inputs[0] literal = "bundled script"
```

```
inputs[1] literal = "manual fallback"
```

```
outputs[0] result: result_004 | inspection_method_choice
```

```
ir_007.constraints: []
```

```
draft_instruction_id = select_method
```

ir_008 dispatch

```
inputs[0] result: result_004 | inspection_method_choice
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_method
```

block_005 Run bundled inspect_pr_checks.py

```
data_source_kind = "external"
```

```
block_005.constraints[0]: Exits non-zero when failures remain so it can be used in automation.
```

ir_009 run_inspect_pr_checks_script

```
inputs[0] external_resource: inspect_pr_checks_script | inspect_pr_checks_script
```

```
inputs[1] external_resource: gh | gh
```

```
inputs[2] external_resource: git | git
```

```
inputs[3] result: result_001 | repo_path
```

```
inputs[4] result: result_002 | pr_input
```

```
outputs[0] result: result_005 | script_inspection_results
```

```
outputs[1] result: result_006 | script_exit_code
```

```
ir_009.constraints: []
```

```
draft_instruction_id = run_inspect_script
```

```
metadata.script_content: 15222 字符; SHA256 9459e5b03f86785d184a83e5cd8d621b832e918ec04158a2962d867785812c6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_script_done
```

block_006 Run manual gh commands to inspect failing checks

```
data_source_kind = "external"
```

```
block_006.constraints[0]: If a field is rejected, rerun with the available fields reported by gh.
```

```
block_006.constraints[1]: If the run log says it is still in progress, fetch job logs directly.
```

```
block_006.constraints[2]: Do not attempt Buildkite or other providers; keep the workflow lean.
```

ir_011 inspect_pr_checks_manually

```
inputs[0] external_resource: gh | gh
```

```
inputs[1] result: result_001 | repo_path
```

```
inputs[2] result: result_002 | pr_input
```

```
outputs[0] result: result_007 | manual_inspection_results
```

```
outputs[1] result: result_008 | manual_failing_checks_found
```

```
ir_011.constraints: []
```

```
draft_instruction_id = run_manual_inspection
```

```
metadata = {"commands": ["gh pr checks <pr> --json name,state,bucket,link,startedAt,completedAt,workflow", "gh run view <run_id> --json name,workflowName,conclusion,status,url,event,headBranch,headSha", "gh run view <run_id>"]}
```

```
--log", "gh api \"/repos/<owner>/<repo>/actions/jobs/<job_id>/logs\" > \"<path>\""]}
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_manual_done
```

block_007 Check whether failing checks were found

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 dispatch

```
inputs[0] result: result_006 | script_exit_code
```

```
inputs[1] result: result_008 | manual_failing_checks_found
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = dispatch_outcome
```

block_008 Report no failing checks

```
data_source_kind = null
```

```
block_008.constraints: []
```

ir_014 return

```
inputs[0] literal = "No failing checks detected."
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = return_no_failures
```

block_009 Summarize failures for the user

```
data_source_kind = null
```

```
block_009.constraints[0]: Provide the failing check name, run URL (if any), and a concise log snippet.
```

```
block_009.constraints[1]: Call out missing logs explicitly.
```

```
block_009.constraints[2]: For non-GitHub Actions checks, only report the details URL.
```

ir_015 summarize_failure_context

```
inputs[0] result: result_005 | script_inspection_results
```

```
inputs[1] result: result_007 | manual_inspection_results
```

```
outputs[0] result: result_009 | failure_summary
```

```
ir_015.constraints: []
```

```
draft_instruction_id = summarize_failures_instr
```

ir_016 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = dispatch_to_plan
```

block_010 Create a fix plan

data_source_kind = null

block_010.constraints[0]: Use the create-plan skill if available; otherwise draft a concise plan inline.

block_010.constraints[1]: Request approval before implementing.

ir_017 draft_fix_plan

inputs[0] result: result_009 | failure_summary

outputs[0] result: result_010 | fix_plan

ir_017.constraints: []

draft_instruction_id = draft_fix_plan

ir_018 request_plan_approval

inputs[0] result: result_010 | fix_plan

outputs = []

ir_018.constraints: []

draft_instruction_id = request_approval

ir_019 dispatch

inputs = []

outputs = []

ir_019.constraints: []

draft_instruction_id = dispatch_to_approval

block_011 Read user approval

data_source_kind = "context"

block_011.constraints: []

ir_020 read_user_approval

inputs[0] context_key: user_approval | user_approval

outputs[0] result: result_011 | approval_status

ir_020.constraints: []

draft_instruction_id = read_approval_instr

ir_021 dispatch

inputs[0] result: result_011 | approval_status

outputs = []

ir_021.constraints: []

draft_instruction_id = dispatch_approval

block_012 Implement approved fix plan

data_source_kind = null

block_012.constraints[0]: Apply the approved plan, summarize diffs/tests, and ask about opening a PR.

ir_022 implement_fix_plan

inputs[0] result: result_010 | fix_plan

outputs[0] result: result_012 | implementation_summary

ir_022.constraints: []

```
draft_instruction_id = implement_plan_instr
```

ir_023 summarize_diffs_and_tests

```
inputs[0] result: result_012 | implementation_summary
```

```
outputs[0] result: result_013 | implementation_report
```

```
ir_023.constraints: []
```

```
draft_instruction_id = summarize_diffs
```

ir_024 ask_about_opening_pr

```
inputs[0] result: result_013 | implementation_report
```

```
outputs = []
```

```
ir_024.constraints: []
```

```
draft_instruction_id = ask_pr
```

ir_025 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_025.constraints: []
```

```
draft_instruction_id = dispatch_to_recheck
```

block_013 Suggest rechecking status

```
data_source_kind = null
```

```
block_013.constraints: []
```

ir_026 suggest_rechecking_tests_and_checks

```
inputs[0] literal = "re-run relevant tests and `gh pr checks`"
```

```
outputs = []
```

```
ir_026.constraints: []
```

```
draft_instruction_id = suggest_recheck
```

ir_027 return

```
inputs[0] literal = "Re-run relevant tests and `gh pr checks` to confirm."
```

```
outputs = []
```

```
ir_027.constraints: []
```

```
draft_instruction_id = return_recheck
```

block_014 Return without implementing

```
data_source_kind = null
```

```
block_014.constraints: []
```

ir_028 return

```
inputs[0] literal = "Implementation not approved."
```

```
outputs = []
```

```
ir_028.constraints: []
```

```
draft_instruction_id = return_no_approval
```

R03 / 第 2 次 实际输出明细

gh-fix-ci

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R03/deepseek-v4-flash-max/2

结构状态: complete | 生成 2 | 修复 1 | HTTP 尝试 2 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

生成 2 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: LICENSE.txt, assets/github-small.svg, assets/github.png, and agents/openai.yaml are non-executable metadata/assets and were not modeled as behavior.

candidate warning: The bundled Python script is represented as a black-box acquisition; internal functions and loops are preserved in metadata.script_content.

candidate warning: Manual fallback commands are summarized as one external acquisition block; internal per-check loops and fallbacks are preserved in constraints/metadata.

Skill 层约束与上下文声明

declared_context_keys = ["repo", "pr", "approval"]

skill.constraints[0]: Prereq: authenticate with the standard GitHub CLI once (for example, run `gh auth login`), then confirm with `gh auth status` (repo + workflow scopes are typically required).

skill.constraints[1]: Use GitHub Actions checks only.

skill.constraints[2]: Treat external providers (for example Buildkite) as out of scope and report only the details URL.

skill.constraints[3]: Do not attempt Buildkite or other providers; keep the workflow lean.

skill.constraints[4]: If a plan-oriented skill (for example `create-plan`) is available, use it; otherwise draft a concise plan inline and request approval before implementing.

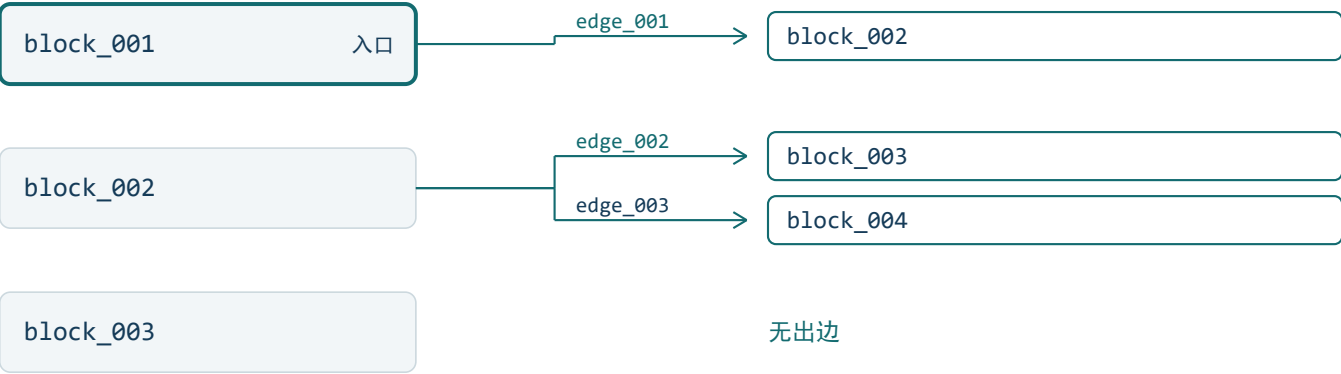
skill.constraints[5]: Implement only after explicit approval.

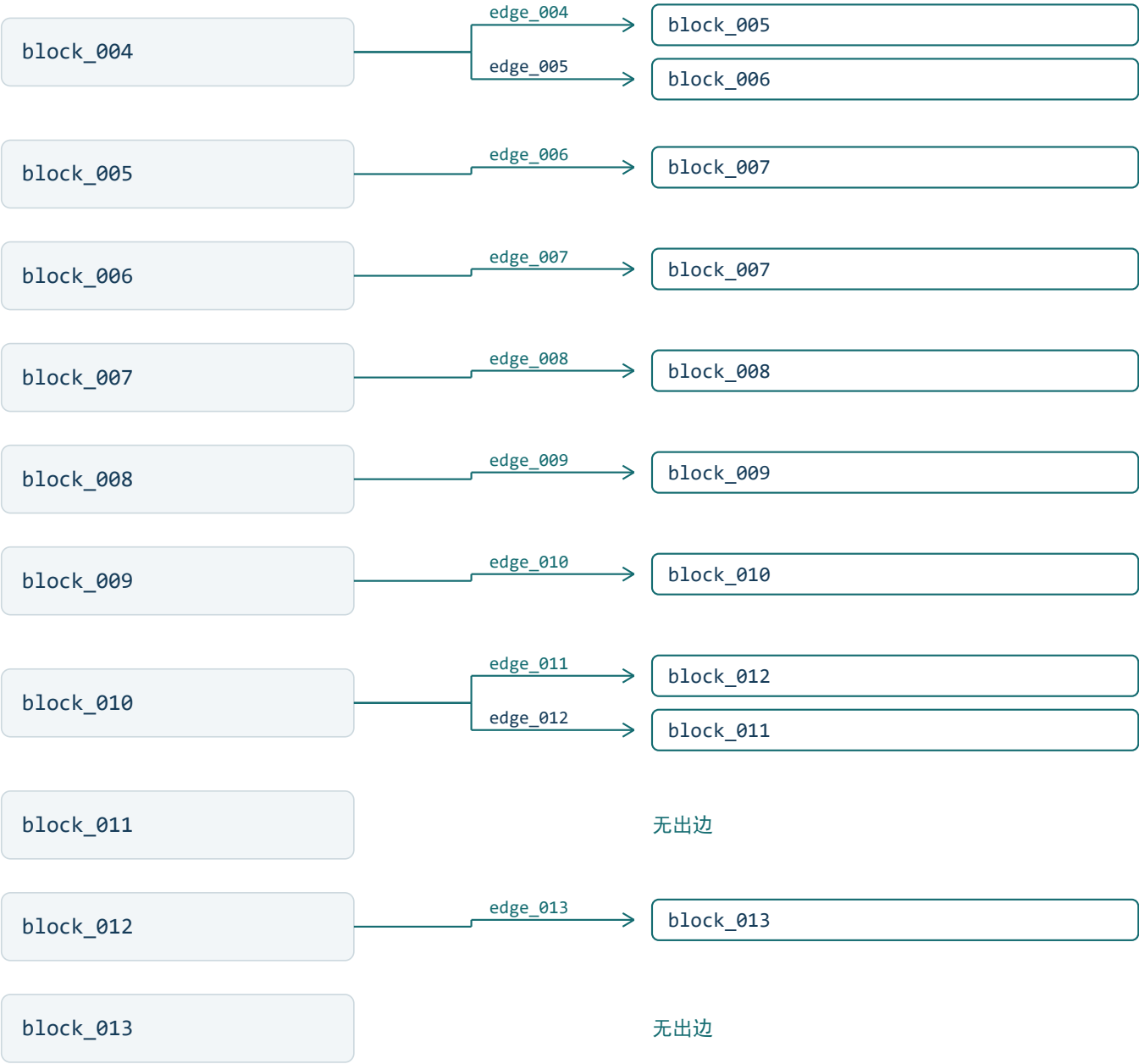
skill.constraints[6]: Provide the failing check name, run URL (if any), and a concise log snippet; call out missing logs explicitly.

skill.constraints[7]: After changes, suggest re-running the relevant tests and `gh pr checks` to confirm.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"not authenticated"
edge_003	block_002 -> block_004	"authenticated"
edge_004	block_004 -> block_005	"preferred: bundled script available"
edge_005	block_004 -> block_006	"manual fallback: bundled script unavailable or preferred method fails"
edge_006	block_005 -> block_007	null
edge_007	block_006 -> block_007	null
edge_008	block_007 -> block_008	null
edge_009	block_008 -> block_009	null
edge_010	block_009 -> block_010	null
edge_011	block_010 -> block_012	"approved"
edge_012	block_010 -> block_011	"not approved"
edge_013	block_012 -> block_013	null

全部操作、数据引用与约束落点

block_001 Read repo and PR inputs

```
data_source_kind = "context"
block_001.constraints: []

ir_001 read_repo_and_pr_inputs

inputs[0] context_key: repo | repo
inputs[1] context_key: pr | pr
outputs[0] result: result_001 | repo_path
outputs[1] result: result_002 | pr_input
ir_001.constraints: []
draft_instruction_id = read_repo_and_pr
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_verify_auth
```

block_002 Verify GitHub CLI authentication

```
data_source_kind = "external"
```

```
block_002.constraints: []
```

ir_003 run_gh_auth_status

```
inputs[0] external_resource: gh | gh
```

```
inputs[1] result: result_001 | repo_path
```

```
outputs[0] result: result_003 | authenticated
```

```
ir_003.constraints[0]: Run `gh auth status` in the repo.
```

```
draft_instruction_id = check_gh_auth
```

ir_004 dispatch

```
inputs[0] result: result_003 | authenticated
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = go_after_auth
```

block_003 Ask user to run gh auth login

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 return

```
inputs[0] literal = "Run `gh auth login` (ensuring repo + workflow scopes) before proceeding."
```

```
outputs = []
```

```
ir_005.constraints: []
```

```
draft_instruction_id = return_login_request
```

block_004 Resolve the PR number or URL

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_006 resolve_pr

```
inputs[0] external_resource: gh | gh
```

```
inputs[1] result: result_001 | repo_path
```

```
inputs[2] result: result_002 | pr_input
```

```
outputs[0] result: result_004 | pr_value
```

```
ir_006.constraints[0]: Prefer the current branch PR: `gh pr view --json number,url`.
```

```
ir_006.constraints[1]: If the user provides a PR number or URL, use that directly.
```

```
draft_instruction_id = resolve_pr_value
```

ir_007 dispatch

```
inputs[0] result: result_004 | pr_value
```

```
outputs = []
```

```
ir_007.constraints: []
```

```
draft_instruction_id = go_inspect
```

block_005 Run bundled inspect_pr_checks.py to inspect failing checks

```
data_source_kind = "external"
```

```
block_005.constraints[0]: Use GitHub Actions checks only.
```

```
block_005.constraints[1]: If `detailsUrl` is not a GitHub Actions run, label it as external and only report the URL.
```

```
block_005.constraints[2]: Do not attempt Buildkite or other providers; keep the workflow lean.
```

ir_008 inspect_pr_checks

```
inputs[0] external_resource: gh | gh
```

```
inputs[1] external_resource: scripts/inspect_pr_checks.py | scripts/inspect_pr_checks.py
```

```
inputs[2] result: result_001 | repo_path
```

```
inputs[3] result: result_004 | pr_value
```

```
outputs[0] result: result_005 | inspection_report
```

```
outputs[1] result: result_006 | inspection_exit_code
```

```
ir_008.constraints[0]: Preferred: run the bundled script (handles gh field drift and job-log fallbacks).
```

```
ir_008.constraints[1]: Add `--json` for machine-friendly output.
```

```
ir_008.constraints[2]: Exits non-zero when failures remain so it can be used in automation.
```

```
draft_instruction_id = run_inspect_pr_checks
```

```
metadata.script_content: 17215 字符; SHA256 0c6c26762073fef8c6a5112b8b9a55a363b3ecc2dfbdb0493003109ed798a308; 计数与摘要依据: JSON 序列化值的 UTF-8 (非原始脚本字节)。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。
```

ir_009 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_009.constraints: []
```

```
draft_instruction_id = go_select_inspection
```

block_006 Manually inspect failing checks with gh

```
data_source_kind = "external"
```

```
block_006.constraints[0]: Use GitHub Actions checks only.
```

```
block_006.constraints[1]: If `detailsUrl` is not a GitHub Actions run, label it as external and only report the URL.
```

```
block_006.constraints[2]: Do not attempt Buildkite or other providers; keep the workflow lean.
```

ir_010 inspect_pr_checks_manually

```
inputs[0] external_resource: gh | gh
```

```
inputs[1] result: result_001 | repo_path
```

```
inputs[2] result: result_004 | pr_value
```

```
outputs[0] result: result_007 | manual_inspection_report
```

```
ir_010.constraints[0]: Manual fallback: `gh pr checks <pr> --json name, state, bucket, link, startedAt, completedAt, workflow`.
```

```
ir_010.constraints[1]: If a field is rejected, rerun with the available fields reported by `gh`.
```

```
ir_010.constraints[2]: For each failing check, extract the run id from `detailsUrl` and run `gh run view <run_id> --json ...` and `gh run view <run_id> --log`.
```

```
ir_010.constraints[3]: If the run log says it is still in progress, fetch job logs directly: `gh api "/repos/<owner>/<repo>/actions/jobs/<job_id>/logs" > "<path>"`.
```

```
draft_instruction_id = run_manual_inspection

metadata = {"manual_commands": ["gh pr checks <pr> --json name,state,bucket,link,startedAt,completedAt,workflow", "
gh run view <run_id> --json name,workflowName,conclusion,status,url,event,headBranch,headSha", "gh run view
<run_id> --log", "gh api \"/repos/<owner>/<repo>/actions/jobs/<job_id>/logs\" > \"/<path>/\""]}

ir_011 dispatch

inputs = []

outputs = []

ir_011.constraints: []

draft_instruction_id = go_select_manual

block_007 Select the inspection result from the chosen method

data_source_kind = null

block_007.constraints: []

ir_012 select_inspection_report

inputs[0] result: result_005 | inspection_report
inputs[1] result: result_007 | manual_inspection_report
outputs[0] result: result_008 | selected_inspection_report
ir_012.constraints: []

draft_instruction_id = select_inspection_result

ir_013 dispatch

inputs = []

outputs = []

ir_013.constraints: []

draft_instruction_id = go_summarize

block_008 Summarize failures for the user

data_source_kind = null

block_008.constraints: []

ir_014 summarize_pr_check_failures

inputs[0] result: result_008 | selected_inspection_report
outputs[0] result: result_009 | failure_summary
ir_014.constraints[0]: Provide the failing check name, run URL (if any), and a concise log snippet.
ir_014.constraints[1]: Call out missing logs explicitly.
draft_instruction_id = summarize_failures

ir_015 dispatch

inputs = []

outputs = []

ir_015.constraints: []

draft_instruction_id = go_plan
```

block_009 Create a fix plan and request approval

```
data_source_kind = null
```

```
block_009.constraints: []
```

ir_016 draft_fix_plan

```
inputs[0] result: result_009 | failure_summary
```

```
outputs[0] result: result_010 | fix_plan
```

```
ir_016.constraints[0]: If a plan-oriented skill (for example `create-plan`) is available, use it; otherwise draft a concise plan inline.
```

```
draft_instruction_id = draft_fix_plan
```

ir_017 request_user_approval

```
inputs[0] result: result_010 | fix_plan
```

```
outputs = []
```

```
ir_017.constraints[0]: Request approval before implementing.
```

```
draft_instruction_id = request_approval
```

ir_018 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = go_read_approval
```

block_010 Read explicit user approval

```
data_source_kind = "context"
```

```
block_010.constraints: []
```

ir_019 read_user_approval

```
inputs[0] context_key: approval | approval
```

```
outputs[0] result: result_011 | approval_status
```

```
ir_019.constraints: []
```

```
draft_instruction_id = read_user_approval
```

ir_020 dispatch

```
inputs[0] result: result_011 | approval_status
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = go_after_approval
```

block_011 Wait for explicit approval before implementing

```
data_source_kind = null
```

```
block_011.constraints: []
```

ir_021 return

```
inputs[0] literal = "Wait for explicit approval before implementing."
```

```
outputs = []
```

```
ir_021.constraints: []
```

```
draft_instruction_id = return_awaiting_approval
```

block_012 Implement the approved plan and summarize diffs/tests

```
data_source_kind = null
```

```
block_012.constraints: []
```

ir_022 implement_approved_plan

```
inputs[0] result: result_010 | fix_plan
```

```
outputs[0] result: result_012 | implementation_summary
```

```
ir_022.constraints[0]: Implement only after explicit approval.
```

```
ir_022.constraints[1]: Apply the approved plan, summarize diffs/tests, and ask about opening a PR.
```

```
draft_instruction_id = implement_approved_plan
```

ir_023 ask_about_opening_pr

```
inputs[0] result: result_012 | implementation_summary
```

```
outputs = []
```

```
ir_023.constraints: []
```

```
draft_instruction_id = ask_open_pr
```

ir_024 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_024.constraints: []
```

```
draft_instruction_id = go_recheck
```

block_013 Suggest rechecking status

```
data_source_kind = null
```

```
block_013.constraints: []
```

ir_025 suggest_rechecking_status

```
inputs[0] literal = "re-run the relevant tests and `gh pr checks` to confirm."
```

```
outputs[0] result: result_013 | recheck_suggestion
```

```
ir_025.constraints: []
```

```
draft_instruction_id = suggest_recheck
```

ir_026 return

```
inputs[0] result: result_013 | recheck_suggestion
```

```
outputs = []
```

```
ir_026.constraints: []
```

```
draft_instruction_id = return_suggestion
```


R03 / 第 3 次 实际输出明细

gh-fix-ci

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R03/deepseek-v4-flash-max/3

结构状态: complete | 生成 3 | 修复 2 | HTTP 尝试 3 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

生成 2 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

生成 3 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: The choice between the bundled script and manual fallback is described as 'preferred' vs 'manual fallback' without an explicit predicate; modeled as a branch.

candidate warning: The manual fallback's per-failing-check loop is represented as list-based operations to avoid splitting the internal iteration as code-like control flow.

candidate warning: The user approval step is modeled as an external interaction returning an approval decision.

Skill 层约束与上下文声明

declared_context_keys = ["repo", "pr", "available_plan_skill"]

skill.constraints[0]: Use only GitHub Actions checks; treat external providers (for example Buildkite) as out of scope and report only the details URL.

skill.constraints[1]: Authenticate with the standard GitHub CLI and confirm with `gh auth status` (repo + workflow scopes are typically required).

skill.constraints[2]: If a plan-oriented skill (for example `create-plan`) is available, use it; otherwise draft a concise plan inline and request approval before implementing.

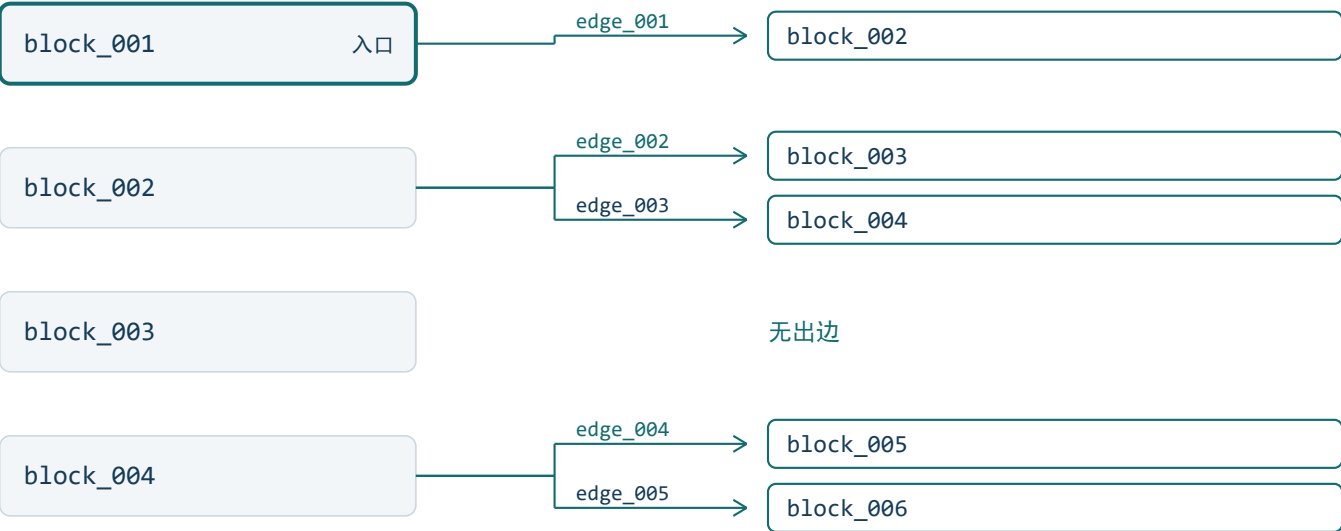
skill.constraints[3]: Implement only after explicit approval.

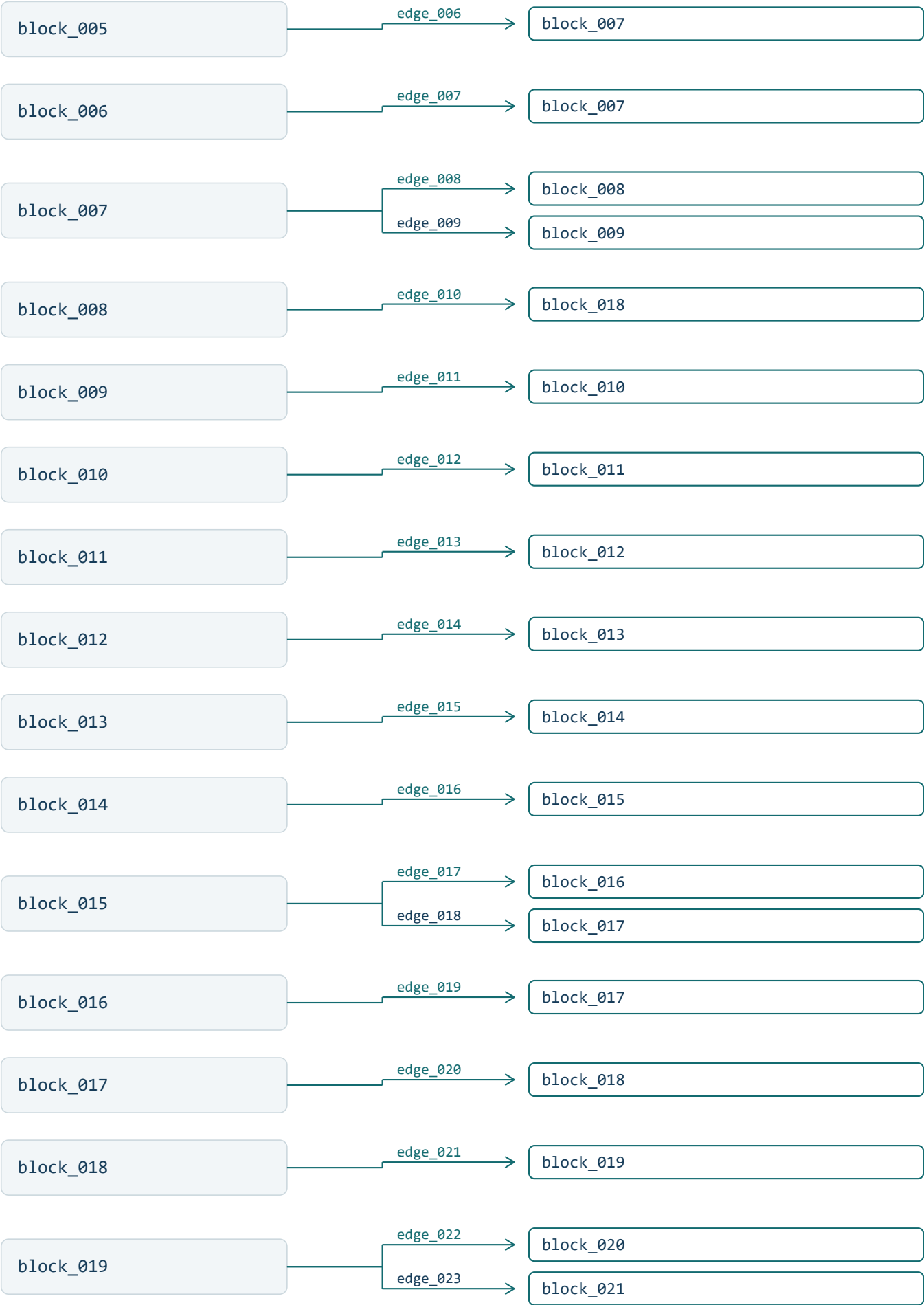
skill.constraints[4]: Call out missing logs explicitly.

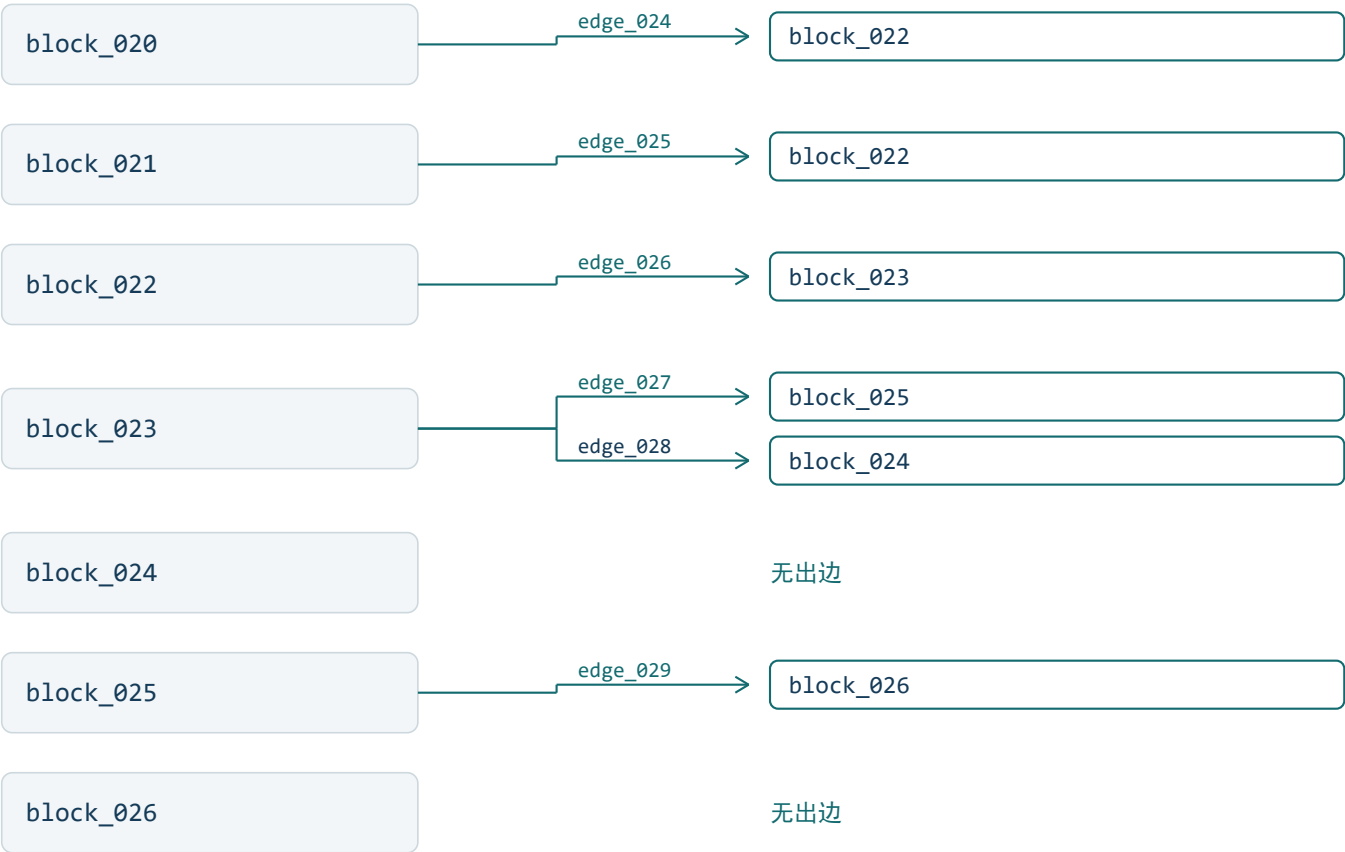
skill.constraints[5]: After changes, suggest re-running the relevant tests and `gh pr checks` to confirm.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。







完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"unauthenticated"
edge_003	block_002 -> block_004	"authenticated"
edge_004	block_004 -> block_005	"PR provided"
edge_005	block_004 -> block_006	"PR not provided"
edge_006	block_005 -> block_007	null
edge_007	block_006 -> block_007	null
edge_008	block_007 -> block_008	"preferred: bundled script"
edge_009	block_007 -> block_009	"manual fallback"
edge_010	block_008 -> block_018	null
edge_011	block_009 -> block_010	null
edge_012	block_010 -> block_011	null
edge_013	block_011 -> block_012	null
edge_014	block_012 -> block_013	null
edge_015	block_013 -> block_014	null
edge_016	block_014 -> block_015	null
edge_017	block_015 -> block_016	"log still in progress"
edge_018	block_015 -> block_017	"logs available"
edge_019	block_016 -> block_017	null
edge_020	block_017 -> block_018	null
edge_021	block_018 -> block_019	null
edge_022	block_019 -> block_020	"plan-oriented skill available"
edge_023	block_019 -> block_021	"no plan skill available"
edge_024	block_020 -> block_022	null
edge_025	block_021 -> block_022	null
edge_026	block_022 -> block_023	null
edge_027	block_023 -> block_025	"approved"

边标识	源块 -> 目标块	condition_text
edge_028	block_023 -> block_024	"not approved"
edge_029	block_025 -> block_026	null

全部操作、数据引用与约束落点

block_001 Read repo and PR inputs

```
data_source_kind = "context"
block_001.constraints: []

ir_001 read_repo_and_pr_inputs

inputs[0] context_key: repo | repo
inputs[1] context_key: pr | pr
outputs[0] result: result_001 | repo_path
outputs[1] result: result_002 | pr_input
ir_001.constraints: []
draft_instruction_id = read_inputs
```

ir_002 dispatch

```
inputs = []
outputs = []
ir_002.constraints: []
draft_instruction_id = go_verify_auth
```

block_002 Verify gh authentication

```
data_source_kind = "external"
block_002.constraints: []

ir_003 run_gh_auth_status

inputs[0] external_resource: gh | gh
inputs[1] result: result_001 | repo_path
outputs[0] result: result_003 | auth_status
ir_003.constraints[0]: repo + workflow scopes are typically required
draft_instruction_id = run_auth_status
```

ir_004 dispatch

```
inputs[0] result: result_003 | auth_status
outputs = []
ir_004.constraints: []
draft_instruction_id = go_auth_decision
```

block_003 Ask user to authenticate gh

```
data_source_kind = null
block_003.constraints: []

ir_005 ask_user_to_run_gh_auth_login
```

```
inputs[0] literal = "gh auth login with repo and workflow scopes"
```

```
outputs = []

ir_005.constraints: []

draft_instruction_id = ask_auth_login

ir_006 return

inputs[0] literal = "User must run gh auth login with repo and workflow scopes before proceeding."
outputs = []

ir_006.constraints: []

draft_instruction_id = return_auth_required

block_004 Decide whether a PR was provided

data_source_kind = null

block_004.constraints: []

ir_007 dispatch

inputs[0] result: result_002 | pr_input
outputs = []

ir_007.constraints: []

draft_instruction_id = choose_pr_source

block_005 Use the provided PR number or URL

data_source_kind = null

block_005.constraints: []

ir_008 dispatch

inputs = []

outputs = []

ir_008.constraints: []

draft_instruction_id = go_merge_pr

block_006 Resolve the current branch PR

data_source_kind = "external"

block_006.constraints: []

ir_009 read_current_branch_pr

inputs[0] external_resource: gh | gh
inputs[1] result: result_001 | repo_path
outputs[0] result: result_004 | fetched_pr
ir_009.constraints: []

draft_instruction_id = fetch_current_pr

ir_010 dispatch

inputs = []

outputs = []

ir_010.constraints: []

draft_instruction_id = go_merge_pr_from_fetch
```

block_007 Select the PR to inspect and choose inspection method

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_011 select_pr_value

```
inputs[0] result: result_002 | pr_input
```

```
inputs[1] result: result_004 | fetched_pr
```

```
outputs[0] result: result_005 | resolved_pr
```

```
ir_011.constraints: []
```

```
draft_instruction_id = select_pr
```

ir_012 select_preferred_inspection_method

```
inputs = []
```

```
outputs[0] result: result_006 | inspection_method_choice
```

```
ir_012.constraints: []
```

```
draft_instruction_id = select_inspection_method
```

ir_013 dispatch

```
inputs[0] result: result_006 | inspection_method_choice
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = choose_inspection_method
```

block_008 Run bundled inspect_pr_checks.py

```
data_source_kind = "external"
```

```
block_008.constraints: []
```

ir_014 inspect_pr_checks

```
inputs[0] external_resource: scripts/inspect_pr_checks.py | scripts/inspect_pr_checks.py
```

```
inputs[1] result: result_001 | repo_path
```

```
inputs[2] result: result_005 | resolved_pr
```

```
inputs[3] literal = true
```

```
outputs[0] result: result_007 | script_report
```

```
outputs[1] result: result_008 | script_exit_code
```

```
ir_014.constraints[0]: Exits non-zero when failures remain so it can be used in automation.
```

```
draft_instruction_id = run_script
```

metadata.script_content: 15222 字符; SHA256 9459e5b03f86785d184a83e5cd8d621b832e918ec04158a2962d867785812c6c; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_015 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_015.constraints: []
```

```
draft_instruction_id = go_select_inspection
```

block_009 Fetch PR checks manually

```
data_source_kind = "external"
```

```
block_009.constraints: []
```

ir_016 list_pr_checks

```
inputs[0] external_resource: gh | gh
```

```
inputs[1] result: result_001 | repo_path
```

```
inputs[2] result: result_005 | resolved_pr
```

```
outputs[0] result: result_009 | checks_json
```

```
ir_016.constraints[0]: If a field is rejected, rerun with the available fields reported by gh.
```

```
draft_instruction_id = fetch_checks_manual
```

ir_017 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_017.constraints: []
```

```
draft_instruction_id = go_filter_failing
```

block_010 Filter failing checks

```
data_source_kind = null
```

```
block_010.constraints: []
```

ir_018 filter_failing_checks

```
inputs[0] result: result_009 | checks_json
```

```
outputs[0] result: result_010 | failing_checks
```

```
ir_018.constraints: []
```

```
draft_instruction_id = filter_failing
```

ir_019 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_019.constraints: []
```

```
draft_instruction_id = go_scope_checks
```

block_011 Separate GitHub Actions checks from external checks

```
data_source_kind = null
```

```
block_011.constraints: []
```

ir_020 separate_github_actions_checks

```
inputs[0] result: result_010 | failing_checks
```

```
outputs[0] result: result_011 | gha_checks
```

```
outputs[1] result: result_012 | external_checks
```

```
ir_020.constraints[0]: Non-GitHub Actions checks are labeled external and only their URL is reported.
```

```
ir_020.constraints[1]: Do not attempt Buildkite or other providers.
```

```
draft_instruction_id = scope_checks
```


ir_021 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_021.constraints: []
```

```
draft_instruction_id = go_extract_run_ids
```

block_012 Extract run IDs from GitHub Actions details URLs

```
data_source_kind = null
```

```
block_012.constraints: []
```

ir_022 extract_run_ids_from_details_urls

```
inputs[0] result: result_011 | gha_checks
```

```
outputs[0] result: result_013 | run_ids
```

```
ir_022.constraints: []
```

```
draft_instruction_id = extract_run_ids
```

ir_023 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_023.constraints: []
```

```
draft_instruction_id = go_fetch_run_metadata
```

block_013 Fetch run metadata for each failing check

```
data_source_kind = "external"
```

```
block_013.constraints: []
```

ir_024 fetch_run_metadata_for_each

```
inputs[0] external_resource: gh | gh
```

```
inputs[1] result: result_001 | repo_path
```

```
inputs[2] result: result_013 | run_ids
```

```
outputs[0] result: result_014 | run_metadata_list
```

```
ir_024.constraints: []
```

```
draft_instruction_id = fetch_run_metadata
```

ir_025 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_025.constraints: []
```

```
draft_instruction_id = go_fetch_run_logs
```

block_014 Fetch run logs for each failing check

```
data_source_kind = "external"
```

```
block_014.constraints: []
```

ir_026 fetch_run_logs_for_each

```
inputs[0] external_resource: gh | gh
```

```
inputs[1] result: result_001 | repo_path
```

```
inputs[2] result: result_013 | run_ids
outputs[0] result: result_015 | run_logs
ir_026.constraints: []
draft_instruction_id = fetch_run_logs

ir_027 dispatch

inputs = []
outputs = []
ir_027.constraints: []
draft_instruction_id = go_check_pending

block_015 Detect logs still in progress

data_source_kind = null
block_015.constraints: []

ir_028 detect_pending_logs

inputs[0] result: result_015 | run_logs
outputs[0] result: result_016 | pending_logs
outputs[1] result: result_017 | completed_logs
ir_028.constraints: []
draft_instruction_id = check_pending

ir_029 dispatch

inputs[0] result: result_016 | pending_logs
outputs = []
ir_029.constraints: []
draft_instruction_id = go_pending_decision

block_016 Fetch job logs directly for pending runs

data_source_kind = "external"
block_016.constraints: []

ir_030 fetch_job_logs_for_pending

inputs[0] external_resource: gh | gh
inputs[1] result: result_001 | repo_path
inputs[2] result: result_016 | pending_logs
outputs[0] result: result_018 | job_logs
ir_030.constraints: []
draft_instruction_id = fetch_job_logs

ir_031 dispatch

inputs = []
outputs = []
ir_031.constraints: []
draft_instruction_id = go_summarize_manual
```

block_017 Summarize manual failure context

data_source_kind = null

block_017.constraints: []

ir_032 summarize_failure_context

inputs[0] result: result_012 | external_checks

inputs[1] result: result_014 | run_metadata_list

inputs[2] result: result_015 | run_logs

inputs[3] result: result_018 | job_logs

outputs[0] result: result_019 | manual_failure_summary

ir_032.constraints[0]: Call out missing logs explicitly.

draft_instruction_id = summarize_manual

ir_033 dispatch

inputs = []

outputs = []

ir_033.constraints: []

draft_instruction_id = go_select_inspection_from_manual

block_018 Select the inspection result

data_source_kind = null

block_018.constraints: []

ir_034 select_inspection_outcome

inputs[0] result: result_007 | script_report

inputs[1] result: result_019 | manual_failure_summary

outputs[0] result: result_020 | inspection_summary

ir_034.constraints: []

draft_instruction_id = select_inspection

ir_035 dispatch

inputs = []

outputs = []

ir_035.constraints: []

draft_instruction_id = go_create_plan_decision

block_019 Decide how to create the fix plan

data_source_kind = "context"

block_019.constraints: []

ir_036 read_plan_skill_availability

inputs[0] context_key: available_plan_skill | available_plan_skill

outputs[0] result: result_021 | plan_skill_available

ir_036.constraints: []

draft_instruction_id = read_plan_skill_availability

ir_037 dispatch

```
inputs[0] result: result_021 | plan_skill_available
```

```
outputs = []
```

```
ir_037.constraints: []
```

```
draft_instruction_id = choose_plan_method
```

block_020 Use create-plan skill

```
data_source_kind = "external"
```

```
block_020.constraints: []
```

ir_038 use_create_plan_skill

```
inputs[0] external_resource: create-plan | create-plan
```

```
inputs[1] result: result_020 | inspection_summary
```

```
outputs[0] result: result_022 | plan_draft
```

```
ir_038.constraints: []
```

```
draft_instruction_id = invoke_create_plan
```

ir_039 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_039.constraints: []
```

```
draft_instruction_id = go_merge_plan
```

block_021 Draft a concise plan inline

```
data_source_kind = null
```

```
block_021.constraints: []
```

ir_040 draft_concise_plan_inline

```
inputs[0] result: result_020 | inspection_summary
```

```
outputs[0] result: result_023 | inline_plan_draft
```

```
ir_040.constraints: []
```

```
draft_instruction_id = draft_plan
```

ir_041 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_041.constraints: []
```

```
draft_instruction_id = go_merge_plan_inline
```

block_022 Select the plan to request approval for

```
data_source_kind = null
```

```
block_022.constraints: []
```

ir_042 select_plan_draft

```
inputs[0] result: result_022 | plan_draft
```

```
inputs[1] result: result_023 | inline_plan_draft
```

```
outputs[0] result: result_024 | plan_to_approve
```

```
ir_042.constraints: []

draft_instruction_id = select_plan

ir_043 dispatch

inputs = []

outputs = []

ir_043.constraints: []

draft_instruction_id = go_request_approval

block_023 Request explicit approval

data_source_kind = "external"

block_023.constraints: []

ir_044 request_plan_approval

inputs[0] external_resource: user | user
inputs[1] result: result_024 | plan_to_approve
outputs[0] result: result_025 | approval_decision
ir_044.constraints[0]: Implement only after explicit approval.
draft_instruction_id = request_approval

ir_045 dispatch

inputs[0] result: result_025 | approval_decision

outputs = []

ir_045.constraints: []

draft_instruction_id = go_approval_decision

block_024 Return without implementing

data_source_kind = null

block_024.constraints: []

ir_046 return

inputs[0] literal = "Plan not approved; no changes made."

outputs = []

ir_046.constraints: []

draft_instruction_id = return_not_approved

block_025 Implement approved plan and summarize

data_source_kind = null

block_025.constraints: []

ir_047 apply_approved_plan

inputs[0] result: result_024 | plan_to_approve
outputs[0] result: result_026 | implementation_summary
ir_047.constraints: []

draft_instruction_id = apply_plan

ir_048 summarize_diffs_and_tests

inputs[0] result: result_026 | implementation_summary
```

```
outputs[0] result: result_027 | diffs_tests_summary
```

```
ir_048.constraints: []
```

```
draft_instruction_id = summarize_diffs
```

ir_049 ask_about_opening_pull_request

```
inputs[0] result: result_027 | diffs_tests_summary
```

```
outputs = []
```

```
ir_049.constraints: []
```

```
draft_instruction_id = ask_open_pr
```

ir_050 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_050.constraints: []
```

```
draft_instruction_id = go_recheck
```

block_026 Suggest rechecking status

```
data_source_kind = null
```

```
block_026.constraints: []
```

ir_051 suggest_rechecking_tests_and_checks

```
inputs[0] result: result_027 | diffs_tests_summary
```

```
outputs[0] result: result_028 | recheck_suggestion
```

```
ir_051.constraints[0]: After changes, suggest re-running the relevant tests and gh pr checks to confirm.
```

```
draft_instruction_id = suggest_recheck
```

ir_052 return

```
inputs[0] result: result_028 | recheck_suggestion
```

```
outputs = []
```

```
ir_052.constraints: []
```

```
draft_instruction_id = return_recheck
```

R04 / 第 1 次 实际输出明细

netlify-deploy

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R04/deepseek-v4-flash-max/1

结构状态: complete | 生成 2 | 修复 1 | HTTP 尝试 2 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

生成 2 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Reference documentation files (LICENSE.txt, references/cli-commands.md, references/deployment-patterns.md, references/netlify-toml.md) were read but treated as non-executable reference material; their command examples and configuration snippets are not expanded into runtime instructions.

candidate warning: agents/openai.yaml and assets/netlify-small.svg are treated as metadata and assets, not runtime behavior.

candidate warning: Branch-specific results (link_result, init_result) are intentionally referenced in choose_deploy_type after merge, per path-sensitive analysis guidance.

candidate warning: The alternative API key authentication is captured as a constraint rather than a separate control-flow branch because the source does not specify a decision predicate for switching from browser OAuth.

Skill 层约束与上下文声明

declared_context_keys = ["user_request"]

skill.constraints[0]: Netlify CLI is installed via npx (no global install required).

skill.constraints[1]: Authentication requires a Netlify account with active login session.

skill.constraints[2]: Project must be a valid web project in current directory.

skill.constraints[3]: When sandboxing blocks the deployment network calls, rerun with `sandbox\_permissions=require\_escalated`.

skill.constraints[4]: The deployment might take a few minutes. Use appropriate timeout values.

skill.constraints[5]: Never commit secrets to Git.

skill.constraints[6]: Set secrets and configuration in Netlify dashboard: Site Settings → Environment Variables.

skill.constraints[7]: Access environment variables in builds via `process.env.VARIABLE\_NAME`.

skill.constraints[8]: Authentication uses either browser-based OAuth (primary) or API key (alternative) via `NETLIFY\_AUTH\_TOKEN`.

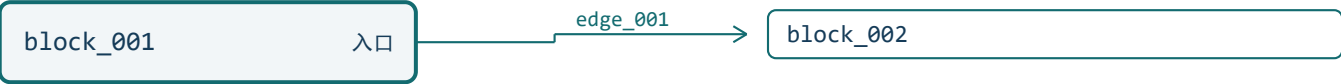
skill.constraints[9]: If a `netlify.toml` file exists, the CLI uses it automatically. If not, the CLI will prompt for build command and publish directory.

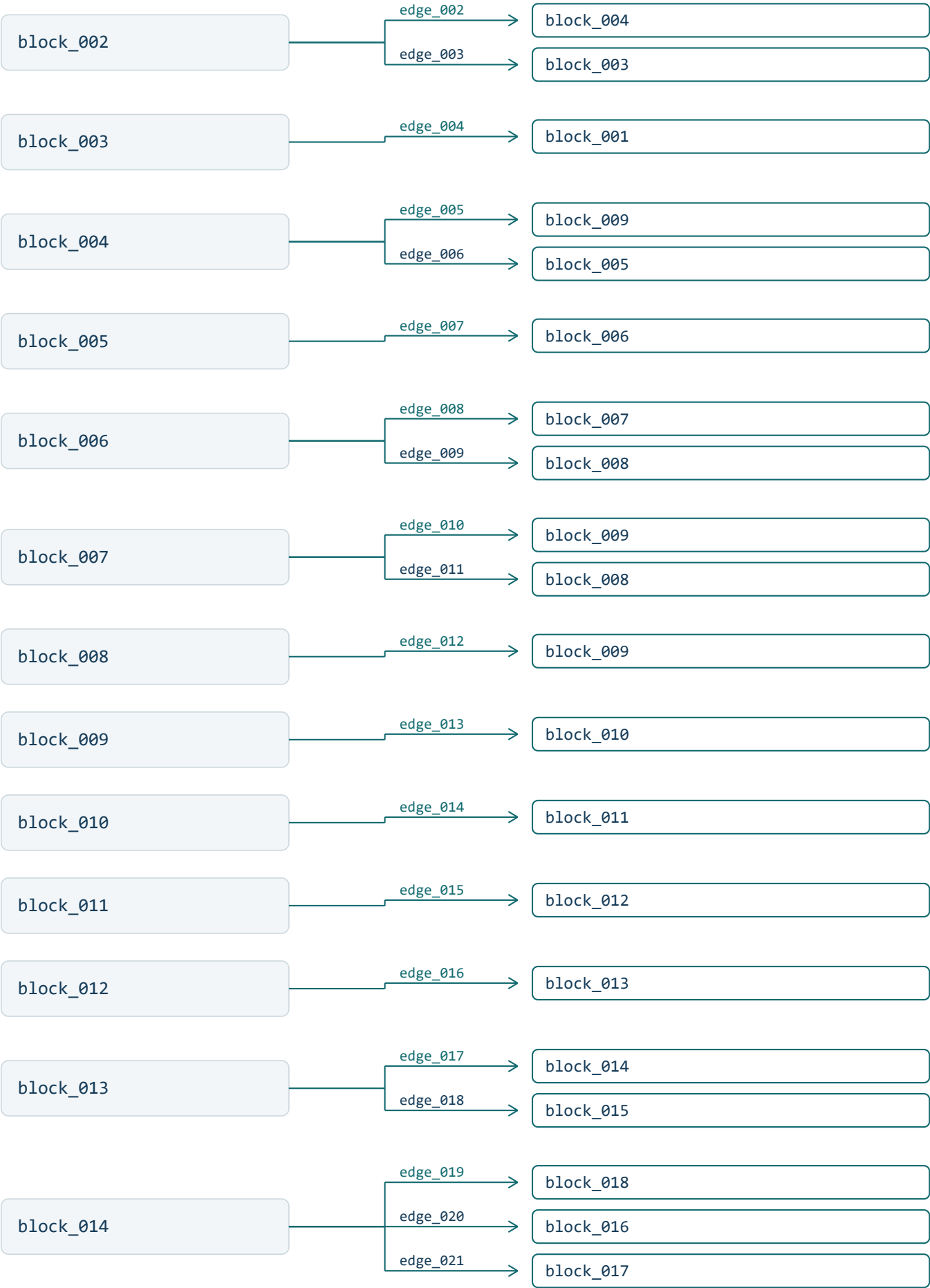
skill.constraints[10]: Common framework defaults: Next.js: build command `npm run build`, publish `.next`; React (Vite): build command `npm run build`, publish `dist`; Static HTML: no build command, publish current directory.

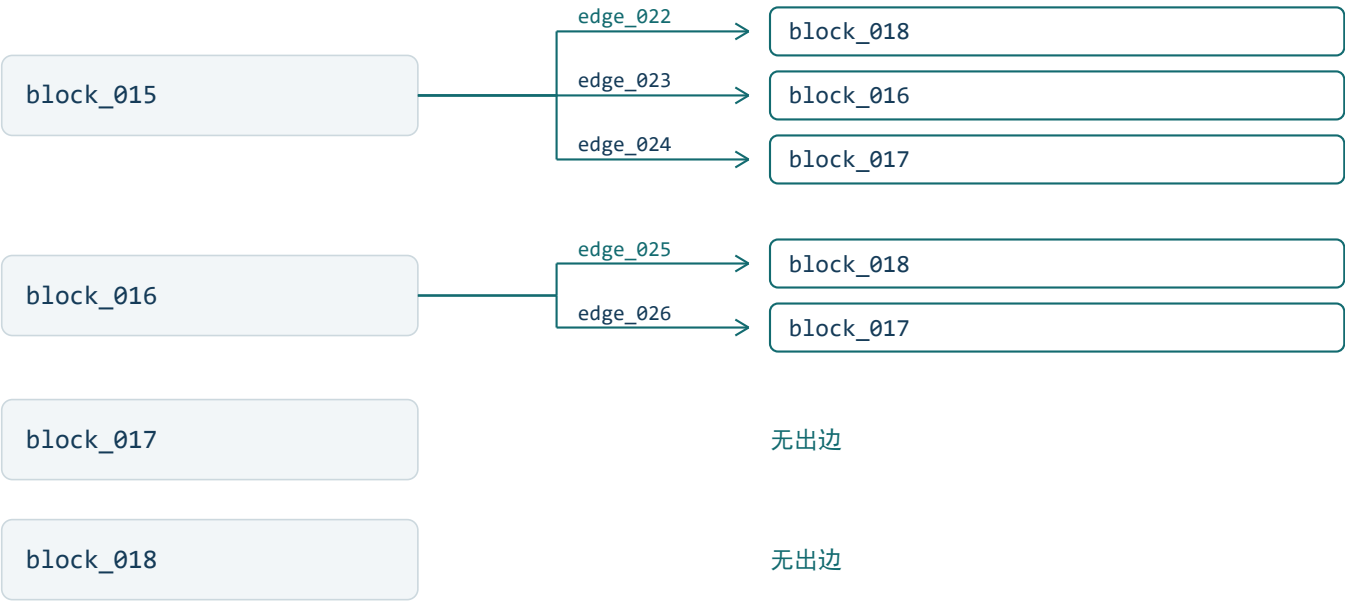
skill.constraints[11]: The skill should detect framework from `package.json` if possible and suggest appropriate settings.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。







完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_004	"authenticated"
edge_003	block_002 -> block_003	"not authenticated"
edge_004	block_003 -> block_001	null
edge_005	block_004 -> block_009	"already linked"
edge_006	block_004 -> block_005	"not linked"
edge_007	block_005 -> block_006	null
edge_008	block_006 -> block_007	"project is Git-based"
edge_009	block_006 -> block_008	"project is not Git-based"
edge_010	block_007 -> block_009	"link succeeded"
edge_011	block_007 -> block_008	"link failed"
edge_012	block_008 -> block_009	null
edge_013	block_009 -> block_010	null
edge_014	block_010 -> block_011	null
edge_015	block_011 -> block_012	null
edge_016	block_012 -> block_013	null
edge_017	block_013 -> block_014	"preview deployment"
edge_018	block_013 -> block_015	"production deployment"
edge_019	block_014 -> block_018	"deploy succeeded"
edge_020	block_014 -> block_016	"network failure"
edge_021	block_014 -> block_017	"other failure"
edge_022	block_015 -> block_018	"deploy succeeded"
edge_023	block_015 -> block_016	"network failure"
edge_024	block_015 -> block_017	"other failure"
edge_025	block_016 -> block_018	"deploy succeeded"
edge_026	block_016 -> block_017	"deploy failed"

全部操作、数据引用与约束落点

block_001 Check Netlify CLI authentication status

data_source_kind = "external"

block_001.constraints[0]: Check authentication status with `npx netlify status`.

ir_001 run_netlify_status

inputs[0] external_resource: netlify_cli | netlify_cli

outputs[0] result: result_001 | netlify_status_output

ir_001.constraints[0]: Expected output patterns: Authenticated shows logged-in user email and site link status; Not authenticated shows 'Not logged into any site' or authentication error.

draft_instruction_id = run_netlify_status

metadata.script_content: 18 字符; SHA256 3d5cb686bf74409f2481c4ba300ad3044bd6bf6ce239c2602f7160a94a527aab; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_002 dispatch

inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = dispatch_to_auth_check

block_002 Determine whether Netlify CLI is authenticated

data_source_kind = null

block_002.constraints: []

ir_003 check_netlify_authentication

inputs[0] result: result_001 | netlify_status_output

outputs[0] result: result_002 | is_authenticated

ir_003.constraints: []

draft_instruction_id = check_netlify_authentication

ir_004 dispatch

inputs[0] result: result_002 | is_authenticated

outputs = []

ir_004.constraints: []

draft_instruction_id = dispatch_on_auth

block_003 Guide user through Netlify login

data_source_kind = null

block_003.constraints[0]: This opens a browser window for OAuth authentication. Wait for user to complete login, then verify with `netlify status` again.

ir_005 run_netlify_login

inputs[0] external_resource: netlify_cli | netlify_cli

outputs = []

ir_005.constraints: []

draft_instruction_id = run_netlify_login

metadata.script_content: 17 字符; SHA256 3d9760ce8f7455d198802519b0128c843ebb23611a471c23f7726a5f92d5ede2; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_006 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = dispatch_to_recheck_auth
```

block_004 Detect site link status from Netlify status output

```
data_source_kind = null
```

```
block_004.constraints: []
```

ir_007 determine_site_link_status

```
inputs[0] result: result_001 | netlify_status_output
```

```
outputs[0] result: result_003 | site_link_status
```

```
ir_007.constraints: []
```

```
draft_instruction_id = determine_site_link_status
```

ir_008 dispatch

```
inputs[0] result: result_003 | site_link_status
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_on_site_link
```

block_005 Read Git remote origin

```
data_source_kind = "external"
```

```
block_005.constraints: []
```

ir_009 git_remote_show_origin

```
inputs[0] external_resource: git | git
```

```
outputs[0] result: result_004 | git_remote_output
```

```
ir_009.constraints: []
```

```
draft_instruction_id = git_remote_show_origin
```

metadata.script_content: 22 字符; SHA256 75a63a96ba570227d61c0bb302d382b9183d5cb4e4fa1041a26757c8f88c8586; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_to_extract_remote_url
```

block_006 Extract Git remote URL and determine if project is Git-based

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 extract_git_remote_url

```
inputs[0] result: result_004 | git_remote_output
```

```
outputs[0] result: result_005 | remote_url
```

```
outputs[1] result: result_006 | is_git_based
```

```
ir_011.constraints: []
```

```
draft_instruction_id = extract_git_remote_url
```

ir_012 dispatch

```
inputs[0] result: result_006 | is_git_based
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_on_git_based
```

block_007 Link project to existing Netlify site by Git remote URL

```
data_source_kind = "external"
```

```
block_007.constraints[0]: If link fails (site doesn't exist on Netlify), create new site interactively.
```

ir_013 netlify_link_git_remote

```
inputs[0] external_resource: netlify_cli | netlify_cli
```

```
inputs[1] result: result_005 | remote_url
```

```
outputs[0] result: result_007 | link_result
```

```
ir_013.constraints: []
```

```
draft_instruction_id = netlify_link_git_remote
```

metadata.script_content: 46 字符; SHA256 c96cc322fee0444948bb31fe510c095b3f025b04f4e28746e28d9e1367ee8933; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_014 dispatch

```
inputs[0] result: result_007 | link_result
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = dispatch_on_link_result
```

block_008 Create and link new Netlify site interactively

```
data_source_kind = "external"
```

```
block_008.constraints[0]: This guides user through choosing team/account, setting site name, configuring build settings, and creating netlify.toml if needed.
```

ir_015 run_netlify_init

```
inputs[0] external_resource: netlify_cli | netlify_cli
```

```
outputs[0] result: result_008 | init_result
```

```
ir_015.constraints: []
```

```
draft_instruction_id = run_netlify_init
```

metadata.script_content: 16 字符; SHA256 b13a47e6d4e34d993041809257f2873f691fd9ddad696078f1b322cd8ed21242; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_016 dispatch

```
inputs = []
outputs = []
ir_016.constraints: []
draft_instruction_id = dispatch_to_read_package_json
```

block_009 Read project package.json

```
data_source_kind = "external"
block_009.constraints: []
```

ir_017 read_project_package_json

```
inputs[0] external_resource: package.json | package.json
outputs[0] result: result_009 | package_json_contents
ir_017.constraints: []
draft_instruction_id = read_project_package_json
```

ir_018 dispatch

```
inputs = []
outputs = []
ir_018.constraints: []
draft_instruction_id = dispatch_to_detect_project_config
```

block_010 Detect framework and package manager from package.json

```
data_source_kind = null
block_010.constraints[0]: Common framework defaults: Next.js: build command `npm run build`, publish `.next`;
React (Vite): build command `npm run build`, publish `dist`; Static HTML: no build command, publish current
directory.
```

ir_019 detect_framework_and_package_manager

```
inputs[0] result: result_009 | package_json_contents
outputs[0] result: result_010 | framework
outputs[1] result: result_011 | package_manager
outputs[2] result: result_012 | suggested_build_settings
ir_019.constraints: []
draft_instruction_id = detect_framework_and_package_manager
```

ir_020 dispatch

```
inputs = []
outputs = []
ir_020.constraints: []
draft_instruction_id = dispatch_to_install_dependencies
```

block_011 Install project dependencies

```
data_source_kind = null
block_011.constraints[0]: Before deploying, ensure project dependencies are installed.
```

ir_021 install_project_dependencies

```
inputs[0] result: result_011 | package_manager
```

```
outputs = []
```

```
ir_021.constraints[0]: For npm projects run `npm install`; for other package managers, detect and use appropriate command.
```

```
draft_instruction_id = install_project_dependencies
```

metadata.script_content: 131 字符; SHA256 8cf3e307ab8bdbde674f770f888277923ce4f78423b7a4a0fac9ed6449c6ffb2; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_022 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_022.constraints: []
```

```
draft_instruction_id = dispatch_to_read_user_request
```

block_012 Read user request for deployment context

```
data_source_kind = "context"
```

```
block_012.constraints: []
```

ir_023 read_user_request

```
inputs[0] context_key: user_request | user_request
```

```
outputs[0] result: result_013 | user_request
```

```
ir_023.constraints: []
```

```
draft_instruction_id = read_user_request
```

ir_024 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_024.constraints: []
```

```
draft_instruction_id = dispatch_to_choose_deploy_type
```

block_013 Choose preview or production deployment

```
data_source_kind = null
```

```
block_013.constraints: []
```

ir_025 determine_deployment_type

```
inputs[0] result: result_013 | user_request
```

```
inputs[1] result: result_003 | site_link_status
```

```
inputs[2] result: result_007 | link_result
```

```
inputs[3] result: result_008 | init_result
```

```
inputs[4] result: result_010 | framework
```

```
inputs[5] result: result_012 | suggested_build_settings
```

```
outputs[0] result: result_014 | deploy_type
```

```
ir_025.constraints[0]: Preview/Draft Deploy is default for existing sites. Production Deploy is for new sites or explicit production deployments.
```

```
draft_instruction_id = determine_deployment_type
```

ir_026 dispatch

```
inputs[0] result: result_014 | deploy_type
```

```
outputs = []
```

```
ir_026.constraints: []
```

```
draft_instruction_id = dispatch_on_deploy_type
```

block_014 Deploy preview to Netlify

```
data_source_kind = "external"
```

```
block_014.constraints[0]: This creates a deploy preview with a unique URL for testing.
```

ir_027 run_netlify_deploy

```
inputs[0] external_resource: netlify_cli | netlify_cli
```

```
outputs[0] result: result_015 | preview_deploy_url
```

```
outputs[1] result: result_016 | preview_site_url
```

```
outputs[2] result: result_017 | preview_deploy_logs
```

```
outputs[3] result: result_018 | preview_deploy_status
```

```
ir_027.constraints: []
```

```
draft_instruction_id = run_netlify_deploy
```

metadata.script_content: 18 字符; SHA256 4c08ad8b00a807a2172e86222b4fc9811b95a1af5dce49a8d46843e7d51727af; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_028 dispatch

```
inputs[0] result: result_018 | preview_deploy_status
```

```
outputs = []
```

```
ir_028.constraints: []
```

```
draft_instruction_id = dispatch_on_preview_deploy_status
```

block_015 Deploy to production on Netlify

```
data_source_kind = "external"
```

```
block_015.constraints[0]: This deploys to the live production URL.
```

ir_029 run_netlify_deploy_prod

```
inputs[0] external_resource: netlify_cli | netlify_cli
```

```
outputs[0] result: result_019 | production_deploy_url
```

```
outputs[1] result: result_020 | production_site_url
```

```
outputs[2] result: result_021 | production_deploy_logs
```

```
outputs[3] result: result_022 | production_deploy_status
```

```
ir_029.constraints: []
```

```
draft_instruction_id = run_netlify_deploy_prod
```

metadata.script_content: 25 字符; SHA256 f5d40113eafa6258e3605b6523bfe8bcffea765b4060cde65e22e38a74223c4d; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_030 dispatch

```
inputs[0] result: result_022 | production_deploy_status
```

```
outputs = []
```

```
ir_030.constraints: []
```

```
draft_instruction_id = dispatch_on_production_deploy_status
```

block_016 Rerun deployment with escalated network permissions

```
data_source_kind = "external"
```

```
block_016.constraints[0]: If deployment fails due to network issues (timeouts, DNS errors, connection resets), rerun the deploy with escalated permissions (use `sandbox_permissions=require_escalated`).
```

ir_031 run_netlify_deploy_escalated

```
inputs[0] external_resource: netlify_cli | netlify_cli
```

```
inputs[1] result: result_014 | deploy_type
```

```
outputs[0] result: result_023 | escalated_deploy_url
```

```
outputs[1] result: result_024 | escalated_site_url
```

```
outputs[2] result: result_025 | escalated_deploy_logs
```

```
outputs[3] result: result_026 | escalated_deploy_status
```

```
ir_031.constraints: []
```

```
draft_instruction_id = run_netlify_deploy_escalated
```

ir_032 dispatch

```
inputs[0] result: result_026 | escalated_deploy_status
```

```
outputs = []
```

```
ir_032.constraints: []
```

```
draft_instruction_id = dispatch_on_escalated_deploy_status
```

block_017 Handle deployment failure

```
data_source_kind = null
```

```
block_017.constraints: []
```

ir_033 check_build_configuration

```
inputs[0] result: result_012 | suggested_build_settings
```

```
outputs = []
```

```
ir_033.constraints[0]: Check build command and publish directory in netlify.toml or CLI prompts.
```

```
draft_instruction_id = check_build_configuration
```

ir_034 verify_dependencies

```
inputs = []
```

```
outputs = []
```

```
ir_034.constraints[0]: Verify dependencies are installed.
```

```
draft_instruction_id = verify_dependencies
```

ir_035 review_build_logs

```
inputs[0] result: result_017 | preview_deploy_logs
```

```
inputs[1] result: result_021 | production_deploy_logs
```

```
inputs[2] result: result_025 | escalated_deploy_logs
outputs = []

ir_035.constraints[0]: Review build logs for specific errors.
draft_instruction_id = review_build_logs

ir_036 verify_publish_directory

inputs[0] result: result_012 | suggested_build_settings
outputs = []

ir_036.constraints[0]: Verify build command ran successfully. Check publish directory path is correct.
draft_instruction_id = verify_publish_directory

ir_037 format_deployment_error_report

inputs = []

outputs[0] result: result_027 | deployment_error_report
ir_037.constraints: []

draft_instruction_id = format_deployment_error_report

ir_038 return

inputs[0] result: result_027 | deployment_error_report
outputs = []

ir_038.constraints: []

draft_instruction_id = return_error

block_018 Report deployment results to user

data_source_kind = null
block_018.constraints: []

ir_039 format_deployment_report

inputs[0] result: result_015 | preview_deploy_url
inputs[1] result: result_019 | production_deploy_url
inputs[2] result: result_016 | preview_site_url
inputs[3] result: result_020 | production_site_url
inputs[4] result: result_017 | preview_deploy_logs
inputs[5] result: result_021 | production_deploy_logs
inputs[6] result: result_023 | escalated_deploy_url
inputs[7] result: result_024 | escalated_site_url
inputs[8] result: result_025 | escalated_deploy_logs
outputs[0] result: result_028 | deployment_report

ir_039.constraints[0]: Report Deploy URL, Site URL (if production deploy), Deploy logs link, and next steps
(suggest `netlify open`).

draft_instruction_id = format_deployment_report
```

ir_040 return

```
inputs[0] result: result_028 | deployment_report
```

```
outputs = []
```

```
ir_040.constraints: []
```

```
draft_instruction_id = return_deployment_report
```

R04 / 第 2 次 实际输出明细

netlify-deploy

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R04/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Non-executable files (LICENSE.txt, agents/openai.yaml, assets/*, references/*.md) were not modeled as executable behavior.

candidate warning: Environment variable setup guidance and troubleshooting recovery steps were not modeled as control flow because they are conditional advice rather than explicit workflow branches.

candidate warning: The deploy command is treated as a black-box operation; its internal build and upload steps are not expanded.

Skill 层约束与上下文声明

declared_context_keys = ["user_request"]

skill.constraints[0]: Netlify CLI is installed via npx (no global install required).

skill.constraints[1]: Authentication requires a Netlify account with an active login session.

skill.constraints[2]: The project must be a valid web project in the current directory.

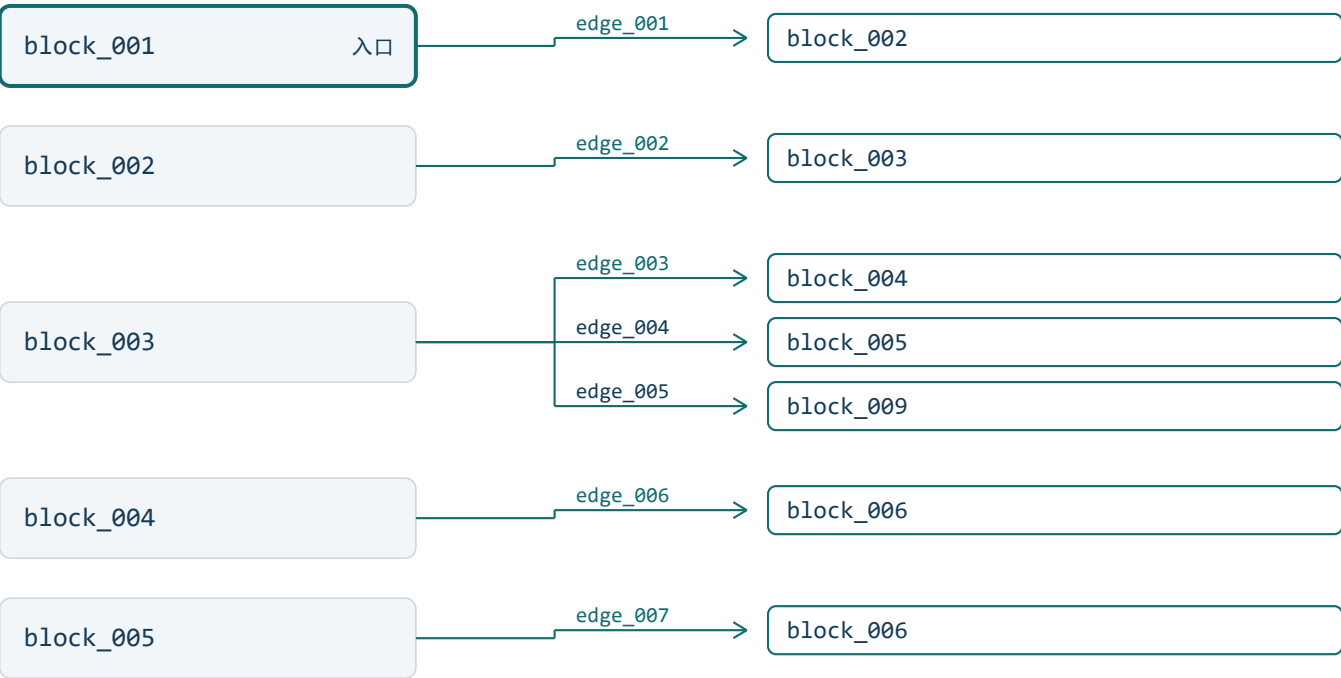
skill.constraints[3]: When sandboxing blocks deployment network calls, rerun with sandbox_permissions=require_escalated.

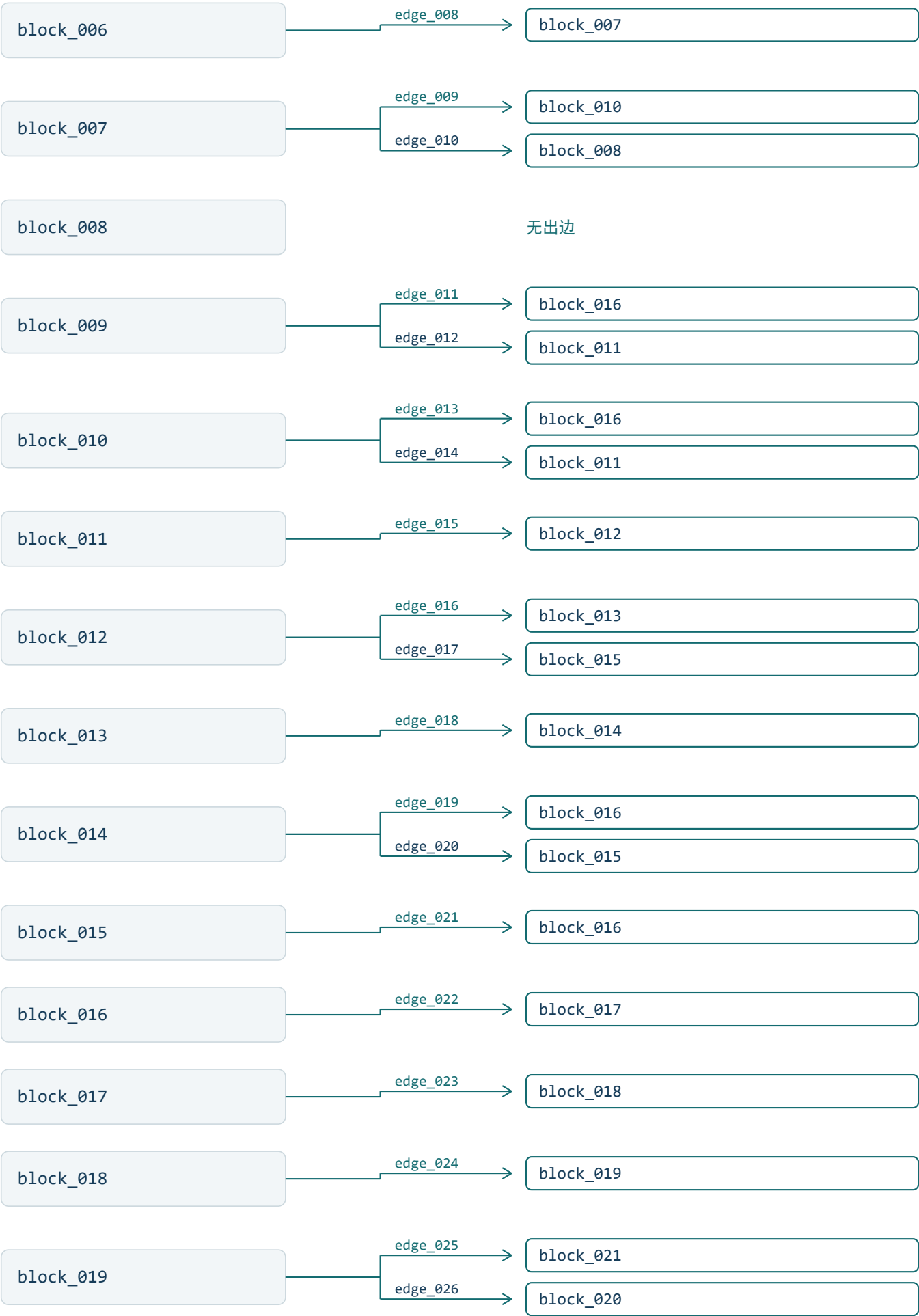
skill.constraints[4]: The deployment might take a few minutes; use appropriate timeout values.

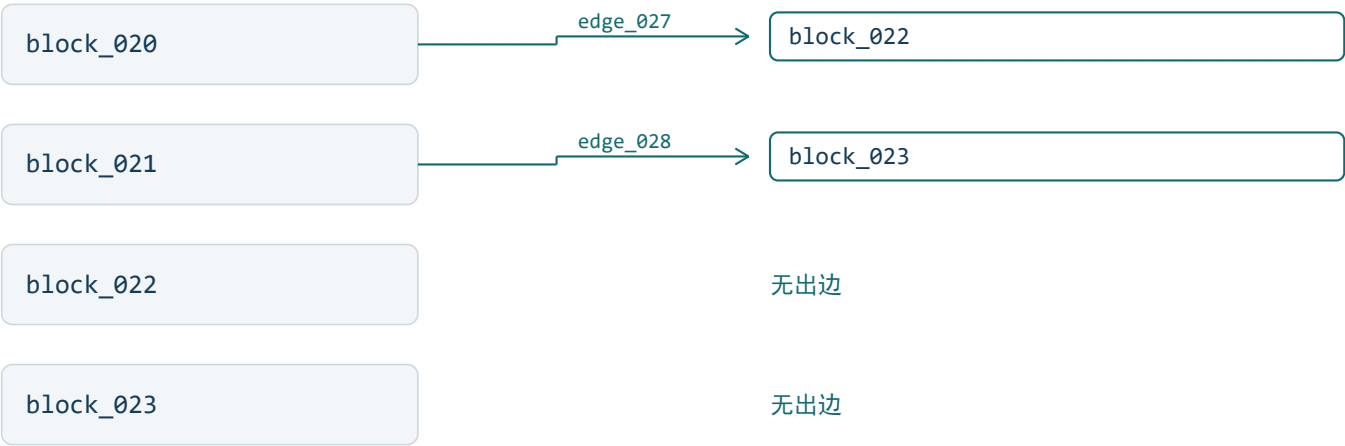
skill.constraints[5]: Never commit secrets to Git.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。







完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"not authenticated and browser authentication available"
edge_004	block_003 -> block_005	"not authenticated and browser authentication not available"
edge_005	block_003 -> block_009	"authenticated"
edge_006	block_004 -> block_006	null
edge_007	block_005 -> block_006	null
edge_008	block_006 -> block_007	null
edge_009	block_007 -> block_010	"authenticated"
edge_010	block_007 -> block_008	"still not authenticated"
edge_011	block_009 -> block_016	"already linked"
edge_012	block_009 -> block_011	"not linked"
edge_013	block_010 -> block_016	"already linked"
edge_014	block_010 -> block_011	"not linked"
edge_015	block_011 -> block_012	null
edge_016	block_012 -> block_013	"Git-based"
edge_017	block_012 -> block_015	"not Git-based"
edge_018	block_013 -> block_014	null
edge_019	block_014 -> block_016	"link succeeded"
edge_020	block_014 -> block_015	"link failed"
edge_021	block_015 -> block_016	null
edge_022	block_016 -> block_017	null
edge_023	block_017 -> block_018	null
edge_024	block_018 -> block_019	null
edge_025	block_019 -> block_021	"new site or explicit production deployment"
edge_026	block_019 -> block_020	"existing site and no explicit production deployment"

边标识	源块 -> 目标块	condition_text
edge_027	block_020 -> block_022	null
edge_028	block_021 -> block_023	null

全部操作、数据引用与约束落点

block_001 Read the user's deployment request

```
data_source_kind = "context"

block_001.constraints: []

ir_001 read_user_request

inputs[0] context_key: user_request | user_request
outputs[0] result: result_001 | deployment_request

ir_001.constraints: []

draft_instruction_id = read_deployment_request__read_request
```

ir_002 dispatch

```
inputs = []

outputs = []

ir_002.constraints: []

draft_instruction_id = read_deployment_request__go_check_status
```

block_002 Check Netlify CLI authentication and site link status

```
data_source_kind = "external"

block_002.constraints: []

ir_003 run_netlify_status

inputs[0] external_resource: netlify_cli | netlify_cli
outputs[0] result: result_002 | netlify_status_output

ir_003.constraints: []

draft_instruction_id = check_netlify_status__run_status

metadata.script_content: 18 字符; SHA256 3d5cb686bf74409f2481c4ba300ad3044bd6bf6ce239c2602f7160a94a527aab; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。
```

ir_004 dispatch

```
inputs = []

outputs = []

ir_004.constraints: []

draft_instruction_id = check_netlify_status__go_parse_status
```

block_003 Determine authentication and site link status from Netlify status output

```
data_source_kind = null

block_003.constraints[0]: Authenticated status shows logged-in user email and site link status. Not authenticated status shows 'Not logged into any site' or authentication error.
```

ir_005 parse_netlify_status

```
inputs[0] result: result_002 | netlify_status_output
```



```
outputs[0] result: result_003 | auth_status
outputs[1] result: result_004 | site_link_status_initial
outputs[2] result: result_005 | logged_in_user
ir_005.constraints: []
draft_instruction_id = parse_netlify_status__parse_status
```

ir_006 dispatch

```
inputs[0] result: result_003 | auth_status
outputs = []
ir_006.constraints: []
draft_instruction_id = parse_netlify_status__dispatch_on_auth
```

block_004 Guide user through Netlify browser login

```
data_source_kind = null
```

block_004.constraints[0]: Opens a browser window for OAuth authentication. Wait for user to complete login, then verify with netlify status again.

ir_007 run_netlify_login

```
inputs[0] external_resource: netlify_cli | netlify_cli
outputs = []
ir_007.constraints: []
draft_instruction_id = run_netlify_login__run_login
```

metadata.script_content: 17 字符; SHA256 3d9760ce8f7455d198802519b0128c843ebb23611a471c23f7726a5f92d5ede2; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_008 dispatch

```
inputs = []
outputs = []
ir_008.constraints: []
draft_instruction_id = run_netlify_login__go_verify_auth
```

block_005 Guide user to set Netlify authentication token

```
data_source_kind = null
```

block_005.constraints[0]: Tokens can be generated at <https://app.netlify.com/user/applications#personal-access-tokens>

ir_009 instruct_user_to_set_netlify_auth_token

```
inputs[0] literal = "export NETLIFY_AUTH_TOKEN=your_token_here"
inputs[1] external_resource: netlify_app_personal_access_tokens | netlify_app_personal_access_tokens
outputs = []
ir_009.constraints: []
draft_instruction_id = guide_api_key_authentication__guide_api_key
```

ir_010 dispatch

```
inputs = []
outputs = []
ir_010.constraints: []
draft_instruction_id = guide_api_key_authentication__go_verify_auth
```

block_006 Re-check Netlify authentication after login or API key setup

```
data_source_kind = "external"
block_006.constraints: []
```

ir_011 run_netlify_status

```
inputs[0] external_resource: netlify_cli | netlify_cli
outputs[0] result: result_006 | netlify_status_output_after_login
ir_011.constraints: []
draft_instruction_id = verify_auth_after_login__run_status_again
metadata.script_content: 18 字符; SHA256 3d5cb686bf74409f2481c4ba300ad3044bd6bf6ce239c2602f7160a94a527aab; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。
```

ir_012 dispatch

```
inputs = []
outputs = []
ir_012.constraints: []
draft_instruction_id = verify_auth_after_login__go_parse_status_again
```

block_007 Determine authentication status after login attempt

```
data_source_kind = null
block_007.constraints: []
```

ir_013 parse_netlify_status

```
inputs[0] result: result_006 | netlify_status_output_after_login
outputs[0] result: result_007 | auth_status_after_login
outputs[1] result: result_008 | site_link_status_after_login
ir_013.constraints: []
draft_instruction_id = parse_auth_after_login__parse_status_again
```

ir_014 dispatch

```
inputs[0] result: result_007 | auth_status_after_login
outputs = []
ir_014.constraints: []
draft_instruction_id = parse_auth_after_login__dispatch_on_auth_after_login
```

block_008 Fail gracefully if authentication cannot be established

```
data_source_kind = null
```

```
block_008.constraints: []
```

ir_015 return

```
inputs[0] literal = "Authentication with Netlify could not be established. Please log in with `npx netlify login` or set NETLIFY_AUTH_TOKEN."
```

```
outputs = []
```

```
ir_015.constraints: []
```

```
draft_instruction_id = return_auth_failure__return_error
```

block_009 Decide whether the project is already linked to a Netlify site

```
data_source_kind = null
```

```
block_009.constraints: []
```

ir_016 dispatch

```
inputs[0] result: result_004 | site_link_status_initial
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = check_site_link_initial__dispatch_on_site_link_initial
```

block_010 Decide whether the project is already linked to a Netlify site after authentication

```
data_source_kind = null
```

```
block_010.constraints: []
```

ir_017 dispatch

```
inputs[0] result: result_008 | site_link_status_after_login
```

```
outputs = []
```

```
ir_017.constraints: []
```

```
draft_instruction_id = check_site_link_after_login__dispatch_on_site_link_after_login
```

block_011 Check if the project is Git-based and read the origin remote URL

```
data_source_kind = "external"
```

```
block_011.constraints: []
```

ir_018 read_git_remote_url

```
inputs[0] external_resource: git_origin | git_origin
```

```
outputs[0] result: result_009 | git_remote_output
```

```
ir_018.constraints: []
```

```
draft_instruction_id = read_git_remote__run_git_remote
```

metadata.script_content: 22 字符; SHA256 75a63a96ba570227d61c0bb302d382b9183d5cb4e4fa1041a26757c8f88c8586; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_019 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_019.constraints: []
```

```
draft_instruction_id = read_git_remote__go_parse_git_remote
```

block_012 Extract Git remote URL and determine if project is Git-based

```
data_source_kind = null
```

```
block_012.constraints: []
```

ir_020 parse_git_remote_output

```
inputs[0] result: result_009 | git_remote_output
```

```
outputs[0] result: result_010 | git_remote_url
```

```
outputs[1] result: result_011 | is_git_based
```

```
ir_020.constraints: []
```

```
draft_instruction_id = parse_git_remote__parse_git_remote
```

ir_021 dispatch

```
inputs[0] result: result_011 | is_git_based
```

```
outputs = []
```

```
ir_021.constraints: []
```

```
draft_instruction_id = parse_git_remote__dispatch_on_git
```

block_013 Link the project to an existing Netlify site using the Git remote URL

```
data_source_kind = "external"
```

```
block_013.constraints: []
```

ir_022 run_netlify_link

```
inputs[0] external_resource: netlify_cli | netlify_cli
```

```
inputs[1] result: result_010 | git_remote_url
```

```
outputs[0] result: result_012 | link_result
```

```
ir_022.constraints: []
```

```
draft_instruction_id = link_site_by_git__run_link
```

metadata.script_content: 46 字符; SHA256 c96cc322fee0444948bb31fe510c095b3f025b04f4e28746e28d9e1367ee8933; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_023 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_023.constraints: []
```

```
draft_instruction_id = link_site_by_git__go_parse_link
```

block_014 Check if linking to the existing site succeeded

```
data_source_kind = null
```

```
block_014.constraints: []
```

ir_024 parse_netlify_link_result

```
inputs[0] result: result_012 | link_result
```

```
outputs[0] result: result_013 | link_success
```

```
ir_024.constraints: []
```

```
draft_instruction_id = parse_link_result__parse_link
```

ir_025 dispatch

```
inputs[0] result: result_013 | link_success
```

```
outputs = []
```

```
ir_025.constraints: []
```

```
draft_instruction_id = parse_link_result__dispatch_on_link
```

block_015 Create a new Netlify site interactively

```
data_source_kind = "external"
```

```
block_015.constraints[0]: Guides user through choosing team/account, setting site name, configuring build settings, and creating netlify.toml if needed.
```

ir_026 run_netlify_init

```
inputs[0] external_resource: netlify_cli | netlify_cli
```

```
outputs[0] result: result_014 | new_site_created
```

```
ir_026.constraints: []
```

```
draft_instruction_id = init_site__run_init
```

```
metadata.script_content: 16 字符; SHA256 b13a47e6d4e34d993041809257f2873f691fd9ddad696078f1b322cd8ed21242; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。
```

ir_027 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_027.constraints: []
```

```
draft_instruction_id = init_site__go_read_package
```

block_016 Read package.json to detect package manager and framework

```
data_source_kind = "external"
```

```
block_016.constraints: []
```

ir_028 read_package_json

```
inputs[0] external_resource: package_json | package_json
```

```
outputs[0] result: result_015 | package_json_contents
```

```
ir_028.constraints: []
```

```
draft_instruction_id = read_package_json__read_pkg
```

```
metadata.script_content: 16 字符; SHA256 c8679ab9e1123536bd0d5a3a8db19ad40dfc4669749ed83b5719a7b37807c7bc; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。
```

ir_029 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_029.constraints: []
```

```
draft_instruction_id = read_package_json__go_parse_pkg
```

block_017 Detect package manager, framework, and suggested build settings

```
data_source_kind = null
```

```
block_017.constraints[0]: Detect framework from package.json if possible and suggest appropriate settings. Common framework defaults: Next.js build command `npm run build`, publish `.next`; React (Vite) build command `npm run build`, publish `dist`; Static HTML no build command, publish current directory.
```

ir_030 detect_project_configuration

```
inputs[0] result: result_015 | package_json_contents
```

```
outputs[0] result: result_016 | package_manager
```

```
outputs[1] result: result_017 | framework
```

```
outputs[2] result: result_018 | suggested_build_settings
```

```
ir_030.constraints: []
```

```
draft_instruction_id = parse_package_json__parse_pkg
```

ir_031 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_031.constraints: []
```

```
draft_instruction_id = parse_package_json__go_install_deps
```

block_018 Install project dependencies with the appropriate package manager

```
data_source_kind = null
```

```
block_018.constraints[0]: For npm projects use `npm install`; for other package managers detect and use the appropriate command (yarn install, pnpm install, etc.).
```

ir_032 install_project_dependencies

```
inputs[0] result: result_016 | package_manager
```

```
outputs = []
```

```
ir_032.constraints: []
```

```
draft_instruction_id = install_dependencies__run_install
```

```
metadata.script_content: 49 字符; SHA256 83b78f9572c8dbb8285c919b33cb6f836aba925d19044aef3e5262f170384749; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。
```

ir_033 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_033.constraints: []
```

```
draft_instruction_id = install_dependencies__go_decide_deploy
```

block_019 Choose preview or production deployment based on site and request

```
data_source_kind = null
```

```
block_019.constraints: []
```

ir_034 determine_deployment_type

```
inputs[0] result: result_001 | deployment_request
```

```
inputs[1] result: result_014 | new_site_created
```

```
outputs[0] result: result_019 | deployment_type
```

```
ir_034.constraints: []
```

```
draft_instruction_id = decide_deployment_type__determine_type
```

ir_035 dispatch

```
inputs[0] result: result_019 | deployment_type
```

```
outputs = []
```

```
ir_035.constraints: []
```

```
draft_instruction_id = decide_deployment_type__dispatch_on_type
```

block_020 Deploy to Netlify preview environment

```
data_source_kind = "external"
```

```
block_020.constraints[0]: Creates a deploy preview with a unique URL for testing.
```

ir_036 run_netlify_deploy

```
inputs[0] external_resource: netlify_cli | netlify_cli
```

```
outputs[0] result: result_020 | deployment_url
```

```
outputs[1] result: result_021 | deploy_logs
```

```
ir_036.constraints: []
```

```
draft_instruction_id = preview_deploy__run_preview_deploy
```

metadata.script_content: 18 字符; SHA256 4c08ad8b00a807a2172e86222b4fc9811b95a1af5dce49a8d46843e7d51727af; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_037 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_037.constraints: []
```

```
draft_instruction_id = preview_deploy__go_report_preview
```

block_021 Deploy to Netlify production environment

```
data_source_kind = "external"
```

```
block_021.constraints[0]: Deploys to the live production URL.
```

ir_038 run_netlify_deploy_prod

```
inputs[0] external_resource: netlify_cli | netlify_cli
```

```
outputs[0] result: result_022 | deployment_url_prod
```

```
outputs[1] result: result_023 | site_url
```

```
outputs[2] result: result_024 | deploy_logs_prod
```

```
ir_038.constraints: []
```

```
draft_instruction_id = production_deploy__run_prod_deploy
```

metadata.script_content: 25 字符; SHA256 f5d40113eafa6258e3605b6523bfe8bcffea765b4060cde65e22e38a74223c4d; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_039 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_039.constraints: []
```

```
draft_instruction_id = production_deploy__go_report_prod
```

block_022 Report preview deployment URL and next steps

```
data_source_kind = null
```

```
block_022.constraints: []
```

ir_040 format_deployment_report

```
inputs[0] result: result_020 | deployment_url
```

```
inputs[1] result: result_021 | deploy_logs
```

```
inputs[2] result: result_017 | framework
```

```
inputs[3] result: result_018 | suggested_build_settings
```

```
inputs[4] literal = "npx netlify open"
```

```
outputs[0] result: result_025 | preview_report
```

```
ir_040.constraints: []
```

```
draft_instruction_id = report_preview_results__format_preview_report
```

ir_041 return

```
inputs[0] result: result_025 | preview_report
```

```
outputs = []
```

```
ir_041.constraints: []
```

```
draft_instruction_id = report_preview_results__return_preview_report
```

block_023 Report production deployment URL, site URL, and next steps

```
data_source_kind = null
```

```
block_023.constraints: []
```

ir_042 format_deployment_report

```
inputs[0] result: result_022 | deployment_url_prod
```

```
inputs[1] result: result_023 | site_url
```

```
inputs[2] result: result_024 | deploy_logs_prod
```

```
inputs[3] result: result_017 | framework
```

```
inputs[4] result: result_018 | suggested_build_settings
```

```
inputs[5] literal = "npx netlify open"
```

```
outputs[0] result: result_026 | production_report
```

```
ir_042.constraints: []
```

```
draft_instruction_id = report_production_results__format_prod_report
```

ir_043 return

```
inputs[0] result: result_026 | production_report
```

```
outputs = []
```

```
ir_043.constraints: []
```

```
draft_instruction_id = report_production_results__return_prod_report
```


R04 / 第 3 次 实际输出明细

netlify-deploy

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R04/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: LICENSE.txt, assets, and agents/openai.yaml were read but treated as non-executable metadata or static resources.

candidate warning: Reference files under references/ were read but not extracted as runtime behavior; they are bundled documentation/guidance.

candidate warning: Error Handling and Troubleshooting sections were treated as non-executable guidance rather than branches of the main deployment workflow.

Skill 层约束与上下文声明

```
declared_context_keys = ["user_request"]
```

skill.constraints[0]: Netlify CLI is installed via npx (no global install required).

skill.constraints[1]: Authentication requires a Netlify account with an active login session.

skill.constraints[2]: Project must be a valid web project in the current directory.

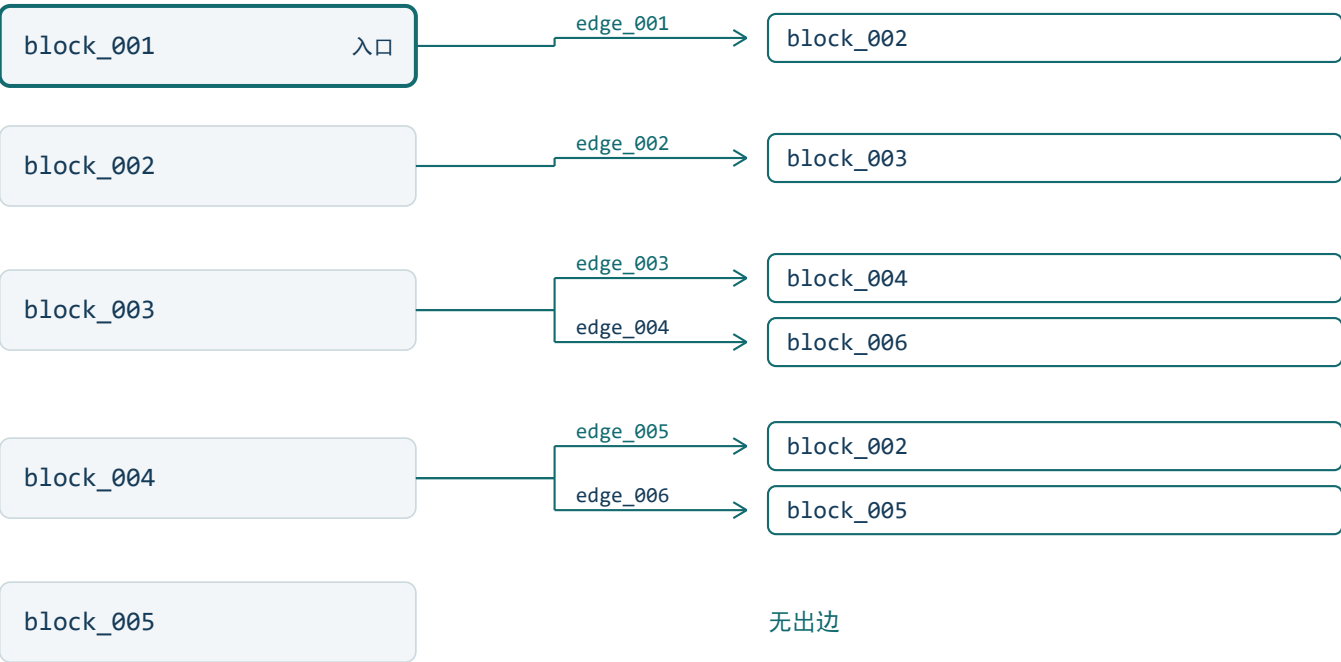
skill.constraints[3]: When sandboxing blocks the deployment network calls, rerun with `sandbox\_permissions=require\_escalated`.

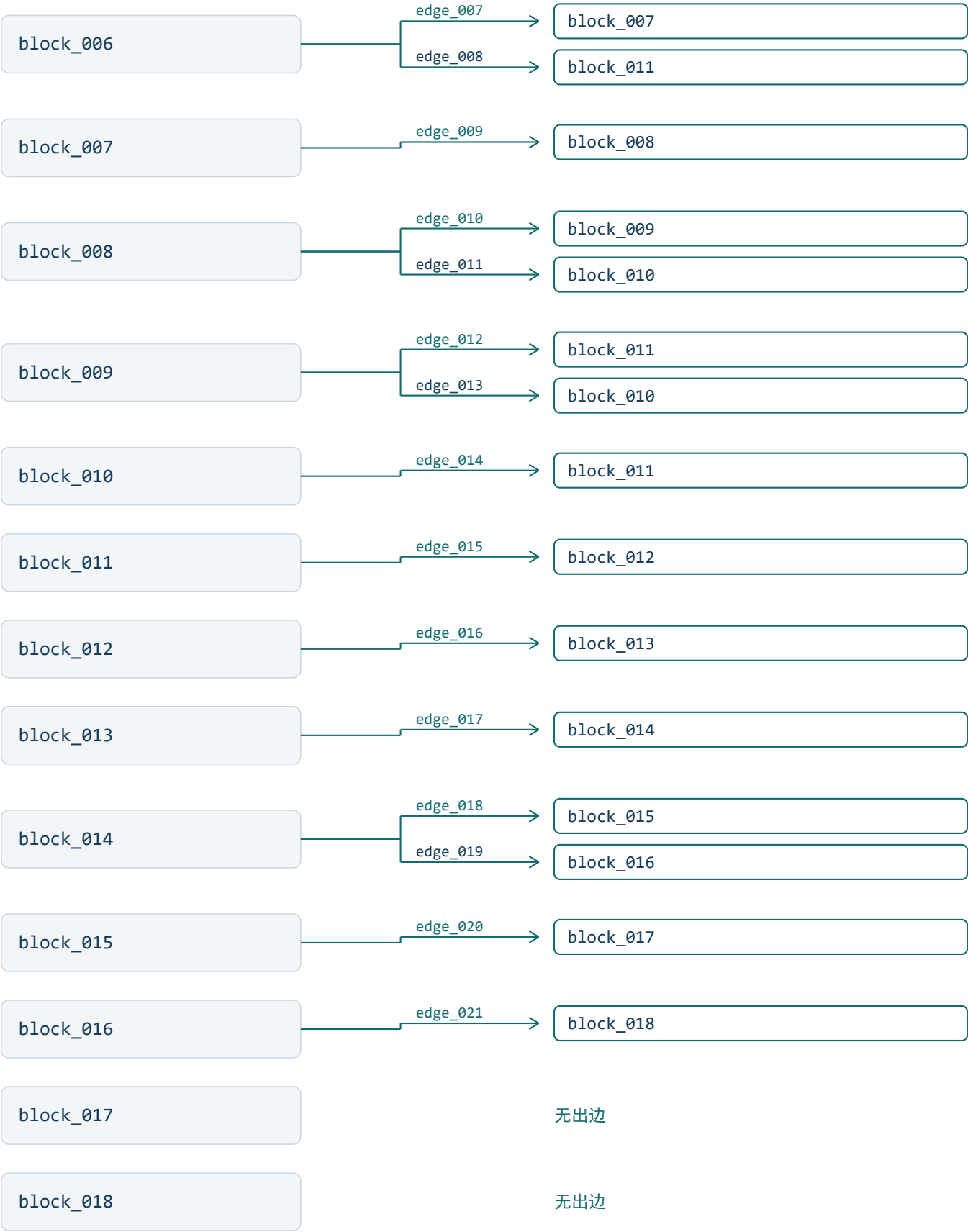
skill.constraints[4]: The deployment might take a few minutes. Use appropriate timeout values.

skill.constraints[5]: Never commit secrets to Git.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	"not authenticated"
edge_004	block_003 -> block_006	"authenticated"
edge_005	block_004 -> block_002	"login completed"
edge_006	block_004 -> block_005	"authentication cannot be established"
edge_007	block_006 -> block_007	"not linked"
edge_008	block_006 -> block_011	"linked"
edge_009	block_007 -> block_008	null
edge_010	block_008 -> block_009	"Git-based"
edge_011	block_008 -> block_010	"not Git-based"
edge_012	block_009 -> block_011	"link succeeds"
edge_013	block_009 -> block_010	"link fails"
edge_014	block_010 -> block_011	null
edge_015	block_011 -> block_012	null
edge_016	block_012 -> block_013	null
edge_017	block_013 -> block_014	null
edge_018	block_014 -> block_015	"preview mode"
edge_019	block_014 -> block_016	"production mode"
edge_020	block_015 -> block_017	null
edge_021	block_016 -> block_018	null

全部操作、数据引用与约束落点

block_001 Read the user's deployment request

data_source_kind = "context"

block_001.constraints: []

ir_001 read_user_request

inputs[0] context_key: user_request | user_request

outputs[0] result: result_001 | User request content

ir_001.constraints: []

```
draft_instruction_id = read_request
```

ir_002 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_002.constraints: []
```

```
draft_instruction_id = go_check_auth
```

block_002 Check Netlify CLI authentication status

```
data_source_kind = "external"
```

```
block_002.constraints: []
```

ir_003 run_netlify_status

```
inputs[0] external_resource: netlify_cli | Netlify CLI
```

```
outputs[0] result: result_002 | Netlify status output
```

```
ir_003.constraints[0]: Check authentication status with `npx netlify status`.
```

```
draft_instruction_id = run_status
```

metadata.script_content: 18 字符; SHA256 3d5cb686bf74409f2481c4ba300ad3044bd6bf6ce239c2602f7160a94a527aab; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_004 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = go_process_auth
```

block_003 Determine whether the user is authenticated

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 parse_netlify_authentication_status

```
inputs[0] result: result_002 | netlify_status_output
```

```
outputs[0] result: result_003 | Authentication status
```

```
ir_005.constraints: []
```

```
draft_instruction_id = parse_auth
```

ir_006 dispatch

```
inputs[0] result: result_003 | auth_status
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = branch_auth
```

block_004 Guide user through Netlify login

```
data_source_kind = "external"
```

```
block_004.constraints: []
```

ir_007 run_netlify_login

```
inputs[0] external_resource: netlify_cli | Netlify CLI
```

```
outputs[0] result: result_004 | Login completion result
```

```
ir_007.constraints[0]: This opens a browser window for OAuth authentication. Wait for user to complete login, then verify with `netlify status` again.
```

```
draft_instruction_id = run_login
```

```
metadata.script_content: 17 字符; SHA256 3d9760ce8f7455d198802519b0128c843ebb23611a471c23f7726a5f92d5ede2; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。
```

ir_008 dispatch

```
inputs[0] result: result_004 | login_result
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = branch_login
```

block_005 Fail gracefully when authentication cannot be established

```
data_source_kind = null
```

```
block_005.constraints: []
```

ir_009 return

```
inputs[0] literal = "Unable to authenticate with Netlify"
```

```
outputs = []
```

```
ir_009.constraints[0]: Fail gracefully if authentication cannot be established.
```

```
draft_instruction_id = return_auth_failure
```

block_006 Detect whether the project is linked to a Netlify site

```
data_source_kind = null
```

```
block_006.constraints[0]: From `netlify status` output, determine whether the project is Linked or Not linked.
```

ir_010 parse_netlify_site_link_status

```
inputs[0] result: result_002 | netlify_status_output
```

```
outputs[0] result: result_005 | Site link status
```

```
ir_010.constraints: []
```

```
draft_instruction_id = parse_link
```

ir_011 dispatch

```
inputs[0] result: result_005 | site_link_status
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = branch_link
```

block_007 Check if the project is Git-based

```
data_source_kind = "external"
```

```
block_007.constraints: []
```

ir_012 run_git_remote_show_origin

```
inputs[0] external_resource: git | Git
```

```
outputs[0] result: result_006 | Git remote output
```

```
ir_012.constraints[0]: Check if project is Git-based.
```

```
draft_instruction_id = run_git_remote
```

metadata.script_content: 22 字符; SHA256 75a63a96ba570227d61c0bb302d382b9183d5cb4e4fa1041a26757c8f88c8586; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_013 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = go_extract_remote
```

block_008 Extract Git remote URL

```
data_source_kind = null
```

```
block_008.constraints: []
```

ir_014 extract_git_remote_url

```
inputs[0] result: result_006 | git_remote_output
```

```
outputs[0] result: result_007 | Git remote URL
```

```
outputs[1] result: result_008 | Whether project is Git-based
```

```
ir_014.constraints[0]: If Git-based, extract remote URL.
```

```
draft_instruction_id = extract_remote
```

ir_015 dispatch

```
inputs[0] result: result_008 | is_git_based
```

```
outputs = []
```

```
ir_015.constraints: []
```

```
draft_instruction_id = branch_git
```

block_009 Link project to existing Netlify site by Git remote

```
data_source_kind = "external"
```

```
block_009.constraints: []
```

ir_016 run_netlify_link

```
inputs[0] external_resource: netlify_cli | Netlify CLI
```

```
inputs[1] result: result_007 | remote_url
```

```
outputs[0] result: result_009 | Netlify link result
```

```
ir_016.constraints[0]: Try to link by Git remote.
```

```
draft_instruction_id = run_link
```

metadata.script_content: 46 字符; SHA256 c96cc322fee0444948bb31fe510c095b3f025b04f4e28746e28d9e1367ee8933; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_017 dispatch

```
inputs[0] result: result_009 | link_result
```

```
outputs = []
```

```
ir_017.constraints: []
```

```
draft_instruction_id = branch_link_result
```

block_010 Create and link a new Netlify site

data_source_kind = "external"

block_010.constraints: []

ir_018 run_netlify_init

inputs[0] external_resource: netlify_cli | Netlify CLI

outputs[0] result: result_010 | Site creation result

ir_018.constraints[0]: This guides user through choosing team/account, setting site name, configuring build settings, and creating `netlify.toml` if needed.

draft_instruction_id = run_init

metadata.script_content: 16 字符; SHA256 b13a47e6d4e34d993041809257f2873f691fd9ddad696078f1b322cd8ed21242; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_019 dispatch

inputs = []

outputs = []

ir_019.constraints: []

draft_instruction_id = go_read_manifest

block_011 Detect package manager and framework from project files

data_source_kind = "external"

block_011.constraints: []

ir_020 read_project_manifest

inputs[0] external_resource: package_json | package.json

outputs[0] result: result_011 | Package manager

outputs[1] result: result_012 | Detected framework

ir_020.constraints[0]: The skill should detect framework from `package.json` if possible and suggest appropriate settings.

draft_instruction_id = read_manifest

ir_021 dispatch

inputs = []

outputs = []

ir_021.constraints: []

draft_instruction_id = go_install

block_012 Install project dependencies

data_source_kind = null

block_012.constraints: []

ir_022 install_project_dependencies

inputs[0] result: result_011 | package_manager

outputs = []

ir_022.constraints[0]: Before deploying, ensure project dependencies are installed.

draft_instruction_id = install_deps

metadata.script_content: 131 字符; SHA256 8cf3e307ab8bdbde674f770f888277923ce4f78423b7a4a0fac9ed6449c6ffb2; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_023 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_023.constraints: []
```

```
draft_instruction_id = go_suggest_build
```

block_013 Suggest Netlify build settings based on framework

```
data_source_kind = null
```

```
block_013.constraints: []
```

ir_024 suggest_netlify_build_settings

```
inputs[0] result: result_012 | framework
```

```
outputs[0] result: result_013 | Suggested build settings
```

```
ir_024.constraints[0]: Common framework defaults: Next.js build command `npm run build`, publish `.next`; React (Vite) build command `npm run build`, publish `dist`; Static HTML no build command, publish current directory.
```

```
draft_instruction_id = suggest_build
```

ir_025 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_025.constraints: []
```

```
draft_instruction_id = go_select_deploy
```

block_014 Choose preview or production deployment

```
data_source_kind = null
```

```
block_014.constraints: []
```

ir_026 choose_netlify_deploy_mode

```
inputs[0] result: result_001 | user_request_content
```

```
inputs[1] result: result_005 | site_link_status
```

```
inputs[2] result: result_010 | site_creation_result
```

```
outputs[0] result: result_014 | Deployment mode
```

```
ir_026.constraints[0]: Preview/Draft Deploy (default for existing sites). Production Deploy (for new sites or explicit production deployments).
```

```
draft_instruction_id = choose_deploy_mode
```

ir_027 dispatch

```
inputs[0] result: result_014 | deploy_mode
```

```
outputs = []
```

```
ir_027.constraints: []
```

```
draft_instruction_id = branch_deploy_mode
```


block_015 Deploy preview to Netlify

```
data_source_kind = "external"
```

```
block_015.constraints: []
```

ir_028 run_netlify_deploy_preview

```
inputs[0] external_resource: netlify_cli | Netlify CLI
```

```
outputs[0] result: result_015 | Preview deployment result
```

```
ir_028.constraints[0]: This creates a deploy preview with a unique URL for testing.
```

```
ir_028.constraints[1]: CLI detects build settings (from `netlify.toml` or prompts user).
```

```
draft_instruction_id = run_deploy_preview
```

metadata.script_content: 18 字符; SHA256 4c08ad8b00a807a2172e86222b4fc9811b95a1af5dce49a8d46843e7d51727af; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_029 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_029.constraints: []
```

```
draft_instruction_id = go_report_preview
```

block_016 Deploy to Netlify production

```
data_source_kind = "external"
```

```
block_016.constraints: []
```

ir_030 run_netlify_deploy_production

```
inputs[0] external_resource: netlify_cli | Netlify CLI
```

```
outputs[0] result: result_016 | Production deployment result
```

```
outputs[1] result: result_017 | Production site URL
```

```
ir_030.constraints[0]: This deploys to the live production URL.
```

```
ir_030.constraints[1]: CLI detects build settings (from `netlify.toml` or prompts user).
```

```
draft_instruction_id = run_deploy_production
```

metadata.script_content: 25 字符; SHA256 f5d40113eafa6258e3605b6523bfe8bcffea765b4060cde65e22e38a74223c4d; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_031 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_031.constraints: []
```

```
draft_instruction_id = go_report_production
```

block_017 Report preview deployment results

```
data_source_kind = null
```

```
block_017.constraints: []
```

ir_032 report_deployment_results

```
inputs[0] result: result_015 | deploy_preview_result
```

```
inputs[1] result: result_013 | build_settings
```

```
inputs[2] literal = "preview"
```

```
outputs = []

ir_032.constraints[0]: After deployment, report to user: Deploy URL, Site URL (if production deploy), Deploy logs,
Next steps.

draft_instruction_id = report_preview_result

ir_033 return

inputs[0] result: result_015 | deploy_preview_result
outputs = []

ir_033.constraints: []

draft_instruction_id = return_preview

block_018 Report production deployment results

data_source_kind = null

block_018.constraints: []

ir_034 report_deployment_results

inputs[0] result: result_016 | deploy_production_result
inputs[1] result: result_017 | production_site_url
inputs[2] result: result_013 | build_settings
inputs[3] literal = "production"

outputs = []

ir_034.constraints[0]: After deployment, report to user: Deploy URL, Site URL (if production deploy), Deploy logs,
Next steps.

draft_instruction_id = report_production_result

ir_035 return

inputs[0] result: result_016 | deploy_production_result
outputs = []

ir_035.constraints: []

draft_instruction_id = return_production
```

R05 / 第 1 次 实际输出明细

linear

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R05/deepseek-v4-flash-max/1

结构状态: complete | 生成 3 | 修复 2 | HTTP 尝试 3 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

生成 2 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

生成 3 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: The source lists available Linear MCP tools but the practical workflows do not explicitly bind each step to a specific tool; opcodes are semantic descriptions of the workflow actions.

candidate warning: The context key 'user_request' is inferred from the skill trigger because the source does not name an explicit runtime context key.

candidate warning: The practical workflows are treated as selectable behavior branches because Step 2 instructs selecting the appropriate workflow from that list.

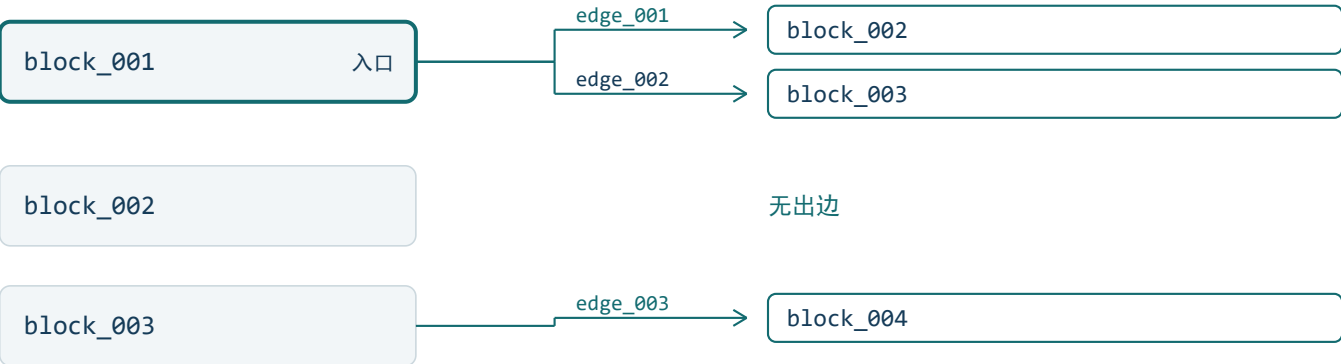
Skill 层约束与上下文声明

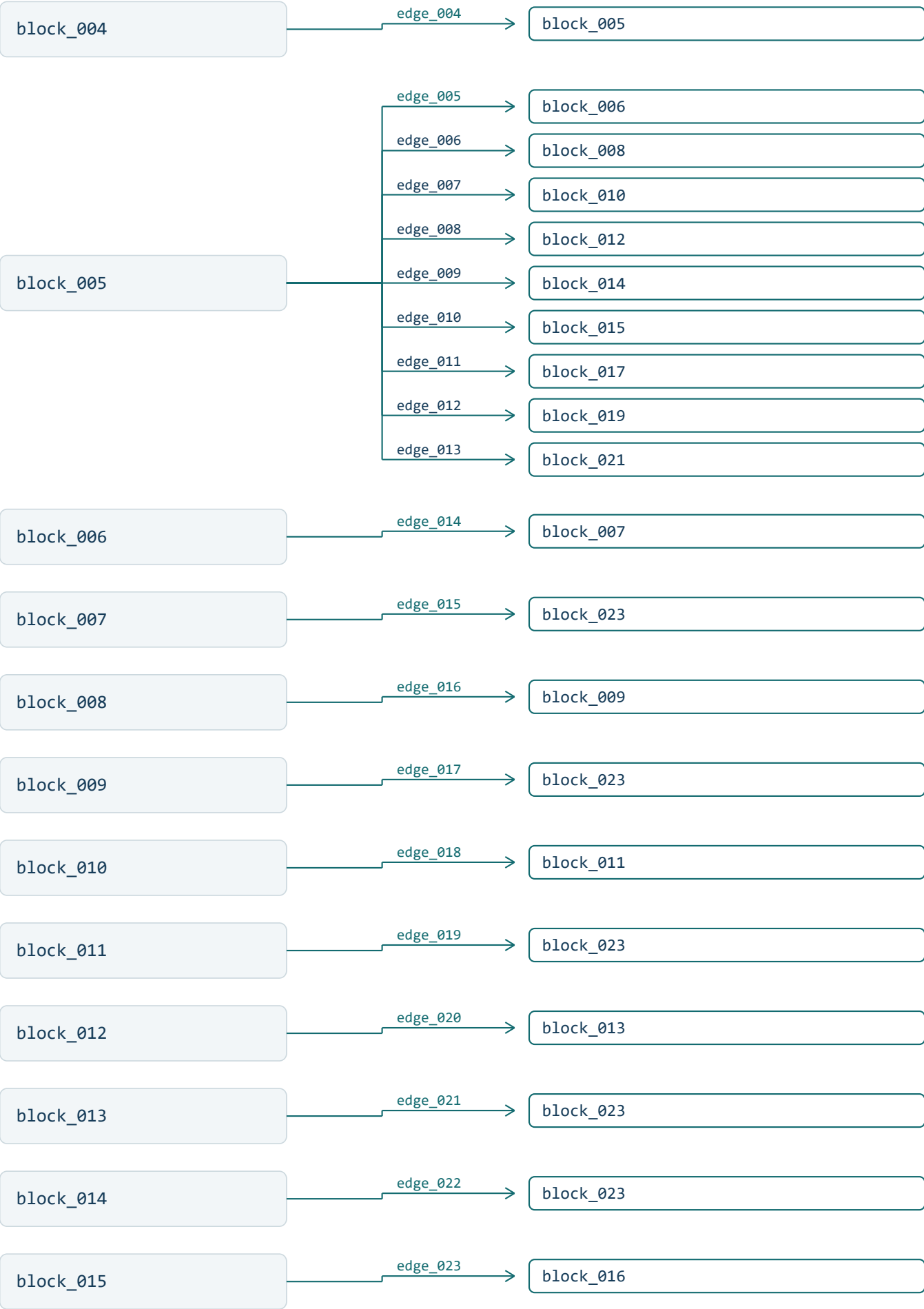
```
declared_context_keys = ["user_request"]

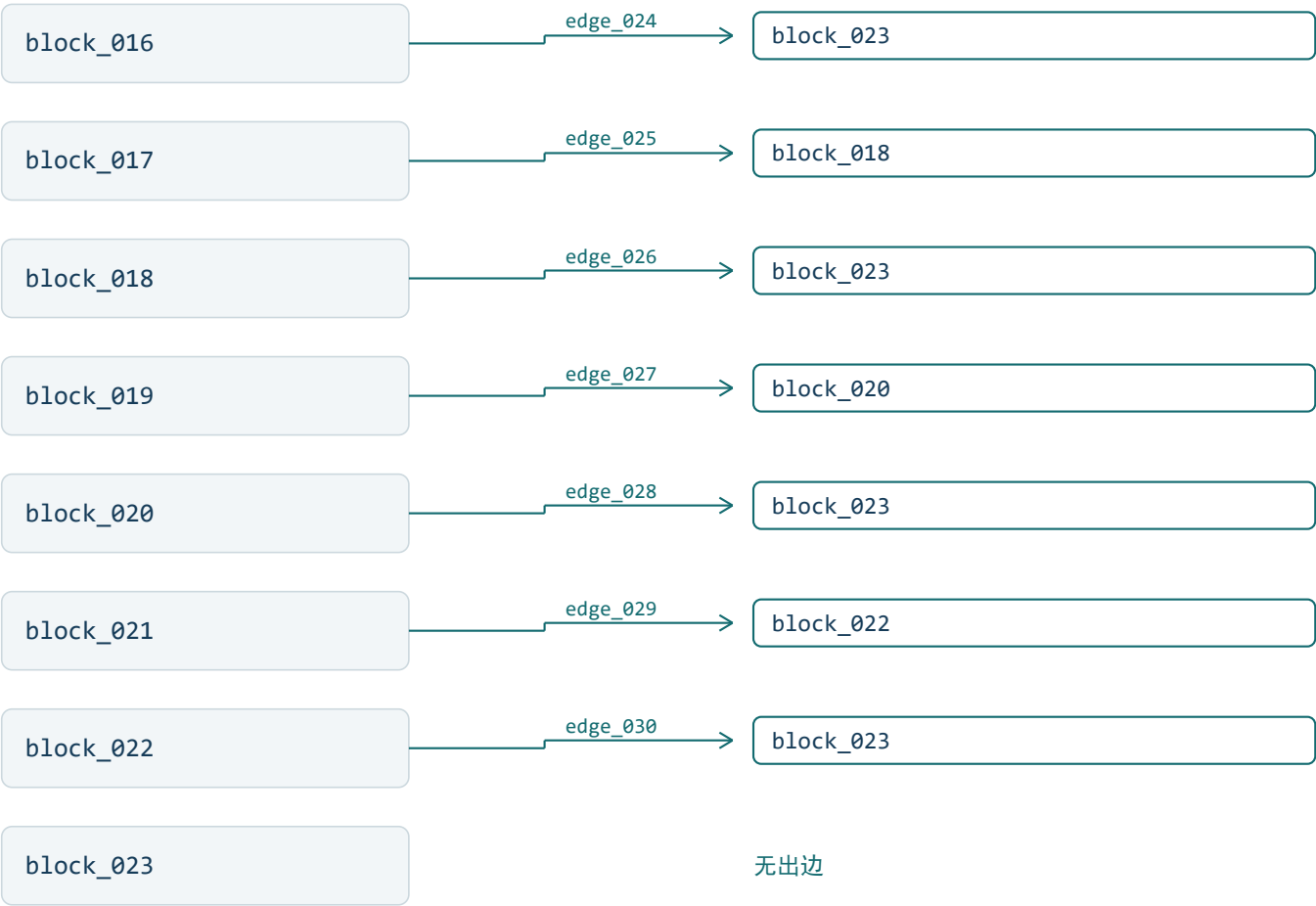
skill.constraints[0]: Linear MCP server must be connected and accessible via OAuth.
skill.constraints[1]: Confirm access to the relevant Linear workspace, teams, and projects.
skill.constraints[2]: Follow these steps in order. Do not skip steps.
skill.constraints[3]: Read first (list/get/search) to build context.
skill.constraints[4]: Create or update next (issues, projects, labels, comments) with all required fields.
skill.constraints[5]: For bulk operations, explain the grouping logic before applying changes.
skill.constraints[6]: Batch operations for related changes; consider smart templates for recurring issue structures.
skill.constraints[7]: Use natural queries when possible ("Show me what John is working on this week").
skill.constraints[8]: Leverage context: reference prior issues in new requests.
skill.constraints[9]: Break large updates into smaller batches to avoid rate limits; cache or reuse filters when listing frequently.
```

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。







完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	"not connected"
edge_002	block_001 -> block_003	"connected"
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	"Sprint Planning"
edge_006	block_005 -> block_008	"Bug Triage"
edge_007	block_005 -> block_010	"Documentation Audit"
edge_008	block_005 -> block_012	"Team Workload Balance"
edge_009	block_005 -> block_014	"Release Planning"
edge_010	block_005 -> block_015	"Cross-Project Dependencies"
edge_011	block_005 -> block_017	"Automated Status Updates"
edge_012	block_005 -> block_019	"Smart Labeling"
edge_013	block_005 -> block_021	"Sprint Retrospectives"
edge_014	block_006 -> block_007	null
edge_015	block_007 -> block_023	null
edge_016	block_008 -> block_009	null
edge_017	block_009 -> block_023	null
edge_018	block_010 -> block_011	null
edge_019	block_011 -> block_023	null
edge_020	block_012 -> block_013	null
edge_021	block_013 -> block_023	null
edge_022	block_014 -> block_023	null
edge_023	block_015 -> block_016	null
edge_024	block_016 -> block_023	null
edge_025	block_017 -> block_018	null
edge_026	block_018 -> block_023	null
edge_027	block_019 -> block_020	null

边标识	源块 -> 目标块	condition_text
edge_028	block_020 -> block_023	null
edge_029	block_021 -> block_022	null
edge_030	block_022 -> block_023	null

全部操作、数据引用与约束落点

block_001 Check Linear MCP connection

```
data_source_kind = "external"

block_001.constraints: []

ir_001 check_linear_mcp_connection

inputs[0] external_resource: linear_mcp_server | Linear MCP server
outputs[0] result: result_001 | MCP connection status

ir_001.constraints: []

draft_instruction_id = check_mcp
```

ir_002 dispatch

```
inputs[0] result: result_001 | mcp_connection_status
outputs = []

ir_002.constraints: []

draft_instruction_id = dispatch_from_mcp_check
```

block_002 Set up Linear MCP

```
data_source_kind = null

block_002.constraints[0]: After successful login, the user will have to restart codex. You should finish your answer and tell them so when they try again they can continue with Step 1.

block_002.constraints[1]: If you see connection errors on Windows, try configuring the Linear MCP to run via WSL.
```

ir_003 add_linear_mcp_server

```
inputs[0] literal = "linear"
inputs[1] literal = "https://mcp.linear.app/mcp"
outputs = []

ir_003.constraints: []

draft_instruction_id = add_linear_mcp

metadata.script_content: 53 字符; SHA256 982f5440da2578106d3e0d9ab1a16b801fa50933976565759ca2c5a0fd0e454b; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。
```

ir_004 enable_codex_remote_mcp_client

```
inputs[0] literal = "rmcp_client = true"
outputs = []

ir_004.constraints: []

draft_instruction_id = enable_remote_mcp

metadata.script_content: 82 字符; SHA256 14354a6bde4646d357d62493ee7035a3a946c7ef9e843f79331f18d6019c4cbb; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。
```

ir_005 login_linear_oauth

inputs[0] external_resource: linear_mcp_server | Linear MCP server

outputs = []

ir_005.constraints: []

draft_instruction_id = login_linear_oauth

metadata.script_content: 22 字符; SHA256 922605f9f096d319f8f2a4d32a26399f27018b8a95b8f2f5be35e5455bc89809; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_006 return

inputs[0] literal = "Linear MCP setup complete. Restart Codex, then continue with Step 1."

outputs = []

ir_006.constraints: []

draft_instruction_id = return_setup_message

block_003 Read the user's request

data_source_kind = "context"

block_003.constraints: []

ir_007 read_user_request

inputs[0] context_key: user_request | user_request

outputs[0] result: result_002 | raw user request

ir_007.constraints: []

draft_instruction_id = read_user_request

ir_008 dispatch

inputs = []

outputs = []

ir_008.constraints: []

draft_instruction_id = dispatch_to_clarify

block_004 Clarify the user's goal and scope

data_source_kind = null

block_004.constraints[0]: Confirm team/project, priority, labels, cycle, and due dates as needed.

ir_009 clarify_user_goal_and_scope

inputs[0] result: result_002 | raw_user_request

outputs[0] result: result_003 | clarified user goal

outputs[1] result: result_004 | confirmed team/project, priority, labels, cycle, and due dates

ir_009.constraints: []

draft_instruction_id = clarify_user_goal

ir_010 dispatch

inputs = []

outputs = []

ir_010.constraints: []

draft_instruction_id = dispatch_to_select

block_005 Confirm identifiers and select the Linear workflow

```
data_source_kind = null
```

```
block_005.constraints[0]: Confirm required identifiers (issue ID, project ID, team key) before calling tools.
```

ir_011 confirm_required_linear_identifiers

```
inputs[0] result: result_003 | clarified_goal
```

```
inputs[1] result: result_004 | confirmed_scope_parameters
```

```
outputs[0] result: result_005 | confirmed issue/project/team identifiers
```

```
ir_011.constraints: []
```

```
draft_instruction_id = confirm_identifiers
```

ir_012 select_linear_workflow_and_tools

```
inputs[0] result: result_003 | clarified_goal
```

```
inputs[1] result: result_005 | confirmed_identifiers
```

```
outputs[0] result: result_006 | selected practical workflow
```

```
outputs[1] result: result_007 | required Linear MCP tools
```

```
ir_012.constraints: []
```

```
draft_instruction_id = select_workflow
```

ir_013 dispatch

```
inputs[0] result: result_006 | selected_workflow
```

```
outputs = []
```

```
ir_013.constraints: []
```

```
draft_instruction_id = dispatch_to_workflow
```

block_006 List open issues for the target team

```
data_source_kind = "external"
```

```
block_006.constraints: []
```

ir_014 list_open_issues_for_team

```
inputs[0] external_resource: linear_mcp_server | Linear MCP server
```

```
inputs[1] result: result_005 | confirmed_identifiers
```

```
outputs[0] result: result_008 | open issues for target team
```

```
ir_014.constraints: []
```

```
draft_instruction_id = list_sprint_issues
```

ir_015 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_015.constraints: []
```

```
draft_instruction_id = dispatch_to_sprint_plan
```

block_007 Pick top issues and create a cycle with assignments

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_016 pick_top_issues_by_priority

```
inputs[0] result: result_008 | sprint_open_issues
```

```
outputs[0] result: result_009 | top priority issues
```

```
ir_016.constraints: []
```

```
draft_instruction_id = pick_top_sprint_issues
```

ir_017 create_linear_cycle

```
inputs[0] result: result_009 | sprint_top_issues
```

```
inputs[1] result: result_005 | confirmed_identifiers
```

```
outputs[0] result: result_010 | created cycle
```

```
ir_017.constraints: []
```

```
draft_instruction_id = create_sprint_cycle
```

ir_018 assign_issues_to_cycle

```
inputs[0] result: result_009 | sprint_top_issues
```

```
inputs[1] result: result_010 | sprint_created_cycle
```

```
outputs[0] result: result_011 | cycle assignments
```

```
ir_018.constraints: []
```

```
draft_instruction_id = assign_sprint_issues
```

ir_019 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_019.constraints: []
```

```
draft_instruction_id = dispatch_sprint_planning_to_summary
```

block_008 List critical and high-priority bugs

```
data_source_kind = "external"
```

```
block_008.constraints: []
```

ir_020 list_critical_high_priority_bugs

```
inputs[0] external_resource: linear_mcp_server | Linear MCP server
```

```
inputs[1] result: result_005 | confirmed_identifiers
```

```
outputs[0] result: result_012 | critical and high-priority bugs
```

```
ir_020.constraints: []
```

```
draft_instruction_id = list_critical_bugs
```

ir_021 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_021.constraints: []
```

```
draft_instruction_id = dispatch_to_bug_triage
```

block_009 Rank bugs and move top items to In Progress

data_source_kind = null

block_009.constraints: []

ir_022 rank_bugs_by_user_impact

inputs[0] result: result_012 | bug_critical_high_bugs

outputs[0] result: result_013 | bugs ranked by user impact

ir_022.constraints: []

draft_instruction_id = rank_bugs

ir_023 move_top_bugs_to_in_progress

inputs[0] result: result_013 | bug_ranked_bugs

outputs[0] result: result_014 | bug status updates

ir_023.constraints: []

draft_instruction_id = move_top_bugs

ir_024 dispatch

inputs = []

outputs = []

ir_024.constraints: []

draft_instruction_id = dispatch_bug_triage_to_summary

block_010 Search Linear documentation

data_source_kind = "external"

block_010.constraints: []

ir_025 search_linear_documentation

inputs[0] external_resource: linear_mcp_server | Linear MCP server

inputs[1] result: result_005 | confirmed_identifiers

outputs[0] result: result_015 | documentation search results

ir_025.constraints: []

draft_instruction_id = search_docs

ir_026 dispatch

inputs = []

outputs = []

ir_026.constraints: []

draft_instruction_id = dispatch_to_doc_audit

block_011 Analyze gaps and open documentation issues

data_source_kind = null

block_011.constraints: []

ir_027 analyze_documentation_gaps

inputs[0] result: result_015 | doc_search_results

outputs[0] result: result_016 | documentation gaps

ir_027.constraints: []

```
draft_instruction_id = analyze_doc_gaps
```

ir_028 create_documentation_issues

```
inputs[0] result: result_016 | doc_gaps
```

```
inputs[1] result: result_005 | confirmed_identifiers
```

```
outputs[0] result: result_017 | created documentation issues
```

```
ir_028.constraints: []
```

```
draft_instruction_id = create_doc_issues
```

ir_029 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_029.constraints: []
```

```
draft_instruction_id = dispatch_documentation_audit_to_summary
```

block_012 List active issues for workload balance

```
data_source_kind = "external"
```

```
block_012.constraints: []
```

ir_030 list_active_issues

```
inputs[0] external_resource: linear_mcp_server | Linear MCP server
```

```
inputs[1] result: result_005 | confirmed_identifiers
```

```
outputs[0] result: result_018 | active issues
```

```
ir_030.constraints: []
```

```
draft_instruction_id = list_active_issues
```

ir_031 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_031.constraints: []
```

```
draft_instruction_id = dispatch_to_workload
```

block_013 Group issues by assignee and suggest redistributions

```
data_source_kind = null
```

```
block_013.constraints: []
```

ir_032 group_active_issues_by_assignee

```
inputs[0] result: result_018 | workload_active_issues
```

```
outputs[0] result: result_019 | active issues grouped by assignee
```

```
ir_032.constraints: []
```

```
draft_instruction_id = group_by_assignee
```

ir_033 flag_high_load_assignees

```
inputs[0] result: result_019 | workload_by_assignee
```

```
outputs[0] result: result_020 | high-load assignees
```

```
ir_033.constraints: []
```

```
draft_instruction_id = flag_high_load
```

ir_034 suggest_or_apply_redistributions

```
inputs[0] result: result_019 | workload_by_assignee
inputs[1] result: result_020 | workload_high_load_assignees
outputs[0] result: result_021 | redistribution results
ir_034.constraints: []
draft_instruction_id = suggest_redistributions
```

ir_035 dispatch

```
inputs = []
outputs = []
ir_035.constraints: []
draft_instruction_id = dispatch_workload_balance_to_summary
```

block_014 Create release project, milestones, and issues

```
data_source_kind = null
block_014.constraints: []
```

ir_036 create_linear_project

```
inputs[0] result: result_005 | confirmed_identifiers
outputs[0] result: result_022 | created release project
ir_036.constraints: []
draft_instruction_id = create_release_project
```

ir_037 create_project_milestones

```
inputs[0] result: result_022 | release_created_project
inputs[1] result: result_005 | confirmed_identifiers
outputs[0] result: result_023 | created milestones
ir_037.constraints: []
draft_instruction_id = create_release_milestones
```

ir_038 create_release_issues_with_estimates

```
inputs[0] result: result_022 | release_created_project
inputs[1] result: result_023 | release_created_milestones
inputs[2] result: result_005 | confirmed_identifiers
outputs[0] result: result_024 | created release issues
ir_038.constraints: []
draft_instruction_id = create_release_issues
```

ir_039 dispatch

```
inputs = []
outputs = []
ir_039.constraints: []
draft_instruction_id = dispatch_release_planning_to_summary
```

block_015 Find blocked issues

```
data_source_kind = "external"
```

```
block_015.constraints: []
```

ir_040 find_blocked_issues

```
inputs[0] external_resource: linear_mcp_server | Linear MCP server
```

```
inputs[1] result: result_005 | confirmed_identifiers
```

```
outputs[0] result: result_025 | blocked issues
```

```
ir_040.constraints: []
```

```
draft_instruction_id = find_blocked_issues
```

ir_041 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_041.constraints: []
```

```
draft_instruction_id = dispatch_to_dependencies
```

block_016 Identify blockers and create linked issues

```
data_source_kind = null
```

```
block_016.constraints: []
```

ir_042 identify_blockers

```
inputs[0] result: result_025 | deps_blocked_issues
```

```
outputs[0] result: result_026 | identified blockers
```

```
ir_042.constraints: []
```

```
draft_instruction_id = identify_blockers
```

ir_043 create_linked_issues_if_missing

```
inputs[0] result: result_026 | deps_identified_blockers
```

```
inputs[1] result: result_025 | deps_blocked_issues
```

```
outputs[0] result: result_027 | created linked issues
```

```
ir_043.constraints: []
```

```
draft_instruction_id = create_linked_issues
```

ir_044 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_044.constraints: []
```

```
draft_instruction_id = dispatch_dependencies_to_summary
```

block_017 Find my issues with stale updates

```
data_source_kind = "external"
```

```
block_017.constraints: []
```

ir_045 list_my_issues_with_stale_updates

```
inputs[0] external_resource: linear_mcp_server | Linear MCP server
```

```
inputs[1] result: result_005 | confirmed_identifiers
```

```
outputs[0] result: result_028 | stale issues
```

```
ir_045.constraints: []
```

```
draft_instruction_id = list_stale_issues
```

ir_046 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_046.constraints: []
```

```
draft_instruction_id = dispatch_to_status_updates
```

block_018 Add status comments based on current state and blockers

```
data_source_kind = null
```

```
block_018.constraints: []
```

ir_047 compute_status_comments

```
inputs[0] result: result_028 | status_stale_issues
```

```
outputs[0] result: result_029 | computed status comments
```

```
ir_047.constraints: []
```

```
draft_instruction_id = compute_status_comments
```

ir_048 create_issue_comments

```
inputs[0] result: result_029 | status_comments
```

```
inputs[1] result: result_028 | status_stale_issues
```

```
outputs[0] result: result_030 | created comments
```

```
ir_048.constraints: []
```

```
draft_instruction_id = create_status_comments
```

ir_049 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_049.constraints: []
```

```
draft_instruction_id = dispatch_status_updates_to_summary
```

block_019 List unlabeled issues

```
data_source_kind = "external"
```

```
block_019.constraints: []
```

ir_050 list_unlabeled_issues

```
inputs[0] external_resource: linear_mcp_server | Linear MCP server
```

```
inputs[1] result: result_005 | confirmed_identifiers
```

```
outputs[0] result: result_031 | unlabeled issues
```

```
ir_050.constraints: []
```

```
draft_instruction_id = list_unlabeled_issues
```

ir_051 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_051.constraints: []
```

```
draft_instruction_id = dispatch_to_labeling
```

block_020 Analyze labels, apply labels, and create missing categories

```
data_source_kind = null
```

```
block_020.constraints: []
```

ir_052 analyze_unlabeled_issues_for_labels

```
inputs[0] result: result_031 | label_unlabeled_issues
```

```
outputs[0] result: result_032 | suggested labels
```

```
ir_052.constraints: []
```

```
draft_instruction_id = analyze_labels
```

ir_053 apply_labels_to_issues

```
inputs[0] result: result_032 | label_suggested_labels
```

```
inputs[1] result: result_031 | label_unlabeled_issues
```

```
outputs[0] result: result_033 | label application results
```

```
ir_053.constraints: []
```

```
draft_instruction_id = apply_labels
```

ir_054 create_missing_label_categories

```
inputs[0] result: result_032 | label_suggested_labels
```

```
outputs[0] result: result_034 | created label categories
```

```
ir_054.constraints: []
```

```
draft_instruction_id = create_missing_labels
```

ir_055 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_055.constraints: []
```

```
draft_instruction_id = dispatch_smart_labeling_to_summary
```

block_021 Get the last completed cycle

```
data_source_kind = "external"
```

```
block_021.constraints: []
```

ir_056 get_last_completed_cycle

```
inputs[0] external_resource: linear_mcp_server | Linear MCP server
```

```
inputs[1] result: result_005 | confirmed_identifiers
```

```
outputs[0] result: result_035 | last completed cycle data
```

```
ir_056.constraints: []
```

```
draft_instruction_id = get_last_cycle
```


ir_057 dispatch

```
inputs = []
outputs = []
ir_057.constraints: []
draft_instruction_id = dispatch_to_retro
```

block_022 Generate retrospective report and open discussion issues

```
data_source_kind = null
block_022.constraints: []
```

ir_058 generate_sprint_retrospective_report

```
inputs[0] result: result_035 | retro_last_cycle_data
outputs[0] result: result_036 | retrospective report
ir_058.constraints: []
draft_instruction_id = generate_retro_report
```

ir_059 open_discussion_issues_for_patterns

```
inputs[0] result: result_036 | retro_report
outputs[0] result: result_037 | discussion issues
ir_059.constraints: []
draft_instruction_id = open_discussion_issues
```

ir_060 dispatch

```
inputs = []
outputs = []
ir_060.constraints: []
draft_instruction_id = dispatch_retrospectives_to_summary
```

block_023 Summarize results and propose next actions

```
data_source_kind = null
block_023.constraints[0]: Summarize results, call out remaining gaps or blockers, and propose next actions
(additional issues, label changes, assignments, or follow-up comments).
```

ir_061 summarize_linear_workflow_results

```
inputs[0] result: result_011 | sprint_cycle_assignments
inputs[1] result: result_014 | bug_status_updates
inputs[2] result: result_017 | doc_created_issues
inputs[3] result: result_021 | workload_redistribution_results
inputs[4] result: result_024 | release_created_issues
inputs[5] result: result_027 | deps_created_linked_issues
inputs[6] result: result_030 | status_comment_results
inputs[7] result: result_033 | label_application_results
inputs[8] result: result_034 | label_created_categories
inputs[9] result: result_037 | retro_discussion_issues
inputs[10] result: result_036 | retro_report
outputs[0] result: result_038 | workflow summary
```

```
ir_061.constraints: []  
draft_instruction_id = summarize_workflow  
  
ir_062 return  
inputs[0] result: result_038 | workflow_summary  
outputs = []  
ir_062.constraints: []  
draft_instruction_id = return_summary
```

R05 / 第 2 次 实际输出明细

linear

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R05/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: The Windows/WSL troubleshooting note is preserved as a block constraint on setup_mcp_commands; it is not modeled as a separate execution path.

candidate warning: Practical Workflows are represented as selection options in the select_linear_workflow metadata; the selected workflow drives the generic read/plan/write/summarize path.

candidate warning: Tips for Maximum Productivity and Troubleshooting sections are non-executable guidance and were not modeled as behavior blocks.

candidate warning: agents/openai.yaml declares the Linear MCP dependency; this is represented by the external_resource identifier linear_mcp_server.

candidate warning: License and asset files are non-executable and were not modeled as behavior blocks.

Skill 层约束与上下文声明

declared_context_keys = ["user_request"]

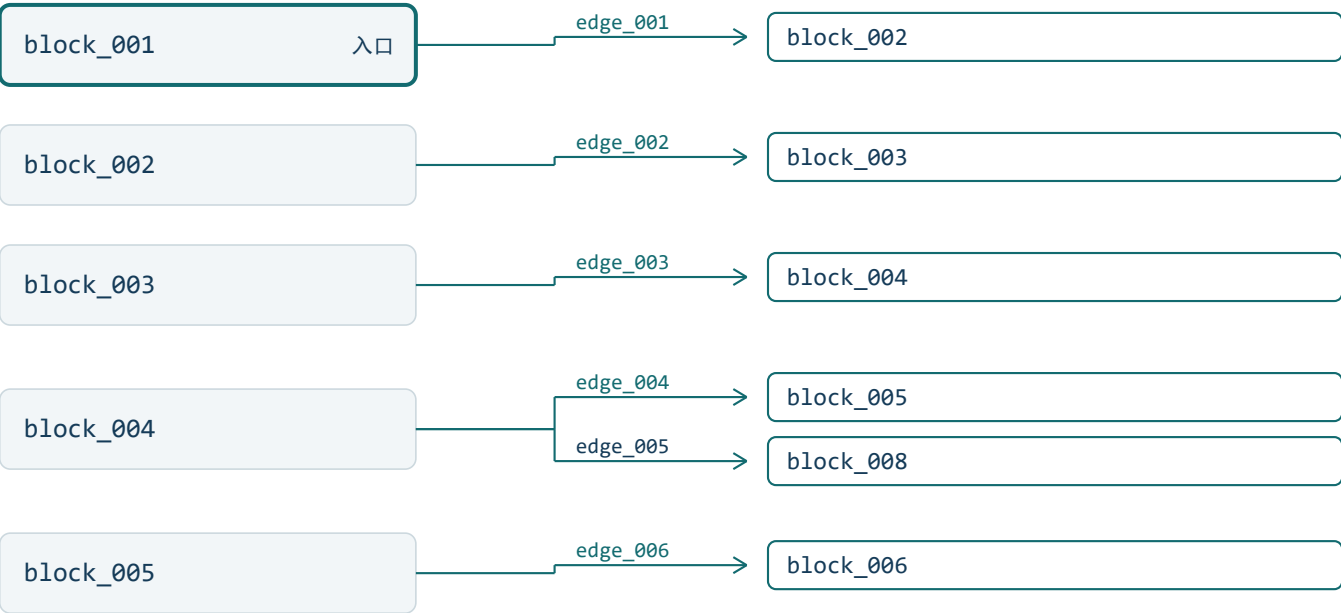
skill.constraints[0]: Linear MCP server must be connected and accessible via OAuth.

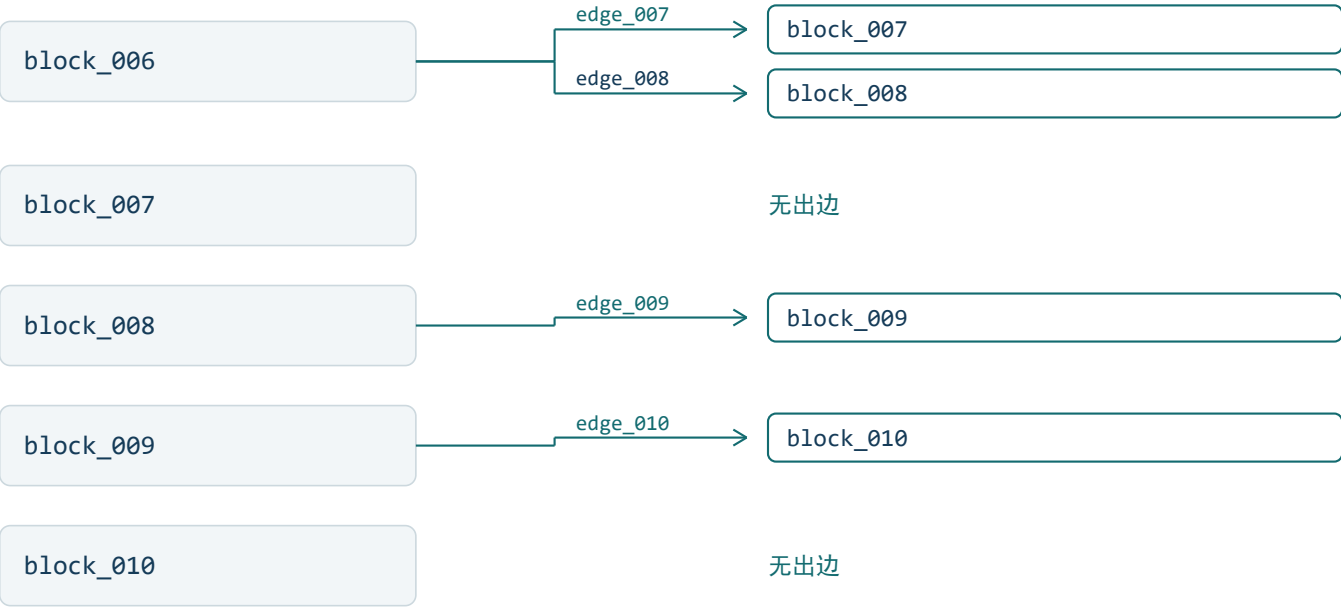
skill.constraints[1]: Confirm access to the relevant Linear workspace, teams, and projects.

skill.constraints[2]: Follow these steps in order. Do not skip steps.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	"Linear MCP read succeeded"
edge_005	block_004 -> block_008	"Linear MCP is not connected"
edge_006	block_005 -> block_006	null
edge_007	block_006 -> block_007	"Linear MCP write succeeded"
edge_008	block_006 -> block_008	"Linear MCP is not connected"
edge_009	block_008 -> block_009	null
edge_010	block_009 -> block_010	"login successful"

全部操作、数据引用与约束落点

block_001 Read the user's Linear request

```
data_source_kind = "context"
block_001.constraints: []

ir_001 read_user_request

inputs[0] context_key: user_request | user_request
outputs[0] result: result_001 | raw_user_request
ir_001.constraints: []
draft_instruction_id = read_user_request
```

ir_002 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_002.constraints: []
```

```
draft_instruction_id = go_clarify_goal
```

block_002 Clarify the user's goal and scope

```
data_source_kind = null
```

```
block_002.constraints[0]: Confirm team/project, priority, labels, cycle, and due dates as needed.
```

ir_003 clarify_user_goal_and_scope

```
inputs[0] result: result_001 | raw_user_request
```

```
outputs[0] result: result_002 | clarified_goal_scope
```

```
ir_003.constraints: []
```

```
draft_instruction_id = clarify_user_goal_and_scope
```

ir_004 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = go_select_workflow
```

block_003 Select the Linear workflow and confirm required identifiers

```
data_source_kind = null
```

```
block_003.constraints[0]: Select the appropriate workflow (see Practical Workflows) and identify the Linear MCP tools you will need.
```

```
block_003.constraints[1]: Confirm required identifiers (issue ID, project ID, team key) before calling tools.
```

ir_005 select_linear_workflow

```
inputs[0] result: result_002 | clarified_goal_scope
```

```
outputs[0] result: result_003 | selected_workflow
```

```
ir_005.constraints: []
```

```
draft_instruction_id = select_linear_workflow
```

```
metadata = {"workflow_options": ["Sprint Planning", "Bug Triage", "Documentation Audit", "Team Workload Balance", "Release Planning", "Cross-Project Dependencies", "Automated Status Updates", "Smart Labeling", "Sprint Retrospectives"]}
```

ir_006 identify_linear_mcp_tools

```
inputs[0] result: result_003 | selected_workflow
```

```
outputs[0] result: result_004 | required_tools
```

```
ir_006.constraints: []
```

```
draft_instruction_id = identify_linear_mcp_tools
```

```
metadata = {"available_tools": {"Documentation & Collaboration": ["list_documents", "get_document", "search_documentation", "list_comments", "create_comment", "list_cycles"], "Issue Management": ["list_issues", "get_issue", "create_issue", "update_issue", "list_my_issues", "list_issue_statuses", "list_issue_labels", "create_issue_label"], "Project & Team": ["list_projects", "get_project", "create_project", "update_project", "list_teams", "get_team", "list_users"]}}
```

ir_007 confirm_linear_identifiers

```
inputs[0] result: result_002 | clarified_goal_scope
outputs[0] result: result_005 | confirmed_identifiers
ir_007.constraints: []
draft_instruction_id = confirm_linear_identifiers
```

ir_008 dispatch

```
inputs = []
outputs = []
ir_008.constraints: []
draft_instruction_id = go_read_linear_context
```

block_004 Read Linear context with list/get/search MCP tools

```
data_source_kind = "external"
block_004.constraints[0]: Read first (list/get/search) to build context.
```

ir_009 execute_linear_mcp_read_calls

```
inputs[0] external_resource: linear_mcp_server | linear_mcp_server
inputs[1] result: result_004 | required_tools
inputs[2] result: result_005 | confirmed_identifiers
outputs[0] result: result_006 | linear_read_results
outputs[1] result: result_007 | linear_read_error
ir_009.constraints: []
draft_instruction_id = execute_linear_mcp_read_calls
metadata = {"mcp_server": {"transport": "streamable_http", "type": "mcp", "url": "https://mcp.linear.app/mcp", "value": "linear"}}
```

ir_010 dispatch

```
inputs[0] result: result_007 | linear_read_error
outputs = []
ir_010.constraints: []
draft_instruction_id = dispatch_after_read
```

block_005 Determine Linear changes from the read context

```
data_source_kind = null
block_005.constraints: []
```

ir_011 determine_linear_changes

```
inputs[0] result: result_006 | linear_read_results
inputs[1] result: result_003 | selected_workflow
inputs[2] result: result_005 | confirmed_identifiers
outputs[0] result: result_008 | planned_linear_changes
ir_011.constraints: []
draft_instruction_id = determine_linear_changes
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_write_linear
```

block_006 Create or update Linear items

```
data_source_kind = "external"
```

```
block_006.constraints[0]: Create or update next (issues, projects, labels, comments) with all required fields.
```

```
block_006.constraints[1]: For bulk operations, explain the grouping logic before applying changes.
```

ir_013 execute_linear_mcp_write_calls

```
inputs[0] external_resource: linear_mcp_server | linear_mcp_server
```

```
inputs[1] result: result_008 | planned_linear_changes
```

```
outputs[0] result: result_009 | linear_write_results
```

```
outputs[1] result: result_010 | linear_write_error
```

```
ir_013.constraints: []
```

```
draft_instruction_id = execute_linear_mcp_write_calls
```

ir_014 dispatch

```
inputs[0] result: result_010 | linear_write_error
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = dispatch_after_write
```

block_007 Summarize Linear results, gaps, and next actions

```
data_source_kind = null
```

```
block_007.constraints[0]: Summarize results, call out remaining gaps or blockers, and propose next actions (additional issues, label changes, assignments, or follow-up comments).
```

ir_015 summarize_linear_results_and_propose_next_actions

```
inputs[0] result: result_003 | selected_workflow
```

```
inputs[1] result: result_006 | linear_read_results
```

```
inputs[2] result: result_009 | linear_write_results
```

```
outputs[0] result: result_011 | linear_workflow_outcome
```

```
ir_015.constraints: []
```

```
draft_instruction_id = summarize_results_and_propose_next_actions
```

ir_016 return

```
inputs[0] result: result_011 | linear_workflow_outcome
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = return_linear_outcome
```

block_008 Set up the Linear MCP server

```
data_source_kind = null
```

```
block_008.constraints[0]: If any MCP call fails because Linear MCP is not connected, pause and set it up.
```

```
block_008.constraints[1]: On Windows, if connection errors occur, try configuring the Linear MCP to run via WSL using the provided JSON configuration: {"mcpServers": {"linear": {"command": "wsl", "args": ["npx", "-y", "mcp-remote", "https://mcp.linear.app/sse", "--transport", "sse-only"]}}}}
```

ir_017 add_linear_mcp_server

```
inputs[0] external_resource: codex_cli | codex_cli
```

```
inputs[1] literal = "linear"
```

```
inputs[2] literal = "https://mcp.linear.app/mcp"
```

```
outputs = []
```

```
ir_017.constraints: []
```

```
draft_instruction_id = add_linear_mcp_server
```

metadata.script_content: 53 字符; SHA256 982f5440da2578106d3e0d9ab1a16b801fa50933976565759ca2c5a0fd0e454b; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_018 enable_codex_remote_mcp_client

```
inputs[0] external_resource: codex_cli | codex_cli
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = enable_codex_remote_mcp_client
```

metadata.script_content: 86 字符; SHA256 995f4d8852720bfde38d33511d332b559a2526f9f226be4408c9edbf75bdd7b1; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_019 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_019.constraints: []
```

```
draft_instruction_id = go_login_linear_mcp
```

block_009 Log in to Linear MCP with OAuth

```
data_source_kind = "external"
```

```
block_009.constraints: []
```

ir_020 login_linear_mcp_oauth

```
inputs[0] external_resource: linear_mcp_server | linear_mcp_server
```

```
outputs[0] result: result_012 | linear_mcp_login_success
```

```
ir_020.constraints: []
```

```
draft_instruction_id = login_linear_mcp_oauth
```

metadata.script_content: 22 字符; SHA256 922605f9f096d319f8f2a4d32a26399f27018b8a95b8f2f5be35e5455bc89809; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_021 dispatch

```
inputs[0] result: result_012 | linear_mcp_login_success
```

```
outputs = []
```

```
ir_021.constraints: []
```

```
draft_instruction_id = dispatch_after_login
```

block_010 Tell the user to restart Codex and end

```
data_source_kind = null
```

```
block_010.constraints[0]: After successful login, the user will have to restart Codex. Finish your answer and tell them so when they try again they can continue with Step 1.
```

ir_022 return

```
inputs[0] literal = "Restart Codex; when you try again, continue with Step 1."
```

```
outputs = []
```

```
ir_022.constraints: []
```

```
draft_instruction_id = return_restart_instruction
```

R05 / 第 3 次 实际输出明细

linear

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R05/deepseek-v4-flash-max/3

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: Step 0 setup is modeled as a conditional prerequisite check because the text says 'if not already configured' and 'must be connected'.

candidate warning: The clarification step is modeled as a computation on the user request; explicit user clarification responses are not separately modeled due to lack of a specified read action.

candidate warning: Practical workflows are modeled as alternative branches selected in Step 2.

candidate warning: Some practical workflows reference tools not listed in Available Tools (e.g., create_cycle, create_milestone). These are modeled with semantic opcodes based on the described actions.

candidate warning: Troubleshooting guidance is not modeled as separate execution blocks; it is treated as advisory for error handling.

candidate warning: Tips for Maximum Productivity are included as Skill-wide constraints.

candidate warning: The default_prompt from agents/openai.yaml is included as a Skill-wide constraint.

candidate warning: The MCP server URL and transport are declared in agents/openai.yaml and referenced by the external_resource 'linear'.

Skill 层约束与上下文声明

declared_context_keys = ["user_request"]

skill.constraints[0]: Follow these steps in order. Do not skip steps.

skill.constraints[1]: Linear MCP server must be connected and accessible via OAuth

skill.constraints[2]: Confirm access to the relevant Linear workspace, teams, and projects

skill.constraints[3]: Read first (list/get/search) to build context.

skill.constraints[4]: Create or update next (issues, projects, labels, comments) with all required fields.

skill.constraints[5]: For bulk operations, explain the grouping logic before applying changes.

skill.constraints[6]: Batch operations for related changes; consider smart templates for recurring issue structures.

skill.constraints[7]: Use natural queries when possible ("Show me what John is working on this week").

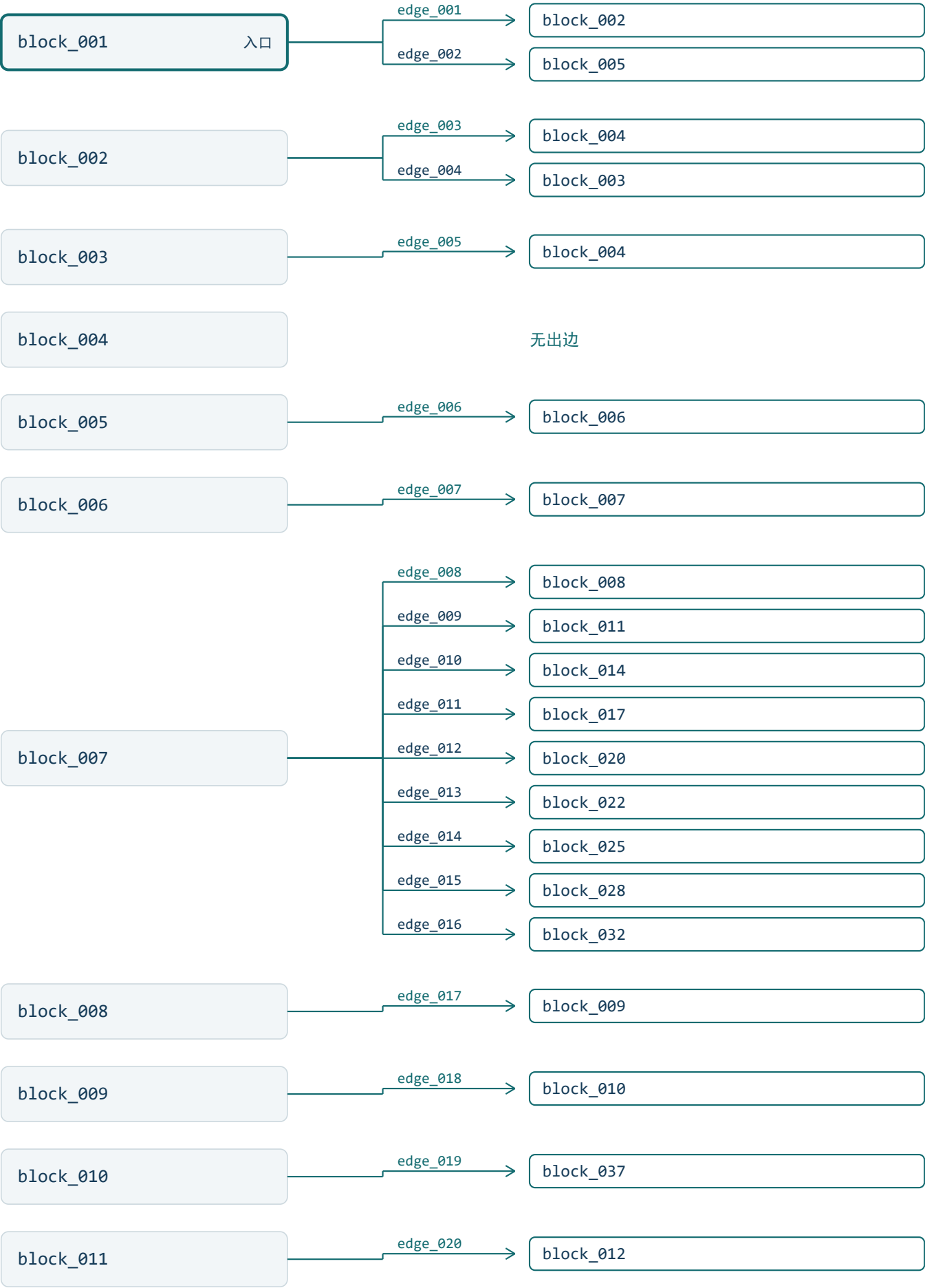
skill.constraints[8]: Leverage context: reference prior issues in new requests.

skill.constraints[9]: Break large updates into smaller batches to avoid rate limits; cache or reuse filters when listing frequently.

skill.constraints[10]: Use Linear context to triage or update relevant issues for this task, with clear next actions.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。







完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	"MCP not connected or not accessible"
edge_002	block_001 -> block_005	"MCP connected and accessible"
edge_003	block_002 -> block_004	"setup_status == success"
edge_004	block_002 -> block_003	"connection errors on Windows"
edge_005	block_003 -> block_004	null
edge_006	block_005 -> block_006	null
edge_007	block_006 -> block_007	null
edge_008	block_007 -> block_008	"Sprint Planning"
edge_009	block_007 -> block_011	"Bug Triage"
edge_010	block_007 -> block_014	"Documentation Audit"
edge_011	block_007 -> block_017	"Team Workload Balance"
edge_012	block_007 -> block_020	"Release Planning"
edge_013	block_007 -> block_022	"Cross-Project Dependencies"
edge_014	block_007 -> block_025	"Automated Status Updates"
edge_015	block_007 -> block_028	"Smart Labeling"
edge_016	block_007 -> block_032	"Sprint Retrospectives"
edge_017	block_008 -> block_009	null
edge_018	block_009 -> block_010	null
edge_019	block_010 -> block_037	null
edge_020	block_011 -> block_012	null
edge_021	block_012 -> block_013	null
edge_022	block_013 -> block_037	null
edge_023	block_014 -> block_015	null
edge_024	block_015 -> block_016	null
edge_025	block_016 -> block_037	null
edge_026	block_017 -> block_018	null
edge_027	block_018 -> block_019	null

边标识	源块 -> 目标块	condition_text
edge_028	block_019 -> block_037	null
edge_029	block_020 -> block_021	null
edge_030	block_021 -> block_037	null
edge_031	block_022 -> block_023	null
edge_032	block_023 -> block_024	null
edge_033	block_024 -> block_037	null
edge_034	block_025 -> block_026	null
edge_035	block_026 -> block_027	null
edge_036	block_027 -> block_037	null
edge_037	block_028 -> block_029	null
edge_038	block_029 -> block_030	null
edge_039	block_030 -> block_031	null
edge_040	block_031 -> block_037	null
edge_041	block_032 -> block_033	null
edge_042	block_033 -> block_034	null
edge_043	block_034 -> block_035	null
edge_044	block_035 -> block_036	null
edge_045	block_036 -> block_037	null

全部操作、数据引用与约束落点

block_001 Check Linear MCP access

```
data_source_kind = "external"

block_001.constraints[0]: Linear MCP server must be connected and accessible via OAuth
block_001.constraints[1]: Confirm access to the relevant Linear workspace, teams, and projects
```

ir_001 check_linear_mcp_access

```
inputs[0] external_resource: linear | linear
outputs[0] result: result_001 | mcp_access_status

ir_001.constraints: []

draft_instruction_id = check_mcp
```

ir_002 dispatch

```
inputs[0] result: result_001 | mcp_access_status
```

```
outputs = []
```

```
ir_002.constraints: []
```

```
draft_instruction_id = dispatch_mcp_check
```

block_002 Set up Linear MCP

```
data_source_kind = null
```

```
block_002.constraints[0]: If any MCP call fails because Linear MCP is not connected, pause and set it up
```

ir_003 codex_mcp_add_linear

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] literal = "https://mcp.linear.app/mcp"
```

```
outputs = []
```

```
ir_003.constraints: []
```

```
draft_instruction_id = add_linear_mcp
```

metadata.script_content: 53 字符; SHA256 982f5440da2578106d3e0d9ab1a16b801fa50933976565759ca2c5a0fd0e454b; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_004 enable_rmcp_client

```
inputs[0] literal = "Set [features] rmcp_client = true in config.toml or run codex --enable rmcp_client"
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = enable_remote_mcp_client
```

metadata.script_content: 86 字符; SHA256 4ef927422c91c1efecde4be26971d9859c8a17d441b6f8c866852054942d6663; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_005 codex_mcp_login_linear

```
inputs[0] external_resource: linear | linear
```

```
outputs[0] result: result_002 | setup_status
```

```
ir_005.constraints: []
```

```
draft_instruction_id = login_linear_oauth
```

metadata.script_content: 22 字符; SHA256 922605f9f096d319f8f2a4d32a26399f27018b8a95b8f2f5be35e5455bc89809; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_006 dispatch

```
inputs[0] result: result_002 | setup_status
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = dispatch_setup
```


block_003 Configure Linear MCP via WSL

```
data_source_kind = null
```

```
block_003.constraints[0]: Windows/WSL note: If you see connection errors on Windows, try configuring the Linear MCP to run via WSL
```

ir_007 configure_linear_mcp_via_wsl

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] literal = {"mcpServers": {"linear": {"command": "wsl", "args": ["npx", "-y", "mcp-remote", "https://mcp.linear.app/sse", "--transport", "sse-only"]}}}
```

```
outputs = []
```

```
ir_007.constraints: []
```

```
draft_instruction_id = configure_wsl_mcp
```

metadata.script_content: 140 字符; SHA256 4906437b9ea2a3b4bf71008b450a2423d55c1876f9527304609335a972b205be; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_008 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = dispatch_wsl
```

block_004 Tell user to restart and finish

```
data_source_kind = null
```

```
block_004.constraints: []
```

ir_009 return

```
inputs[0] literal = "After successful login, the user will have to restart codex. You should finish your answer and tell them so when they try again they can continue with Step 1."
```

```
outputs = []
```

```
ir_009.constraints: []
```

```
draft_instruction_id = return_restart_message
```

block_005 Read the user's request

```
data_source_kind = "context"
```

```
block_005.constraints: []
```

ir_010 read_user_request

```
inputs[0] context_key: user_request | user_request
```

```
outputs[0] result: result_003 | user_request
```

```
ir_010.constraints: []
```

```
draft_instruction_id = read_user_request
```

ir_011 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_011.constraints: []
```

```
draft_instruction_id = dispatch_user_request
```

block_006 Clarify the user's goal, scope, and required fields

```
data_source_kind = null
```

```
block_006.constraints[0]: Clarify the user's goal and scope (e.g., issue triage, sprint planning, documentation audit, workload balance). Confirm team/project, priority, labels, cycle, and due dates as needed.
```

ir_012 clarify_user_goal_and_scope

```
inputs[0] result: result_003 | user_request
```

```
outputs[0] result: result_004 | clarified_goal_scope
```

```
ir_012.constraints: []
```

```
draft_instruction_id = clarify_goal
```

ir_013 confirm_required_fields

```
inputs[0] result: result_004 | clarified_goal_scope
```

```
outputs[0] result: result_005 | confirmed_identifiers
```

```
ir_013.constraints: []
```

```
draft_instruction_id = confirm_fields
```

ir_014 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = dispatch_clarify
```

block_007 Select workflow and confirm identifiers

```
data_source_kind = null
```

```
block_007.constraints[0]: Select the appropriate workflow (see Practical Workflows below) and identify the Linear MCP tools you will need. Confirm required identifiers (issue ID, project ID, team key) before calling tools.
```

```
block_007.constraints[1]: Execute Linear MCP tool calls in logical batches: Read first (list/get/search) to build context. Create or update next (issues, projects, labels, comments) with all required fields. For bulk operations, explain the grouping logic before applying changes.
```

ir_015 select_workflow

```
inputs[0] result: result_004 | clarified_goal_scope
```

```
outputs[0] result: result_006 | selected_workflow
```

```
ir_015.constraints: []
```

```
draft_instruction_id = select_workflow
```

ir_016 identify_mcp_tools

```
inputs[0] result: result_006 | selected_workflow
```

```
outputs[0] result: result_007 | required_tools
```

```
ir_016.constraints: []
```

```
draft_instruction_id = identify_mcp_tools
```

ir_017 confirm_required_identifiers

```
inputs[0] result: result_005 | confirmed_identifiers
```

```
outputs[0] result: result_008 | validated_identifiers
```

```
ir_017.constraints: []
```

```
draft_instruction_id = confirm_identifiers
```

ir_018 dispatch

```
inputs[0] result: result_006 | selected_workflow
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = dispatch_workflow
```

block_008 List open issues for the target team

```
data_source_kind = "external"
```

```
block_008.constraints: []
```

ir_019 list_issues

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] result: result_008 | validated_identifiers
```

```
inputs[2] literal = "open"
```

```
outputs[0] result: result_009 | open_issues
```

```
ir_019.constraints: []
```

```
draft_instruction_id = sp_list_issues
```

ir_020 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = sp_dispatch_list
```

block_009 Pick top issues by priority

```
data_source_kind = null
```

```
block_009.constraints: []
```

ir_021 rank_issues_by_priority

```
inputs[0] result: result_009 | open_issues
```

```
outputs[0] result: result_010 | top_priority_issues
```

```
ir_021.constraints: []
```

```
draft_instruction_id = sp_rank
```

ir_022 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_022.constraints: []
```

```
draft_instruction_id = sp_dispatch_rank
```

block_010 Create a new cycle with assignments

```
data_source_kind = "external"
```

```
block_010.constraints: []
```

ir_023 create_cycle_with_assignments

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] result: result_010 | top_priority_issues
```

```
inputs[2] literal = "Q1 Performance Sprint"
```

```
outputs[0] result: result_011 | created_cycle
```

```
ir_023.constraints: []
```

```
draft_instruction_id = sp_create_cycle
```

ir_024 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_024.constraints: []
```

```
draft_instruction_id = sp_dispatch_create
```

block_011 List critical/high-priority bugs

```
data_source_kind = "external"
```

```
block_011.constraints: []
```

ir_025 list_issues

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] literal = "critical/high-priority bugs"
```

```
outputs[0] result: result_012 | critical_bugs
```

```
ir_025.constraints: []
```

```
draft_instruction_id = bt_list_bugs
```

ir_026 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_026.constraints: []
```

```
draft_instruction_id = bt_dispatch_list
```

block_012 Rank bugs by user impact

```
data_source_kind = null
```

```
block_012.constraints: []
```

ir_027 rank_bugs_by_user_impact

```
inputs[0] result: result_012 | critical_bugs
```

```
outputs[0] result: result_013 | ranked_bugs
```

```
ir_027.constraints: []
```

```
draft_instruction_id = bt_rank
```

ir_028 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_028.constraints: []
```

```
draft_instruction_id = bt_dispatch_rank
```

block_013 Move top bugs to In Progress

```
data_source_kind = "external"
```

```
block_013.constraints: []
```

ir_029 update_issue_status

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] result: result_013 | ranked_bugs
```

```
inputs[2] literal = "In Progress"
```

```
outputs[0] result: result_014 | updated_bugs
```

```
ir_029.constraints: []
```

```
draft_instruction_id = bt_update_status
```

ir_030 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_030.constraints: []
```

```
draft_instruction_id = bt_dispatch_update
```

block_014 Search documentation

```
data_source_kind = "external"
```

```
block_014.constraints: []
```

ir_031 search_documentation

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] literal = "API auth"
```

```
outputs[0] result: result_015 | doc_search_results
```

```
ir_031.constraints: []
```

```
draft_instruction_id = da_search_docs
```

ir_032 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_032.constraints: []
```

```
draft_instruction_id = da_dispatch_search
```

block_015 Identify documentation gaps or outdated sections

```
data_source_kind = null
```

```
block_015.constraints: []
```

ir_033 identify_documentation_gaps

```
inputs[0] result: result_015 | doc_search_results
```

```
outputs[0] result: result_016 | documentation_gaps
```

```
ir_033.constraints: []
```

```
draft_instruction_id = da_identify
```

ir_034 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_034.constraints: []
```

```
draft_instruction_id = da_dispatch_identify
```

block_016 Open labeled documentation issues

```
data_source_kind = "external"
```

```
block_016.constraints: []
```

ir_035 create_issue

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] result: result_016 | documentation_gaps
```

```
inputs[2] literal = "documentation"
```

```
outputs[0] result: result_017 | created_doc_issues
```

```
ir_035.constraints: []
```

```
draft_instruction_id = da_create_issues
```

ir_036 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_036.constraints: []
```

```
draft_instruction_id = da_dispatch_create
```

block_017 List active issues

```
data_source_kind = "external"
```

```
block_017.constraints: []
```

ir_037 list_issues

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] literal = "active"
```

```
outputs[0] result: result_018 | active_issues
```

```
ir_037.constraints: []
```

```
draft_instruction_id = tw_list_active
```

ir_038 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_038.constraints: []
```

```
draft_instruction_id = tw_dispatch_list
```

block_018 Group active issues by assignee and flag high load

```
data_source_kind = null
```

```
block_018.constraints: []
```

ir_039 analyze_team_workload

```
inputs[0] result: result_018 | active_issues
```

```
outputs[0] result: result_019 | workload_analysis
```

```
ir_039.constraints: []
```

```
draft_instruction_id = tw_analyze
```

ir_040 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_040.constraints: []
```

```
draft_instruction_id = tw_dispatch_analyze
```

block_019 Suggest or apply redistributions

```
data_source_kind = "external"
```

```
block_019.constraints: []
```

ir_041 suggest_or_apply_redistributions

```
inputs[0] result: result_019 | workload_analysis
```

```
outputs[0] result: result_020 | redistribution_result
```

```
ir_041.constraints: []
```

```
draft_instruction_id = tw_handle
```

ir_042 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_042.constraints: []
```

```
draft_instruction_id = tw_dispatch_handle
```

block_020 Create release project with milestones

```
data_source_kind = "external"
```

```
block_020.constraints: []
```

ir_043 create_project

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] literal = "v2.0 Release"
```

```
inputs[2] literal = ["feature freeze", "beta", "docs", "launch"]
```

```
outputs[0] result: result_021 | created_project_id
```

```
ir_043.constraints: []
```

```
draft_instruction_id = rp_create_project
```

ir_044 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_044.constraints: []
```

```
draft_instruction_id = rp_dispatch_project
```

block_021 Generate issues with estimates

```
data_source_kind = "external"
```

```
block_021.constraints: []
```

ir_045 create_issue

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] result: result_021 | created_project_id
```

```
inputs[2] literal = "estimates"
```

```
outputs[0] result: result_022 | created_release_issues
```

```
ir_045.constraints: []
```

```
draft_instruction_id = rp_create_issues
```

ir_046 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_046.constraints: []
```

```
draft_instruction_id = rp_dispatch_issues
```

block_022 Find all blocked issues

```
data_source_kind = "external"
```

```
block_022.constraints: []
```

ir_047 list_issues

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] literal = "blocked"
```

```
outputs[0] result: result_023 | blocked_issues
```

```
ir_047.constraints: []
```

```
draft_instruction_id = cd_list_blocked
```

ir_048 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_048.constraints: []
```

```
draft_instruction_id = cd_dispatch_list
```

block_023 Identify blockers

```
data_source_kind = null
```

```
block_023.constraints: []
```

ir_049 identify_blockers

```
inputs[0] result: result_023 | blocked_issues
```

```
outputs[0] result: result_024 | blockers
```

```
ir_049.constraints: []
```

```
draft_instruction_id = cd_identify
```


ir_050 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_050.constraints: []
```

```
draft_instruction_id = cd_dispatch_identify
```

block_024 Create linked issues if missing

```
data_source_kind = "external"
```

```
block_024.constraints: []
```

ir_051 create_linked_issues_if_missing

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] result: result_024 | blockers
```

```
outputs[0] result: result_025 | created_linked_issues
```

```
ir_051.constraints: []
```

```
draft_instruction_id = cd_create_linked
```

ir_052 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_052.constraints: []
```

```
draft_instruction_id = cd_dispatch_create
```

block_025 Find your issues with stale updates

```
data_source_kind = "external"
```

```
block_025.constraints: []
```

ir_053 list_my_issues

```
inputs[0] external_resource: linear | linear
```

```
outputs[0] result: result_026 | my_issues
```

```
ir_053.constraints: []
```

```
draft_instruction_id = su_list_my_issues
```

ir_054 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_054.constraints: []
```

```
draft_instruction_id = su_dispatch_list
```

block_026 Filter stale issues

```
data_source_kind = null
```

```
block_026.constraints: []
```

ir_055 filter_stale_issues

```
inputs[0] result: result_026 | my_issues
```

```
outputs[0] result: result_027 | stale_issues
```

```
ir_055.constraints: []
```

```
draft_instruction_id = su_filter
```

ir_056 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_056.constraints: []
```

```
draft_instruction_id = su_dispatch_filter
```

block_027 Add status comments based on current state/blockers

```
data_source_kind = "external"
```

```
block_027.constraints: []
```

ir_057 create_comment

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] result: result_027 | stale_issues
```

```
inputs[2] literal = "status update based on current state/blockers"
```

```
outputs[0] result: result_028 | added_comments
```

```
ir_057.constraints: []
```

```
draft_instruction_id = su_add_comments
```

ir_058 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_058.constraints: []
```

```
draft_instruction_id = su_dispatch_comment
```

block_028 Analyze unlabeled issues

```
data_source_kind = "external"
```

```
block_028.constraints: []
```

ir_059 list_issues

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] literal = "unlabeled"
```

```
outputs[0] result: result_029 | unlabeled_issues
```

```
ir_059.constraints: []
```

```
draft_instruction_id = sl_list_unlabeled
```

ir_060 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_060.constraints: []
```

```
draft_instruction_id = sl_dispatch_list
```

block_029 Suggest labels and identify missing categories

```
data_source_kind = null
```

```
block_029.constraints: []
```

ir_061 analyze_issues_for_labels

```
inputs[0] result: result_029 | unlabeled_issues
```

```
outputs[0] result: result_030 | label_suggestions
```

```
outputs[1] result: result_031 | missing_labels
```

```
ir_061.constraints: []
```

```
draft_instruction_id = sl_analyze
```

ir_062 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_062.constraints: []
```

```
draft_instruction_id = sl_dispatch_analyze
```

block_030 Apply suggested labels

```
data_source_kind = null
```

```
block_030.constraints: []
```

ir_063 update_issue_labels

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] result: result_029 | unlabeled_issues
```

```
inputs[2] result: result_030 | label_suggestions
```

```
outputs = []
```

```
ir_063.constraints: []
```

```
draft_instruction_id = sl_apply_labels
```

ir_064 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_064.constraints: []
```

```
draft_instruction_id = sl_dispatch_apply
```

block_031 Create missing label categories

```
data_source_kind = "external"
```

```
block_031.constraints: []
```

ir_065 create_issue_label

```
inputs[0] external_resource: linear | linear
```

```
inputs[1] result: result_031 | missing_labels
```

```
outputs[0] result: result_032 | created_labels
```

```
ir_065.constraints: []
```

```
draft_instruction_id = sl_create_labels
```

ir_066 dispatch

```
inputs = []
outputs = []
ir_066.constraints: []
draft_instruction_id = sl_dispatch_create
```

block_032 Find the last completed cycle

```
data_source_kind = "external"
block_032.constraints: []
```

ir_067 list_cycles

```
inputs[0] external_resource: linear | linear
inputs[1] literal = "completed"
outputs[0] result: result_033 | completed_cycles
ir_067.constraints: []
draft_instruction_id = sr_list_cycles
```

ir_068 dispatch

```
inputs = []
outputs = []
ir_068.constraints: []
draft_instruction_id = sr_dispatch_list
```

block_033 Select the last completed cycle

```
data_source_kind = null
block_033.constraints: []
```

ir_069 select_last_completed_cycle

```
inputs[0] result: result_033 | completed_cycles
outputs[0] result: result_034 | last_completed_cycle
ir_069.constraints: []
draft_instruction_id = sr_select_last
```

ir_070 dispatch

```
inputs = []
outputs = []
ir_070.constraints: []
draft_instruction_id = sr_dispatch_select
```

block_034 List issues for the last completed cycle

```
data_source_kind = "external"
block_034.constraints: []
```

ir_071 list_issues

```
inputs[0] external_resource: linear | linear
inputs[1] result: result_034 | last_completed_cycle
outputs[0] result: result_035 | cycle_issues
```

```
ir_071.constraints: []  
draft_instruction_id = sr_list_cycle_issues  
  
ir_072 dispatch  
  
inputs = []  
outputs = []  
ir_072.constraints: []  
draft_instruction_id = sr_dispatch_issues  
  
block_035 Generate sprint report noting completed vs. pushed work  
  
data_source_kind = null  
block_035.constraints: []  
  
ir_073 generate_sprint_report  
  
inputs[0] result: result_035 | cycle_issues  
outputs[0] result: result_036 | sprint_report  
ir_073.constraints: []  
draft_instruction_id = sr_generate  
  
ir_074 dispatch  
  
inputs = []  
outputs = []  
ir_074.constraints: []  
draft_instruction_id = sr_dispatch_generate  
  
block_036 Open discussion issues for patterns  
  
data_source_kind = "external"  
block_036.constraints: []  
  
ir_075 create_issue  
  
inputs[0] external_resource: linear | linear  
inputs[1] result: result_036 | sprint_report  
outputs[0] result: result_037 | discussion_issues  
ir_075.constraints: []  
draft_instruction_id = sr_create_discussion  
  
ir_076 dispatch  
  
inputs = []  
outputs = []  
ir_076.constraints: []  
draft_instruction_id = sr_dispatch_create
```

block_037 Summarize results and propose next actions

```
data_source_kind = null
```

```
block_037.constraints[0]: Summarize results, call out remaining gaps or blockers, and propose next actions  
(additional issues, label changes, assignments, or follow-up comments).
```

ir_077 summarize_results

```
inputs[0] result: result_011 | created_cycle  
inputs[1] result: result_014 | updated_bugs  
inputs[2] result: result_017 | created_doc_issues  
inputs[3] result: result_020 | redistribution_result  
inputs[4] result: result_022 | created_release_issues  
inputs[5] result: result_025 | created_linked_issues  
inputs[6] result: result_028 | added_comments  
inputs[7] result: result_032 | created_labels  
inputs[8] result: result_037 | discussion_issues  
outputs[0] result: result_038 | summary  
ir_077.constraints: []  
draft_instruction_id = summarize
```

ir_078 return

```
inputs[0] result: result_038 | summary  
outputs = []  
ir_078.constraints: []  
draft_instruction_id = return_summary
```

R06 / 第 1 次 实际输出明细

transcribe

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R06/deepseek-v4-flash-max/1

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: The bundled CLI's internal control flow (e.g., dry-run, stdout, multi-file loop) is encapsulated in the run_transcribe_diarize_cli black-box operation per code-file extraction rules.

candidate warning: The exact job-id generation for output/transcribe/<job-id>/ is not specified; the analysis uses the base directory output/transcribe/ as the output directory.

candidate warning: OPENAI_API_KEY is also checked inside the CLI script; the analysis includes an explicit agent-level context read for the key before running the CLI.

Skill 层约束与上下文声明

```
declared_context_keys = ["user_request", "OPENAI_API_KEY"]
```

skill.constraints[0]: Transcribe audio using OpenAI, with optional speaker diarization when requested.

skill.constraints[1]: Prefer the bundled CLI for deterministic, repeatable runs.

skill.constraints[2]: OPENAI_API_KEY must be set for live API calls.

skill.constraints[3]: If the key is missing, instruct the user to create one in the OpenAI platform UI and export it in their shell.

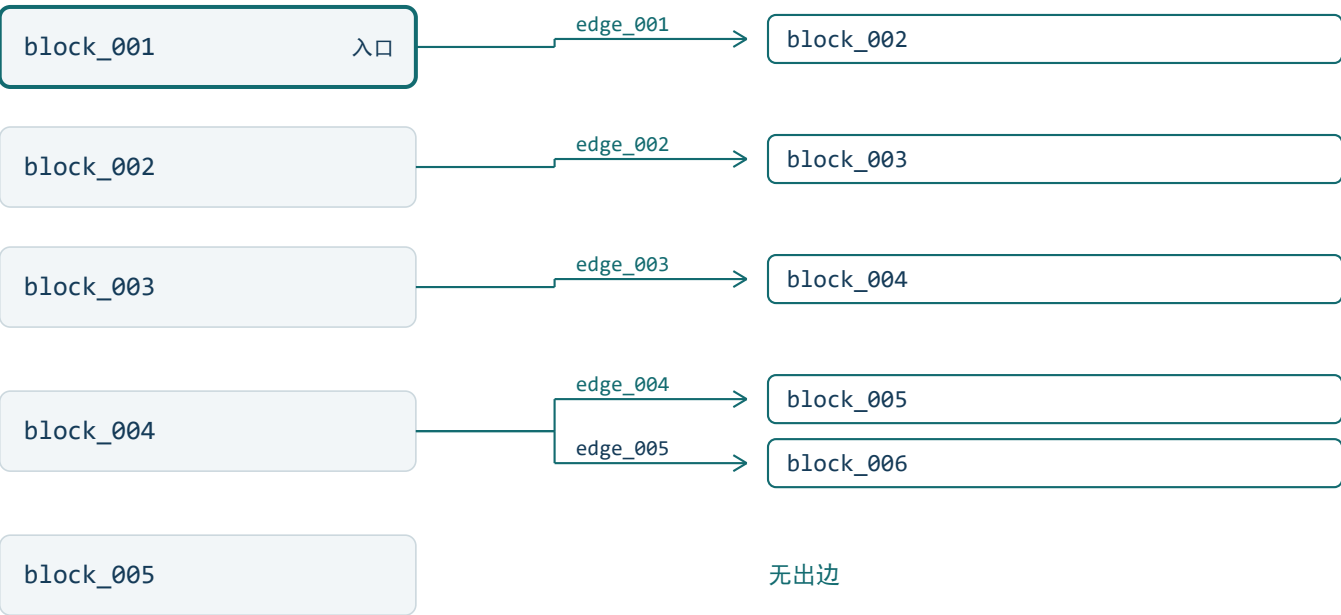
skill.constraints[4]: Never ask the user to paste the full key in chat.

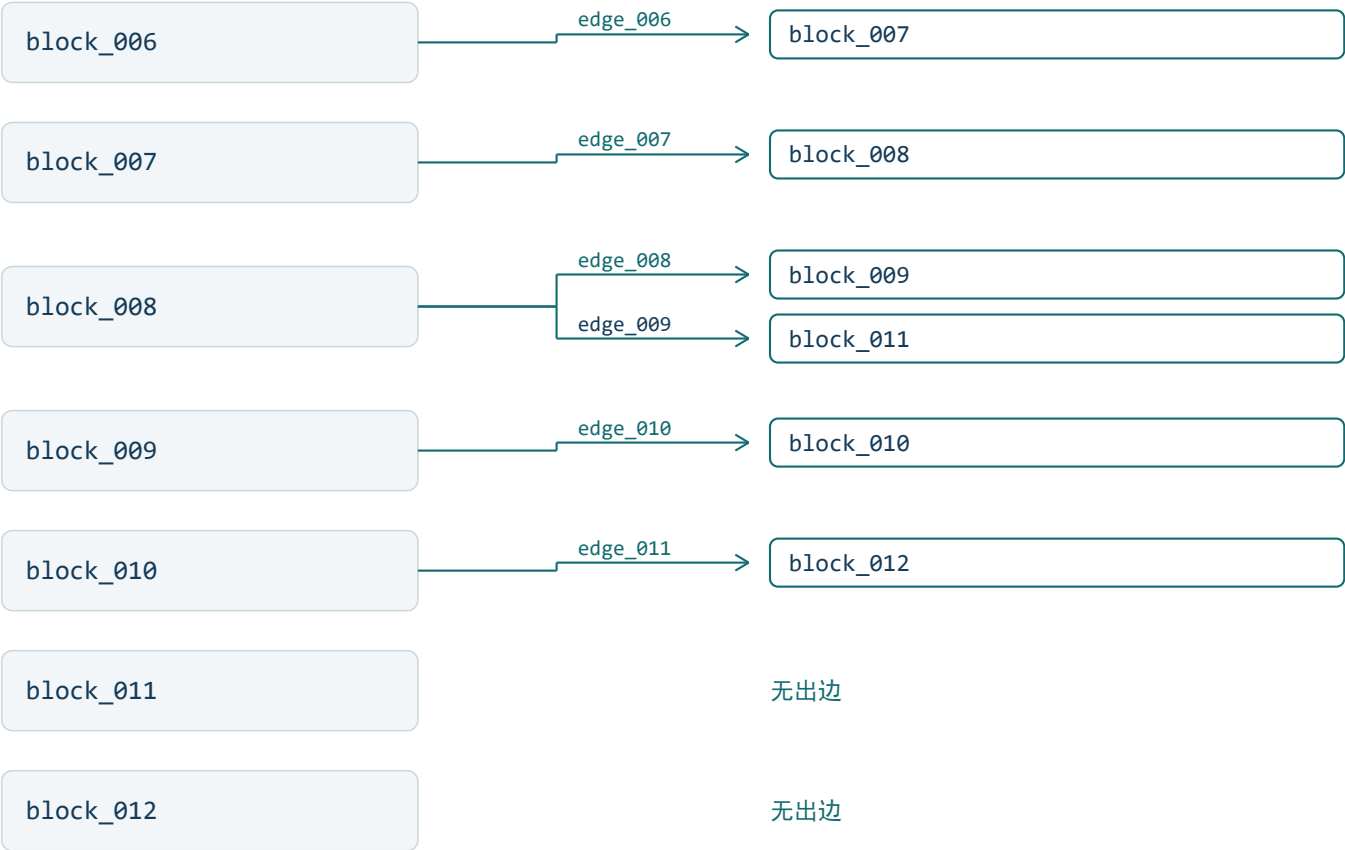
skill.constraints[5]: Use output/transcribe/<job-id>/ for evaluation runs.

skill.constraints[6]: Use --out-dir for multiple files to avoid overwriting.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	null
edge_003	block_003 -> block_004	null
edge_004	block_004 -> block_005	"OPENAI_API_KEY is missing or empty"
edge_005	block_004 -> block_006	"OPENAI_API_KEY is set"
edge_006	block_006 -> block_007	null
edge_007	block_007 -> block_008	null
edge_008	block_008 -> block_009	"needs adjustment"
edge_009	block_008 -> block_011	"no adjustment needed"
edge_010	block_009 -> block_010	null
edge_011	block_010 -> block_012	null

全部操作、数据引用与约束落点

block_001 Read the user's transcription request

```
data_source_kind = "context"
```

```
block_001.constraints: []
```

ir_001 read_user_request

```
inputs[0] context_key: user_request | user_request
```

```
outputs[0] result: result_001 | user_request_text
```

```
ir_001.constraints: []
```

```
draft_instruction_id = read_request
```

ir_002 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_002.constraints: []
```

```
draft_instruction_id = go_extract
```

block_002 Extract transcription inputs from the request

```
data_source_kind = null
```

```
block_002.constraints: []
```

ir_003 extract_transcription_inputs

```
inputs[0] result: result_001 | user_request_text
```

```
outputs[0] result: result_002 | audio_paths
```

```
outputs[1] result: result_003 | requested_response_format
```

```
outputs[2] result: result_004 | language_hint
```

```
outputs[3] result: result_005 | known_speaker_refs
```

```
outputs[4] result: result_006 | diarization_requested
```

```
ir_003.constraints: []
```

```
draft_instruction_id = extract
```

ir_004 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = go_select_model
```

block_003 Select model and response format based on diarization request

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 select_transcription_model_and_format

```
inputs[0] result: result_003 | requested_response_format
```

```
inputs[1] result: result_006 | diarization_requested
```

```
outputs[0] result: result_007 | selected_model
```

```
outputs[1] result: result_008 | selected_response_format
```

```
ir_005.constraints[0]: Default to gpt-4o-mini-transcribe with --response-format text for fast transcription.  
ir_005.constraints[1]: If the user wants speaker labels or diarization, use --model gpt-4o-transcribe-diarize  
--response-format diarized_json.  
ir_005.constraints[2]: If audio is longer than ~30 seconds, keep --chunking-strategy auto.  
ir_005.constraints[3]: Prompting is not supported for gpt-4o-transcribe-diarize.  
draft_instruction_id = select_model_format
```

ir_006 dispatch

```
inputs = []  
outputs = []  
ir_006.constraints: []  
draft_instruction_id = go_check_key
```

block_004 Read the OpenAI API key from the environment

```
data_source_kind = "context"  
block_004.constraints: []
```

ir_007 read_env_openai_api_key

```
inputs[0] context_key: OPENAI_API_KEY | OPENAI_API_KEY  
outputs[0] result: result_009 | openai_api_key_value  
ir_007.constraints: []  
draft_instruction_id = read_api_key
```

ir_008 dispatch

```
inputs[0] result: result_009 | openai_api_key_value  
outputs = []  
ir_008.constraints: []  
draft_instruction_id = dispatch_key_check
```

block_005 Ask the user to set the OpenAI API key locally

```
data_source_kind = null  
block_005.constraints: []
```

ir_009 ask_user_to_set_openai_api_key_locally

```
inputs[0] literal = "Please set OPENAI_API_KEY in your shell environment. Do not paste the key in chat."  
outputs[0] result: result_010 | api_key_setup_message  
ir_009.constraints: []  
draft_instruction_id = ask_user
```

ir_010 return

```
inputs[0] result: result_010 | api_key_setup_message  
outputs = []  
ir_010.constraints: []  
draft_instruction_id = return_setup_message
```

block_006 Ensure the OpenAI Python SDK is installed

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 ensure_openai_sdk_installed

```
inputs = []
```

```
outputs = []
```

```
ir_011.constraints[0]: Prefer uv for dependency management. Install with `uv pip install openai`.
```

```
ir_011.constraints[1]: If uv is unavailable, use `python3 -m pip install openai`.
```

```
draft_instruction_id = ensure_openai_sdk
```

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_run_cli
```

block_007 Run the bundled transcribe_diarize.py CLI

```
data_source_kind = "external"
```

```
block_007.constraints: []
```

ir_013 run_transcribe_diarize_cli

```
inputs[0] external_resource: scripts/transcribe_diarize.py | scripts/transcribe_diarize.py
```

```
inputs[1] result: result_002 | audio_paths
```

```
inputs[2] result: result_007 | selected_model
```

```
inputs[3] result: result_008 | selected_response_format
```

```
inputs[4] result: result_004 | language_hint
```

```
inputs[5] result: result_005 | known_speaker_refs
```

```
inputs[6] literal = "auto"
```

```
inputs[7] literal = "output/transcribe/"
```

```
outputs[0] result: result_011 | transcript_output_initial
```

```
ir_013.constraints[0]: Input formats: mp3, mp4, mpeg, mpga, m4a, wav, webm.
```

```
ir_013.constraints[1]: Max file size: 25 MB per request.
```

```
ir_013.constraints[2]: response_format options: text, json, diarized_json.
```

```
ir_013.constraints[3]: For audio longer than ~30 seconds, pass chunking_strategy (use "auto" to split into chunks).
```

```
ir_013.constraints[4]: Known speakers: up to 4 references via extra_body known_speaker_names +  
known_speaker_references (data URLs).
```

```
ir_013.constraints[5]: Prompting is not supported for gpt-4o-transcribe-diarize.
```

```
draft_instruction_id = run_cli
```

```
metadata.script_content: 8684 字符; SHA256 d623b1fed1491b7ca4b7b5b393089165e3581a83d90d5a57868241f1b2a5d951; 计数与  
摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。
```

ir_014 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = go_validate
```

block_008 Validate the initial transcription output

```
data_source_kind = null
```

```
block_008.constraints: []
```

ir_015 validate_transcription_output

```
inputs[0] result: result_011 | transcript_output_initial
```

```
outputs[0] result: result_012 | validation_result_initial
```

```
outputs[1] result: result_013 | needs_adjustment_initial
```

```
ir_015.constraints[0]: Validate transcription quality, speaker labels, and segment boundaries.
```

```
draft_instruction_id = validate_output
```

ir_016 dispatch

```
inputs[0] result: result_013 | needs_adjustment_initial
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = dispatch_validation
```

block_009 Apply a single targeted change for the next attempt

```
data_source_kind = null
```

```
block_009.constraints: []
```

ir_017 apply_single_targeted_transcription_change

```
inputs[0] result: result_012 | validation_result_initial
```

```
inputs[1] result: result_002 | audio_paths
```

```
inputs[2] result: result_007 | selected_model
```

```
inputs[3] result: result_008 | selected_response_format
```

```
inputs[4] result: result_004 | language_hint
```

```
inputs[5] result: result_005 | known_speaker_refs
```

```
outputs[0] result: result_014 | adjusted_model
```

```
outputs[1] result: result_015 | adjusted_response_format
```

```
outputs[2] result: result_016 | adjusted_language_hint
```

```
outputs[3] result: result_017 | adjusted_known_speaker_refs
```

```
ir_017.constraints[0]: Iterate with a single targeted change if needed.
```

```
draft_instruction_id = apply_change
```

ir_018 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_018.constraints: []
```

```
draft_instruction_id = go_retry
```

block_010 Re-run the CLI with the targeted change

```
data_source_kind = "external"
```

```
block_010.constraints: []
```

ir_019 run_transcribe_diarize_cli

```
inputs[0] external_resource: scripts/transcribe_diarize.py | scripts/transcribe_diarize.py
```

```
inputs[1] result: result_002 | audio_paths
```

```
inputs[2] result: result_014 | adjusted_model
```

```
inputs[3] result: result_015 | adjusted_response_format
```

```
inputs[4] result: result_016 | adjusted_language_hint
```

```
inputs[5] result: result_017 | adjusted_known_speaker_refs
```

```
inputs[6] literal = "auto"
```

```
inputs[7] literal = "output/transcribe/"
```

```
outputs[0] result: result_018 | transcript_output_final
```

```
ir_019.constraints[0]: Input formats: mp3, mp4, mpeg, mpga, m4a, wav, webm.
```

```
ir_019.constraints[1]: Max file size: 25 MB per request.
```

```
ir_019.constraints[2]: response_format options: text, json, diarized_json.
```

```
ir_019.constraints[3]: For audio longer than ~30 seconds, pass chunking_strategy (use "auto" to split into chunks).
```

```
ir_019.constraints[4]: Known speakers: up to 4 references via extra_body known_speaker_names +  
known_speaker_references (data URLs).
```

```
ir_019.constraints[5]: Prompting is not supported for gpt-4o-transcribe-diarize.
```

```
draft_instruction_id = run_cli_retry_instr
```

```
metadata.script_content: 8684 字符; SHA256 d623b1fed1491b7ca4b7b5b393089165e3581a83d90d5a57868241f1b2a5d951; 计数与  
摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。
```

ir_020 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_020.constraints: []
```

```
draft_instruction_id = go_return_final
```

block_011 Return the validated transcript

```
data_source_kind = null
```

```
block_011.constraints: []
```

ir_021 return

```
inputs[0] result: result_011 | transcript_output_initial
```

```
outputs = []
```

```
ir_021.constraints: []
```

```
draft_instruction_id = return_initial
```

block_012 Return the adjusted transcript

```
data_source_kind = null
```

```
block_012.constraints: []
```

ir_022 return

```
inputs[0] result: result_018 | transcript_output_final
```

```
outputs = []
```

```
ir_022.constraints: []
```

```
draft_instruction_id = return_final
```

R06 / 第 2 次 实际输出明细

transcribe

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R06/deepseek-v4-flash-max/2

结构状态: complete | 生成 1 | 修复 0 | HTTP 尝试 1 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: The Python CLI script is treated as a black-box external operation; its internal file reads, API calls, and output writing are not split into separate instructions.

candidate warning: The run_transcribe_diarize_cli instruction consumes targeted_change, which is only produced on the loop backedge from apply_targeted_transcription_change after validation failure.

candidate warning: The output directory uses the literal 'output/transcribe' because no job-id is specified in the package.

Skill 层约束与上下文声明

```
declared_context_keys = ["user_request", "OPENAI_API_KEY"]
```

skill.constraints[0]: Prefer the bundled CLI for deterministic, repeatable runs.

skill.constraints[1]: Collect inputs: audio file path(s), desired response format (text/json/diarized_json), optional language hint, and any known speaker references.

skill.constraints[2]: Verify `OPENAI\_API\_KEY` is set. If missing, ask the user to set it locally (do not ask them to paste the key).

skill.constraints[3]: Run the bundled `transcribe\_diarize.py` CLI with sensible defaults (fast text transcription).

skill.constraints[4]: Validate the output: transcription quality, speaker labels, and segment boundaries; iterate with a single targeted change if needed.

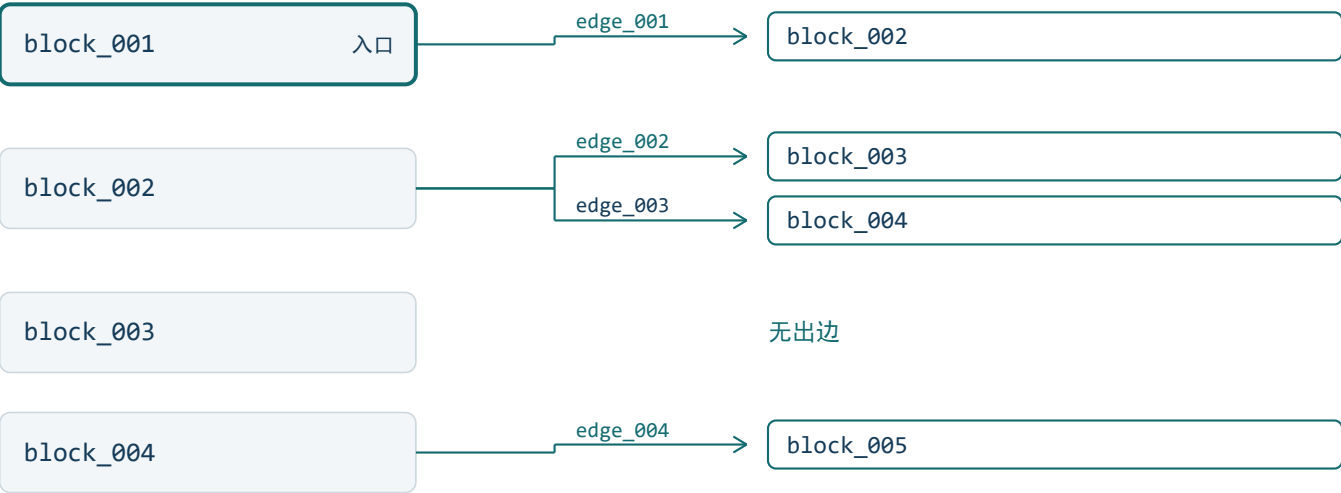
skill.constraints[5]: Save outputs under `output/transcribe/` when working in this repo.

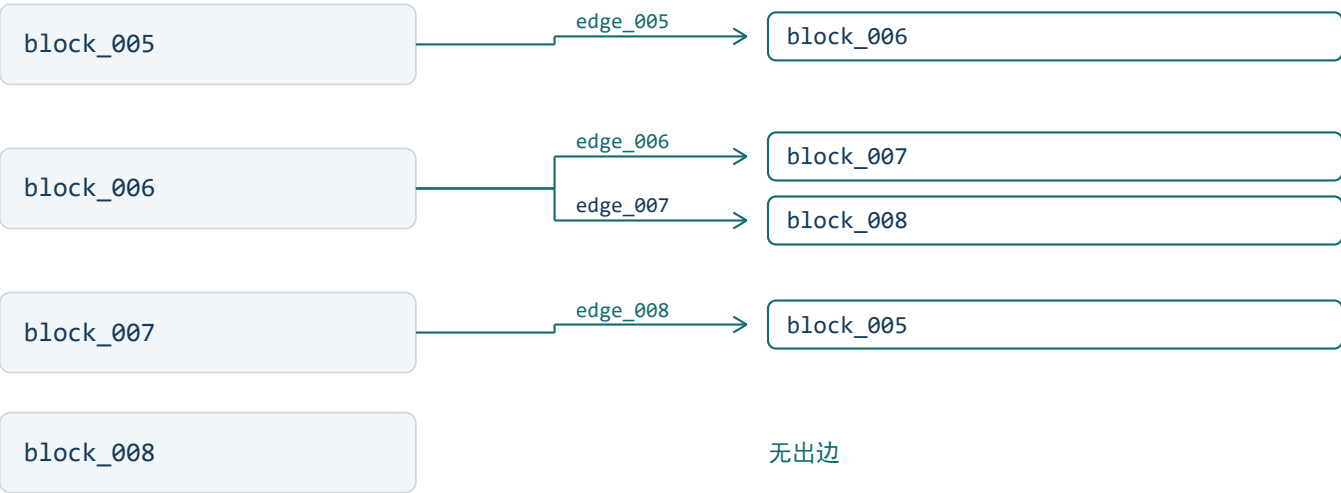
skill.constraints[6]: Prefer `uv` for dependency management.

skill.constraints[7]: If `uv` is unavailable: `python3 -m pip install openai`.

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"OPENAI_API_KEY is not set"
edge_003	block_002 -> block_004	"OPENAI_API_KEY is set"
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	"validation failed"
edge_007	block_006 -> block_008	"validation passed"
edge_008	block_007 -> block_005	null

全部操作、数据引用与约束落点

block_001 Read the user's transcription request

```
data_source_kind = "context"
block_001.constraints: []

ir_001 read_transcription_request

inputs[0] context_key: user_request | user_request
outputs[0] result: result_001 | audio_paths
outputs[1] result: result_002 | requested_response_format
outputs[2] result: result_003 | language_hint
outputs[3] result: result_004 | known_speaker_references
outputs[4] result: result_005 | wants_diarization
ir_001.constraints: []
draft_instruction_id = read_transcription_request
```


ir_002 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_002.constraints: []
```

```
draft_instruction_id = dispatch_to_api_key_check
```

block_002 Verify OPENAI_API_KEY is set

```
data_source_kind = "context"
```

```
block_002.constraints[0]: If the key is missing, instruct the user to create one in the OpenAI platform UI and export it in their shell.
```

```
block_002.constraints[1]: Never ask the user to paste the full key in chat.
```

ir_003 read_openai_api_key

```
inputs[0] context_key: OPENAI_API_KEY | OPENAI_API_KEY
```

```
outputs[0] result: result_006 | openai_api_key_present
```

```
ir_003.constraints[0]: OPENAI_API_KEY must be set for live API calls.
```

```
draft_instruction_id = read_openai_api_key
```

ir_004 dispatch

```
inputs[0] result: result_006 | openai_api_key_present
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = dispatch_on_api_key_presence
```

block_003 Ask the user to set OPENAI_API_KEY locally

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 ask_user_to_set_openai_api_key_locally

```
inputs = []
```

```
outputs = []
```

```
ir_005.constraints[0]: Do not ask the user to paste the key in chat.
```

```
draft_instruction_id = request_api_key_setup
```

ir_006 return

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = return_api_key_missing
```

block_004 Select transcription model and response format

```
data_source_kind = null
```

```
block_004.constraints: []
```

ir_007 select_transcription_model_and_format

```
inputs[0] result: result_005 | wants_diarization
```

```
inputs[1] result: result_002 | requested_response_format
```

```
outputs[0] result: result_007 | selected_model
outputs[1] result: result_008 | selected_response_format
ir_007.constraints[0]: Default to `gpt-4o-mini-transcribe` with `--response-format text` for fast transcription.
ir_007.constraints[1]: If the user wants speaker labels or diarization, use `--model gpt-4o-transcribe-diarize --response-format diarized_json`.
ir_007.constraints[2]: If audio is longer than ~30 seconds, keep `--chunking-strategy auto`.
ir_007.constraints[3]: Prompting is not supported for `gpt-4o-transcribe-diarize`.
draft_instruction_id = select_transcription_options
```

ir_008 dispatch

```
inputs = []
outputs = []
ir_008.constraints: []
draft_instruction_id = dispatch_to_run_cli
```

block_005 Run the bundled transcribe_diarize.py CLI

```
data_source_kind = "external"
block_005.constraints[0]: Use `output/transcribe/<job-id>` for evaluation runs.
block_005.constraints[1]: Use `--out-dir` for multiple files to avoid overwriting.
```

ir_009 run_transcribe_diarize_cli

```
inputs[0] external_resource: transcribe_diarize.py | transcribe_diarize.py
inputs[1] result: result_001 | audio_paths
inputs[2] result: result_007 | selected_model
inputs[3] result: result_008 | selected_response_format
inputs[4] result: result_003 | language_hint
inputs[5] result: result_004 | known_speaker_references
inputs[6] result: result_013 | targeted_change
inputs[7] literal = "auto"
inputs[8] literal = "output/transcribe"
outputs[0] result: result_009 | transcript_output
outputs[1] result: result_010 | written_output_paths
ir_009.constraints[0]: Input formats: mp3, mp4, mpeg, mpga, m4a, wav, webm.
ir_009.constraints[1]: Max file size: 25 MB per request.
ir_009.constraints[2]: response_format options: text, json, diarized_json.
ir_009.constraints[3]: For audio longer than ~30 seconds, pass chunking_strategy (use "auto" to split into chunks).
ir_009.constraints[4]: Known speakers: up to 4 references via extra_body known_speaker_names + known_speaker_references (data URLs).
ir_009.constraints[5]: Prompting is not supported for gpt-4o-transcribe-diarize.
draft_instruction_id = run_cli
```

metadata.script_content: 8684 字符; SHA256 d623b1fed1491b7ca4b7b5b393089165e3581a83d90d5a57868241f1b2a5d951; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = dispatch_to_validate
```

block_006 Validate transcription quality, speaker labels, and segment boundaries

```
data_source_kind = null
```

```
block_006.constraints: []
```

ir_011 validate_transcription_output

```
inputs[0] result: result_009 | transcript_output
```

```
inputs[1] result: result_007 | selected_model
```

```
inputs[2] result: result_008 | selected_response_format
```

```
outputs[0] result: result_011 | validation_passed
```

```
outputs[1] result: result_012 | validation_feedback
```

```
ir_011.constraints[0]: Validate the output: transcription quality, speaker labels, and segment boundaries.
```

```
draft_instruction_id = validate_transcription_output
```

ir_012 dispatch

```
inputs[0] result: result_011 | validation_passed
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = dispatch_validation_result
```

block_007 Apply a single targeted change and re-run transcription

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 apply_targeted_transcription_change

```
inputs[0] result: result_012 | validation_feedback
```

```
inputs[1] result: result_007 | selected_model
```

```
inputs[2] result: result_008 | selected_response_format
```

```
outputs[0] result: result_013 | targeted_change
```

```
ir_013.constraints[0]: Iterate with a single targeted change if needed.
```

```
draft_instruction_id = apply_targeted_change
```

ir_014 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = dispatch_rerun
```

block_008 Return the transcription result

```
data_source_kind = null
```

```
block_008.constraints: []
```

ir_015 return

```
inputs[0] result: result_009 | transcript_output
```

```
outputs = []
```

```
ir_015.constraints: []
```

```
draft_instruction_id = return_transcription
```

R06 / 第 3 次 实际输出明细

transcribe

产物目录: runs/baseline-deepseek-v4-flash-max-20260910/trials/R06/deepseek-v4-flash-max/3

结构状态: complete | 生成 2 | 修复 1 | HTTP 尝试 2 | HTTP 重试 0

请求模型: deepseek-v4-flash。以下返回名称逐项来自 trace，不用请求名称补齐。

生成 1 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

生成 2 返回模型: deepseek-flash | HTTP 1 (200): deepseek-flash

诊断与失败边界

candidate warning: LICENSE.txt, assets/transcribe-small.svg, and assets/transcribe.png are non-executable and not modeled.

candidate warning: agents/openai.yaml provides interface metadata and is not modeled as runtime behavior.

candidate warning: The transcribe_diarize.py CLI is treated as one external black-box operation; its internal control flow and file reads are preserved in metadata.script_content.

Skill 层约束与上下文声明

declared_context_keys = ["audio_file_paths", "response_format", "language_hint", "known_speaker_references", "OPENAI_API_KEY"]

skill.constraints[0]: Transcribe audio using OpenAI, with optional speaker diarization when requested.

skill.constraints[1]: Prefer the bundled CLI for deterministic, repeatable runs.

skill.constraints[2]: Default to gpt-4o-mini-transcribe with --response-format text for fast transcription.

skill.constraints[3]: If the user wants speaker labels or diarization, use --model gpt-4o-transcribe-diarize --response-format diarized_json.

skill.constraints[4]: If audio is longer than ~30 seconds, keep --chunking-strategy auto.

skill.constraints[5]: Prompting is not supported for gpt-4o-transcribe-diarize.

skill.constraints[6]: Use output/transcribe/<job-id>/ for evaluation runs.

skill.constraints[7]: Use --out-dir for multiple files to avoid overwriting.

skill.constraints[8]: OPENAI_API_KEY must be set for live API calls.

skill.constraints[9]: Never ask the user to paste the full key in chat.

skill.constraints[10]: Known speakers: up to 4 references via extra_body known_speaker_names + known_speaker_references (data URLs).

skill.constraints[11]: Input formats: mp3, mp4, mpeg, mpga, m4a, wav, webm.

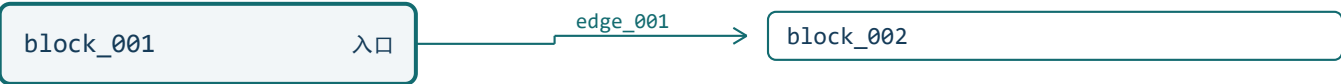
skill.constraints[12]: Max file size: 25 MB per request.

skill.constraints[13]: response_format options: text, json, diarized_json.

skill.constraints[14]: For audio longer than ~30 seconds, pass chunking_strategy (use 'auto' to split into chunks).

实际 CFG：分页邻接图

每个左框是实际 block ID；箭头连接到右框标明的目标 ID。目标可位于其他行 / 页，回边和独立组件保留原貌。edge_001 等按 analysis.cfg.edges 原数组顺序编号为 1 起的检索标识；完整条件见紧随的边表。节点名称与操作见后续逐块明细。





完整控制流边与条件

边标识	源块 -> 目标块	condition_text
edge_001	block_001 -> block_002	null
edge_002	block_002 -> block_003	"OPENAI_API_KEY is missing"
edge_003	block_002 -> block_004	"OPENAI_API_KEY is set"
edge_004	block_004 -> block_005	null
edge_005	block_005 -> block_006	null
edge_006	block_006 -> block_007	null
edge_007	block_007 -> block_009	"output accepted"
edge_008	block_007 -> block_008	"needs targeted change"
edge_009	block_008 -> block_005	"re-run with adjusted arguments"

全部操作、数据引用与约束落点

block_001 Read transcription request inputs

```
data_source_kind = "context"
```

```
block_001.constraints: []
```

ir_001 read_transcription_request_inputs

```
inputs[0] context_key: audio_file_paths | audio_file_paths
```

```
inputs[1] context_key: response_format | response_format
```

```
inputs[2] context_key: language_hint | language_hint
```

```
inputs[3] context_key: known_speaker_references | known_speaker_references
```

```
outputs[0] result: result_001 | audio_paths
```

```
outputs[1] result: result_002 | requested_response_format
```

```
outputs[2] result: result_003 | language_hint
```

```
outputs[3] result: result_004 | known_speaker_refs
```

```
ir_001.constraints: []
```

```
draft_instruction_id = read_inputs
```

ir_002 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_002.constraints: []
```

```
draft_instruction_id = go_check
```

block_002 Check OPENAI_API_KEY environment variable

```
data_source_kind = "context"
```

```
block_002.constraints: []
```

ir_003 read_openai_api_key

```
inputs[0] context_key: OPENAI_API_KEY | OPENAI_API_KEY
```

```
outputs[0] result: result_005 | api_key_value
```

```
ir_003.constraints: []
```

```
draft_instruction_id = read_api_key
```

ir_004 dispatch

```
inputs[0] result: result_005 | api_key_value
```

```
outputs = []
```

```
ir_004.constraints: []
```

```
draft_instruction_id = dispatch_on_api_key
```

block_003 Ask the user to set OPENAI_API_KEY locally

```
data_source_kind = null
```

```
block_003.constraints: []
```

ir_005 ask_user_to_set_api_key

```
inputs[0] literal = "OPENAI_API_KEY"
```

```
outputs = []
```

```
ir_005.constraints[0]: Do not ask the user to paste the full key in chat.
```

```
draft_instruction_id = ask_user
```

```
ir_006 return
```

```
inputs = []
```

```
outputs = []
```

```
ir_006.constraints: []
```

```
draft_instruction_id = return_after_ask
```

```
block_004 Select transcription model and response format
```

```
data_source_kind = null
```

```
block_004.constraints: []
```

```
ir_007 select_transcription_settings
```

```
inputs[0] result: result_002 | requested_response_format
```

```
inputs[1] result: result_004 | known_speaker_refs
```

```
outputs[0] result: result_006 | selected_model
```

```
outputs[1] result: result_007 | selected_response_format
```

```
ir_007.constraints[0]: Default to gpt-4o-mini-transcribe with --response-format text for fast transcription.
```

```
ir_007.constraints[1]: If the user wants speaker labels or diarization, use --model gpt-4o-transcribe-diarize --response-format diarized_json.
```

```
draft_instruction_id = select_settings
```

```
ir_008 dispatch
```

```
inputs = []
```

```
outputs = []
```

```
ir_008.constraints: []
```

```
draft_instruction_id = go_prepare_args
```

```
block_005 Prepare transcribe_diarize.py CLI arguments
```

```
data_source_kind = null
```

```
block_005.constraints: []
```

```
ir_009 prepare_transcribe_cli_arguments
```

```
inputs[0] result: result_001 | audio_paths
```

```
inputs[1] result: result_006 | selected_model
```

```
inputs[2] result: result_007 | selected_response_format
```

```
inputs[3] result: result_003 | language_hint
```

```
inputs[4] result: result_004 | known_speaker_refs
```

```
inputs[5] result: result_012 | targeted_change
```

```
outputs[0] result: result_008 | cli_arguments
```

```
ir_009.constraints[0]: Keep --chunking-strategy auto for audio longer than ~30 seconds.
```

```
ir_009.constraints[1]: Prompting is not supported for gpt-4o-transcribe-diarize.
```

```
ir_009.constraints[2]: Use output/transcribe/<job-id>/ for evaluation runs.
```

```
ir_009.constraints[3]: Use --out-dir for multiple files to avoid overwriting.
```

```
ir_009.constraints[4]: Known speakers are limited to 4 references.
```



```
draft_instruction_id = prepare_args
```

ir_010 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_010.constraints: []
```

```
draft_instruction_id = go_run_cli
```

block_006 Run the bundled transcribe_diarize.py CLI

```
data_source_kind = "external"
```

```
block_006.constraints: []
```

ir_011 run_transcribe_diarize_cli

```
inputs[0] external_resource: scripts/transcribe_diarize.py | scripts/transcribe_diarize.py
```

```
inputs[1] result: result_008 | cli_arguments
```

```
outputs[0] result: result_009 | transcript_output
```

```
outputs[1] result: result_010 | saved_output_path
```

```
ir_011.constraints[0]: openai SDK must be installed; install with uv pip install openai if missing, or python3 -m pip install openai if uv is unavailable.
```

```
ir_011.constraints[1]: Max file size: 25 MB per request.
```

```
ir_011.constraints[2]: Supported input formats: mp3, mp4, mpeg, mpga, m4a, wav, webm.
```

```
draft_instruction_id = run_cli
```

metadata.script_content: 8684 字符; SHA256 d623b1fed1491b7ca4b7b5b393089165e3581a83d90d5a57868241f1b2a5d951; 计数与摘要依据: 原文 UTF-8。完整值保留在本次 analysis.json; 此处保留黑盒接口及全部数据引用。

ir_012 dispatch

```
inputs = []
```

```
outputs = []
```

```
ir_012.constraints: []
```

```
draft_instruction_id = go_validate
```

block_007 Validate transcription quality, speaker labels, and segment boundaries

```
data_source_kind = null
```

```
block_007.constraints: []
```

ir_013 validate_transcription_output

```
inputs[0] result: result_009 | transcript_output
```

```
inputs[1] result: result_004 | known_speaker_refs
```

```
outputs[0] result: result_011 | validation_outcome
```

```
ir_013.constraints[0]: Validate transcription quality, speaker labels, and segment boundaries.
```

```
ir_013.constraints[1]: Iterate with a single targeted change if needed.
```

```
draft_instruction_id = validate_output
```

ir_014 dispatch

```
inputs[0] result: result_011 | validation_outcome
```

```
outputs = []
```

```
ir_014.constraints: []
```

```
draft_instruction_id = dispatch_validation
```

block_008 Apply a single targeted change and re-run transcription

```
data_source_kind = null
```

```
block_008.constraints: []
```

ir_015 apply_single_targeted_change

```
inputs[0] result: result_011 | validation_outcome
```

```
outputs[0] result: result_012 | targeted_change
```

```
ir_015.constraints: []
```

```
draft_instruction_id = apply_change
```

ir_016 dispatch

```
inputs[0] result: result_011 | validation_outcome
```

```
outputs = []
```

```
ir_016.constraints: []
```

```
draft_instruction_id = go_prepare_args_again
```

block_009 Return the saved transcript path

```
data_source_kind = null
```

```
block_009.constraints: []
```

ir_017 return

```
inputs[0] result: result_010 | saved_output_path
```

```
outputs = []
```

```
ir_017.constraints: []
```

```
draft_instruction_id = return_saved_path
```